

Systematics and Biogeography of the Neotropical “Polymorphic Earless Praying Mantises” (Mantodea: Acanthopoidea)

by

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A thesis submitted in conformity with the requirements
for the degree of PhD in Ecology and Evolutionary Biology

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Abstract

The Mantodea, or praying mantises, are a lineage of cryptic predatory insects with a history of rampant evolutionary convergence in their cryptic strategies. Historical emphasis on multiply-evolved character states related to crypsis resulted in largely phenetic classification schemes. Although praying mantises are the subject of renewed interest in both basic and applied research, the lack of a phylogenetic framework encumbers research on these organisms. This dissertation uses phylogenetic methods to investigate relationships in a unique branch of the mantodean tree of life: the polymorphic earless praying mantises (PEPM), a Neotropical assemblage comprising ~55% of mantodean diversity in the region. The overarching goals of this dissertation are to: (a) construct a new higher-level classification of the PEPM; (b) establish the morphological limits of all supraspecific taxa, providing updated diagnoses for each one; and (c) reconstruct an evolutionary timeline of the PEPM. Using molecular phylogenetic analyses, relationships among higher-level taxonomic groups were resolved with high levels of nodal support. These analyses also revealed numerous inconsistencies with current classifications. Reconciling phylogeny and classification required eleven new synonymies at the species-group level, and a further four synonymys and 40 new combinations at the genus-group level. The PEPM, now assigned to the

superfamily Acanthopoidea, include 8 families, 15 subfamilies, 18 tribes, 63 genera (all morphologically redefined), and 315 valid species. Historical biogeographic analyses revealed that higher-level diversification occurred in mesic biomes throughout the Cretaceous Period, Crown diversification occurred during the Cenozoic (Paleogene/Neogene) Period, when punctuated evolutionary shifts to more seasonal, drier, biomes likely occurred in response to cyclical climatic fluctuations. The implications of this pattern of diversification, and potential drivers of mantodean diversity, are discussed.

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Chapter 1

General Introduction

Among the most charismatic, yet least understood, insects are the cryptic predators known as praying mantises. With approximately 2300–2500 species worldwide (Ehrmann 2002, Otte and Spearman 2005), praying mantises constitute the monophyletic order Mantodea, a member of the Polyneoptera — an ancient lineage of insects that includes the familiar grasshoppers, walking sticks, earwigs and their allies (Grimaldi and Engel 2005). Praying mantises are notable for their highly specialized raptorial forelegs used to capture and subdue living prey and also for their exuberant diversity of morphological and behavioural adaptations. These adaptations have enabled praying mantises to access arthropod prey in a great variety of ecological settings. For example, lichen mantises have cuticular outgrowths and colour patterns that render them invisible when resting motionless amongst lichen. Others, like grass mantises, have elongated bodies that match the outline of grass blades and conveniently express two colour morphs, green and light brown. Green leaves, dead leaves, flowers, sticks, and bark are among the plant structures used as hunting grounds, the evolutionary arena in which natural selection worked to shape their plastic and seemingly custom-built morphology or “ecomorph” (Svenson and Whiting 2009).

Praying mantises have been the subject of renewed interest after a long history of intermittent research (Rivera 2010, Rivera and Svenson 2014). Recent developments in their taxonomy and systematics, and their newly acquired popularity as biological models in ecological and evolutionary research (e.g., Allen *et al.* 2014, Barry 2013, 2015, Barry *et al.* 2008, Barry *et al.* 2015, Brown and Barry 2016, O'Hanlon 2014, O'Hanlon *et al.* 2014, O'Hanlon *et al.* 2015) have drawn an increased number of researchers to these fascinating organisms, including those engaged in applied research. For example, the protein used in egg case (ootheca) production has attracted attention for its potential application in the biomaterial industry (Kameda *et al.* 2014, Walker *et al.* 2012, Walker *et al.* 2015). A fundamental constraint of any research program is lack of taxonomic knowledge, which greatly limits interpretation of biological phenomena. Similarly, knowledge of clade-level relationships is critical for testing evolutionary hypotheses

of diversity patterns and biological processes within the Mantodea. Without a sound phylogenetic framework such fundamental lines of inquiry simply cannot be undertaken. Accordingly, the overarching goal of my dissertation is to contribute towards the expansion of Mantodea research by providing a sounder taxonomic- and phylogenetic framework from which to launch future studies.

This thesis is focused on one particular branch of the Mantodea tree of life: the polymorphic earless praying mantises (hereafter referred to as the PEPM). The PEPM are endemic to the Neotropical Region, where they represent ~55% of the total mantodean diversity in that region. The PEPM represent an early clade of the extant Mantodea, comprising the sister group of a much more diversified clade that evolved hearing adaptations — a lineage that includes virtually all other mantodean species worldwide. The PEPM include a disparate assemblage of lineages with a remarkably wide range of ecological and morphological adaptations. Consequently, the PEPM were not recognized as a natural group until molecular phylogenetic tools were implemented (Yager and Svenson 2008). The Mantodea include many other clades that exhibit a similar range of morphological disparity, and therefore, the PEPM provide a case study of how to disentangle the taxonomic- and phylogenetic problems associated with rampant phenotypic plasticity — a problem that continues to obscure the mantodean tree of life.

This dissertation consists of three principal components:

In Chapter 2, I implement phylogenetic methods to explore evolutionary relationships within the PEPM. The Mantodea were not the subject of formal phylogenetic inquiry until the turn of the 21st Century, considerable later than most Polyneoptera and other insect orders. The milestone molecular phylogenetic study of Svenson and Whiting (2009) showed that rampant evolutionary convergence existed throughout the order. One historical consequence was misinterpretation of relationships based on overall morphological similarity of otherwise unrelated taxa, resulting in largely phenetic and biologically misleading classifications, which persist today. Molecular phylogenetic methods provide an effective means to overcome the limitations imposed by the highly homoplastic nature of mantodean ecomorphs, and is therefore the underlying methodological approach used in this investigation. The study of Svenson and Whiting (2009) also revealed that a molecular phylogenetic approach will require a sea change in Mantodea

systematics, where para- and even polyphyletic taxonomic groups are currently commonplace. Accordingly, this chapter sets the stage for a reevaluation of higher-level classification to reflect new insights about evolutionary relationships within this fascinating branch of the Mantodea tree of life.

A fundamental aspect of biodiversity research is the construction of natural classifications, which serve as an information storage and retrieval system. The molecular phylogeny of Svenson and Whiting (2009) revealed multiple inconsistencies in current higher-level classifications of praying mantises. Solving nomenclatural challenges created by such inconsistencies requires a reappraisal of the morphological definition of not only all higher-level taxa, but also of the 60 or so genera of PEPM so far recognized. Accordingly, establishing revised morphological criteria that incorporate novel character systems (i.e. male genitalia and egg cases) into the diagnosis of recognized genera is the main goal of Chapter 3.

Chapter 4 concerns one of the most distinct features of Mantodea — their uncanny morphological plasticity and the parallel evolution of ecomorphs. Determining an evolutionary timeline of PEPM could provide important new insights about potential drivers of this ubiquitous pattern, which occurs repeatedly throughout the Mantodea. Towards this end, this final data chapter brings together the phylogenetic and biological information generated in Chapters 2 and 3 to investigate patterns of historical biogeography of PEPM throughout the Neotropical Region.

In summary, the primary objectives of this research are to: (a) use a molecular phylogenetic approach to investigate evolutionary relationships among the PEPM, and to construct a more natural higher-level classification; (b) revise the morphological definition of all supraspecific taxa within the PEPM providing new diagnoses and keys; and (c) reconstruct an evolutionary timeline of the PEPM to investigate potential drivers of the mode of diversification within the Neotropical Region.

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Chapter 2

The Neotropical ‘polymorphic earless praying mantises’ – molecular phylogeny and revised higher-level systematics (Insecta: Mantodea, Acanthopoidea)

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2.1 Summary

I perform phylogenetic analyses of the ‘polymorphic earless praying mantises’, a heterogeneous assemblage comprising *c.* 55% of mantodean diversity in the Neotropics. Bayesian and maximum-likelihood were implemented on a DNA dataset of 9949 aligned nucleic acid characters comprising ten mitochondrial and nuclear genes. My analyses largely resolved congruent relationships with high levels of support for higher-level taxonomic groups, but revealed extensive inconsistencies between the resolved topology and morphology-based classification systems. The polymorphic earless praying mantises, or superfamily Acanthopoidea, comprises 8 families, 15 subfamilies and 18 tribes. My revised organization is as follows: (i) Thespidae, including subfamilies Pseudopogonogastrinae, Pseudomiopteryginae, Bantiinae, Miobantiinae and Thespinae (tribes Musoniellini and Thespini); (ii) Angelidae; (iii) Coptopterygidae; (iv) Liturgusidae; (v) Photinaidae, including Macromantinae, Cardiopterinae, Photiomantinae and Photinainae (tribes Microphotinini, Orthoderellini and Photinaini); (vi) Stenophyllidae; (vii) Acontistidae; and (viii) Acanthopidae. This system also resulted in the reassignment of various genera to new and existing higher-level taxa, the exclusion of old world genera otherwise traditionally classified among the Thespidae, Liturgusidae and Angelidae, the confirmation of *Stenophylla* Westwood as member of this clade, and the revalidation of *Paradiabantia* Piza. I provide diagnoses for all suprageneric taxa using external morphological characters and male genitalia. A key to higher-level groups is provided. I incorporate egg case structural variation as a novel approach for taxon delineation.

2.2 Introduction

Praying mantises (Order Mantodea) are predatory, polyneopteran insects distributed in tropical and subtropical habitats across the globe, from the rainforest canopy to the desert ground.

Together with cockroaches and termites, they constitute the monophyletic superorder Dictyoptera (e.g. Hennig, 1981; Klass, 1997; Klass and Meier, 2006; Ware *et al.*, 2008; Misof *et al.*, 2014; Legendre *et al.*, 2015). Mantodea is a meso-diverse order with c. 2300–2500 species described, according to most recent estimates (Ehrmann, 2002; Otte and Spearman, 2005).

Praying mantises exhibit a remarkable range of behavioural and morphological adaptations associated with their habitat-specific predatory lifestyle. Many mantises use strategies including mimicry or special aggressive resemblance for avoiding their own natural enemies while securing their prey (Kumar, 1973; Edmunds and Brunner, 1999). Besides the familiar green leaf-mimicking praying mantises often depicted in popular media, such as the widespread *Mantis religiosa* (Linnaeus, 1758), a great variety of lesser-known forms exist. For example, the orchid mantis, *Hymenopus coronatus* (Olivier, 1792) (Hymenopodidae), from the tropical rainforests of Southeast Asia have features and colouration to resemble large petalled flowers. Many species of praying mantis also mimic dead leaves, sticks, bark, lichen, moss, pebbles and blades of grass, and some even ants or wasps (e.g. Jackson and Drummond, 1974; Milledge, 1990; Salazar, 2003; Agudelo and Rafael, 2014). These cryptic adaptations and the rarity of most praying mantis species make them difficult to observe in nature as well as scarce in museum collections. Coupled with a historical dearth of praying mantis specialists, our understanding of their ecological significance, evolutionary relationships and geographic patterns of global diversity has been inhibited (Hurd, 1999; Rivera, 2010c; Rivera and Svenson, 2014).

The monophyly of Mantodea has been well-established through extensive research on Dictyoptera (e.g. Hennig, 1981; Kristensen, 1995; Klass, 1997; Lo *et al.*, 2000; Deitz *et al.*, 2003; Klass and Meier, 2006; Inward *et al.*, 2007; Ware *et al.*, 2008; Klass *et al.*, 2009; Djernæs *et al.*, 2012; Legendre *et al.*, 2015). However, relationships within Mantodea are not thoroughly understood, because finite phylogenetic studies are rare (Svenson *et al.*, 2015, 2016). Numerous classifications have been proposed (e.g. Audinet Serville, 1831; Burmeister, 1838; De Haan, 1842; Saussure, 1869, 1870; Stål, 1873, 1877; Kirby, 1904; Giglio-Tos, 1919, 1927; Handlirsch, 1925; Chopard, 1949; Beier, 1964, 1968), but none are framed in an explicit phylogenetic

framework. The complexity and plasticity of praying mantis external morphology have been proposed as major obstacles for recognizing natural groups (Svenson and Whiting, 2009; Wieland, 2013). Consequently, all previous classification schemes – including the most recent and widely used ones by Ehrmann (2002) and Otte and Spearman (2005) – remain phenetic in their formulation.

The first phylogenetic test of mantodean relationships revealed that many of the currently recognized higher-level groups are not monophyletic (Svenson and Whiting, 2004). Although the primary aim of this study was to explore the evolution of hunting strategies, it did raise numerous concerns about the organization of the classification. Svenson and Whiting (2009) considerably expanded their previous dataset by including nine genes and a more thorough taxon sampling across the order. Monophyly was recovered for fewer than half of all recognized mantodean higher-level taxa, whereas the remaining ones were either para- or polyphyletic. They concluded that incongruence between phylogeny and classification was due to the profusion of homoplastic characters. They further hypothesized that completely unrelated lineages occupying similar habitats – and using similar hunting and cryptic strategies in geographically distant locations – were subject to adaptive convergent evolution. More specifically, comparable selective pressures are exerted for both predator avoidance and maximizing hunting efficiency, thus resulting in convergently evolved ‘ecomorphs’. Many taxonomic characters associated with such ecomorphs were found to be convergent, further obscuring natural affinities among mantodean lineages. For instance, members of the family Liturgusidae, otherwise known as ‘bark mantises’, represent a rather homogeneous assemblage of taxa that hunt on the vertical surfaces of tree trunks (Svenson, 2014b). All liturgusid taxa share similar cryptic and disruptive colouration, flattened bodies, agile displacement, and several other behavioural and morphological traits. Although always classified together (Beier, 1935b, 1964, 1968; Terra, 1995; Ehrmann, 1997, 2002; Otte and Spearman, 2005), Svenson and Whiting (2009) demonstrated conclusively that Liturgusidae is polyphyletic and the bark-dwelling ecomorph evolved multiple times. This pattern is repeated in other higher-level taxa, corroborating earlier concerns about the limitations of praying mantis morphology for systematic inference (e.g. Roy, 1987, 1999; Ehrmann, 2002).

Svenson and Whiting (2009) recovered a large assemblage of Neotropical Mantodea near the base of the phylogeny. This lineage comprises numerous higher-level taxa, including a number of paraphyletic lineages that reflect the overall pattern of nonmonophyly in convergent groups distributed over two or more continents. Neotropical members of the globally distributed Liturgusidae, Thespidae and Angelinae (Mantidae), as well as the exclusively Neotropical lineages Photinae (Mantidae) and Acanthopidae, were included in this clade (Table 2.1). Based on the shared characteristic of deafness (Yager and Svenson, 2008), Svenson and Whiting (2009) dubbed the clade ‘earless mantises’ – representing *c.* 55% of total mantodean diversity in the region (Agudelo *et al.*, 2007). The discovery of the earless mantis clade using molecular tools was significant because members of this lineage exhibit extensive, and often extreme, morphological and ecological variation (Figs 2.1–2.3), thus precluding their recognition as members of a single lineage, even within a phylogenetic context (Wieland, 2013). Although this feature was found to be a symplesiomorphy (Yager and Svenson, 2008), the lack of a functional ear across this entire clade is a compelling characteristic considering the context of its evolution.

As important elements of the total Neotropical praying mantis fauna, their extensive ecomorphological diversity and largely inadequate systematic treatment, earless mantises constitute an ideal case-study for formulating a methodological approach to resolving other equally mistreated lineages within the mantodean tree of life. Therefore, the goal of this study is to contribute towards the long-needed reformulation of the higher-level classification system of the Mantodea. Specifically, I aim to: (i) expand both taxon and molecular sampling to build a robust phylogeny for the ‘earless praying mantises’; (ii) revise the higher-level classification system for the group; and (iii) establish new morphology-based criteria for taxon delimitation and diagnosis for members of the clade. In addition, I define new character systems derived from male genitalia and egg case morphology that are capable of diagnosing natural groups.

2.3 Material and methods

2.3.1 Taxon sampling and taxonomic identification

I expanded on the taxon sampling of Svenson and Whiting (2009) to include at least one representative of all recognized higher-level taxa of ‘*polymorphic earless praying mantises*’ (PEPM), exclusive of Chaeteessidae and Mantoididae. Based on the external anatomy of the metasternum (the physical location of the cyclopean ear), Yager and Svenson (2008) identified four anatomical patterns associated with the functionality of the auditory system across Mantodea: the DK morphology (deep groove and prominent knobs), its two less common and sexually dimorphic derivatives (DO type – deep groove opened, and DNK type – deep groove without knobs) and the MSMT morphology (‘mesothoracic and metathoracic segments similar’). The MSMT type does not indicate a particular morphological pattern, but rather the lack of external evidence of auditory capabilities and thus deafness. Therefore, I restricted my sampling to PEPM taxa according to the morphological, phylogenetic and biogeographic interpretations of Yager and Svenson (2008) and Svenson and Whiting (2009) under the following two criteria: (i) taxa that lack metasternal anatomy associated with hearing capabilities (i.e. MSMT), and (ii) taxa that are distributed in the New World. Taxa currently classified among Liturgusidae, Thespidae and Angelinae (Mantidae), but phylogenetically nested within Old World lineages in Svenson and Whiting (2009), were not considered as ingroup taxa in this study (see Table 2.2). I used the classification schemes of Ehrmann (2002) and Otte and Spearman (2005) – which incorporate elements from Terra (1995) with regards to Thespidae – as working hypotheses for my phylogenetic reconstruction (Table 2.1). Non-Mantodea outgroup taxa include four species of cockroaches (including *Cryptocercus* Scudder) and two species of termites (including *Mastotermes* Froggatt). I also included *Chaeteessa* Burmeister (Chateessidae, two species) and *Mantoida* Newman (Mantoididae, four species) given their position as sister of all other Mantodea (Klass, 1997; Grimaldi, 2003; Klass and Meier, 2006; Lo *et al.*, 2000; Svenson and Whiting, 2004; Wieland, 2006; Ware *et al.*, 2008; Djernæs *et al.*, 2012). My sampling included 135 terminal taxa, encompassing 84 species distributed in 45 of the 64 genera of PEPM (i.e. 75% of generic representation). Sampled taxa were identified to at least genus level. I included undescribed, yet relevant, taxa among my study material; these are labelled as such in the phylogeny and Appendix 1. For taxonomic identification I used the available literature on

Neotropical Mantodea listed in Rivera (2010b), in addition to more recently published contributions (Rivera *et al.*, 2011; Roy and Stiewe, 2011; Lombardo *et al.*, 2013; Menezes and Bravo, 2013; Agudelo, 2014; Agudelo and Rafael, 2014; Rivera, 2014; Scherrer, 2014; Svenson, 2014b; Agudelo and Rivera, 2015; François and Roy, 2015; Roy, 2015). I studied type specimens when literature-based identification was inconclusive. Collections where type specimens were examined included the Natural History Museum (London, U.K.), Museu de Zoologia, Universidade de São Paulo (São Paulo, Brazil), Museu de Ciências e Tecnologia, Pontifícia Universidade Católica do Rio Grande do Sul (Porto Alegre, Brazil), National Museum of Natural History, Smithsonian Institution (Washington, DC, U.S.A.), Muséum national d'Histoire naturelle (Paris, France), Muséum d'histoire naturelle de la Ville de Genève (Geneva, Switzerland), Museum für Naturkunde der Humboldt Universität (Berlin, Germany), Zoologische Staatssammlung München (Munich, Germany), Zoologischen Museums Greifswald (Greifswald, Germany), Oxford University Museum of Natural History (Oxford, U.K.), Museo Nacional de Historia Natural (Madrid, Spain), Museo Nacional de Historia Natural, Universidad Nacional Mayor de San Marcos (Lima, Peru), California Academy of Sciences (San Francisco, U.S.A.), Naturhistorisches Museum Wien (Wien, Austria), Biozentrum Grindel und Zoologisches Museum, Hamburg (Hamburg, Germany), Museo de Entomología Klaus Raven Büller, Universidad Nacional Agraria La Molina (Lima, Peru), Academy of Natural Sciences of Drexel University (Philadelphia, U.S.A.), Muzeum i Instytut Zoologii Polskiej Akademii Nauk (Warsaw, Poland), and Escola Superior de Agricultura Luiz de Queiroz, Universidade de São (Piracicaba, Brazil).

2.3.2 Data acquisition, processing and phylogenetic analysis

Tissues for molecular analyses were obtained from the ethanol-preserved Mantodea collection at the Cleveland Museum of Natural History (CMNH). A small incision was made in the pleural region of the thorax to remove fresh wing muscle for DNA extraction. In some instances, additional tissue was obtained from the prothoracic coxae and femora. Except in rare cases where an entire specimen was used for DNA extraction, such as early instar nymphs, all vouchers were preserved in 95% ethanol and maintained at -80°C . I targeted the same gene

complement as Svenson and Whiting (2004, 2009): mitochondrial ribosomal 12S rRNA and 16S rRNA; mitochondrial protein-coding Cytochrome Oxidase I (*COI*), Cytochrome Oxidase II (*COII*) and NADH dehydrogenase subunit 4 (*ND4*); nuclear protein-coding Histone 3 (*H3*) and Wingless (*wnt-1*); nuclear ribosomal genes 18S rRNA and 28S rRNA. PCR protocols and primers were taken from Whiting (2002), and Svenson and Whiting (2004, 2009). In addition, I optimized one other gene, the nuclear protein-coding Histone 2A (*H2A*), thus expanding the final molecular dataset to ten genes. *H2A* was amplified with H2AF1 forward primer (5'-TGT CTG GYC GCG GCA ARG G-3') and H2AR1 reverse primer (5'-ACG GCC TGG ATG TTG GGC A-3') with a thermocycler protocol as follows: 94(45 s), 48(45 s), 72(1m) for 40 cycles. Excess genomic extract was preserved in labelled cryotube vials and kept at -80°C for long-term storage at CMNH. Amplicons for all loci were purified via Illustra ExoProStar (GE Healthcare Life Sciences™, Pittsburgh, PA, U.S.A.) protocol and submitted to the High Throughput Genomics Center at the University of Washington for sequencing on an ABI 3730XL. Sequence data were imported into Geneious v7.1.4 (<http://www.geneious.com>, Kearse *et al.*, 2012) for nucleotide editing and contig assembly. Gene regions were sequenced with complements and sufficient overlap with adjacent regions to ensure the accuracy of sequence data (see Appendix S1 for Genbank accession numbers). Sequences from each locus were aligned with MAFFT v7 (Katoh *et al.*, 2002; Katoh and Standley, 2013), except ribosomal genes which were aligned in Geneious using MUSCLE (Edgar, 2004), an iterative algorithm that has been shown to outperform other methods (Wilm *et al.*, 2006). Mesquite v3.01 (Maddison and Maddison, 2014) was used to delimiting codon positions, delete introns and concatenate alignments. The resulting aligned matrix included 9949 characters. For phylogenetic analyses, the concatenated dataset was divided into ten partitions that corresponded to the ten included gene loci. I used RaxML v8 (Stamatakis, 2014) to find the maximum-likelihood (ML) topology. Nucleotide substitution parameters were estimated independently from each data partition. One thousand nonparametric bootstrap (BV) pseudoreplicates were performed under a GTR model with CAT approximation of Gamma-distributed among-site rate heterogeneity. Every fifth bootstrapped tree was used as a starting tree for more thorough optimization of the real data under GTR+Gamma. I also conducted a mixed model Bayesian analysis (BI) in MrBayes v3.2.1 (Huelsenbeck and Ronquist, 2001; Ronquist and Huelsenbeck, 2003; Altekar *et al.*, 2004) with each of two runs started from a random tree and subsequently monitored for convergence using the program Tracer v1.3

(Rambaut and Drummond, 2003). All sampled generations (every 1000) prior to stationary were discarded (burn-in). The trees sampled from the stationary distribution were summarized as a 50% majority rule consensus tree to find posterior probabilities (PP; Huelsenbeck and Imennov, 2002; Huelsenbeck *et al.*, 2002). The program jModelTest (Darriba *et al.*, 2012) selected GTR+G+I best-fit model of nucleotide substitution for all ten gene partitions.

2.3.3 Examined morphology

I examined all 64 PEPM genera for morphological diagnoses of all suprageneric taxa, including those not available for molecular analyses but nonetheless relevant to classification. I rely on the extensive morphological phylogenetic treatment of Wieland (2013) in addition to Beier (1968) as a main source for labelling external, anatomical structures, whereas I follow La Greca (1955) for male genital structures. To visualize genital parts, I removed the last two or three abdominal segments and placed them into separate vials containing hot 10% KOH solution until all remains of soft tissue were dissolved. Time in KOH was estimated and adjusted by eye according to the specimen size. After clearing, genitalic parts were first rinsed in distilled water, then 95% ethanol, and finally stored along with the original specimen inside a glass microvial containing glycerin. All relevant morphological features were illustrated using a camera lucida mounted on a dissecting microscope with male genitalia placed in a drop of glycerin in a depression slide but left uncovered in order to retain natural shape. Egg case morphology was incorporated as a complementary character system for taxon diagnosis. Terms for egg case structures follow Breland and Dobson (1947), the only authoritative and integrative account about the significance of egg cases for mantodean taxonomy. However, this study scarcely captured the full range of structural diversity and cryptic adaptations in mantodean egg cases, which prompted me to introduce additional terms that are useful for facilitating taxon diagnosis. With few exceptions, egg case samples were originally associated with their corresponding taxon by: (i) rearing individuals hatched from wild-caught egg cases; (ii) obtaining egg cases directly from wild-caught gravid females, and/or; (iii) collecting both female and egg case together in the field (rarely). Most pinned praying mantis specimens used for morphological diagnoses are deposited at CMNH. Egg cases are housed largely in the Museum of Entomology *Klaus Raven Büller*, Universidad Nacional Agraria La Molina (Lima, Peru), with additional samples examined at the

Museu de Zoologia, Universidade de São Paulo (São Paulo, Brazil). Donations from praying mantis breeders and hobbyists helped to capture the extent of natural variation in egg cases.

2.3.4 Revision

Phenotypic diagnosis is fundamental for taxon delineation and recognition by end-users (Wiley, 1979; Wiley and Lieberman, 2011). I revised the classification of the PEPM in view of relationships revealed by the molecular phylogeny. Three principles were followed in revising the classification: (i) preference for historical family group names (Roy, 2014) and ranks when possible to preserve taxonomic stability; (ii) assignments of formal names only to monophyletic groups (Hennig, 1966); and (iii) strict adherence to the International Code of Zoological Nomenclature (ICZN, 1999). Family-group ranks were assigned based on relative phylogenetic position to neighboring groups in order to minimize the number of ranks and name changes in my classification. Each diagnosis is composed of a description of morphological traits that apply to all constituent species it contains. All higher-level taxa are defined using a combination of external morphological characters and features of the male genitalia, the latter used only in those cases where external morphology was ambiguous. A new key to higher-level taxa was produced using these morphological characters.

2.4 Results

The two phylogenies (ML and BI) are largely congruent and generally well-supported. I present the ML topology and highlight only incongruences with the BI analysis (Fig. 2.4). The ML analysis recovered a topology (likelihood score: -165295.11) with both low (60–80) and high (≥ 90) BS values across deep backbone nodes. Major higher-level clades representing evident morphological discontinuities, and most nodes (80.5%) subtending such clades had all high BS values (> 90). The Bayesian analysis (mean likelihood across runs: -163705.59) had high posterior probabilities (PP > 90) across all backbone, major higher-level clades, and in 96% of nodes subtending the latter. Chaeteessidae and Mantoididae were recovered at the base of the

tree with high values (PP=1; BS=100). PEPM were recovered as monophyletic (PP=1; BS=100) and subdivided into two clades. Clade 1 comprises the Neotropical Thespidae (PP=1; BS=98) and Clade 2 includes all other PEPM including Acanthopidae, Neotropical Liturgusidae, and the Mantidae subfamilies Photinainae (Coptopterygini and Photinaini) and Neotropical Angelinae (genus *Angela* Audinet Serville) (PP=1; BS=66).

Within the Thespidae, *Pseudopogonogaster kanjaris* Rivera and Yagui, 2011 was recovered as sister to all other Thespidae (PP=1; BS=95). Subsequent, deeper dichotomies defining distinct sub-clades were all resolved with similar high support values (PP $\geq$ 0.99; BS $\geq$ 95). None of the putative subfamilies suggested by Terra (1995); Ehrmann (2002) and Otte and Spearman (2005) including Thespiinae, Oligonicinae, Pseudomiopteryginae and Miopteryginae (now Miobantiinae *sensu* Roy, 2013) were recovered as monophyletic in Clade 1. Among the Thespidae, *Bantia* Stål, *Musoniella* Giglio-Tos, *Oligonyx* Saussure and *Carrikerella* Hebard were found to be nonmonophyletic.

The topology of Clade 2 (PP=1; BS=100) is largely congruent in both the ML and BI analyses. Clade 2 exhibited shallow backbone nodes, some with relatively low BS (>60) but all with high PP (<0.90). Clade 2 contains six nodes subtending seven main lineages (coloured in Fig. 2.4). All lineages (Angelinae, Coptopterygini, Liturgusidae, Photinaini, Aconstistinae and Acanthopinae) exhibited high PP (1) and BS (98–100). The Photinaini represents the only topological incongruence between the ML and BI analyses (Fig. 2.4, BI topology in lower right inset). Both ML and BI analyses resolved *Macromantis* Saussure as sister to all other Photinaini (PP=1; BS=96), whereas the remaining genera formed five distinct subgroups in both analyses: (i) *Photiomantis*; (ii) *Cardioptera* Burmeister; (iii) *Microphotina* Beier, *Chromatophotina* Rivera; (iv) *Orthoderella* Giglio-Tos, and *Paraphotina* Giglio-Tos; (v) *Metriomantis* Saussure and Zehntner, *Photina* Burmeister, *Photinella* Giglio-Tos and *Hicetia* Saussure and Zehntner. The backbone nodes resolving the positions of these five subgroups in the ML topology have low BS values (<50) and very short branch lengths. In the BI topology, the same clade recovering these five subgroups exhibited high PP values (>0.80) with longer branch lengths. Despite these topological differences, internal clades corresponding to the five subgroups are recovered with high PP and BS in both analyses (Fig. 2.4). The Photinaini was also confirmed as paraphyletic relative to Coptopterygini (Svenson and Whiting, 2009), thus at odds with their

assumed monophyly (Terra, 1995; Ehrmann, 2002; Otte and Spearman, 2005). In addition, *Microphotina* is paraphyletic relative to Microphotinini gen.n. sp.n., *Orthoderella* is paraphyletic relative to *Paraphotina*, and *Metriomantis* is paraphyletic relative to *Photinella* and *Photina*. The clade corresponding to the Acanthopidae (PP=1; BS=90) is subdivided into three lineages: Stenophyllinae, Acontistinae and Acanthopinae, which is congruent with current classification (Terra, 1995; Ehrmann, 2002; Otte and Spearman, 2005). *Stenophylla* Westwood was recovered as sister to Acontistinae and Acanthopinae (PP=1; BS=100). The Acontistinae+Acanthopinae (both with PP=1; BS=100) were recovered as monophyletic (PP=1; BS=100). Within the Acontistinae, *Acontista* Saussure is nonmonophyletic, as *Acontista bimaculata* Saussure, 1870 was sister to *Raptrix* Terra in both analyses (PP=1; BS=100). *Callibia* Stål is sister to the remaining species of *Acontista*, but values for this sister-group relationship were relatively low (PP=0.81; BS<50). Acanthopinae was recovered as monophyletic, congruent with Svenson and Whiting (2009). *Acanthops erosula* Stål, 1877 was recovered as sister to the clade *Pseudacanthops* Saussure+remaining *Acanthops* Audinet Serville (PP=1; BS=100), thus rendering the latter genus as nonmonophyletic.

2.5 Discussion

2.5.1 Discordances between molecular phylogeny and morphology-based classifications

Thespidae was considered as a distinct lineage early on by Saussure (1892), and subsequently treated as a family group of various ranks (Giglio-Tos, 1919, 1927; Beier, 1935a). Chopard (1949) conferred this lineage with family status, and subsequent contributions assembled Thespidae classification to its current form (Beier, 1964, 1968). The current concept of Thespidae includes the Oriental Haaninae, the African Hoplocoryphinae, and the Neotropical Pseudomiopteryginae, Thespiinae, Oligonicinae and Miobantiinae (formerly Miopteryginae, emended by Roy (2013)). Terra (1995) provided a classification scheme for the Neotropical Thespidae, which formed the foundation of subsequently proposed systems by Ehrmann (2002) and Otte and Spearman (2005) (Table 2.1). Terra (1995) used a combination of characters to define the four Neotropical subfamilies, but none was identified as a definite synapomorphy.

This is reflected in his key to genera (Terra, 1995), where couplets leading to each subfamily include no definitive characters but instead consist of combinations of ‘exceptions’. My genetic data are clearly at odds with Terra’s classification, in that none of the subfamilies he proposed were recovered as monophyletic (Table 2.1). Terra tried to capture the variation he observed to provide diagnoses for each subfamily, while underestimating the great variability of external morphology. My own examination of specimens in light of my phylogeny revealed that the characters that Terra considered relevant are actually all markedly variable (e.g. degree of development of the vertex and its associated juxtaocular bulges, number of spines in the raptorial forelegs, shape of the supracoxal dilation, and chromatic features of male wings, among others). This is true even among sister genera, thus rendering the character system of Terra inadequate for diagnosing higher-level taxa in Thespidae.

Using only male external characters, Terra (1995) classified *Calopteromantis* Terra, 1982 (now a junior synonym of *Pseudopogonogaster* Beier; see Rivera *et al.* (2011)) within the Miopteryginae (i.e. Miobantiinae), along with *Miobantia* Giglio-Tos, *Chloromiopteryx* Giglio-Tos and other genera (Table 2.1); Ehrmann (2002) and Otte and Spearman (2005) and more recently Rivera *et al.* (2011) followed this arrangement. My analyses recovered *Pseudopogonogaster* as sister to all other Thespidae, suggesting instead the Miobantiinae as nonmonophyletic. Moreover, *Pseudomiopteryx* Saussure did not show a close relationship to any of the other members of Pseudomiopteryginae *sensu* Terra (1995) here tested, such as *Anamiopteryx* Giglio-Tos, *Eumiopteryx* Giglio-Tos and *Leptomiopteryx* Chopard (*Pizaia* Terra was not included in my analyses) (Table 2.1). Taxa such as *Pseudomiopteryx*, *Anamiopteryx* and *Eumiopteryx* all exhibit various pronotal tubercles and other cuticular outgrowths for cryptic purposes, features that Terra (1995) presumably assumed to be synapomorphic for these three genera altogether; however, my phylogenetic analyses suggest that this is not the case. Instead, members of Pseudomiopteryginae are scattered throughout the phylogeny, where *Anamiopteryx* is sister to *Eumiopteryx* and together sister to *Miobantia*, but neither is sister to *Pseudomiopteryx* nor *Leptomiopteryx*.

I also found evidence for the rejection of the subfamily Oligonicinae. Giglio-Tos (1919) defined this taxon on the basis of apico-dorsolateral tibial spines, a unique feature among the Mantodea. According to Wieland (2013) these spines resulted from either one or two of the distal-most

anteroventral spines shifting to a more dorsolateral position. My observations of *Pseudomiopteryx* and *Macromusonia* Hebard revealed the presence of these apico-dorsolateral tibial spines in very early instars (Fig. 2.5), but not in adults. This contrasts with the condition observed in other taxa, such as *Thesprotia* Stål and *Thesprotiella* Giglio-Tos, where these are present in both juveniles and adults (J. Rivera, personal observation). This suggests that, at least in some thespids, the dorsolateral positioning of tibial spines is lost during post-embryonic development. Genera such as *Bantia*, *Thrinaconyx* Saussure and *Mantillica* Westwood bear these apico-dorsolateral spines, but form a clade sister to *Pseudomiopteryx*. The remaining sampled genera of Oligonicinae, such as *Oligonyx*, *Oligonicella* Giglio-Tos, *Thesprotia* Stål, *Thesprotiella*, *Carrikerella* and *Musoniola* Giglio-Tos form a well-defined clade along with few other taxa classified in Thespinae *sensu* Terra (1995) (i.e. *Macromusonia*, *Musonia* Saussure, *Galapagia* Scudder). Clearly, the loss of juvenile features in the configuration of tibial spines of adults occurred at various grades among the Neotropical Thespidae. The occurrence of such apico-dorsolateral spines in the unrelated Haaniinae (see figs 253–254 in Wieland, 2013, for comparison with Neotropical Thespidae) also contradicts the single origin hypothesis proposed by Roy (1999). The elimination of the subfamily Oligonicinae (along with its two subordinated tribes, the Oligonicini and Pogonogasterini), coupled with an expanded concept of Thespinae, is therefore necessary to solve the nomenclatural incongruences revealed in this study.

The bark-dwelling Liturgusidae is unquestionably polyphyletic in its current nomenclatorial sense (Yager and Svenson, 2008; Svenson and Whiting, 2009). Historically, a total of 13 non-Neotropical genera has been assigned to the family, mostly on the basis of their remarkable superficial similarity (see Table 2.2). Given the current evidence, the monophyly of its Neotropical members seems conclusive. Nomenclatorial concepts of Mantidae have changed considerably through their taxonomic history (Roy, 2014). As currently conceived, Mantidae includes 20 subfamilies exhibiting extensive morphological and ecological variation. Accordingly, Mantidae cannot be considered a natural group (Svenson and Whiting, 2009; Rivera, 2010c), because of its historical role as a ‘taxonomic dumping ground’ for the Mantodea (Svenson and Whiting, 2009). Several former subfamilies were elevated to family rank in recent years, such as Tarachodidae, Iridopterygidae, Toxoderidae and Sibyllidae (Ehrmann, 2002). However, these actions were not couched in an explicit phylogenetic framework (Wieland, 2013). Members of the PEPM currently assigned to Mantidae include the Angelinae and

Photinae. Based on current evidence, retention of these two taxa within Mantidae is no longer justified.

The Angelinae includes only a single Neotropical genus, *Angela*, and seven other genera distributed in the Ethiopian, Australian and/or Oriental regions (Table 2.2). Among these latter, the Oriental *Euchomenella* Giglio-Tos, *Indomenella* Roy, and the Ethiopian *Leptocola* Gerstaecker and *Stenopyga* Karsch, were shown to have no close relationship with *Angela* (Yager and Svenson, 2008; Svenson and Whiting, 2009) as previously supposed (Ehrmann, 2002; Otte and Spearman, 2005). In fact, Yager and Svenson (2008) recovered *Angela* within the earless mantises, a relationship herein confirmed. The hypothesis of Terra (1995), where *Angela* is member of the family Thespidae is not supported by my analysis. The remaining Old World taxa (*Agrionopsis* Werner, *Cotigaonopsis* Vyjayandi and *Tagalomantis* Hebard) require removal from Angelinae (Table 2.2). Consequently, the name Angelinae is now restricted only to *Angela*. Previous analyses (Yager and Svenson, 2008; Svenson and Whiting, 2009) suggested that earlier concepts of the Photinae were paraphyletic. This finding is supported with inclusion of all genera currently assigned to this exclusively Neotropical lineage. The Photinae includes two tribes, the Coptopterygini and the Photinini. The Coptopterygini consists of *Coptopteryx* Saussure and *Brunneria* Saussure, which together have long been recognized as a natural group. Giglio-Tos (1915a) considered the ‘Coptopteriges’ to be related to the ‘Photinae’ and ‘Orthodorellae’ (i.e. Photinini of most authors). This assumption was later followed in part by Beier (1968), who recognized two tribes within the Photinae: Photinini and Coptopterygini. However, he expanded the concept of Coptopterygini to include taxa otherwise attributed to the Photinini, such as *Cardioptera*, *Orthoderella* and *Paraphotina*. Terra (1995) followed a more conservative approach and, perhaps influenced by an even older arrangement (Beier, 1935b), omitted tribal distinctions for the Photinae. Ehrmann (2002) and Otte and Spearman (2005) presented a similar arrangement, with Coptopterygini including only *Coptopteryx* and *Brunneria*, whereas the other genera were included in Photinini. Based on my results, Coptopterygini constitutes a monophyletic lineage distinct and separate from Photinini (Fig. 2.4). The Acanthopidae includes three subfamilies: Acanthopinae, Acontistinae and Stenophyllinae. In its current concept, Acanthopidae is arguably the most polymorphic family of Neotropical praying mantises. My analyses resolved all three groups as monophyletic. However, considering the morphological diversity among these three lineages and the lack of definite autopomorphies

(Wieland, 2013), including these taxa as members of a single family is suspect. The concept of Acanthopidae has gone through numerous modifications since it was proposed by Burmeister (1838) – originally cited as Acanthopsidae and later emended to Acanthopidae by Travassos Filho (1945). The group initially included what is now recognized as the Acanthopinae, but was later expanded to include the Acontistinae. Because of their superficial similarities with members of Epaphroditinae, Acromantinae and Hymenopodinae, all within the Hymenopodidae – an assemblage of Old World plant-mimicking mantises (Svenson *et al.*, 2015) – representatives of Acanthopinae and Acontistinae have been historically assigned to this family (e.g. Giglio-Tos, 1927; Beier, 1934, 1968), or considered subfamilies on their own within the Acanthopidae (Travassos Filho, 1945; Terra, 1995). Terra (1995) expanded the concept of Acanthopidae to include also the monotypic subfamily Stenophyllinae. The Acanthopinae has received considerable attention by taxonomists in recent years (La Greca and Lombardo, 1997; Lombardo, 2000b; Roy, 2002a,c, 2004; Lombardo and Ippolito, 2004; Rivera, 2005; Ippolito, 2007; Lombardo *et al.*, 2013). However, little is known about its affinities. Svenson and Whiting (2009) included exemplars of *Acanthops* and *Metilia* Stål in their study, recovering both genera as reciprocally paraphyletic. My expanded analyses and reassessment of original voucher specimens revealed that only *Acanthops* is paraphyletic. Apparent paraphyly in *Metilia* was due to misidentification of what was identified as *Metilia amazonica* (Beier, 1930) by Svenson and Whiting (2009), called ‘Acanthopidae gen.n. sp.n.’ in this study. Similarly, *Acanthops* sp. in Svenson and Whiting (2009) actually corresponds with an unidentified species of *Decimiana* Uvarov (*Decimiana* sp. in my study). Lombardo and Ippolito (2004) performed a morphology-based phylogenetic analysis of *Acanthops*, resolving it as monophyletic, but using only *Pseudacanthops* as outgroup. This limited sampling outside *Acanthops* might have influenced their topology by forcing monophyly on *Acanthops*. More extensive molecular sampling within the well-diversified *Acanthops* is needed to test the phylogenetic hypothesis of Lombardo and Ippolito (2004). Even though members of *Lagrecacanthops* Roy and *Miracanthops* Roy were not available for study, their distinct morphology (e.g. compound eyes each with a distinct apical spine, the conformation of the prothoracic legs, forewings resembling dead leaves and male genital structures) clearly places them as members of this group (Roy, 2004). The Acontistinae is a distinct Neotropical lineage of small-sized and seemingly well-diversified praying mantises (Roy, 2006). My analysis revealed *Acontista* to be paraphyletic, with *Acontista bimaculata*

resolved as sister to *Raptrix*. *Acontista bimaculata* is widely distributed south of the Amazon and is the only known member of *Acontista* with brachypterous females. Unfortunately, I was unable to include *Tithrone* Stål and *Paratithrone* Lombardo in my phylogenetic analyses. Preliminary assessment of these two genera, including comparative analyses of male genitalia (see Lombardo, 1996), unquestionably places them among the Acontistinae, likely with *Tithrone* as sister to all other Acontistinae. Finally, the Stenophyllinae and its single genus *Stenophylla*, has been a historically contentious taxon regarding its placement in the classification. Likely due to its aberrant morphology, its systematic status (ranked either as tribe or subfamily) has remained unstable, having been assigned to either the Oriental Toxoderidae (e.g. Giglio-Tos, 1914, 1927; Beier, 1934, 1968; Roy, 2009) or the Vatinae (Lombardo, 2000b). Terra (1995) was the first to recognize a likely relationship between Acanthopinae, Acontistinae and Stenophyllinae, placing the group within Acanthopidae. Yager and Svenson (2008) supported Terra's assumptions based on the MSMT metasternal anatomy of *Stenophylla*. More recently, Svenson *et al.* (2015) included *Stenophylla* in their phylogenetic study of the Empusidae and Hymenopodidae, recovering it within their outgroup sampling of PEPM taxa, although as sister of *Macromantis nicaraguae* Saussure and Zehntner, 1894 (Photinae). My analysis supports Terra's (1995) and Yager and Svenson's (2008) hypotheses, and further elaborates on Svenson *et al.* (2015), because *Stenophylla* was robustly and consistently resolved as sister to the clade Acanthopinae+Acontistinae.

2.5.2 Revised new higher-level classification system for polymorphic earless praying mantises

Molecular and morphological evidence presented by Svenson and Whiting (2009) and Wieland (2013) points to the same conclusion: homoplasy is rampant in Mantodea. Convergent evolution of form and function has misled historical interpretations of morphological traits that are associated with functionality. This seems to be true at all hierarchical levels across the order, sometimes even within apparently monophyletic and well-delimited taxa. Incongruences between morphology-based classifications and my molecular phylogeny support this pattern. Considering the profusion of convergences in Mantodea, any system formulated under phenetic criteria can no longer be considered a reasonable approximation to the mantodean tree of life.

Current classification systems are thus inadequate for storage and retrieval of biological information. They also lack a framework for the formulation and testing of evolutionary hypotheses. Diagnostic features presented below should not be assumed to be definite autapomorphies. The extensive homoplasy in morphological characters in Mantodea suggests that though a given character may prove to be useful for taxon delimitation, it may also have evolved independently in unrelated taxa. Accordingly, I use multiple lines of evidence in addition to morphology (Figs 2.6–2.10), such as egg case structure (Figs 2.11–2.14), biogeography and ecology, to both facilitate successful identification of PEPM at the supra-generic level, and to fulfil the diagnosability criterion of Linnaean ranks (Vences *et al.*, 2013).

The polymorphic earless praying mantises, or superfamily Acanthopoidea, comprises 8 families, 15 subfamilies and 18 tribes (Table 2.3). My revised organization is as follows: (i) Thespidae, including subfamilies Pseudopogonogastrinae, Pseudomiopteryginae, Bantiinae, Miobantiinae and Thespinae (tribes Musoniellini and Thespini); (ii) Angelidae; (iii) Coptopterygidae; (iv) Liturgusidae; (v) Photinaidae, including Macromantinae, Cardiopterinae, Photiomantinae and Photinainae (tribes Microphotinini, Orthoderellini and Photinaini); (vi) Stenophyllidae; (vii) Acontistidae; and (viii) Acanthopidae.

2.5.3 Taxonomy

2.5.3.1 Superfamily Acanthopoidea Burmeister, 1838

I apply the family group name Acanthopoidea to all PEPM. The Acanthopoidea, as well as the primitive Chaeteessidae and Mantoididae, are the only primitively earless mantises in the neotropics. PEPM can be further distinguished from the also earless but morphologically more conservative Chaeteessidae and Mantoididae by exhibiting forefemora with 3–4 discoidal spines, **AND** either of the following: (i) membrane of both pair of wings of males densely setose (as seen under backlight), and wingless females (i.e. all Thespidae), **or** (ii) pronotum at least twice as long as wide (except in the recently described *Ovalimantis* Roy, where it is 1.5× as long as wide and strongly oval in shape), and females with **at least** forewings present (i.e. all non-Thespidae taxa). The presence of 3–4 discoidal spines distinguishes members of Acanthopoidea

from Chaeteessidae (which has 2), whereas the densely setose wings and pronotum longer than wide rules out Mantoididae (as it has glabrous wings and a square-like pronotum).

Key to separate PEPM from other New World Mantodea:

1. Prothoracic metatarsi inserted apically to the tibiae, apical tibial spur seemingly missing (Fig. 2.7A)**CHAETEESIDAE**
- 1'. Prothoracic metatarsi inserted laterally to the tibiae, apical tibial spur always present (Fig. 2.6B)**2**
- 2(1'). Lower half of anterior surface of forefemora with long, scattered setae accompanying the femoral brush, the latter not forming a well delimited and compact structure (Fig. 2.7C); anterior abdominal segments of males strongly constricted and waist-like (Fig. 2.7D), whereas in females the abdomen as a whole tapers posteriorly (i.e. anterior segments are not constricted, Fig. 2.7E).....**MANTOIDIDAE**
- 2'. Lower half of anterior surface of forefemora usually without long, scattered setae, femoral brush well-defined (Fig. 2.7B); abdomen not as above (i.e. never anteriorly constricted in males, and often enlarged and/or fusiform in females).....**3**
- 3(2'). Males and some females with DK metasternal anatomy (Fig. 2.7F; see also Yager and Svenson, 2008).....**other Neotropical Mantodea**
(i.e. Stigmatopterinae, Vatinae, Stagmomantinae, Antemninae, Mellierinae, Choeradodinae, *Epaphrodita* Audinet Serville and *Gonatista* Saussure).
- 3'.Metasternum different from Fig. 2.7F (both sexes).....
.....Polymorphic Earless Praying Mantises (PEPM), superfamily **ACANTHOPOIDEA**

The Acanthopoidea are restricted to the Neotropical region. The great majority of species are concentrated in warmer environments (particularly in the Amazon), and at mid-elevations within the tropical Andes, with *Pseudopogonogaster hebardi* (Terra, 1982) reaching the highest elevation (3500m a.s.l.) in southern Ecuador (Lombardo and Ayala, 1998; Rivera *et al.*, 2011). My examination of museum specimens shows that the northernmost records correspond to

Brunneria borealis Scudder, 1896 around the 35°N parallel in eastern U.S.A., whereas the southernmost limits correspond to *Brunneri subaptera* Saussure, 1869, around the 37°S parallel, within the northern boundaries of Patagonia, Argentina. Interestingly, Chile exhibits a depauperated praying mantis fauna (Agudelo *et al.*, 2007); no endemic species are known thus far.

Key to families, subfamilies and tribes reflecting the new organization of the PEPM

The key applies to adults of both sexes to family level. Some subordinate taxa (i.e. subfamily and tribe) can be distinguished only through male genitalia.

1. Forefemora with 5 or more posteroventral spines (Figs 2.6B; 2.8A, I)2
- 1'. Forefemora with 4 or fewer posteroventral spines (Figs 2.6C; 2.7J).....6
- 2(1). Compound eyes with an apical spur of variable size and shape (Figs 2.6B; 2.8K).....**Acanthopidae**
- 2'. Compound eyes smooth, without processes (Figs 2.6A; 2.7G; 2.8D–G, J, L–M).....3
- 3(2'). Head with three long projections, one above each compound eye and a third one in the middle of vertex (Fig. 2.8G), the same are larger and often of different shape in females (Fig. 2.8H); mid- and hindfemora with a large, ventral foliaceous lobe with irregular edges.....**Stenophyllidae**
- 3'. Head normally without projections (Figs 2.7G; 2.8D, M) but, if present, these are different than those described above for Stenophyllidae (e.g. Fig. 2.8L); mid- and hindfemora normally without lobes but, if present, they are small, with smooth edges and confined to the apical tip of femora.....4
- 4(3'). Males: *processo ventrale sclerificato* of right phallomere large, hook-like and sharp (Fig. 2.10N); Females: forewings reduced, extending as far as the distal edge of the first abdominal tergite or shorter, and hindwings always absent (Fig. 2.7S, T).....**Coptopterygidae**
- 4'. Males: *processo ventrale sclerificato* of right phallomere shaped differently (e.g. Fig. 2.10T–U); Females: forewings usually extending beyond the distal edge of the second abdominal tergite, and hindwings always present.....5

- 5(4'). Smaller species (pronotum length 8mm or less in both sexes); ventral phallomere of male genitalia with a specialized and conspicuously sclerotized distal projection of various sizes and shapes (Fig. 2.10V–Z); both pairs of wings frequently patterned and/or pigmented with colour in both sexes, rarely completely hyaline and colourless in males.....**Acontistidae, Acontistinae**
- 5'. Larger species (pronotum length 9mm or more in both sexes); ventral phallomere of male genitalia without specialized, distal projections, edge simple and smooth (Fig. 2.10S); hindwings rarely pigmented with colour, if so, only in females.....**Photinaidae, 8**
- 6(1'). Cerci broadly flattened (Fig. 2.7P–R); proximal discoidal spine of forefemora substantially longer than subsequent discoidal spine**Angelidae**
- 6'. Cerci cylindrical, never broadly flattened; proximal discoidal spine of forefemora shorter or as long as the subsequent discoidal spine7
- 7(6'). Forefemora with a well-defined, circular pit between the first and second posteroventral spine (the pit accommodates the distal-most posteroventral tibial spine when flexed against the femur). Antennomeres of males bare or only with minute, sparse setae; both sexes fully winged; both pair of wings bare or only with minute, sparse setae along the costal vein (Fig. 2.7I).....**Liturgusidae**
- 7'. Forefemora lacking pits. Antennomeres of males with a preapical crown of long setae that are as long or longer than the corresponding antennomere (Figs 2.6A; 2.7G); females always wingless (Fig. 2.1A, B, D–H, J); wings of males: costal vein of both pairs of wings densely setose (Fig. 2.7H), wings membrane in general bearing conspicuous pilosity (clearly seen under backlight).....**Thespididae, 13** (key continues for males only)
- 8(5'). Large insects, body length (measured from the tip of the abdomen to the top of the vertex) ranging from 80 to 110mm.....**Macromantinae, Macromantini**
- 8'. Smaller insects, body length 70mm or less.....9
- 9(8'). Forecoxae with a well-defined row of strong, conical or lamellar, spines along both their ventral and dorsal edge (Fig. 2.8C); mid- and hindfemora of females with a small, distal ventral lobe; subgenital plate of males with a large, polished, black central spot.....**Cardiopterinae, Cardiopterini**

- 9'. Forecoxae with a single, well-defined row of small (rarely strong), conical spines, along their dorsal edge only (ventral edge might exhibit granules and/or tubercles, but without forming a well-defined row of spines); subgenital plate of males without a black, central spot.....10
- 10(9'). Head strongly compressed (flattened), compound eyes elongated in lateral view (Fig. 2.8E), and juxtaocular bulges strongly reduced to minute, inconspicuous elevations (Fig. 8D).....**Photiomantinae, Photiomantini**
- 10'. Head not particularly compressed, compound eyes either spherical (Fig. 2.8F) or only slightly elongated in lateral view, and juxtaocular bulges evident, sometimes large and prominent (as in Fig. 2.8M).....**Photinainae**, 11 (key continues for males only)
- 11(10'). Right margin of ventral phallomere exhibiting a proximal, well-sclerotized and strongly recurved lateral process (e.g., Fig. 2.10S).....**Photinaini**
- 11'. Right margin of the ventral phallomere with proximal process weakly or only partially sclerotized process that is never recurved (Fig. 2.10O–R).....12
- 12(11'). Right apodeme of right phallomere bearing numerous spine-like processes on its ventral aspect (Fig. 2.10T).....**Orthoderellini**
- 12'. Right apodeme of right phallomere lacking spine-like processes on its ventral aspect.....**Microphotinini**
- 13(7'). Dorsal lamina of the left phallomere completely fused to, and indistinct from, the ventral phallomere, altogether appearing to be a single, U-shaped structure with a large central notch (Fig. 2.9F–K)**Bantiinae, Bantiini**
- 13'. Dorsal lamina of the left phallomere well differentiated from the ventral phallomere (as in Fig. 2.9A or similar), not forming a single, U-shaped structure.....14
- 14(13'). Distal margin of the ventral phallomere with a tapering, and caudally projected, distal process (Fig. 2.9A,C).....**Pseudomiopteryginae, Pseudomiopterygini**
- 14'. Distal margin of the ventral phallomere without such process but, if present, not tapering.....15
- 15(14'). Apofisi falloide simple, its surface exhibiting longitudinal striae (Fig. 2.9D); right margin of ventral phallomere with a proximal, caudally-projected, short process (Fig. 2.9E).....**Pseudopogonogastrinae, Pseudopogonogastrini**

- 15'. Apofisi falloide simple, forked or multi-denticulated (Figs. 2.9G, H, J, L, N, P; 2.10H, L), and lacking longitudinal striae; right margin of ventral phallomere without caudally-projected process.....16
- 16(15'). Ventral phallomere with a distinct, strongly sclerotized process (or region) of variable shape near its right, distal corner (Fig. 2.9L–P); foretibiae with 6–9 posteroventral spines (if 6, then pronotum surface has tubercles).....**Miobantiinae, Miobantiini**
- 16'. Ventral phallomere without a well-defined and strongly sclerotized lateral process on its right, distal corner (Fig. 2.10A–M); foretibiae with 6 or fewer posteroventral spines (if 6, pronotum surface lacks tubercles).....**Thespinae, 17**
- 17(16'). Wings in resting position do not reach tip of subgenital plate (exceptionally, wings may extent to or only slightly beyond this point; if so, then foretibiae bear 3 or fewer posteroventral spines); foretibiae frequently exhibiting 1 or 2 of the distal-most anteroventral spines displaced to a decidedly dorsal or dorsolateral position (Fig. 2.7K–N). General habitus more commonly filiform and of delicate appearance (e.g. Fig. 2.1G–J).....**Thespini**
- 17'. Wings in resting position always and substantially extend beyond the tip of the subgenital plate; foretibiae never with displaced anteroventral spines (Fig. 2.7O). Habitus not particularly filiform, but relatively stouter (e.g. Fig. 2.1E, F).....**Musoniellini**

2.5.3.1.1 Thespidae Saussure, 1869

Type genus: *Thespis* Audinet Serville 1831

Diagnosis. Members of this family can be identified using the following combination of characters: (i) males with dense pilosity on their body (mostly ventrally), limbs, antennae (Figs 2.6A; 2.7G), and all four wings, particularly along the costal veins (Fig. 2.7H); pilosity is inconspicuous and much more scattered in females; (ii) adult females always apterous, but traces of venation can be seen through the cuticle of seemingly rudimentary primordial wing pads; (iii) forefemora with 5–15 anterolateral spines, four posteroventral spines (Fig. 2.7J) (exceptionally only one), and normally four discoidal spines (rarely three); (iv) foretibiae with 1–9 posterolateral and 1–12 anterolateral spines respectively; the foretibiae of several genera exhibit

one or two of the distal most anteroventral spines displaced to a more dorsolateral or decidedly dorsal position (indicated by black arrows in Fig. 2.7K–N).

Egg cases. Rectangular in shape, and ventrally attached to flat surfaces (Fig. 2.12A–M), such as leaves, bark and rocks (Figs 2.1B; 2.12G). Females coat their egg cases with a layer of whitish, easily removable, spongy material of variable thickness. The outer layer covers most of the egg case and emergence area (e.g. Fig. 2.12J) when they are freshly formed, slowly deteriorating when exposed to environment; consequently, it is not unusual to find partially covered egg cases. Eggs in contact with external walls, the latter normally light brown or straw-coloured, but sometimes with a reddish tone (Fig. 2.12K, L). A distal, residual process is sometimes present. The emergence area may be located on a longitudinal, dorsal depression (Fig. 2.12B), and in some taxa part of the egg case walls delimiting the chamber openings project dorsally, conferring the egg case with a crested outline, thus making the depression somewhat more conspicuous, (Fig. 2.12L).

Natural history. Thespidae is the most speciose and ecologically diversified lineage of Neotropical praying mantises. Many species seem to be microhabitat specialists including grass, undergrowth, open ground, tree bark and epiphytic vegetation, many exhibiting specialized – and often remarkable – cryptic adaptations. Noteworthy examples are *Pseudopogonogaster*, *Thesprotiella*, *Pogonogaster* Rehn (lichen), *Eumusonia* (grass) and *Anamiopteryx* (rough bark) (Rivera *et al.*, 2011; J. Rivera, personal observation). The same female can produce egg cases of varying length and volume (which correlates with the number of eggs contained), but shape and overall morphology is otherwise well conserved (J. Rivera, unpublished data). Thespids can be found in various ecological situations, either in dry, seasonal and humid environments. They are widespread across the entire Neotropic and adjacent areas in the southern Nearctic Region. Thespids are also commonly found on islands (e.g. Galapagos, Antilles). The highest elevations with praying mantises are also occupied by species of thespids (3500m a.s.l.).

Remarks. The Thespidae exhibit a great variety of body size and shapes, from short and compact (as in *Pseudopogonogaster*, *Pseudomiopteryx* and *Eumiopteryx*; Fig. 2.1A, B, D), to long and slender (as in *Thesprotiella* and *Oligonyx*; Fig. 2.1I, J). Sexual dimorphism varies across taxa, where males tend to be relatively smaller and slimmer than females, the latter often exhibiting

more elaborate camouflage. This family includes some of the smallest species in the New World, such as *Diabantia* Giglio-Tos (9 mm), but also much larger taxa such as *Macromusonia* (66 mm). Body colouration is usually in tones of brown and grey, sometimes black, with some species even presenting both brown and green morphs; members of *Pseudopogonogaster* are polychromatic (Rivera *et al.*, 2011). Tibial spines often exhibit conspicuous sexual dimorphism, with smaller, blunter and more reclined spines in males. The displacement of one or two of the distal-most anteroventral spines into a more dorsolateral or decidedly dorsal position, is a common occurrence among the Thespidae (Fig. 2.7K–N). When such spines occur, ventral microserrulations in these and other associated spines in both tibiae and femora can be seen at high magnification. Important variation in the number and arrangement of foretibial spines exists and is informative for diagnosis of genera. Male genitalia constitute a reliable character system for interspecific differentiation, whereas congeneric females are often difficult to tell apart and associate with conspecific males. Egg case features are well conserved across Thespidae; most variation observed concerns structural details of the emergence area, colouration of the external walls, relative height and thickness. I have examined egg cases produced by members of *Bantia*, *Carrikerella*, *Chloromiopteryx*, *Eumusonia*, *Galapagia*, *Macromusonia*, *Musonia*, *Oligonicella*, *Oligonyx*, *Pizaia*, *Pseudomiopteryx*, *Pseudopogonogaster*, *Thesprotiella* and *Thrinaconyx*, all exhibiting strong resemblance in shape, type of attachment and other features (Fig. 2.12A–M). The literature reports the egg cases of few other species, such as *Musonia surinama* (Saussure, 1869), *Thesprotia filum* (Lichtenstein, 1796), *Bantiella trinitatis* Giglio-Tos, 1915 (cited as *Promiopteryx granadensis* (Saussure, 1870)) (Beebe *et al.*, 1952) and *Miobantia fuscata* Giglio-Tos, 1915 (Scherrer, 2014), all conforming with my observations.

Pseudopogonogastrinae Rivera and Svenson, 2016

Type genus: *Pseudopogonogaster* Beier, 1942

Diagnosis. Body size small (23–27 mm). *Pseudopogonogaster* is the only representative of this subfamily. For the time being, the diagnostic features of *Pseudopogonogaster* outlined by Rivera *et al.* (2011), apply to the subfamily. These include: (i) pronotal prozona with double protuberance (proximal and distal, visible in the lateral view); (ii) forefemora with four

posteroventral spines, 12 anterolateral spines, and four discoidal spines; (iii) foretibiae with 6 and 8–11 posterolateral and anterolateral spines, respectively; (iv) wings of males in resting position extending beyond the tip of abdominal subgenital plate; (v) abdomen of females with various lobes and lichen-like cuticular outgrowths (Fig. 2.1A); (vi) left phallomere of male genitalia with apofisi falloide exhibiting striae (Fig. 2.9D); and, (vii) right margin of ventral phallomere with a proximal, caudally-projected, short process (Fig. 2.9E)

Natural history. Members of *Pseudopogonogaster* live among epiphytic vegetation in mountainous forest areas (as high as 3500m a.s.l.) in the northern tropical Andes of South America (Peru, Ecuador and Colombia). They exhibit a highly derived and specialized lichen-like morphology (Fig. 2.1A). Lombardo and Ayala (1998), Ayala and Onore (2001) and Salazar (2006b) provided information concerning their natural history, whereas Rivera *et al.* (2011) presented a summary of current knowledge, taxonomy and distribution.

Remarks. According to the analysis of Burks (2012), *gastr-* the syncopated stem of -gaster (meaning ‘stomach’), which is the correct and most widely used form of the two singular genitives (the other being *gaster-*) attributed to this classic word commonly used in zoological nomenclature. Therefore, I choose *gastr-* over *gaster-* for the formation of the family group name Pseudopogonogastrinae.

Included Genera:

1. *Pseudopogonogaster* Beier, 1942

Pseudomiopteryginae Giglio-Tos, 1915

Type genus: *Pseudomiopteryx* Saussure, 1870

Diagnosis. Members of this subfamily are small-sized (14–25 mm) with dark to light brown colouration (Fig. 2.1B). They exhibit the following combination of characters: (i) foretibiae with 7–9 and 8–10 posteroventral and anteroventral spines, respectively; all spines steadily growing in size distad; (ii) wings of males in resting position extending beyond the tip of abdominal subgenital plate; (iii) ventral phallomere with a distal, tapering projection (Fig. 2.9A, C); (iv)

dorsal lamina of left phallomere narrowly articulated to the ventral phallomere, so that both structures are clearly differentiated (Fig. 2.9A); and (v) apofisi falloide curved and laterally projected (Fig. 2.9A).

Natural history. Anecdotal records found in the literature suggest that members of this lineage occupy different levels within the vegetation and generally exhibit a wide range of altitudinal distribution. For instance, *Pseudomiopteryx infuscata* Saussure and Zehntner, 1894 has been found at ground level as well as on epiphytic bromeliads (Rehn, 1935), where egg cases have been found (J. Rivera, personal observation), and also through rainforest canopy fogging (Smithsonian Collection, J. Rivera, personal observation). In Colombia, Agudelo and Chica (2002) reported *Pseudomiopteryx bogotensis* Saussure, 1870 in Paramo formations around 3000m a.s.l. in the tropical Andes, whereas Arteaga *et al.* (2014) reported *Pseudomiopteryx spinifrons* Saussure, 1870 in dry forest formations in the northern part of the country. Other species can be found at lower elevations within the eastern slopes of the Andes and into the amazonian lowlands, such as *Pseudomiopteryx amazonensis* Piza, 1968, originally described from the vicinity of Manaus (Brazil). *Pseudomiopteryx* is the most widespread and diversified member of this subfamily and Thespidae in general, found from central Mexico south into the Amazon and tropical Andes, whereas *Promiopteryx* is currently known from Andean habitats in Colombia and the Coastal Cordillera in Venezuela. Males of *Pseudomiopteryx* are active fliers, often caught in malaise traps (J. Rivera, personal observation).

Remarks. My concept of Pseudomiopteryginae greatly differs from its classical treatment. Giglio-Tos (1915b, 1919) hypothesized close affinities between *Pseudomiopteryx* and genera such as *Bantiella*, *Diabantia*, *Mantillica*, *Mantellias* Wetwood (all herein transferred to a new subfamily, the Bantiinae), *Miobantia* (in part) and *Anamiopteryx* (herein transferred to Miobantiinae), grouping them under his subfamily Pseudomiopteryginae. Beier (1935a) further elaborated on Giglio-Tos' system and added *Promiopteryx* Giglio-Tos, *Miobantia* (in part, cited as *Antimiopteryx* Giglio-Tos), *Eumiopteryx* and *Chloromiopteryx*, but this time under tribe Pseudomiopterygini, within the Thespinae. He followed this system in later contributions (Beier, 1964, 1968). Beier's concept of Pseudomiopterygini thus included genera here treated separately under Pseudomiopteryginae, Bantiinae and Miobantiinae. Terra (1995) proposed a completely different system justified by characters found in the head, forelegs, pronotum and chromatic

features of male wings. Terra modified the prevailing concept of Pseudomiopteryginae (now under Thespidae) to include *Pseudomiopteryx*, *Pizaia*, *Leptomiopteryx*, *Anamiopteryx* and *Eumiopteryx*. Terra's Pseudomiopteryginae was adopted and expanded recently by Ehrmann (2002) to include *Palaeothespis* Tinkham and *Sinomiopteryx* Tinkham, two Asian genera that I am here excluding from Thespidae (see Table 2.2). Based on my comparative analyses of specimens coupled to the phylogenetic evidence, my concept of Pseudomiopteryginae is much more restricted and includes only two genera.

Included Genera:

1. *Pseudomiopteryx* Saussure, 1870
2. *Promiopteryx* Giglio-Tos, 1915

Bantiinae Rivera and Svenson, 2016

Type genus: *Bantia* Stål, 1977

Diagnosis. Members of this subfamily include very small to small species (9–25 mm), which are light to dark brown in colour (Fig. 2.1C). They exhibit the following combination of characters: (i) foretibiae with 2–7 and 1–9 posteroventral and anteroventral spines, respectively; (ii) foretibiae with posteroventral and anteroventral spines in various arrangements (i.e. spines of different sizes normally occur within the same series); (iii) all genera but *Bantiella* and *Diabantia*, exhibit 1 or 2 distal-most anteroventral spines shifted to a dorsolateral or decidedly dorsal position (Fig. 2.7K); (iv) wings of males in resting position extending beyond the tip of abdominal subgenital plate; and (v) dorsal lamina of left phallomere completely fused to the ventral phallomere and thus both structures appear as a single, U-shaped structure of various shapes, but always with a large central notch with irregular inner edges (Fig. 2.9F–K).

Natural history. Little is known about their habits. *Thrinaconyx fumosa* Saussure and Zehntner, 1894 and *Mantillica nigricans* Westwood, 1889 have been collected at ground level in Costa Rica (Rehn, 1935) and the Brazilian Amazon (Agudelo and Rafael, 2014), respectively; the latter species is remarkable for its strong resemblance to the Ponerinae ant *Pachycondyla apicalis* (Latreille, 1802), a common mimic model for other Amazonian insects and arachnids (Agudelo

and Rafael, 2014). Label data of museum specimens suggest that members of this clade have wide geographical and ecological distribution in tropical, humid habitats in Central and South America, including tropical Andes at mid-elevations and through the Amazon (where the highest diversity is found), but at least one species of *Bantia* is found in the Atlantic rainforest of southern Brazil. Males of *Thrinaconyx*, *Bantia* and *Mantillica* are active fliers, often caught in malaise traps (Agudelo and Rafael, 2014; J. Rivera, personal observation).

Remarks. My concept of Bantiinae has no equivalent in the literature. Except for *Bantiella* and *Diabantia*, most members of this lineage exhibit dorsoapical spines in the foretibiae. This unusual modification is found also in some members of Thespinae **sensu n.** (see below) and the Oriental Haaninae, which evolved virtually identical foretibial morphology (see Wieland, 2013). Terra (1995) grouped all taxa presenting these spines, in addition to *Diabantia*, under the Oligonicinae. My phylogenetic evidence suggests such spines evolved more than once in Thespidae, thus rendering Oligonicinae (and its two tribes, Oligonicini and Pogonogasterini) as nonmonophyletic.

Included Genera:

1. *Bantia* Stål, 1877
2. *Bantiella* Giglio-Tos, 1915
3. *Diabantia* Giglio-Tos, 1915
4. *Mantillica* Westwood, 1889
5. *Mantellias* Westwood, 1889
6. *Thrinaconyx* Saussure, 1892

Miobantiinae Roy, 2013

Type genus: *Miobantia* Giglio-Tos, 1917

Diagnosis. Members of this subfamily include very small to mid-sized species (9–36 mm), which exhibit disparate body shapes. Colouration usually includes various tones of grey, brown and sometimes green (Fig. 2.1D). They exhibit the following combination of characters: (i) foretibiae with 6–9 and 7–11 posteroventral and anteroventral spines, respectively; (ii) foretibiae with posteroventral and anteroventral spines steadily increasing in size distad; (iii) wings of males in resting position extending beyond the tip of abdominal subgenital plate; (iv) ventral

phallomere with its right laterodistal edge often exhibiting a well-sclerotized process (exceptionally two) that varies from short and blunt to long and sharp (Fig. 2.9L–P); and (v) apofisi falloide hook-like, short or long, curved or strongly recurved at its base and laterally projecting (Fig. 2.9L, N, P).

Natural history. Label data of museum specimens suggest that members of this clade exhibit disparate ecologies. With few known exceptions, species are typically distributed south of the Amazon, with the bulk of diversity found in the Atlantic forest of southern Brazil and neighbouring seasonal to dry habitats (Cerrado and Caatingas). For instance, *Miobantia* prefers lower vegetation in humid forest (Scherrer, 2014), whereas *Anamiopteryx* and *Eumiopteryx* (Fig. 2.1D) are bark dwellers, inhabiting mostly seasonal and dry habitats.

Remarks. Roy (2013) recently created the taxon name Miobantiinae as a *nomen novum* in replacement of Miopteryginae Kirby, 1904, as the nominotypical taxon *Miopteryx* de Sausure, 1869 was in conflict with its primary homonym *Miopteryx* Blanchard, 1842 (Coleoptera: Cerambycidae). Terra (1995) outlined the history of the Miopteryginae, as well as other genus-group names (prominently *Miobantia*) that resulted from several nomenclatorial procedures conducted by earlier authors in their attempts to solve this homonymy. Roy (2013) addressed the issue and ultimately resolved it by proposing the name Miobantiinae, an action that Scherrer (2014) adopted in his revision of *Miobantia*. Terra (1995) considered *Paradiabantia* Piza a synonym of *Diabantia* Giglio-Tos, (Bantiinae). Although superficially very similar, my analysis of type material and examination of additional specimens of both genera, all indicated that *Paradiabantia* is distinct from *Diabantia*. *Paradiabantia* differs from

Diabantia as follows: (i) forefemora with 3 discoidal spines (4 in *Diabantia*); (ii) cell membrane of meso- and hindwings exhibiting both micro- and macrosetae (only microsetae in *Diabantia*); and (iii) distribution south of the Amazon (*Diabantia* is found north of the Amazon). Further, the most compelling evidence for distinguishing both genera pertains to male genitalia. The ventral phallomere of *Paradiabantia* is distinct from the left phallomere and conforms with the Miobantiinae type (Fig. 2.9N), whereas *Diabantia* exhibits the dorsal lamina of the left phallomere completely fused to the ventral phallomere (Fig. 2.9H), wherein the distinct central

notch typical of Bantiinae can also be observed. Thus, I herein remove *Paradiabantia* **stat.r.** from synonymy and accept it as valid and distinct from *Diabantia*.

Included Genera:

1. *Anamiopteryx* Giglio-Tos, 1915
2. *Chloromiopteryx* Giglio-Tos, 1915
3. *Eumiopteryx* Giglio-Tos, 1915
4. *Miobantia* Giglio-Tos, 1917
5. *Paradiabantia* Piza, 1973

Thespinae Saussure, 1869

Diagnosis. This subfamily includes small to medium-sized species (18–66 mm) with a wide array of morphological and ecological adaptations. Most species exhibit body forms that are decidedly elongated and slim (e.g. Fig. 2.1E, I, J), although few species may also exhibit a relatively stouter, more compact build (e.g. Fig. 2.1F–H); these, nevertheless, show a general trend towards pronotal and foreleg elongation. Colouration varies from tones of brown and grey, to black and rarely green, sometimes in combinations. Intraspecific chromatic polymorphisms (e.g. brown and green morphs) are not uncommon. Members of this subfamily can be identified by the following combination of characters: (i) foretibiae with 1–5 (exceptionally 6) and 2–12 posteroventral and anteroventral spines, respectively; (ii) most genera with one or two distal-most anteroventral spines shifted to a more dorsolateral or decidedly dorsal position (Fig. 2.7L–N); and (iii) ventral phallomere usually subrectangular or square-like in shape, with various marginal irregularities and processes but never forming a well-defined and strongly sclerotized projection on its right laterodistal edge (as seen in the Miobantiinae) (Fig. 2.10A–M).

Remarks. My new concept of Thespinae includes those genera previously included in both Thespinae and Oligonicinae (except for those taxa here included in Bantiinae) *sensu* Terra (1995). I subdivide the Thespinae into two tribes.

Musoniellini Rivera and Svenson, 2016

Type genus: *Musoniella* Giglio-Tos, 1916

Diagnosis. Body size small (18–34 mm), habitus relatively elongated, but not slim (e.g. Fig. 2.1E), or with a more compact build (Fig. 2.1F). Colouration is light to dark brown, but green morphs may also occur. They exhibit the following combination of characters: (i) foretibiae with 5 (exceptionally 6) posteroventral spines, typically steadily increasing in size distad (as in Fig. 2.7O); and (ii) wings of males in resting position extending beyond the tip of the abdominal subgenital plate.

Natural history. The bulk of species are found associated mostly with seasonal habitats south of the Amazon (e.g. Costa-Pereira *et al.*, 2010; Rodrigues and Cancellato, 2013), except for *Leptomiopteryx*, which is known from the northeastern Amazon. However, Rodrigues and Cancellato (2013) cited an unidentified species of *Leptomiopteryx* from eastern Brazil, south of the Amazon.

Included Genera:

1. *Eumusonia* Giglio-Tos, 1916
2. *Leptomiopteryx* Chopard, 1911
3. *Musoniella* Giglio-Tos, 1916
4. *Pizaia* Terra, 1982

Thespini Saussure, 1869

Diagnosis. Body size small to mid-sized (20–66 mm), habitus from compact and relatively elongated (Fig. 2.1G, H) to long and decidedly slim (e.g. Fig. 2.1I, J). Colouration is highly variable, most commonly in tones of brown or grey, sometimes black, and rarely green. They exhibit the following combination of characters: (i) foretibiae with 1–5 (exceptionally 6) posteroventral spines; (ii) about half of the genera exhibit one or two of the distal-most anteroventral spines shifted to a more dorsolateral or decidedly dorsal position (Fig. 2.7L–N); and (iii) wings of males in resting position normally not reaching the tip of the abdomen (in a

few genera, wings may exceptionally reach the tip of the subgenital plate; if so, then foretibiae bear 3 or fewer posteroventral spines).

Natural history. Members of this tribe are ground level and/or low vegetation dwellers, both in humid and semi-arid habitats. Species adapted to epiphytic habitats may also occur at higher levels within the vegetation. For instance, on the one hand, genera such as *Musonia*, *Oligonicella*, *Musoniella* and *Thespis* all exhibit rather generalist morphology, and are often associated with arid and seasonal ecosystems, such as riparian vegetation along valleys in the Pacific coastal dessert, grasslands, xerophytic scrubs and savannah-like habitats across the neotropics (Blatchley, 1920; Rehn, 1935; Salazar, 2002; Agudelo and Chica, 2003; Rivera, 2003b; Battiston *et al.*, 2005; Battiston and Picciau, 2008; Arteaga *et al.*, 2014). On the other hand, members of *Macromusonia* are found in the Amazon rainforest, at ground level and/or lower vegetation, whereas *Carrikerella*, *Pogonogaster* and *Thesprotiella* inhabit epiphytic vegetation and other plant outgrowths (Rehn, 1935; Ayala and Onore, 2001; Blanco and Salazar, 2015). Members of Thespiinae are strictly Neotropical, with few species reaching the boundaries of the Nearctic region and Florida; taxa with marginal, Nearctic distributions (southeastern U.S.A.) include only members of *Oligonicella* and *Thesprotia*, the latter exhibiting the most extensive (but patchy) distribution. With plenty of undescribed and morphologically similar species, *Thesprotia* ranges from southern Florida, Mexico, through Central and South America and south of the Amazon, and is often associated with open grasslands (Gurney, 1951), but also on the ground among grass in pine woods (Blatchley, 1920; Peck and Beninger, 1989). Some species of *Oligonyx*, *Thesprotia*, *Musonia*, *Thespis* Audinet Serville are endemic to islands, whereas *Galapagia* and *Liguanea* Rehn and Hebard are strictly insular groups.

Remarks. My own analyses of male genital morphology suggest that a number of taxa within Thespiinae, marked with an (*) in Fig. 4, are either misclassified in the wrong genera, or will eventually demonstrate to be invalid. For instance, *Paramusonia* Rehn seems to be a synonym of *Thespis*, whereas *Carrikerella* and *Pogonogaster* are most likely synonyms of a rather variable and highly diversified *Thesprotiella*, to which *Oligoyx insularis* Bonfils, 1967 likely also belongs. Similarly, noninsular members of *Galapagia* – namely *G. peruana* Beier, 1935 (Fig. 2.1G) and *G. amazonica* Terra, 1982 – more than likely do not belong to this genus or any other so far recognized. Also, *Bactromantis virga* Scudder, 1896, a taxon usually associated to

Oligonyx and *Oligonicella*, was actually described on the basis of male juveniles of *Stagmomantis carolina* (Linné, 1763) as Rehn and Hebard (1916) and Blatchley (1920) clearly pointed out and my own examination of type specimens confirmed. Finally, *Musoniella margheritae* Battiston and Picciau, 2008 is most likely a member of *Musonia*. Clearly, the Thespinae (and Thespidae in general) is in need of a major taxonomic revision.

Included Genera:

1. *Carrikerella* Hebard, 1922
2. *Galapagia* Scudder, 1893
3. *Liguanea* Rehn and Hebard, 1938
4. *Macromusonia* Hebard, 1922
5. *Musonia* Saussure, 1869
6. *Musoniola* Giglio-Tos, 1917
7. *Oligonicella* Giglio-Tos, 1915
8. *Oligonyx* Saussure, 1869
9. *Paramusonia* Rehn, 1904
10. *Pogonogaster* Rehn, 1918
11. *Pseudomusonia* Werner, 1909
12. *Thespis* Audinet Serville, 1831
13. *Thesprotia* Stål, 1877
14. *Thesprotiella* Giglio-Tos, 1915

2.5.3.1.2 Angelidae Beier, 1935

Type genus: *Angela* Audinet Serville, 1839

Diagnosis. Stick-like insects (Fig. 2.2A) with body lengths ranging from mid-size to very large (60–141 mm); *Angela maxima* (Chopard, 1910) stands out as arguably the longest praying mantis in the NewWorld so far recorded (up to 141mm for females) (Roy, 2011). Body colouration is often brown in both sexes, but females tend to exhibit lighter pigmentation compared to males, sometimes attaining yellowish tones. They exhibit the following combination of characters: (i) forefemora with first discoidal spine substantially longer than the

second; (ii) male wings are shortened but otherwise functional, in the resting position they extend beyond the mid abdominal tergites but without reaching the supraanal plate; (iii) female wings in resting position do not extend beyond the mid abdominal tergites; and (iv) cerci foliaceous (i.e. broad and flattened) (Fig. 2.7P–R); this latter trait allows for quick separation of any instar of Angelidae from other stick-like taxa in the neotropics, such as *Macromusonia* and *Thespis* (Thespidae) or *Brunneria* (Coptopterygidae).

Egg case. Availability was limited to a few exemplars and is likely not representative of all Angelidae. Egg cases have drop-like (rounded proximally, tapering distally) shape and are attached by their ventral aspect (Figs 2.11F; 2.12N, O). These are entirely and tightly covered with a whitish secretion, whereas the external walls are reddish brown. The egg case is peculiar in having a layer of extra material, immediately next to the emergence area, projecting over each side of the egg case. These projections act as a sort of ‘supporting roof’ to the external coating that females apply to their egg cases. Both the supporting roof and the outer coating form a secondary wall, which is the part of the egg case in direct contact to the environment; there is a hollow space between the outer layer and the egg case body. The external coating projects forward to form a hollow, air-filled, tapering process. Roy (2011) depicted the egg case of *Angela maxima*; I also examined this egg case (Fig. 2.12P), which differs from the type above described by its greater size, oblong shape and thicker outer coating, the latter more densely packed with tiny air-filled bubbles (Fig. 2.12O). No dissections were done and thus internal architecture could not be observed. More sampling will be necessary to determine the extent of egg case variation within this family.

Natural history. Biology is largely unknown. Members of *Angela* are normally collected in dense tropical rainforests, where they hang from branches and leaves, much like stick insects (Fig. 2.2A) (e.g. Rehn, 1935; Robinson, 1969b; J. Rivera and G.J. Svenson, personal observation), although Arteaga *et al.* (2014) recorded *Angela trifasciata* Stål, 1877 in dry forest formations of northern Colombia. *Angela* is distributed from southern Mexico through Central America into the Amazon Basin and the island of Trinidad.

Remarks. Females exhibit considerable intraspecific variation, making species delimitation problematic. This contrasts with males, where obvious variation in the chromatic patterning of

hindwing exists, thus making species distinction more feasible. Considerable sexual dimorphism and paucity of females make conspecific sexes exceedingly difficult to associate, unless caught *in copula* or reared from the same egg case (males however tend to be common occurrence in light traps and thus in collections). As a whole, Angelidae is in need of a thorough taxonomic treatment.

Included Genera:

1. *Angela* Audinet Serville, 1839

2.5.3.1.3 Coptopterygidae Giglio-Tos, 1915

Type genus: *Coptopteryx* Saussure, 1869

Diagnosis. This family includes mid-sized to large praying mantises (30–120 mm). The only two recognized genera exhibit extremely different morphological appearances: *Coptopteryx* as a typical, generalist species, whereas *Brunneria* is stick-like (Fig. 2.2B). The typical colouration is green in both cases, although brown morphs do exist, especially in *Brunneria*. They exhibit the following combination of characters: (i) forefemora with 5 posteroventral and 3 discoidal spines; (ii) females lacking hindwings, whereas forewings are highly reduced, not extending beyond the distal edge of the first abdominal tergite (Fig. 2.7S, T); males are fully winged; and (iii) *processo ventrale sclerificato* of right phallomere of males well-developed, large and hook-like, posteriorly projected and sharp (Fig. 2.10N).

Egg cases. Females attach egg cases by their ventral aspect, typically to branches, stems or ramification points (Fig. 2.13B, C) (Jantsch, 1984). The cases can be either barrel-like or rectangular in shape (Fig. 2.13A–H). Egg cases of *Brunneria* (Fig. 2.13F–H) exhibit an external coating firmly cemented to the external walls. One distinct feature is the modification of the emergence area, which is closed and reduced to a single opening on a tubular or spine-like, distal projection (e.g. Fig. 2.13H). Zapata (1964) describes the egg case of *Coptopteryx gayi* (Blanchard, 1851), Rivera and Novoa (2006) depicted that of *Coptopteryx brevipennis* Beier, 1958 and so did Breland and Dobson (1947) for *Brunneria borealis*.

Natural history. Members of this lineage inhabit open, seasonal ecosystems, where they are found in bushes, trees and grassy habitats. For instance, in Peru *Coptopteryx brevipennis* is arboricolous and is found in dry forest formations at mid-elevations within the southern Andes (Rivera and Novoa, 2006), whereas *Brunneria orinocesis* Agudelo and Chica, 2002 is found in grass and herbaceous vegetation in northeastern Colombia (Agudelo and Chica, 2002). Members of both genera exhibit wide but disjunct distributions. *Brunneria* constitutes an extreme example of such distributional pattern, as *B. borealis* is found only in southern U.S.A.; no species are known from Mexico to Panama, but reports exist for Colombia (*Brunneria orinocesis*) and Venezuela (*Brunneria* sp.) (Cerdá, 1997; Agudelo and Chica, 2002). *Coptopteryx* is restricted to South America only and is mostly concentrated in semi-arid, seasonal habitats south of the Amazon, although small populations of both genera seem to persist in enclaves of suitable habitats outside this area (Rivera and Novoa, 2006). No coptopterygids are known from the Amazon.

Included Genera:

1. *Brunneria* Saussure, 1869
2. *Coptopteryx* Saussure, 1869

2.5.3.1.4 Liturgusidae Giglio-Tos, 1915

Type genus: *Liturgusa* Saussure, 1869

Diagnosis. Small to medium-sized species (14–52 mm) with a characteristic flattened body and disruptive colouration, often consisting of a mottled pattern combining tones of green, brown, grey, pale pink and black (Svenson, 2014b) (Fig. 2.2C). Most species exhibit little variation in their external morphology, and main differences are mostly allometric in nature. However, species differentiation through male genitalia is feasible. Sexual dimorphism is moderate, mostly based on size and robustness. The ocelli are highly reduced in males, usually as large as or only slightly larger than those of females, and do not sit on large cuticular protuberances as in males of other taxa. Liturgusids exhibit the following combination of characters: (i) forefemora with a circular pit in the vicinity of the proximal-most posteroventral spine (pit accommodates the

distal-most posteroventral spine of the corresponding tibia when flexed against the femur); and (ii) mid- and hindlegs very long, their corresponding femora flattened, basally thickened but steadily tapering distally.

Egg cases. Usually spherical (‘maraca-like’), single chambered and with a single emergence opening located at the distal end of a tubular projection of variable size (Fig. 2.13I–Q). Females of some species may add a thin layer of whitish, frothy secretion that partially covers the egg case (as in *Liturgusa nubeculosa* Gerstaecker, 1889; Fig. 2.13N), whereas a similar secretion seals the single emergence opening of unhatched egg cases. Females commonly attach egg cases to bark by its posterior or posterodorsal aspect, sometimes also on relatively flat surfaces within their habitat (stems or leaves) (Figs 2.11J; 2.13I). The external wall is brown to chestnut-coloured, normally thick and rigid, but sometimes appearing thinner, reticulated and translucent in ‘inflated’ morphs, so that the egg mass can be seen within, sitting at the core of the egg case (Fig. 2.13L). It is unknown if the same female can produce both the normal and ‘inflated’ forms. Liturgusid egg cases were depicted in the literature as early as Shelford (1909), and more recently by Svenson (2014b).

Natural history. All species live on tree trunks (Fig. 2.2C), often preferring bare bark or with little epiphytic coverage (Svenson, 2014b). My new concept of Liturgusidae is restricted to Neotropical species only, ranging from central Mexico south to the Atlantic rainforest of southern Brazil. Most species are found in humid, tropical environments, although *Liturgusa maya* Saussure and Zehntner, 1894) is also found in drier, seasonal habitats (Arteaga *et al.*, 2014; Svenson, 2014b).

Remarks. The name *Liturgusa* Saussure, 1869 is in prevailing use over its original spelling, *Liturgousa*. Svenson (2014a) provided a detailed justification for the conservation of the name *Liturgusa* (derived from an unjustified emendation attributed to Stål, 1877), thus making it the nominotypical genus for the taxon name Liturgusidae.

Included Genera:

1. *Corticomantis* Svenson, 2014
2. *Fuga* Svenson, 2014

3. *Hagiomantis* Saussure and Zehntner, 1894

4. *Liturgusa* Saussure, 1869

5. *Velox* Svenson, 2014

2.5.3.1.5 Photinaidae Giglio-Tos, 1915

Type genus: *Photina* Burmeister, 1838

Diagnosis. Rather variable in body form and colouration, with green as the most common and widespread colour among species of the group (Fig. 2.2D–F); a few species observed as yellowish brown (Fig. 2.2G), often exhibiting contrasting dark markings on the body, limbs and wings (Fig. 2.2H). Size variation is also extreme, ranging from small to very large, with members of *Microphotina* (min. 21 mm) and *Macromantis* (max. 110 mm) representing extremes. They exhibit the following combination of characters: (i) forefemora normally exhibit 5–6 (exceptionally 4, 7 or 8) posteroventral spines (Fig. 2.8A) and 3–4 discoidal spines (intrageneric variation in the number of these spines is common); (ii) forewings of males normally colourless and hyaline, iridescent and delicate-looking (Fig. 2.2E), whereas the same in females are normally green (rarely yellowish brown) and opaque (Fig. 2.2D, F, H); (iii) hindwings of males hyaline and colourless, whereas in females (except *Macromantis*) the same exhibit pigmented areas (e.g. yellow or green cross-veins, or large areas in crimson, dark brown, yellow or orange); and (iv) ventral phallomere of male genitalia with the distal margin smooth (i.e. without processes or specialized appendages) (indicated by a black arrow in Fig. 2.10S), and the lateroproximal margin of the same bears a well-developed process of various shapes and lengths, straight to strongly recurved, projecting along the right margin of the phallomere, sometimes associated with a membranous lobe (Fig. 2.10O–S).

Egg cases. Photinaidae egg cases exhibit more within-clade differences than any other PEPM, and three main morphs can be observed. Despite such differences, a commonality to all Photinaidae egg cases so far examined (unknown only in *Photiomantis*) relates to the type of attachment. Females typically attach their egg cases by their proximal or ventroproximal aspect in such way that its ventral surface remains away from the substrate (i.e. unattached) (Fig. 2.14A–K). Egg cases are fixed by a ring-like process that encircles the substrate (indicated by a

black arrow in Figs 2.11G; 2.14C, E–G, I, K), the only exception being *Macromantis*. In the latter such a structure is not so precisely defined; instead, the substrate is deeply embedded within the egg case itself (Fig. 2.14B). The three identified morphs are:

Macromantis-type (Fig. 2.14A, B). Greenish or yellowish brown, more lightly-coloured distally; oblong in shape, distally tapering, sometimes with a short, filiform residual process; external wall layered, coriaceous and rough; each side of the egg case with a conspicuous, longitudinal suture that converge distally; egg chambers and the external wall is separated by a layer of material containing multiple air-filled bubbles (Fig. 2.11B) (this layer can be detected, without dissection, by gently squeezing the egg case to perceive slight resistance to compression and a distinct crackling noise). Egg case forms a 90–130 degrees angle with the substrate; emergence area well-defined along the dorsal aspect, but the emergence openings are not easily visible as they are concealed by flexible, flap-like triangular projections of the surrounding egg case walls; these projections seal the chamber openings until hatching occurs. Roy (2002b) depicted the egg case of *Macromantis hyalina* (De Geer, 1773).

Cardioptera-type (Fig. 2.14C). Green or yellowish brown; oblong in shape, dorsal edge prolonged into a very long and tapering, residual process; external wall smooth; internally, egg chambers are surrounded by a thin layer of material containing multiple air-filled bubbles, altogether separated from the external wall by a gap of empty space (Fig. 2.11C); emergence area well delimited and chamber openings visible along the mid dorsal axis. Females attach their egg case by its proximal or ventroproximal end by means of a ring-like process that encircles the substrate, typically a tree branch, twig or similar structures (Fig. 2.14C). da Costa-Lima (1938) depicted an unidentified egg case that seemingly corresponds to *Cardioptera*, whereas Heitzmann-Fontenelle (1969) reported and described the egg case of *Cardioptera brachyptera* Burmeister, 1838.

Photinae-type (Fig. 2.14D–K). Colouration dark to light chestnut brown, sometimes greenish; shape sub-spherical to barrel-like; external walls smooth. Internally, the egg chambers can be either separated from the external walls by an intermediate layer of fibrous, loose material, or in direct contact with it (Fig. 2.11D, E). Internal configuration of egg case seems to depend on the thickness of the external walls: thinner walls are lighter and not in contact with the egg chambers

(Fig. 2.11D) (in this latter case, egg cases appear ‘inflated’ and tend to be spherical, as in Fig. 2.14K), whereas thicker walls are darker and in contact with the egg chambers (Fig. 2.11E). It is unknown if both types of egg cases can be produced by the same female. Emergence area well defined, chamber openings fully visible and sometimes slightly elevated relative to the egg case surface, projecting into a short and rigid residual process (Fig. 2.14H). Egg cases without external coating, but unhatched egg cases exhibit a layer of whitish material along the emergence area sealing the chamber openings (remains of this sealing layer are visible in Fig. 2.14H). Females attach egg cases to the substrate by means of a small, ring-like process that encircles it (Fig. 2.14C, E–G, I, K), typically twigs, branches or similar, sometimes on leaves (Jantsch, 1984; Rivera, 2010a).

Natural history. Members of Photinaidae are more commonly distributed in tropical, humid rainforest, but a number of species occur in seasonal habitats east and south (Caatinga, Cerrado) (Menezes and Bravo, 2013), as well as north (llanos orientales) (Agudelo and Chica, 2001) of the Amazon. *Macromantis* is the only genus of the family that ranges outside South America, reaching as far north as Nicaragua (Roy, 2002b). The great majority of species exhibit generalist morphology, suggesting that they are mostly vegetation dwellers. (Fig. 2.2D–F). However, members of *Paraphotina* and *Orthoderella* are branch-dwellers (Fig. 2.2G, H), and thus show a more specialized morphology and behaviours that depart from ‘typical’ Photinaidae (Brunner and Gandolfo, 1990; Roy and Stiewe, 2011).

Remarks. My proposed concept of Photinaidae is restricted to include only the Photinaini *sensu* Ehrmann (2002) and Otte and Spearman (2005). My phylogenetic analyses resulted in two conflicting topologies, where the only common relationship was *Macromantis* as sister to the remaining Photinaidae. Considering the morphological differences in egg case structure and male genital configuration among *Macromantis*, *Cardioptera* and *Photiomantis*, and among these and the remaining photinaids (namely *Orthoderella*, *Paraphotina*, *Microphotina*, *Chromatophotina*, *Hicetia*, *Photina*, *Metriomantis* and *Photinella*), I establish higher-level groups (subfamily) for these four groups under the guidance of the ML topology (Fig. 2.4): Macromantinae, Photiomantinae, Cardiopterinae, and Photinainae. My analyses of egg case structure revealed at least three morphs (see *Diagnosis* above and Fig. 2.14A–K). With the sole exception of *Photiomantis* (whose egg cases remain unknown), such discrete, structural differences are

reflected in the groupings recovered in the ML analysis. A similar situation is derived from male genitalia. The Macromantinae (Fig. 2.10O), Cardiopterinae (Fig. 2.10P) and Photiomantinae (not shown, but comparable to Fig. 2.10O and 2.10P; see also Agudelo and Rivera, 2015: 348) all exhibit a less-specialized lateral process on the right margin of the ventral phallomere, where this appears to be a continuation of the phallomere itself in terms of both its level of differentiation and sclerotization. This contrasts with the condition observed in Photinainae, where the same structure is clearly differentiated, showing higher structural variability and various degrees of sclerotization and specialization (Fig. 2.10Q–S). Thus, I favour the ML over the BI topology for the formulation of the Photinainae classification, as it best conforms to my morphological observations. Svenson and Branham (2007) recently addressed a long-standing nomenclatorial issue existing between Photinae Olivier, 1907 (Coleoptera, Lampyridae) and Photinae Giglio-Tos, 1915 (Mantodea, Mantidae). The authors proposed changing the root Photina- to Photina- so that the subfamily name will be emended to ‘Photinainae’ Giglio-Tos, 1915. Although prevailing usage of names is to be maintained until the ruling of the Commission is published (ICZN, 1999; Art. 82), I consider that the proposed resolution for this case of homonymy is otherwise straightforward. In order to prevent further nomenclatorial actions in the future, I choose to apply the emended root to family group taxa within the context of my new classification.

Macromantinae Brunner von Wattenwyl, 1893

Type genus: *Macromantis* Saussure, 1871

Diagnosis. The following diagnostic features identify this taxon: (i) large size (80–110 mm); (ii) lateral edges of pronotal prozona expanded, forming a rudimentary lamella, wider along the prozona than along the lateral margins of the metazona (Fig. 2.8B); (iii) forefemora with 5–6 posteroventral (exceptionally as few as 4 or as many as 8) (Fig. 2.8A) and 4 discoidal spines; (iv) ventral phallomere of male genitalia with its right, lateroproximal margin exhibiting a well-developed, flat and blunt process (Fig. 2.9O), which is similar to the rest of the phallomere in texture and level of sclerotization; and (v) dorsal lamina of left phallomere strongly sclerotized medially, cuticle darkened.

Natural history. These are large, vegetation-dwelling praying mantises (Fig. 2.2D). Label data of museum specimens indicate that they are distributed in wet, tropical environments from Nicaragua through Central America into the Amazon basin.

Remarks. Brunner von Wattenwyl (1893) applied the taxo name ‘Macromantides’ to group both *Macromantis* and *Ardesca* Stål, the latter accepted as a synonym of *Parastagmatoptera* (Mantidae: Stagmatopterinae) (Lombardo *et al.*, 2015).

Included Genera:

1. *Macromantis* Saussure, 1871

Photiomantinae Rivera and Svenson, 2016

Type genus: *Photiomantis* Piza, 1968

Diagnosis. The following combination of characters identify this subfamily: (i) head strongly flattened; (ii) juxtaocular bulges reduced to small, un conspicuous elevations (Fig. 2.8D); (iii) compound eyes appearing elongated in the lateral view (Fig. 2.8E); (iv) lower frons (head’s frontal shield of authors) large, 1.5× as wide as long, its upper-most angle pointy; and (v) ventral phallomere of male genitalia with its right, lateroproximal margin exhibiting a well-developed, flat and blunt process, similar to the rest of the phallomere in terms of texture and level of sclerotization (similar to Fig. 2.10O; see also Agudelo and Rivera, 2015: 348).

Natural history. Restricted to northeastern Brazil, associated with seasonal, semi-arid habitats (Menezes *et al.*, 2013; Menezes and Bravo, 2015).

Remarks. This is the first time that *Photiomantis* is included in a phylogenetic analysis. Its only two known species superficially resemble members of *Photina* (Agudelo and Rivera, 2015), but my phylogenetic analyses did not recover a sister-group relationship between these two genera. Distinct features of male genital morphology and other characters found in head and pronotum, in addition to the phylogenetic evidence, clearly set these two lineages apart. Rivera (2010b) and

Agudelo and Rivera (2015) summarized the complicated taxonomic and nomenclatorial history of this genus.

Included Genera:

1. *Photiomantis* Piza, 1968

Cardiopterinae Rehn, 1911

Type genus: *Cardioptera* Burmeister, 1838

Diagnosis. The following diagnostic features identify this subfamily: (i) head with strongly concave vertex; (ii) forecoxae bearing large spines along dorsal and ventral edges (Fig. 2.8C); (iii) forefemora with 5 posteroventral spines and 3 discoidal spines; (iv) mid- and hindfemora of females with an evident longitudinal, ventral carina, distally forming a lobe (carina and lobes inconspicuous in males); (v) subgenital plate of males ventrally with a large, shiny black spot; and (vi) ventral phallomere of male genitalia exhibiting a well-developed, flat and blunt process on its right, lateroproximal margin, similar to the rest of the phallomere in terms of texture and level of sclerotization (Fig. 2.10P).

Natural history. These are vegetation-dwelling praying mantises (Terra, 1992). *Cardioptera* is a distinct genus distributed in north and southern South America, in both rainforest (such as *Cardioptera squalodon* Werner, 1925 in French Guiana) and seasonal habitats (such as *Cardioptera parva* Beier, 1942 in northeastern Brazil) (Heitzmann-Fontenelle, 1969; Menezes *et al.*, 2013). Terra (1992) reported egg case-guarding behaviour in females of *Cardioptera brachyptera* Burmeister, 1838 in eastern Brazil.

Remarks. Historically, the placement of *Cardioptera* has been controversial due to a number of unique morphological features not seen in other Photinaidae. For instance, some authors used the presence of a vague, yet noticeable, ventral carina on the mid- and hind tibiae to justify its placement within the Neotropical Vatinae (Mantidae), which exhibit multiple, well-developed carinae (e.g. Brunner von Wattenwyl, 1893; Kirby, 1904; Rehn, 1911; Giglio-Tos, 1914, 1927; Heitzmann-Fontenelle, 1969). However, Beier (1935b) correctly recognized its relationship to

the Photinainae, an opinion more recently followed by Terra (1995), Ehrmann (2002) and Otte and Spearman (2005). The taxon Cardiopterinae is thus based on the ‘Cardiopterae’ group of Rehn (1911) (Roy, 2014), who presumably considered it equivalent to tribal rank.

Included Genera:

1. *Cardioptera* Burmeister, 1838

Photinainae Giglio-Tos, 1915

Type genus: *Photina* Burmeister, 1838

Diagnosis. These are small to mid-sized mantises (30–55 mm) with the following combination of characters: (i) forefemora with 5–6 posteroventral spines and 3–4 anteroventral spines; and (ii) ventral phallomere of male genitalia with its right lateroproximal margin exhibiting a sharp, well-developed process that is either straight, curved or strongly recurved, normally well differentiated from the rest of the phallomere in terms of texture and degree of sclerotization (Fig. 2.10Q–S). On the basis of the variation of the lateral process of the ventral phallomere of males, I further justify the subdivision of Photinainae into three tribes: Microphotinini, Orthoderellini and Photinaini.

Microphotinini Rivera and Svenson, 2016

Type genus: *Microphotina* Beier, 1935

Diagnosis. These are the smallest members of the Photinaidae (21–38 mm), with green body colouration. The following combination of characters distinguish the tribe: (i) male genitalia with apofisi falloide of the left phallomere either short or long, recurved and also with sharp apex; (ii) right, lateroproximal margin of the ventral phallomere with partially sclerotized process that is either short or long, and with sharp apex (Fig. 2.10R); and (iii) hindwings of females opaque, brightly coloured in yellow and/or orange.

Natural history. Little is known about these seemingly vegetation-dwelling and rarely collected praying mantises. The

available data, which are meagre, suggest a distribution within the northwestern Amazon Basin (Rivera, 2010a; Medellín and Salazar, 2011; Ippolito and Lombardo, 2012). Members of both known genera were found among canopy fogging samples from Peru (Smithsonian Institution, J. Rivera, personal observation) and males are occasionally caught in light traps (J. Rivera and G.J. Svenson, personal observation).

Remarks. During the course of my investigation, I found evidence of two undescribed genera. On the one hand, although only one of these new genera was sampled for DNA (labelled as ‘*Microphotinini* **gen.n. sp.n.**’ in Fig. 2.4), the diagnosis above incorporates the variation observed in their lateral process of the ventral phallomere to assist the recognition of these undescribed taxa as members of this tribe. On the other, François and Roy (2015) suggested that *Microphotina viridescens* could represent a yet unrecognized genus related to true *Microphotina* (on the basis of *Microphotina vitripennis*). My phylogenetic evidence shows that *M. cf. viridescens* is paraphyletic relative to *Microphotina* **sp.n.**, thus adding support to François and Roy’s interpretation.

Included Genera:

1. *Chromatophotina* Rivera, 2010
2. *Microphotina* Beier, 1935

Orthoderellini Giglio-Tos, 1917

Type genus: *Orthoderella* Giglio-Tos, 1897

Diagnosis. Members of this tribe include small to mid-size species (35–50 mm), with mainly light brown body colouration and yellowish/green tones (often with scattered, dark undulating marks in pronotum and prothoracic legs) (Fig. 2.2G, H). The following combination of characters distinguishes the tribe: (i) male genitalia with ventral phallomere exhibiting a well differentiated lateral process, often with a longitudinal region that is clearly more sclerotized

(and thus darker) than the rest of the phallomere (Fig. 2.10Q); (ii) right apodeme of right phallomere of males ventrally bearing conspicuous spiniform setae (Fig. 2.10T); (iii) cross-veins of forewings of females darkly pigmented (Fig. 2.2H) (to a lesser extent in males, often unpigmented); and (iv) hindwings of females exhibit crimson and black pigmentation (the same are largely hyaline and unpigmented in males).

Natural history. Members of this lineage are branch-dwellers, placing their body tightly aligned to branches and twigs (Brunner and Gandolfo, 1990) (Fig. 2.2G, H). They are mostly distributed in seasonal habitats in northern and southern South America (Agudelo and Chica, 2001; Roy and Stiewe, 2011; Menezes and Bravo, 2013; Arteaga *et al.*, 2014), with some species found in tropical rainforest (Lombardo, 1998; Rivera, 2003a). da Costa-Lima (1938) depicted an unidentified egg case that seems to correspond to *Orthoderella* and Roy and Stiewe (2011) did the same for *Orthoderella ornata* Giglio-Tos, 1897.

Remarks. The species *Orthoderella caatingaensis* Menezes and Bravo, 2013 included in my phylogeny renders *Orthoderella* paraphyletic relative to *Paraphotina*. However, this species seems to be misclassified and a member of *Paraphotina* instead (A. Agudelo, personal communication).

Included Genera:

1. *Orthoderella* Giglio-Tos, 1897
2. *Paraphotina* Giglio-Tos, 1915

Photinaini Giglio-Tos, 1915

Type genus: *Photina* Burmeister, 1838

Diagnosis. Members of this tribe include mid-sized species (40–60 mm) with entirely green body colouration. The following combination of characters distinguish the tribe: (i) ventral phallomere of male genitalia with a well-sclerotized and strongly recurved process of variable shape on the right lateroproximal margin, the same is associated with a conspicuous membranous lobe (Fig. 2.10S); and (ii) hindwings of females largely colourless and hyaline, except for the cross-veins

and the area immediately around them, which exhibit either green or yellow pigmentation (absent in males).

Natural history. Label data of museum specimens and literature records indicate that *Hicetia*, and species attributed to *Photinella* and *Photina* occur east and south of the Amazon (Menezes *et al.*, 2013), whereas species attributed to *Metriomantis* exhibit a more typical (northern) Amazonian distribution (Lombardo, 1999). Terra (1996) reported female egg case-guarding behaviour in a species of *Photina* from northeastern Brazil (cited as *P. amplipennis* Stål, 1877, but likely a misidentification). Terra (1996) regarded this species as arboricolous, with individuals having preference for perching among the vegetation, flattening their body against the underside for cryptic effect. A similar behaviour was observed in *Metriomantis boliviana* Lombardo, 1999 in Peru (J. Rivera, unpublished data).

Remarks. My concept of Photinaini has no equivalent in the historical literature. Although no critical assessment of the taxonomic characters justifying the validity of *Photina*, *Photinella* and *Metriomantis* exists in the recent literature (although see Lombardo, 1999), my phylogenetic analyses and morphological examinations (including type material) revealed a complex taxonomic and nomenclatorial scenario. Several species currently assigned to *Photinella* in Ehrmann (2002) have navigated through *Photina* (e.g. *brevis* Rehn, 1907, *magna* Giglio-Tos, 1915), *Metriomantis* (*biramosa* Saussure and Zehntner, 1894) and *Photiomantis* (*silvai* Piza, 1868, currently a synonym of *Photiomantis planicephala* (Rehn and Hebard, 1916), see Agudelo and Rivera, 2015). A similar scenario occurred with *Metriomantis*, which at times has been considered a subgenus of *Photina* (Beier, 1935b; Terra, 1995) or a genus on its own (Lombardo, 1999; Agudelo *et al.*, 2007). It is possible that *Photinella* and *Metriomantis* were created on the basis of characters reflecting interspecific and/or regional variation within *Photina*. The variability observed in the number of discoidal and posteroventral spines in the forefemora, now confirmed as a widespread feature of Photinaidae in general, might be the origin of the confusion. Such disparity is often useful for generic diagnosis in other Mantodea taxa, but the Photinaidae seems to be an exception in this respect. It is now evident to me that a thorough taxonomic treatment to define the taxonomic boundaries among members of this tribe is urgently needed.

Included Genera:

1. *Hicetia* Saussure and Zehntner, 1894
2. *Metriomantis* Saussure and Zehntner, 1894
3. *Photina* Burmeister, 1838
4. *Photinella* Giglio-Tos, 1915

2.5.3.1.6 Stenophyllidae Saussure, 1869

Type genus: *Stenophylla* Westwood, 1845

Diagnosis. This is one of the most easily recognizable groups, with its members showing an aberrant morphology (Fig. 2.3A). Body colouration ranges from dark to reddish brown. The following combination of characters distinguishes the family: (i) head with three projections, one central and two lateral, such structures are also sexually dimorphic in shape and size (all more developed in females) (Fig. 2.8G, H); (ii) forefemora strongly triangular in cross-section, unusually large relative to body size compared to other mantises; (iii) ventral aspect of the forefemora and tibiae wide, entirely flat and smooth; (iv) forefemora 6–10 posteroventral spines, and tibiae with 25–41 posteroventral and 20–21 anteroventral spines; (v) mid- and hindfemora with large, foliaceous lobes; and (vi) cerci with their proximal-most segments cylindrical, whereas the distal segment is considerably enlarged, flattened and foliaceous.

Egg case. Unknown.

Natural history. Overall habitus and colouration suggest that *Stenophylla* falls within the dead-leaf cryptic strategists. Salazar (2004) mentioned that *Stenophylla* is found in leaf litter, although no further observations exist to confirm this. The scant literature records suggest an Amazonian lowland distribution for two of the three known species. Salazar (2007) reported *S. lobivertex* Lombardo, 2000 for Moca, Putumayo in Colombia at 500 m a.s.l., whereas *S. cornigera* Westwood, 1843 is endemic to the Atlantic rainforest of southern Brazil. Lombardo (2000a) and Roy (2005) provided a detailed taxonomic treatment of *Stenophylla*.

Remarks. Saussure (1869) first applied the name ‘*Stenophyllites*’ to *Stenophylla* Westwood, 1845 in the family group concept.

Included Genera:

1. *Stenophylla* Westwood, 1845

2.5.3.1.7 Acontistidae Giglio-Tos, 1915

Type genus: *Acontista* Saussure, 1872.

Diagnosis. The Acontistidae includes small-sized species (15–31 mm), usually with a short and compact body (Fig. 2.3B–F). Although brown forms exist (Fig. 2.3D, F), most species tend to be green, sometimes in combination with yellow, brown and even white (Fig. 2.3B, C, E). The following combination of characters distinguishes the family: (i) forefemora with 5 posteroventral and 3 discoidal spines; (ii) hindwings of females extensively pigmented (single coloured or polychromatic), colours include crimson, orange, dark brown (sometimes with purplish reflections) or pink, often in combination with hyaline regions, whereas in males the same vary from hyaline and colourless to decidedly pigmented as in females. Members of *Tithrone* are remarkable for having asymmetrically-coloured forewings, one green and the other pinkish (Barabás and Hancock, 1999); and (iii) ventral phallomere of male genitalia is rhomboidal in shape, normally with a proximal lateral process on its right margin associated with a large membranous area (Fig. 10V), in addition to a noticeable and well-sclerotized distal process of various sizes and shapes (simple or forked) (Fig. 2.10W–Z).

Egg cases. Usually guttiform (i.e. ‘drop-like’) in shape, less commonly rectangular or barrel-like; in all cases a filiform, residual process projects distally (Fig. 2.14L–P). Egg cases exhibit a conspicuous outer coating consisting of a layer of whitish to intensely sulphur yellow-coloured material (coating fades to pale yellow or brown in specimens exposed to the elements for long periods of time). External walls of egg cases are brown to reddish brown (Fig. 2.14M). The attachment is normally by its posterior end, so that the emergence area is either perpendicular to (Fig. 2.14M) or forming an angle relatively with the substrate (Figs 2.11H; 2.14L, N–P). In the

latter case, the ootheca can be slightly tilted to decidedly reclined, so that the emergence area comes closer to the substrate itself. The most extreme example is in *Callibia diana* (Stoll, 1813), where the egg case virtually rests on its dorsal aspect (i.e. upside-down), so what appears to be the dorsal surface is actually its ventral side (Figs 2.11I; 2.14N, O). Females usually fix their egg cases to flat surfaces, commonly leaves (Fig. 2.14O), but sometimes to bark or twigs. The egg case of *Tithrone* and *Paratithrone* were not available for study. However, Beebe *et al.* (1952) depicted the egg case of *Tithrone roseipennis* (Saussure, 1870) from Trinidad; this fits the above description well except that the attachment is ventral (they also did not mention any external coating).

Natural history. Early instars of *Acontista multicolor* Saussure, 1870 exhibit ant mimicry, both in overall appearance, coloration and behaviour, possibly using *Pseudomyrmex* Lund (Hymenoptera: Formicidae, Pseudomyrmecinae) as a model (Salazar, 2003). Adults are often found on bushes and lower vegetation, sitting on leaves, branches, and near or among inflorescences (Beebe *et al.*, 1952; Salazar, 2003; J. Rivera, personal observation). The Acontistidae are widely distributed across the neotropics, mostly in tropical, humid habitats from central Mexico through Central America, such as the Amazon and the Atlantic rainforest (Roy, 2006), and occasionally in seasonal habitats (Arteaga *et al.*, 2014). Salazar (2002) reported *A. multicolor* and *A. cordillerae* Saussure, 1869 between 1000 and 1800m a.s.l. for Andean habitats in Colombia. Only *A. multicolor* is known from the Caribbean area, specifically from Trinidad (Beebe *et al.*, 1952) and Guadeloupe (Kirby, 1904), distribution I confirmed on the basis of museum specimens; the record of *Acontista cubana* de Zayas, 1974, described from Cuba, seems to correspond to a misidentified member of *Callimantis* Stål, 1877 (Mantidae: Stagmomantinae), a Caribbean endemic.

Remarks. Although only members of *Acontista*, *Raptrix* and *Callibia* were available for molecular analyses, my morphological examinations inclusive of male genitalia confirmed *Tithrone* and *Paratithrone* as members of this family. On the basis of male genital structures, I also confirmed *Ovalimantis* as member of this family (Roy, 2015). My analyses recovered *Acontista bimaculata* as sister to *Raptrix* with high support (Fig. 2.4), rendering *Acontista* as paraphyletic relative to *Raptrix*. This finding is not particularly surprising, because this species departs from typical *Acontista* in many respects. For instance, *A. bimaculata* (Fig. 2.3C) differs

from *Acontista* spp. in having brachypterous females (macropterous in *Acontista* spp.) (Fig. 2.3B) and ventral phallomere of males with a secondary, spine-like distal process (Fig. 2.10W); see also Heitzmann-Fontenelle, 1968), which is absent in *Acontista* spp. (Fig. 2.10X–Z; see also Heitzmann-Fontenelle, 1965; Lombardo and Agabiti, 2001). Label data of museum specimens suggest that *A. bimaculata* is ecologically associated with seasonal habitats south of the Amazon, whereas *Acontista* spp. are essentially restricted to humid environments. My phylogenetic evidence, in addition to observations of morphology and geographic distribution, all point to *A. bimaculata* as distinct from the remaining *Acontista*. In fact, my examination of the type specimens of *A. bimaculata* (deposited at Muséum d’histoire naturelle de la Ville de Genève, Switzerland) and *Metaphotina piracicabensis* Piza, 1964 (deposited at Escola Superior de Agricultura Luiz de Queiroz, Universidade de São Paulo, Brazil), in addition to numerous other specimens from northern Argentina, southern Bolivia, southern Brazil and Paraguay, reveal that they all consistently exhibit the character states described above for *A. bimaculata*, supporting that they are members of the same lineage. I anticipate that the revalidation of *Metaphotina* Piza (currently a synonym of *Acontista*) would thus be necessary to accommodate, in addition to *M. piracicabensis* and *A. bimaculata*, two other species that also clearly belong to this distinct group, namely *Acontista brevipennis* Saussure, 1872 and *Acontista rehni* Giglio-Tos, 1927. This question will be further elaborated in a forthcoming publication (J. Rivera and G.J. Svenson, in preparation).

Included Genera:

1. *Acontista* Saussure, 1872
2. *Callibia* Stål, 1877
3. *Ovalimantis* Roy, 2015
4. *Paratithrone* Lombardo, 1996
5. *Raptrix* Terra, 1995
6. *Tithrone* Stål, 1877

2.5.3.1.8 Family Acanthopidae Burmeister, 1838

Type genus: *Acanthops* Audinet Serville, 1831

Diagnosis. The Acanthopidae includes mid-sized species (25–55 mm) with highly specialized, cryptic morphology (Fig. 2.3G–I) and conspicuous sexual dimorphism. Colouration normally occurs in tones of brown (Fig. 2.3G), but sometimes tones of green predominate as in *Pseudacanthops* (Fig. 2.3H). They exhibit the following combination of characters: (i) compound eyes with a conspicuous apical process of variable size (Figs 2.6B; 2.8K); (ii) forefemora with 6–7 posteroventral spines (Figs 2.6B; 2.8I) and 4 discoidal spines; and (iii) forewings resembling dry or partially dried leaves, with irregular edges and/or distal lobes of various sizes (Fig. 2.3H, 2.I).

Egg cases. Greatly elongated and stalked (Figs 2.11K; 2.14Q–U). The stalk arises from the anterior end; it can be either short and thick (Fig. 2.14Q, U), or very long (Fig. 2.14R, S) (as much as twice as long as the egg case itself) and thread-like (Fig. 2.14T). As a result, the egg case dangles from the substrate, typically a branch or bark, somewhat resembling a dried seedpod or a curled, dead leaf. Egg case colour varies from brown to reddish brown, the external walls are coriaceous, smooth and flexible, and they lack an external coating. Egg cases documented in the literature include those of *Decimiana bolivari* (Chopard, 1916) (Travassos Filho, 1945) *Acanthops falcata* Stål, 1877 (Beebe *et al.*, 1952) and *Acanthops centralis* Lombardo and Ippolito, 2004 (Salazar, 2006a).

Natural history. The Acanthopidae are mostly dead-leaf mimics. Females (Fig. 2.3G) hang upside down from twigs and branches resembling dry, curled leaves (Robinson, 1969a; Robinson and Robinson, 1979; Edmunds and Brunner, 1999). *Pseudacanthops* females and juvenile males bear strong resemblance to moss (Lombardo *et al.*, 2013) (Fig. 2.3H). Adult males of all genera tend to rest both on top of vegetation or tree branches, where they resemble fallen or hanging dry to partially dried leaves (Lombardo *et al.*, 2013; J. Rivera, personal observation) (Fig. 2.3I). The Acanthopidae are widely distributed from central Mexico south to the Atlantic rainforest of southern Brazil (Lombardo, 2000b; Agudelo *et al.*, 2007; Lombardo and Ippolito, 2004; Roy, 2004; Lombardo *et al.*, 2013). They are more commonly found in humid, tropical areas, although *Acanthops falcata* and most members of *Decimiana* are associated with more seasonal habitats north, and south and east of the Amazon basin, respectively (Lombardo, 2000b; Menezes and Bravo, 2012; Arteaga *et al.*, 2014).

Included Genera:

1. *Acanthops* Audinet Serville, 1831
2. *Decimiana* Uvarov, 1940
3. *Lagrecacanthops* Roy, 2004
4. *Metilia* Stål, 1877
5. *Miracanthops* Roy, 2004
6. *Pseudacanthops* Saussure, 1870

2.6 Conclusions

The higher-level systematics of Mantodea historically has been hampered by the unusual profusion of homoplastic characters, as well as the evolution of radically divergent ecomorphs even in closely related taxa (e.g. Thespidae). Numerous taxonomic and systematic problems have stemmed from the historical misinterpretation of such characters. My comprehensive molecular phylogeny, in combination with comparative morphological data from Yager and Svenson (2008), confirmed the existence of a mainly Neotropical and plesiomorphically ‘deaf’ clade, consistent with Svenson and Whiting (2009). I have re-dubbed this lineage as ‘polymorphic earless praying mantises’ (PEPM) and granted it superfamily rank, the Acanthopoidea. This is the first study to propose a new classification for several higher-level groups of New World mantodeans on the basis of a rigorous analysis of molecular data using formal phylogenetic methods. As a result, a considerable number of taxonomic changes and nomenclatorial actions were required to reflect our new understanding of PEPM relationships. With a number of genera recovered as paraphyletic (e.g. *Acanthops*, *Acontista*, *Musonia*, *Thesprotiella*, *Metriomantis*, etc.) more in-depth systematic treatments will be required in the PEPM group. The phylogenetic framework also facilitated the detection of potentially homoplastic characters (e.g. dorsal tibial spines in Thespidae), and revealed diagnostic attributes in egg cases and male genitalia. Figure 15 summarizes the findings of this study.

Breland and Dobson’s (1947) hypothesis that egg case morphology is species-specific is only partially confirmed. Instead, my results indicate that egg case traits are clade-specific, with utility for diagnosing taxa above the genus-level. Male genitalic characters also proved useful for

delineating taxa within the taxonomically difficult Thespidae and Photinainidae. Even though our knowledge of male genitalic diversity has increased considerably over the last 20 years (Cerdá, 1993; Rivera, 2010c; Roy, 2014), little has been done to investigate the phylogenetic signal of this character system (e.g. Klass, 1997; Klass and Meier, 2006; Jensen *et al.*, 2009; Svenson *et al.*, 2016). Nevertheless, both egg cases and male genitalia were revealed as promising character systems for taxon delineation and reasonably good proxy indicators of clade coherence in a phylogenetic context. I also agree with Wieland (2013) that data on postembryonic development is likely to be fundamental to understanding evolutionary patterns of adult morphology. This is particularly relevant in the Thespidae, where juvenile features (e.g. dorsal tibial spines, shortened tibiae, smaller metazona/prozona ratios, etc.) seem to be inconsistently retained in the adults. This suggests that some form of heterochronic process, such as neoteny, could be at work. Further analyses of molecular datasets using phylogenetic methods will unquestionably pave new roads for evolutionary and ecological studies of the Mantodea. As the molecular phylogeny of Mantodea becomes more populated and better resolved, it will also point to new research directions by revealing the relative importance of particular morphological characters and their utility for resolving relationships at various hierarchical levels (Svenson *et al.*, 2015, 2016).

Tables and Figures for Chapter 2

Table 2.1. Current classification systems for polymorphic earless praying mantises, according to Ehrmann (2002) and Otte & Spearman (2005). Classification of Thespidae follows Terra (1995). Modifications from original schemes resulted from subsequent nomenclatorial procedures introduced in recent times (see corresponding footnotes for details). Only New World taxa are included. Old world species classified under any of the higher-level taxa herein listed were excluded according to criteria presented in Table 2.2.

Ehrmann (2002) system		Otte & Spearman (2005) system	
ACANTHOPIDAE	<i>Macromusonia</i>	ACANTHOPIDAE	<i>Macromusonia</i>
Acanthopinae	<i>Musonia</i>	Acanthopinae	<i>Musonia</i>
Acanthopini	<i>Musoniola</i>	Acanthopini	<i>Musoniola</i>
<i>Acanthops</i>	<i>Musoniella</i>	<i>Acanthops</i>	<i>Musoniella</i>
<i>Decimiana</i>	<i>Paramusonia</i>	<i>Decimiana</i>	<i>Paramusonia</i>
<i>Lagrecacanthops</i>	<i>Thespis</i>	<i>Lagrecacanthops</i>	<i>Thespis</i>
<i>Metilia</i>	Oligonicinae	<i>Metilia</i>	Oligonicinae
<i>Miracanthops</i>	Oligonicini	<i>Miracanthops</i>	Oligonicini
<i>Pseudacanthops</i>	<i>Bactromantis</i>	<i>Pseudacanthops</i>	<i>Bactromantis</i>
Acontistinae ¹	<i>Bantia</i>	Acontistinae ¹	<i>Bantia</i>
Acontistini	<i>Bantiella</i>	Acontistini	<i>Bantiella</i>
<i>Acontista</i>	<i>Diabantia</i>	<i>Acontista</i>	<i>Diabantia</i>
<i>Callibia</i>	<i>Oligonicella</i>	<i>Callibia</i>	<i>Oligonicella</i>
<i>Ovalimantis</i> ²	<i>Oligonyx</i>	<i>Ovalimantis</i> ²	<i>Oligonyx</i>
<i>Paratithrone</i>	<i>Pseudomusonia</i>	<i>Paratithrone</i>	<i>Pseudomusonia</i>
<i>Raptrix</i>	<i>Thesprotiella</i>	<i>Raptrix</i>	<i>Thesprotiella</i>
<i>Tithrone</i>	<i>Thrinaconyx</i>	<i>Tithrone</i>	<i>Thrinaconyx</i>
Stenophyllinae	Pogonogasterini	Stenophyllinae	Pogonogasterini
<i>Stenophylla</i>	<i>Carrikerella</i>	<i>Stenophylla</i>	<i>Carrikerella</i>
LITURGUSIDAE	<i>Liguanea</i>	LITURGUSIDAE	<i>Liguanea</i>
Liturgusinae	<i>Mantillica</i>	Liturgusinae	<i>Mantillica</i>
Liturgusini	<i>Mantellias</i>	Liturgusini	<i>Mantellias</i>
<i>Corticomantis</i> ³	<i>Pogonogaster</i>	<i>Corticomantis</i> ³	<i>Pogonogaster</i>
<i>Fuga</i> ³	<i>Thesprotia</i>	<i>Fuga</i> ³	<i>Thesprotia</i>
<i>Hagiomantis</i>	MANTIDAE	<i>Hagiomantis</i>	MANTIDAE
<i>Liturgusa</i>	Angelinae	<i>Liturgusa</i>	Angelinae
<i>Velox</i> ³	Angelini	<i>Velox</i> ³	Angelini
THESPIDAE	<i>Angela</i>	THESPIDAE	<i>Angela</i>
Pseudomiopteriginae	Photinae	Pseudomiopteriginae	Photinae
Pseudomiopterigini	Photinaini	Pseudomiopterigini	Photinaini
<i>Anamiopteryx</i>	<i>Cardioptera</i>	<i>Anamiopteryx</i>	<i>Cardioptera</i>
<i>Eumiopteryx</i>	<i>Chromatophotina</i> ⁷	<i>Eumiopteryx</i>	<i>Chromatophotina</i> ⁷
<i>Leptomiopteryx</i>	<i>Hicetia</i>	<i>Leptomiopteryx</i>	<i>Hicetia</i>
<i>Pizaia</i>	<i>Macromantis</i>	<i>Pizaia</i>	<i>Macromantis</i>
<i>Pseudomiopteryx</i>	<i>Metriomantis</i>	<i>Pseudomiopteryx</i>	<i>Metriomantis</i>
Miobantiinae ⁴	<i>Microphotina</i>	Miobantiinae ⁴	<i>Microphotina</i>
Miobantini	<i>Orthoderella</i>	Miobantini	<i>Orthoderella</i>
<i>Chloromiopteryx</i> ⁵	<i>Paraphotina</i>	<i>Chloromiopteryx</i> ⁵	<i>Paraphotina</i>
<i>Miobantia</i>	<i>Photina</i>	<i>Miobantia</i>	<i>Photina</i>
<i>Promiopteryx</i>	<i>Photinella</i>	<i>Promiopteryx</i>	<i>Photinella</i>
<i>Pseudopogonogaster</i> ⁶	<i>Photiomantis</i> ⁸	<i>Pseudopogonogaster</i> ⁶	<i>Photiomantis</i> ⁸
Thespiinae	Coptopterygini	Thespiinae	Coptopterygini
Thespini	<i>Brunneria</i>	Thespini	<i>Brunneria</i>
<i>Galapagia</i>	<i>Coptopteryx</i>	<i>Galapagia</i>	<i>Coptopteryx</i>
<i>Eumusonia</i>		<i>Eumusonia</i>	

¹ Originally cited as Acontiothespiinae/Acontiothespini in Ehrmann (2002), and modified according to Roy (2004)

² Assigned to Acontistinae by Roy (2015).

³ Assigned to Liturgusidae by Svenson (2014b)

⁴ Originally cited as Miopteryginae/ Miopterygini, and modified according to Roy (2013)

⁵ Agudelo & Rivera (2015) recently recognized *Emboicy* as synonym of *Chloromiopteryx* and thus the former is not listed here.

⁶ Transferred from Pogonogasterini (Oligonicinae) to Miobantiinae by Rivera *et al.*, (2011), with *Calopteromantis* as its synonym.

⁷ Subsequently assigned to Photinae/Photinaini by Rivera (2010a)

⁸ Initially transferred from Eremiaphilidae to Photinaini as *Margaromantis* Piza, 1982 by Rivera (2010b), subsequently established as synonym of *Photiomantis* Piza, 1968 by Agudelo & Rivera (2015).

Table 2.2. List of taxa excluded from new classification and here considered *insertae sedis*. Justification for their exclusion is provided in all cases and according to criteria specified in *Material and Methods*. Classification follows both Ehrmann (2002) and Otte & Spearman (2005), whereas discrepancies between both classifications, and systematics below family-level, are properly highlighted in the footnotes. Taxa classified after 2005 are marked with (*). Taxa formerly included among polymorphic earless praying mantises (PEPM) in Ehrmann (2002) and Otte & Spearman (2005), but reclassified recent times, are listed at the bottom of the table. Abbreviations for the type of external anatomy of the metasternum (from Yager & Svenson, 2008) are provided at the bottom of the table.

Taxa herein considered as <i>insertae sedis</i>	Current Classification	Reasons for exclusion from analyses
<i>Haania</i> spp. and <i>Astape denticollis</i> (Haaninae)	Thespidae ¹	Oriental lineage. Both exhibit DK metasternal anatomy (Yager & Svenson, 2008); females are winged but brachypterous
<i>Hoplocorypha</i> spp., <i>Apterocorypha</i> spp. and <i>Hoplocoryphella grandis</i>	Thespidae ²	Ethiopian lineage. Both sexes exhibit DO metasternal anatomy (Yager & Svenson, 2008)
<i>Pseudothespis meghalayensis</i>	Thespidae ³	Species from India. Svenson & Withing (2009) resolved it as sister to <i>Haania</i> (Haaninae) and thus unrelated to new world Thespidae
<i>Thespis dissimilis</i>	Thespidae ³	Species from India. Females are winged
<i>Parathespis humbertiana</i>	Thespidae ⁴	Species from Sri Lanka. Svenson & Withing (2009) resolved this species virtually at the other end of their phylogeny, far away from Thespidae (at node 284, see Fig. 3 in Svenson & Withing, 2009). This species is sister to a clade comprising both old and new world lineages (Mantinae, Paramantinae, Stagmatopterinae, Vatinae, Miomantinae and Antemninae)
<i>Sinomiopteryx</i> spp.	Thespidae ⁵	Oriental lineage. Exhibits DK metasternal anatomy (Yager & Svenson, 2008)
<i>Palaeothespis</i> spp.	Thespidae ⁵	Oriental lineage. Revised in Ge & Chen (2008). Remarkably similar to Neotropical <i>Pseudopogonogaster</i> , but unlikely to be a member of Thespidae
<i>Agrionopsis</i> spp.	Mantidae ⁶	Ethiopian lineage
<i>Cotigaonopsis providenceae</i> *	Mantidae ⁶	Oriental species. Schwarz & Helmkamp (2014) suggested a possible placement among the Deroplatyinae (Mantidae)
<i>Euchomenella</i> spp. & <i>Indomenella indica</i> **	Mantidae ⁶	Oriental lineages. Svenson & Withing (2009) resolved <i>Euchomenella</i> and <i>Indomenella</i> as sister genera, and together as sister to Oriental <i>Deroplatys</i> (Mantidae: Deroplatyinae). Schwarz & Helmkamp (2014) suggested a better placement among the Deroplatyinae (Mantidae). <i>Euchomenella</i> exhibits DNK metasternal anatomy (Yager & Svenson, 2008)

Taxa herein considered as <i>insertae sedis</i>	Current Classification	Reasons for exclusion from analyses
<i>Leptocola</i> spp. & <i>Stenopyga</i> spp.	Mantidae ⁶	Ethiopian lineages. Svenson & Withing (2009) resolved <i>Leptocola</i> and <i>Stenopyga</i> as sister genera, both sister to a clade comprising the also Ethiopian <i>Danuria</i> and <i>Popa</i> , the latter currently classified in the polyphyletic Vatiniae (Mantidae). In Svenson & Withing (2009), these four genera form a clade sister to another including <i>Euchomenella</i> , <i>Indomenella</i> and <i>Deroplatys</i> . <i>Stenopyga</i> exhibits DNK metasternal anatomy (Yager & Svenson, 2008)
<i>Tagalomantis</i> spp.	Mantidae ⁶	Oriental and Australian lineage. Schwarz & Helmkampf (2014) suggested a possible placement among the Deroplatyinae (Mantidae)
<i>Ciulfina</i> spp. & <i>Stenomantis novaeguineae</i>	Liturgusidae ⁷	Australian lineage. Svenson & Withing (2009) found <i>Ciulfina</i> to be sister to <i>Stenomantis</i> , which together are sister to members of the Nanomantinae (Iridopterygidae). Wieland (2013) found additional evidence for the dichotomy <i>Ciulfina</i> + <i>Stenomantis</i>
<i>Dactylopteryx</i> spp. & <i>Theopompella</i> spp.	Liturgusidae ⁷	Ethiopian lineage. Svenson & Withing (2009) resolved these two genera as sister to a clade comprising, <i>Leptocola</i> , <i>Stenopyga</i> (Mantidae, Angelinae), <i>Danuria</i> and <i>Popa</i> (Mantidae, Vatiniae) all Ethiopian. In Svenson & Withing (2009), these four genera form a clade sister to another including <i>Euchomenella</i> , <i>Indomenella</i> and <i>Deroplatys</i> . Both <i>Dactylopteryx</i> and <i>Theopompella</i> exhibit DNK metasternal anatomy (Yager & Svenson, 2008)
<i>Gonatista</i> spp.	Liturgusidae ⁷	Neotropical lineage (Caribbean area and Florida-USA). Not related to other Neotropical Liturgusidae. Exhibits DK and DNK metasternal anatomy (Yager & Svenson, 2008, JR & GJS pers. obs.). Egg cases also notably depart from those herein described for the Liturgusidae. The systematic placement of <i>Gonatista</i> remains unresolved
<i>Gonatistella nigropicta</i>	Liturgusidae ⁷	Australian species
<i>Liturgusella malagassa</i>	Liturgusidae ⁷	Ethiopian species
<i>Humbertiella</i> spp.	Liturgusidae ⁷	Oriental and Australian lineage. Svenson & Withing (2009) showed this genus to be related to Oriental members of Amelinae (Mantidae). Exhibits DNK metasternal anatomy (Yager & Svenson, 2008)
<i>Majanga</i> spp.	Liturgusidae ⁷	Oriental lineage
<i>Mellierella</i> spp.	Liturgusidae ⁷	Australian lineage
<i>Pseudogousa</i> spp.	Liturgusidae ⁷	Oriental lineage
<i>Scolodera</i> spp.	Liturgusidae ⁷	Australian lineage
<i>Theopompa</i> spp.	Liturgusidae ⁷	Oriental lineage
<i>Zouza radiosa</i>	Liturgusidae ⁷	Ethiopian lineage
<i>Epaphrodita</i> spp.	Hymenopodidae ⁸	Neotropical lineage (Caribbean area). Exhibits DK metasternal anatomy (GJS pers. obs.). Wieland (2013) and Svenson <i>et al.</i> (2015) consistently resolved <i>Epaphrodita musarum</i> outside both Hymenopodidae and PEPM. Therefore, the proposal of Terra (1995) suggesting <i>Epaphrodita</i> as a member of Acanthopinae (Acanthopidae) has no support. The systematic placement of <i>Epaphrodita</i> remains unresolved

Taxa reclassified in recent times	Former systematic placement	Current systematic placement and references
<i>Thespis dissimilis</i>	Thespidae ¹⁰	Mukherjee <i>et al.</i> (2014) recognized <i>Thespis dissimilis</i> Westwood, 1889 as a synonym of <i>Parathespis humberiana</i> , species from India and Sri Lanka. Svenson & Whiting (2009) resolved <i>P. humberiana</i> virtually at the other end of their phylogeny, far away from Thespidae (at node 284; see fig. 3 in Svenson & Whiting (2009)). This species is sister to a clade comprising both old and new world lineages (Mantinae, Paramantinae, Stagmatopterinae, Vatinae, Miomantinae and Antemninae). Females are winged
<i>Thespis disparilis</i>	Thespidae ³	Milledge & Ehrmann (2014) recently demonstrated its misplacement among the Thespidae, confirmed its identity as <i>Rhodomantis</i> and placed it in among the Mantinae (Mantidae). Distributed in Australia.
<i>Mythomantis</i> spp.	Mantidae ⁹	Schwarz & Helmkamp (2014) placed it in Deroplatyinae (Mantidae). Distributed in southeast Asia.
<i>Thespoides bolivari</i>	Mantidae ⁶	Rivera (2014) synonymized this species with <i>Angela guianensis</i> , and placed it among the Angelinae (Mantidae). Found in South and Central America
<i>Majangella</i> spp.	Liturgusidae ⁷	Svenson & Vollmer (2014) transferred this genus from Liturgusidae to Acromantinae (Hymenopodidae) (see also Svenson <i>et al.</i> (2015)). Distributed in southeast Asia.

¹ Haaninae in Ehrmann (2002), Thespinae-Haaniini in Otte & Spearman (2005); ² Hoplocoryphinae; ³ Thespinae-Thespini in Ehrmann (2002), Thespinae-Parathespini in Otte & Spearman (2005); ⁴ Thespinae: Parathespini; ⁵ Pseudomiopteryginae: Pseudomiopterygini; ⁶ Angelinae: Angelini; ⁷ Liturgusinae: Liturgusini; ⁸ Hymenopodidae: Epaphroditinae; ⁹ Deroplatyinae; ¹⁰Thespidae: Parathespinae in Mukherjee *et al.* (2014); * Vyjayandi *et al.*, (2009); ** Roy (2008). Abbreviations: DK, deep groove and prominent knobs; DNK, deep groove without knobs; DO, deep groove opened; MSMT, mesothoracic and metathoracic segments similar.

Table 2.3. New higher-level classification system for Acanthopoidea or ‘polymorphic earless praying mantises’. New higher-level taxa are underscored. Genera marked with (*) were not included in the molecular phylogeny and were classified on the basis of morphology and/or egg case features.

ORDER MANTODEA	
SUPERFAMILY ACANTHOPOIDEA	
Family THESPIDAE	
Subfamily <u>Pseudopogonogastrinae</u>	<i>Pseudopogonogaster</i>
Subfamily <u>Pseudomiopteryginae</u>	<i>Pseudomiopteryx</i> , <i>Promiopteryx</i> *
Subfamily <u>Bantiinae</u>	<i>Bantia</i> , <i>Bantiella</i> *, <i>Diabantia</i> *, <i>Mantellias</i> *, <i>Mantillica</i> , <i>Thrinaconyx</i>
Subfamily <u>Miobantiinae</u>	<i>Anamioteryx</i> , <i>Chloromiopteryx</i> *, <i>Eumiopteryx</i> , <i>Miobantia</i> , <i>Paradiabantia</i> *
Subfamily <u>Thespinae</u>	
Tribe <u>Musoniellini</u>	<i>Eumusonia</i> , <i>Leptomiopteryx</i> , <i>Musoniella</i> , <i>Pizaia</i> *
Tribe <u>Thespini</u>	<i>Carrikerella</i> , <i>Galapagia</i> , <i>Liguanea</i> *, <i>Macromusonia</i> , <i>Musonia</i> , <i>Musoniola</i> , <i>Oligonicella</i> , <i>Oligonyx</i> , <i>Paramusonia</i> *, <i>Pogonogaster</i> *, <i>Pseudomusonia</i> , <i>Thespis</i> *, <i>Thesprotia</i> , <i>Thesprotiella</i>
Family ANGELIDAE	<i>Angela</i>
Family COPTOPTERYGIDAE	<i>Brunneria</i> , <i>Coptopteryx</i>
Family LITURGUSIDAE	<i>Corticomantis</i> *, <i>Fuga</i> *, <i>Hagiomantis</i> , <i>Liturgusa</i> , <i>Velox</i> *
Family PHOTINAINIDAE	
Subfamily <u>Macromantinae</u>	<i>Macromantis</i>
Subfamily <u>Cardiopterinae</u>	<i>Cardioptera</i>
Subfamily <u>Photiomantinae</u>	<i>Photiomantis</i>
Subfamily <u>Photinainae</u>	
Tribe <u>Microphotinaini</u>	<i>Chromatophotina</i> , <i>Microphotina</i>
Tribe <u>Orthoderellini</u>	<i>Orthoderella</i> , <i>Paraphotina</i>
Tribe <u>Photinaini</u>	<i>Hicetia</i> , <i>Photina</i> , <i>Photinella</i> , <i>Metriomantis</i>
Family STENOPHYLLIDAE	<i>Stenophylla</i>
Family ACONTISTIDAE	<i>Acontista</i> , <i>Callibia</i> , <i>Ovalimantis</i> *, <i>Paratithrone</i> *, <i>Raptrix</i> , <i>Tithrone</i> *
Family ACANTHOPIDAE	<i>Acanthops</i> , <i>Decimiana</i> , <i>Lagrecacanthops</i> *, <i>Metilia</i> , <i>Miracanthops</i> *, <i>Pseudacanthops</i>

Genus inquirenda

***Astollia* Kirby, 1904**

[Note: this genus includes a single species described from Surinam, *Astollia chloris* (Olivier, 1792), which Roy (2006) considered a likely member of Acontistinae. The available taxonomic information is insufficient to determine its actual identity or systematic placement, and no types are available. See Roy (2006) for a detailed account on this genus.]

Figure 2.1. Live habitus of representative Thespidae genera. (A) *Pseudopogonogaster* Beier (Pseudopogonogastrinae), female from Azuay, Ecuador (photo by Sebastián Padrón); (B) *Pseudomiopteryx* Saussure (Pseudomiopteryginae), female with egg case from Pará, Brazil (photo by César Favacho); (C) *Thrinacoxys* Saussure (Bantiinae), male from El Castillo, Nicaragua (photo by Gavin J. Svenson); (D) *Eumiopteryx* Giglio-Tos (Miobantiinae), female from Santa Cruz, Bolivia (photo by Gavin J. Svenson); (E) *Eumusonia* Giglio-Tos (Thespinae – Musoniellini), female (green morph) from São Paulo, Brazil (photo by Julio Rivera); (F) *Musoniella* Giglio-Tos (Thespinae – Musoniellini), female from Pará, Brazil (photo by César Favacho); (G) *Galapagia peruana* (Thespinae-Thespini), female from Huarochirí, Peru (photo by Julio Rivera); (H) *Musoniola* Giglio-Tos (Thespinae-Thespini), female from Matagalpa, Nicaragua (photo by Gavin J. Svenson); (I) *Oligonyx* Saussure (Thespinae-Thespini), male from El Castillo, Nicaragua (photo by Gavin J. Svenson); (J) *Thesprotiella* Giglio-Tos (Thespinae-Thespini) female from Madre de Dios, Peru (photo by Gavin J. Svenson).

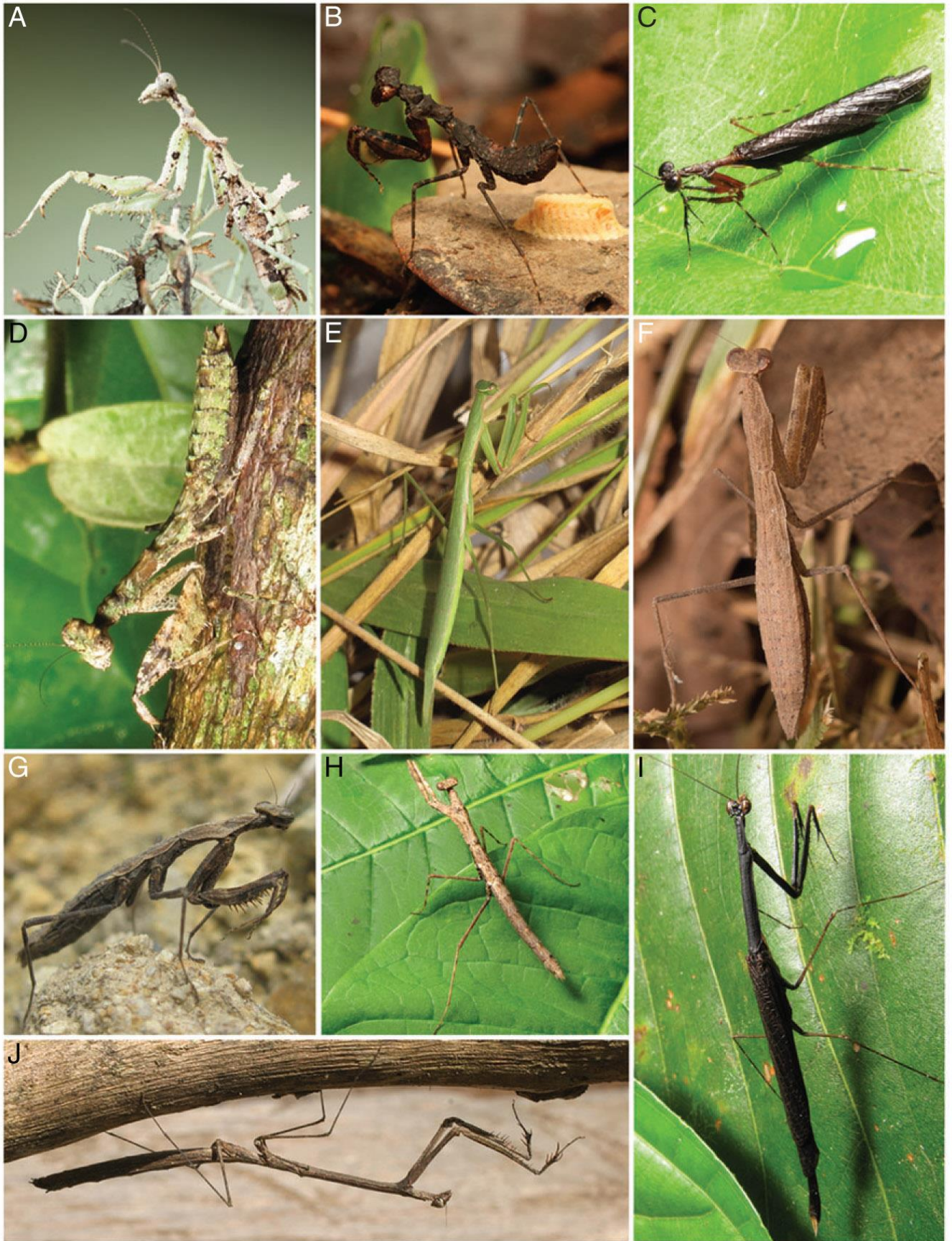


Figure 2.1

Figure 2.2. Live habitus of representative Angelidae, Coptopterygidae, Liturgusidae and Photinaidae genera. (A) *Angela* Audinet Serville, male from Santa Cruz, Bolivia (photo by Gavin J. Svenson); (B) *Brunneria* Saussure, female from Venezuela (photo by Reinhard Ehrmann, courtesy of Matthias Borer); (C) *Liturgusa* Saussure, female from Madre de Dios, Peru (photo by Christopher Darling); (D) *Macromantis* Saussure (Macromantinae), female from Iquitos, Peru (photo by Robert Oelman); (E) *Metriomantis* Saussure & Zehntner (Photinae – Photinini), male from Pará, Brazil (photo by César Favacho); (F) *Photinella* Giglio-Tos (Photinae – Photinini), female from Santa Cruz, Bolivia (photo by Gavin J. Svenson); (G) *Orthoderella* Giglio-Tos (Photinae – Orthoderellini), female (juv.) from São Paulo, Brazil (photo by Mateus Pepinelli); (H) *Paraphotina* Giglio-Tos (Photinae – Orthoderellini), female from Pará, Brazil (photo by César Favacho).

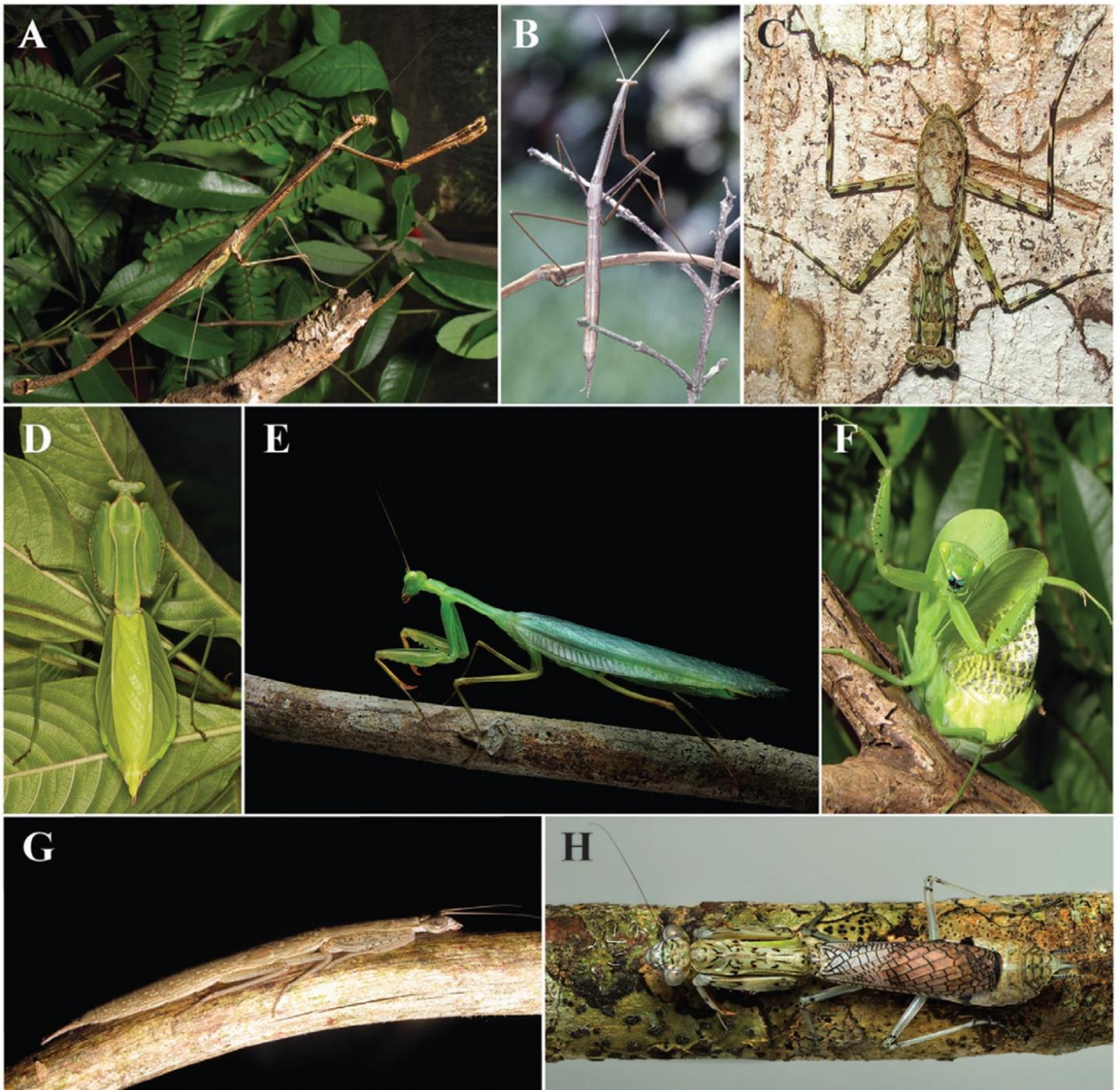


Figure 2.2

Figure 2.3. Live habitus of representative Stenophyllidae, Acontistidae and Acanthopidae genera. (A) *Stenophylla* Westwood, male from Orellana, Ecuador (photo by Robert Oelman); (B) *Acontista* Saussure, male and female from Pará, Brazil (photo by César Favacho); (C) *Acontista* Saussure (*A. bimaculata*), female from Santa Cruz, Bolivia (photo by Gavin J. Svenson); (D) *Callibia* Stål, male from Pará, Brazil (photo by César Favacho); (E) *Callibia* Stål, female from Pará, Brazil (photo by César Favacho); (F) *Raptrix* Terra, female from Pará, Brazil (photo by César Favacho); (G) *Acanthops* Burmeister, female from Iquitos, Peru (photo by Robert Oelman); (H) *Pseudacanthops* Saussure, female from Napo River, Ecuador (copyright Mark W. Moffett/Minden Pictures, all rights reserved); (I) *Metilia* Stål, male from Pará, Brazil (photo by César Favacho).

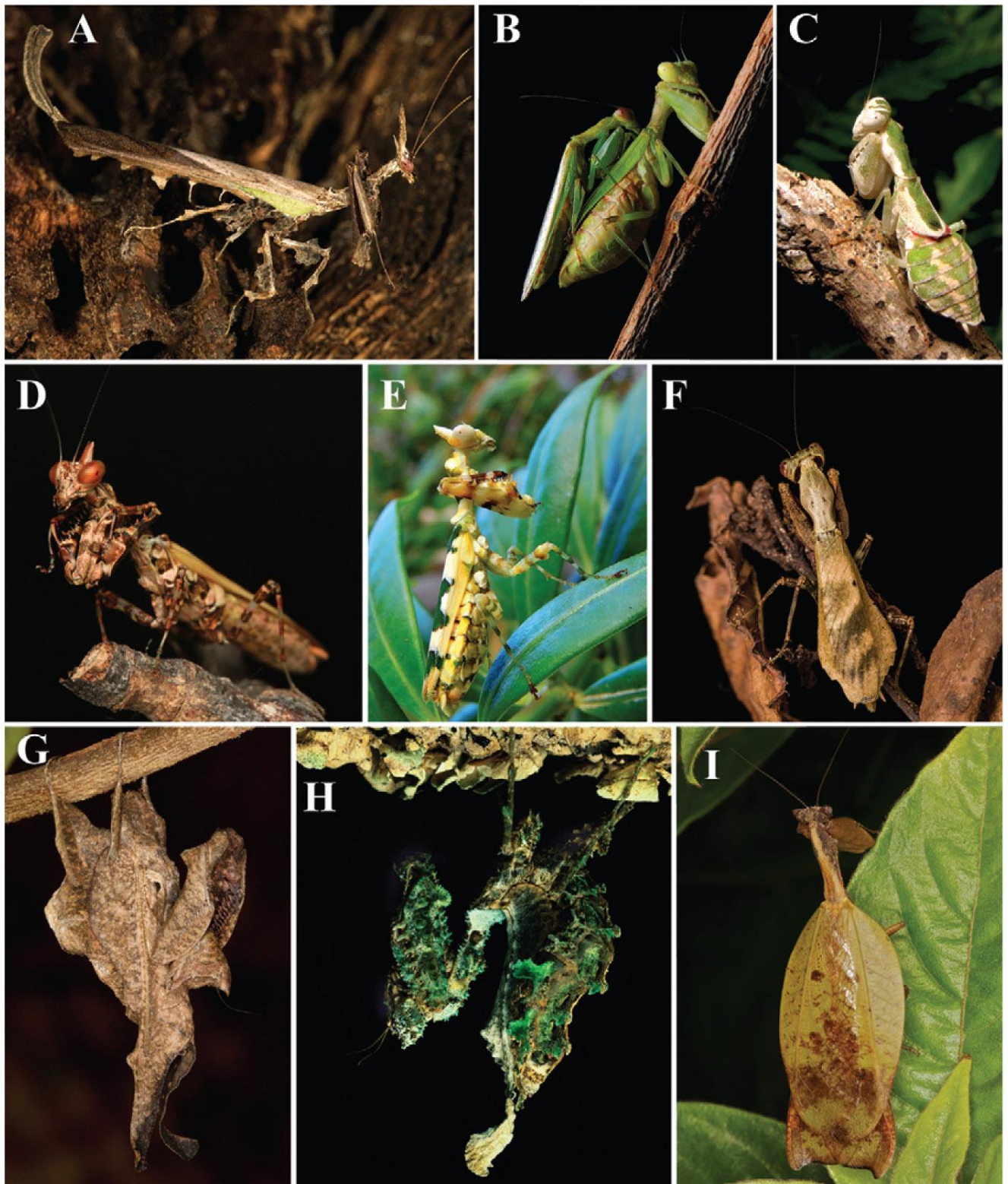


Figure 2.3

Figure 2.4. Molecular phylogeny of Acanthopoidea or ‘polymorphic earless praying mantises’ (PEPM). Numbers on branches represent the posterior probabilities (PP) and bootstrap support (BS) of the Bayesian inference (BI) and maximum-likelihood (ML) analyses, respectively (PP/BS). Branches are colour-coded to represent the eight families herein proposed. Taxa marked with (*) indicate that current taxonomy requires reconsideration, either because those species were originally classified in the wrong genus, or synonymy in the genus-group has yet to be recognized (see text for details). Inset on the lower left is the simplified phylogeny of Mantodea highlighting the relative position of PEPM, and the various hypotheses concerning the phylogenetic placement of the also earless Metallyticidae according to Svenson & Whiting (2009) [S&W2009], Wieland (2013) [W2013] and Legendre *et al.* (2015) [L2015]. The Cernomantodea, which evolved hearing capabilities, is sister to PEPM. Inset on the lower right represents the Bayesian topology for Photinaidae, the only topological discordance between BI and ML phylogenies (discordant clade indicated with a red circle).

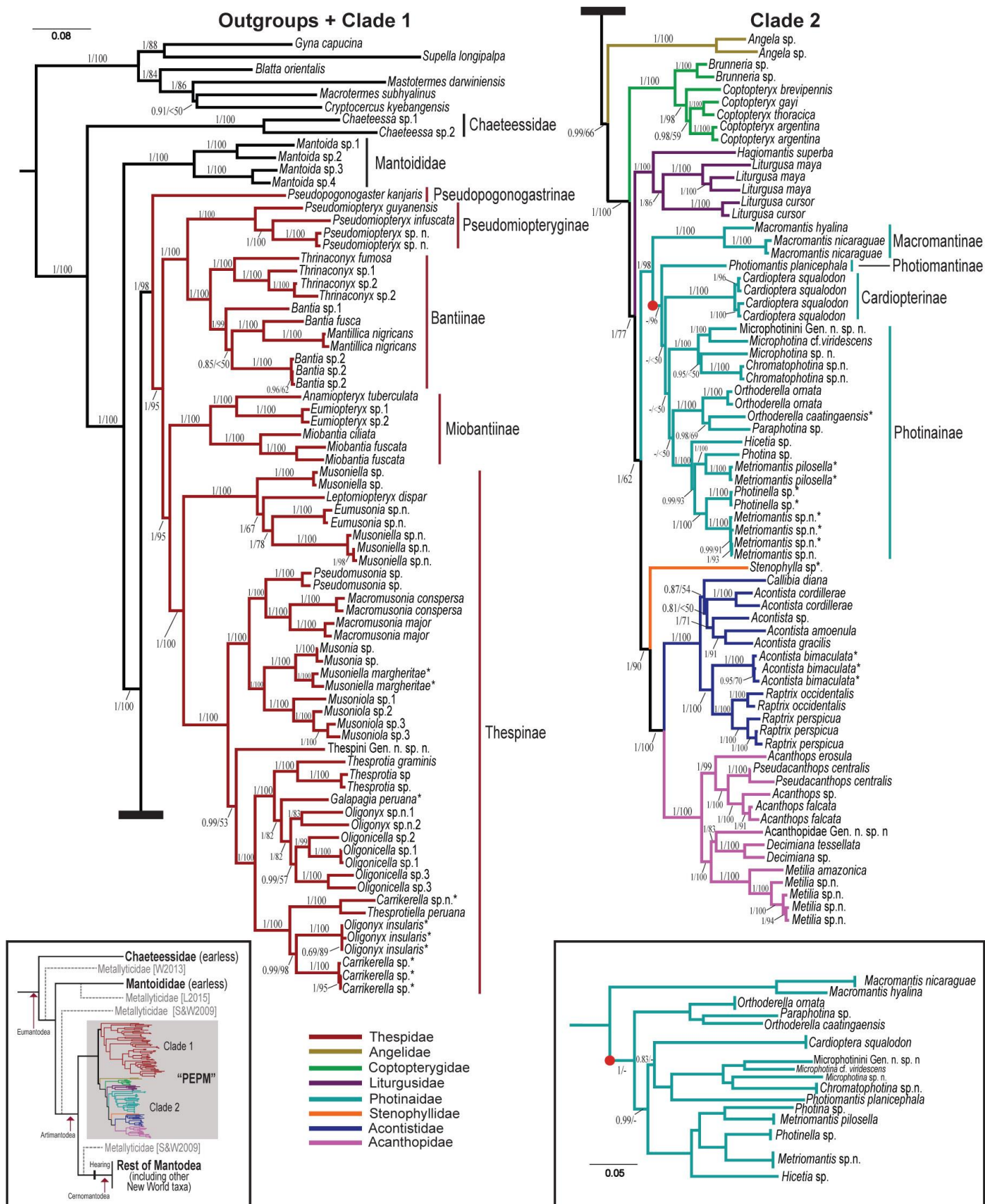


Figure 2.4

Figure 2.5. Detail of tibial spination in juvenile Thespidae (first instar). (A) *Macromusonia* sp., from Pará, Brazil; (B) *Pseudomiopteryx* sp., from Pará, Brazil. Black arrows point to the distal-most anteroventral spine in dorsolateral position. These spines are absent in the adults of these two genera, in which all tibial spines are ventrally positioned. Most members of Bantiinae and several Thespinae retain such spines into adulthood. Photos courtesy of César Favacho.

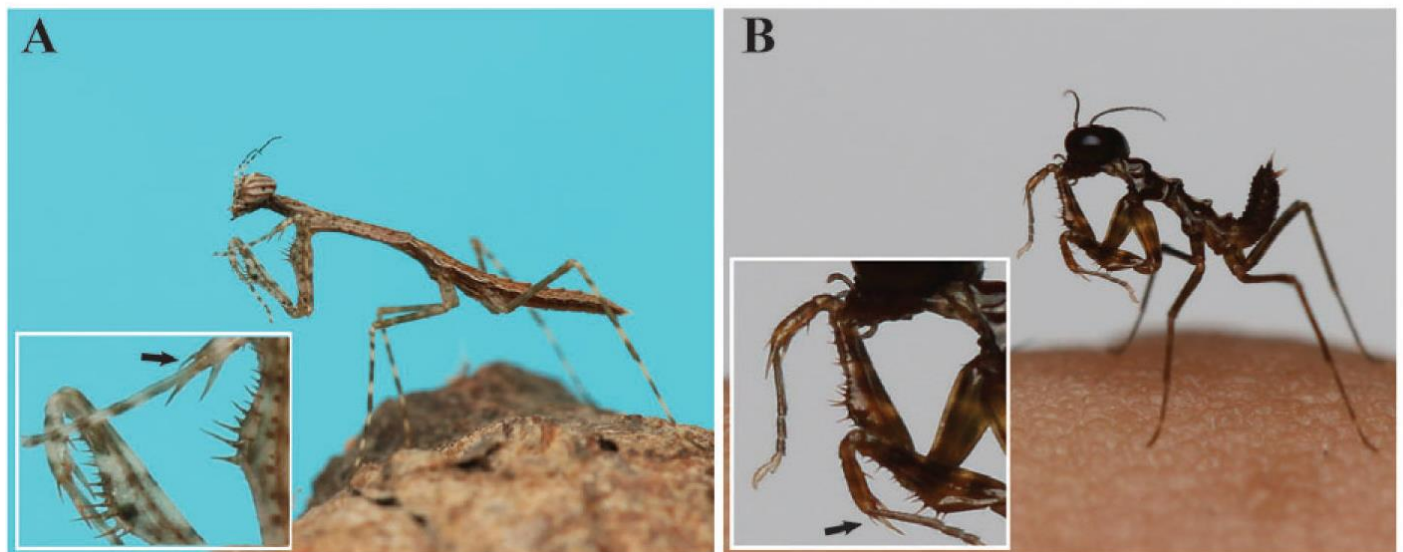


Figure 2.5

Figure 2.6. External morphology. (A) *Miobantia ciliata* (male), head (lower frons perspective) and detail of vertex (ocelli perspective); (B) *Acanthops erosula* (male), prothorax (dorsal view); (C) *Macromusonia conspersa* (female), prothoracic femur (anteroventral view, discoidal spines highlighted in black); (D) generalized terminalia, male (dorsal view); (E) female, same as (D). Abbreviations: AvS, anteroventral spines; DS, discoidal spines; PvS, posteroventral spines.

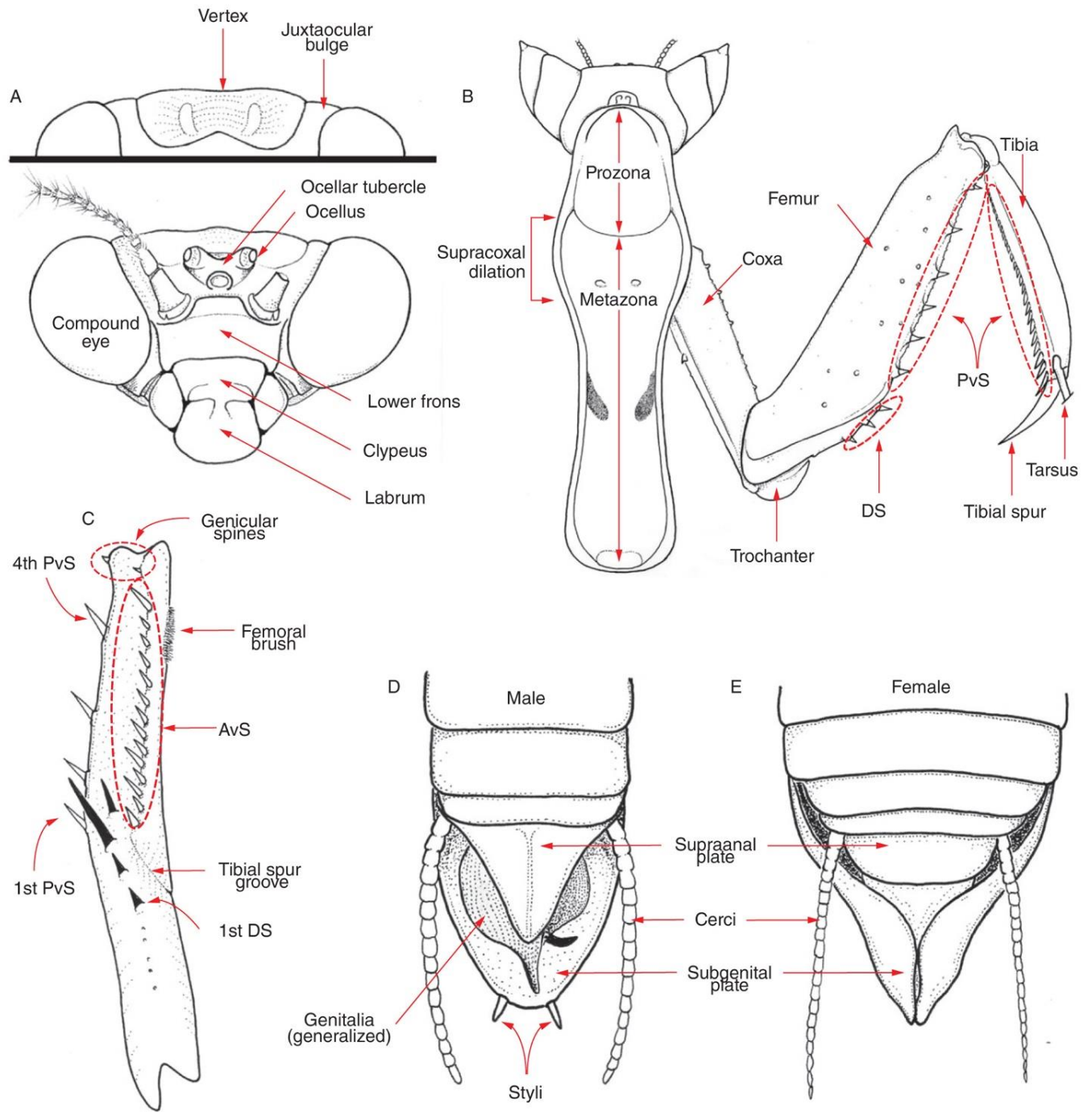


Figure 2.6

Figure 2.7. External morphology. (A) *Chaeteessa* sp. (female), right prothoracic tibia and partial view of metatarsus (dorsal view); (B) generalized, distal section of right prothoracic femur showing the location of the femoral brush (partial, anterior view); (C) *Mantoida maya* (male), same as (B) (notice the lack of a well-defined femoral brush); (D) *Mantoida brunneriana* (male), abdomen (ventral view) (redrawn from La Greca & Lombardo, 1989); (E) *Mantoida argentinae* (female) same as (D); (F) *Parastagmatoptera vitreola* (Mantidae, Stagmatopterinae) (male), metasternal morphology associated with the presence of functional cyclopean ear corresponding to the DK-type (i.e. ‘Deep groove with Knobs’) of Yager & Svenson, 2008) (posteroventral view); (G) *Mantellias pubicornis* (male), head (frontal view); (H) *Miobantia ciliata* (male), costal vein (section, dorsal view); (I) *Hagiomantis superba* (male), same as (H); (J) *Pseudomiopteryx* sp. (female), prothoracic femur (posterior view); (K) *Thrinaconyx fumosa* (male), left prothoracic tibia (posterior view); (L) *Thesprotiella peruana* (female), right prothoracic tibia (anterior view); (M) *Oligonyx bidens* (female), same as in (L); (N) *Pseudomusonia lineativentris* (female), same as in (L); (O) *Lepomiopteryx dispar* (male), same as in (L); (P) *Angela subhyalina* (male), cercus (lateral view); (Q) *Angela armata* (female), same as (P); (R) *Angela guianensis* (male), same as (P); (S) *Brunneria subaptera* (female), mesoand metathorax (dorsal view); (T) *Coptopteryx gayi* (female), same as (D). Black arrows point to distal-most anteroventral tibial spines shifted to a dorsolateral positioning. Abbreviations: AO, anterior opening of auditory chamber; AvS, anteroventral spines; CV, costal vein; DS, discoidal spines; LB, lateral bulges; MT, metathorax; MTC, metacoxa; PO, posterior opening of auditory chamber; PvS, posteroventral spines.

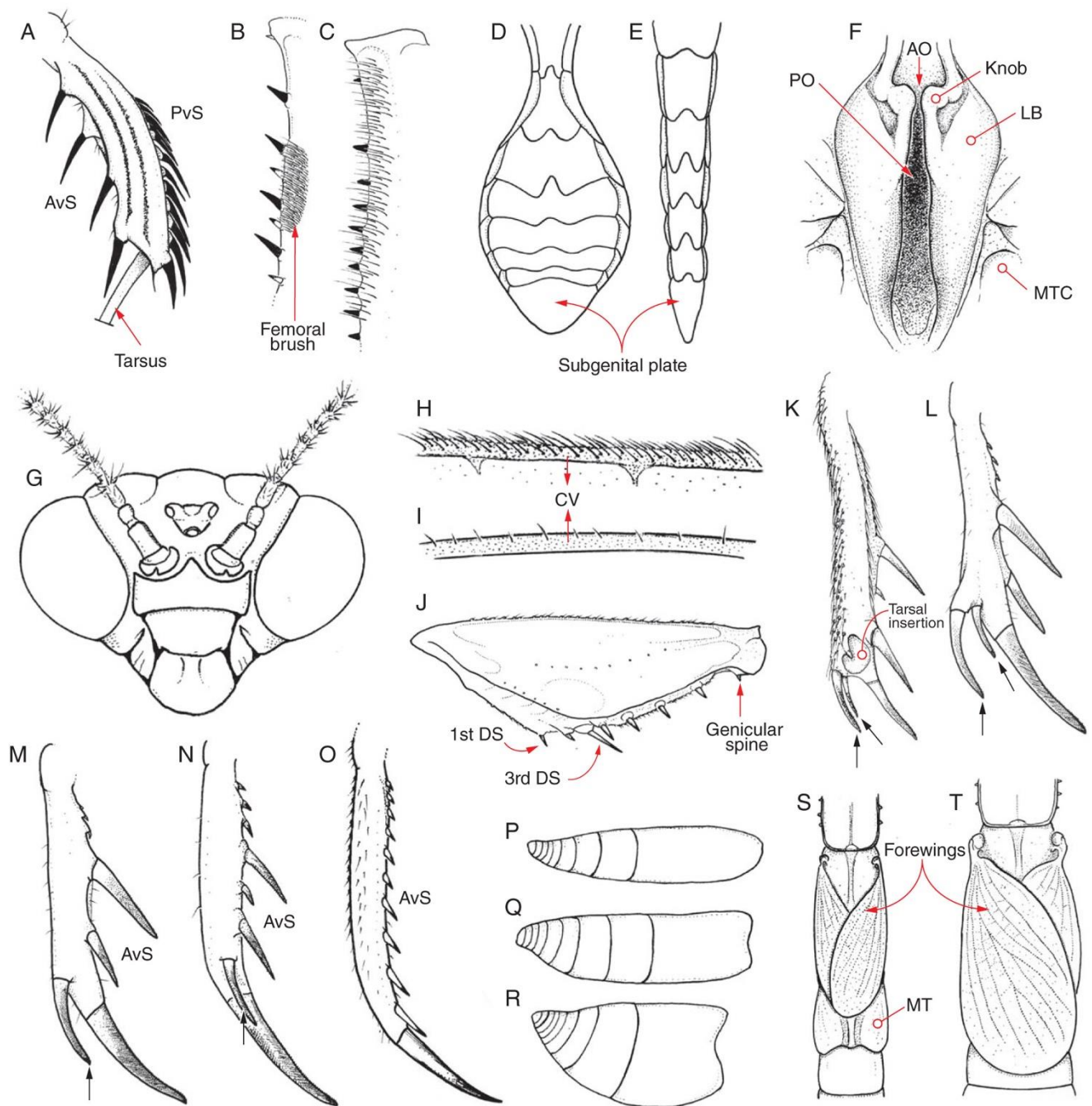


Figure 2.7

Figure 2.8. External morphology. (A) *Macromantis hyalina* (male), right prothoracic femur, detail of posteroventral margin (posterior view); (B) *Macromantis hyalina* (male), head and pronotum (dorsal view); (C) *Cardioptera squalodon* (male), right prothoracic coxa (anterior view); (D) *Photiomantis planicephala* (female), head (frontal view); (E) *Photiomantis planicephala* (male), head (posterolateral view); (F) *Photina vitrea* (male), head (partial, posterolateral view); (G) *Stenophylla lobivertex* (male), head (frontal *Stenophylla lobivertex* (male): head (frontal view); (H) *Stenophylla lobivertex* (female), detail of central process of vertex (frontal view); (I) *Metilia adusta* (male), right prothoracic femur (posterior view); (J) *Raptrix perspicua* (male), head (frontal view); (K) *Acanthops fuscifolia* (male), head (frontal view); (L) *Callibia diana* (female): head (frontal view); (M) *Orthoderella delucchii* (male), head (frontal view). Abbreviations: PvS, posteroventral spines.

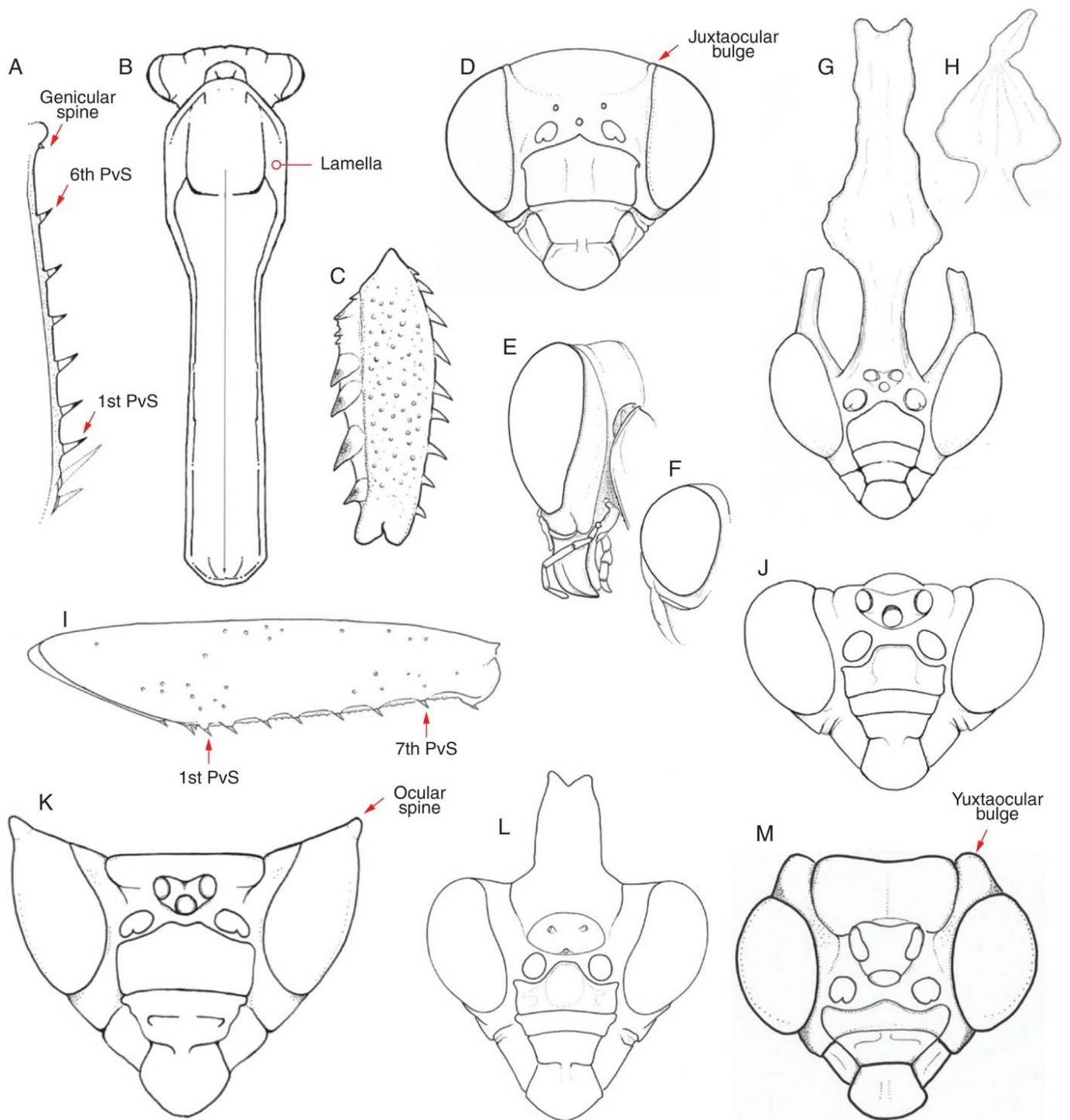


Figure 2.8

Figure 2.9. Male genitalia of Thespidae – Pseudomiopteryginae (A–C), Pseudopogonogastrinae (D–E), Bantiinae (F–K) and Miobantiinae (L–P). (A) *Pseudomiopteryx* sp., ventral (ventral view) and left dorsal phallomere (dorsal view) complex; (B) *Pseudomiopteryx* sp., right phallomere (ventral view); (C) *Promiopteryx* sp., ventral phallomere (ventral view); (D) *Pseudopogonogaster kanjaris*, left phallomere (dorsal view; from Rivera *et al.*, 2011); (E) *Pseudopogonogaster kanjaris*, ventral phallomere (dorsal view; from Rivera *et al.*, 2011); fused ventral phallomere and dorsal lamina of left dorsal phallomere (all in ventral view); (F) *Bantia* sp.; (G) *Bantiella trinitatis*; (H) *Diabantia* sp., (I) *Mantellias* sp.; (J) *Mantillica nigricans*; (K), *Thrinaconyx fumosa*, ventral phallomere (all in ventral view); (L) *Anamiopteryx tuberculata*; (M) *Chloromiopteryx thalassina* and a partial view of *Chloromiopteryx* sp.; (N) *Paradiabantia* sp.; (O) *Eumiopteryx* sp.; (P) *Miobantia ciliata*. The corresponding apofisi falloide (outlined) is shown for some taxa.

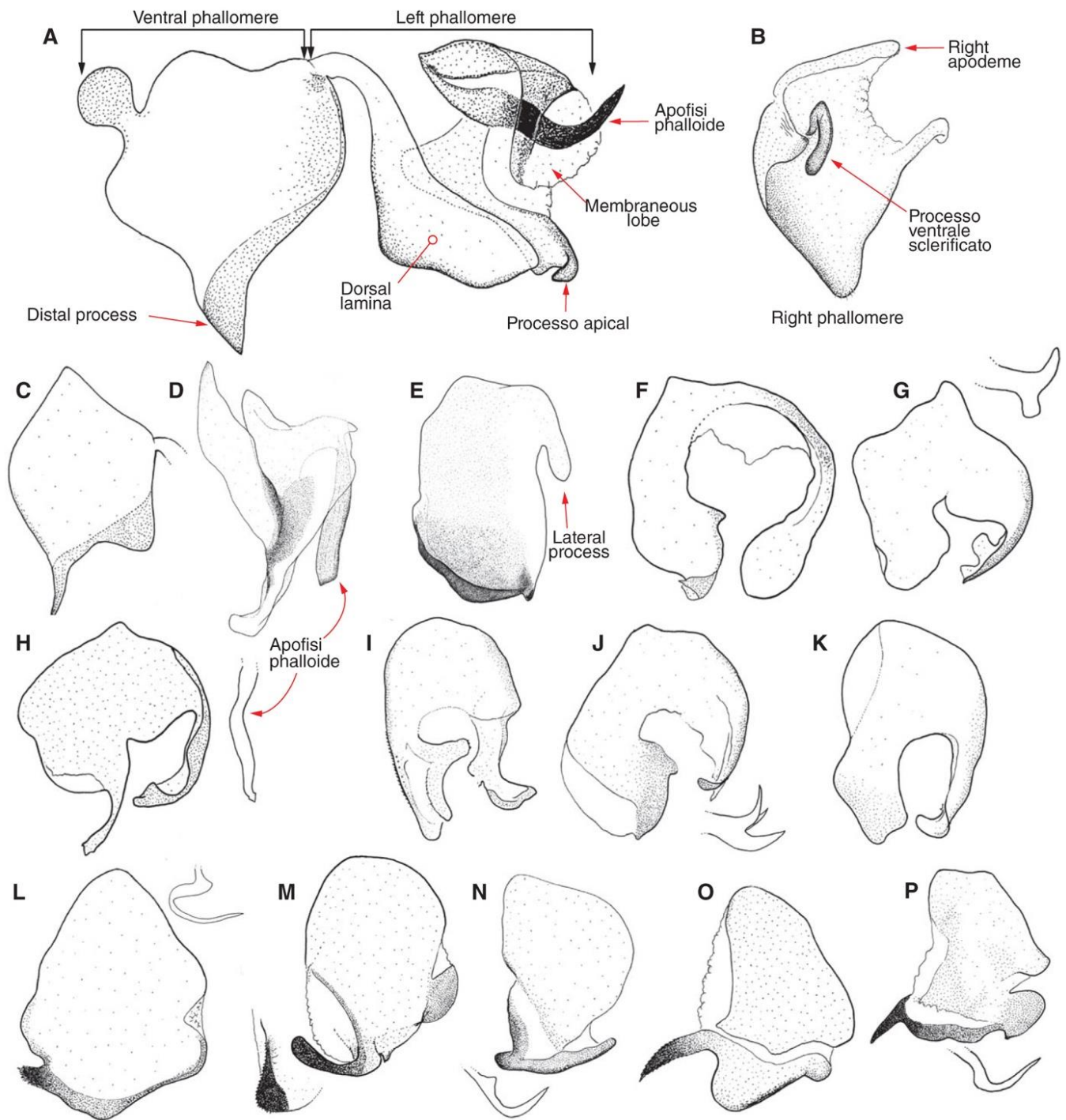


Figure 2.9

Figure 2.10. Male genitalia of Thespidae – Thespinae (A–M), Coptopterygidae (N), Photinaidae (O–T), and Acontistidae (U–Z). Ventral phallomere (ventral view): (A) *Eumusonia* sp.; (B) *Musoniella argentina*; (C) *Leptomiopteryx dispar*; (D) *Pizaia* sp.; (E) *Galapagia peruana*; (F) *Galapagia solitaria*; (G) *Macromusonia major*; (H) *Musonia surinama*; (I) *Musoniola* sp.; (J) *Oligonicella scudderi*; (K) *Oligonyx bicornis*; (L) *Pseudomusonia lineativentris*; (M) *Thespis media*; (N) *Coptopteryx argentina*, processo ventrale sclerificato of right phallomere. Ventral phallomere and variation of the lateral process (dorsal view): (O) *Macromantis hyalina*; (P) *Cardioptera brachyptera*; (Q) *Paraphotina reticulata*; (R) *Chromatophotina awajun*; (S) *Metriomantis pilosa*; (T) *Orthoderella delucchii*, right dorsal phallomere (ventral view); (U) *Raptrix perspicua*, processo ventrale sclerificato of right phallomere; (V) *Raptrix perspicua*, ventral phallomere (dorsal view). Variation of the distal process of the ventral phallomere (dorsal view): (W) *Acontista bimaculata*; (X) *Acontista amoenula*; (Y) *Callibia diana*; (Z) *Tithrone roseipennis*. The corresponding apofisi falloide (outlined) is shown for some taxa.

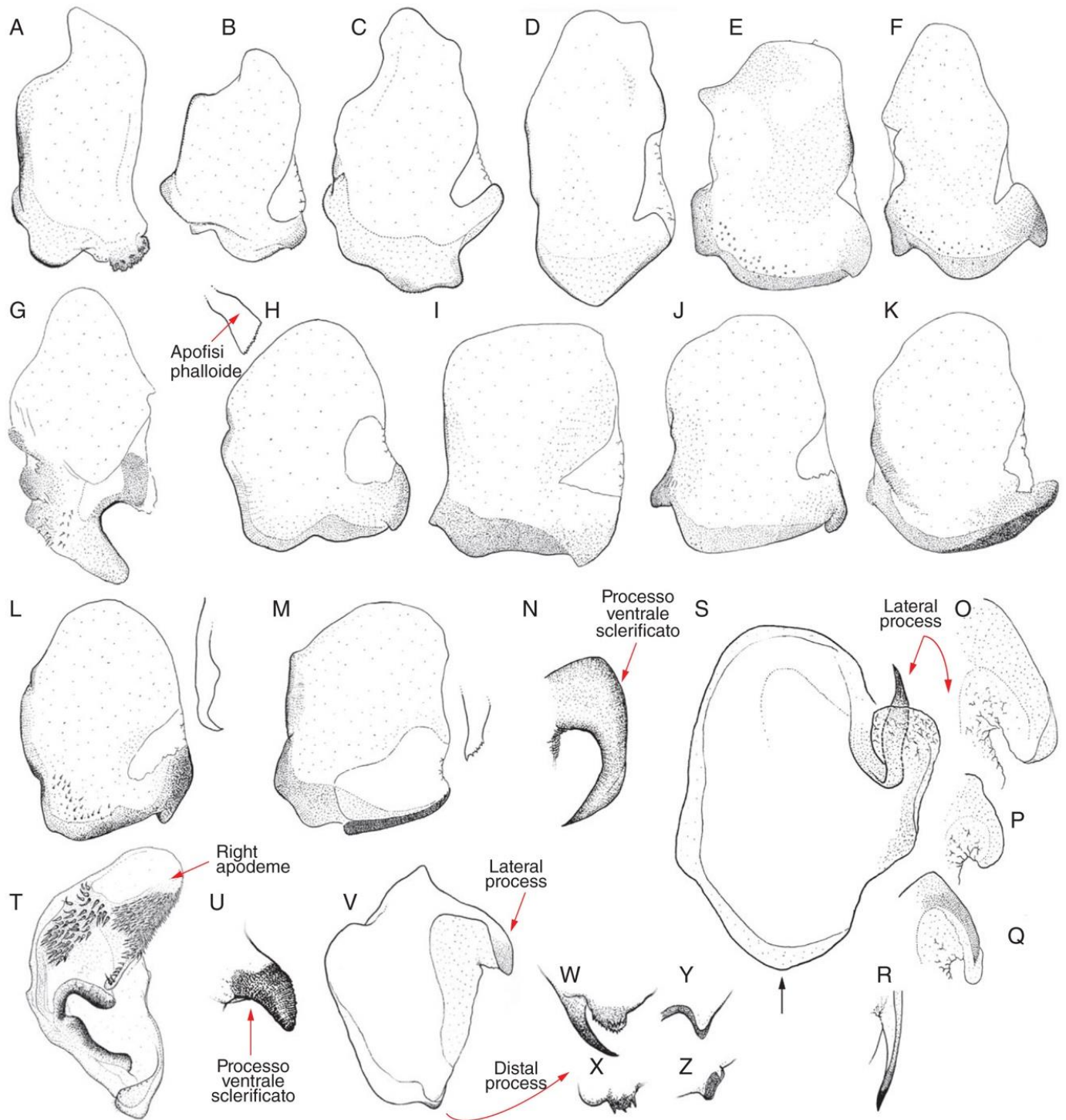


Figure 2.10

Figure 2.11. Egg case morphology. (A) Generalized morphology (schematic). Cross section showing internal arrangement: (B) *Macromantis hyalina*; (C) *Photinella* sp.; (D) *Photina* sp.; (E) *Metriomantis boliviana*. Types of attachment (schematic): (F) ventral; (G) posteroventral; (H) posterior; (I, J) posterodorsal; (K) stalked posterior.

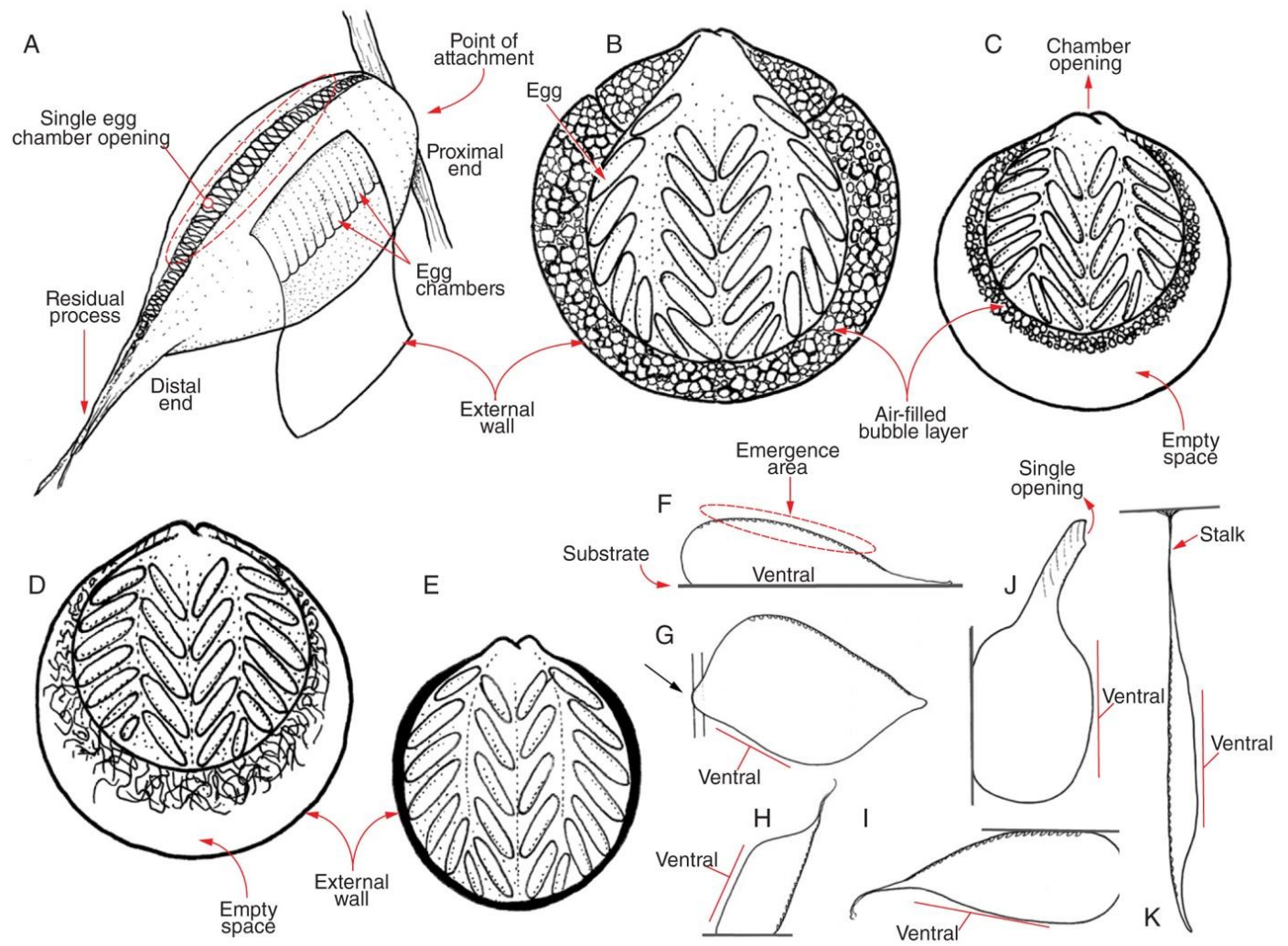


Figure 2.11

Figure 2.12. Egg cases of Thespidae (A–M) and Angelidae (O–P). (A) *Pseudomiopteryx* sp. (lateral); (B) *Pseudomiopteryx* sp. (dorsal); (C) *Thrinaconyx* sp. (lateral); (D) *Chloromiopteryx mirim* (lateral); (E) *Musonia margheritae* (lateral); (F) *Galapagia peruana* (lateral); (G) *Galapagia peruana* (attached to a rock, photo taken by Julio Rivera in Huarochirí, Peru); (H) *Oligonicella scudderi* (lateral); (I) *Macromusonia conspersa* (lateral, fully coated); (J) *Macromusonia conspersa* (dorsal, fully coated); (K) *Macromusonia conspersa* (lateral, external coating removed); (L) *Carrikerella* sp. (lateral); (M) *Thesprotia* sp. (lateral); (N) *Angela* sp. (lateral, fully coated); (O) *Angela* sp., (lateral), and detail of emergence area; (P) *Angela maxima* (lateral).

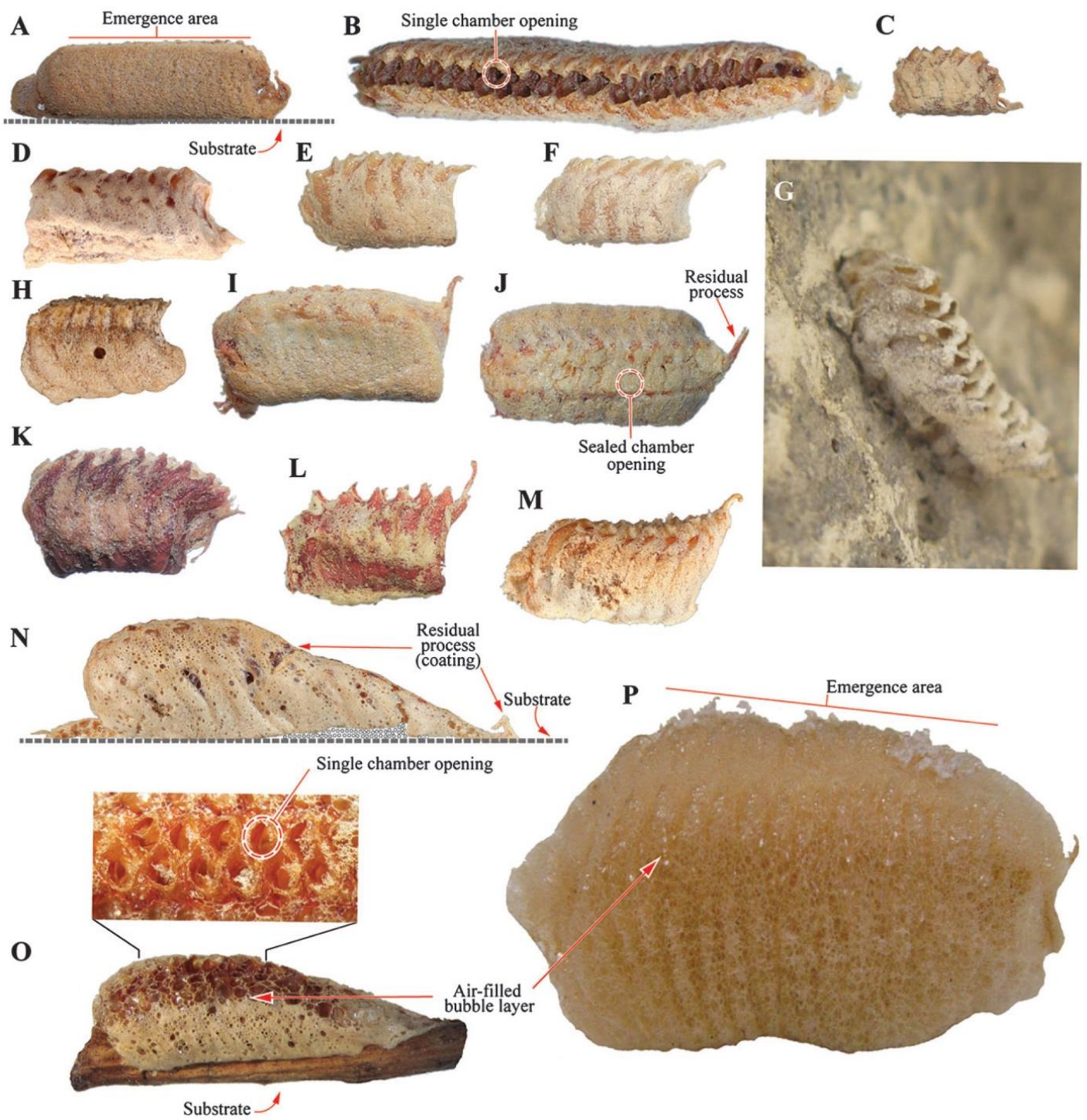


Figure 2.12

Figure 2.13. Egg cases of Coptopterygidae (A–H) and Liturgusidae (I–Q). (A) *Coptopteryx* sp. (lateral); (B) *Coptopteryx* sp. (lateral); (C) *Coptopteryx* sp. (lateral, photo taken by Julio Rivera in Santa Catarina, Brazil); (D) *Coptopteryx* sp. (lateral); (E) *Coptopteryx brevipennis* (lateral); (F) *Brunneria subaptera* (lateral); (G) *Brunneria borealis* (lateral); (H) *Brunneria borealis* (dorsal); (I) *Fuga annulipes* (lateral), and detail of emergence opening; (J) *Liturgusa maya* (lateral); (K) *Liturgusa maya* (lateral), showing emerging parasitic wasps; (L) undetermined liturgusid (lateral), inflated morph; (M) undetermined liturgusid (lateral); (N) *Liturgusa nubeculosa* (dorsal), showing remains of outer coating; *Liturgusa nubeculosa* (lateral); (P) undetermined liturgusid (lateral); (Q) undetermined liturgusid (lateral).

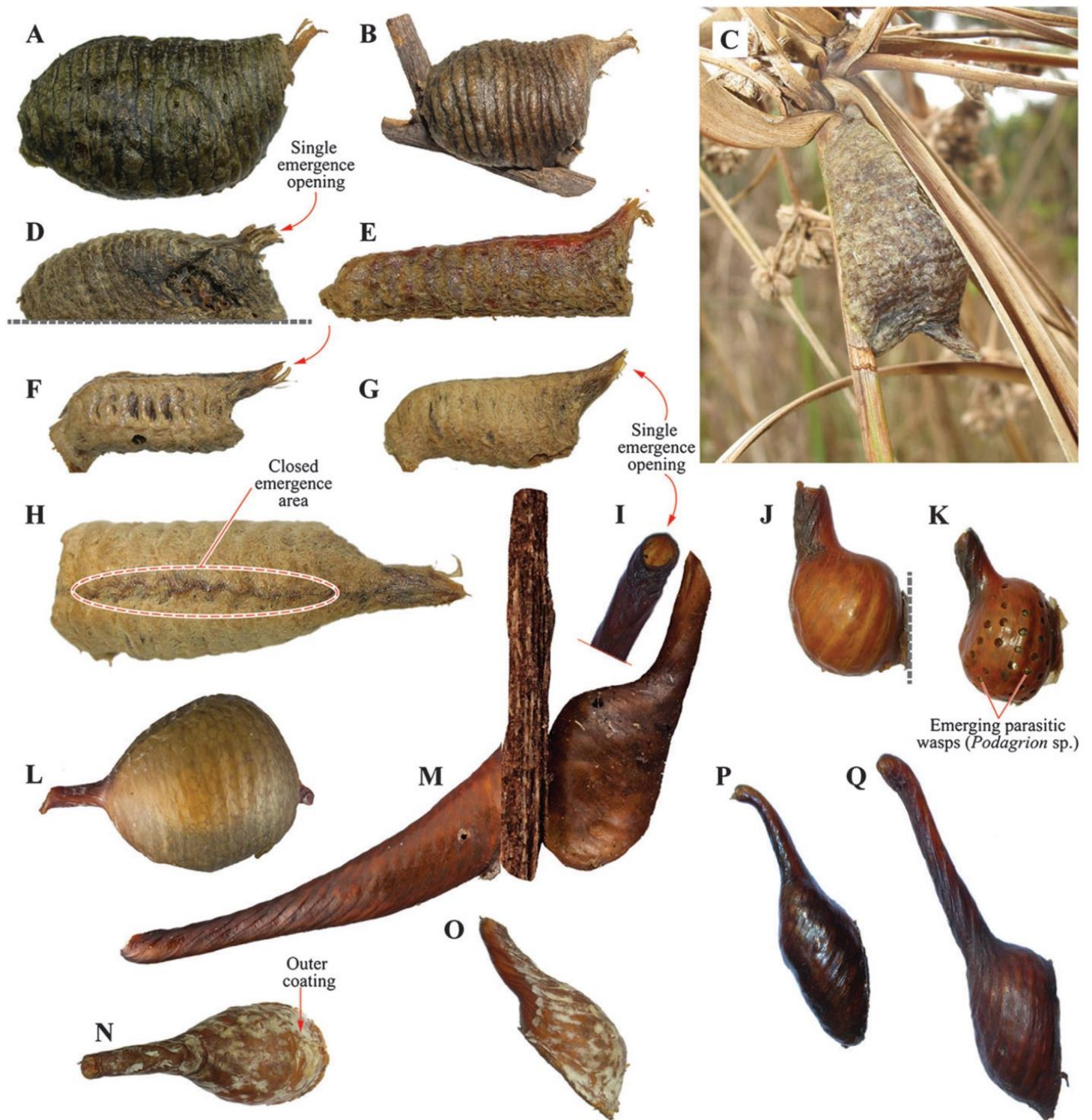


Figure 2.13

Figure 2.14. Egg cases of Photinaidae, Acontistidae and Acanthopidae. (A) *Macromantis hyalina* (lateral); (B) *Macromantis hyalina* (posterior); (C) *Cardioptera brachyptera* (lateral); (D) *Chromatophotina awajun* (lateral); (E) *Orthoderella ornata* (lateral); (F) *Paraphotina* sp. (lateral); (G) *Photina* sp. (lateral); (H) *Metriomantis boliviana* (dorsal); (I) *Metriomantis* sp., female forming egg case (photo by Reinhard Ehrmann, courtesy of Matthias Borer; specimen from French Guiana); (J) *Metriomantis boliviana* (lateral); (K) *Hicetia* sp. (lateral); (L) *Acontista cordillerae* (lateral); (M) *Acontista* sp. (external coating removed); (N) *Callibia diana* (lateral); (O) *Callibia diana* (egg case attached to a leaf's ventral lamina; photo taken by Fernando Carvalho in Pará, Brazil); (P) *Raptrix occidentalis* (lateral); (Q) *Pseudacanthops* sp. (lateral); (R) *Acanthops erosula* (lateral); (S) *Acanthops erosula* (dorsal); (T) undetermined acanthopid (lateral); (U) *Metilia* sp. (lateral).

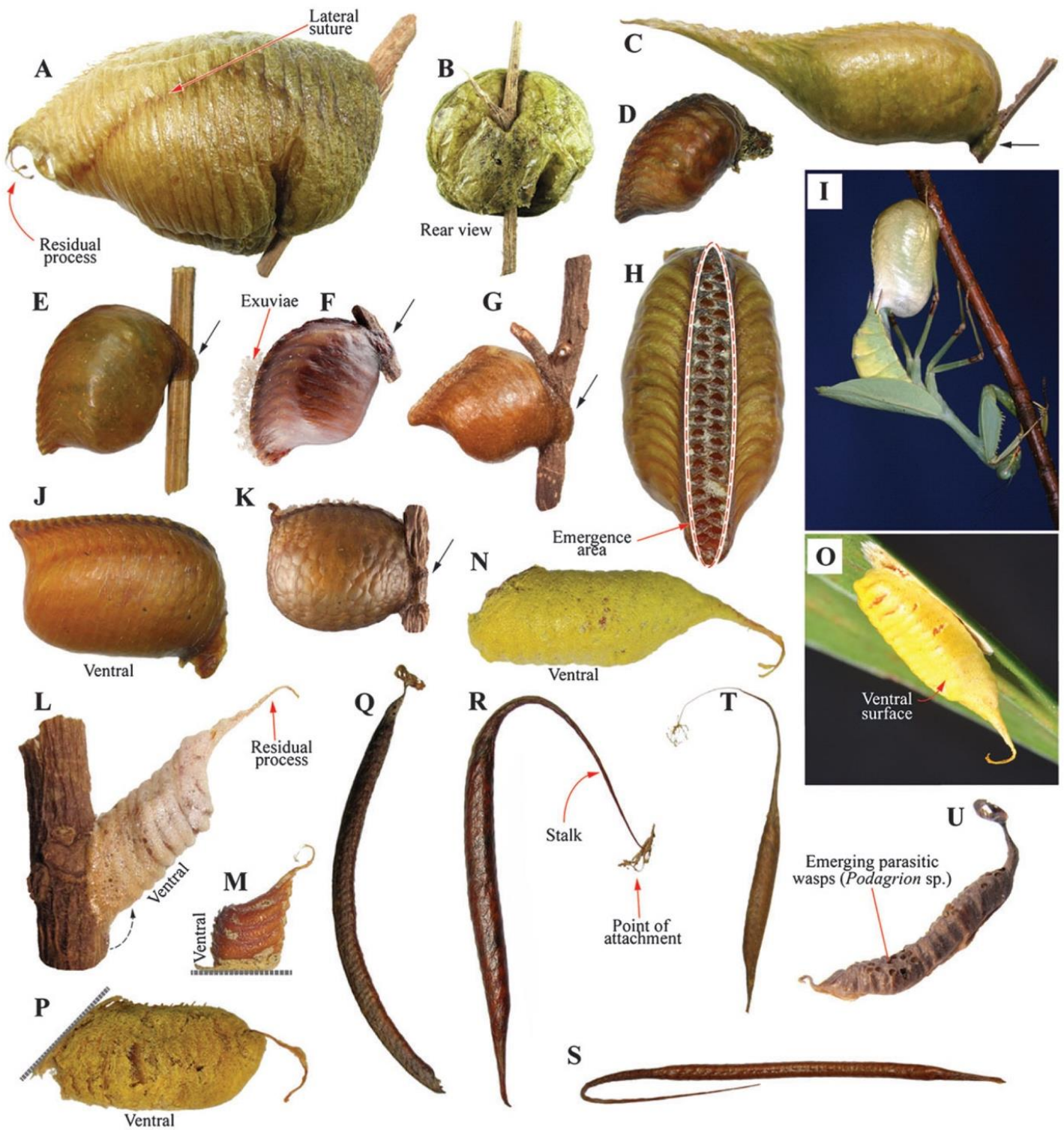


Figure 2.14

Figure 2.15. Summary of new higher-level classification of Acanthopoidea or ‘polymorphic earless praying mantises’ (PEPM), including representative taxa and their corresponding egg cases (photo credits as in Figs 1–3).

Phylogeny of Acanthopoidea
The Polymorphic Earless
Praying Mantises

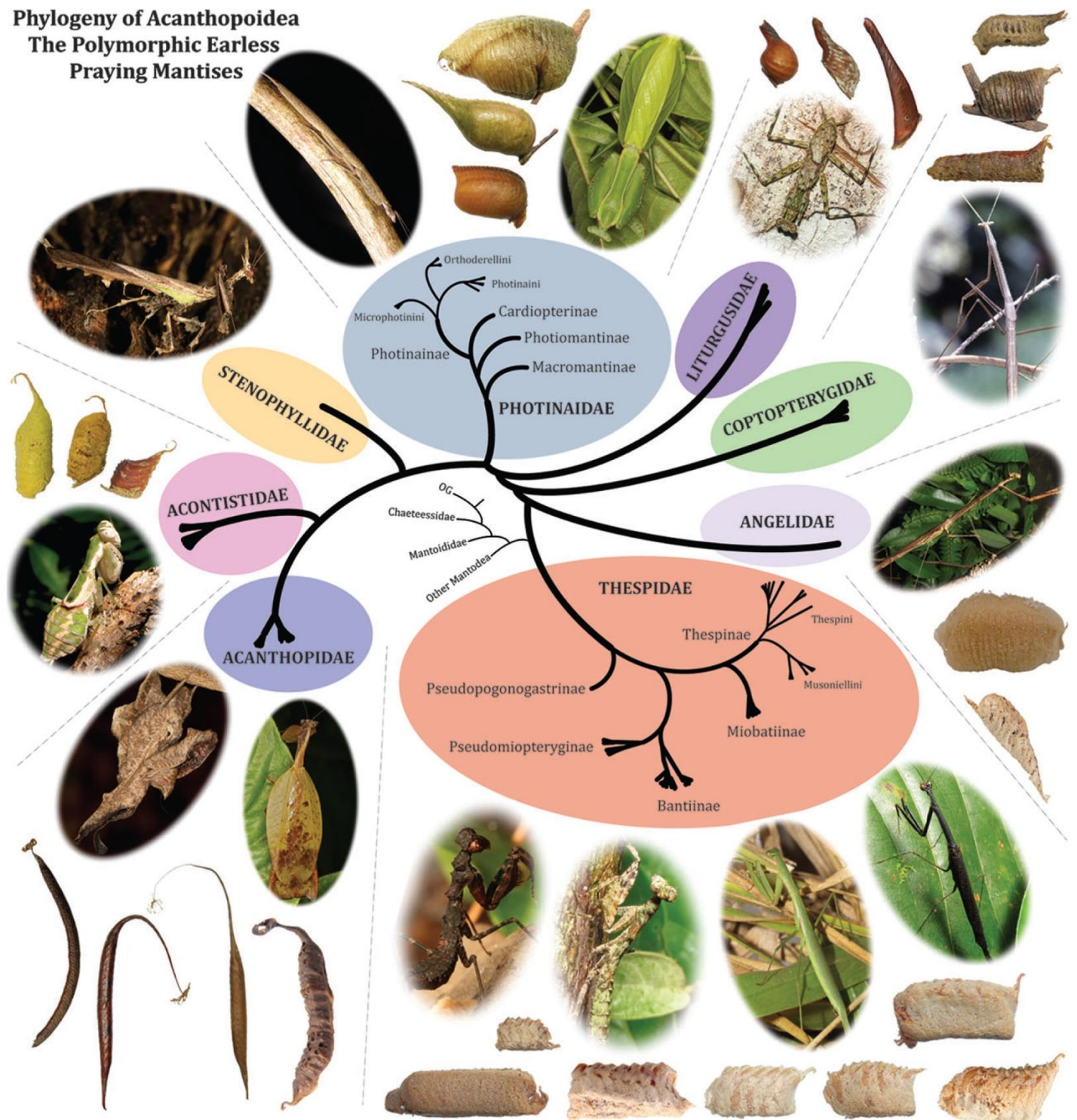


Figure 2.15

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Chapter 3

A Review of the Genera of the Acanthopoidea, the “Polymorphic Earless Praying Mantises” (Insecta: Mantodea)

3.1 Summary

The Acanthopoidea (Insecta: Mantodea) constitute a monophyletic and largely Neotropical lineage of praying mantises comprising eight families: Thespidae, Angelidae, Coptopterygidae, Liturgusidae, Photinainidae, Stenophyllidae, Acontistidae and Acanthopidae. Recent phylogenetic studies that explored evolutionary relationships within this clade, otherwise known as “polymorphic earless praying mantises”, revealed numerous issues with the taxonomic status and morphological boundaries of many of the included genera and species. I conduct a thorough morphological assessment across the superfamily on the basis of museum specimens and the re-examination of type material, totalling 1336 specimens, 238 of which were primary types (158 holotypes, 72 syntypes, 8 lectotypes). The goals of this study are to: i) re-define the morphological delineation of all Acanthopoidea genera; ii) verify the generic identity of all nominal species within the superfamily, and; iii) establish taxonomic and nomenclatural foundations for future revisionary studies of acanthopoid genera. My investigation resulted in numerous modifications in the taxonomy of this lineage. In Thespidae: *Miopteryx simoni* Chopard, 1916 (currently assigned to *Bantia*) is transferred to *Bantiella* as *B. simoni* (Chopard, 1916) (new combination); *Bantiella hyalina* Beier, 1942 is transferred to *Mantellias* as *M. hyalina* (Beier, 1942) (new combination); *Oligonicella agudelo* Salazar and Ariza, 2005, and *Oligonyx brunneri* Saussure, 1871 (currently assigned to *Oligonicella*) are both transferred to *Thrinaconyx*, respectively as *T. agudelo* (Salazar and Ariza, 2005) (new combination) and *T. brunneri* (Saussure, 1871) (new combination); *Promiopteryx punctata* Giglio-Tos, 1917 is transferred to *Miobantia* as *M. punctata* (Giglio-Tos, 1917) (new combination); *Musonia costalis* Rehn, 1920 is transferred to *Musoniella* as *M. costalis* (Rehn, 1920) (new combination), whereas *Mionyx lineatus* Chopard, 1911 (currently assigned to *Musonia*) is a new synonym of *Musonia surinama* (Saussure, 1869); *Mantis stigmatica* Burmeister, 1838 (currently assigned to *Promiopteryx*) is transferred to *Musoniella* as *M. stigmatica* (Burmeister, 1838); *Galapagia*

amazonica Terra, 1982, *Musonia maculata* Beier, 1942, *Mionyx maculosus* Chopard, 1911 (currently assigned to *Pseudomusonia*), and *Musonia sexdentata* Beier, 1942 are all transferred to *Dougonyx* (new genus), respectively as *D. amazonicus* (Terra, 1982) (new combination), *D. maculatus* (Beier, 1942) (new combination), *D. maculosus* (Chopard, 1911) (new combination), and *D. sexdentatus* (Beier, 1942) (new combination); *Musoniella margharethae* Battiston and Picciau, 2008 is transferred to *Musonia* as *Musonia margharethae* (Battiston and Picciau, 2008) (new combination); *Mionycoides* Rehn, 1935 is removed from synonymy under *Pseudomusonia* Werner, 1909 and considered instead a new synonym of *Musoniola* Giglio-Tos, 1917, thus *Mionyx ferus* Saussure and Zehntner, 1894 and *Mionyx rapax* Saussure and Zehntner, 1894 (both currently assigned to *Pseudomusonia*), are herein transferred to *Musoniola* as *Musoniola fera* (Saussure and Zehntner, 1894) (new combination) and *Musoniola rapax* (Saussure and Zehntner, 1894) (new combination); *Galapagia peruana* Beier, 1935 is transferred to *Piscomantis* (new genus) as *Piscomantis peruana* (Beier, 1935) (new combination); *Olygonyx insularis* Bonfils, 1967 is transferred to *Thesprotiella* as *T. insularis* (Bonfils, 1967) (new combination); *Paramusonia* Rehn, 1904 is a new synonym of *Thespis* Audinet Serville, 1831, *Paramusonia seclusa* Rehn, 1913 and *Musonia boliviana* Beier, 1930 are both transferred to *Thespis*, respectively as *T. seclusa* (Rehn, 1913) (new combination), and *T. boliviana* (Beier, 1930) (new combination), whereas *Leptomiopteryx argentina* Beier, 1930 is a new synonym of *T. seclusa* (Rehn, 1913); *Pogonogaster* Rehn 1918 and *Carrikerella* Hebard, 1921 are both new synonyms of *Thesprotiella* Giglio-Tos, 1915, thus *Pogonogaster tristani* Rehn, 1918, *Pogonogaster latens* Hebard, 1919, *Carrikerella ceratophora* Hebard, 1922, and *Carrikerella empusa* Rehn, 1935 are transferred to *Thesprotiella*, respectively as *T. tristani* (Rehn, 1918) (new combination), *Thesprotiella latens* (Hebard, 1919) (new combination), *T. ceratophora* (Hebard, 1922) (new combination), *T. empusa* (Rehn, 1935) (new combination). In Photinaidae: *Orthoderella caatingaensis* Menezes and Bravo 2013 is transferred to *Paraphotina caatingaensis* (Menezes and Bravo, 2013); *Photinella* Giglio-Tos 1915 and *Metriomantis* Saussure and Zehntner, 1894 are both new synonyms of *Photina* Burmeister 1838, thus *Photinella magna* Giglio-Tos, 1915, *Photinella media* Giglio-Tos, 1915 *Metriomantis boliviana* Lombardo, 1999, *Cardioptera cupido* Saussure, 1869 (currently assigned to *Metriomantis*), *Metriomantis occidentalis* Saussure and Zehntner, 1894, *Metriomantis ovata* Saussure and Zehntner, 1894, *Metriomantis paraensis* Giglio-Tos, 1915, and *Metriomantis pilosella* Giglio-Tos, 1915 are all transferred, respectively to

Photina as *Photina magna* (Giglio-Tos, 1915) (new combination), *Photina media* (Giglio-Tos, 1915) (new combination), *Photina boliviana* (Lombardo, 1999) (new combination), *Photina cupido* (Saussure, 1869) (new combination), *Photina occidentalis* (Lombardo, 1999) (new combination), *Photina ovata* (Saussure and Zehntner, 1894) (new combination), *Photina paraensis* (Giglio-Tos, 1915) (new combination), *Photina pilosella* (Giglio-Tos, 1915) (new combination), whereas *Parastagmatoptera amazonica* Werner, 1928 is also transferred to *Photina* as *Photina amazonica* (Werner, 1928), and *Metriomantis biramosa* Saussure and Zehntner, 1894 (currently assigned to *Photinella*) is considered a new synonym of *Photina vitrea* (Burmeister, 1838). In Coptopterygidae: *Brunneria gracilis* Giglio-Tos, 1915 is a new synonym of *Brunneria subaptera* Saussure, 1869; *Coptopteryx claraziana* Saussure, 1869 (*partim*-male) and *Coptopteryx viridis* Giglio-Tos, 1915 (*partim*-male), are both new synonyms of *Coptopteryx argentina* (Burmeister, 1838), whereas *Coptopteryx fallax* Giglio-Tos, 1917 and *Coptopteryx spinosa* Giglio-Tos, 1915 are both new synonyms of *Coptopteryx thoracica* Rehn, 1913. In Acontistidae: *Acontista maroniensis* Chopard, 1911 is a new synonym of *Acontista cayennensis* Saussure and Zehntner, 1894, *Acontiothespis iriodes* (Hebard 1919) (currently assigned to *Acontista*) is a new synonym of *Acontista cordillerae* Saussure, 1869, *Acontiothespis ecuadorica* Hebard, 1924 (currently assigned to *Acontista*) is a new synonym of *Acontista fraterna* Saussure and Zehntner, 1894, and *Acontista chopardi* Giglio-Tos, 1927 is a new synonym of *Acontista gracilis* Chopard, 1911; *Acontista cubana* de Zayas, 1974 is transferred to *Callimantis* Stål, 1877 (Mantidae: Stagmomantinae) as *C. cubana* (de Zayas, 1974) (new combination); *Metaphotina* Piza, 1964 (status revalidated) is removed from synonymy with *Acontista* Saussure, 1869 and now includes, in addition to *M. pracicabensis* (its type species), the following three species, all currently assigned to *Acontista*: *M. bimaculata* (Saussure, 1870) (new combination), *M. brevipennis* (Saussure, 1872) (new combination), and *M. rehni* (Giglio-Tos, 1927) (new combination). In Acanthopidae: *Acanthops coloradensis* González, Miller and Salazar, 2012 and *Acanthops septemspinosa* Ippolito, 2007, are both transferred to *Metilia* Stål, 1877, respectively as *Metilia coloradensis* (González, Miller and Salazar, 2012) (new combination) and *Metilia septemspinosa* (Ippolito, 2007) (new combination); *Plesiacanthops* Chopard, 1913 (status revalidated) is removed from synonymy with *Acanthops* Burmeister, 1838, whereas *Miracanthops* Roy 2004 is a new synonym of *Plesiacanthops*, which now includes, in addition to *Plesiacanthops tuberculata* (its type species), the following species, all currently assigned to

Miracanthops: *P. lombardoi* (Roy, 2004) (new combination), *M. occidentalis* Lombardo and Ippolito, 2004 (new combination), and *M. poulaini* (Roy, 2004) (new combination).

Nomenclatorial procedures conducted include 11 new synonyms in the species-group, four new synonyms in the genus-group, 40 new combinations, and four lectotype designations.

Collectively, the Acanthopoidea now includes eight families, 15 subfamilies, 18 tribes, 63 genera (including one *genus inquirenda*), and 315 species considered valid. I also present dichotomous keys for the identification of all acanthopoid genera, annotated checklists of species for each genus, and abundant visual material to aid taxonomic identification.

3.2 Introduction

The Acanthopoidea is here regarded as monophyletic assemblage of Neotropical praying mantises comprising eight families: Thespidae, Angelidae, Coptopterygidae, Liturgusidae, Photinainidae, Stenophyllidae, Acontistidae and Acanthopidae. Although this eight-family scheme was only recently established by Rivera and Svenson (2016), Yager and Svenson (2008) were the first to recognize that members of these taxa formed a natural group near the base of the mantodean tree of life. This clade was characterized by their presumed lack of auditory capabilities, forming a sister-group with the rest of mantodean taxa that do exhibit this capability — an adaptation which is regarded as an evolutionary response to ultrasonic biosonar signalling of insectivorous bats during nocturnal flight, see Yager *et al.* (1990). This “deaf” clade was again resolved with high support and in the same position in Svenson and Whiting’s (2009) large-scale phylogenetic analysis of the Mantodea; accordingly, Svenson and Whiting (2009) dubbed this clade “earless praying mantises”. Rivera and Svenson’s (2016) exploration of phylogenetic relationships within this lineage resulted in a major rearrangement of higher-level classification, notably: i) elevation of several subfamilies to family-level (Angelidae, Coptopterygidae, Photinaidae, Stenophyllidae, Acontistidae), ii) complete reorganization of the Thespidae, including the recognition of several new family-group taxa, such as Pseudopogonogastrinae, Bantiinae, Musoniellini (Thespinae), the rejection of Oligonicinae, and numerous re-assignments of genera to new and existing subfamilies and tribes; iii) reorganization of the Photinaidae, including the recognition of new family-group taxa such as Photiomantinae, and Microphotinini

(Photinae), and ; iv) exclusion of many old world genera previously classified among Thespidae, Angelidae, and Liturgusidae. Rivera and Svenson (2016) assigned members of this “deaf” clade to the superfamily Acanthopoidea, dubbing them “polymorphic earless praying mantises” (PEPM hereafter). These polymorph earless mantises are distinguished from the also earless (though morphologically more conservative), Neotropical families Chaeteessidae and Mantoididae — two primitive lineages that form a sister-group relationship with Acanthopoidea + Cernomantodea (i.e. the hearing mantises *sensu* Svenson and Whiting, 2009) and symplesiomorphically sharing the lack of the hearing apparatus. Figure 3.1 is a summary of the higher-level relationships and classification of Rivera and Svenson (2016), which is followed here.

Although Rivera and Svenson (2016) brought the higher-level systematics of Acanthopoidea into a more organized state, it raised questions about morphological boundaries and taxonomic status of many genera and species. For instance, genera such as *Bantia*, *Musoniella*, *Oligonyx*, *Carrikerella* (Thespidae), *Metriomantis* (Photinaidae), *Acontista* (Acontistidae) and *Acanthops* (Acanthops) were found not to be monophyletic. Rivera and Svenson (2016) suggested that instances of non-monophyly resulted from misassignment at the time of description. Recently published studies have revealed that misinterpretation of intrageneric variation (including sexually dimorphic characters), poor morphological definitions of taxa, and lack of phylogenetic studies, have resulted in extensive synonymies at the genus-group level, both within the Acanthopoidea and other Neotropical lineages (e.g. Terra 1995, Rivera 2001, 2005, 2010b, 2014, Rivera *et al.* 2011, Agudelo 2014, Agudelo and Rivera 2015, Ippolito and Lombardo 2015, Svenson *et al.* 2016).

No taxonomic treatment of the Acanthopoidea has so far being undertaken, largely because the PEPM were only recently recognized as monophyletic by molecular phylogenetic analyses (Yager and Svenson 2008, Svenson and Whiting 2009). Instead, fragmentary studies involving description of new taxa, distributional records (both local and regional) and, more recently, taxonomic revisions of individual genera, dominate in the literature — a historical account of praying mantis taxonomy in the neotropics has been presented elsewhere (Rivera 2010c). From all these studies, Terra (1995) stands out as the first genus-level revision of Neotropical praying

mantises, including what now represents the Acanthopoidea. Terra (1995) provided morphological diagnoses for all supraspecific taxa (up to family-level), as well as for most of the 74 Neotropical genera he accepted as valid. Also included were identification keys and checklist of species with known geographical distribution. Terra (1995) also presented a new classification for the six families he considered to be represented in the region, namely Chaeteessidae, Mantoididae, Acanthopidae, Vatidae, Liturgusidae, and Thespidae, with special emphasis in the latter (Rivera and Svenson 2015). Although Terra's (1995) classification was not developed in an explicitly phylogenetic framework, his contribution was nonetheless a major milestone to our understanding of otherwise little known Neotropical Mantodea (Rivera 2010c). Taxonomic treatments of various Acanthopoidea genera followed in the wake of Terra's (1995) landmark study (e.g., Lombardo 1996, 1998, 1999, 2000a, 2000b, 2000c, Lombardo and Ippolito 2004, Lombardo and Marletta 2004, Lombardo *et al.* 2013, Roy 2002a,b,c, 2004, 2005a,b, 2006, 2015, Roy and Stiewe 2011, Rivera 2003, 2005, 2010a, Rivera *et al.* 2011, Scherrer 2014, Svenson 2014b). Studies addressing nomenclatural matters, or clarifying the taxonomic status of little known taxa, have been a major focus of inquiry in recent years (e.g. Agudelo 2014, Agudelo and Rafael, 2014, Agudelo and Rivera 2015, Maldaner *et al.* 2015; Rivera 2010b, 2014, Roy 2011, 2013, 2014; Svenson 2014a, 2014c, Svenson and Branham, 2007), spurred largely by ongoing and forthcoming developments in Mantodea phylogenetics (Roy 2014, Agudelo and Rivera 2015). With this wealth of information now available, the timing is opportune for a revision of the Acanthopoidea, a clade that represents at least 55% of praying mantis diversity in the Neotropical Region.

In this paper I supplement the phylogenetic classification of Rivera and Svenson (2016) by revisiting all acanthopoid genera to assess their taxonomic status and morphological boundaries. The specific goals of this study are to: i) morphologically diagnose all acanthopoid genera; ii) validate the generic assignment of all nominal species, and; iii) build a more solid nomenclatorial framework from which to launch future revisionary studies.

3.3 Material and Methods

3.3.1 Examined material

This study is based largely on material held by the National Museum of Natural History, Smithsonian Institution (USNM - Washington D.C., USA). Additional specimens, including reported type material, were studied in the following collections (listed in alphabetic order by acronym): American Museum of Natural History (AMNH - New York, USA), Academy of Natural Sciences of Philadelphia (ANSP - Pennsylvania, USA), The Natural History Museum (BM - London, United Kingdom), California Academy of Sciences (CAS - San Francisco, USA), Carnegie Museum of Natural History (CM - Pittsburgh, USA), Charles Darwin Research Station (CDRS - Galapagos Islands, Ecuador), Cleveland Museum of Natural History (CMNH - Cleveland, USA), Escola Superior de Agricultura “Luiz de Queiroz” Universidade de São Paulo (ESALQ - Piracicaba, Brazil), Instituto Nacional de Pesquisas da Amazônia (INPA - Manaus, Brazil), Instituto Fundación Miguel Lillo (IFML - Tucumán, Argentina), Field Museum (FMNH - Chicago, USA), Muséum national d'Histoire naturelle (MNHN - Paris, France), Museo Nacional de Ciencias Natural (MNCN - Madrid, Spain), Martin Stiewe Private Collection (MSMC - London, UK), Museu de Zoologia da Universidade de São Paulo, (MZUSP - São Paulo, Brazil), Muséum d'histoire naturelle de la Ville de Genève, (MHNG - Geneva, Suisse), Museo Nacional de Historia Natural, Universidad Nacional Mayor de San Marcos, (MUSM - Lima, Peru), Museu de Ciências e Tecnologia, Pontifícia Universidade Católica do Rio Grande do Sul, (MCTP - Porto Alegre, Brazil), Muzeum i Instytut Zoologii Polskiej Akademii Nauk (MIZ - Warsaw, Poland), Naturhistorisches Museum Wien (NHMW - Vienna, Austria), Oxford University Museum of Natural History (OUMNH - Oxford, UK), Royal Ontario Museum (ROM – Toronto, Canada), Senckenberg Deutsches Entomologisches Institute (SDEI - Müncheberg, Germany), University of Minnesota Insect Collection (UMSP - Minneapolis, USA), Museo de Entomología *Klaus Raven Büller*, Universidad Nacional Agraria La Molina, (UNALM - Lima, Peru), Museum für Naturkunde der Humboldt Universität (ZMB - Berlin, Germany), Zoologischen Museums Greifswald (ZMG - Greifswald, Germany), Biozentrum Grindel und Zoologisches Museum (ZMUH - Hamburg, Germany), Zoologische Staatssammlung München (ZSM - Munich, Germany). A total of 1336 specimens, 238 of which represented

primary types (158 holotypes, 72 syntypes, 8 lectotypes), totalling 315 species considered valid, were examined. All material examined (including type specimens and depository) are listed in Appendix 1.

3.3.2 Morphology, taxonomic identification and nomenclature

Terms for structures and methods follow Rivera and Svenson (2016). Additional morphological structures not treated in that study are herein labeled for clarity. I identified specimens using one or more of the following procedures: i) by consulting original descriptions; ii) by comparing specimens against type material, and; iii) by consulting modern taxonomic revisions. Original descriptions vary greatly in quality depending on author(s) (Rivera 2010c), and thus these were considered with reservations unless the corresponding type material was also available. In most cases comparison with type material relied on external morphology as dissection of male genitalia was not possible in most instances. To compensate, I dissected topotypical material whenever possible (i.e. non-type material collected from the vicinity of the type locality), in order to enhance accuracy of identification. Modern taxonomic treatments, including individual species descriptions and/or generic revisions, when available, were favoured for taxonomic identification. Nomenclatorial actions (e.g. synonymies) were made with reference to the International Code of Zoological Nomenclature (ICZN 1999). Each diagnosis includes a new version of the spination formula originally introduced in Rivera (2010a), and modified in Brannoch and Svenson (2016) according to the morphological conventions of Wieland (2013). The spination formula specifies the number and variation in the number of spines in both femur (F) and tibiae (T). Other abbreviation in the formula: DS, discoidal spines; AvS, anteroventral spines; PvS, posteroventral spines (see Rivera and Svenson 2016).

3.3.3 Scientific illustration and digital imaging

Line artwork was executed with the aid of a camera lucida attached to a stereomicroscope. Standard views (Figs. 3.2–3.13) are presented of various genus-level diagnostic features of the head and prothorax, with particular emphasis on foreleg spination in Thespidae. Illustrations of wings, cursorial legs, terminalia and male genitalia are presented for some taxa. In most cases,

representatives of both sexes are figured to account for differences attributed to sexual dimorphism. Most type specimens figured (Figs. 3.14–3.47) were examined under magnification with a Nikon SMZ645[®] stereomicroscope, and photographed using a Passport Storm[©] portable photographic system (Visionary Digital[™], 2012). This latter system includes a Stackshot z-stepper, a Canon 5D SLR, macro lenses (50mm, 100mm, and MP-E 65mm), three Speedlight 580EX II flash units, and a computer with Canon utility and Adobe Lightroom 3.6 software. The z-stepper was controlled through Zerene Stacker 1.04 and images were processed using the P-Max protocol. Specimens were placed on an 18% grey card background to facilitate adjusting white balance standards during image processing. All line artwork and photographic images were processed using Adobe Photoshop CS6; plates were assembled using Adobe Illustrator CS6.

3.3.4 Generic diagnosis

Morphological diagnoses are provided for all genera considered valid. To simplify taxonomic identification and minimize wording, I emphasize only relevant and readily recognizable characters that apply to both sexes; however, in instances of marked sexual dimorphism, characters that apply to each gender are properly highlighted in the diagnoses. Also reported in each generic diagnosis is the observed body size range (in mm), and details of colour variation based on observations of preserved specimens. A checklist of species is provided for each genus (see below), with comments on controversially classified species discussed in the text.

3.3.5 Distributional records

One important caveat pertaining to geographical distribution of Neotropical Mantodea is the high number of species misidentifications in the literature. For instance, Agudelo and Rivera (2015) gave a detailed account about the contributions of L.J. Jantsch, whose questionable taxonomic practices resulted in numerous erroneous records on mantodeans in Brazil. Many of these records can be traced back to Jantsch's partially unpublished PhD thesis (Jantsch 1999), in addition to other contributions by the same author (e.g. Jantsch and Corseuil 1988; Dornels *et al.* 2005).

These records were included in the latest Checklist of Neotropical Mantodea (Agudelo *et al.* 2007), but many have been clarified in recent times (e.g., Roy and Ehrmann 2009; Roy and Stiewe 2011; Rivera *et al.* 2011; Scherrer 2014; Agudelo and Rivera 2015). Nevertheless, it is likely that many more spurious records remain to be recognized. Consequently, current geographical records of Neotropical Mantodea warrant a critical assessment from future researchers. In order to avoid perpetuating suspect records, I report only the following: i) type locality by country (also by state for Brazil and USA), ii) label data of specimens examined, iii) localities reported in the most recent and peer-reviewed taxonomic treatments. Sources of distributional data other than the aforementioned are properly cited in the text (e.g. Ehrmann, 2002, Agudelo *et al.* 2007). Distributional records for Nearctic species complementary to those herein reported are from Gurney (1951). All label data is transcribed in Appendix 2.

3.3.6 Annotated checklist

A checklist of species is presented for each genus. Each species entry includes the following information in annotated form:

- All taxonomic and nomenclatural modifications are highlighted in bold (e.g., “**comb. n.**”, “**syn. n.**”).
- Type specimens examined are indicated with an asterisk (*).
- Only the original generic assignment and known synonyms in their original form are listed under each species. For example, *Photiomantis planicephala* (current name) was originally described as *Metriomantis planicephala* and has two known synonyms: *Photiomantis silvai* and *Margaromantis margaritaria*; all these names are listed in the checklist. However, former combinations under other generic names (now synonyms of *Photiomantis*), such as *Colombiella* (as *C. planicephala*) or *Lombardoa* (as *L. planicephala*), are not included in the checklist. For combinations under genera other than original and not listed here, refer to Ehrmann (2002), Otte and Spearman (2005), and Agudelo *et al.* (2007)

- Geographical distribution is cited in abbreviated form by country and in square brackets; Brazilian and USA states are cited in parentheses. Abbreviations for countries: **AR**-Argentina; **AB**-Antigua and Barbuda; **BH**-Bahamas; **BE**-Belize; **CI**-Cayman Islands; **CO**-Bolivia; **BR**-Brazil; **CH**-Chile; **CO**-Colombia; **CR**-Costa Rica; **CU**-Cuba; **DO**-Dominica; **DR**-Dominican Republic; **EC**-Ecuador; **ES**-El Salvador; **FG**-French Guiana; **GA**-Guadeloupe; **GR**-Grenada; **GT**-Guatemala; **GY**-Guyana; **HA**-Haiti; **HO**-Honduras; **JA**-Jamaica; **MA**-Martinique; **MX**-Mexico; **NI**-Nicaragua; **PA**-Panama; **PY**-Paraguay; **PE**-Peru; **SU**-Suriname; **TT**-Trinidad and Tobago; **US**-United States; **UY**-Uruguay; **VE**-Venezuela. Abbreviations for cited Brazilian states: **AP**-Amapá, **AM**-Amazonas, **BA**-Bahia, **CE**-Ceará, **DF**-Distrito Federal, **ES**-Espírito Santo, **GO**-Goiás, **MA**-Maranhão, **MT**-Mato Grosso, **MS**-Mato Grosso do Sul, **MG**-Minas Gerais, **PA**-Pará, **PB**-Paraíba, **PR**-Paraná, **PE**-Pernambuco, **PI**-Piauí, **RJ**-Rio de Janeiro, **RN**-Rio Grande del Norte, **RS**-Rio Grande del Sur, **SC**-Santa Catarina, **SP**-São Paulo. Abbreviations for USA states: **AR**-Arkansas, **AZ**-Arizona, **FL**-Florida, **KS**-Kansas, **OK**-Oklahoma, **GA**-Georgia, **HI**-Hawaii, **NC**-North Carolina, **TX**-Texas.
- Countries (and Brazilian and USA states) where primary type (i.e., holotype, syntypes, lectotype, or neotype) localities are located are underlined (e.g. [BR (MT), BO, AR] or [AR, BR (GO, MT), PY]). In instances where more than one country is underlined, it means that the indicated species is typologically represented by two or more syntypes from different localities (e.g. [BO, BR (MT), AR, PY]). See Appendix 2 for details of type status.

3.4 Results

Rivera and Svenson (2016) presented diagnoses and identification keys to all higher-level categories within the Acanthopoidea. This remains the primary taxonomic resource for recognition of families, subfamilies and tribes. In this contribution, I present diagnoses and identification keys for all valid genera, except for monogeneric taxa: *Pseudopogonogaster* (Thespidae-Pseudopogonogastrinae), *Angela* (Angelidae), *Stenophylla* (Stenophyllidae), and *Macromantis*, *Photiomantis*, *Cardioptera* (in the photinaid subfamilies Macromantinae,

Photiomantinae, and Cardiopterinae, respectively). In such instances the generic diagnosis corresponds within its higher-level rank as described by Rivera and Svenson (2016).

3.4.1 Family Thespidae Saussure, 1869

On the basis of phylogenetic evidence and male genitalic morphology, Rivera and Svenson (2016) subdivided the Thespidae into five subfamilies: Pseudopogonogastrinae (monogeneric), Pseudomiopteryginae, Bantiinae, Miobantiinae and Thespinae (including tribes Musoniellini and Thespini). They also presented a dichotomous key to distinguish among subfamilies and tribes of Thespidae, also using mainly male genitalic characters. The genus-level dichotomous keys in the present work are essentially expansions of those keys, but also included is a separate key for females. The rationale for this approach is threefold: i) thespids often exhibit marked sexual dimorphism; ii) subfamilies and tribes cannot be distinguished based on female external morphology, and; iii) females exhibit a wide array of morphological modifications for crypsis that are not represented in males. Accordingly, the females of most thespid genera can be easily recognized whereas the males cannot. The following key is for the identification of genera based on females. For identification of suprageneric thespid taxa based on male specimens, readers are referred to the keys of Rivera and Svenson (2016). Keys to genera based on males are also included here in the section for each subfamily.

Key to the genera of Thespidae (adult females)

Note: Females of *Promiopteryx* (Pseudomiopteryginae), *Diabantia* (Bantiinae) and *Paradiabantia* (Miobantiinae) remain unknown. However, members of *Paradiabantia* exhibit three discoidal spines in the forefemora, an uncommon condition among the Thespidae (the usual number of discoidal spines observed is four). Although this character is known only for *Paradiabantia* males, the number of discoidal spines is a well-conserved, non-sexually dimorphic character at the genus level in this family. Accordingly, I provisionally add this character state to include the female of *Paradiabantia* in the key. Subfamily abbreviations within the key: **Bant** = Bantiinae; **Miob** = Miobantiinae; **Pseg** = Pseudopogonogastrinae; **Psem** = Pseudomiopteryginae; Thespinae = **Thes**.

1. Foretibiae with 1 or 2 of its distal-most anteroventral spines slightly to strongly displaced from the ventral margin of the tibia to a more dorsolateral or decidedly dorsal position, in the lateral view either clearly projecting over the apical tibial spur or overlapping with it (Figs. 3.2n, 3.3j, 3.3m, 3.7j, 3.7m-n, 3.8d, 3.9b, 3.9k)2
- 1'. Foretibiae with all its anteroventral spines clearly situated near or along the ventral margin of the tibia, in the lateral view not projecting over the tibial apical spur nor overlapping with it (Figs. 3.2f, 3.5j, 3.6j, 3.6m, 3.8j).....13
2. Foretibiae with 1 or 2 of its distal-most anteroventral spines strongly displaced from the ventral margin of the tibia, in the lateral view clearly projecting over the apical tibial spur (Figs. 3.2n, 3.3m, 3.7j, 3.9b, 3.9k).....3
- 2'. Foretibiae with only its distal-most anteroventral spine less strongly displaced from the ventral margin of the tibia, in the lateral view clearly overlapping with the tibial apical spur (e.g. Fig. 3.8d).....10
- 3(2). Foretibiae only with its distal-most anteroventral spine strongly displaced from the ventral margin and more dorsally-positioned (Figs. 3.2n, 3.3j, 3.7m).....4
- 3'. Foretibiae with its two distal-most anteroventral spines strongly displaced from the ventral margin and more dorsally-positioned 3.3m, 3.9b, 3.9k).....8
- 4(3). Foretibiae with 4–7 posteroventral spines.....5
- 4'. Foretibiae with 1 posteroventral spine.....7
- 5(4). Meso- and metanotum each with a prominent, spine-like, dorsal projection.....*Mantillica* [Bant]
- 5'. Meso- and metanotum without projections.....6
- 6(5'). Forefemora with its dorsal margin expanded in the form of a rudimentary but evident lamella (as in Fig. 3.3h).....*Mantellias* [Bant]
- 6'. Forefemora with its dorsal margin not forming a lamella.....*Bantia* (in part) [Bant]
- 7(4'). Juxtaocular bulges well developed, conical in shape, forming a strong concave outline with the vertex (Fig. 3.7k); forefemora with 3 discoidal spines.....*Oligonyx* [Thes]

- 7'. Juxtaocular bulges moderately developed, forming a somewhat weak, concave outline with the vertex (Fig. 3.7d); forefemora with 4 discoidal spines.....***Oligonicella*** [Thes]
- 8(3'). Forecoxae considerably longer than pronotal metazona (Fig. 3.3k); forefemora with 4 discoidal spines; body short and compact (body length no more than 25 mm)...***Thrinaconyx*** [Bant]
- 8'. Forecoxae as long or shorter than pronotal metazona (Fig. 3.9a, 3.9j); prothoracic femora with 3 discoidal spines; body more elongated (body length no less than 35 mm).....9
- 9(8'). Forefemora with 1 posteroventral spine..... ***Thesprotia*** [Thes]
- 9'. Forefemora with 4 posteroventral spines.....***Thesprotiella*** [Thes]
- 10(2'). Smaller mantises (body length less than 25 mm).....***Bantia*** (in part) [Bant]
- 10'. Larger mantises (body length 35 mm or longer).....11
11. Pronotal surface conspicuously and densely granulated, some granules are conspicuously larger, resembling pearl-like tubercles, often paired on each side of the medial carina (although not always at the same level) (Fig. 3.7c).....***Musoniola*** [Thes]
- 11'. Pronotum smooth.....12
12. Pronotal metazona with straight, parallel margins (Fig. 3.8c); pronotal surface lacking maculae***Pseudomusonia*** [Thes]
- 12'. Pronotal metazona medially narrowed, not parallel (Fig. 3.6c); pronotal surface with a variable number of maculae, often paired on each side of the medial carina.....***Dougonyx*** gen. n. [Thes]
- 13(1'). Forefemora with 3 discoidal spines (Fig. 3.4t).....***Paradiabantia*** [Miob]
- 13'. Forefemora with 4 discoidal spines.....14
- 14(13'). Foretibiae with 5 or more posteroventral spines.....15
- 14'. Foretibiae with 1–3 posteroventral spines.....28
- 15(14). Costal edge of wing pads serrated.....16
- 15'. Costal edge of wing pads not serrated.....17
- 16(15). Head with a well developed carina that runs across the head and behind the ocelli,

- extending backwards along the upper edges of compound to reach the vertex, delimiting a flattened area on top of the head (sometimes this area exhibits minute tubercles) (Fig. 3.5r); pronotum short and stocky (Fig. 3.5t).....**Pizaia** [Thes]
- 16'. Head without such carina (Fig. 3.5g); pronotum elongated (Fig. 3.5i).**Leptomiopteryx** [Thes]
- 17(15'). Foretibiae with 5 posteroventral spines.....18
- 17'. Foretibiae with 6 or more posteroventral spines.....22
- 18(17). Foretibiae less than half as long as forefemora (i.e., tibial spur groove is closer to the tip of the femur than to its base).....**Macromusonia** [Thes]
- 18'. Foretibiae half as long as forefemora or longer (i.e., tibial spur groove is either centrally located or closer to the base of the femur).....19
- 19(18'). Forecoxae as long as or shorter than pronotal metazona.....20
- 19'. Forecoxae longer than pronotal metazona.....21
- 20(19). Cerci twice as long as supraanal plate, considerably projecting beyond the tip of the latter (as in Fig. 3.8k).....**Thespis** [Thes]
- 20'. Cerci as long as supraanal plate or only slightly longer, projecting scarcely beyond the tip of the latter.....**Musonia** [Thes]
- 21(19'). Supraanal plate at most twice as long as its basal width (Fig. 3.5p-q).....**Musoniella** [Thes]
- 21'. Supraanal plate more than twice as long as its basal width (Fig. 10d-e) (supraanal plate sometimes as long pronotum).....**Eumusonia** [Thes]
- 22(17'). Foretibiae with 6 posteroventral spines.....23
- 22'. Foretibiae tibiae with 7 or more posteroventral spines.....24
- 23(17). Juxtaocular bulges conical and prominent, rising well above the vertex and the imaginary line connecting the dorsal margins of both compound eyes Fig. 3.4b); pronotum surface with multiple and prominent tubercles (Fig. 3.4d).....**Anamiopteryx** [Miob]
- 23'. Juxtaocular bulges poorly developed, rising only to level of vertex, barely more elevated than the imaginary line connecting the dorsal margins of both compound eyes; pronotum surface smooth or at most with swollen areas, never forming prominent

tubercles.....	<i>Pseudopogonogaster</i> [Pseg]
24(22'). Pronotal surface with tubercles (Fig. 3.2d).....	<i>Pseudomiopteryx</i> [Psem]
24'. Pronotal surface without tubercles.....	25
25(24'). Forefemora with 10 or more anteroventral spines; foretibiae with anteroventral spines gradually increasing size distad.....	26
25'. Forefemora with 9 or less anteroventral spines; foretibiae with anteroventral spines not gradually increasing size distad (i.e., one or more spines might be substantially shorter than a preceding spine).....	<i>Bantiella</i> [Bant]
26(25). Anterolateral margins of pronotal prozona with a small triangular process, and a longitudinal, proximal carina on each side of the metazoan (Fig. 3.4m); dorsal margin of forefemora slightly expanded and curved, appearing convex in lateral view.....	<i>Eumiopteryx</i> [Miob]
26'. Lateral margins of pronotal prozona and metazona smooth; dorsal margin of forefemora straight or only slightly sinuous.....	27
27(26'). Foretibiae with proximal-most posteroventral spine distinctly separated from remaining spines by at least twice the distance between any other two spines in the same series.....	<i>Miobantia</i> [Miob]
27'. Prothoracic tibiae with all posteroventral spines more or less equally spaced	<i>Chloromiopteryx</i> [Miob]
28(14'). Foretibiae with its 3 large, distal-most anteroventral spines equally spaced (Fig. 3.6j)..	29
28'. Foretibiae with its 3 large, distal-most anteroventral spines not equally spaced (as in Fig. 3.6m); distributed in Jamaica.....	<i>Liguanea</i> [Thes]
29(28). Pronotum elongated (total body length 29 mm or more) (Fig. 3.6h-i); distributed on the Galápagos Island.....	<i>Galapagia</i> [Thes]
29'. Pronotum more compact (total body length 22 mm or less) (Fig. 3.7q); distributed in central Peru (Pacific coast and occidental slopes of the Andes at mid elevations)	<i>Piscomantis</i> gen. n. [Thes]

3.4.1.1 Subfamily Pseudopogonogastrinae Rivera and Svenson, 2016

Rivera and Svenson (2016) established this subfamily to accommodate *Pseudopogonogaster* Beier. Therefore, the morphological diagnosis of the Pseudopogonogastrinae also applies to the genus (see also Rivera *et al.* 2011). Parada (2015) recently described the male of *P. iguaquensis*.

Pseudopogonogaster Beier, 1942

Pseudopogonogaster Beier, 1942: 132–133. Type species: *Pseudopogonogaster mirabilis* Beier, 1942 (by original designation and monotypy).

=*Calopteromantis* Terra, 1982: 330. Type species: *Calopteromantis hebardi* Terra, 1982 (by monotypy).

Distribution. Mountainous rainforests of the tropical Andes, from Central Peru, to Ecuador, Colombia and Venezuela, in elevations as high as 3500m.

Species checklist

Pseudopogonogaster hebardi (Terra, 1982) [EC] – **Fig. 3.14a**

=*Calopteromantis hebardi* Terra, 1982*

Pseudopogonogaster iguaquensis Carrejo and Salazar, 2002 [CO]

Pseudopogonogaster kanjaris Rivera and Yagui, 2011* [PE]

Pseudopogonogaster marulandae (Salazar, 2002) [CO]

=*Calopteromantis marulandae* Salazar, 2002

Pseudopogonogaster mirabilis Beier, 1942* [EC]

=*Calopteromantis otongica* Lombardo and Ayala, 1998

Pseudopogonogaster muscosa Salazar, 2000 [CO]

3.4.1.2 Subfamily Pseudomiopteryginae Giglio-Tos, 1915

Key to genera (males)

1. Lower ocelli situated on a spine-like, cuticular projection (Fig. 3.2a). Apex of mesothoracic wings rounded..... ***Pseudomiopteryx***
- 1'. Lower ocelli not situated on a spine-like, cuticular projection (Fig. 3.2g). Apex of mesothoracic wings with a small, curved projection..... ***Promiopteryx***

Pseudomiopteryx Saussure, 1870 (Fig. 3.2a-f)

Pseudomiopteryx Saussure, 1870: 225. Type species *Pseudomiopteryx spinifrons* Saussure, 1870 (subsequent designation: Kirby, 1904: 275).

=*Myrcinellus* Giglio-Tos, 1898: 15. Type species: *Myrcinellus festae* Giglio-Tos, 1898 (by monotypy).

Distribution. Southern Mexico south through Central America to the Guiana Shield and Amazon Basin.

Diagnosis. *Size range:* 14–25 mm.

Body colouration: light to dark brown, forecoxae reddish brown anteriorly.

Head: Vertex well developed, slightly sinuous, juxtaocular bulges rising above level of vertex (Figs. 3.2a-b). Lower ocellus of males borne on the base of a well-developed, pointed, cuticular process (Fig. 3.2a); the latter absent in females (Fig. 3.2b).

Thorax: pronotum short and stocky (Figs. 3.2c-d); longitudinal carina conspicuous along metazona, slightly elevated distally in lateral view; supracoxal dilation well developed, often with angular outline; surface with various tubercles and granulations, a pair of these at the supracoxal dilation section of the metazona are markedly more developed and always present in both sexes (Figs. 3.2c-d). Proximal tubercle of metazona well developed in both sexes.

Forecoxae longer than metazona of pronotum (Fig. 3.2c). Spination formula: F=4DS/11–12AvS/4PvS; T=8–10AvS/7–9PvS. Anterior aspect of female forefemora almost always with a

large, purplish-black mark with metallic reflections, whereas that of males with less conspicuous colouration, with little to no metallic reflections. Tibial posteroventral spines evenly increasing in size distad (Fig. 3.2l), with the anteroventral series subdivided into 2 groups (g1 and g2 in Fig. Xe-f)): g1 contains 3–4 spines that are conspicuously shorter than the distal group, g2 consists of 5 larger spines (the second spine slightly but distinctly shorter than the preceding one); this two-group spine arrangement of spines is more evident in females, whose tibial spines are relatively larger and more erect than those of males (Fig. 3.2f). Males: wings broad, smoky brown, hyaline and iridescent, their apex extend beyond tip of subgenital plate in resting position.

Abdomen: slender and cylindrical in males, dilated and trapezoidal in females.

Remarks. *Pseudomiopteryx* is a widely distributed, relatively diverse, yet morphologically conservative genus of Pseudomiopteryginae mantises. Considerable confusion about species identity is evident in the literature. All species are difficult to distinguish using original descriptions, which in all cases rely exclusively on external morphology. Pronotal shape, body size and proportions, and wing length, all exhibit marked intra- and interspecific variation. The shape of the lower frons (frontal shield of authors) — previously proposed as a mean to distinguish among members of this genus — exhibit considerable variation and thus is an unreliable character for identification (Rehn, 1935, Ariza *et al.*, 2012). In contrast, interspecific variation is reliably expressed in male genitalia, in particular the shape of the apofisi falloidi, and the ventral and left phallomeres. Most species are known only from males, and knowledge of females remain scant, making literature-based species identification difficult. Conspicuous sexual dimorphism also makes it difficult to associate conspecific males and females. Females are rare in entomological collections; in contrast, males are much more common and are often collected in great numbers using Malaise traps. Given the widespread distribution of *Pseudomiopteryx* it is possible that an important number of localized, undescribed species remain to be discovered.

Species checklist

Pseudomiopteryx amazonensis Piza, 1968* [[BR](#) ([AM](#))]

Pseudomiopteryx bogotensis Saussure, 1870* [[CO](#)] – **Fig. 3.14b**

Pseudomiopteryx bogotensis gorgonae Ariza and Salazar, 2005 [CO]

Pseudomiopteryx columbica Giglio-Tos, 1915* [CO] – **Fig. 3.14c**

Pseudomiopteryx decipiens Giglio-Tos, 1915* [CO] – **Fig. 3.14d–f**

Pseudomiopteryx festae (Giglio-Tos, 1898) [EC]

Myrcinellus festae Giglio-Tos, 1898

Pseudomiopteryx guyanensis Chopard, 1911* [FG] – **Fig. 3.14g, 3.15a**

Pseudomiopteryx infuscata Saussure and Zehntner, 1894* [MX, GU, HO, BE, NI, CR, PN] – **Fig. 3.15b–c**

Pseudomiopteryx maculata Beier, 1942 [VE]

Pseudomiopteryx meridana Giglio-Tos, 1915* [FG] – **Fig. 3.15d**

Pseudomiopteryx spinifrons Saussure, 1870* [BR?] – **Fig. 3.15e**

Taxonomic notes: Hebard (1933) suggested that *P. columbica* and *P. decipiens* could represent variants of *P. bogotensis*. My analyses of type specimens suggest that Hebard’s proposal could well hold true; future studies are needed to confirm this possibility. Ariza and Salazar (2005) proposed the subspecies *P. bogotensis gorgonae* from Gorgona Island, off the coast of Colombia near the Ecuadorian border. Ariza and Salazar (2005) justified their new subspecies by contrasting its morphology with the original description and subsequent redescription of *P. bogotensis* by Giglio-Tos (1927). However, they did not explain how this subspecies differed from the nominative form. Considering the level of intra- and interspecific variation in *Pseudomiopteryx*, the subspecific status of *Pseudomiopteryx bogotensis gorgonae* is in need of re-assessment.

Ehrmann (2002) reported *P. guyanensis* for Peru, but comparisons of Peruvian populations with the type specimen of *P. guyanensis* revealed them to be heterospecific. *Pseudomiopteryx* populations from Peru and Bolivia that I examined most likely represent undescribed species.

Pseudomiopteryx infuscata is the most widely distributed species of the genus, occurring from central Mexico south to northern Panama. Ariza *et al.* (2012) illustrated the purported male genitalia of *infuscata*; however, it does not correspond with this species. I have examined specimens from central Panama, Colombia (Anolaima, near Bogotá) and Ecuador (Puyo) that correspond well with Ariza *et al.*’s depiction. However, they cannot be attributed to this species

because in this the apophysi falloid has a forked apex, whereas the same is unforked in *P. infuscata*. Given its superficial resemblance to *P. infuscata* and its distribution in northwestern South America, this unidentified species might account for some of the southern-most literature records of *P. infuscata*.

Promiopteryx Giglio-Tos, 1915 (Fig. 3.2g-h)

Promiopteryx Giglio-Tos, 1915: 137 (nomen novum pro *Miopteryx* Saussure, nec Blanchard, 1842-Coleoptera, Cerambycidae). Type species: *Miopteryx granadensis* Saussure, 1870 (original designation).

=*Miopteryx* Saussure, 1870: 55, 64 (nec Blanchard, 1842-Coleoptera, Cerambycidae). Type species: *Miopteryx granadensis* Saussure, 1870 (subsequent designation: Rehn, 1904: 566).

Distribution: northern South America.

Diagnosis (males only): *Size range:* 20–25 mm.

Body coloration: light to dark brown.

Head: Vertex elevated and concave (Fig. 3.2g), juxtaocular bulges poorly developed yet rising above the upper edges of compound eyes.

Thorax: pronotum relatively slender (Fig. 3.2h), metazona 1.3–1.8x as long a prozona, the latter markedly elongated, lateral margins with small denticles and longitudinal carina well developed along metazona, only scarcely expressed or vestigial along prozona. Forecoxae only slightly longer than metazona. Spination formula: F=4DS/12AvS/4PvS; T=9AvS/7PvS. In the foretibiae, the two basal-most spines of the posteroventral series are small and apart from the remaining ones. Males: meso- and metathoracic wings pale brown and hyaline, with scattered brown spots in the middle of some cells; apex of both pair of wings with a small projection that is strongly curved backwards (more accentuated in the metathoracic wings); wing apex extend beyond tip of subgenital plate in resting position.

Abdomen: slender and cylindrical.

Remarks. Species originally assigned to *Miopteryx* were variously assigned to *Promiopteryx*, *Miobantia* and other genera by early specialists, resulting in a complex nomenclatural history. Different authors simultaneously proposed a series of nomenclatorial remedies to address issues of generic affinity, until Terra (1995) considered only *Promiopteryx* and *Miobantia* as valid. Five species of *Promiopteryx* are recognized in the most recent checklists and catalogues of praying mantises (i.e. Ehrmann, 2002; Otte and Spearman, 2005; Agudelo *et al.*, 2007); however, *P. punctata* Giglio-Tos, 1915 and *P. stigmatica* (Burmeister, 1838) are here assigned to *Miobantia* and *Musoniella* respectively, thus reducing to three the number of valid species names assigned to *Promiopteryx*.

Species checklist

Promiopteryx fallax (Giglio-Tos, 1915)* [CO] – **Fig. 3.16a**

Miopteryx fallax Giglio-Tos, 1915

Promiopteryx granadensis (Saussure, 1870)* [CO] – **Fig. 3.16b**

Promiopteryx simplex Giglio-Tos, 1915 [VE]

Taxonomic notes: *Promiopteryx fallax* and *P. simplex* are very similar species. Giglio-Tos (1915b) distinguished *P. fallax* and *P. simplex* on the basis of differences he observed in the colour patterns of wings (e.g. the number and density of dark maculations). Hebard (1919) questioned the diagnostic value of such characters, considering them as mere intraspecific variation of *P. granadensis*. Hebard's opinion is reinforced by the fact that both species share the same type locality in Bogotá, Colombia. I nonetheless consider all three species valid pending study of additional material.

3.4.1.3 Subfamily Bantiinae Rivera and Svenson, 2016

Key to genera (males)

1. Foretibiae with all anteroventral spines clearly situated along the ventral margin.....2
- 1'. Foretibiae with 1 or 2 of its distal-most anteroventral spines strongly displaced from the ventral

- margin and more dorsal- or dorsolaterally positioned, in lateral view clearly projecting over the apical tibial spur or overlapping with it (Figs. 3.2n, 3.3j, 3.3m)3
- 2(3). Metazona of pronotum markedly short (1.2x as long as prozona) (Fig. 3.3e). Forefemora with 10 or more anteroventral spines. Foretibiae with anteroventral spines gradually increasing size distad.....**Diabantia**
- 2'. Metazona of pronotum relatively longer (1.4–1.6x as long prozona) (Fig. 3.3c). Forefemora with 8–9 anteroventral spines. Foretibiae with anteroventral spines not gradually increasing size distad (i.e., one or more spines is conspicuously shorter than a preceding one).....**Bantiella**
- 3(1'). Forefemora broad, its dorsal margin slightly expanded along its proximal $\frac{3}{4}$ to form a subtle but otherwise notorious lamella that is distally delimited by a sudden inflexion (Fig. 3.3h).....**Mantellias**
- 3'. Forefemora narrow, dorsal margin without lamella.....4
- 4(3'). Foretibiae with 2 distal-most anteroventral spines strongly displaced from the ventral margin and more dorsally positioned (Fig. 3.3m).....**Thrinaconyx**
4. Foretibiae with 1 distal-most anteroventral spine displaced from the ventral margin and more dorsal- or dorsolaterally positioned (Figs. 3.2n, 3.3j).....5
- 5(4). Head length (measured from top of vertex to tip of labrum) about as long as pronotum, the latter markedly short and stocky (metazona = 1.4–1.5x the length of prozona) (Fig. 3.3i).....**Mantillica**
- 5'. Head length (measured from top of vertex to tip of labrum) conspicuously shorter than pronotum, the latter elongated and not particularly stocky (metazona = 2.3–2.5x the length of prozona) (Fig. 3.2j-m)**Bantia**

Bantia Stål, 1877 (Fig. 3.2i-n)

Bantia Stål, 1877: 44. Type species: *Oligonyx pygmaea* Saussure, 1872 (by monotypy).

=*Eubantia* Giglio-Tos, 1915: 189. Type species: *Bantia fusca* Chopard, 1911 (by monotypy).

=*Mionicella* Giglio-Tos, 1915: 189. Type species: *Bantia wernerii* Chopard, 1913 (by original designation).

Distribution. Northern tropical Andes at mid elevation, Amazon basin, and Atlantic forest of southern Brazil.

Diagnosis. *Size range:* 13–18 mm.

Body colouration: light to dark brown.

Head: vertex variable, straight in males but slightly convex in females; juxtaocular bulges rising to level of vertex or only slightly more elevated; compound eyes prominent (Fig. 3.2i).

Thorax: pronotum relatively elongated (Figs. 3.2j–m), metazona 2.3–2.5x as long as prozona, its lateral edges bearing denticles; supracoxal dilation distinct; metazona sometimes slightly constricted medially. Forecoxae longer than metazona. Spination formula: F=4DS/9–10AvS/4PvS; T=5–7AvS/4–7PvS. Prothoracic tibiae (Fig. 3.2n) straight, the largest spines with microsculptures along surface; distal-most spine of the posteroventral series is largest, all other spines anterior to this one are smaller; tibial spines in the anteroventral series exhibit more variation in their size, but at least five of them are large (including the distal-most spine that is dorsally positioned), whereas other, much smaller spines, may be present proximally or in between any two larger spines. Forefemora elongated and curved (Fig. 3.2j), distal half conspicuously tapered; anterior genicular spine absent in some species. Males: wings iridescent, hyaline, either smoky or mostly colorless, sometimes with brown spots on cells.

Abdomen: cylindrical in both sexes, slightly dilated in females.

Remarks. My examination of the type specimen of *Miopteryx simoni* Chopard, 1916 revealed it as a member of *Bantiella*, and not *Bantia*, as Terra (1995) suggested. Accordingly, I treat *simoni* under *Bantiella* (see below).

Species checklist

Bantia chopardi (Giglio-Tos, 1915) [CO] – **Fig. 3.16c**

Mionicella chopardi Giglio-Tos, 1915*

Bantia fusca Chopard, 1911* [FG] – **Fig. 3.16d**

Bantia metzi Beier, 1935* [BR (ES)] – **Fig. 3.16e**

Bantia michaelesi Beier, 1935* [BR (ES)] – **Fig. 3.16f**

Bantia pygmaea (Saussure, 1872) [BR (RJ)]

Oligonyx pygmea Saussure, 1872

Bantia marmorata Saussure and Zehntner, 1894* [BR (RJ)] – **Fig. 3.17a**

Bantia nana (Piza, 1969) [BR (RJ)] – **Fig. 3.17b**

Musoniella nana Piza, 1969*

Bantia yotocoensis Salazar, 2004 [CO]

Bantia wernerii Chopard, 1913 [PE]

Taxonomic notes. Saussure (1872) cited the type locality of *B. pygmaea* simply as “Brasilia” (i.e., Brazil); later on, Stål (1877) mentioned that Rio de Janeiro was the type locality. These two species, along with *B. marmorata*, *B. metzi*, *B. michaelesi* and *B. nana*, were all described from neighboring localities within the Atlantic forest of southern Brazil. Although I was unable to study the type specimen of *B. pygmaea*, I observed slight but conspicuous variation in pronotal proportions among all other type specimens examined (i.e. *B. marmorata*, *B. metzi*, *B. michaelesi* and *B. nana*), as well as in the few specimens available for study from the same general area (southern Brazil). Because of the dearth of material it was difficult to assess if such variation reflected intra- or interspecific differences. This question will remain open until additional, topotypical material, becomes available. Saussure and Zehntner (1894) further suggested that *B. marmorata*, also described from Rio de Janeiro, probably represented the female of *B. pygmaea* (which remains known from males only). This question also needs to be addressed.

Bantiella Giglio-Tos, 1915 (Fig. 3.3a-d)

Bantiella Giglio-Tos, 1915: 186. Type species: *Bantiella trinitatis* Giglio-Tos, 1915 (original designation).

Distribution. Tropical Andes of northern South America (up to 2000 m), and Trinidad and Tobago.

Diagnosis. *Size range:* 14–16 mm.

Body colouration: light to dark brown, often with contrasting, darker spots in body and limbs.

Head: vertex straight, in males (Fig. 3. 3a) the same barely rising above level of compound eyes, whereas juxtaocular bulges are barely more elevated than the latter; females (Fig. 3.3b) exhibit a similar head configuration, except that both juxtaocular bulges and vertex conspicuously rise above compound eyes. Ocelli of males relatively small, approximately twice as big as those of females.

Thorax: pronotum (Figs. 3.3c-d) with distinct supracoxal dilation, lateral edges bearing small denticles, longitudinal carina marked along metazona and proximal half of prozona; metazona 1.4–1.6x as long as prozona, the latter relatively elongated. Forecoxae longer than metazona (Fig. 3. 3c). Spination formula: F=4DS/8–9AvS/4PvS; T=8–9AvS/7PvS. Forefemora strongly triangular (Fig. 3. 3c). Foretibial spines of the posteroventral series gradually but scarcely increasing size distad, except for the distal-most spine which is conspicuously longer; foretibial spines of the anteroventral series exhibiting various sizes, the 2 or 3 proximal-most spines are smallest and are followed by a larger spine, the latter is followed by one short and three large spines. Males: wings iridescent, brown and hyaline.

Abdomen: slender and cylindrical in males, slightly dilated in females.

Species checklist

Bantiella pallida Giglio-Tos, 1915* [CO] – **Fig. 17c**

Bantiella simoni (Chopard, 1916) (**comb. n.**) [VE] – **Fig. 17d**

=*Miopteryx simoni* Chopard, 1916* [*Bantia*]

Bantiella trinitatis Giglio-Tos, 1915* [GY, TT, VE] – **Fig. 17e–i**

=*Bantiella columbina* Giglio-Tos, 1915* (**syn. n.**)

=*Bantiella fusca* Giglio-Tos, 1915* (**syn. n.**)

Taxonomic notes: *Bantiella simoni* Chopard (1916) was originally described in *Miopteryx*. After a long and checkered taxonomic history Terra (1995) assigned it to *Bantia*; however, my analysis of the holotype reveals that it actually belongs in *Bantiella*.

Giglio-Tos (1915b) described *B. trinitatis* and *B. fusca* from the same type locality — Port of Spain, Trinidad — assuming that the former represented a light-coloured species and the latter a darker one. Beebe *et al.* (1952) made several observations of these species in Trinidad and commented on their extensive colour variation, suggesting that *B. trinitatis* and *B. fusca* could be considered synonyms of each other and of *Promiopteryx granadensis*. I examined the types of *B. trinitatis* and *B. fusca*, in addition to male genitalia of topotypical specimens from Port of Spain, plus populations of light and dark specimens from other localities in Trinidad and Tobago, Venezuela and Guyana. The similarities observed in male genitalic structures revealed them all to be conspecific, thus confirming Beebe *et al.* (1952) assessment of synonymy between *B. trinitatis* and *B. fusca*. However, *B. trinitatis* is by no means referable to *P. granadensis* — a separate species that clearly belongs to another valid genus. Examination of the type of *Bantiella columbina* Giglio-Tos, 1915 from Caracas, Venezuela, reveals that this species is also conspecific with *B. trinitatis*. Therefore, both *B. fusca* and *B. columbina* are here considered new synonyms of *B. trinitatis*.

Diabantia Giglio-Tos, 1915 (Fig. 3.3e–f)

Diabantia Giglio-Tos, 1915: 182. Type species: *Diabantia minima* Giglio-Tos, 1915 (by monotypy).

Distribution. Rare in collections. Available locality information suggests that the genus is restricted to northern South America.

Diagnosis (males only). *Size range*: 10–10.5 mm

Body colouration: light to dark brown.

Head: vertex straight, rising only to level of juxtaocular bulges, all together slightly more elevated than dorsal margins of compound eyes.

Thorax: pronotum short and compact (Fig. 3.3e), lateral margins with small, sparse denticles, supracoxal dilation distinct, slightly expanded laterally, its curved outline takes most of the metazona except along its narrower, proximal-most section (where it appears to be slightly constricted), metazona markedly short, ca. 1.2x as long as the prozona. Forecoxae almost twice as long as metazona (Fig. 3. 3e). Spination formula: F=4DS/10AvS/4PvS; T=6–7AvS/7PvS. Forefemora triangular in shape. Foretibial spines from the anteroventral series variable in size, where the 1st, 2nd and 4th are shortest; foretibial spines from the posteroventral series about the same size and scarcely increasing size distad, except for the distal-most spine which is conspicuously larger. Forefemoral spines from the posteroventral series with its 2 proximal-most spines closer to each other than any other two spines (by about 1/3 of the distance between the two distal-most spines); 3rd forefemoral discoidal spine is longest (Fig. 3.3f). Males: wings iridescent, smoky brown and hyaline.

Abdomen: slender and cylindrical.

Remarks. *Diabantia* is currently known only from Colombia. I also examined a possibly undescribed species of this genus from Venezuela.

Species checklist

Diabantia minima Giglio-Tos, 1915* [CO] – **Fig. 3.18a**

Taxonomic notes. I consider the report of *Diabantia minima* from southern Brazil (Mato Grosso and São Paulo) by Agudelo *et al.* (2007) as dubious, as this genus seems to be restricted to northern South America.

Mantellias Westwood, 1889 (Fig. 3.3g-h)

Mantellias Westwood, 1889: 38. Type species: *Mantellias pubicornis* Westwood, 1889 (by monotypy).

Distribution. French Guiana, eastern and central Amazon.

Diagnosis. *Size range:* 20–25 mm.

Body colouration: dark brown, with multiple contrasting darker spots across the body.

Head: vertex almost straight or with a slight central undulation (Fig. 3.3g), as elevated as juxtaocular bulges, altogether more elevated than dorsal margin of compound eyes. Male antennomeres narrowed proximally and with two types of setae: one type are thin and more abundant, the other are less numerous but much thicker, longer (each seta twice as long as the corresponding antennomere) and form a well differentiated pre-apical “crown”. Lower frons with its upper angle pointy.

Thorax: pronotum (Fig. 3.3h) with comparatively elongated prozona, lateral edges with small denticles (relatively larger in females), supracoxal bulges poorly developed, metazona 1.2–1.5x as long as prozona. Forecoxae slender, longer than metazona. Spination formula: F=4DS/10AvS/4PvS; T=5–7AvS/5–7PvS. Dorsal aspect of forefemora slightly expanded to form a rudimentary but notorious lamella, thus conferring the forefemora (in the lateral view) with a curved dorsal margin with a distinct, pre-apical inflexion (Fig. 3.3h); dorsal edge with serrations; anterior aspect of forefemora almost completely darkened (with bluish, metallic reflections in females). Foretibial spines of the posteroventral series scarcely increasing in size distad (distal-most spines are longest); foretibial spines of the anteroventral series of various sizes, with its distal-most spine slightly displaced from the ventral margin and more dorso-laterally positioned (more clearly so in females). Males: wings brown and hyaline, anal edge of metathoracic wings relatively straight, conferring the wing a more rectangular shape.

Abdomen: cylindrical, distal edge of proximal terga with a small, median lobe.

Remarks: Maldaner *et al.* (2015) clarified the validity and taxonomic identity of *Mantellias*.

Species checklist

Mantellias hyalina (Beier, 1942) (**comb. n.**) [BR (PA)]

Bantiella hyalina Beier, 1942

Mantellias pubicornis Westwood, 1889* [BR (PA), FG] – **Fig. 3.18b**

Taxonomic notes: Beier (1942) described *Bantiella hyalina* on the basis of a single male holotype. Although this specimen was not available for study, the original description includes sufficient detail to draw conclusions about the identity of this species. In his description, Beier (1942) explicitly mentioned the peculiar dorsal expansion of the dorsal edge of the prothoracic femora, in addition to colour details of the anterior surface of the same structure: “*Vorderfemora medial größtenteils Schwarz, nur basal und apikal hell, der Dorsalrand leicht, aber deutlich konvex gebogen, fein gezähnt und bewimpert*” (“forefemora mostly black medially, but distally and proximally paler, dorsal margin slightly, but significantly convex/curved, finely serrated and ciliated”). These characters are diagnostic of *Mantellias* and absent from any species of *Bantiella* I have examined, which instead exhibit strongly triangular prothoracic femora with an straight dorsal edge (Maldaner *et al.* 2015). Therefore, I herein transfer *B. hyalina* to *Mantellias*. This species is distinct from *M. pubicornis* and I consider both to be valid. A distinctive feature of *M. hyalina* is the large, dark area covering most of the inner side of the prothoracic coxae. In *M. pubicornis*, the inner side of the prothoracic coxae has small, scattered brown spots.

Mantillica Westwood, 1889 (Fig. 3.3i-j)

Mantillica Westwood, 1889: 39. Type species: *Mantillica nigricans* Westwood, 1889 (subsequent designation: Kirby, 1904: 225).

Distribution: Northern Amazon.

Diagnosis (female characters and measurements from Agudelo and Rafael, 2014). *Size range:* 10.7–11.6 mm.

Body colouration: Body dark brown, almost black.

Head: vertex convex, rising slightly above compound eyes and juxtaocular bulges, the latter poorly developed. Male antennae longer than body, densely covered with long, black setae.

Thorax: pronotum markedly short and stocky (Fig. 3.3i), rhomboidal in shape, metazona 1.4-1.5x as long as prozona, lateral margins bearing denticles. Forecoxae robust, narrowing distally. Spination formula: F=4DS/9–10AvS/4PvS; T=5–6AvS/5–6PvS. Forefemora slender and curved. Foretibiae (Fig. 3.3j) with its distal-most posteroventral spine much larger than any other spine from the same series; foretibial spines of the anteroventral series with its 1st and 3rd spines shortest, whereas the distal-most spine is displaced to a more latero-dorsal position (Fig. 3. 3j); most tibial spines with microserrulations. Mesonotum and metanotum of both females and juveniles of both sexes, with a dorsal, spine-like projection, the same reduced to a small, barely indicated dilation in adult males. Male: wings brown, subhyaline, and iridescent. Meso- and metathoracic metatarsi considerably longer than remaining tarsomeres altogether.

Abdomen: ovoid (formicoid) in both female and juveniles of both sexes; cylindrical in adult males.

Remarks: Agudelo and Rafael (2014) discussed the taxonomic identity of *Mantillica nigricans* and provided information on their natural history of the ant-mimicry behaviour.

Species checklist

Mantillica nigricans Westwood, 1889* [BR (AM)] – **Fig. 3.18c**

Thrinaconyx Giglio-Tos, 1915 (Fig. 3.3k-m)

Thrinaconyx Saussure, 1892: 122. Type species: *Thrinaconyx fumosus* Saussure and Zehntner, 1894 (subsequent designation: Kirby, 1904: 279).

=*Costaricella* Sjöstedt, 1930: 25. Type species: *Costaricella fasciata* Sjöstedt, 1930 (by monotypy).

Distribution. Central America south into the Amazon basin, and at mid-elevations along the tropical Andes.

Diagnosis. *Size range:* 12–20 mm.

Body colouration: light to dark brown, often with scattered, lighter spots.

Head: vertex rising only to level of the prominent compound eyes, and juxtaocular bulges only slightly more elevated than the latter, central ocellus enlarged and spherical in frontal view, lateral ocelli appearing elliptical, cuticle around the central ocellus might exhibit a small conical process that dorsally projected.

Thorax: pronotum (Fig. 3.3k–l) with supracoxal bulges well-developed (slightly laterally expanded in females), lateral edges with small denticles that are larger in females (Fig. 3.3l); metazona of pronotum 1.3–1.5x the length of prozona. Forecoxae longer than metazona (Fig. 3.3k). Spination formula: F=4DS/6–10AvS/4PvS; T=2–4AvS/1–2PvS. Forefemora slender, basal half thicker than distal half (less evident in *T. kirschianus*). Foretibiae cylindrical and slightly curved (Fig. 3.3k), dorsal surface of male foretibiae with numerous small tubercles, each bearing a strong seta (Fig. 3.3k). Foretibiae with a single and large distal posteroventral spine (exceptionally, a second, markedly smaller spine might occur). Foretibial spines of the anteroventral series with at least 4 spines always present: two distal (both strongly displaced to a more dorsal position (Fig. 3.3m), their size comparable to the apical tibial spur), followed by two large spines (a medium-sized spine may occur in between), and a variable number of minute spines (usually one but as many as three) may also occur more proximally along the posteroventral margin. Forefemora of some species lacking genicular spines. All spines, especially those on tibiae, with multiple microserrulations. Males: wings smoky, hyaline; crossveins on discoidal area of mesothoracic wings usually lightly pigmented, wavy and almost equally spaced.

Abdomen: cylindrical, slightly dilated in females.

Species checklist

Thrinaconyx agudelo (Salazar and Ariza, 2005) (**comb. n.**) [CO]

Oligonicella agudelo Salazar and Ariza, 2005

Thrinaconyx brunneri (Saussure, 1871) (**comb. n.**) [VE]

Oligonyx brunneri Saussure, 1871 [*Oligonicella*]

Thrinaconyx fumosa Saussure and Zehntner, 1894* [CR, PA] – **Fig. 3.18f**

=*Costaricella fasciata* Sjöstedt, 1930

Thrinaconyx kirschianus Saussure and Zehntner, 1894 [CO, CR]

Thrinaconyx sialidea (Westwood, 1889)* [BR (AM)] – **Fig. 3.18d–e**

=*Mantillica sialidea* Westwood, 1889

Taxonomic notes: Salazar and Ariza (2005) described *Oligonicella agudelo* from multiple localities within the Colombian Amazon. However, the original description and accompanying figures show that this species is actually a member of *Thrinaconyx*. Accordingly, I here transfer *agudelo* to the latter genus. The type locality of *T. agudelo* is about 500 km west of the type locality of *T. sialidea* — the latter originally described under *Mantillica*, and recently transferred to *Thrinaconyx* by Agudelo and Rafael (2014). Both *T. agudelo* and *T. sialidea* represent lowland tropical rainforest species within the upper Amazon basin, thus it is possible that both names are referable to the same species.

Saussure (1871) described *Oligonyx brunneri*. Giglio-Tos (1915b) later established *Oligonicella* to accommodate this and another species of *Oligonyx* he considered misplaced. No critical assessment of Giglio-Tos' procedure has been done since. Saussure's (1871) description of *Oligonyx brunneri* is detailed enough to conclude that it is actually a member of *Thrinaconyx* — as evidenced by the conformation of its prothoracic tibiae. Consequently, I here transfer *brunneri* to *Thrinaconyx*. I have examined three male specimens with distinct genital morphology from Rancho Grande, Aragua, Venezuela, ca. 60 km southeast of Puerto Cabello — the type locality of *brunneri* — in Venezuela, not Colombia, as stated in the original description). I tentatively assigned these specimens to *T. brunneri*.

Agudelo *et al.* (2007) compiled distributional data of *Thrinaconyx fumosa*, but I consider their records from USA and Bolivia as dubious. These records actually trace back to Beier (1935a) who, in addition to Panama (the type locality), cited *T. fumosa* for Bolivia and “Colorado”, presumably in the USA (no specimens were cited). On the one hand, the “Colorado” record

likely refers to “Barro Colorado” Island, in Panama, instead of Colorado State in the USA. On the other, the Bolivian record seems too far south for this species. My own examination of specimens shows that *T. fumosa* is distributed in Costa Rica and Panama and no specimens south of Barro Colorado Island (Panama) were found, suggesting that the Bolivian record likely resulted from a misidentification. I have not seen specimens of *Thrinaconyx* from Bolivia, but having examined at least two undescribed species from southeastern Peru, the presence of *Thrinaconyx* in northeastern Bolivia cannot entirely be ruled out. Summarizing, *T. fumosa* is restricted to Central America, and thus Bolivian and USA records are equivocal.

4.3.1.4 Subfamily Miobantiinae Roy, 2013

Key to genera (males)

1. Forefemora with 3 discoidal spines (Fig. 3.4t); cell membrane of both pair of wings of males with, in addition to microsetae, numerous and differentiated macrosetae (more abundant across the distal half of wings) (Fig. 3.4u)..... ***Paradiabantia***
- 1'. Forefemora with 4 discoidal spines; cell membrane of both wings of males with only microsetae.....2
2. Juxtaocular bulges prominent and conical, rising well above vertex (Fig. 3.4a, 3.4j). Pronotum surface at least with proximal longitudinal carinae on each side of the metazona, often accompanied by various other irregularities (e.g., granules, tubercles) (Fig. 3. 4c, 3.4l).....3
- 2'. Juxtaocular bulges poorly developed, not rising above level of vertex or only scarcely so (Fig. 4j, 4n). Pronotum surface smooth (Fig. 3.4h, 3.4p).....4
3. Foretibiae tibiae with 7 or more posteroventral spines. Ventral phallomere with a preapical, transversal membranous strip (see Fig. 9O in Rivera and Svenson 2016).....***Eumiopteryx***
- 3'. Foretibiae with 6 or less posteroventral spines. Ventral phallomere lacking a preapical, transversal membranous strip (see Fig. 9L in Rivera and Svenson 2016).....***Anamiopteryx***
4. Foretibiae with proximal-most posteroventral spine separated from subsequent spine by at least twice the distance between any other two spines within the same series (Fig. 3.4r); male

colouration in shades of brown and grey; longitudinal veins of wings light brown with various darker and contrasting spots***Miobantia***

4'. Foretibiae with all posteroventral spines more or less equally spaced; male body, limbs, wings and longitudinal veins uniformly green, the latter without darker spots.....***Chloromiopteryx***

Anamiopteryx Giglio-Tos, 1915 (Fig. 3.4a-e)

Anamiopteryx, Giglio-Tos, 1915: 184. Type species: *Anamiopteryx borellii* Giglio-Tos, 1915 (by monotypy).

=*Trachymiopteryx* Rehn, 1920: 220. Type species: *Trachymiopteryx tuberculata* Rehn, 1920 (by monotypy).

Distribution. Found within and around the Paraná River basin: southern Bolivia, northeastern Argentina, and through southeastern Brazil as far Espírito Santo state (one species known from eastern Pará).

Diagnosis. *Size range:* 19–35 mm.

Body colouration: males light to dark brown, females greenish-brown to almost black, both sexes with numerous contrasting darker spots scattered on body and limbs (living specimens are likely to exhibit greener tones).

Head: vertex strongly concave (Fig. 3.4a–b), juxtaocular bulges prominent, conical, rising well above the dorsal margins of compound eyes and vertex, postocellar region of vertex with a central swell; clypeus with a short, but conspicuous, central tubercle (best seen in lateral view).

Thorax: pronotum with supracoxal bulges well-defined, lateral edges conspicuously expanded (expansion runs into prozona); metazona 1.7–2x as long as prozona, lateral edge of the latter forming a subtle but noticeable angular projection. Pronotum with a characteristic pattern ridges and tubercles that are more prominent in females (Fig. 3.4c–d). Forecoxae longer than metazona. Spination formula: F=4DS/11–12AvS/4PvS; T=10–11AvS/6PvS. Foretibial posteroventral

spines almost always 6 in number and distributed along the distal half of the tibia, whereas basal half of the same exhibit only minute spine-like, cuticular processes (vestigial spines?), each bearing a small seta. Males: wings from pale grey to dark brown, longitudinal veins often with darker spots of varied intensity.

Abdomen: slender and cylindrical in males, oval-shaped and flattened in females. Females exhibit two small, fimbriated lobes distally on their pleural membrane (Fig. 3.4e).

Species checklist

Anamiopteryx borellii Giglio-Tos, 1915 [BR (MT), BO, AR]

Anamiopteryx grandis Beier, 1935* [BR (ES)] – **Fig. 3.19a–b**

Anamiopteryx magna (Jantsch, 1991) [BR (PA)]

Eumiopteryx magna Jantsch, 1991

Anamiopteryx tuberculata (Rehn, 1920) [BR (DF, GO), PY] – **Fig. 3.19c**

Trachymiopteryx tuberculata Rehn, 1920*

Chloromiopteryx Giglio-Tos, 1915 (Fig. 3.4f-i)

Chloromiopteryx Giglio-Tos, 1915: 141. Type species: *Mantis (Photina) thalassina* Burmeister, 1838 (by monotypy).

=*Metathespis* Piza, 1968: 1. Type species: *Metathespis modesta* Piza, 1968 (original designation).

=*Emboicy* Terra, 1982: 330. Type species: *Emboicy mirim* Terra, 1982 (by monotypy).

Distribution: Southern Brazil.

Diagnosis. *Size range*: 20–30 mm.

Body colouration: males are uniformly green, females light to dark brown, sometimes with greenish tones, body and limbs in general with multiple and scattered, darker spots.

Head: Vertex with a distinct, central concavity (Fig. 3.4f–g), juxtaocular bulges rising above dorsal margin of compound eyes, yet lower than vertex.

Thorax: pronotum slender, lateral edges smooth in males (Fig. 3.4h), with small denticles in females (Fig. 3.4i), metazona 1.7–1.8x as long as prozona, the latter relatively elongated. Forecoxae longer than metazona. Spination formula: F=4DS/10–12AvS/4PvS; T=8–10AvS/6–9PvS. Males: Fore- and hindwings colourless and hyaline, longitudinal veins mostly green (not always evident in preserved specimens); wings extend beyond tip of abdomen in resting position.

Abdomen: slender in males and slightly dilated in females, the latter with a small, central lobe distally on each tergite.

Species checklist

Chloromiopteryx mirim (Terra, 1982) [BR (SP)] – **Fig. 19d**

Emboicy mirim Terra, 1982*

Chloromiopteryx modesta (Piza, 1968) [BR (RJ)]

Metathespis modesta Piza, 1968*

Chloromiopteryx plurilobata (Mello-Leitão, 1937) [BR (RJ)]

Musoniola plurilobata Mello-Leitão, 1937

Chloromiopteryx thalassina (Burmeister, 1838) [BR (RJ, SC)] – **Fig. 19e**

Mantis (Photina) thalassina Burmeister*

= *Miopteryx brunneri* Saussure, 1871

Taxonomic notes. *Chloromiopteryx* was considered as monotypical (*C. thalassina*) until Agudelo and Rivera (2015) transferred three additional species from other genera (original genus in parenthesis): *mirim* (*Emboicy* Terra; now synonym of *Chloromiopteryx*), *modesta* (*Metathespis* Piza; now synonym of *Chloromiopteryx*) and *plurilobata* (*Musoniola* Giglio-Tos). There is one additional name associated to *Chloromiopteryx* not mentioned in Agudelo and Rivera (2015): *Miopteryx brunneri* Saussure, 1871. This species remains known only from its original description (single male holotype, not female as originally stated), and type locality in Nova Friburgo (Rio de Janeiro, Brazil). Giglio-Tos (1927) considered *M. brunneri* a synonym of *C. thalassina*, then known just from Santa Catarina Island (Brazil); no justification was given for this synonymy. Beier (1935a) also listed *Miopteryx brunneri* as synonym of *C. thalassina* but as *Miopteryx brunneri* Chopard, 1914. Apparently, Beier (1935a) mistakenly attributed authority of

this name to Chopard (1913) instead to Saussure (1871), even though Chopard (1913) cited this species as “*Miopteryx Brunneri* **Sauss.**” in his key to *Miopteryx* known to him at the time (now largely outdated). This wrong citation likely influenced Terra (1995), who also cited this species as *Miopteryx brunneri* Chopard, 1914 in his synonymic list of *C. thalassina*. Later records cited Saussure (1871) as the correct source (Ehrmann, 2002; Agudelo, *et al.*, 2007).

Whether *Miopteryx brunneri* is a valid species or a synonym of *C. thalassina* cannot be determined from the literature alone. Saussure (1871) mentioned renowned Swiss orthopterologist C. Brunner de Wattenwyl as the owner of the only known specimen of *M. brunneri*, but its whereabouts remain unknown. The lack of topotypical material also prevented further inquiring on the identity of this species. I here opt to retain *M. brunneri* as synonym of *C. thalassina* until more specimens become available.

Eumiopteryx Giglio-Tos, 1915 (Fig. 3.4j-m)

Eumiopteryx Giglio-Tos, 1915: 140. Type species: *Eumiopteryx laticollis* Giglio-Tos, 1915 (by monotypy).

=*Piracicaba* Piza, 1967: 42:135. Type species: *Piracicaba bicentenaria* Piza, 1967 (by monotypy).

Distribution. Found within and around the Paraná River basin (southern Bolivia, Paraguay southern western Brazil). I examined a few specimens representing undescribed species from rainforest areas of Peru and French Guiana, suggesting a much wider distribution.

Diagnosis. *Size range:* 20–30 mm.

Body colouration: various tones of yellowish to greenish brown, often with shades of grey (living specimens are likely to be predominantly greyish green), with scattered darker markings on body and limbs.

Head: vertex slightly but conspicuously concave (Fig. 3.4j–k), lateral tubercles conical, rising above the dorsal margin of compound eyes; postocellar region of vertex with a central bump.

Thorax: pronotum (Fig. 3.4l–m) with well-defined supracoxal bulges, metazona 1.5–2x as long as prozona, lateral margin of the latter forming a discreet but noticeable angular projection (small in males, more evident in females). Pronotal surface of both sexes lacking granules or tubercles (except for the proximal tubercle of the metazona), and proximal half of metazona often with a longitudinal, lateral carina on each side of the midline (poorly expressed in males, but well developed in females). Forecoxae longer than metazona. Spination formula: F=4DS/11–12AvS/4PvS; T=8–10AvS/8–7PvS. Forefemora with its dorsal edge either straight or weakly curved proximally. Foretibial posteroventral spines almost always six in number, distributed along the distal half of the tibia, whereas basal half of the same exhibit minute spine-like, cuticular processes (exceptionally, one or more of the latter sometimes developed into a fully recognizable spine). Males: forewings slightly iridescent, from almost colorless to pale grey and dark brown, longitudinal veins often with darker spots of varied intensity; hindwings mostly colorless, iridescent and mostly hyaline.

Abdomen: slender in males, slightly dilated in females, in the latter the basal-most tergite exhibit a small, tooth-like, median projection.

Species checklist

Eumiopteryx bicentenaria (Piza, 1967) [BR (SP)] – **Fig. 3.19f**

Piracicaba bicentenaria Piza, 1967*

Eumiopteryx laticollis Giglio-Tos, 1915* [BO, PY] – **Fig. 3.19g–h**

Eumiopteryx terrai (Jantsch, 1994) [BR (MT)]

Calopteromantis terrai Jantsch, 1994 *

Taxonomic notes. Piza (1967) described *Piracicaba bicentenaria* on the basis of a single male “type” (i.e. holotype). My examination of Piza’s collection at ESALQ revealed a female specimen attributed to this species and labeled as “alotypus” (i.e. allotype), but not reported in the original publication. Label data however shows that this female cannot be considered as such

because it is not even part of a type series, as it was collected in August 1973, several years after Piza's description of *P. bicentenaria*.

Giglio-Tos (1915b) described *E. laticollis* on the basis of a male holotype from Paraguay, and a female allotype from Bolivia (Sara), both at ZMB. However, the conspecificity of both specimens could not be assessed at this time; future studies must resolve this question as well as ruling out the conspecificity of this species and *E. terrai*.

Miobantia Giglio-Tos, 1917 (Fig. 3.4n-r)

Miobantia Giglio-Tos, 1917: 48. Type species: *Miobantia aptera* Giglio-Tos, 1917 (by monotypy).

=*Miopteryx* Kirby, 1904: 274 (nec Saussure, 1869: 55). Type species: *Mantis rustica* Fabricius, 1781 (subsequent designation: Kirby, 1904: 274).

=*Cnephomantis* Rehn, 1920: 225. Type species: *Miopteryx fuscata* Giglio-Tos, 1915 (original designation).

=*Antimiopteryx* Giglio-Tos, 1927: 210. *Mantis rustica* Fabricius, 1781 (original designation).

Distribution. Found mostly along the southern Atlantic coast of Brazil, as far north as Bahia state, but also northeastern Argentina (Misiones) and Southern Paraguay (Scherrer, 2014).

Miobantia rustica (Fabricius, 1781) was originally described from coastal Patagonia, Argentina; this latter record needs verification.

Diagnosis. *Size range:* 16–26 mm.

Body colouration: greyish brown to pale yellow, often lighter ventrally, with scattered darker markings on body and limbs.

Head: vertex (Fig. 3.4n–o) ranging from straight to markedly convex, rising above the dorsal margin of compound eyes, juxtaocular bulges poorly developed.

Thorax: pronotum (Fig. 3.4p–q) with supracoxal bulges weakly to decidedly marked, lateral margins with denticulation in females and smooth in males; metazona about 1.6–2.0x as long as prozona. Forecoxae longer than metazona. Spination formula: F=4DS/12AvS/4PvS; T=9–11AvS/7PvS. Foretibial spines of the posteroventral (Fig. 3.4r) series with its proximal-most spine distinctly separated from the subsequent one by at least twice the distance between any other two spines. Males: forewings about same length or shorter than hindwings, colouration varies from light grey to blackish brown and in various shades, longitudinal veins with dark spots of varied intensity, crossveins sometimes with dark spots of different sizes.

Abdomen: slender in males, slightly fusiform in females.

Remarks. Scherrer (2014) conducted a detailed revision of *Miobantia*.

Species checklist

Miobantia aptera Giglio-Tos, 1917 [BR (ES, RJ)]

Miobantia arctissima Scherrer, 2014 [BR (ES)]

Miobantia ciliata (Stål, 1860) [BR (ES, RJ, SC)]

=*Mantis ciliata* Stål, 1860

Miobantia fuscata (Giglio-Tos, 1915) [BR (ES, SC)]

=*Miopteryx fuscata* Giglio-Tos, 1915

Miobantia immanis Scherrer, 2014 [BR (ES)]

Miobantia nordestina Scherrer, 2014 [BR (BA)]

Miobantia phryganea (Saussure, 1869) [BR]

=*Miopteryx phryganea* Saussure, 1869

Miobantia punctata (Giglio-Tos, 1917) (**comb. n.**) [PY] – **Fig. 3.20a**

=*Promiopteryx punctata* Giglio-Tos, 1917*

Miobantia rustica (Fabricius, 1781) [AR? BR (RJ)]

=*Mantis rustica* Fabricius, 1781

=*Miopteryx perlodes* Saussure, 1869: 64; 1870: 237.

=*Miopteryx nebulosa* Giglio-Tos, 1915

=*Metathespis precaria* Piza, 1968

Miobantia sulista Scherrer, 2014 [AR, BR (PR, SC, RS), PY]

Taxonomic notes. My examination of the holotype of *Promiopteryx punctata* revealed it as a member of *Miobantia*. I here transfer *punctata* to *Miobantia*, but its status as a valid species needs further verification.

Paradiabantia Piza, 1973 (Fig. 3.4s-u)

Paradiabantia Piza, 1973: 117. Type species: *Paradiabantia perparva* Piza, 1973 (by monotypy).

Distribution. Southern Brazil.

Diagnosis (males only). *Size range*: 11–12 mm.

Body colouration: light to dark brown, often with darker spots scattered over the body and limbs.

Head: vertex straight, rising to level of juxtaocular bulges, all together rising slightly above the dorsal margin of compound eyes.

Thorax: pronotum short and compact (Fig. 3.4s), supracoxal bulges accentuated, lateral margins with black marks and only with few, markedly small denticles; metazona 1.1–1.2x as long as prozona. Spination formula: F=3DS/8–10AvS/4PvS; T=7–8AvS/8–9PvS. Forefemora triangular. Distal-most foretibial spine large of the posteroventral series is largest, other spines anterior to this one are smaller and barely increase size distad; anteroventral series similarly arranged but third spine sometimes longer than the others (except for the larger, distal-most spine).

Forefemoral spines of the posteroventral series with its two proximal-most spines separated by a distance that is approximately 1/3 of the distance between the two distal-most spines of the same series; forefemoral posteroventral genicular spine as long as the 3rd spine of the same series; within the discoidal spines, the second is longest (Fig. 3.4t). Males: wings iridescent and hyaline, colourless to smoky brown, with dark marks along longitudinal veins; two types of differentiated setae (micro- and macrosetae) are clearly visible within the wing cells under backlight (Fig. 3.4u), the macrosetae are relatively more abundant towards the tip of the wings.

Abdomen: slender and cylindrical.

Remarks. Rivera and Svenson (2016) revalidated *Paradiabantia*, formerly a synonym of *Diabantia*.

Species checklist

Paradiabantia perparva (Piza, 1973)* [BR (SP)] – **Fig. 3.20b**

4.3.1.5 Subfamily Thespinae Saussure, 1869

4.3.1.5.1 Tribe Musoniellini Rivera and Svenson, 2016

Key to genera (males)

1. Wings of males densely reticulated (i.e., with more than two rows of cells between longitudinal veins), more conspicuously so on the distal portion of the wing membrane.....2
- 1'. Wings of males not densely reticulated (i.e. with two or less rows of cells between longitudinal veins).....3
- 2(1). Head with a strong, well-developed carina that runs behind the ocelli and across the head, extending along the upper edges of compound eyes and backwards into the vertex, clearly delimiting a flattened area on top of the head (sometimes this area exhibits minute tubercles) (Fig. 3.5r). Pronotum stocky (Fig. 3.5s).....***Pizaia***
2. Head without such carina. Pronotum relatively slim (Fig. 3.5h).....***Leptomiopteryx***
- 3(1'). Supraanal plate at most twice as long as its basal width (Fig. 3.5p).....***Musoniella***
- 3'. Supraanal plate more than twice as long as its basal width, sometimes as long the pronotum (Fig. 3.5d).....***Eumusonia***

Eumusonia Giglio-Tos, 1916 (Fig. 3.5a-e)

Eumusonia Giglio-Tos, 1916: 8. Type species: *Thespis livida* Audinet Serville, 1839 (subsequent designation: Giglio-Tos, 1916).

Distribution. Found within the Paraná River Basin (Bolivia, Northern Argentina through Paraguay), and from southern to northeastern Brazil.

Diagnosis. *Size range:* 25–35 mm (supraanal plate excluded).

Body colouration: two colour morphs exist, light to dark brown, and green.

Head: vertex only slightly convex, juxtaocular bulges slightly lower than vertex, the latter bearing minute, spine-like setae (inconspicuous in males, but notorious in females under higher magnification).

Thorax: pronotum relatively short and stocky (Figs. 3.5a–c), lateral margins bearing denticles, cuticle with minute and scattered, spine-like setae (more evident in females), supracoxal bulges almost inconspicuous, lateral margins of the metazona almost parallel, especially in females where the pronotum is comparatively stouter; metazona 1.2–1.25x as long as prozona. Spination formula: $F=4DS/12-15AvS/4PvS$; $T=10-11AvS/5PvS$. Forecoxae longer than the metazona. Foretibial anteroventral spines gradually and barely increasing size distad (distal-most spines is largest), all spines arranged mostly along the distal half of the tibia. Forefemora elongated. Males: wings slightly iridescent, from light to dark brown or green, mostly hyaline; in brown specimens crossveins are darkened at their point of insertion to longitudinal veins (such feature is absent in green specimens). Wings in resting position extend beyond the tip of the subgenital plate.

Abdomen: slender in males, slightly dilated in females. Supraanal plate of males (Fig. 3.5d) triangular, its length twice as long or more than as its basal width (one male from an undescribed species from Bolivia has a supraanal plate that is longer than the pronotum); cerci of males of various lengths (either as long or longer than supraanal plate), cercomeres cylindrical or flattened, varying in number across species. Supraanal plate of females (Figs. 3.5e) triangular, more than twice as long than as its basal width, and ordinarily longer than that of conspecific

males; cerci of females shorter than supraanal plate. Abdominal surface of females with small, scattered spine-like setae.

Remarks. *Eumusonia* is a widespread and diversified genus whose species are difficult to tell apart using external characters alone. However, I was able to recognize at least seven undescribed species among the studied material on the basis of male genital structures. Some species of *Eumusonia* superficially resemble members of *Musoniella*. Females are seldom collected and thus are in collections. Both genera can be distinguished in function of differences in the relative length of the supraanal plate, and the shape of the lateral margins of the metazona (see diagnosis of *Musoniella* for comparison).

Species checklist

Eumusonia intermedia Piza, 1973* [AR, BR (RS), PY] – **Fig. 3.20c**

Eumusonia livida (Audinet Serville, 1839) [BR]

Thespis livida Audinet Serville, 1839

=*Thespis?* *caudata* Saussure, 1870

Eumusonia viridis Giglio-Tos, 1916* [BR (SP)] – **Fig. 3.20d**

Taxonomic notes. Much confusion exists in the literature regarding the actual number of species of *Eumusonia* and their distribution. The rather superficial original description of Audinet Serville (1839) (as *Thespis livida*) does not allow for literature-based recognition. After its original description this species remained virtually unmentioned in the literature until the turn of the 20th Century. Subsequent studies reported *livida* for Mato Grosso, Brazil (Giglio-Tos, 1900; as *Paramusonia cubensis*), Paraguay (Caudell 1904, as *Musonia livida*; Rehn 1907, as *Paramusonia livida*) and Argentina (Rehn 1913, as *Paramusonia livida*). Giglio-Tos (1916) recognized the distinctiveness of *livida* and thus established *Eumusonia* to accommodate it (as the type species) in addition to another, newly described species, *Eumusonia viridis*. Giglio-Tos (1927) considered all distributional records of *Eumusonia* before 1916 to correspond to *E. livida*, adding a new record from Bolivia. Moreover, Giglio-Tos (1927) added yet another species, *Thespis?* *caudata* Saussure, 1871 from Goias, Brazil (clearly a member of *Eumusonia*) to the synonymic list of *Eumusonia livida*.

More recent accounts considered *livida* as extending into other southeastern Brazilian states and even into the Amazon (Agudelo *et al.* 2007). There is no evidence in the literature that the original specimen of *livida* was studied by anyone else besides Audinet Serville. Consequently, all subsequent references to this species in the literature likely used a “best-guess” approach, attributing to *livida* every single specimen matching Audinet Serville’s description. The exact type locality of *livida* also remains unknown. On the basis of a single male specimen, Audinet Serville (1839) reported type locality simply as “Brasil”. According to his own accounts, the specimen Audinet Serville (1839) used to describe *livida* belonged to the collection of Pierre André Latreille, who himself obtained it from Agustin de Saint-Hilaire (1779–1853), a renowned French Botanist and explorer who collected extensively across northeastern Brazil south to Uruguay and Central Argentina during the first half of the 19th Century. Because *Eumusonia* is widespread across the territories Saint-Hilaire surveyed, it is difficult to determine the approximate locality from which he obtained the specimen Audinet Serville (1839) studied. Ehrmann (2002) cited the MNHN as depository of this specimen, but I did not find it there; this specimen is probably lost. Matters further complicate as several species (mostly undescribed) occur within the presumed distribution of *livida*. Audinet Serville (1839) mentioned a dark spot in the mid section of the inner side of the forefemora, a colour feature I also observed in some individuals. However, this character is inconsistent as it is not equally expressed in conspecific individuals. Further, I observed this character in more than one species. Consequently, the taxonomic identity of *Eumusonia livida* could not be assured in this study.

Another species, *Eumusonia intermedia* Piza, 1973, remains known from two “types”. I examined these specimens (at ESALQ) and noted that labels make no indication whatsoever of any particular specimen being the name-bearing holotype. Consequently, both specimens are to be considered syntypes. A taxonomic revision of *Eumusonia* is urgently needed.

Leptomiopteryx Chopard, 1911 (Fig. 3.5f-j)

Leptomiopteryx Chopard, 1911: 329. Type species: *Leptomiopteryx dispar* Chopard 1911 (monotypy).

=*Navimons* Piza, 1970: 1. Type species: *Navimons amapaensis* Piza, 1970.

Distribution. Northeastern South America.

Diagnosis. *Size range:* 29–34 mm.

Body colouration: light to dark brown.

Head: vertex convex (Fig. 3.5f), juxtaocular bulges only slightly produced, altogether rising above the level of their prominent compound eyes, more conspicuously so in females (Fig. 3.5g).

Thorax: pronotum elongated (Fig. 3.5h–i), its lateral margins bearing denticles, metazona ca. 2.1x as long as prozona; pronotal surface with quitinous nodules that superficially appear to be tubercles but barely rise above the level of the cuticle, each accompanied by a small, thick seta (such nodules are more conspicuous in females (Fig. 3.5i), whereas in males these are barely visible). Forecoxae slightly longer than metazona. Spination formula: F=4DS/11–12AvS/4PvS; T=9–10AvS/5PvS. Foretibial anteroventral spines (Fig. 3.5j) gradually increasing size distad, basal-most spines of the same series slightly but conspicuously displaced towards the ventral side of the tibia. Foretibial posteroventral spines well spaced (Fig. 3.5j), the gap separating the two proximal-most spines is greater than that between any other two contiguous spines. Males: wings hyaline, slightly iridescent and golden brown in colour, relatively large and densely reticulated, apex of both pairs with a short projection, the one on the hindwings curved backwards. Wings in resting position extend beyond the tip of subgenital plate. Females: costal edge of all four wingpads serrated.

Abdomen: slender in males, dilated and fusiform in females.

Species checklist

Leptomiopteryx dispar Chopard, 1911* [BR (AM, AP), FG] – **Fig. 3.20e–f, 3.21a**

= *Navimons amapaensis* Piza, 1970*

Taxonomic notes. Terra (1995) established *Navimons amapaensis* Piza, 1970 as a synonym of *L. dispar*. My own examination of type material confirmed Terra's procedure. *Leptomiopteryx*

argentina Beier, 1930, a species listed under this genus (Terra, 1995; Ehrmann, 2002; Agudelo *et al.*, 2007), is treated under *Thespis*.

Musoniella Giglio-Tos, 1916 (Fig. 3.5k-q)

Musoniella Giglio-Tos, 1916: 4. Type species: *Miopteryx argentina* Saussure, 1870

Distribution. Found within the Paraná River Basin (Bolivia, Northern Argentina through Paraguay), and from southern to northeastern Brazil.

Diagnosis. *Size range:* 29–34 mm.

Body colouration: light to dark brown.

Head: vertex (Fig. 3.5k–l) straight or only slightly convex, juxtaocular bulges rising only to level of vertex, the latter bearing small, spine-like setae (inconspicuous in males, but evident in females under higher magnification).

Thorax: pronotum relatively elongated (Fig. 3.5m–o), lateral margins bearing denticles, surface with scattered, small, spine like setae, (more evident in females), supracoxal bulges poorly developed and moderately curved, lateral margins of metazona narrowed medially; metazona 1.3–1.5x as long as prozona. Spination formula: F=4DS/12AvS/4PvS; T=9–10AvS/5PvS. Forecoxae longer than metazona. Foretibial anteroventral spines gradually but barely increasing size distad (distal-most spine is largest), all spines arranged mostly along the distal half of the tibia. Forefemora elongated. Males: wings slightly iridescent, from light to dark brown, mostly hyaline; crossveins darkened at their point of insertion with longitudinal veins. Wings in resting position extend beyond the tip of the subgenital plate.

Abdomen: slender in males, slightly dilated in females. Supraanal plate triangular, less than twice as long as its basal width; cerci of males (Fig. 3.5p) at least twice as long than supraanal plate, and as long or slightly longer in females (Fig. 3.5q). Abdominal cuticle of females with scattered small, spine-like setae.

Species checklist

Musoniella affinis Piza, 1961* BR [SP] – **Fig. 3.21f**

Musoniella argentina (Saussure, 1870) [AR, BR (MG, SP), PY] – **Fig. 3.21b**

Miopteryx argentina Saussure, 1870*

Musoniella brasiliensis Giglio-Tos, 1916 [BR (MT)]

Musoniella chopardi Giglio-Tos, 1916* [unknown distribution] – **Fig. 3.21d**

Musoniella costalis (Rehn, 1920) (**comb. n.**) [BR (GO)] – **Fig. 3.21g**

Musonia costalis Rehn, 1920*

Musoniella fragilis (Piza, 1965) [BR (MT)] – **Fig. 3.21h**

Eumusonia fragilis Piza, 1965*

Musoniella ipiranga Rehn, 1918* [BR (SP)] – **Fig. 3.22a**

Musoniella laevithorax (Chopard, 1916) [PY] – **Fig. 3.21c**

Miopteryx laevithorax Chopard, 1916*

Musoniella longicauda Piza, 1969* [BR (MG)] – **Fig. 3.21i**

Musoniella parva Beier, 1935* [PY] – **Fig. 3.21e**

Musoniella precaria Piza, 1983* [BR (PE)] – **Fig. 3.22c**

Musoniella stigmatica (Burmeister, 1838) (**comb. n.**) [BR] – **Fig. 3.22b**

Mantis stigmatica Burmeister, 1838* [*Promiopteryx*]

Taxonomic notes. *Musoniella argentina* is a small-sized species that ranges from northern Argentina to Paraguay and southern Brazil. This species exhibits important variation across its range, particularly in pronotal proportions (e.g. length of metazona relative to prozona, degree of elongation). However, only slight variation in the conformation of male genitalia was observed — otherwise comparable to typical intraspecific variation. Several other species, namely *M. chopardi*, *M. parva*, *M. laevithorax* and *M. affinis*, all strongly resemble *M. argentina*. I examined the types of all these species and, at least in terms external morphology and distribution, they all conform well within the variation and range observed among the examined material of *M. argentina*. Phylogenetic evidence presented in Rivera and Svenson (2016) suggested that *Musoniella* is paraphyletic relative to *Eumusonia*. Terminal taxa labeled as “*Musoniella* sp.” in Rivera and Svenson (2016, Fig. 4) conforms to the habitus of *M. argentina*,

whereas another group, represented in the same phylogeny by *Musoniella* sp. n., represent larger forms, somehow similar in general habitus and size to members of *Eumusonia*, to which they are sister. The morphological variation across *Musoniella* needs to be assessed to determine whether it should be subdivided into two genera.

Additionally, my analysis of the types of *Musonia costalis* Rehn, 1920 and *Promiopteryx stigmatica* Burmeister, 1838 revealed them to be members of *Musoniella* and thus I transferred these two species accordingly. Further studies are needed to clarify the taxonomic status of most species within this genus.

Pizaia Terra, 1982 (Fig. 3.5r-t)

Pizaia Terra, 1982: 332. Type species: *Musoniella seabrai* Piza, 1961 (by monotypy).

Distribution. Northeastern Brazil.

Diagnosis. *Size range:* 24–33 mm.

Body colouration: light to dark brown.

Head: vertex almost straight (Fig. 3.5r), juxtaocular bulges more posteriorly located, a well-developed carina runs across the head and behind the ocelli, this carina may prolong backwards along the dorsal margin of compound eyes to reach the vertex, clearly delimiting a flattened area on top of the head that sometimes exhibit minute tubercles.

Thorax: pronotum (Fig. 3.5s–t) relatively robust, specially in females, lateral margins bearing denticles, supracoxal bulges well developed, metazona ca. 1.5x as long as prozona. Spination formula: F=4DS/12AvS/4PvS; T=8–9AvS/5–6PvS. Forecoxae longer than metazona, often with small, black denticles along their dorsal edges. Forefemora relatively large and robust, area between postero- and anteroventral spines depressed, with a row of small tubercles than run parallel to the posteroventral margin and between 1st and 3rd spine of this series. Foretibial spines arranged mostly along the distal half of the tibia. Foremetatarsi distally with 5–7 small, spine-like, ventral setae (observed in at least one species). Males: wings hyaline, pale brown, scarcely

iridescent, with dense reticulation between longitudinal veins. Wings in resting position extending beyond tip of subgenital plate. Female: costal edge of all four wingpads serrated.

Abdomen: cylindrical in males, tergites laterally expanded; dilated in females, tergite edges serrated, slightly expanded laterally and strongly sclerotized.

Remarks. Terra (1982) established *Pizaia* to accommodate *Musoniella seabrai* Piza, 1961, a species from northeastern Brazil. I examined Piza's type specimen and confirmed Terra's procedure as valid. This specimen exhibits various unique character states that clearly justify its placement in its own genus, such as the distinctive carina across the vertex and the shape of the pronotum. *Pizaia* shares with *Leptomiopteryx* the shape of the head, the conspicuous reticulation of fore- and hindwings, and the serrated costal margin of the wing pads of females.

Species checklist

Pizaia seabrai (Piza, 1961) [BR(PE)] – **Fig. 3.22d**

Musoniella seabrai Piza, 1961*

4.3.1.5.1 Tribe Thespini Saussure, 1869

Key to genera (males)

1. Foretibiae with 1 or 2 of its distal-most anteroventral spines slightly to strongly displaced from the ventral margin of the tibia, in such way that, in the lateral view, these clearly project over the apical tibial spur or overlap with it (Figs. 3.7m, 3.8d, 3.9b, 3.9k).....2
- 1'. Foretibiae with all its anteroventral spines clearly located near or along the ventral margin of the tibia, not projecting over the tibial apical spur nor overlapping with it (Figs. 3.6j, 3.6m, 3.8j).....8
- 2(1). Forefemora with 3 discoidal spines.....3
- 2'. Forefemora with 4 discoidal spines.....5
- 3(2). Foretibiae with its distal-most anteroventral spine dorsally positioned (Fig. 3.7n, 3.7m)

.....	<i>Oligonyx</i>
3'. Foretibiae with its two distal-most anteroventral spines dorsally positioned (Fig. 3.9b, 3.9k).....	4
4(3'). Forefemora with 1 posteroventral spine.....	<i>Thesprotia</i>
4. Forefemora with 4 posteroventral spines.....	<i>Thesprotiella</i>
5(2'). Foretibiae with only 1 posteroventral spine, distal-most anteroventral spine strongly displaced from the ventral margin of the tibia, in such way that, in the lateral view, it clearly projects over the apical tibial spur (Fig. 3.7j).....	<i>Oligonicella</i>
5'. Foretibiae with 3 or more posteroventral spines (as many as 6), and distal-most anteroventral spine only slightly displaced from the ventral margin of the tibia, in such way that, in the lateral view, it overlaps with the apical tibial spur (Fig. 3.8d).....	6
6(5'). Pronotal surface scarcely but conspicuously granulated; some granules are larger, resembling pearl-like tubercles often paired on each side of the medial carina (although not always at the same level), number and density of granules vary across individuals, but at least one pair of larger, pearl-like, tubercles occur either on the metazona, prozona or both (Fig. 3.7b).....	<i>Musoniola</i>
6'. Pronotum smooth.....	7
7(6'). Distal and right margins of ventral phallomere forming an almost straight angle (see Fig. 3.10L in Rivera and Svenson 2016); distal margin of dorsal lamina of left phallomere deeply incised (i.e., bilobed).....	<i>Pseudomusonia</i>
7'. Distal and right margins of ventral phallomere forming an acute angle (Fig. 3.6e); distal margin of dorsal lamina of left phallomere not incised	<i>Dougonyx</i> gen. n.
8(1'). Foretibiae with 1–3 posteroventral spines.....	9
8'. Foretibiae with 5 posteroventral spines.....	11
9(8). Foretibiae with its 3 larger, distal-most anteroventral spines, equally spaced (Fig. 3.6j).....	10
9'. Foretibiae with its 3 larger, distal-most anteroventral spines, not equally spaced (Fig. 3.6m); found in Jamaica.....	<i>Liguanea</i>

- 10(9). Body more stick-like (metazona = 1.9–2.2x the length of prozona) (Fig. 3.6g); found in Galápagos Island.....***Galapagia***
- 10'. Body not particularly stick-like (metazona = 1.3x the length of the prozona) (Fig. 3.7p); found in central Peru (Pacific coast and occidental slopes of Andes at mid elevations).....***Piscomantis*** gen. n.
- 11(8'). Metazona of pronotum 3.5x or more the length of prozona; juxtaocular bulges well developed, conspicuously projecting above the vertex (Fig. 3.6n).....***Macromusonia***
- 11'. Metazona of pronotum 2.6x or less the length of prozona; juxtaocular bulges weakly developed, not projecting above vertex (Figs. 3.6p, 3.8e).....12
- 12(11'). Cerci only slightly longer than supraanal plate; ventral phallomere with a small, membranous region (see Fig. 10H in Rivera and Svenson 2016); body length 34 mm or less.....***Musonia***
- 12'. Cerci approximately twice as long as supraanal plate (Fig. 3.8k); ventral phallomere with a large, membranous region (see Fig. 10M in Rivera and Svenson 2016); body length 35 mm or more..... ***Thespis***

Dougonyx Rivera **gen. n.** (Fig. 3.6a-e)

Type species: *Mionyx maculosus* Chopard, 1911 (by present designation)

Distribution. Northern South America (French Guyana and northern Amazon).

Diagnosis. *Size range:* 31–42 mm.

Body colouration: light to dark brown, sometimes with sparse, darker spots.

Head: vertex (Fig. 3.6a) slightly sinuous, rising only to level of juxtaocular bulges, altogether rising above the dorsal margin of compound eyes, particularly in females.

Thorax: pronotum long and narrow (Figs. 3.6b–c), surface smooth, lateral margins not parallel, bearing small denticles, supracoxal bulges relatively well developed, metazona 2.7–3.1x as long

as prozona, the former with 2–4 small, black to brown, maculae paired on each side of the medial carina. Forecoxae slightly shorter than metazoan (Figs. 3.6b). Spination formula: F=4DS/9–12AvS/4PvS; T=7–12AvS/3–6PvS. Forefemora narrow, elongated, and markedly curved at their distal 1/3. Foretibial spines of the anteroventral series forming two groups (g1 and g2 in Fig. 3.6d), g1 contains 2–8 spines, whereas g2 contains 4, distal-most spine of each group is largest, whereas distal-most g2 spine is slightly displaced from the ventral margin of the tibia into a slightly more lateral position. Foretibial spines of the posteroventral (Fig. 3.6d) series gradually increasing size distad, one or more of the more centrally located spines are often missing, thus forming a spine-less gap in the middle of the tibia (Fig. 3.6d) (a single, isolated spine in the middle of the gap may also occur); all foretibial spines with ventral microserrulations (barely visible in males). Males: wings iridescent, hyaline, with smoky, irregular spots along longitudinal veins.

Abdomen: slender and cylindrical in both sexes.

Etymology. A name in the patronymic case, this new taxon is named after Doug Currie, my academic supervisor, mentor and long time collaborator. Gender is masculine.

Remarks. Chopard (1911) described *Mionyx maculosus* on the basis of three males and one female from Saint-Laurent du Maroni, French Guiana (see remarks under *Pseudomusonia* for a summary of the nomenclatural history of the name *Mionyx*). Later on, Giglio-Tos (1927) transferred *maculosus* to *Eumionyx*, whereas Beier (1935a) finally relocated it to *Pseudomusonia*, its current accepted placement (Terra 1995, Ehrmann 2002, Otte and Spearman 2005; Agudelo *et al.* 2007). In his review of the praying mantises of Costa Rica, Rehn (1935) had already anticipated that this species could be assigned in its own genus. My analysis of the type specimens of *Mionyx maculosus*, which I compared to true *Pseudomusonia*, confirmed Rehn's speculations and thus I established *Dougonyx* to accommodate Chopard's *maculosus* (type species) in addition to the following species: *Galapagia amazonica* Terra, 1982, *Musonia sexdentata* Beier, 1942 and *Musonia maculata* Beier, 1942.

Dougonyx superficially resembles *Pseudomusonia*. However, *Dougonyx* has a relatively slender and a more elongated pronotum compared to *Pseudomusonia*. The metazona is 2.7–3.1x as long

as the prozona in *Dougonyx*, but 2.2–2.3x in *Pseudomusonia*. The forelegs of *Dougonyx* overall tend have more spines (compare spination formulas of both genera). *Dougonyx* sometimes exhibits 6 posteroventral spines in the foretibiae, making it one of the few exceptions among the Thespinae, where five spines or less is the usual condition among members of the subfamily. Although some individuals of *Dougonyx* with fewer spines could match those of *Pseudomusonia* in number and arrangement, the particular arrangement of the foretibial posteroventral spines (see details in their respective diagnoses) could also assist the distinction of these two genera. Male genitalia of both genera also show conspicuous differences in their conformation. For instance, the dorsal lamina of the right dorsal phallomere of *Dougonyx* has its distal margin entire, whereas the same exhibits a deep, submedian notch in *Pseudomusonia*. Also, the lateral, membranous portion of the ventral phallomere of *Dougonyx* (Fig. 3.6e) is comparatively smaller than that of *Pseudomusonia*, where the same projects obliquely and more deeply into the phallomere, almost reaching its midline (see Fig. 10L in Rivera and Svenson 2016). The lobe immediately below the membranous area is small and rounded in *Dougonyx* (Fig. 3.6e), whereas the same is larger and subrectangular in *Pseudomusonia* (Fig. 10L in Rivera and Svenson 2016).

Lectotype designation. Chopard (1911) did not select a holotype. Therefore, I herein designate as lectotype one male syntype (with genitalia still preserved) bearing the following three labels: (1) pale pink label: “GUYANA FRANÇ^{se}/St-Laurent du Maroni/COLLECTION LE MOULT”; (2) pale green label: “MUSEUM PARIS/Coll. L. CHOPARD 1919”; (3) red label: “SYNTYPE” (MNHN).

Species checklist

Dougonyx amazonicus (Terra, 1982) (**comb. n.**) [BR (PA or AP)] – **Fig. 3.22f–g**

Galapagia amazonica Terra, 1982*

Dougonyx maculatus (Beier, 1942) (**comb. n.**) [BR (AM)]

Musonia maculata Beier, 1942

Dougonyx maculosus (Chopard, 1911) (**comb. n.**) [FG, GY] – **Fig. 3.23a–c**

Mionyx maculosus Chopard, 1911* [*Pseudomusonia*]

Dougonyx sexdentatus (Beier, 1942) (**comb. n.**) [PE]

Musonia sexdentata Beier, 1942

Taxonomic notes. My examination of the female holotype of *Galapagia amazonica* revealed its similarities to Chopard's *maculosus*, clearly departing from true insular *Galapagia* (see remarks under *Galapagia*). Therefore, I regard *amazonica* as member of *Dougonyx* instead. Terra (1982) omitted a few details of the holotype of *G. amazonica* that I believe appropriated to highlight here. Terra's description pointed to 5 posteroventral spines in the foretibiae, three distal and two proximal, both groups separated by a clear gap. This arrangement is shown in Terra (1982: Fig. 2), which depicts the left prothoracic leg. My own examination of the holotype revealed that such arrangement is instead observed on the right prothoracic tibia rather than on the left one, which actually has an extra spine located at the gap between spine sets. This arrangement falls within the variation observed in *Dougonyx*.

Musonia maculata and *Musonia sexdentata*, both described by Beier (1942) in the same publication, are also attributable to *Dougonyx*. Although I was unable to study the holotypes, I reached to this conclusion after careful analyses of the original descriptions. Key diagnostic characters in *M. maculata* include (Beier 1942): "...die Metazona etwa dreimal so lang wie Prozona, ebenso wie diese mit einigen dunklen Punkten und außerdem noch mit 4 schwarzen Makeln jederseits, von denen die beiden vorderen punktförmig, die beiden hinteren aber als Schrägwische ausgebildet sind ..." ("...the metazona about three times as long as prozona, as well as with some dark spots and four smeared macula on each side, the two in the front are punctiform, the two in the back are oblique ..."), and; "Vorder tibia etwa zwei Fünftel so lang wie die Femora, die Krallenfurche der letzteren daher distal der Gliedmitte gelegen, mit 5 bis 6 Außendornen, von denen die beiden proximalen isoliert stehen, und 7 bis 8 Innerdornen, davon der 5. (bzw. 4.) von der Spitze verlängert" ("Frontal tibia about two-fifths as long as the femora, the claws groove of the latter therefore distal to the limb's center, with 5 to 6 outer spines of which the two proximal are isolated, and 7 to 8 inner spines, of which the 5th (or 4th) from the tip is the longest"). Key diagnostic characters in *Musonia sexdentata* (Beier 1942): "...die Metazona fast dreimal so lang wie die Prozona, mit feinem Mittelkiel, jederseits auf der Scheibe mit einer Reihe von 3 schwarzen Punkten..." ("...the metazona almost three times as long as the prozona, with a fine center keel, on each side of the disk with a series of 3 black dots..."), and; "Vorder tibien nur etwas mehr als ein Drittel so lang wie die Femora, mit 6 Außendornen und 8 Innerdornen, von letzteren der 5th (von der Spitze) ziemlich stark verlängert" (Front tibiae only

slightly more than one-third as long as the femora, with 6 outer and 8 inner spines, the 5th from the tip is the largest). Beier (1942) further provided figures of the supraanal plate of both species, these being triangular in *M. sexdentata*, and with a broadly rounded apex in *M. maculata*. I also observed this variation in examined individuals. Beier (1942) further considered *M. maculata* close to *G. amazonica*, but I could not confirm this at this time. Subsequent treatments should rule out the possibility of synonymy between the latter two species.

Galapagia Scudder, 1893 (Fig. 3.6f-j)

Galapagia Scudder, 1893: 7. Type species: *Galapagia solitaria* Scudder, 1893 (by monotypy).

Distribution. Galápagos Islands.

Diagnosis. *Size range:* 25–44 mm.

Body colouration: highly variable, from pale yellowish brown to blackish grey and dark brown, often with darker spots scattered over body and limbs.

Head: vertex straight (Fig. 3.6f), juxtaocular bulges conical, altogether rising above the dorsal margin of compound eyes.

Thorax: pronotum long as narrow (Figs. 3.6g–i), lateral margins with small denticles; metazona 1.9–2.2x as long as prozona. Forecoxae slightly shorter than metazona. Spination formula: F=4DS/7–10AvS/4PvS; T=3–6AvS/1–2PvS. Foretibial spines of the anteroventral series subdivided into two groups (g1 and g2 in Fig. 3.6j): g1 consists of 3 or less, much smaller, basal spines (sometimes absent altogether), whereas g2 consists of three large, distal spines of about the same size (Fig. 3.6j). Foretibial spines of the posteroventral series as in Fig. 3.6j. Tibial spines with weak, ventral micro-serrations, the same are somewhat more conspicuous at the tip of the tibial apical spur of females only. Males: wings iridescent, hyaline, either colourless or light brown, crossveins darkened at their point of insertion.

Abdomen: cylindrical in both sexes.

Remarks. Scudder (1893) described *Galapagia* to accommodate his new species *G. solitaria*, the only praying mantis so far known from the Galápagos archipelago. Scudder (1893) described his new genus and species on the basis of unique male and female specimens, both collected in Isabela Island (the largest island of the Archipelago) during two separated collecting events: the first, by the “Albatross Expedition”, lead by the renowned American Naturalist Alexander Agassiz in early 1891 and the second by the “Clark University Expedition” lead by Georg Baur later that year. After scudder’s description, *Galapagia* remained unmentioned in the literature until Hebard (1920) provided one additional distributional record of *G. solitaria* for Charles Island, as well as relevant morphological details and illustrations of the head and prothoracic tibiae. Beier (1935a) briefly redescribed *Galapagia* and named a second species, *G. peruana* Beier, 1935, on the basis of five females specimens, citing simply “Peru” as type locality. Terra (1982) described a third species, *G. amazonica*, from the Brazilian Amazon. Terra’s species showed considerable divergence in its morphology relative to the other two members of the genus, prompting Terra (1995) to modify the generic diagnosis of *Galapagia* in order to include the variation he observed in his new species. After Terra (1995) no additional species were assigned to *Galapagia* and few publications referring to this genus followed. These include Peck (2001), who provided additional distributional and ecological data for *G. solitaria* within the Galapagos, and Agudelo *et al.* (2007), who recorded the same species for Peru. My analyses of the species currently included in *Galapagia* suggest that *peruana* and *amazonica* cannot be attributed to this genus, as they exhibit important divergences from the genotype of *Galapagia*. These two species are thus removed from *Galapagia* and each treated under two different genera, both described as new in the present study. Consequently, *G. solitaria* is now the only species in the genus.

Species checklist

Galapagia solitaria Scudder, 1863* [EC: Galápagos Islands (Baltra, Charles, Española, Fernandina, Floreana, Genovesa, Isabela, San Cristóbal, Santa Cruz, Santiago, and Seymour)]

Taxonomic notes. *Galapagia solitaria* is the only praying mantis so far known from the Galápagos Islands. There are records of this species for all major islands within the archipelago and I have examined specimens from most of them. Although I did not observe conspicuous differences in the conformation of male genitalia accross specimens, I did observe important variation in external morphology and colouration among and within populations. Specimens observed ranged from 27 to 44 mm, and extreme sizes may occur within the same island. For instance, I observed adult males from Santa Cruz island ranging from 27 to 42 mm, whereas a pre-adult male from that same island reached 43 mm, suggesting that males from this population are easily attaining body lengths of at least 45 mm. Females ranged from 29 to 41 mm. Relative proportions of body parts also exhibited important variation. I have also observed males that are unusually larger than females, even within the same population. Body colouration varies from a pale yellowish brown to blackish grey and dark brown. Summarizing, *G. solitaria* exhibit important variation within and among islands, whereas male genitalia exhibits inconspicuous variation. Whether this variability represents one or more species/subspecies cannot be resolved at this time; genetic data will likely be necessary to solve this question.

Agudelo *et al.* (2007) recorded *G. solitaria* for Peru, but without mentioning any specific locality. This record is actually based on a single male specimen Jantsch (1999) previously examined for his PhD thesis and collected at Quistococha, Ucayali, in the Peruvian Amazon. Because the occurrence of *G. solitaria* in Peru, especially east of the Andes, seemed unlikely, I re-examined this specimen (at UMSP) and concluded that it is a misidentified member of *Macromusonia* (genitalia was missing, but likely referable to *M. major*). The record of *G. solitaria* for Peru is therefore erroneous and thus I remove *G. solitaria* from the Peruvian praying mantis fauna.

Liguanea Rehn and Hebard, 1938 (Fig. 3.6k-m)

Liguanea Rehn and Hebard, 1938: 34. Type species: *Liguanea pediodromia* Rehn and Hebard, 1938 (by original designation).

Distribution. Known only from the island of Jamaica, in the Greater Antilles.

Diagnosis. *Size range:* 27–28.3 mm.

Body colouration: light to dark brown.

Head: both vertex and juxtaocular bulges forming an almost straight outline, barely rising above level of of compound eyes (Fig. 3.6k).

Thorax: pronotum relatively slender (Fig. 3.6l), lateral edges bearing denticles, metazona 1.8–2x as long as prozona. Forecoxae about as long as metazona (Fig. 3.6l). Spination formula: $F=4DS/10-11AvS/4PvS$; $T=3-5AvS/1-2PvS$. Forefemora with curved dorsal margin (more strongly curved in females). Foretibial spines of the anteroventral series (Fig. 3.6m) consisting in 3 large, distal spines, the gap between the 1st and 2nd is about twice as long as the same between the 2nd and 3rd (sometimes one or more additional and smaller spines may occur at the gap between the 1st and 2nd spines, and/or closer to the base of the tibiae). Foretibial spines of the posteroventral series consisting mostly in two distal spines (proximal-most one is sometimes absent) (Fig. 3.6m). All foretibial spines with ventral microserrulations. Males: wings iridescent, hyaline and colourless, longitudinal veins with minute dark brown marks.

Abdomen: slender and cylindrical in both sexes.

Remarks. Rehn and Hebard (1938) assigned this species to the Thespinae. Although they suggested a closer relationship to *Musonia*, but also acknowledged its similarities to *Galapagia* given their similar arrangement of spines in the foretibiae. Terra (1995) questioned the validity of *Liguanea*. Although he did not examine any specimen, he considered *Liguanea* a likely synonym of *Galapagia*. My examination of the type series (ANSP) and additional specimens confirmed *Liguanea* as a valid taxon and distinct from *Galapagia*.

Species checklist

Liguanea pediodromia Rehn and Hebard, 1938* [JA] – **Fig. 3.23d–e**

Macromusonia Hebard, 1922 (Fig. 3.6n-o)

Macromusonia Hebard, 1922: 329. Type species: *Musonia major* Saussure and Zehntner, 1894 (by original designation).

=*Catamusonia* Giglio-Tos, 1927: 223. Type species: *Musonia major* Saussure and Zehntner, 1894 (by original designation).

Distribution. Northern South America and Amazon basin.

Diagnosis. *Size range:* 43–66 mm.

Body colouration: from pale to dark brown, often completely black and rarely green, frequently with various darker spots over body and limbs; legs often darker than the body.

Head: vertex straight, juxtaocular bulges developed, rising above vertex and dorsal margin of compound eyes altogether (Fig. 3.6n).

Thorax: pronotum elongated (Fig. 3.6o) and narrow, metazona 3.1–4.1x as long as prozona, lateral margins bearing denticles, supracoxal dilation weakly developed. Forecoxae shorter than metazona. Spination formula: F=4DS/11–13AvS/4PvS; T=8–9AvS/5PvS. Foretibial spines of the antero- and posteroventral series gradually increasing size distad (the 4th or 5th spine of the anteroventral series is sometimes slightly larger than the subsequent spine). Males: wings long and narrow, iridescent and hyaline, from colorless to darkly pigmented, often with darker spots along longitudinal veins; in resting position wings do not extend beyond tip of subgenital plate.

Abdomen: slender and cylindrical in both sexes.

Species checklist

Macromusonia conspersa (Saussure, 1870) [BR, PE] – **Fig. 3.23f**

Thespis conspersa Saussure, 1870

Macromusonia major (Saussure and Zehntner, 1894) [BR (PA), CO, FG, GY, PR] – **Fig. 3.23g–h**

Musonia major Saussure and Zehntner, 1894

Musonia Stål, 1877 (Fig. 3.6p-u)

Musonia Stål, 1877: 45. Type species: *Thespis surinama* Saussure, 1869 (subsequent designation: Rehn, 1904).

=*Promusonia* Giglio-Tos, 1916: 6. Type species: *Thespis surinama* Saussure, 1869 (by original designation).

Distribution. Nicaragua south through Central America into the Pacific and Atlantic lowlands of Western and Northern South America respectively.

Diagnosis. *Size range:* 21–34 mm.

Body colouration: light to dark brown or grey, exceptionally green.

Head: vertex almost straight (Fig. 3.6p–q), rising only to level of juxtaocular bulges, altogether only slightly more elevated than dorsal margin of compound eyes.

Thorax: pronotum slender (Fig. 3.6r–t), metazona 1.8–2.2x as long as prozona, lateral margins bearing denticles. Forecoxae about as long as metazona (Fig. 3.6r). Spination formula: F=4DS/12AvS/4PvS; T=7–9AvS/5PvS. Foretibial spines of the anteroventral series normally eight in number, forming two groups (g1 and g2 in Fig. 3.6u), g1 contains 3–5 spines, whereas g2 contains 4, distal-most spine of each group is largest. Foretibial spines of the posteroventral series gradually increasing size distad (distal-most spine is largest), gap between the 1st and 2nd basal-most spines is larger than between any other two spines within the same series (Fig. 3.6u). Males: wings iridescent, hyaline and normally colourless, darker individuals exhibit fuscous wigs.

Abdomen: slender and cylindrical in both sexes.

Remarks. *Musonia* has received exceedingly poor taxonomic treatment since originally established by Stål (1877). Most species listed in the latest catalogues and checklists (i.e. Ehrmann 200; Agudelo et al 2007, Otte and Spearman 2005) do not belong to this genus. For instance, *M. boliviana* Beier, 1930 and *M. seclusa* (Rehn, 1913) are here treated under *Thespis*,

M. costalis Rehn, 1920 is transferred to *Musoniella*, and *M. sexdentata* Beier, 1942 and *M. maculata* Beier, 1942 are transferred to *Dougonyx*, a new genus herein described.

Species checklist

Musonia chocoensis Salazar, 2002 [CO]

Musonia margharethae (Battiston and Picciau, 2008) (**comb. n.**) [EC, PE, CH]

Musoniella margharethae Battiston and Picciau, 2008

Musonia surinama Saussure, 1869 [BB, CR, FG, GY, NI, PA, SU, VE, TT] – **Fig. 3.24a–d**

=*Musonia femorata* Saussure and Zehntner, 1894*

=*Mionyx fuscescens* Chopard, 1911*

=*Mionyx lineatus* Chopard, 1911* [*Musonia*] (**syn. n.**)

Taxonomic notes. The characters provided in the original description of *Musonia chocoensis* Salazar, 2002 are diagnostic at genus level, and thus I consider it correctly classified in *Musonia*. However, this description is insufficient for accurate species-level identification, as male genitalia, the most useful structure to distinguish among species of *Musonia*, remains undescribed — the holotype was not available for study. The validity of this species needs verification.

Battiston and Picciau (2008) originally described *Musonia margharethae* under *Musoniella*. However, its identity as a member of *Musonia* was revealed after contrasting its original description against the available material — it is common to find specimens of *Musonia* misidentified as *Musoniella* among museum specimens (JR, pers. obs.). This species is the same cited as *M. surinama* in Rivera (2003b) and found along the Peruvian Pacific coast. I am also attributing specimens from northern Chile to this species (see Appendix 2).

Hebard (1923) established *M. fuscescens* as synonym *M. surinama*, a procedure Rehn (1935) later confirmed. *Musonia lineata* Chopard (1911), originally described under *Mionyx*, corresponds to a unique individual with a characteristic pattern of dark, longitudinal strips. Although apparently distinct, this unusual colour variation seems to correspond to an otherwise conspecific member of *M. surinama*. Chopard (1911) described *lineata* and *fuscescens* from a locality where, as Chopard himself pointed out, *surinama* “occurred abundantly”. My own

examination of the pertaining type material (including a few stripped individuals otherwise clearly referable to *M. surinama*) corroborated Hebard's conclusions about *fuscescens* and the invalidity of *lineata*, the latter here establish as a new synonym of *surinama*. Further, the wide distribution attributed to *M. surinama*, from Nicaragua through Central America to northern South America (including some Caribbean islands) and along the Pacific coast as far south as Lima (Rehn, 1904, 1935; Hebard, 1921, 1923, 1924; Salazar, 1999, 2000a, 2000b, 2001, 2002; Lombardo y Agabiti, 2001; Rivera, 2003b; Battiston and Picciau, 2008) possibly resulted from misidentifications and equivocal taxonomic procedures. For instance, Saussure and Zehntner (1894) described *Musonia femorata* from Nicaragua, which later on Hebard (1923) considered a synonym of *M. surinama*, thus extending its geographical distribution to Central America. My analysis of multiple representative populations from Central America and northern South America, including comparative analyses of male genital features, suggest that *femorata* and *surinama* might actually be distinct. If further demonstrated, the name *M. femorata* would apply to Central America populations, whereas *M. surinama* would be restricted to northern South America and Trinidad and Barbados.

Finally, Lombardo and Perez-Gelabert (2004) reported *Paramusonia cubensis* (Saussure, 1869) for Dominican Republic (Hispaniola); however, my examination of these specimens revealed them to be an undescribed species of *Musonia*. Interestingly, I found a female with green colouration among these specimens, an uncommon colour morph among Thespini not previously reported for *Musonia*. I also examined a few similarly colored individuals from Colombia and Surinam (all unidentified).

Musoniola Giglio-Tos, 1917 (Fig. 3.7a-c)

=*Mionycoides* Rehn, 1935: 181. Type species: *Mionyx ferus* Saussure and Zehntner, 1894 (by original designation) (**n. syn.**).

Distribution. Central America.

Diagnosis. *Size range:* 25–41 mm.

Body colouration: light to dark brown, sometimes with sparse, dark spots.

Head: vertex (Fig. 3.7a) straight or slightly sinuous, juxtaocular bulges well developed and strongly conical, altogether rising above the dorsal margin of compound eyes; lower ocellus of males often conical, lateral ocelli elliptical.

Thorax: pronotum relatively elongated (Fig. 3.7b–c), supracoxal bulges developed, lateral margins with blunt denticles, surface often with tubercles and granules, some are larger and pearl-like (specially in females), paired on either side of the medial carina; both granules and tubercles vary in number and density, but at least one pair of the largest, pearl-like tubercles occur either on the metazona, prozona or both (tubercles are often darkly pigmented in lighter specimens); metazona 1.9–3.3x as long as prozona, distal 1/5 of the latter depressed. Forecoxae shorter than metazoan (Fig. 3.7c). Spination formula: F=4DS/10–12AvS/4PvS; T=4–7AvS/3–5PvS. Forefemora elongated, and strongly curved at their distal 1/5. Foretibial spines of the anteroventral series subdivided into two groups: g1 group consisting in as many as 3 small, proximal spines (sometimes absent altogether), whereas g2 consists of 4 spines, where the second proximal-most is smallest, the distal-most spine has conspicuously moved to a more lateral position so that, in the lateral view, this spine overlaps the apical tibial spur. Foretibial spines of the posteroventral series gradually increasing size distad, sometimes medially interrupted by a spine-less gap; all tibial spines with only faint microserrulations ventrally, not visible in males but present only at the tip of the tibial apical spur and some of the antero- and posteroventral foretibial spines of females. Males: wings hyaline, smoky, often with contrasting, paler spots on cells; in resting position wings do not extend beyond tip of subgenital plate.

Abdomen: cylindrical and slender in both sexes, although slightly dilated in females, the latter with a small, median triangular lobe on the distal margin of each tergite (smaller in distal-most terga).

Remarks. See remarks under *Pseudomusonia*.

Species checklist

Musoniola fera (Saussure and Zehntner, 1894) [CR, GT, PA] – **Fig. 3.24e–g**

Mionyx fera Saussure and Zehntner, 1894* [*Pseudomusonia*]

=*Harpagonyx carlottae* Rehn, 1904*

=*Catamusonia minor* Beier, 1931*

Musoniola rapax (Saussure and Zehntner, 1894) [CR]

Mionyx rapax Saussure and Zehntner, 1894 [*Pseudomusonia*]

Musoniola dohrniana (Saussure and Zehntner, 1894) [GT]

Mionyx dohrnianus Saussure and Zehntner, 1894

Musoniola vicina Giglio-Tos, 1917* [NI] – **Fig. 3.24h**

Taxonomic notes. Rehn (1935) established the synonymy herein listed under *Musoniola fera*, to which I concur after my own examination of types specimens. Rehn (1935) further suggested that Saussure and Zehntner (1894) formulated their description of *Mionyx rapax* on the basis of a juvenile female, and that *fera* could be a synonym of *rapax*. My analysis of the holotype of *M. rapax* corroborated Rehn's suggestion regarding the immature nature its holotype of *rapax*, but I was unable to confirm the synonymy of *fera* at this time. Future treatments of *Musoniola* must rule out this possible synonym.

Terra (1995) and Ehrmann (2002) respectively reported *M. fera* and *M. rapax* for Venezuela (as *Pseudomusonia fera*). However, Cerdá (1996) did not mention their occurrence in this country. These two Venezuelan records need verification.

Two additional species, *Musoniola conservatrix* Terra, 1982 and *Musoniola venezuelana* Terra, 1982, do not belong to this genus. See remarks on these two species under “Thespidae *insertae sedis*” at the end of this section.

Oligonicella Giglio-Tos, 1915 (Fig. 3.7d-j)

Oligonicella Giglio-Tos, 1915: 190. Type species: *Oligonyx scudleri* Saussure, 1870 (by original designation).

Distribution. Southern USA (Florida and southern-most states, as far west as Texas), south through Central America to Costa Rica.

Diagnosis. *Size range:* 25–40 mm.

Body colouration: light to dark brown or greyish brown.

Head: Vertex typically straight, either slightly convex or concave, as high (males) or slightly more elevated (females) (Fig. 3.7d) than the dorsal margin of compound eyes, juxtaocular bulges produced and rising above level of vertex.

Thorax: pronotum ranging from long and slender to short and stocky (Fig. 3.7e–i), lateral margins bearing small denticles, metazona 1.2–1.3x as long as prozona in most species (twice as long in *O. mexicana*). Forecoxae as long or longer than metazona. Spination formula: F=4DS/7–10AvS/4PvS; T=4–6AvS/1PvS. Foretibial spines of the anteroventral series subdivided into two groups (g1 and g2 in Fig. 3.7j): g1 group consists of 2–4 spines where the distal most of this group is largest and the rest markedly small, whereas g2 consists in two much larger spines, its distal-most one clearly displaced and more dorsally positioned so that, in the lateral view, this spine projects over the apical tibial spur (Fig. 3.7j). Single posteroventral spine of males moderately curved (Fig. 3.7j). All spines, especially those on tibiae, with multiple microserrulations over most of their surface. Prothoracic femora slender and elongated. Males: wings slightly iridescent, light to dark brown and hyaline, often with darker spots on longitudinal wings; in resting position wings do not project beyond tip of subgenital plate.

Abdomen: cylindrical in both sexes, sometimes slightly fusiform in females.

Remarks. Members of *Oligonicella* exhibit important intra- and interspecific variation in body proportions and shape. For instance, *O. tesellata* and *O. mexicana* respectively exhibit short and long pronotum, whereas *O. scudleri* represents an intermediate form. Although relative elongation varies, the metazona is usually 1.2–1.3x longer than the prozona. I recognize five valid species in *Oligonicella*, including *O. mexicana* though with reservations (see below); two species previously treated under of this genus, i.e. *O. brunneri* and *O. agudelo*, are not members of this lineage; both are herein treated under *Thrinaconyx* Saussure and Zehntner, 1894.

Species checklist

Oligonicella mexicana (Saussure and Zehntner, 1894) [MX, US (AZ, TX)] – **Fig. 3.25a–b**

Oligonyx mexicanus Saussure and Zehntner, 1894* [*Bactromantis*]

=*Oligonyx toltecus* (Saussure and Zehntner, 1894)* [*Bactromantis*]

Oligonicella punctulata Saussure and Zehntner, 1894 [MX] – **Fig. 25c**

Oligonyx punctulatus Saussure and Zehntner, 1894*

Oligonicella scudderi Saussure, 1870 [MX, USA (AR, FL, KS, OK, TX)] – **Fig. 3.25d–e**

Oligonyx scudderi Saussure, 1870

=*Mantis? missouriensis* Glover, 1872

=*Oligonyx uhleri* Stål (1877)

=*Oligoxyx bollianus* Saussure and Zehntner, 1894*

Oligonicella striolata Saussure and Zehntner, 1894 [GT] – **Fig. 3.25f, 26a**

Olygonyx striolatus Saussure and Zehntner, 1894*

Oligonicella tesellata Saussure and Zehntner, 1894 [MX] – **Fig. 3.26b**

Olygonyx tesellatus Saussure and Zehntner, 1894*

Taxonomic notes. Scudder (1896) established *Bactromantis* to accommodate a new species from Sanford, Florida (USA): *Bactromantis virga*. Scudder (1896) considered his new genus and species as related to *Oligonyx*. In their revision of the Orthoptera of southeast USA, Rehn and Hebard (1916) recognized that the type specimens of *B. virga* were actually male juveniles of *Stagmomantis carolina* (Linné, 1763) (and not females, as stated by Scudder). Rehn and Hebard (1916) concluded that *Bactromantis* was a synonym of *Stagmomantis*, and *virga* a synonym of *carolina*. Later on, Hebard (1922) correctly established *Oligonyx tolteca* as synonym of *Oligonyx mexicana*, placing the latter within *Oligonicella*. Even though subsequent remarks on the actual identity of *Bactromantis virga* exist in the literature (e.g. Hebard 1932, 1943), this name remained surprisingly elusive in its condition as a synonym of *Stagmomantis carolina*. Giglio-Tos (1927) introduced further modifications after considering both *Oligonyx tolteca* and *Oligonyx mexicana* as valid, transferring them to *Bactromantis*. As a result, modern treatments still consider *Bactromantis tolteca*, *B. mexicana* and *B. virga* as valid species and member of the Thespidae (e.g. Terra, 1995; Ehrmann, 2002; Agudelo *et al.*, 2007). I examined the type series of *B. virga* (consisting in four syntypes, all at ANSP) and compared it with extensive holdings of *S. carolina* from the same institution, including large series of immature individuals of different instars. I concluded that Rehn and Hebard's accounts were correct, and *Bactromantis* is indeed a

synonym of *Stagmomantis*. Although *mexicana* exhibit a relatively more elongated body compared to other *Oligonicella* species, I choose to follow Hebard's (1922) opinion at this time and assign *mexicana* to *Oligonicella*, but further studies are needed to confirm its generic classification.

Oligonicella scudderi exhibits conspicuous disparity in colour, general body size and pronotal proportions. I analyzed male genital structures of various specimens comprising the entire range of *O. scudderi* and I did not find substantial differences suggesting that more than one species is involved. This variation prompted Hebard (1931) to consider another species, *Oligonyx bollianus*, a synonym of *Oligonicella scudderi* (something that Scudder also hinted as early as 1896); later on, Hebard (1943) further elaborated on the distribution and pronotal variation. Nevertheless, *O. scudderi* and *O. bolliana*, remained as distinct and valid species (e.g Terra 1995, Ehrman 2002, Agudelo *et al.* 2007). Battiston *et al.* (2005) partially illustrated the male genitalia of a single specimen they identified (probably based on distribution only) as *O. bolliana*, collected in León, Monterrey state, northern Mexico; this locality lays just south of the southern most record of *O. scudderi* within the USA, as reported by Hebard (1943) for Laredo, Texas. There is strong resemblance of the image presented by Battiston *et al.* (2005) and the material I studied, including one male from the original type series also from Dallas (type locality of *O. bolliana*). The type locality of *O. scudderi* was simply given as “America borealis” (i.e. North America) in Saussure (1870), whereas Saussure and Zehntner (1894) described *O. bolliana* from Dallas (Texas) and “Northern Mexico”, regarding it as a slenderer form than *O. scudderi*. All these observations concur with Hebard (1931) regarding the synonymy of *O. bolliana* and *O. scudderi*. These species, in addition to *Oligonyx uhleri* Stal, 1877, currently a synonym of *O. scudderi* (see Rehn and Hebard, 1916) were all established on the basis of conspicuous variation existing across the various populations of *O. scudderi*, which I here regard as the valid name for this species.

Oligonyx Saussure, 1869 (Fig. 3.7k-n)

Oligonyx Saussure, 1869: 58. Type species: *Oligonyx bicornis* Saussure, 1869 (subsequent designation: Giglio-Tos, 1915).

=*Spanionyx* Saussure, 1892: 122. Type species: *Spanionyx bidens* Saussure and Zehntner, 1894 (subsequent designation: Rehn, 1904).

=*Harpagonyx* Saussure, 1892: 122. Type species: *Harpagonyx gryps* Saussure and Zehntner, 1894 = *Oligonyx bicornis* Saussure, 1869 (subsequent designation: Rehn, 1904).

Distribution. Central Mexico to Costa Rica.

Diagnosis. *Size range (mm):* 28–54.

Body colouration: light to dark brown, sometimes with darker spots scattered over body and limbs.

Head: vertex (Fig. 3.7k) concave, with prominent and strongly conical juxtaocular bulges (particularly so in females).

Thorax: pronotum elongated and slender (Fig. 3.7l), supracoxal bulges marked, lateral margins with small denticles, metazona 2–2.7x as long as prozona. Forecoxae as long or shorter than metazona. Spination formula: F=3–4DS/7AvS/4PvS; T=3–6AvS/1PvS. Forefemora markedly slender, normally with 3 discoidal spines, exceptionally 4 (in which case the same is markedly small and vestigial), and with a long spine-less gap between the 6th and 7th anteroventral spines. Foretibial spines of the anteroventral series subdivided into two groups (g1 and g2 in Fig. 3.7n): g1 group consists in 4 spines, where the distal most of the group is largest and the rest are markedly small (sometimes missing altogether), whereas g2 consists of two much larger spines, its distal-most one more dorsally positioned so that, in the lateral view, this spine projects over the apical tibial spur (Fig. 3.7m–n). Foretibiae shortened, tip of apical spur not projecting beyond the second discoidal spines of the forefemora when both structures are flexed against each other. Single posteroventral spine of males strongly curved and recumbent (Fig. 3.7m); the same are larger and upright in females (Fig. 3.7n). Most foretibial spines with multiple microserrulations

on most of their surface. Males: wings iridescent, light to dark brown and hyaline; in resting position wings do not project beyond tip of subgenital plate.

Abdomen: long and cylindrical in both sexes.

Remarks. The species *Oligonyx insularis* Bonfils, 1967 is herein treated under *Thesprotiella*.

Species checklist

Oligonyx bicornis Saussure, 1869 [MX] – **Fig. 3.26c–e**

Spanionyx (Oligonyx) bicornis (Saussure, 1869)*

=*Harpagonyx gryps* Saussure and Zehntner, 1894*

Oligonyx bidens Saussure and Zehntner, 1894 [CR, GT, HO, MX] – **Fig. 3.26f**

Spanionyx bidens Saussure and Zehntner, 1894*

Oligonyx dohrnianus Saussure and Zehntner, 1894 [GT]

Harpagonyx dohrnianus Saussure and Zehntner, 1894

Oligonyx maya Saussure and Zehntner, 1894 [MX] – **Fig. 3.26g**

Harpagonyx maya Saussure and Zehntner, 1894*

Piscomantis Rivera gen. n. (Fig. 3.7o–r)

Type species: *Galapagia peruana* Beier, 1935 (by present designation and monotypy).

Distribution. Western slopes of the central Andes of Peru and adjacent pacific lowlands.

Description (on the basis of *G. peruana* Beier, 1935). *Size range*: 20–25 mm.

Body colouration: light to dark brown or greyish brown.

Head: vertex (Fig. 3.7o) straight to slightly concave, juxtaocular tubercles conical, well-developed and rising above vertex and dorsal margin of compound eyes.

Thorax: pronotum short and narrow in males (Fig. 3.7p), relatively stocky in females (Fig. 3.7q), supracoxal bulges developed, lateral margins with small denticles, metazona 1.3x as long as prozona. Forecoxae as long as metazona. Spination formula: F=4DS/8–10AvS/4PvS; T=5–

6AvS/1–3PvS. Foretibial spination similar to that of *Galapagia* (Fig. 3.7j). Foretibial spines of the anteroventral series subdivided into two groups (g1 and g2): g1 group consists in 2–3 markedly small, proximal spines, whereas g2 consists of three much larger and equally spaced spines. Tibial spines with weak, ventral microserrulations, the latter are somewhat more conspicuous along the tip of the tibial apical spur of females. Males: forewings narrow, hyaline, smoky grey with scattered darker spots along longitudinal veins and cells, crossveins often darkened at their point of insertion. Hindwings hyaline and colourless, often with a few darker spots near costal margin.

Abdomen: slender and cylindrical in both sexes, although slightly dilated in females. Distal margin of abdominal terga with a small, medial process, the same slightly larger in distal tergites (especially in females) (Fig. 3.7r).

Etymology. The generic epithet combines the word “Pisco”, and the word “mantis”, meaning “prophet” in ancient Greek, a common termination in many praying mantis genera. The new genus name makes reference to Pisco, the famous Peruvian national spirit native of Pisco, a town located the central Peruvian coast. Gender is feminine.

Remarks. Beier (1935a) described *Galapagia peruana* on the basis of one holotype and four paratypes (all females) from an unknown locality in Peru. This species exhibits important differences that depart from the morphology of true *Galapagia*. Overall, *Piscomantis* differs from *Galapagia* in having a much stouter and compact built; as a result, pronotal proportions are radically different. *Galapagia peruana* exhibit a metazona that is 1.3x as long as the prozona, whereas the same is 1.9–2.2x true insular *Galapagia*, the latter having a more stick-like habitus, typical of most Thespini. Differences among females include the presence of small tergal lobes in *G. peruana*, which are lacking in insular *Galapagia*. Given these evident morphological differences and the biogeographic separation, I proposed the new genus *Piscomantis* to accommodate *G. peruana*.

Species checklist

Piscomantis peruana (Beier, 1935) (**comb. n.**) [PE] – **Fig. 3.27a**

= *Galapagia peruana* Beier, 1935*

Pseudomusonia Werner, 1909 (Fig. 3.8a-d)

Pseudomusonia Werner, 1909: 78 (*nomen novum* pro *Mionyx* Saussure, 1892; pre-occupied: *Mionyx* Cope, 1886-Reptilia).

=*Mionyx* Saussure, 1892: 122 (nec *Mionyx* Cope, 1886-Reptilia). Type species: *Musonia lineativentris* Stål, 1877 (subsequent designation: Kirby, 1904).

=*Eumionyx* Giglio-Tos, 1927: 262 (unnecessary new name *pro Mionyx* Saussure, 1892).

Distribution. Southern Central America and North Eastern South America.

Diagnosis. *Size range:* 28.5–44 mm.

Body colouration: dark to light brown, sometimes yellowish green (solid or in combination), often with contrasting, darker spots over body and limbs.

Head: vertex (Fig. 3.8a) sinuous, occasionally slightly swollen medially, both vertex and juxtaocular bulges as high or rising above the dorsal margins of compound eyes.

Thorax: pronotum long and slender (Fig. 3.8b–c), surface smooth, lateral margins almost parallel and with small denticles, supracoxal bulges poorly developed but otherwise conspicuous, metazona 2.2–2.3x as long as prozona. Forecoxae conspicuously shorter than metazona.

Spination formula: F=4DS/8–10AvS/4PvS; T=5–9AvS/4PvS. Forefemora narrow, elongated, and slightly curved. Foretibial spines of the anteroventral series (Fig. 3.8d) forming two groups (g1 and g2), g1 is more proximal and contains 1–7 spines, whereas g2 contains 3 spines where the proximal-most is smallest and the distal-most one of the same group is conspicuously displaced from the ventral margin of the tibia, more dorsolaterally positioned and clearly overlapping with tibial apical spur in lateral view. Foretibial spines of the posteroventral series

(Fig. 3.8d) gradually increasing size distad, its proximal-most spine removed from the remaining spines. Tibial spines of males small and blunt, those of females pointy and sharper; all tibial spines, specially the largest ones, with ventral microserrulations (more clearly visible in females). Males: wings hyaline, iridescent, smoky dark brown to amber.

Abdomen: slender and cylindrical in both sexes.

Remarks. *Pseudomusonia* and other associated names (prominently *Musoniola*) have a complex taxonomic history; for clarity sake, I summarize it as follows:

- Saussure (1892) first established *Mionyx* without including any species.
- Saussure and Zehntner (1894) elaborated on the description of *Mionyx*, and added six species: i) four described as new: *rapax*, *ferus*, *saevus* and *dohrnianus*, and; ii) two transferred from *Musonia* Stål, 1877: *surinamus* Saussure, 1969 and *lineativentris* Stål, 1877. Note: *surinamus* eventually returned to *Musonia*, its current generic placement, and thus it will not be further discussed here.
- Kirby (1904) subsequently selected *Musonia lineativentris* Stål, 1877 as the type species of *Mionyx*.
- Werner (1909) recognized that *Mionyx* was not available, as *Mionyx* Cope, 1886, was a name already in use under the Reptilia. To solve this case of homonymy, Werner (1909) correctly proposed a new replacement name: *Pseudomusonia* Werner, 1909.
- Giglio-Tos (1917) described *Musoniola* to accommodate both *Mionyx dohrnianus* (which he designated as the type species) and *M. vicina*, a newly described species from Nicaragua.
- Hebard (1923) established *saevus* as synonym of *lineativentris*.
- Giglio-Tos (1927), unaware of Werner's (1909) procedure, established a new, albeit unnecessary, replacement name for *Mionyx*: *Eumionyx*. He included the following species: *rapax*, *ferus*, *lineativentris*, and *maculosus*, the latter previously described under *Mionyx* by Chopard (1911). Giglio-Tos (1927), also unaware of Hebard's (1923) nomenclatural procedure, continued to consider *saevus* as valid, placing it also under *Eumionyx*.
- Rehn (1935) accepted *Pseudomusonia* as a valid name, and correctly stated that both *Mionyx* (*sensu* Saussure, 1892) and *Eumionyx* were its synonyms. In the same publication, he further transferred both *rapax* and *ferus* (as *ferum*) from *Pseudomusonia* to a newly established

genus: *Mionycoides* (with *Mionyx ferus* as type species), rendering *lineativentris* as the only one member of *Pseudomusonia*. I regard Rehn's recognition of *rapax* and *ferus* as members of a lineage distinct from *lineativentris* as correct; however, the creation of *Mionycoides* was unnecessary, as *rapax* and *ferus* could have been simply transferred to *Musoniola* instead, where they actually belong. Not having analyzed the holotype of *Mionyx dohrnianus* (the type species of *Musoniola*) made Rehn (1935) introducing nomenclatural instability.

- Finally, Terra (1995) considered both *Musoniola* and *Pseudomusonia* as valid, and considered *Mionycoides* as a synonym of the latter (no justification was given). Following both Giglio-Tos (1927) and Hebard (1923), Terra (1995) included the following species under *Pseudomusonia*: *rapax*, *fera*, *lineativentris* (= *saevus*) and Chopard's (1911) *maculosus* (as *P. maculosa*). Most recent catalogues and checklists followed this opinion (e.g. Ehrmann, 2002; Agudelo *et al.*, 2007). I consider the placement of *Mionycoides* as synonym of *Pseudomusonia* as an additional source of nomenclatural instability.

Summarizing, the following conclusions are drawn to procure nomenclatorial stability:

- 1) *Pseudomusonia* Werner, 1909 is valid and both *Mionyx* Saussure, 1892 (nec Cope, 1886), and *Eumionyx* Giglio-Tos, 1927, are its synonyms. The only species included is *lineativentris*, and *saevus* is its synonym.
- 2) *Musoniola* Giglio-Tos, 1917 is valid and *Mionycoides* Rehn, 1935 is its synonym. The following species are members of this genus: *rapax*, *fera*, *dohrniana* and *vicina*.
- 3) *Mionyx maculosus* Chopard, 1911 is not a member of either *Musoniola* or *Pseudomusonia*. This species is here treated under a newly described genus (*Dougonyx*, see above).

Species checklist

Pseudomusonia lineativentris Stål, 1877 [CO, PA] – **Fig. 3.27c–b**

Musonia lineativentris Stål, 1877

= *Mionyx saevus* Saussure and Zehntner, 1894 *

Thespis Audinet Serville, 1831 (Fig. 3.8e-k)

Thespis Audinet Serville, 1831: 54 (original description). Type species: *Mantis parva* Drury, 1773 (subsequent designation: Rehn, 1904).

=*Paramusonia* Rehn. 1904: 567. Type species: *Thespis cubensis* Saussure, 1869 (subsequent designation: Rehn, 1904) (**syn .n.**).

=*Diamusonia* Giglio-Tos, 1916: 6. Type species: *Mantis parva* Drury, 1773 (subsequent designation: Giglio-Tos, 1916)

Distribution. Central Argentina, Northern South America, and Caribbean area.

Diagnosis. *Size range:* 35–60 mm.

Body colouration: usually light to dark brown, sometimes green.

Head: vertex and juxtaocular bulges rising above dorsal margin of compound eyes (sometimes barely so in males), lower frons markedly narrow and strongly transversal, 5–6x as wide as long (Fig. 3.8e–f).

Thorax: pronotum long and slender (Fig. 3.8g–i), lateral margins bearing denticles, supracoxal bulges relatively well marked, metazona 2.4–2.6x as long as prozona. Forecoxae shorter or about as long as metazona. Spination formula: F=4DS/12–13AvS/4PvS; T=8–11AvS/5PvS. All foretibial spines gradually increasing size distad, the posteroventral series restricted to the distal half of the tibiae (Fig. 3.8j). Males: wings hyaline and iridescent, darker individuals exhibit somewhat smoky brown mesothoracic wings; crossveins of both mesothoracic wing and discoidal area of metathoracic wings partially or fully pigmented in brown.

Abdomen: slender and cylindrical in both sexes, cerci approximately twice as long as supraanal plate, projecting far beyond the tip of the latter (Fig. 3.8k).

Remarks. Rehn (1904) proposed *Paramusonia* to accommodate *Thespis cubensis* Saussure, 1869, species he selected as the type species. Kirby (1904) immediately subsumed *Paramusonia* under *Musonia*; however, his procedure was not broadly accepted, as Rehn (1913) described a

second species, *P. seclusa*, from Argentina. Giglio-Tos (1916) transferred *seclusa* to *Promusonia* (= *Musonia*) and later on (Giglio-Tos 1927) accepted *Paramusonia* (and its only species, *cubensis*) as valid, remaining as such as for today (e.g. Terra, 1995; Ehrmann, 2002; Agudelo *et al.*, 2007).

My analysis of the syntypes of *cubensis*, in addition to conspecific individuals from Cuba, Grand Cayman and Bahamas, revealed no differences of generic value among this species and members of *Thespis*. Distinct diagnostic characters for *Thespis* include the unusually narrowed frontal shield (5–6x as wide as long) and cerci ca. twice as long as the supraanal plate. Male genital features of *cubensis* were found to be remarkably similar to those of other species of the genus from the main general area, such as *T. media* (see Rivera and Svenson 2016). Accordingly, I here consider *Paramusonia* as a new synonym of *Thespis*.

Species checklist

Thespis boliviana (Beier, 1930) (**comb. n**) [AR, BO] – **Fig. 3.27d–e**

Musonia boliviana Beier, 1930*

=*Leptomiopteryx argentina* Beier, 1930* (**syn. n.**)

Thespis cubensis Saussure, 1869* [BH, CI, CU] – **Fig. 3.27f**

Thespis exposita Beier, 1963 [AR]

Thespis major (Giglio-Tos, 1916) [CO, VE]

Diamusonia major Giglio-Tos, 1916

Thespis media Giglio-Tos, 1916 [TT, VE] – **Fig. 3.28a**

Diamusonia media Giglio-Tos, 1916*

Thespis metae Hebard, 1922* [CO] – **Fig. 3.28b**

Thespis pacifica Salazar, 2002 [CO]

Thespis parva (Drury, 1773) [CO]

Mantis parva Drury, 1773

=*Mantis minuta* Goeze, 1778

=*Thespis vicina* Saussure, 1872

Thespis seclusa (Rehn, 1913) (**comb. n**) [AR] – **Fig. 3.28c**

=*Paramusonia seclusa* rehn, 1913*

Taxonomic notes. Beier (1930) described both *Musonia boliviana* and *Leptomiopteryx argentina* in the same publication. However, my analyses of the type specimens and additional Argentinian and Bolivian material, revealed them to be conspecifics, and thus I consider *Leptomiopteryx argentina* as synonym of *Musonia boliviana*. Further, *Musonia boliviana* exhibits all diagnostic features of *Thespis*; accordingly, I also transfer this species to *Thespis*.

Lombardo and Perez-Gelabert (2004) reported *Thespis cubensis* (as *Paramusonia cubensis*) for a few localities in Dominican Republic, and provided a figure of the male genitalia. I had the chance to study the specimens reported in Lombardo and Perez-Gelabert (2004) and contrasted them with true members of *Thespis cubensis*. I concluded that the record of *Thespis cubensis* is equivocal, and that those specimens Lombardo and Perez-Gelabert (2004) reported as such actually correspond to an undescribed species of *Musonia*. *Thespis cubensis* is therefore removed from the praying mantis fauna of Dominican Republic.

Although I was unable to examine the female type of *Thespis exposita* Beier, 1963, the analysis of the original description clearly points to key character states diagnostic of *Thespis*. For instance, Beier (1963) mentioned the characteristic and conspicuously narrow frontal shield (“*Frontschild fast leistenförmig*”), and the cerci longer than the supraanal plate (“*Supraanalplatte sehr schmal und lang dreieckig, dachförmig gekielt, spitzig, fast 3mal länger als an der Basis breit, das Ende der Cerci nicht erreichend*”). Further, the only specimen of *Thespis seclusa* available for study was the male holotype deposited at ANSP. The analyses of the supraanal plate (glued to a piece of cardboard) and the strongly transversal frontal shield, clearly places this species within *Thespis*. It is also very likely that *T. exposita* represents the female of *T. seclusa*, as their respective type localities (Villa Merlo for *exposita*, and Alto Pencoso for *seclusa*) both occur within San Luis province in Argentina, and are only a few hundred of kilometers apart. Future treatments of *Thespis* must rule out this possible synonym.

Marshall (1975) reported one syntype of *Diamusonia media* (= *Thespis media*) deposited in BM (male) and two additional syntypes at ZMB (females), all of which I examined. One of the presumed syntypes at the ZMB bears an holotype label, clearly identifying it as such. However, the omission of type designation in the original description renders all specimens of the type series as syntypes (ICZN 1999, Art. 73.2).

I was unable to examine the holotype of *Thespis pacifica*. Described from western Colombia, this species differs from other members of *Thespis* by the unusual conformation of their forelegs, with three (instead of four) discoidal, and five (instead of four) posteroventral spines in the forefemora, and four (instead of five) posteroventral spines in the foretibia. Another feature of *T. pacifica* that departs from typical *Thespis* is its much more elongated pronotum, with a metazona 3.5x as long as prozona, instead of 2.4–2.6 as here documented. Further, Salazar (2002) reported this species for humid environments along the pacific coast of Colombia, corresponding to the Chocóan rainforest biome, whereas all other species of *Thespis* are found in seasonal and open habitats. I provisionally retain this species within *Thespis*, pending examination of additional material.

***Thesprotia* Stål, 1877 (Fig. 3.9a-b)**

Thesprotia Stål, 1877: 45 (original description). Type species: *Oligonyx filiformis* Saussure, 1869 (= *Thesprotia infumata* Audinet Serville, 1839) (by monotypy).

Distribution. Widespread but patchily distributed across the American continent. Found in Southeastern USA (east of the Mississippi River), Costa Rica, northern South America, within the Paraná River basin and adjacent areas, including southeastern Brazil; also common in oceanic islands (Caribbean area and Juan de Noronha Island off the coast of eastern Brazil). Apparently absent in Mexico, most of Central America and the Amazon. One undescribed species from Peru occurs at high elevations (ca. 2500), suggesting a possibly broader, vertical distribution along at least the eastern slope of the central Andes.

Diagnosis. *Size range:* 35–65 mm.

Body colouration: light to dark brown or grey, sometimes with contrasting, darker spots.

Head: vertex typically straight, sometimes slightly concave, as high as (males), or slightly more elevated (females) than the dorsal margin of compound eyes, whereas the juxtaocular bulges are well developed and clearly rising above the vertex.

Thorax: pronotum elongated and markedly narrow (Fig. 3.9a), lateral margins bearing small denticles, dorsolateral flanks of pronotum often with dense pilosity, in some species this pilosity

is spine-like, metazona 2.5–3.8x as long as prozona. Forecoxae long and slender, shorter than metazona, often with a strong undulation near its mid-section (Fig. 3.9a). Spination formula $F=3DS/5-6AvS/1PvS$; $T=2AvS/1PvS$. Foretibiae considerably shorter than forefemora (when flexed against the femur, the tibial apical spur reaches as far as the 2nd discoidal spine). The only two foretibial spines of the anteroventral series (Fig. 3.9b) have strongly shifted to a more dorso-lateral position, whereas the single spine of the posteroventral series is strongly recurved and recumbent in males (i.e. resting against the tibial edge), but larger, straighter and upright in females (Fig. 3.9b). Forefemora markedly narrow, lacking genicular spines (Fig. 3.9a). Males: wings iridescent, light to dark brown and hyaline; wings in resting position do not extend beyond tip of abdomen.

Abdomen: slender and cylindrical in both sexes.

Remarks. *Thesprotia* is one of the most diversified and widespread thespid lineages, but also among the least studied ones. Such diversity is in great part evident in the variation observed in body metrics across species. Unfortunately, the taxonomic literature hardly captures such variability, thus making literature-based identification difficult. This is aggravated by the fact that male genital morphology remains unknown for most species. Several undescribed species certainly exist, as well as a yet undetermined number of synonyms. *Thesprotia* is in need of revision.

Species checklist

Thesprotia alvarengai Bregante, 1969 [BR (PE)]

Thesprotia brasiliensis Chopard, 1916* [BR] – **Fig. 3.29a**

Thesprotia brevis Giglio-Tos, 1915 [PY]

Thesprotia caribea Rehn and Hebard, 1938 [AB]

Thesprotia filum (Lichtenstein, 1796) [SU]

Mantis filum Lichtenstein, 1796

Thesprotia fuscipennis Saussure and Zehntner, 1894 [BR (RJ)]

Thesprotia gigas Giglio-Tos, 1915* [BR] – **Fig. 3.28d**

Thesprotia graminis (Scudder, 1877) [USA (FL, GA)]

Oligonyx graminis Scudder, 1877

Thesprotia infumata (Audinet Serville, 1839) [BR]

Mantis infumata Audinet Serville, 1839

= *Thesprotia filiformis* Saussure, 1869*

Thesprotia insolita Rehn, 1935* [CR] – **Fig. 3.28f**

Thesprotia macilenta Saussure and Zehntner, 1894* [BR (RJ)] – **Fig. 3.28e, g–h**

= *Thesprotia vidua* Saussure and Zehntner, 1894*

Thesprotia maculata Giglio-Tos, 1915* [BR] – **Fig. 3.29b**

Thesprotia pellucida Giglio-Tos, 1915* [BR] – **Fig. 3.29c–d**

Thesprotia simplex Giglio-Tos, 1915* [BR] – **Fig. 3.29e**

= *Thesprotia brasiliensis* Chopard, 1916*

Thesprotia subhyalina (Saussure, 1870) [BR]

Oligonyx subhyalina Saussure, 1870

Thesprotiella Giglio-Tos, 1915 (Fig. 3.9c–n)

Thesprotiella Giglio-Tos, 1915b: 193. Type species: *Spanionyx festae* Giglio-Tos, 1898 (by monotypy).

= *Pogonogaster* Rehn, 1918: 326. Type species: *Pogonogaster tristani* Rehn, 1918 (by monotypy) (**syn. n.**).

= *Carrikerella* Hebard, 1921: 157. Type species: *Carrikerella ceratophora* Hebard, 1922 (by monotypy) (**syn. n.**).

Distribution. Nicaragua south into the Amazon basin, and the Lesser Antilles.

Diagnosis. *Size range:* 27–45 mm.

Body colouration: dark to light brown in preserved specimens, but in tones of green with contrasting brown spots when alive; pronotum often with 2–3 contrasting, oblique brown bands on each side of the metazona (the same are often green in living specimens).

Head: vertex straight, juxtaocular bulges well marked and often conical, altogether rising above dorsal margin of compound eyes, postocellar region either smooth or with some form of cuticular projection of variable length and shape, either single or paired (Fig. 3.9c–e).

Thorax: pronotum (Fig. 3.9f–j) with supracoxal bulges normally well marked, lateral margins bearing denticles, metazona with lateral margins medially narrowed or parallel, metazona 1.7–3.1x as long as prozona; pronotum of most species conspicuously sinuous in lateral view (straight in *T. peruana*) and often with a proximal, ventral swell. Forecoxae as long or slightly shorter than metazona. Spination formula: F=3DS/8–9AvS/4PvS; T=4–6AvS/1PvS. Forefemora long and narrow, both femora and tibiae dorsally curved (less conspicuously so in males). Foretibial spines of the anteroventral series (Fig. 3.9k) with a characteristic arrangement: three (or less) small, proximal spines of about the same size, followed by a larger spine positioned in the middle of the tibiae (strongly curved and recumbent in males; much larger and upright in females), and two distal spines clearly displaced and more dorsally positioned. Single foretibial spines of the posteroventral series (Fig. 3.9k) positioned distally (strongly curved and recumbent in males; much larger and upright in females). All foretibial spines with microserrulations. Meso- and metanotum distally with a small, median lobe. Males: wings dark to greenish brown, sometimes colorless, iridescent, with scattered brown spots, veins darkly pigmented.

Abdomen: cylindrical in both sexes. Males with highly reduced ornamentation or completely absent, whereas females often exhibit distal, lobe-like cuticular extensions in their abdominal terga (usually one central and one on each postero-lateral corner of each tergum); these lobes often have entire edges (Figs. 3.9l–m), but in some species these have fimbriated projections of variable sizes (Fig. 3.9n) (lobes on terga 2–4 tend to be much larger in some species). Some sternites may also exhibit a central, distal lobe but these are smaller than the dorsal ones and often darkly pigmented.

Remarks. Giglio-Tos (1915b) proposed *Thesprotiella* to accommodate *Spanionyx festae* Giglio-Tos, 1898, a species he previously described on the basis of a single male specimen from Ecuador. Following Giglio-Tos (1915b), a few other species were attributed to *Thesprotiella* (e.g. Chopard, 1916; Beier, 1935a, 1942), all these on the basis of male specimens. Rehn (1918c) and Hebard (1921) respectively described *Pogonogaster* and *Carrikerella*, this time on the basis

of female specimens. My examination of specimens of both sexes of *Thesprotiella*, *Carrikerella* (including conspecific sexes in some cases), the reexamination of *Pogonogaster*, in addition to most type specimens involved, made me conclude that *Carrikerella* and *Pogonogaster* were unnecessarily formulated as a result of marked interspecific and sexually dimorphic variation within *Thesprotiella*. This conclusion finds further support in Rivera and Svenson (2016), who found phylogenetic evidence for *Carrikerella* forming sister-group relationships with both *Thesprotiella* and *Oligonyx insularis*, altogether monophyletic — *Pogonogaster* was not tested, but future studies will most likely resolve it within this clade. All these taxa share several morphological traits, such as the characteristic foreleg spination pattern, the shape of the head, and various cryptic adaptations consistent with an epiphyte-dwelling life-style, such as lichen-like cuticular outgrowths on head and abdomen. The abdominal lobes of females exhibit great variation across species, from large and fimbriated to markedly small and even absent. Similarly, the postocellar projection exhibits great variability in both sexes, from long and strongly bifurcated to small and simple, sometimes absent. Therefore, I herein consider *Carrikerella* and *Pogonogaster* to be synonyms of *Thesprotiella*, to which *Oligonyx insularis* also belongs.

Thesprotiella seems to be a well-diversified and widespread lineage; nevertheless, specimens are rare in collections.

Species checklist

Thesprotiella bicorniculata Beier, 1942 [BR (AM)]

Thesprotiella ceratophora (Hebard, 1922) (**comb. n**) [CO, CR?] – **Fig. 3.29f**

Carrikerella ceratophora Hebard, 1922*

Thesprotiella empusa (Rehn, 1935) (**comb. n**) [CR] – **Fig. 3.29g**

Carrikerella empusa (Rehn, 1935)*

Thesprotiella festae (Giglio-Tos, 1898) [EC] – **Fig. 3.29h**

Spanionyx festae Giglio-Tos, 1898*

=*Thesprotiella similis* Giglio-Tos, 1915*

Thesprotiella fronticornis Chopard, 1916* [EC] – **Fig. 3.29i**

Thesprotiella insularis (Bonfils, 1967) (**comb. n**) [DO, GA, MA] – **Fig. 3.30a–b**

Oligonyx insularis Bonfils, 1967*

Thesprotiella latens (Hebard, 1919) (**comb. n**) [CO]

Pogonogaster latens Hebard, 1919*

Thesprotiella peruana Beier, 1935* [EC, PE] – **Fig. 3.30d**

Thesprotiella tristani (Rehn, 1918) [CR] (**comb. n**) – **Fig. 3.30c**

Pogonogaster tristani Rehn, 1918*

Thespidae *insertae sedis*

Musoniola conservatrix Terra, 1982

Musoniola venezuelana Terra, 1982

Remarks. Terra (1982) described these two species on the basis of unique male specimens collected in high-altitude (1500–2000m) habitats within the Coastal Cordillera of northern Venezuela. The original description of Terra (1982) and accompanying illustrations (*M. conservatrix* only) provide enough detail to recognize that both species are not members of *Musoniola*. For instance, both *conservatrix* and *venezuelana* exhibit prominent tubercles distally on the metazona near the supracoxal bulges (absent in *Musoniola*), and seven foretibial posteroventral spines (3–5 in *Musoniola*). The generic placement of Terra’s species cannot be corroborated at this time because of the lack of specimens among the studied material — holotypes were not available. However, having seven foretibial posteroventral spines, and distribution at high-elevations within northern South America, suggest instead a closer affinity to Pseudopogonogastrinae or Pseudomiopteryginae. According to Terra, *conservatrix* and *venezuelana* are very similar, differing only in details of colouration, slightly different body metrics and different (yet near) type localities. Therefore, it is likely that both names represent the same species in a yet, undescribed genus.

3.4.2 Family Angelidae Beier, 1935

Rivera and Svenson (2016) restricted the taxonomic concept of Angelidae to include only the Neotropical members of in this lineage (formerly a subfamily of Mantidae). This family is only represented by *Angela* Audinet Serville. Therefore, the morphological diagnosis of this

subfamily also applies to this genus (Rivera and Svenson 2016).

Angela Audinet Serville, 1839

Angela Audinet Serville, 1839: 171. Type species: *Mantis purpurascens* Oliver, 1792 (subsequent designation: Rehn, 1904).

=*Thespoides* Chopard, 1916: 166. Type species: *Thespoides bolivari* Chopard, 1916 (by monotypy).

Species checklist (distributional records from Ehrmann 2002)

Angela armata (Haan, 1842) [EC, FG, PE]

Mantis (Thespis) armata Haan, 1842

=*Angela fulgida* Saussure, 1872

Angela championi Saussure and Zehntner, 1894 [GU]

=*Mantis brachyptera* Stoll, 1813

Angela decolor (Chopard, 1914) [BO]

Thespis decolor Chopard, 1914

Angela guianensis Rehn, 1906 [BO, BR, CO, CR, FG, GY, SU, VE]

=*Thespis infuscata* Chopard, 1911

=*Thespoides bolivari* Chopard, 1916

Angela lemoulti (Chopard, 1910) [BR, FG, VE]

Thespis lemoulti Chopard, 1910

Angela maxima (Chopard, 1910) [FG]

Thespis maxima Chopard, 1910

Angela minor (Giglio-Tos, 1916) [VE]

Thespis minor Giglio-Tos, 1916

Angela miranda Saussure, 1871 [MX]

Thespis miranda (Saussure, 1871)

Angela perpulchra Westwood, 1889 [CR, NI, PA]

Thespis perpulchra (Westwood, 1889)

Angela peruviana (Giglio-Tos, 1916) [BR, PE]

Thespis peruviana Giglio-Tos, 1916

Angela purpurascens (Olivier, 1792) [BR, FG, PE, SU]

Mantis purpurascens Olivier, 1792

=*Mantis brachyptera* Stoll, 1813

=*Mantis versicolor* Stoll, 1813

Angela quinquemaculata (Olivier, 1792) [BR, FG, SU, VE, TR]

Mantis quinquemaculata Olivier, 1792

=*Mantis picta* Lichtenstein, 1802

Angela saussurii Giglio-Tos, 1927 [BR (AM, PA), FG, SU, VE]

Angela subhyalina (Chopard, 1914) [BO, BR, PE]

Thespis subhyalina Chopard, 1914

Angela trifasciata Stål, 1877 [BR (BA), CO, VE]

Angela wernerii (Chopard, 1914) [FG, SU, VE]

Thespis wernerii Chopard, 1914

Taxonomic notes. The taxonomic status of most species of *Angela* needs to be assessed. This is a taxonomically challenging genus with highly dimorphic sexes. Conspecific sexes are difficult to associate, and considerably intra- and interspecific variation exists. Additional diagnostic characters are shown in Figs. 3.10a–d), including examples of the variation in the colour pattern of the hindwings of males (Figs. 3.10e–h).

3.4.3 Family Coptopterygidae Giglio-Tos, 1915

Key to genera (both sexes)

1. Proximal antennal flagellomeres swollen (Fig. 3.10i), turning moniliform distally; wings of males do not reach tip of abdomen in resting position. Praying mantises exhibiting stick-like habitus.....***Brunneria***

1'. Antennae filiform; wings of males with its apex extending beyond tip of abdomen in resting position (except in Peruvian endemic *C. brevipennis*). Praying mantises with relatively stouter habitus, never stick-like.....***Coptopteryx***

Brunneria Saussure, 1869 (Fig. 3.10i-j)

Brunneria Saussure, 1869, 3: 58, 71. Type species: *Brunneria subaptera* Saussure, 1869 (by monotypy).

Diagnosis. *Size range:* 60–98 mm.

Body colouration: light to dark green or brown.

Head: vertex (Fig. 3.10i) slightly swollen and somewhat curved, rising above upper margins of compound eyes, and juxtaocular bulges poorly developed. Lower frons markedly narrow, virtually inconspicuous. Antennae in both sexes with the proximal-most section of flagellum swollen (Fig. 3.10i); this is particularly evident in females, where the proximal antenomeris are disc-like and laterally compressed, but becoming moniliform distally. Antennae of males as long or longer than mesothoracic wings, in females almost as long as the metazona of pronotum.

Thorax: pronotum in both sexes markedly slender and elongated (Fig. 3.10j), its margins bearing denticles, the latter often with dark basal pigmentation; metazona 3.2–3.9x as long as prozona. Forecoxae 1/3–1/2 the length of the metazona, dorsal margin with minute denticles. Spination formula: F=3DS/15–17AvS/5PvS; T=14–18AvS/9–11PvS. Anteroventral foretibial spines more numerous and more tightly arranged than those from the posteroventral series; posteroventral margin lacking spines along its proximal-most section — this spine-less section of the posteroventral margin may extend for up to 2/5 of the whole margin. Males: forewings well-developed, predominantly hyaline and greenish, not iridescent; base of costal field and area comprised between the subcostal and medial veins, coriaceous in appearance, with contrasting green, yellow (but more often) brown to reddish brown pigmentation along the area delimited by such veins; hindwings colored as mesothoracic wings, distal portion of costal area sometimes coriaceous and slightly pigmented in yellow and/or brown. Females: forewings reduced, not extending beyond the distal edge of the metanotum, thick and with coriaceous texture, fully green, or fully or partially darkened; metathoracic wings undeveloped.

Abdomen: Slender and cylindrical in both sexes, slightly dilated in females.

Species checklist

Brunneria borealis Scudder, 1869* [USA (FL, GA, HI, NC, TX)] – **Fig. 3.31b**

Brunneria brasiliensis Saussure, 1870 [AR, BR, PY]

Brunneria longa Giglio-Tos, 1915* [AR, BO, PY] – **Fig. 3.31c–d**

Brunneria orinicensis Agudelo and Chica, 2002 [CO]

Brunneria subaptera Saussure, 1869* [AR, BR (PR, RS), UY] – **Fig. 3.31e, 3.32a–b**

=*Brunneria gracilis* Giglio-Tos, 1915* (**syn n.**)

Taxonomic notes. I was able to associate the holotype of *B. subaptera* and three male syntypes of *B. gracilis* to several additional and clearly conspecific specimens of both sexes, ranging from eastern Argentina to southern-most Brazil, south of the Paraná River basin. My analyses suggest that *B. gracilis* is a synonym of *B. subaptera*. It is worth mention, however, that although one male specimen of *B. gracilis* (at ZMB) bears a red holotype label, this is not a holotype but a syntype instead because Giglio-Tos (1915a) did not designate any particular specimen of the type series as the holotype (ICZN 1999, Art. 74).

Coptopteryx Saussure, 1869 (Fig. 3.10k)

Coptopteryx Saussure, 1869: 56. Type species: *Mantis crenaticollis* Blanchard, 1867 (subsequent designation: Kirby, 1904).

=*Brachypteromantis* Piza, 1960. Type species: *Brachypteromantis bonariensis* Piza, 1960 (original designation and monotypy).

Distribution. Found in seasonal habitats across South America, mainly within the Paraná River basin and adjacent areas, southern Bolivia, northeastern and southern-most Brazil, northern South America and smaller enclaves of similar habitats, like dry forest and savanna-like habitats along the Andes (Rivera and Novoa 2006). Also found in central Chile.

Diagnosis. *Size range:* 45–80 mm.

Body colouration: usually green; some species also exhibit brown morphs.

Head: vertex slightly swollen, and somewhat curved, rising above upper edges of compound eyes, juxtaocular bulges poorly developed. Antennae filiform in both sexes, in females as long as the pronotum, as long as the abdomen in males.

Thorax: pronotum slender with lateral margins normally smooth in males, but stouter and with small to large marginal spines in females (Fig. 3.10k); metazona 2–3x as long as the prozona. Forecoxae as long, or slightly shorter, than metazona, dorsal and posterior margins bearing spines, the latter larger in females; in females, the anterior aspect of the forecoxae may also exhibit protuberances and spine-like tubercles. Spination formula: F=3DS/14–16AvS/5PvS; T=7–11AvS/13–15PvS. Anteroventral foretibial spines more numerous and more tightly arranged than those from the posteroventral series; posteroventral margin lacking spines along its proximal-most section — this spine-less section of the posteroventral margin may extend for up to 2/5 of the whole margin. Males: wings of males with its apex extending beyond tip of abdomen in resting position (except in *C. brevipennis*, where they attain the fifth abdominal segment). Wing colouration varies according to colour morph: a) Green morph: forewings hyaline and greenish, not iridescent, base of costal area, and section comprised between the subcostal and medial veins coriaceous and with contrasting green, yellow (but more often) brown to reddish brown pigmentation, hindwings colored as forewings, distal portion of costal area sometimes coriaceous, slightly pigmented in yellow and/or brown; b) Brown morph: forewings exhibit a pattern similar to that observed in green specimens, except for having brown veins, and a tinge of brown on the membrane, hindwings subhyaline, conspicuously brown and darker than forewings, crossveins on discoidal and anal fields pale, forming a contrasting network pattern. Females: forewings reduced, extending as far as the distal edge of the first abdominal tergite or shorter, thick and of coriaceous texture, fully green, sometimes with its basal half darkened; metathoracic wings undeveloped.

Abdomen: slender and cylindrical in males, dilated in females; cerci at most four times as long as supraanal.

Species checklist

Coptopteryx argentina (Burmeister, 1838) [AR, BR (PR, RS), PY] – **Fig. 3.32c, e**

Mantis argentina Burmeister, 1838

=*Coptopteryx claraziana* Saussure, 1869* (**syn. n.**)

=*Coptopteryx affinis* Giglio-Tos, 1915*

=*Coptopteryx viridis* Giglio-Tos, 1915* (**syn. n.**)

=*Paraphotina precaria* Piza, 1966*

=*Coptopteryx gigliotosi* Piza, 1973*

=*Coptopteryx ermannoi* Jantsch and Corseuil, 1988*

Coptopteryx brevipennis Beier, 1958* [PE] – **Fig. 3.33b**

Coptopteryx constricta Rehn, 1913* [AR] – **Fig. 3.33c–d**

Coptopteryx gayi (Blanchard, 1851) [AR, BR (RS), CL, UY] – **Fig. 3.33e–f**

Mantis gayi Blanchard, 1851

=*Mantis crenaticollis* Blanchard, 1851

=*Mantis grisea* Philippi, 1863

=*Coptopteryx rehni* Giglio-Tos, 1915*

=*Coptopteryx minuta* Giglio-Tos, 1915*

=*Brachypteromantis bonariensis* Piza, 1960*

Coptopteryx gracilis Giglio-Tos, 1915 [AR, BO]

Coptopteryx gymnopyga (Burmeister, 1838)* [BR (PA)]

Coptopteryx inermis Werner, 1925* [PY] – **Fig. 3.34a**

Coptopteryx magna Giglio-Tos, 1915 [PY]

Coptopteryx parva Giglio-Tos, 1915* [AR] – **Fig. 3.34b**

Coptopteryx platana Giglio-Tos, 1915 [UY]

Coptopteryx pumila Werner, 1935* [AR] – **Fig. 3.34c**

Coptopteryx thoracica Rehn, 1913* [AR, BR, PY] – **Fig. 3.34d–g, 3.35a**

=*Coptopteryx fallax* Giglio-Tos, 1917* (**syn. n.**)

=*Coptopteryx spinosa* Giglio-Tos, 1915* (**syn. n.**)

Coptopteryx thoracoides Giglio-Tos, 1915* [UY] – **Fig. 3.35c–d**

Taxonomic notes. Beier (1935b) and Agudelo and Rivera (2015) established the currently accepted synonyms of *C. argentina*. My examination of additional type material within *Coptopteryx* allowed the discovery of two additional names attributable to this species.

First, five syntypes (three males and two females) of *Coptopteryx claraziana*, presumably from the same type locality (Bahía Blanca, Argentina) and preserved at MNHG, are not all conspecific. The distinct pronotum of the males revealed them to be *C. argentina* (Fig. 3.32c), whereas the females likely correspond to a robust form of *C. gayi* (Fig. 3.32d), something that Saussure (1970) himself later hinted by attributing these females to *C. crenaticollis*, name formerly given to females of *C. gayi* and currently considered a *nomen nudum* (see Pfäuti and Hollier, 2012).

Lectotype designation. I herein opt to designate one of the males preserved at MNHG as the lectotype of *C. claraziana*. This specimen bears the three following labels (Fig. 3.32c): (1) white label (lined paper): MALE SYMBOL/“BAHIA BLANCA”/“REP.ARG. “/“ENV. G. CLARAZ”; (2) green paper: “COPTOPTERYX”/“CRENATICOLLIS”/“BLANCH.”; (3) red label: “SYNTYPUS”. All other syntypes, including the females I identified as *C. gayi*, now become paralectotypes without name-bearing status. Moreover, I also subsume *C. claraziana* as a new synonym of *C. argentina*.

Secondly, the type series of *C. viridis* includes three syntypes: one male (from Argentina) and one female (from Bolivia) were examined at ZMB, and one additional male at BM (from Argentina). At least the male specimen at ZMB is conspecific with *C. argentina* (Fig. 3.33a), whereas the BM male corresponds to a different, yet unidentified species (Fig. 3.32f).

Lectotype designation. I herein opt to designate the male specimen deposited at ZMB as the lectotype of *C. viridis*. This specimen bears the three following labels (Fig. 3.32e): (1) green label: “3.4.05” [handwrite]/Argentinien [printed]/ “SALTA CACHI” [handwrite]”J. STEINBACH V.” [printed]/ “2500” [handwrite, vertically oriented]; (2) red label: “COTYPUS”/”Nr.”; (3) white label: “COPTOPTERYX”/“VIRIDIS”/“GIGL.TOS”/“TYPUS E. GIGLIO-TOS *det*”. The remaining two syntypes now become paralectotypes without name-bearing status. Because the names *C. argentina* and *C. viridis* apply to the same species, I here

consider the latter as a new synonym of *C. argentina*.

The type series of *C. fallax* includes two conspecific individuals of each sex (syntypes), both with type locality in Paraguay (Fig. 3.34e, g). I compared these specimens to the type series of *C. thoracica*, which included the holotype female (Fig. 3.34d), a male allotype (Fig. 3.34f), plus five paratypes (all at ANSP), and found them to be conspecific with *C. fallax*; accordingly, I herein establish this new synonymy (note: the allotype of *C. thoracica* shown in Fig. 3.34f bears one additional label indicating its status as the paratype of *C. constricta*; however, this label is misplaced).

Giglio-Tos (1915a) described *C. spinosa* on the basis of one specimen of each sex, both from Buenos Aires, Argentina (MZB). My examination of these specimens revealed them to be syntypes and heterospecific: the male (Fig. 3.35a) is attributable to *C. thoracica*, whereas the female (Fig. 3.35b) corresponds to *C. gayi*.

Lectotype designation. Because the type series of *C. spinosa* consists of heterospecific syntypes (no holotype was designated in the original description), I herein designate as the lectotype of *C. spinosa* the male specimen preserved at MZB and portrayed in Fig. 3.35a. The remaining specimens are now paralectotypes. Also, I also establish *C. spinosa* as a new synonym of *C. thoracica*.

4.3.4 Family Liturgusidae Giglio-Tos, 1915

Rivera and Svenson (2015) restricted the taxonomic concept of Liturgusidae to include only the Neotropical bark mantises from the Acanthopoidea clade. Members of this family conform to a distinct body habitus (Fig. 3.10o). In his revision of *Liturgusa*, Svenson (2014b) also treated in detail three newly described genera: *Corticomantis*, *Fuga* and *Velox*. The genus *Hagiomantis* however remains understudied. Below is a modified version of the key to genera of Liturgusidae devised by Svenson (2014b), which now includes features of male genitalia.

Key to genera of Liturgusidae (both sexes)

1. Prominent tubercles distributed across the posterior (upper) surface of the mid- and hindfemora.....**Hagiomantis**
- 1'. Posterior (upper) surface of the mid- and hindfemora smooth.....2
2. Foretibiae with 7 posteroventral spines (females may exceptionally exhibit 8 spines on one or both tibiae), where the second spine is larger than others (except for the distal-most spine).....**Liturgusa**
- 2'. Foretibiae with 8 posteroventral spines, where the third spine is larger than others (except the most distal spine).....3
3. Forefemoral pit (a cuticular depression that accommodates the distal-most posteroventral tibial spine when flexed against the femur) positioned between the two proximal-most posteroventral spines, extending laterally to reach the posterior margin of the femur; subgenital plate of males with pronounced styli; distal margin of ventral phallomere smooth, lacking projections.....**Velox**
- 3'. Forefemoral pit positioned between the proximal-most posteroventral spine and the distal-most discoidal spine; subgenital plate of males lacking styli; distal margin of ventral phallomere with a prominent process.....4
4. Dorsal surface of pronotum with tubercles. Ratio *pronotum width at the supra-coxal bulge/total length of pronotum* of male and female 0.51 or greater (observed range 0.51–0.55); pronotum short and broad; distal process of ventral phallomere strongly recurved. Distributed entirely in Central America or at most in northern South America (Colombia).....**Corticomantis**
- 4'. Dorsal surface of the pronotum smooth or at most with markedly small tubercles widely dispersed. Ratio *pronotum width at the supra-coxal bulge/total length of pronotum* of male and female 0.49 or less (observed range 0.33–0.49); pronotum more elongated; distal process of ventral phallomere only moderately curved. Distributed in southeastern through eastern Brazil, not found into the Amazon basin.....**Fuga**

Corticomantis Svenson, 2014

Corticomantis Svenson, 2014b: 180. Type species. *Liturgusa atricoxata* Beier, 1931 (subsequent designation: Svenson, 2014b).

Distribution. Known only from Costa Rica.

Diagnosis. *Size range:* 18–20 mm.

Body colouration: a characteristic mottled pattern with contrasting colouration including shades of brown, black and green; all limbs exhibit dark bands.

Head: transverse, compound eyes large, rounded and forwardly projected; juxtaocular bulges present, small in males and well-developed in females; vertex straight. Ocelli reduced in size in both sexes.

Thorax: pronotum short and broad, lateral margins somewhat expanded to form a ridge, dorsal surface with evident tubercles of various sizes and density; metazona ca. 2x as long as prozona, its lateral margins almost parallel proximally but becoming divergent towards the relatively little pronounced but broadly rounded supracoxal bulges. Forecoxae longer than metazona. Spination formula: F=4DS/15–16AvS/4PvS; T=9–10AvS/8PvS. Forefemora robust, with a slightly concave dorsal margin distally (more evident in females); forefemoral pit that accommodates the distal-most posteroventral tibial spine well-developed, positioned between the proximal-most posteroventral spine and the distal-most discoidal spine. Foretibial spines robust; in the posteroventral series, the 1st and 2nd proximal-most and 5th to 7th spines, are shorter than the much longer proximal 3rd, 4th, and apical spines; in the anteroventral series, all spines steadily increase in size towards the tip of the tibiae. Wings: fully developed in both sexes; proximal ¼ of forewings dark, the remaining turning to a mottled pattern with contrasting regions of brown, white, green, and black (this colour pattern may be restricted to one forewing only, whereas the other is blackened). Hindwings darkened.

Abdomen: dilated in both sexes (broadest at segments 5–6), tergites with pointed posterolateral projections in the distal half of the abdomen of females. Cerci cylindrical, long and setose, tapering to a point. Subgenital plate of males lacking styli. Ventral phallomere with a prominent,

and strongly recurved, distal process; left phallomere with a fringe of long setae near the apofisi falloide.

Species checklist

Corticomantis atricoxata (Beier, 1931) [CR]

Liturgusa atricoxata Beier, 1931*

Fuga Svenson, 2014

Fuga Svenson, 2014b: 151. Type species. *Mantis annulipes* Audinet Serville, 1838 (subsequent designation: Svenson, 2014b).

Distribution. Southeastern through eastern Brazil.

Diagnosis. *Size range:* 18–30 mm.

Body colouration: a characteristic mottled pattern that combines various shades of green with black, brown, grey, pale turquoise and white; all limbs exhibit dark bands.

Head: transversal, compound eyes large, rounded and forwardly projected. Vertex straight, concave or slightly irregular; juxtaocular bulges usually prominent (especially in females), rising above level of vertex and dorsal edge of compound eyes. Ocelli reduced in size in both sexes.

Thorax: pronotum elongated to stocky, its dorsal surface smooth or at most with minute, widely dispersed, tubercles. Prozona with lateral margins parallel, tapering distally. Metazona 1.5–2x as long as prozona, its lateral margins always concave and medially narrowed; supracoxal bulges little pronounced but broadly rounded. Forecoxae longer than metazona. Spination formula: F=4DS/14–17AvS/4PvS; T=9–10AvS/8PvS. Forefemora robust, dorsal margin either straight or only slightly concave; forefemoral pit that accommodates the distal-most posteroventral tibial spine well-developed, positioned between the proximal-most posteroventral spine and the distal-most discoidal spine. Foretibial spines robust; in the posteroventral series, the 1st and 2nd proximal-most, and 5th to 7th posteroventral spines, are shorter than proximal 3rd, 4th, and apical

spines; in the anteroventral series, spines increasing in size towards the tip of the tibiae, but 6th and 7th spines longer than adjacent spines. Wings: fully developed in both sexes; mottled pattern and contrasting regions of forewings in various tones of brown, green and black (this colour pattern may be restricted to one forewing only, whereas the other is blackened, sometimes with traces of mottled pattern). Hindwings darkened.

Abdomen: dilated in both sexes (broadest at segments 5–7), tergites simple, lacking pointed posterolateral projections. Cerci cylindrical, long and setose, tapering to a point. Subgenital plate of males lacking styli. Ventral phallomere with a prominent, moderately curved, distal process; left phallomere with or without a fringe of long setae near the apofisi falloide.

Species checklist

Fuga annulipes (Audinet Serville, 1838) [BR (BA, RJ, SP, ES)]

Mantis annulipes Audinet Serville, 1838*

=*Hagiomantis parva* Piza, 1966*

=*Liturgusa sinvalnetoi* Piza, 1982*

=*Liturgusa parva* Giglio-Tos, 1914*

Fuga fluminensis (Piza, 1965) [BR (RJ, ES, PA, PR, SP)]

*Hagiomantis fluminensis**

Fuga grimaldii Svenson, 2014* [BR (ES, MG, SC, RJ)]

Hagiomantis Saussure and Zehntner, 1894 (Fig. 10l-n)

Hagiomantis Saussure and Zehntner, 1894: 158. Type species. *Mantis ornata* Stoll, 1813 (by monotypy).

Distribution. Amazon basin and Guiana shield.

Diagnosis. *Size range*: 31–52 mm.

Body colouration: a characteristic mottled pattern that combines various shades of black, brown and grey on a predominantly pinkish-brown background; all limbs exhibit dark bands.

Head: transverse, lower frons strongly narrowed and often fully black; compound eyes prominent and forwardly projected, juxtaocular bulges well-developed; vertex straight or only slightly concave, the same is lower dorsal margins of compound eyes. Ocelli reduced in size in both sexes.

Thorax: pronotum (Fig. 10l) relatively elongated, supracoxal bulges inconspicuous, appearing more as moderate undulations rather than broadly rounded protrusions; metazona ca. 2x as long as prozona, the former being long, depressed, and bearing small denticles along their concave lateral margins (more conspicuous in females); dorsal surface with numerous tubercles. Forecoxae longer than metazona, its surface often with darkened regions of variable size (Fig. 10m). Spination formula: F=4DS/14–15AvS/4PvS; T=8–9AvS/10PvS. Forefemora normally robust, spines well-developed, femoral pit that accommodates the distal-most posteroventral tibial spine well-delimited, positioned medial to the two proximal-most posteroventral spines; posterior surface with numerous tubercles; anteroventral surface often partially and strikingly pigmented in black or brown, the corresponding spines always black (Fig. 3.10n). Posterior surface of mid- and hindfemora with numerous and prominent tubercles. Wings: fully developed in both sexes; forewings mottled with contrasting regions in various tones of black, brown and grey on a predominantly pinkish-brown background (this colour pattern may be restricted to one forewing only, whereas the other is much darker). Hindwings often with either brown or yellow pigmentation, sometimes in combination to form contrasting patterns (i.e. reticulations, bands) on most of the anal and discoidal areas.

Abdomen: cylindrical in males, dilated in females; tergites sometimes with small, pointed, posterolateral projections. Subgenital plate of males without styli.

Species checklist

Hagiomantis mesopoda (Westwood, 1889) [FG]

Liturgousa mesopoda Westwood, 1889*

Hagiomantis ornata (Stoll, 1813) [FG?]

=*Mantis lutescens* Guérin-Mén. and Percheron, 1835

Hagiomantis pallida Beier, 1942 [BR]

Hagiomantis superba (Gerstaecker, 1889) [PE] – **Fig. 3.31a**

Liturgusa superba Gerstaecker, 1889*

Hagiomantis surinamensis (Saussure, 1872) [SU]

Liturgusa surinamensis Saussure, 1872

Taxonomic notes. Their scarcity in scientific collections, and the few literature records available, make of *Hagiomantis* one of the least known bark mantises from the Acanthopoidea clade. Most species listed above share several character states that suggest a close relationship. Commonalities include having conspicuous tubercles on pronotum and all six femora, anterior aspect of forelegs with distinct colour pattern (often with large, blackened areas), and large females (ca. 50mm, although these are sensible smaller in *H. mesopoda* with ca. 38mm). Original descriptions also suggest a wide range of variation in the colour patterns of the hindwings, these being either yellow and/or brown, sometimes with contrasting bands and/or reticulations (e.g *H. superba*). Unfortunately, published data on this genus is largely insufficient for conclusive literature-based identification and most types were not available. A taxonomic revision of *Hagiomantis* and a critical assessment of all names attributed to this genus are in order.

Liturgusa Saussure, 1869

Liturgousa Saussure, 1869: 55. Type species: *Liturgusa cayennensis* Saussure, 1869 (Subsequent designation: Kirby, 1904).

Liturgusa: Stål, 1877: 3 (emend.). Note: unjustified emendation, but in prevailing usage; see Svenson (2014a).

Distribution. From southern Mexico, through Central America, the Guiana shield and central Amazon basin.

Diagnosis. *Size range (mm):* 18–52.

Body colouration: all species exhibit a characteristic mottled pattern that combines various shades of green with contrasting black, brown and grey marks; all limbs exhibit dark bands.

Head: transversal, compound eyes large, rounded and forwardly projected; juxtaocular bulges usually prominent (especially in females), rising above level of vertex and dorsal edge of compound eyes; vertex straight, concave or slightly irregular. Ocelli reduced in size in both sexes.

Thorax: pronotum highly variable, elongated to stocky. Metazona 2.1–2.5x as long as prozona; dorsal surface typically smooth or with scattered tubercles, particularly in the posterior half. Supracoxal bulge well defined. Forecoxae as long or slightly longer than metazona. Spination formula: $F=4DS/12-17AvS/4PvS$; $T=7-11AvS/7PvS$. Femoral spines well-developed; femoral pit that accommodates the distal-most posteroventral tibial spine is well-defined, positioned medial to the two proximal-most posteroventral spines. Tibial spines robust; in the posteroventral series, the 1st is smallest, the 2nd and/or 3rd longer, the 3rd or 4th through 6th of similar length; in the anteroventral series, the longest are at the distal end and shortening proximally (although two proximal spines might be slightly longer than the adjacent spines). Meso- and metathoracic legs long, with dark banding on femora and tibiae. Wings: fully developed in both sexes; forewings mottled with dark marks on a predominantly greenish background (this colour pattern may be restricted to one forewing only, whereas the other is much darker). Hindwings hyaline to opaque, smoky or darkened, sometimes with rusty, yellow, or orange regions.

Abdomen: males and females with varying degrees of gradual widening (abdomen is broadest medially). Some species exhibit a pointed projection in the posterolateral corner of the distal-most tergites (in either or both sexes); other species lack such tergal projections. Cerci cylindrical, long and setose, tapering to a point. Subgenital plate of males without styli. Ventral phallomere with or without distal process (small if present); left phallomere without fringes of long setae near the apofisi falloide.

Species checklist

Liturgusa actiosa Rehn, 1950 [PA]

Liturgousa actiosa Rehn, 1950*

Liturgusa cayennensis Saussure, 1869 [BR (MT, AM) EG, GY, SU]

Liturgousa cayennensis Saussure, 1869*

- Liturgusa algorei* Svenson, 2014* [EC, PE]
- Liturgusa cursor* Rehn, 1950 [CR, NI, PA]
- Liturgousa cursor* Rehn, 1950*
- Liturgusa cameroni* Svenson, 2014* [GY, VN]
- Liturgusa dominica* Svenson, 2014* [DO]
- Liturgusa krattorum* Svenson, 2014* [EC, CO, PE]
- Liturgusa lichenalis* Gerstaecker, 1889* [CO, EC, PE, VE]
- Liturgusa guyanensis* La Greca, 1939* [BR (AM), GY]
- Liturgusa neblina* Svenson, 2014* [VE]
- Liturgusa bororum* Svenson, 2014* [BR (AM), PE]
- Liturgusa cura* Svenson, 2014* [VE]
- Liturgusa fossetti* Svenson, 2014* [CR, NI, PA]
- Liturgusa kirtlandi* Svenson, 2014* [BO]
- Liturgusa manausensis* Svenson, 2014* [BR (AM)]
- Liturgusa maroni* Svenson, 2014* [FG]
- Liturgusa maya* Sauss. and Zehntner, 1894 [CO, CR, EC, GT, HO, MX, NI, PA, PE, VE]
- Liturgousa cayennensis* [var.] *maya* Saussure and Zehntner, 1894*
- Liturgousa maya* Saussure and Zehntner, 1894
- =*Liturgusa charpentieri* Giglio-Tos, 1927*
- Liturgusa milleri* Svenson, 2014* [FG]
- Liturgusa nubeculosa* Gerstaecker, 1889 [BO, BR (AM, PA), EC, PE, VN]
- =*Liturgusa peruviana* Giglio-Tos, 1914*
- Liturgusa purus* Svenson, 2014* [BR (AM)]
- Liturgusa stiewei* Svenson, 2014* [CO]
- Liturgusa tessae* Svenson, 2014* [BO, BR, PE]
- Liturgusa trinidadensis* Svenson, 2014* [TT]
- Liturgusa zoeae* Svenson, 2014* [BE, GT, HO, PN]

Velox Svenson, 2014

Velox Svenson, 2014b: 171. Type species: *Velox wielandi* Svenson, 2014b (by original designation and monotypy)

Distribution. Southeastern Brazil.

Diagnosis. *Size range (mm):* 26–34.

Body colouration: a characteristic mottled pattern that combines various shades of black, brown and grey on a pinkish-brown background; all limbs exhibit dark bands.

Head: transverse, compound eyes large, rounded and forwardly projected; juxtaocular bulges present, and vertex straight or barely concave. Ocelli reduced in size in both sexes.

Thorax: pronotum elongated, lateral margins somewhat expanded to form a ridge, the latter sometimes bearing few denticles, dorsal surface with tubercles; metazona ca. 2.5x as long as prozona; distal portion of prozona and proximal half of metazone almost parallel, supracoxal bulges present but poorly developed. Forecoxae about as long as the metazona. Spination formula: F=4DS/16AvS/4PvS; T=10AvS/8PvS. Forefemora relatively slender, dorsal margin straight, and proximal third slightly convex; femoral pit that accommodates the distal-most posteroventral tibial spine well-developed, positioned on the lateral margin, pushing the margin outward between the two proximal-most posteroventral spines. Foretibial spines robust; in the posterventral series, the 1st and 2nd proximal-most and 4th through 7th spines, are shorter than the proximal 3rd and apical spines; in the anteroventral series, spines steadily increase size distad. Wings: fully developed in both sexes; forewings mottled with contrasting regions of brown, white, and black on a pinkish-brown background (this colour pattern may be restricted to one forewing only, whereas the other is blackened). Hindwings darkened.

Abdomen: elongated in males and dilated in females. Tergites without posterolateral projections. Cerci cylindrical, long and setose, tapering to a point. Subgenital plate of male elongate, bearing two prominent styli. Ventral phallomere without distal process; left phallomere without fringes of long setae near the apofisi falloide.

Species checklist

Velox wielandi Svenson, 2014* [BR (BA, ES)]

3.4.5 Family Photinaidae Giglio-Tos, 1915

Rivera and Svenson (2016) modified the concept of Photinainae, formerly assigned to Mantidae, and granted it family status. The Photinaidae is subdivided into four subfamilies. The three less diversified subfamilies are relatively well studied; their main references include Roy (2002b) (Macromantinae), Rivera (2010a), Agudelo and Rivera (2015) and Menezes and Bravo (2015) (Photiomantinae), and Heitzmann-Fontenelle (1968b) (Cardiopterinae). Morphological diagnoses for these three subfamilies can be found in Rivera and Svenson (2016). The relatively more diversified Photinainae (including tribes Orthoderellini, Microphotinini and Photinaini) are poorly studied; only *Orthoderella* (Orthoderellini) has been subject of taxonomic inquire (Rivera, 2003a; Roy and Stiewe, 2011, Agudelo and Rivera, 2015). Although additional studies treating individual taxa and nomenclatural matters have also been produced (Lombardo, 1998, 1999; Rivera, 2003a, 2010a; Medellin and Salazar, 2011; Ippolito and Lombardo, 2012; Agudelo, 2014; Agudelo and Rivera, 2015). The Photinainae are in need of a more in-depth revision.

3.4.5.1 Subfamily Macromantinae Brunner von Wattenwyl, 1893

Macromantis Saussure, 1871

Macromantis Saussure, 1871: 77. Type species: *Mantis hyalina* Degeer, 1773 (subsequent designation: Kirby, 1904).

=*Pentachanta* Stål, 1871. Type species: *Mantis hyalina* Degeer, 1773 (by monotypy).

Species checklist

Macromantis hyalina (De Geer, 1773) [BR, CO, EC, FG, GY, PE, SU, VE]

Macromantis nicaraguae Sauss. and Zehntner, 1894 [BE, BR (AM), CO, CR, GU, NI, PE]

=*Macromantis ovalifolia* var. *nicaraguae* Saussure and Zehntner, 1894

Macromantis ovalifolia (Stoll, 1813) [BR (PA), FG, PE, SU]

Mantis ovalifolia Stoll, 1813

Macromantis saussurei Roy, 2002* [BO, FG, GY?, SU, VE]

3.4.5.2 Subfamily Photiomantinae Rivera and Svenson, 2016

Photiomantis Piza, 1968

Photiomantis Piza, 1968: 49. Type species: *Photiomantis silvai* Piza, 1968 (original designation).

=*Margaromantis* Piza, 1982: 98. Type species: *Margaromantis margaritaria* Piza, 1982 (original designation).

=*Rehniella* Lombardo, 1999: 402 (*non* Hebard, 1928: Orthoptera, Gryllidae). Type species: *Metriomantis planicephala* Rehn, 1916 (original designation).

=*Colombiella* Koçak and Kemal, 2008 (unnecessary new name for *Rehniella* sensu Lombardo, 1999). Type species: *Metriomantis planicephala* Rehn, 1916 (original designation).

=*Lombardoa* Özdikmen, 2008: 765 (unnecessary new name for *Rehniella* sensu Lombardo, 1999). Type species: *Metriomantis planicephala* Rehn, 1916 (original designation).

Species checklist

Photiomantis nigrolineata (Menezes and Bravo, 2015) [BR (BA)]

Margaromantis nigrolineata Menezes and Bravo, 2015

Photiomantis planicephala (Rehn, 1916) [BR (BA, PB, PE, RN)]

Metriomantis planicephala Rehn, 1916*

=*Photiomantis silvai* Piza, 1968*

=*Margaromantis margaritaria* Piza, 1982*

Taxonomic notes. Lombardo *et al.* (2015) suggested that *Parastagmatoptera glauca* Rehn, 1920 could actually be a member of *Margaromantis* (= *Photiomantis*). I examined the female type specimen of *P. glauca* and concluded that it is not a member of Photinaidae; in fact, its inclusion in *Parastagmatoptera* also seems dubious. The generic placement of *P. glauca* requires further investigation. The characteristic pronotal shape of *Photiomantis* is shown in Figs. 3.10p–q.

3.4.5.3 Subfamily Cardiopterinae Rehn, 1911

Cardioptera Burmeister, 1838

Mantis (Cardioptera) Burmeister, 1838: 540. Type species: *Mantis (Cardioptera) brachyptera* Burmeister, 1838 (by monotypy).

Species checklist

Cardioptera brachyptera Burmeister, 1838* [BR (ES, RJ, SP)] – **Fig. 3.36a–b**

= *Mantis pilipes* Audinet Serville, 1839

= *Mantis sublobata* Audinet Serville, 1839

Cardioptera minor Rehn, 1916* [BR (BA, PB)] – **Fig. 3.36c**

Cardioptera nigridens Werner, 1925 [CO]

Cardioptera parva Beier, 1942* [BR (BA, CE, MG, PB)] – **Fig. 3.36d–e**

= *Tithrone signata* Piza, 1961

Cardioptera squalodon Werner, 1932 [FG, SU, GY]

Cardioptera viridipennis Beier, 1931* [BR (GO, MG, MT)] – **Fig. 3.36f**

= *Cardioptera paraopeba* Piza, 1967

Taxonomic notes. The characteristic pronotal shape and spined coxae of *Cardioptera* are shown in Figs. 3.10r–s

3.4.5.4 Subfamily Photinainae Giglio-Tos, 1915

3.4.5.4.1 Tribe Microphotinini Rivera and Svenson, 2016

Rivera and Svenson (2016) established this tribe to accommodate two distinct genera of photinaids: *Microphotina* Beier, 1935 and *Chromatophotina* Rivera, 2010. Members of Microphotinini are rarely found and thus are rare in collections, where male specimens predominate. Females of this lineage remained elusive until Rivera (2010a) described *Chromatophotina* on the basis of male and female specimens. Subsequent contributions characterizing additional females of *Chromatophotina* (Medellín and Salazar, 2011; Ippolito and Lombardo, 2012) helped to expand our knowledge of sexually dimorphic variation in this genus.

Examined males of this tribe are all externally very similar. Noteworthy, yet subtle variation in the shape of antenomeres, pronotal proportions and general body shape, foreleg shape and spine count, and cerci morphology, all seem potentially useful to differentiate among taxa. The analyses of male genital structures in the available material also revealed important variation suggesting uncovered genus-level diversity, thus concurring with François and Roy (2015). Because of the limits imposed by the dearth of specimens, I herein restrict the concept of Microphotinini to the only two known genera; however, below I present characters from what I believe are unrecognized genera for future reference. Additional collections are needed to gain a better understanding of this apparently well-diversified and enigmatic group of praying mantises.

I observed at least four genital types:

Type 1 (*Microphotina*). Ventral phallomere with a long, narrow, but poorly sclerotized, process that is born near the proximal end of the phallomere, projecting backwards and parallel to its right margin; the right, lateral margin of the same phallomere exhibits an elongated patch of dark setae. The left phallomere has a short, curved and unforked distal process; the apofisi falloide is long, acute and strongly recurved forwardly. François and Roy (2015) recognized this genital type as representative of true *Microphotina* (on the basis of *M. vitripennis*), interpretation I concur with. I have observed this genital type in material from French Guiana.

Type 2 (*Chromatophotina*). Ventral phallomere exhibits a long, narrow and well-sclerotized, lateral process. The left phallomere is characterized by having a long, curved and forwardly projected apofisi falloide, and a simple distal process. This morphological pattern is similar to Type 1 (*Microphotina*), but the left phallomere lacks the elongated patch of setae as described above (see also François and Roy 2015). I observed this genital type in material from Ecuador and Peru, including one undescribed species.

Type 3. The left phallomere exhibits a hammerhead-like distal process, whereas the apofisi falloide is reduced to a markedly small, sclerotized process. This genital type is clearly related to what Rivera (2010a) and François and Roy (2015) determined, respectively, as *Microphotina* sp. (from Peru) and *Microphotina viridescens* (from French Guiana). I also observed this genital type in a likely undescribed species from Bolivia. I concur with François and Roy (2015) in that this genital type probably represents an unrecognized lineage distinct from true *Microphotina*.

Type 4 (Genus A). Ventral phallomere with a short and relatively well sclerified process that originates near the proximal end of the phallomere; this process is broad at its base, but it exhibits a short and distally sharp projection (Fig. 3.10z). The left phallomere exhibits a short, curved and unforked distal process, whereas the apofisi falloide is well sclerotized and strongly recurved (Fig. 3.10z). The pronotum is relatively more robust than other males of tribe (Fig. 3.10y). This species is distinct and likely represents an unrecognized lineage. I have observed this genital type in material from French Guiana.

The variability in the number of forefemoral posteroventral and discoidal spines also complicates generic identification. In their redescription of *Microphotina*, François and Roy (2015) proposed 4 discoidal and 5 posteroventral (external) spines as a distinct feature of this genus; however, my own examination of specimens showed that 3–4 and 5–6 spines, respectively, represent the range of interspecific variability not only in *Microphotina*, but also in *Chromatophotina*. Variation may occur within the same individual as well, as reported for *Chromatophotina cofan* (Ippolito and Lombardo 2012). These characters seem to be more variable in Photinainae than in other acanthopoids, a fact that needs to be accounted for when dealing with the taxonomy of this group.

Key to genera of Microphotinini (males)

1. Lateral, right margin of the ventral phallomere with an elongated patch of dark setae.....*Microphotina* (in part)
- 1'. Ventral phallomere without an elongated patch of dark setae.....2
2. Left phallomere with hammerhead-like distal process; apofisi falloide short.....*Microphotina* (in part)
- 2'. Left phallomere with simple and curved distal process; apofisi falloide long and strongly recurved forward.....3
3. Lateral process of ventral phallomere long, narrow and well sclerotized.....*Chromatophotina*
- 3'. Lateral process of ventral phallomere with a short projection, the same is broad proximally, but narrows to a point distally (Fig. 3.10z).....**Genus A**

Chromatophotina Rivera, 2010 (Fig. 3.10t-w)

Cromatophotina Rivera, 2010: 23. Type species: *Chromatophotina awajun* Rivera, 2010 (original designation).

Distribution. Amazon rainforest of Colombia, Ecuador and Peru.

Diagnosis. *Size range:* 29–38 mm.

Body colouration: green.

Head: strongly transversal, vertex somewhat rising to level of juxtaocular bulges and slightly arched, altogether only slightly more elevated than the upper edges of compound eyes (Fig. 3.10t–u); lower frons markedly narrow, approximately 4.6–6x as wide as long.

Thorax: pronotum slender, supracoxal dilation poorly developed (Fig. 3.10v–w); lateral margins of prozona almost parallel, subtly converging distally; metazona 2.4–3.1x as long as prozona. Forecoxae as long or slightly shorter than metazona of pronotum. Spination formula: F=3–4DS/15–17AvS /5–6PvS; T=16–22AvS/18–21PvS. Males: fore- and hindwings iridescent, hyaline

and colorless, extending beyond tip of abdomen when at rest. Forewings opaque; costal area covered with white, dusty pigmentation, rest of wing green; hindwings square-like, costal area greenish, rest of wing opaque and brightly colored in yellow or orange; wings do not reach supraanal plate in resting position, leaving last 3 or 4 terga exposed.

Abdomen: slender in males, dilated in females, supraanal plate trapezoidal to tongue-like, cerci longer than supraanal plate, inner aspect of cercomeres darker.

Species checklist

Chromatophotina awajun Rivera, 2010* [PE]

Chromatophotina cofan Rivera, 2010* [CO, EC]

Microphotina Beier, 1935 (Fig. 3.10x)

Microphotina Beier, 1935: 122. Type species: *Cardioptera vitripennis* (original designation and monotypy).

Distribution. Amazon basin.

Diagnosis (males only). *Size range*: 21–36 mm.

Body colouration: green.

Head: vertex somewhat rising to level of juxtaocular bulges and slightly arched, altogether only slightly more elevated than the upper edges of compound eyes (Fig. 3.10x); lower frons markedly narrow, approximately 4x as wide as long.

Thorax: pronotum slender, supracoxal dilation poorly developed; lateral margins of prozona parallel; metazona 2.2–2.4x as long as prozona. Forecoxae as long or slightly shorter than metazona. Spination formula: F=3–4DS /13–15AvS/5–6PvS; T=14–18AvS/15–21PvS. Middle tibiae shorter than corresponding femora. Males: fore- and hindwings iridescent, hyaline and colorless, longer than abdomen when at rest. Female: forewings opaque; costal area covered with white, dusty pigmentation, rest of wing green; hindwings with discoidal area yellow; in resting

position wings do not reach supraanal plate, leaving last 3 or 4 terga exposed.

Abdomen: slender in males, dilated in females, supraanal plate tongue-like, cerci longer than supraanal plate.

Remarks. The redescription above incorporates elements of François and Roy (2015) and A. Agudelo (pers. comm.).

Species checklist

Microphotina vitripennis (Saussure, 1872) [FG]

Cardioptera vitripennis Saussure, 1872

Microphotina viridescens Chopard, 1911 [FG]

Photina viridescens Chopard, 1911*

Taxonomic notes. The holotype of *M. vitripennis*, presumably deposited at MNHG, according to Saussure (1872), could not be located. François and Roy (2015) reported that this type is most likely lost, thus designating a neotype (at MNHN).

Another species, *Microphotina viridescens*, has remained known from its single male holotype. According to François and Roy (2015), this type specimen exhibit definite male genital structures, but also a number of other traits that appear to be “feminized”: opaque, green pigmentation on the distal-most cells of the costal and discoidal area in both pair of wings, and a relatively more robust pronotum. My own examination of this specimen corroborates François and Roy’s observations. Recent taxonomic studies have uncovered a few cases of “feminized” (i.e. intersexual) male type specimens with similar characteristics (e.g. Lombardo and Umbriaco 2011, Agudelo 2014). Infection by parasitic Nematomorpha worms has been suggested as the cause of intersexuality in praying mantises (Schmidt-Rhaesa and Ehrmann 2001), and the finding of these parasites within the body cavity of types confirmed this. Whether such intersexual features seen in the type of *Microphotina viridescens* resulted from a parasitic load warrants further confirmation. The misleading morphology of intersexual individuals prompted François and Roy (2015) to redescribed *Microphotina viridescens* on the basis of multiple and presumably healthy male specimens. François and Roy (2015) also removed this species from synonymy

with *M. vitripennis*, a procedure I concur with. However, the eventual discovery of the female, and a more thorough sampling of this and other closely related species, could result in the assignment of *viridescens* to its own genus; until then, I follow the more conservative approach of François and Roy (2015), retaining this species under *Microphotina*.

3.4.5.4.2 Tribe Orthoderellini Giglio-Tos, 1917

Key to genera of Orthoderellini

1. Head elongated and strongly pentagonal in shape, vertex and juxtaocular bulges prominent (Fig. 3.10aa; see also Fig. 8M in Rivera and Svenson 2016). Body elongated.....***Orthoderella***
- 1'. Head not particularly elongated and more triangular in shape, vertex and juxtaocular bulges weakly developed. Body not particularly elongated.....***Paraphotina***

Orthoderella Giglio-Tos, 1897 (Fig. 3.10 aa-ab)

Orthoderella Giglio-Tos, 1897: 13. Type species: *Orthoderella ornata* Giglio-Tos, 1897 (by monotypy).

=*Euryderes* Brancsik, 1897: 63. Type species: *Euryderes anisitsi* Brancsik, 1897 (by monotypy).

Distribution. Found mainly within the Paraná River basin and neighbouring areas, including southern Bolivia, and southern through northeastern and Brazil, usually associated to seasonal, savanna-like habitats. At least one species (*Orthoderella delucchii*) is found in the humid eastern tropical slopes of the Andes of Peru and Bolivia). An undescribed species from Venezuela (ANSP) suggests a possible wider range in northern South America.

Diagnosis. *Size range:* 35–50 mm.

Body colouration: normally pale yellow or brown, exceptionally green, sometimes with dark markings on thorax; proximal tergites with lustrous, black bands.

Head: strongly pentagonal in shape (Fig. 3.10aa), as long as wide or conspicuously elongated (often preserved in “prognathous” position), vertex and juxtaocular bulges well-developed and prominent, altogether rising well above upper edges of compound eyes.

Thorax: pronotum elongated, surface smooth, lateral edges of prozona straight and nearly parallel, supracoxal bulge inconspicuous (Fig. 3.10ab). Metazona 2.4–2.8x as long as prozona. Forecoxae as long, or only slightly longer than metazona. Spination formula: F=3DS/13–14AvS/5PvS; T=14–18AvS/11–18PvS. Mid- and hind femora and tibiae of males with dense, whitish pubescence that extends into the coxae, thoracic sterna and abdominal terga; pilosity is absent or inconspicuous in females. Cursorial legs relatively short (i.e. mid- and hind- femora shorter than prothoracic femora). Males: fully-winged, wings of males with its apex extending beyond tip of abdomen at rest; forewings hyaline and iridescent, membrane colorless, longitudinal veins greenish brown, crossveins sometimes light brown; hindwings similarly colored but crossveins less conspicuously marked in brown. Females: brachypterous, in resting position wings do not extend beyond mid abdominal segments; forewings opaque, greenish brown, crossveins darkened; hindwings with discoidal and anal area reddish brown, distal margin bordered in dark brown, crossveins on anal area pale.

Abdomen: slender in both sexes; cerci markedly long and flattened, especially distal-most segments.

Species checklist

Orthoderella delucchii Rivera, 2003* [BR (AM), PE]

Orthoderella elongata Roy and Stiewe, 2009* [AR, BR (GO), UR]

Orthoderella major (Piza, 1962) [AR, BR (MG, MT, RS, SC, SP,) UY]

Tithrone major Piza, 1962*

=*Orthoderella brasiliensis* Roy and Stiewe, 2011*

Orthoderella ornata Giglio-Tos, 1897 [AR, BO, PY]

=*Euryderes anisitsi* Brancsik, 1897

Paraphotina Giglio-Tos, 1915 (Fig. 10ac)

Paraphotina Giglio-Tos, 1915: 72. Type species: *Cardioptera reticulata* Saussure, 1871 (subsequent designation: Giglio-Tos, 1915).

=*Guaraunia* Rehn, 1941: 1. Type species: *Guaraunia insolitum*, 1941 (designación original y monotipia).

Distribution. According to my own examination of specimens and literature records (Cerdá 1992, Lombardo 1998, Agudelo and Chica (2001), *Paraphotina* seems to be more commonly associated to seasonal habitats in the periphery of the Amazon. However, a few records also indicate an Amazonian distribution. Found in Colombia, Venezuela, eastern Peru and through eastern Brazil.

Diagnosis. *Size range:* 35–47 mm.

Body colouration: predominantly yellowish brown, but sometimes greenish, often with multiple contrasting dark markings on head, pronotum (Fig. 3.10ac) and legs.

Head: triangular in shape and not particularly elongated, vertex and juxtaocular bulges weakly developed. Compound eyes strongly spherical.

Thorax: pronotum surface smooth, lateral margins smooth in males, bearing small denticles in females; prozona with lateral edges straight and nearly parallel, supracoxal bulges inconspicuous (Fig. 3.10ac). Metazona 2.5–2.8x as long as prozona. Forecoxae as long as metazona. Spination formula: F=3–4DS/13–15AvS/5PvS; T=15–17AvS/11–16PvS. Mid- and hind femora and tibiae of males with dense, whitish pubescence, extending into the coxae, thoracic sterna and abdominal terga; pilosity is absent or inconspicuous in females. Males: fully-winged, wings of males with its apex extending beyond tip of abdomen at rest; forewings hyaline and colorless, iridescent, longitudinal veins green or brown, crossveins brown, strongly contrasting against the hyaline membrane, forming a reticulated pattern; hindwings similarly colored but crossveins mostly colorless (except more distally). Females: wings shortened, in resting position extending between mid abdominal segments and supraanal plate; forewings opaque, membrane greenish brown, crossveins brown; hindwings with yellow costal and anal areas (the former with brown

crossveins), discoidal area reddish brown, distal margin of wing bordered in light brown with pale crossveins.

Abdomen: slender in males, dilated in females, first three terga with a pair of black, shiny marks. Cercomeres square-like, flattened.

Species checklist

Paraphotina caatingaensis (Menezes and Bravo, 2013) (**comb. n.**) [BR (BA, PI)]

Orthoderella caatingaensis Menezes and Bravo 2013

Paraphotina occidentalis Lombardo, 1998 [PE]

Paraphotina insolita Rehn, 1941* [VE]

Paraphotina reticulata (Saussure, 1871) [BR (PA), VE] – **Fig. 3.37a–b**

Cardioptera reticulata Saussure, 1871*

=*Photina gracilipes* Kirby, 1904

=*Paraphotina venezuelana* Beier, 1963*

Remarks. Menezes and Bravo (2013) described *Orthoderella caatingaensis* from localities in Northeastern Brazil. The phylogenetic analysis of Rivera and Svenson (2016), in addition to the examination of specimens from the original type series (one paratype and a photograph of the holotype), raised concerns about the inclusion of this species in *Orthoderella*. Key characters that set this species apart from *Orthoderella* are: less elongated body, strongly contrasting brown forewing crossveins and markings on pronotum and limbs, poor development of vertex and juxtaocular bulges, and compound eyes more spherical (Menezes and Bravo 2013). All these features are referable to *Paraphotina* and thus I transferred this species accordingly.

3.4.5.4.3 Tribe Photinaini Giglio-Tos, 1915

Key to genera of Photinaini

1. Head with lower frons strongly narrowed, ca. 5x as wide and long. Pronotum relatively narrow (Fig. 3.10ad), lateral margins of prozona parallel. Cerci markedly long, more than 4x as long as supraanal plate, distal cercomeres flattened.....***Hicetia***

1'. Head with lower frons 2–2.5x as wide and long. Pronotum relatively stout (Fig. 3.10ae-ah), lateral margins of prozona not parallel. Cerci less than twice as long as supraanal plate, distal cercomeres cylindrical.....***Photina***

Hicetia Saussure and Zehntner, 1894 (Fig. 3.10ab)

Hicetia Saussure and Zehntner, 1894:152. Type species: *Hicetia goeldiana* Saussure and Zehntner, 1894 (by monotypy).

Distribution. Atlantic rainforest of southern Brazil.

Diagnosis. *Size range:* 40–48 mm.

Body colouration: green, limbs often yellowish.

Head: triangular, much wider than long, vertex smooth and weakly convex, juxtaocular bulges weakly developed; lower frons narrow, its width ca. 5x its length.

Thorax: pronotum long and narrow (Fig. 3.10ad), lateral edges of pronotum smooth in males, with minute denticles in females, supracoxal bulges poorly developed, and lateral edges of prozona almost parallel; metazona 3–3.5x as long as prozona. Spination formula:

F=4DS/14AvS/6PvS; T=14–17AvS/14–17PvS. Forecoxae shorter than prozona. Forefemora relatively elongated, dorsal edge serrated in females. Males: forewings hyaline and colorless, crossveins in both costal and discoidal areas simple and parallel; hindwings hyaline and colorless. Females: wings do not extend beyond mid abdominal terga in resting position; forewings green, costal area opaque, discoidal area subhyaline; hindwings hyaline, crossveins yellow.

Abdomen: long and cylindrical in males, slightly fusiform in females; supraanal plate triangular, cerci considerably longer than supraanal plate and with at least the distal-most cercomeres flattened.

Species checklist

Hicetia breviceps (Stål, 1877) [BR (RJ)]

Photina breviceps Stål, 1877

Hicetia goeldiana Saussure and Zehntner, 1894* [BR (RJ, SP)] – **Fig. 3.37c–e**

=*Photina (Metriomantis) gracilicollis* Beier, 1931*

Taxonomic notes. Both species of *Hicetia* were originally described from unspecified localities within Rio de Janeiro, Brazil (Stål 1877, Saussure and Zehntner 1894). Beier (1931) described both sexes of *Photina (Metriomantis) gracilicollis* from the vicinity of São Paulo, Brazil, but soon later (Beier 1935b) he recognized it as conspecific with *H. goeldiana* and thus its synonym. Given that *H. goeldiana* is found in Atlantic forest formations both in São Paulo and Rio de Janeiro, it could well represent yet another synonym of *H. breviceps*. Future studies must rule out this possibility.

Photina Burmeister 1838 (Fig. 3.10ad–ah)

Photina Burmeister, 1838: 531. Type species: *Mantis (Photina) vitrea* Burmeister, 1838 (subsequent designation: Kirby, 1904).

=*Metriomantis* Saussure and Zehntner, 1894: 151. Type species: *Mantis cupido* Saussure, 1869 (subsequent designation: Kirby, 1904) (**syn n.**)

=*Photinella* Giglio-Tos 1915: 73. Type species: *Photina brevis* Rehn, 1907 (original designation) (**syn n.**)

Distribution. Guiana shield, south through the Amazon and Paraná river basins, as far south as southern Brazil.

Diagnosis. *Size range:* 40–60 mm.

Body colouration: green.

Head: triangular, vertex convex and curved to various degrees, rising only to level of compound eyes, juxtaocular bulges poorly developed (i.e., not protuberant) (Figs. 3.10ag–ah); lower frons 2–2.5x as wide as long; compound eyes spherical to conical.

Thorax: pronotum elongated and variable in width (Figs. 3.10ae–ah), lateral edges smooth in males, with denticles in females, supracoxal bulges developed, prozona and distal part of metazona forming an ellipsoidal outline, more strongly so in females; metazona 1.8–3x as long as prozona. Spination formula: F=4DS/12–16AvS/5–6PvS; T=13–15AvS/14–18PvS. Forecoxae as long as metazona. Forefemora of males often with dark spots on or near spines, and/or more proximally on anterior aspect (such spots are uncommon in females); dorsal edge serrated in females. Males may exhibit a dense, whitish, golden or dark pubescence on meso- and metasternum, and the corresponding coxae; such pubescence often spread into mid- and hind legs in some species (sometimes a dark pubescence occurs on legs instead of whitish or golden); such pubescence is absent or inconspicuous in females. Males: forewings iridescent and shimmering, delicate-looking and with glassy appearance, normally completely hyaline with green veins, although the costal area of few species is sub-opaque with a tinge of green, costal area with parallel veins, sometimes anastomosed and appearing forked in some species; hindwings coloured as mesothoracic wings; wings extend beyond tip of abdomen in resting position. Females: forewings elliptical, opaque green, costal area wide, with multiple parallel or partially anastomosed crossveins; hindwings mostly colorless and hyaline, except for the tip of the discoidal area, crossveins and the area immediately around the latter, which can be either green or yellow.

Abdomen: slender in males, wide and oval in females.

Remarks. The phylogenetic analysis of Rivera and Svenson (2016) revealed problems with the taxonomic and nomenclatural definition of *Metriomantis*, *Photina* and *Photinella*. They suggested that the number of discoidal and posteroventral spines of the forefemora, formerly used to distinguish among these three genera, cannot be a reliable character because they are not entirely consistent among congeners — as it is often the case in other photinaids taxa. Additional morphological similarities among these three genera raised additional concerns about their distinctiveness. For instance, the conformation of the head of *Photina*, *Photinella* and *Metriomantis* is very similar. They all exhibit poorly developed juxtaocular bulges flanking a straight to relatively convex vertex. The shape of the compound eyes, which ranges from spherical to conical, also influences the overall appearance of the head's dorsal outline across

members of these genera. Variation in eye shape within *Metriomantis* has already been documented (Lombardo 1999), and my own examination of specimens revealed a similar gradation. Males of *M. pilosella* and *M. boliviana*, respectively with spherical and conical eyes, exemplify both ends of this range of variation, with *M. pilosa* representing an intermediate form — and comparable to *Photinella*. Conspecific males and females also exhibit variation in this respect, but there is no evident trend towards a particular shape in either sex. Giglio-Tos (1915a) justified the creation of *Photinella* on the basis of anastomosed crossveins on a slightly pigmented (green) costal area of male forewings. This character states presumably distinguishes *Photinella* from *Photina* and *Metriomantis*, where crossveins are predominantly straight and parallel, on a hyaline and unpigmented costal area. Whether this character alone can justify a whole genus is questionable, especially when similar within genus variation have been shown to occur in other groups, as in *Macromantis* (Roy 2002b) and, outside Photinaidae, in *Parastagmatoptera* (Mantidae: Stagmatopterinae) (Lombardo and Umbriaco 2015). Having denser reticulation on a more pigmented costal area could also be a functional, adaptive response to specific environmental selective pressures acting on some species (Dudley 1999). Finally, the conformation of male genitalia in *Metriomantis*, *Photina* and *Photinella* do show comparable morphology, with the ventral phallomere with a well-sclerotized and strongly recurved lateral process, and a relatively elongated and bifurcated apofisi phalloid — except in *Photina vitrea*, where the latter it is short and unforked (Agudelo 2014). Variation observed in male genitalia thus cannot be ruled out as representing intrageneric variation.

Summarizing, the available evidence justifying the taxonomic distinction of *Metriomantis*, *Photina* and *Photinella* is insufficient. Therefore, I choose following a more conservative approach and consider all species involved to be included under a more broadly defined genus *Photina*, the oldest name. A more comprehensive phylogenetic study, including wider taxon sampling, would be necessary to gain a better understanding of character distribution and the significance of their subtle variability for the taxonomy of the group.

Species checklist

Photina amazonica (Werner, 1928) (**comb. n.**) [BR (AM?)] – **Fig. 3.37f**

Parastagmatoptera amazonica Werner, 1928*

- Photina amplipennis* Stål, 1877 [distribution unknown]
Photina (Metriomantis) amplipennis Stål, 1877
- Photina brevis* Rehn, 1907* [*Photinella*] [PY] – **Fig. 3.38a**
- Photina boliviana* (Lombardo, 1999) (**comb. n.**) [BO, PE]
Metriomantis boliviana Lombardo, 1999
- Photina cupido* (Saussure, 1869) (**comb. n.**) [unknown distribution] – **Fig. 3.38b**
Cardioptera cupido Saussure, 1869* [*Metriomantis*]
- Photina magna* (Giglio-Tos, 1915) (**comb. n.**) [PY]
Photinella magna Giglio-Tos, 1915
- Photina media* (Giglio-Tos, 1915) (**comb. n.**) [BR (PR), PY]
Photinella media Giglio-Tos, 1915
- Photina occidentalis* (Lombardo, 1999) (**comb. n.**) [EC, PE]
Metriomantis occidentalis Saussure and Zehntner, 1894
- Photina ovata* (Saussure and Zehntner, 1894) (**comb. n.**) [FG] – **Fig. 3.38c**
Metriomantis ovata Saussure and Zehntner, 1894*
- Photina paraensis* (Giglio-Tos, 1915) (**comb. n.**) [BR (PA)] – **Fig. 3.38d–e**
Metriomantis paraensis Giglio-Tos, 1915*
- Photina pilosa* Chopard, 1911* [FG, VE] – **Fig. 3.38f**
- Photina pilosella* (Giglio-Tos, 1915) (**comb. n.**) [FG, GY, VE] – **Fig. 3.39a–b**
Metriomantis pilosella Giglio-Tos, 1915*
- Photina reticulata* (Burmeister, 1838) [BR (ES)] – **Fig. 3.39c–d**
Mantis (Cardioptera) reticulata Burmeister, 1838*
 =*Cardioptera burmeisteri* Westwood, 1889
 =*Metriomantis biramosa* Saussure and Zehntner, 1894* [*Photinella*] (**syn. n.**)
- Photina vitrea* (Burmeister, 1838) [AR, BR (PR, RJ, RS, SC)] – **Fig. 3.39e**
Mantis (Photina) vitrea Burmeister, 1838*
 =*Mantis pellucida* Audinet Serville, 1839
 =*Mantis longicornis* Charpentier, 1841*
 =*Cardioptera translucida* Saussure, 1869
 =*Photina laevis* Giglio-Tos, 1915*
 =*Photina gracilis* Giglio-Tos, 1915*
 =*Tithrone catharinensis* Piza, 1961*

Taxonomic notes.

The taxonomic and nomenclatural history of several of the species above listed remain exceedingly confusing. One of the most complex situations involves *P. vitrea*, *P. amplipennis* and, to a lesser extent, *P. reticulata*.

Saussure (1871b) presented, on the basis of separated sexes, a redescription of two species previously described by Burmeister (1838): *Mantis (Photina) vitrea* (males) and *Mantis (Cardioptera) reticulata* (females). Saussure (1871b) reported *vitrea* for both Brazil and Guyana but without attributing specific localities to or depository. Stål (1877) accepted Saussure's (1871b) redescription of the male of *vitrea* but he (correctly) considered that the redescription of the female did not correspond to this species. Accordingly, Stål (1877) formulated the new name *amplipennis* to accommodate the female — no type was designated. Giglio-Tos (1927) described the presumed male of *P. amplipennis*, extending its distribution to Bolivia. Beier (1935b) later cited this species as *P. (Metriomantis) amplipennis*, a subgeneric distinction that Terra (1995) later followed. Lombardo (1999) granted *Metriomantis* with genus status and arrived to two main conclusions about *amplipennis*: 1) *Metriomantis* and *Photina* had, respectively 6 and 5 posteroventral spines in the forefemora, and thus *amplipennis* sensu Stål (1877) was a member of the latter; 2) the presumed male of *P. amplipennis* that Giglio-Tos (1927) was most likely a member of *Photinella*.

As a result of all of the above, two nomenclatural scenarios for *amplipennis* are thus possible. If Saussure (1871b) mistakenly assigned the female of an undescribed species to *C. vitrea*, then Stål (1877) was correct in assigning the new name *amplipennis* to these females and thus the name is available under ICZN (1999) Art. 12.2.1 (for names published before 1931): “*the proposal [i.e. Stål (1877)] of a new genus-group name or of a new species-group name [i.e. amplipennis] in association with an illustration of the taxon being named, or with a bibliographic reference to such an illustration [i.e. Saussure, 1871b, Plate I, Fig. 3] even if the illustration is contained in a work published before 1758...*”. However, if at the time Saussure (1871b) unknowingly assigned the new name *amplipennis* to the female of a species already named, then *amplipennis* will turn out to be a synonym of the earliest name. Stål (1877) did not designate a type specimen for *amplipennis* but in agreement to ICZN (1999) Art. 72.4.2. “*If a*

new nominal species-group taxon is based, in whole or in part, on a published misidentification by an earlier author, the type series consists of or includes the specimen or specimens which had been misidentified, whether the later author refers to them directly or through an illustration or a description". Saussure (1871b) did not specify what specimens he used to redescribe and illustrate the female he presumed to be *vitrea*. This material could not be tracked for this study. Stål (1877) mentioned that he studied material provided by Saussure from his collection in Geneva. However, the name *amplipennis* is not listed in the Mantodea type collection of the MNHG (Roy and Cuche 2008, Pfauti and Hollier 2015). Stål (1877) did mention that he examined specimens of *amplipennis* from "Brussels", but it is unknown whether he contrasted these specimens to those Saussure believed to be the female of *vitrea*. The distribution of *P. amplipennis* is also another contentious issue that adds to the confusion. Lombardo (1999) stated that Stål (1877) established this name on the basis of the Brazilian material reported by Saussure's (1871b). However, Saussure's (1871b) did not specify a Brazilian origin for the female of what he considered *vitrea*, and Stål (1877) further stated that the locality of *amplipennis* was unknown ("*patria ignota*"). Therefore, the Brazilian origin of *amplipennis* that Lombardo (1999) suggested is controversial, because there is no such precision in the historical literature, and it is not clear whether Lombardo examined Saussure's specimens for locality validation. Although it would be important to verify the validity of Stål's name *amplipennis*, resolving the complex taxonomic and nomenclatural history associated to this name is clearly beyond the scope of this study. However, I suggest that one first step towards its resolution would be finding and re-examining of the female specimens that Saussure (1871b) believed to be the female of *vitrea*, and/or a correct taxonomic identification of specimens on the basis of Saussure's (1871b) description.

Whereas the identity of *Photina vitrea* has been recently well established (Agudelo 2014), and the validity of the name *amplipennis* is debatable, the identity of Burmeister's *Mantis* (*Cardioptera*) *reticulata* now seems better defined. Saussure (1871b) and Stål (1877) presumed that *Mantis* (*Cardioptera*) *reticulata* represented the female of *Mantis* (*Photina*) *vitrea* (discussed above). Recent studies (Agudelo 2014, Agudelo and Rivera, 2015), in addition to my own analysis of Burmeister's types and other material from museum collections, confirmed that these types are not conspecific. The female of *Photina vitrea* exhibit a comparatively narrower

pronotum, a more elongated prozona and shorter wings (see Agudelo 2014, Agudelo and Rivera 2015), whereas the type of *P. reticulata* exhibits a relatively wider and a more broadly rounded prozona, and longer wings (Fig. 3.39c). Furthermore, the analysis of the type of *Metriomantis biramosa* Saussure and Zehntner, 1894 (Fig. 3.39d) was found to be conspecific with *P. reticulata*, and thus I also establish this new synonym.

In their revision of *Parastagmatoptera* (Mantidae: Stagmatopterinae), Lombardo *et al.* (2015) suggested that *P. amazonica* should be transferred to *Metriomantis*. I examined the female holotype of this species and corroborated Lombardo *et al.*'s (2015) observations. Although the validity of this species could not be assessed at this time, I nevertheless choose to formally transfer this species to *Photina*.

I also examined the presumed type of *Metriomantis ovata* at MHNG. Roy and Cuche (2008) found that this unique specimen does not conform to the measurements Saussure and Zehntner (1894) provided in the original description, although label data does coincide. It is unknown whether the original type series consisted of several syntypes or this unique, possibly holotype specimen.

Giglio-Tos (1915a) described *M. pilosella* on the basis of a unique male holotype from “Guiana Inglese”, i.e. Guyana. Giglio-Tos stated that the “Imperial Boureau of Entomology” was the owner of this specimen. Originally with headquarters at the British Museum of Natural History, the Imperial Boureau of Entomology was a British organization in commad of conducting entomological surveys across British colonies (Blight, 2011). Marshall (1975) reported that the holotype of *M. pilosella* was not deposited in the BM collection, whereas Ehrmann (2002) listed two “holotypes” of *this species*: one at BM and another at ZMB. I examined the ZMB specimen mentioned in Ehrmann (2002) and was able to corroborate its identity as the actual holotype of *M. pilosella*. However, I was unable to locate the presumed BM “holotype” Marshall (1975) reported as missing. I believe that such specimen probably never existed. The confusion likely originated by the way Giglio-Tos cited the owner of the specimen: “Imp. Bureau of Entomol. Londra”, thus giving the impression that the specimen was deposited at BM, as most collections conducted by the Bureau were (Blight, 2011). It is unknown how this specimen ended up at ZMB, but it is likely that Giglio-Tos took it for examination and mistakenly deposited there.

Finally, the female type specimens of *Metriomantis ovata* from Cayenne (French Guiana) and *M. paraensis* from Pará (Brazil), herein treated as valid species, must be ruled out as conspecific with *Photina pilosella*.

3.4.6 Family Stenophyllidae Saussure, 1869

Granted family-rank by Rivera and Svenson (2016), the Stenophyllidae constitutes a bizarre-looking group of little known, dead-leaf mimic praying mantises. Lombardo (2000c) and Roy (2005) both treated the only three known species of *Stenophylla*. Key to species in Roy (2005).

Stenophylla Westwood, 1843

Stenophylla Westwood, 1843: 52. Type species: *Stenophylla cornigera* Westwood, 1843 (by monotypy).

Species checklist

Stenophylla cornigera Westwood, 1843* [BR (ES, PR, RJ, SC, SP)] – **Fig. 3.40a**

Stenophylla lobivertex Lombardo 2000 [EC, PE]

Stenophylla gallardi Roy, 2005* [FG] – **Fig. 3.40b**

3.4.7 Family Acontistidae Giglio-Tos, 1915

Granted family status by Rivera and Svenson (2016), the Acontistidae includes seven genera (including one herein revalidated). Modern taxonomic treatments are available for *Tithrone* and *Paratithrone* (Lombardo 1996), and *Raptrix* (Lombardo and Marletta 2004). Roy (2006) presented a detailed historical account of the taxonomy and nomenclature of all genera Acontistidae then known, whereas Agudelo and Rivera (2015) clarified the identity of a number of species historically attributed to *Tithrone* that were not treated in Lombardo (1996). The latest addition to this group was *Ovalimantis* (Roy 2015). The remaining genera are in need of review. The identity of the genus *Astollia* Kirby, often attributed to this lineage, remains unknown; see Roy (2006) for an overview of this genus.

Key to genera (both sexes); adapted from Roy (2006)

1. Mid- and hindfemora ventrally with a small, pre-apical lobe (Fig. 3.12c).....*Callibia*
- 1'. Mid- and hindfemora without lobes.....2
2. Vertex with a median, short, bifid projection flanked by one spine-like projection on each side; metazona of pronotum shorter than prozona.....*Ovalimantis*
- 2'. Vertex smooth or at most with a median bump (Fig. 3.12k-l); metazona of pronotum longer than prozone.....3
3. Pronotum with supracoxal bulges scarcely pronounced or null (Fig. 3.12s-t); posteroventral foretibial spines erect and separated (i.e. there is a clear gap between any two contiguous spines).....4
- 3'. Pronotum with distinct and rounded supracoxal bulges (e.g. Figs. 3.11c, 3.11m, 3.11s, 3.12g, 3.12p); posteroventral foretibial spines largely reclined and tightly arranged (i.e. there is no visible gap between any two contiguous spines).....5
4. Discoidal area of forewings exhibiting chromatic asymmetry: one forewing is green, while the other is pink.....*Tithrone*
- 4'. Discoidal area of forewings similarly coloured in brown.....*Paratithrone*
5. Body colouration predominantly brown. Wings of males: hyaline, mostly colorless, often with a few scattered brown marks. Wings of females: largely brown and opaque, except for a contrasting network of paler crossveins on the anal area of the hindwings.....*Raptrix*
- 5'. Body colouration predominantly green. Wings in both sexes often with opaque to semi-opaque regions that combine tones of green, yellow, red, orange and/or brown, the latter presenting purplish reflections in at least one species; overall, wings are rarely predominantly hyaline and colorless, if so, only in males.....6
6. Ratio metazona/prozone 1.5 or less. Males: distal process of ventral phallomere accompanied by a large, sharp, spine-like process. Females: wings reduced, about as long as pronotum. Distributed in seasonal habitats within the Paraná River basin (northeastern Argentina, Paraguay, Uruguay and southern Brazil) and adjacent areas of southern Bolivia.....*Metaphotina*

6'. Ratio metazona/prozone 1.6 or more. Males: ventral phallomere with a single distal process, if accompanied by a secondary process, the same is not spine-like. Females: wings longer, about as long as abdomen. Distributed mostly in humid habitats from Central Mexico, through Central America, south into the Amazon basin, and Atlantic rainforest of Southern Brazil; found also in the Lesser Antilles.....**Acontista**

Acontista Saussure, 1869 (Fig. 3.11a–s)

Acontistes Burmeister, 1838: 542 (pre-occupied: Sundevall, 1836 - Aves).

Acontista Saussure and Zehntner, 1894: 130 (emend.). Type species: *Mantis concinna* Perty, 1833 (subsequent designation: Rehn, 1904).

=*Acontiothespis* Rehn, 1916: 258 (injustified *nomen novum* pro *Acontistes sensu* Burmeister, 1838).

=*Acontistella* Beier, 1929: 246. Type species: *Acontistella amazonica* Beier, 1929 (by monotypy).

Distribution. Distributed mostly in humid tropical and subtropical habitats, from Central Mexico, through Central America, south into the Amazon basin, and Atlantic rainforest of Southern Brazil. A few records also indicate a distribution in the Lesser Antilles.

Diagnosis. *Size range:* 16–29 mm.

Body colouration: usually completely green, sometimes with a more yellowish colouration with contrasting green to brownish spots, legs may also exhibit a contrasting banded pattern.

Abdomen normally green, often with contrasting bands and even reddish pigmentation dorsally.

Head: vertex with well-marked sutures, without central projections or well-delimited protuberances (Fig. 3.11a–b, 3.11g–h, 3.11 n). Lower frons flat, sometimes with a median circular depression (the latter is more conspicuous in some females).

Thorax: pronotum variable (Fig. 3.11c–f, 3.11 i–m, 3.11o–s), from long and slender, as in *Acontista amoenula* (Fig. 3.11a–b), to short and stout as in *Acontista concinna* (Figs. 3.11e–f); surface and lateral edges always smooth, supracoxal bulges conspicuous, protruding and rounded; metazona often medially narrowed, its lateral edges sometimes with a conspicuous, median undulation. Spination formula: $F=3DS/11-13AvS/5PvS$; $T=11-16AvS/12-21PvS$. Males: forewings ranging from completely colorless and hyaline, to having some darker spots, green patches of various sizes, or even completely pigmented, sometimes with purplish reflections; hindwings ranging from completely colorless and hyaline, to having darker spots and bands of various sizes (from a single small dot to an entire infusate wing) and/or with patches of crimson or orange. Female: forewings usually opaque (colorless and hyaline fenestrations may occur), entirely or partially pigmented in green and sometimes with contrasting transversal bands in yellow; hindwings distally darkly pigmented and usually with crimson or orange pigmentation along the costal area, and at least in part of the discoidal and anal areas.

Abdomen: slender in males and dilated in females. Distal process of ventral phallomere variable in shape but simple, i.e. not accompanied by a large, hook-like process (See Rivera and Svenson, 2016).

Remarks. *Acontista* includes small, forest dwelling species whose members often exhibit conspicuous sexual dimorphism, especially in regards of wing colouration. Intraspecific, non-sexual variation further complicates the already-difficult task of associating conspecific sexes in a comparative morphology context. Several synonyms have resulted from the historical misinterpretation of colour patterns in otherwise conspecific individuals. A clear example of this is *A. cordillerae*, a widespread species with broad colour variation for which at least 6 other names apply (see details below). Beier (1929) used the wide variation in body proportions and colouration existing across *Acontista* to justify the creation of *Acontistella*, genus where he accommodated his new species: *A. amazonica* Beier, 1929. Beier (1929) assumed that body proportions (e.g. elongated pronotum) and colour features (e.g. darkened wings) he observed in the single male holotype he studied, were distinct enough to warrant its own generic placement, even though other species with similar features (e.g. *Acontista amoenula* Gerstaecker, 1889; *Acontista gracilis* Chopard, 1911; *Acontiothespis ecuadorica* Hebard, 1924) were already in

place within *Acontista* (or, at times, in *Acontiothespis*; see above for a list of synonyms under *Acontista*). Later on, Beier (1931) added another species with similar features, *A. violacea* Beier, 1931, also on the basis of two male specimens. Terra (1995) transferred both *A. amazonica* and *A. violacea* to *Acontista* with *Acontistella* as its synonym, although he did not provide any justification for his procedure. Because of the wide inter- and intraspecific morphological variation observed in *Acontista* spp., and the fact that *Acontistella* was established only on the basis of (no less-variable) male features, I consider that the character system suggested by Beier (1929) is inadequate for diagnosis at the genus level, and thus cannot support the distinctiveness of *Acontistella*. Therefore, I opt to accept Terra's (1995) procedure and retain *Acontistella* as synonym of *Acontista*. Despite its limitations, colour can sometimes be useful as a diagnostic character of species with unique colour particularities (e.g. *A. fraterna*). Characters with important amount of variation but much more useful to distinguish among species include head shape, relative proportions of prothoracic structures, and male genitalia.

Recent catalogues and checklists cited either 25 (Ehrmann, 2002; Agudelo *et al.*, 2007) or 26 (Otte and Spearmann, 2005) species of *Acontista*, whereas the total number of names assigned to this taxon ascends to 33. I examined type material representative of sixteen of these names. As a result of these observations, I have been able to reduce the number of valid species down to eleven. Interestingly, the great majority of synonyms listed below were already suggested or *de facto* established in the past; nevertheless, subsequent authors have ignored some. Some of these procedures taken with anteriority are herein confirmed, and only 4 new synonyms are newly introduced. See Roy (2006) for a more detailed historical account on all names under *Acontista*.

Overall, a critical assessment of the taxonomic value of sexual and non-sexual intraspecific variation (including male genitalia), and its connection to the taxonomic diversity of this lineage, are necessary.

Species checklist

Acontista amazonica (Beier, 1929) [BR (AM)] – **Fig. 3.41c**

Acontistella amazonica Beier, 1929*

Acontista amoenula Gerstaecker, 1889* [EC, BR (AM), PE, SU] – **Fig. 3.41d–e**

=*Acontista semirufa* Westwood, 1889*

Acontista cayennensis Saussure and Zehntner, 1894* [FG] – **Fig. 3.41f, 3.41a–c**

=*Acontista maroniensis* Chopard, 1911* (**syn. n.**)

Acontista concinna (Perty, 1833) [BR (MG, PR, RJ, SC, SP), UR] – **Fig. 3.41d**

Mantis concinna Perty, 1833*

=*Mantis tricolor* Burmeister, 1838

=*Mantis aurantica* Burmeister, 1838

=*Mantis quadrimaculata* Audinet Serville, 1839

=*Acontista bolivarii* Giglio-Tos, 1915

Acontista cordillerae Saussure, 1869* [CO, CR, EC, ES, GU, NI, MX, PN] – **Fig. 3.41e–g, 3.42a–f**

=*Acontista inquinata* Saussure and Zehntner, 1894*

=*Acontista mexicana* Saussure and Zehntner, 1894*

=*Acontista mexicana* var. *quadrimaculata* Saussure and Zehntner, 1894*

=*Acontista vitrea* Saussure and Zehntner, 1894*

=*Acontista championi* Kirby, 1904*

=*Acontiothespis iriodes* Hebard 1919* (**syn. n.**)

Acontista eximia (Pascoe, 1882) [BR (PA)] – **Fig. 3.42g**

Callimantis eximia Pascoe, 1882

=*Acontista paraensis* Saussure and Zehntner, 1894*

Acontista festae Giglio-Tos, 1915 [EC]

Acontista fraterna Saussure and Zehntner, 1894* [CR, EC] – **Fig. 3.43a–e**

=*Acontiothespis ecuadorica* Hebard, 1924* (**syn n.**)

=*Acontistella violacea* Beier, 1931*

Acontista gracilis Chopard, 1911* [FG] – **Fig. 3.43f–g**

=*Acontista chopardi* Giglio-Tos, 1927 (**syn n.**)

Acontista minima Giglio-Tos, 1915* [CO] – **Fig. 3.44a**

Acontista multicolor Saussure, 1870* [GA, VE, TT] – **Fig. 3.44b**

Taxonomic notes. My analyses of type material of most species attributed to *Acontista* allowed me to draw several conclusions about their taxonomic status. These are summarized as follows:

***Acontista cayennensis* Saussure and Zehntner, 1894.** This species was described on the basis of a single female specimen collected from Cayenne, French Guiana. Later, Chopard (1911) described *A. maroniensis* from a type series consisting of one male and three females from various localities along northern French Guiana. In his description, Chopard (1911) stated that his new species resembled “*A. guianensis* Sauss. et Zehnt” — a *lapsus* in reference to *A. cayennensis* (see Roy, 2006). Chopard pointed out the similarities between these two species, but justified their distinctiveness because, unlike *cayennensis*, *maroniensis* exhibited, in the mesothoracic wings, a brown spot under the stigma, and a dilated abdomen. A darkened spot does occur right below the stigma (a swollen, calloused area usually found between the RP+M and CuA veins; see Wieland 2013 for a detailed description) in *A. maroniensis*. This dark spot is a distinct feature of those *Acontista* species whose females exhibit banded mesothoracic wings (i.e. yellowish bands on a green background, sometimes with a distal hyaline area and more or less defined smoky brownish areas). Examples of this character state are *A. cordillerae*, *A. fraterna*, and *A. multicolor*; the same darkened spot is also present in males. Chopard considered the presence of a dark spot below the stigma as a character of diagnostic value to distinguish his species from *cayennensis*. However, this condition alone cannot be considered reliable because of intraspecific variation. Material of *A. cordillerae* examined in this study reveals that this character exhibits various levels of pigmentation, from absent, to presence of a minute speck, to an extensively-pigmented and highly contrasting spot that extends onto the stigma, which is otherwise pale green or yellowish. Variation in the degree of pigmentation also exists in the type series of *A. maroniensis*. The holotype of *cayennensis* exhibits a very weak spot below the stigma (although somehow more visible in the right mesothoracic wing). I consider the brown spot in the stigma of little diagnostic importance relative, for example, to pronotal proportions. Similarly, a dilated abdomen, a character also considered by Chopard (1911) to be of diagnostic value in *maroniensis*, is also highly variable and cannot be considered reliable for species delimitation. For example, several factors may influence the state of preservation and general appearance of the abdomen (e.g., gut contents, number of fertilized eggs). Finally, the type localities of *cayennensis* and *maroniensis* are within ca. 200 kilometers of each other, support a similarly structured rainforest, and the range of colour expressed in *A. maroniensis* falls within the range of variation exhibited by *A. cayennensis*. Consequently, I consider *A. maroniensis* a new synonym of *A. cayennensis*.

***Acontista concinna* (Perty, 1833)** (Fig. 3.11e–f). Considerable confusion persists about the identity of this species, likely the result of historical misidentification. The type locality was reported simply as “Rio Negro”, Brazil; however, there are many such named geographical areas in that country. Label data of specimens I identified as *A. concinna* suggest that this species is restricted to southern-most Brazil and adjacent areas, where it occurs rather commonly. Therefore, Rio Negro likely refers to a yet undetermined locality within southern Brazil, perhaps in either Paraná or Mato Grosso states. However, literature reports of *A. concinna* include localities throughout much of South America, from southern Brazil north to Colombia and French Guiana (cf. Agudelo *et al.* 2007). Records of *A. concinna* Ecuador (Lombardo and Agabiti 2001) do not correspond with this species, as confirmed by my examination of the male genitalia and previously published work (Heitzmann-Fontenelle 1965). Similarly, I have been unable to confirm the presence of this species in Peru, despite extensive prospections in that country over the last decade (Personal Observation). The distribution of *A. aurantica* and *A. bolivarii* — both currently considered junior synonyms of *A. concinna* — may account for why the latter species is assumed to occur in the Amazon Rainforest (Giglio-Tos 1927). Roy (2006) provided a summary of all the names currently in synonymy with *A. concinna*, and it is conceivable that some of these may eventually proved to be valid.

***Acontista cordillerae* Saussure, 1869** (Fig. 3.11g–m). The synonymys of Hebard (1919, 1932) and Rehn (1935) are upheld in the current study. Hebard (1924) recognized two subspecies: *A. cordillerae cordillerae* (distributed mainly in Mexico) and *A. cordillerae vitrea* (distributed mainly in Central America and northern South America). Rehn (1935) provided a rather detailed analysis of colour variation in this species supporting Hebard’s views. However, I consider such subdivision is invalid because it is supported by slight morphological and colour disparity, otherwise characters of little interspecific value. For instance, body colouration in males is normally entirely green, although the tips of both cursorial femora and adjacent part of the corresponding tibiae are almost always darkened; some individuals often have darkened head and forelegs. The forewings of males are frequently hyaline and colorless — except for the green, and fully or partially opaque costal area and darkly pigmented stigma — whereas hindwings show extensive variation, from hyaline and colourless to darkly pigmented, often with proximal reddish areas on or near the anal and costal areas. Females exhibit less variation in

wing colouration and patterning. I have examined the genitalia of several males ranging from Mexico, Costa Rica, Ecuador and French Guiana and could not determine any consistent difference to justify the subspecific subdivision of *cordillerae*. In fact, the discrepancies observed in the colouration of wings across all species considered synonyms of *cordillerae* exemplifies the range of variation observed across examined material. Therefore, I consider the observed disparities as intraspecific, and thus I opt for a more conservative approach and simply apply the name *A. cordillerae* in its most restricted sense, to this species. I also examined the holotype of *A. iriodes* (Hebard, 1919). This specimen (ANSP) resembles *A. cordillerae* in every respect, except for its lighter colouration and unpigmented limbs, although the forewing stigma is dark, as usual for *A. cordillerae*. I examined additional specimens (six males and four females) referable to *A. iriodes* and collected in the vicinity of its type locality (Santa Marta, Colombia). This series contains both pale and dark males, the latter with the distal black marks on the cursorial femora, typical of *A. cordillerae*. Consequently, I also consider *A. iriodes* as new junior synonym of *A. cordillerae*.

***Acontista eximia* Pascoe (1882).** Originally assigned to *Callimantis*, *A. eximia* was described from Nazaré (now a district of the city of Belém) in Pará, Brazil. Saussure and Zehntner (1894) described *Acontista paraensis* from “Pará”. According to Rehn (1918), Kirby (1904) examined the type specimens of *Callimantis eximia* and considered it a synonym of *A. paraensis*. Rehn (1918) accepted Kirby’s procedure and considered *Acontista eximia* as the valid name for this species, providing also a brief description of what he considered to be the male. I contrasted the original description of both *eximia* and *paraensis* against the female type specimen of the latter (the type of *eximia* could not be located at the BM), in addition to the specimens reported by Rehn (1918), and one additional male from Benevides, a locality just 30 Km. east of Belém, the type locality of *eximia*. I concluded that *eximia* is indeed a valid species and *paraensis* is its synonym, as Kirby (1904) originally proposed. This species exhibits long, golden pilosity on body (particularly denser in the pronotum) and limbs, a distinct character state also observed in the type of *A. paraensis*.

***Acontista fraterna* Saussure and Zehntner, 1894** (Fig. 3.11n–p). This is one of the most distinctive species of *Acontista*. The males have a rather slender and elongated pronotum,

completely infusate forewings (except for a colourless, hyaline pre-apical spot), blackish purple, metallic reflections on the hindwings, and banded fore-, mid- and hindlegs. Females superficially resemble *A. cordillerae* but the pronotum is much narrower, the metazona is medially constricted and elongated, and hindwings with a different colour pattern. Rehn (1935) established *Acontistella violacea* as synonym of this species, procedure I concur with after my own examination of the types. I also examined the male holotype and female allotype of *Acontiothespis ecuadorica* Hebard, 1924. Except for their smaller body size, *A. ecuadorica* resembles *A. fraterna* in all respects, i.e., wing colour pattern, pronotal shape and proportions, and males with banded legs. Male genitalia showed only slight variation in the apical process of the ventral phallomere, differences I regard merely as intraspecific variation, as observed in other *Acontista* species. Consequently, I here consider *A. ecuadorica* a new synonym of *A. fraterna*. Lombardo and Agabiti (2001) illustrated the genitalia of *A. fraterna* (cited as *A. ecuadorica*).

***Acontista gracilis* Chopard, 1911/ *A. chopardi* Giglio-Tos, 1927.** Roy (2006) stated that *A. chopardi* is an unnecessary replacement name for *A. gracilis*. Giglio-Tos (1927) apparently considered the male and female of the original type series of *A. gracilis* (syntypes) as non-conspecifics, despite their obvious similarities. Giglio-Tos (1927) wrongly assigned the female of *gracilis* to *A. cordillerae*, formulating a new replacement name for the male, which became the holotype. Roy (2006) further reported a male and female deposited at MNHN caught in copula and whose morphological characters correspond to those diagnostic for *A. gracilis*. Accordingly, I here follow Roy's (2006) suggestion and formally establish *Acontista chopardi* as a new objective synonym of *Acontista gracilis*.

***Acontista minima* Giglio-Tos, 1915.** This species, known only from its male holotype, superficially resembles the also relatively small males of *A. multicolor*. Future studies must rule out their possible synonymy.

One more species warrants a separated mention. *Acontista cubana* de Zayas, 1974. This species, described in the third volume of the series "Entomofauna Cubana" by de Zayas (1974), is unlikely to belong to *Acontista*. The original description refers to characters of both sexes, but de Zayas did not mention number of specimens, nor type designation or depository. The gender of the figured specimen is stated to be female; however, the length of the antennae and the

conspicuous styli revealed it as a male instead. Character states observed in this figure cast doubts about its generic identity. For instance, *A. cubana* exhibits small denticles along the pronotal edges, a character not seen in *Acontista*. My detailed analysis of the original description indicates that this species is referable instead to *Callimantis*, especially in regards to the colour pattern of the hindwings. *Callimantis* is a monotypic genus represented by *C. antillarum* (Saussure, 1859), a widespread species found in the Caribbean region (Ehrmann 2002). Interestingly, de Zayas (1974) himself considered his new species as related to “*A[contista]. antillana* Sauss.” clearly in allusion to *Mantis antillarum* Saussure, 1859, name under which *C. antillarum* was originally described. Saussure (1859) himself considered his *Mantis antillarum* close to *Mantis concinna* (i.e. *Acontista concinna*). This interpretation likely influenced de Zayas (1974) (not a praying mantis specialist) at the time of classifying its new species, prompting him to place it under *Acontista*. The type locality of *A. cubana* (Baracoa), is located in the far east part of Cuba, which lays less than 100 kilometers from the west end of Hispaniola island (comprising both Haiti and Dominican Republic). Interestingly, Saussure (1870) already recorded *C. antillarum* for Cuba, but as *Iris antillarum*. Ehrmann (2002) reported *Callimantis antillarum* from Haiti, Puerto Rico, as well as Mona, Culebras and Virgin Islands, all within the Greater Antilles. I have examined specimens of *Callimantis antillarum* from all these localities, thus confirming Ehrmann’s accounts. The lack of records of other *Acontista* species in the Greater Antilles also strongly suggests that *A. cubana* is a member of *Callimantis*. Whether *cubana* is a good species, or a synonym of *C. antillarum*, cannot be determined at this time, as the type series was not available for study. Therefore, I opt for a more conservative approach and I simply refer to this species as *Callimantis cubana* (de Zayas, 1974) **comb. n.** This question will remain open until de Zayas’ specimens can be located or topotypic material of *A. cubana* can be analyzed and compared to *C. antillarum*.

***Callibia* Stål, 1877 (Fig. 3.12a-c)**

Callibia Stål, 1877: 79. Type species: *Harpax pictipennis* Audinet Serville, 1839 = *Mantis diana* Stoll, 1813 (by monotypy).

=*Anastira* Gerstäcker, 1883: 97. Type species: *Mantis diana* Stoll, 1813 (by monotypy).

Distribution. Northern South America, south to northern Bolivia, central and eastern Brazil.

Diagnosis. *Size range:* 18–31 mm.

Body colouration: females predominantly yellow, with green markings on thorax, abdomen and legs; males mostly dark brown, with light brown markings on thorax, abdomen and legs.

Head: Vertex with a large, conical and bifid, central projection (Fig. 3.12a–b). Lower frons with a central, circular depression.

Thorax: pronotum smooth, prozona protuberant (on lateral view) and supracoxal bulges broad and well marked. Spination formula: F=5DS/12–13AvS/5PvS; T=15–20AvS/16PvS. Forecoxae longer than metazona. Femoral posteroventral spines siting on socket-like extensions of the cuticle. Mid- and hind femora with a small, distal lobe ventrally (Fig. 3.12c). Male: meso- and metathoracic wings strongly iridescent, hyaline and with golden reflections (except for the opaque green costal area of forewings). Female: forewings with a characteristic yellow background crossed by three large, irregular, green bands (pre-apical, central and proximal); discoidal area of hindwings with a red and black, pre-apical mark, distal portion of discoidal field and most of the anal field darkened, with contrasting white veins.

Abdomen: dilated and with lateral projections on each tergite in females, slender and simple in males.

Species checklist

Callibia diana (Stoll, 1813) [BO, BR (AM, PE), CO, EC, FG, PE, VE]

Mantis diana Stoll, 1813

=*Harpax pictipennis* Audinet Serville, 1839

=*Oxypilus? vigilax* Westwood, 1889

Taxonomic notes. *Callibia diana* was originally described by Stoll (1813) from the “East Indies” and under the genus *Mantis*; no type specimen was designated. Later on, Audinet Serville (1839) described *Harpax pictipennis* from Cayenne (French Guiana). Stål (1877) noticed the generic misplacement of *pictipennis* and thus established *Callibia* to accommodate Audinet Serville’s species, thus becoming *Callibia pictipennis*. After studying specimens from Ega and Pernambuco (Brazil), Gerstaecker (1883) recognized that *Harpax pictipennis* and *Mantis diana* both referred to the same species, thus giving priority to *diana*, the oldest name. However, Gerstaecker (1883) ignored the procedure of Stål (1877) and established yet another genus, *Anastira*, to accommodate *Mantis diana*. Saussure (1898) ultimately brought all together and, having *Callibia* priority over *Anastira*, he considered *Callibia diana* as the valid name, with *Harpax pictipennis* as its synonym. Later on, Giglio-Tos (1927) added *Oxypilus? vigilax* Westwood, 1889 (also from Ega, Brazil) to the synonymic list of *C. diana*.

I examined one female specimen at ZMB bearing the following three labels: [1] “EGA, BRAZIL” (green label, handwritten); [2] “TYPUS” (read label, printed); [3] “CALLIBIA (ANASTIRA) DIANA SERV.” (white label, handwritten). It is likely that this is one of the specimens Gerstaecker (1883) used to justify the distinctiveness of *Anastira* (ibíd., p. 96). Gerstaecker (1883) himself identified the ZMB as depository of this and three other specimens he studied at the time of establishing his genus *Anastira*; these other three additional specimens were not found at the ZMB. Summarizing, despite of being labeled as such, the referred ZMB specimen cannot be considered to be a type specimen of any nominal species now synonyms of *Callibia diana*.

Metaphotina Piza, 1964 **stat. rev.** (Fig. 3.12d-j)

Metaphotina Piza, 1964: 125. Type species: *Metaphotina piracicabensis* Piza, 1964 (by monotypy).

Distribution. Found mainly in seasonal habitats within the Paraná River basin (northeastern Argentina, Paraguay, southern Brazil and Uruguay) and adjacent areas, including southern Bolivia. Recorded as far East as Northeastern Brazil (Rehn 1916).

Diagnosis. *Size range:* 13–23 mm.

Body colouration: rather variable, both when alive and preserved. According to Heitzmann-Fontenelle (1968a), living specimens vary from reddish black to bright green. Males are more commonly green, whereas females combine whitish areas with green marks across the entire body. Such variation in colouration is more or less evident in preserved specimens, where colour intensity fades considerably and patterning becomes diffuse, often perceived as pale yellow and/or greenish.

Head: vertex convex (Fig. 3.12d–e), sutures well defined. Antennae moniliform along its proximal 1/3, becoming filiform on its distal 2/3.

Thorax: pronotum short and stout (Fig. 3.12f–j), supracoxal bulges well marked. Lateral margins of pronotum expanded, forming a narrow ledge (particularly around the prozona). Prozona and distal 2/3 of metazona broad, forming an oval outline. Metazona 1.5x as long as prozona, the former with its proximal 1/3 slightly constricted, and its proximal end appearing elevated in the lateral view. Spination formula: F=3DS/11–13AvS/5PvS; T=11–14AvS/10–12PvS. Forecoxae longer than prozona. Males: wings of males with its apex extending beyond tip of abdomen; forewings hyaline and iridescent, often colourless but sometimes with a brownish tinge, appearing smoky; hindwings hyaline, mostly colorless but sometimes with a dark, circular spot of variable intensity in the middle of the anal area, sometimes expanding into the discoidal area, and with pale crossveins. Females: wings shortened, about as long as the pronotum; forewings oval, membrane with a marbled, greenish brown pattern over a whitish background, stigma and veins whitish, the latter with a tinge of green in some areas; hindwings short, costal and proximal portion of anal areas reddish orange, the latter becoming dark brown marginally and with pale crossveins. As with the rest of the body, wing colouration and patterning becomes considerably faint and diffuse in preserved specimens of both sexes.

Abdomen: cylindrical in males; ventral phallomere with distal process accompanied by a large, spine-like projection; broadly dilated and oval in females.

Remarks. Piza (1964) established *Metaphotina* to accommodate his new species *M. piracicabensis*, with type locality in Piracicaba, São Paulo, Brazil. Piza originally classified his

new genus and species among the Photinaidae. Terra (1995) recognized its wrong placement, transferred to Acontistinae, and considered it a synonym of *Acontista*. Rivera and Svenson (2016) included *Acontista bimaculata* in their phylogenetic analyses where it was found to be sister to *Raptrix*, thus rendering *Acontista* as paraphyletic. The phylogenetic evidence, in addition to various unique morphological and ecological features shared by members of this lineage (e.g. brachypterous females, distinct male genitalia, distribution in seasonal habitats), clearly set them apart from *Acontista*. Consequently, I remove *Metaphotina* from synonymy with *Acontista* and consider it as valid. The diagnosis above summarizes relevant characters from the original description, my own examination of type material and museum specimens — including those reported by Heitzmann-Fontenelle (1968a).

Species checklist (note: distributional records represent type locality only)

Metaphotina bimaculata (Saussure, 1870) (**comb. n.**) [BR (GO)] – **Fig. 3.44c–d**

Acontista bimaculata Saussure, 1870*

Metaphotina brevipennis (Saussure, 1872) (**comb. n.**) [BR]

Acontista brevipennis Saussure, 1872

Metaphotina piracicabensis Piza, 1964* [BR (RS)] – **Fig. 3.44e**

Metaphotina rehni (Giglio-Tos, 1927) (**comb. n.**) [PY] – **Fig. 3.44f**

Acontista rehni Giglio-Tos, 1927*

Taxonomic notes. Although morphological similarities shared by all four species clearly make them attributable to *Metaphotina*, I could not confirm the validity of all these names at this time. For instance, Rehn (1916) considered *A. brevipennis* and *A. bimaculata* as distinct species, remaining as such as for today. Rehn provided a number of anatomical differences to justify the valid status of both species, such as pronotal proportions, relative body size and presence/absence of the dark maculation of the hind wings of males, concluding that populations from northern Argentina and southern Bolivia corresponded to *bimaculata*, and those from Paraguay, and southern Brazil to *brevipennis*. I have examined specimens attributable to *Metaphotina* from virtually its entire distribution, ranging from southern Bolivia, Paraguay, Argentina and Brazil (BA, CE, ES, GO, MG, PR, RJ, RS, SP, SC) (see Appendix 1), and found extensive variation in both body metrics and degree of pigmentation of the dark maculations of

the hindwings of males (e.g. Heitzman-Fontenelle 1968). I was unable to find consistent patterns of variation associated to male genitalia, which showed only subtle disparity. Therefore, I could not corroborate Rehn's observations regarding *A. brevipennis* and *A. bimaculata*. Distributional records that followed original descriptions of these two species thus need verification.

Piza (1964) described *Metaphotina piracicabensis* on the basis of a single male specimen, which was associated to a female specimen labeled as the "allotype" of *M. piracicabensis* at ESALQ. However, this female was collected more than a decade after the description of *M. piracicabensis* and thus cannot be considered to be as part of the type series. Finally, *Acontista rehni* was a new name Giglio-Tos (1927) proposed to accommodate the female of what Rehn (1907) believed to be the female of *A. bimaculata*; however, Giglio-Tos's procedure was not justified and thus requires verification.

Ovalimantis Roy, 2015

Ovalimantis Roy, 2015: 143. Type species: *Ovalimantis maculata* Roy, 2015 (by monotypy).

Distribution. Known only from Montagnes des Chevaux (Chevaux Mountains), in French Guiana.

Diagnosis (males only; modified from Roy 2015). *Size range*: 38 mm.

Body colouration: predominantly brown, with multiple darker spots or areas across the body and limbs, especially ventrally.

Head: vertex with a median, short, bifid projection flanked by one spine-like projection on each side.

Thorax: pronotum remarkably shortened, ovoidal in shape; metazona marginally but conspicuously shorter (and wider) than prozona. Spination formula: F=3DS/11AvS/5PvS; T=11AvS/11PvS. Forecoxae longer than metazona. Males: wings hyaline and colorless, with

multiple brown spots along longitudinal veins and a large and darker preapical spot on the discoidal area of both wings.

Abdomen: cylindrical, supraanal plate distally with a short, bifid projection.

Remarks. *Ovalimantis* exhibits an unusual combination of characters not seen in other members of the family, such as the ornamentation of the head and the unusual shape and proportions of the pronotum. Despite its aberrant external morphology, the shape of male genital structures, as reported in Roy (2015), unquestionably places *Ovalimantis* among the Acontistidae.

Species checklist

Ovalimantis maculata Roy, 2015 [EG]

Paratithrone Lombardo, 1996

Paratithrone Lombardo, 1996: 47. Type species: *Paratithrone royi* Lombardo, 1996 (by monotypy).

Distribution. Restricted to northern South America.

Diagnosis. *Size range*: 21.5–36 mm.

Body colouration: thorax and abdomen predominantly dark brown, with multiple pale marks and spots, prothoracic coxae orange to turquoise green.

Head: vertex smooth without tubercles and slightly arched.

Thorax: Pronotum smooth, supracoxal bulges absent. Spination formula: F=3DS/14AvS/5PvS; T=12–13AvS/13–14PvS. Posteroventral tibial spines erected and separated from each other.

Males: forewings dark, reddish brown with two slightly darker, preapical, transversal bands; hindwings mostly bi-colored, proximal half yellow-colored, distal half dark brown, with a pale preapical circular spot on the discoidal area, whereas the costal area becomes reddish distally.

Wings reach tip of abdomen in the resting position. Females: colouration patterning as in males

but darker and more opaque, crossveins colorless, especially in the anal area; wings do not reach tip of abdomen in the resting position, terminal tergites exposed.

Abdomen: dilated in females, slender in males.

Species checklist

Paratithrone royi Lombardo, 1996* [FG, BR (AM)] – **Fig. 3.45a–b**

Raptrix Terra, 1995 (Fig. 3.12k–p)

Raptrix Terra, 1995: 25. Type species: *Mantis perspicua* Fabricius, 1787 (by original designation).

Distribution. Widely distributed across the Amazon basin and northwestern South America, from Colombia to Costa Rica.

Diagnosis. *Size range*: 20–30 mm.

Body colouration: light to dark brown, with darker spots scattered all over the body, and banded cursorial legs.

Head: vertex with well-marked sutures, delimiting a tripartite area, the middle section slightly elevated in males but conspicuously protuberant in females, forming a central bump (better seen in rear view) (Fig. 12k–l). Lower frons with a circular, central depression (shallow in males, deeper in females).

Thorax: Pronotum smooth, supracoxal dilation widely curved (Fig. 3.12m–p). Spination formula $F=3dS/13-15AvS/5PvS$; $T=14-17AvS/13-15PvS$. Posteroventral tibial spines short and reclined. Males: forewings hyaline and iridescent, with a pair of brown spots or entirely spotless, hindwings hyaline and colorless; in resting position wings extend slightly beyond the tip of the abdomen. Females: forewings opaque, dark to light brown, often with 2 oblique and contrasting darker bands, but sometimes they can be uniformly colored; hindwings opaque, sub-hyaline,

dark brown in colour, crossveins pale, especially in the anal area; in resting position wings reach tip of abdomen. Mesothoracic wings of both sexes with a dark brown, shiny stigma.

Abdomen: slender in males, dilated in females.

Species checklist

Raptrix intermedia Lombardo and Marletta, 2004 [BR (MG)]

Raptrix occidentalis Lombardo and Marletta, 2004 [CO, EC, PE]

Raptrix perspicua (Fabricius, 1787) [BR (PA), FG, GY, VE]

Mantis perspicua Fabricius 1787

=*Mantis fusca* Olivier, 1792

=*Mantis truncata*, Lichtenstein 1802

=*Mantis fuscata* Stoll, 1813

=*Mantis multistriata* Audinet Serville, 1839

=*Acontista elegans* Saussure 1869

=*Acontista travassosi* Jantsch, 1986

Raptrix westwoodi (Saussure and Zehntner, 1894) [BR [AM, PA], CR, CO]

Acontista westwoodi Saussure and Zehntner, 1894*

Tithrone Stål, 1877 (Fig. 3.12q-t)

Tithrone Stål, 1877: 42. Type species: *Acontista roseipennis* Saussure, 1870 (by monotypy).

Distribution. Restricted to northern South America and Central America north to Costa Rica.

Diagnosis. *Size range*: 19–25 mm.

Body colouration: predominantly green (preserved specimens might turn yellowish), abdominal tergites with crimson to pink pigmentation, often visible only in traces in preserved specimen.

Head: vertex almost straight (Fig. 3.12q–r) lower frons transversal, juxtaocular bulges slightly developed, vertex somewhat arched.

Thorax: pronotum smooth, supracoxal bulges scarcely expressed (Fig. 3.12s–t). Spination formula: F=3DS/13–15AvS/5PvS; T=12–16AvS/14–16PvS; posteroventral tibial spines erected and separated from each other. Wings: forewings exhibiting colour asymmetry in both sexes, where one is opaque green (except for a pink anal area) and the other is opaque pink (except for green costal area); hindwings hyaline and pink in both sexes.

Abdomen: dilated in females, slender in males.

Species checklist

Tithrone laeta Lombardo, 1996 [VE]

Tithrone latipennis Lombardo, 1996* [CO, EC] – **Fig. 3.45c–d**

Tithrone roseipennis (Saussure, 1869) [CO, CR, FG, GY, TT, VE] – **Fig. 3.45e**

Acontista roseipennis Saussure 1870*

=*Tithrone trinitatis* Giglio-Tos, 1927

3.4.8 Family Acanthopidae Burmeister, 1838

Rivera and Svenson (2016) restricted the concept of Acanthopidae to include only the Acanthopinae. The dead leaf-mimicking Acanthopidae is one of the best-studied families from the acanthopoid clade. Fully revised genera, such as *Decimiana* (Lombardo 2000a), *Acanthops* (Lombardo and Ippolito 2004), and *Pseudacanthops* (Lombardo *et al.* 2013), in addition to other shorter treatments dealing with individual species or little known genera (Roy 2002a,c 2004, Rivera 2005, Ariza and Salazar 2005, Ippolito 2007, Ippolito and Lombardo 2004, González *et al.* 2012, Menezes and Bravo 2012), have considerably added to our understanding of acanthopid diversity.

A key setback in the taxonomy of acanthopids is that 50% of its species remain known from one sex only. This includes *Lagrecacanthops*, whose females have not yet been recognized. This situation is patently clear in *Acanthops*, where incomplete assessment of sexually dimorphic characters has made its morphological delimitation uncertain, resulting in the historical treatment of otherwise congeneric sexes under separated genera (for further discussion see “taxonomic notes” under *Plesiacanthops* below). Certainly, a more inclusive phylogenetic analysis of

Acanthopidae is required to resolve the taxonomic gap left by unsampled sexually dimorphic variation.

Key to genera (both sexes; females of *Lagrecacanthops* unknown)

1. Post-ocellar area of vertex with a rectangular projection bearing tubercles (Fig. 3.12ab-ac).....***Pseudacanthops***
- 1'. Post-ocellar area without projections, smooth, or only with few, scattered granulations.....2
2. Cerci with distal cercomere conical, not bilobed.....3
- 2'. Cerci with distal cercomere flattened, normally with a well-defined central notch (thus appearing bilobed).....4
3. Males: forewings with costal area broad, costal vein evenly curved (Fig. 3.13q). Females: forewings lacking a defined apical lobe (Fig. 3.13s).....***Metilia***
- 3'. Males: forewings with costal area relatively broad, costal vein bi-sinuous (Fig. 47c). Females unknown.....***Lagrecacanthops***
4. Male.....5
- 4'. Female.....7
5. Costal area of forewings only slightly broadened proximally, costal vein describing a moderated undulation.....***Decimiana***
- 5'. Costal area of forewings much wider proximally, costal vein describing a broad, proximal undulation, sometimes followed by secondary but smaller pre-apical one (Fig. 3.13i).....6
6. Fore- and hindwings with apical, rounded lobe; proximal section of costal area broadly curved, followed by a small, pre-apical sinuosity (thus, costal edge appears bi-sinuous) (Fig. 3.13f, 3.13g).....***Acanthops***
- 6'. Fore- and hindwings without apical lobe; proximal section of costal area broadly curved but lacking a pre-apical sinuosity (Fig. 3.13z).....***Plesiakanthops***
7. Mid- and hind femora each with both genicular lobes elongated and triangular (Fig. 3.13u); forefemora with 6–7 posteroventral spines.....***Plesiakanthops***

7'. Mid- and hind femora each with at least the genicular lobe associated to the genicular spine shorter, rounded or only subtriangular (Fig. 3.13v-y); forefemora with 6 posteroventral spines....8

8. Forewings with costal margin strongly bi-sinuous, with a deep, central excavation (Fig. 3.13h). Distributed mostly in warm, humid habitats from southern Mexico through Central America, and south into the Amazon basin (one species, *A. falcataria*, is found further south, mainly within the Atlantic rainforest of Southern Brazil).....***Acanthops***

8'. Forewings with costal margin only slightly bi-sinuous (Fig. 3.13j). Distributed mostly in warm to temperate, seasonal habitats, within and around the Paraná River basin and through south to eastern Brazil.....***Decimiana***

Acanthops Audinet Serville, 1831 (Fig. 3.13a-i, 3.13v-w)

Acanthops Audinet Serville, 1831: 46. Type species: *Mantis fuscifolia* Olivier, 1792 (by monotypy).

Distribution. Found mostly in warm, humid habitats from southern Mexico, Central America south into the Amazon basin; one species (*A. falcataria*) occurs in the Atlantic rainforest of Southern Brazil.

Diagnosis. *Size range:* 38–50 mm.

Body colouration: body and limbs light to brown, often with reddish or yellowish tones, particularly ventrally, and with scattered darker marks.

Head: eyes ovoid or spherical (Fig. 3.13a–b), bearing a conical, ocular spine of variable size (relatively larger in males). Vertex smooth, sometimes with a few tubercles that may spread into the juxtaocular bulges; lower frons with bifid upper angle.

Thorax: pronotum variable, from slender to relatively stocky in males (Fig. 3.13c–d), relatively more robust in females (Fig. 3.13e), surface smooth or with a few small granules or tubercles (the latter restricted to the prozona and the portion of metazona associated to the supracoxal dilation); metazona usually 2–2.5x as long as prozona; supracoxal bulges well developed.

Spination formula $F=4DS/13-15AvS/6PvS$; $T=15-18AvS/18-20PvS$. Proximal, dorsal margin of forefemora with a small, lobe-like protrusion (rarely absent). Genicular lobes of mid- and hindfemora, rounded or subtriangular, apex blunt and not projected (Fig. 3.13v–w). Male: forewings brown, opaque or subhyaline, usually with hyaline, colorless regions distally and/or along its posterior margin, costal margin with a proximal sinuosity that is broadly curved, encompassing $\frac{3}{4} - \frac{1}{2}$ of the costal margin, and another smaller, pre-apical sinuosity, thus making the costal margin appear strongly bi-sinuuous (Fig. 3.13f–g); hindwings exhibiting two main patters: largely colourless and hyaline with a distal brown patch, or completely fuscous; both fore- and hindwings with a rounded, apical lobe. Females: forewings brown, opaque, costal margin bi-sinuuous, and with an elongated and falciform apical lobe (Fig. 3.13h); hindwings brown, all veins and adjacent membrane yellow (thus forming a reticulated pattern), and no apical lobe present.

Abdomen: lateral margins of segments 5–6 with conspicuous lobe-like, cuticular extentions in both sexes (relatively larger in females). Supraanal plate with distal, central notch. Cerci with their distal cercomere flattened, normally with a well-defined central notch (thus appearing bilobed).

Remarks. See “Remarks” under *Plesiacanthops*.

Species checklist

Acanthops bidens Hebard, 1922* [MX] – **Fig. 3.46a**

Acanthops centralis Lombardo and Ippolito, 2004 [CO, CR, PA]

Acanthops chocoensis Salazar, 2005 [CO]

Acanthops contorta Gerstaecker, 1889 [PE]

Acanthops elegans Lombardo and Ippolito, 2004 [CR, GT, MX]

Acanthops erosa Audinet Serville, 1839 [BR (PA), FG]

Acanthops erosula Stål, 1877 [BO, EC, PE]

Acanthops falcata Stål, 1877 [BR (PA), CO, GY, PA, VE] – **Fig. 3.46b**

=*Acanthops angulifera* Westwood, 1889

=*Acanthops griffinii* Giglio-Tos, 1915*

Acanthops falcataria (Goeze, 1778) [BR (MG, MT, RJ, SC, SP)]

Mantis falcataria Goeze, 1778

=*Acanthops mortuifolia* Brunner von Wattenwyl, 1897

Acanthops fuscifolia (Olivier, 1792) [EG, SU]

Mantis fuscifolia Olivier 1792

=*Mantis angulata* Lichtenstein, 1802

=*Mantis sinuata* Stoll 1813

Acanthops godmani Saussure and Zehntner, 1894* [BE, MX] – **Fig. 3.46c**

Acanthops onorei Lombardo and Ippolito, 2004 [EC]

Acanthops parafalcata Lombardo and Ippolito, 2004* [TT] – **Fig. 3.46d**

Acanthops parva Beier, 1942 [BR (PA)]

Acanthops royi Lombardo and Ippolito, 2004* [EC, PE] – **Fig. 3.46e**

Acanthops soukana Roy, 2002* [EG] – **Fig. 3.46f**

Taxonomic notes. *Acanthops soukana*, known from a single female specimen, is an aberrant member of *Acanthops*. This species differs from its congeners in having prominent vertex and juxtaocular tubercles, a long and relatively slender pronotum, forewings with costal margin strongly sinuated and with a markedly short apical lobe, and abdominal tergites with remarkably large foliaceous lobes. Discovering the male is critical to confirm the generic identity of this species.

Decimiana Uvarov, 1940 (Fig. 3.13i-3.13j, y)

Decimia Stål, 1877: 84 (non *Decimia* Walker, 1856: 1717 - Lepidoptera). Type species:

Acanthops tessellata Charpentier, 1841 (by monotypy).

Decimiana Uvarov, 1940: 173 (nom. nov. pro *Decimia* sensu Stål, 1877).

Distribution. *Decimiana* is distributed in warm to temperate, seasonal habitats, within and around the Paraná River basin, mostly through northeastern Argentina, southern Bolivia, and Paraguay. Also found from southern to eastern Brazil, where *D. bolivari* also occurs in more

humid and forested habitats (Menezes and Bravo 2012). Not yet recorded for Uruguay but likely present.

Diagnosis. *Size range:* 30–45 mm.

Body colouration: body and limbs light to brown, often with reddish or yellowish tones, particularly ventrally, and with scattered darker marks.

Head: vertex smooth in males, with scattered granules in female; juxtaocular bulges marked; upper angle lower frons bifid.

Thorax: pronotum slender and mostly smooth in males, but relatively stouter in females, with marginal denticles and surface finely granulated; metazona 2–3x as long as prozona; supracoxal bulges weakly developed. Spination formula F=4DS/14–16AvS/6PvS; T=15–17AvS/17–24PvS. Proximal, dorsal margin of forefemora lacking a small, lobe-like protrusion, although sometimes only weakly expressed in some females. Genicular lobes of mid- and hindfemora, rounded, apex blunt (Fig. 3.13y). Male: forewings brown, opaque or subhyaline, uniformly coloured but some individuals may exhibit lighter patches, costal area only slightly wider proximally, its costal vein describing a weak but evident sinuosity along its proximal half, in addition to a small, pre-apical sinuosity (Fig. 3.13i); hindwings yellowish, basal-most cells and area near apex darkened; both fore- and hindwings with a short, apical lobe. Females: forewings, brown and opaque, costal margin with proximal sinuosity widely curved, pre-apical sinuosity only weakly indicated, apical lobe relatively elongated and falciform (Fig. 3.13j); mesothoracic wings short, yellow-coloured, with brown marking on distal-most cells, forming a reticulated pattern.

Abdomen: lateral margins of segments 5–6 with conspicuous foliaceous extensions in both sexes (relatively larger in females). Cerci with their distal cercomeres flattened, square-like and slightly bilobed.

Species checklist

Decimiana bolivari (Chopard, 1916)* [BR (BA, CE, SP), PY] – **Fig. 3.46g, 47a**

Decimiana clavata Ippolito and Lombardo, 2004 [BR]

Decimiana elliptica Menezes and Bravo, 2012 [BR (BA)]

Decimiana hebardi Lombardo and Ippolito, 2000 [BO]

Decimiana rehni (Chopard, 1913)* [AR, PY] – **Fig. 3.47b**

Plesiacanthops rehni Chopard, 1913

Decimiana tessellata (Charpentier, 1841) [BR (GO, MT, PR, SC), PY]

Acanthops tessellata Charpentier, 1841

Decimia tessellata (Charpentier, 1841)

Lagrecacanthops Roy, 2004 (Fig. 3.47c)

Lagrecacanthops Roy, 2004: 491 (original description). Type species: *Lagrecacanthops guyanensis* Roy, 2004 (by original designation).

Distribution. Distributed in Northern Brazil, Guyana and French Guiana.

Diagnosis (males only; adapted from Roy 2004). *Size range*: 48–53 mm.

Body colouration: dark brown, with paler patches scattered over body and limbs.

Head: vertex convex, with multiple small tubercles; upper angle of lower frons bifid.

Thorax: pronotum relatively short, finely granulated and without marginal denticles; metazone about twice as long as prozona. Forecoxae longer than metazona, with multiple, whitish, callous spots on its anterior aspect. Spination formula F=4DS/14–16AvS/6PvS; T=14–15AvS/18–19PvS. Genicular lobes of cursorial femora rounded, their corresponding tibiae with a central swelling. Male: forewings opaque or subhyaline, uniformly brown (some sparse, paler patches could be present), its costal margin strongly bi-sinuous; metathoracic wings brownish, with clear spots in some of the cells within the discoidal and anal areas; apical lobe of both pairs of wings present, relatively large and strongly curved backwards (Fig.3.47c).

Abdomen: lateral margins of segments 5–6 with foliaceous extensions. Cerci with their distal cercomere conical, not bilobed, apex rounded. Supraanal plate with a shallow, distal emargination; subgenital plate distally with a weak, median suture.

Species checklist

Lagrecacanthops brasiliensis Roy, 2004 [BR (RO)]

Lagrecacanthops guyanensis Roy, 2004* [FG, GY] – **Fig. 3.47c**

Metilia Stål, 1877 (Fig. 3.13k–t, 3.13w)

Metilia Stål, 1877: 84. Type species: *Metilia integra* Stål, 1877 = *Acanthops brunnerii* Saussure, 1871 (by monotypy).

Acanthogaster Werner, 1927: 281. Type species: *Acanthops brunnerii* Saussure, 1871 (by monotypy).

Distribution. Found from Costa Rica, south into the Amazon basin and the Guiana Shield.

Diagnosis. *Size range*: 25–40 mm.

Body colouration: males greenish to yellowish brown, rarely entirely brown; females light to dark brown.

Head: vertex smooth or with few granulations, normally flat (Fig. 3.13k, m) but slightly convex in *M. amazonica* (Fig. 3.13l). Compound eyes normally ovoid, rarely spherical.

Thorax: Pronotum normally smooth (Fig. 3.13n), sometimes with small, scattered granulations in females (Fig. 3.13o); metazona 1.5–2.5x as long as prozona. Forecoxae smooth in males, with marginal spines in females. Spination formula: F=4DS/15–18AvS/6–7PvS; T=20–28AvS/14–19PvS. Proximal, dorsal margin of forefemora smooth, without a lobe-like protrusion. Mid- and hind legs without lobes. Male: forewings conferring a distinct habitus (Fig. 3.13p) typically resembling partially dried leaf, i.e. yellowish green on its proximal 1/3–1/2, turning brown distally (but rarely entirely brown), costal area broad, its costal vein evenly curved, its apex

exhibiting various degrees of development, sometimes forming a small lobe that is curves backwards (Fig. 3.13q); hindwings often hyaline in great part, colorless and rarely fuscous, distally with an opaque patch of yellowish brown, apex always with a small and triangular, well-developed, distal lobe that conspicuously curves backwards (Fig. 3.13r). Females: forewings short, costal edge ample and curved as in males, apex without lobe (Fig. 3.13s); hindwings short, yellow-coloured, cells with brown marking, apex without lobe (Fig. 3.13t).

Abdomen: lateral margins with conspicuous lateral lobes on segment 5th, segments 4th and 6th may also exhibit lobes in both sexes, but these are much smaller; abdominal lobes of females considerable more developed than in males. Cerci with last cercomere conical, not bilobed.

Remarks. Having seven posteroventral forefemoral spines was long considered a distinctive character of *Metilia* (e.g. Beier 1934a, Terra 1995). The transference of *Acanthops amazonica* to *Metilia*, which has six spines instead, modified the variability of this character state (now 6–7) in the taxonomic definition of this genus (Roy 2002a). However, having six posteroventral spines is not the only character that makes *Metilia amazonica* distinct from other conspecifics, making its assignment to *Metilia* contentious. For instance, the male of *M. amazonica* exhibits a distinct fully dark brown habitus; this contrasts with the condition observed in most species of *Metilia* examined, where males are greenish to yellowish brown, with forewings resembling partially dried leafs (but see below). A convex and somewhat protuberant vertex, and spherical compound eyes (Fig. 3.13l), are also distinct features of *M. amazonica* that depart *Metilia* spp. — where vertex is flat and low, and compound eyes ellipsoidal (Fig. 3.13k). These differences suggest that *M. amazonica* could well be assigned in its own genus. The discovery of the female of *M. amazonica* is necessary to elucidate its actual generic identity.

Species checklist

Metilia amazonica (Beier, 1930) [BR (AM), FG] – **Fig. 3.47d**

Acanthops amazonica Beier, 1930*

Metilia boliviana (Werner, 1927) [BO]

Decimia boliviana Werner, 1927

Metilia brunnerii (Saussure, 1871) [SU] – **Fig. 3.47e**

Acanthops Brunnerii Saussure, 1871

=*Metilia integra* Stål, 1877

=*Acanthops adusta* Gerstäcker, 1889*

Metilia coloradensis (González, Miller and Salazar, 2012) (**comb. n**) [CO]

Acanthops coloradensis González, Miller and Salazar, 2012

Metilia septemspinosa (Ippolito, 2007) (**comb. n**) [EC]

Acanthops septemspinosa Ippolito, 2007

Taxonomic notes. The studied material revealed important variation in male genital morphology, suggesting uncovered diversity within this genus. This contrasts with the discrete interspecific variability of their external morphology, which further difficult species identification. This situation possibly resulted in numerous misidentifications in the existing taxonomic literature. In fact, it is likely that a yet undetermined number of both described and undescribed species have historically been attributed to *M. brunnerii*, as this name has been applied to almost every reported member of this genus. As a result, in addition to its type locality in Surinam, *M. brunnerii* has also been reported from Costa Rica, Nicaragua, Colombia, Venezuela, French Guiana, Peru, Ecuador, Bolivia, and Brazil (Amazonas, Mato Grosso, Pará, Roraima, Rondônia, and Rio Grande do Sul) (Lombardo and Agabiti 2001, Ehrmann 2002, Agudelo *et al.* 2007). For instance, Rehn (1935) considered members of *Metilia* from Central America conspecific with *M. brunnerii*; however, closer examination of these populations suggests the possibility of an undescribed species. A similar situation occurs with material from Peru currently assigned to *M. brunnerii*, but for which the name *M. adusta* — currently in synonym — should instead apply (at least, to some populations from that country). Consequently, the wide geographical distribution of *M. brunnerii* is most likely in great part equivocal.

I also recognized *Acanthops coloradensis* and *Acanthops septemspinosa* as members of *Metilia*. González *et al.* (2012) described *Acanthops coloradensis* on the basis of a single female specimen from Colombia. The original description and accompanying figures clearly show characters associated to *Metilia*, such as the absence of the apical lobe in the forewings, forefemora with 7 posteroventral spines, and large lateral lobes on the fifth abdominal segment. Whether this is a valid species remains to be determined after the eventual recognition of

conspecific males. On the other hand, Ippolito (2007) described *Acanthops septemspinosa* on the basis of a single male specimen from Napo, Ecuador. The conformation of male genitalia, the characteristic shape of wings, and cerci with their last cercomere conical (not bilobed), reveal it as a member of *Metilia* instead — although its fully brown habitus is unique among *Metilia*. Given the present evidence I transfer both *A. coloradensis* and *A. septemspinosa* to *Metilia*.

Plesiacanthops Chopard, 1913 **stat. rev.** (Fig. 3.13u, 3.13z, 3.13aa)

Plesiacanthops Chopard, 1913: 60. Type species: *Acanthops tuberculata* Saussure, 1870 (by monotypy)

=*Miracanthops* Roy, 2004: 494. Type species: *Miracanthops poulaini* Roy, 2004 (by original designation) (**syn. n.**)

Distribution. Currently known for Ecuador, Peru, Guyana and French Guiana.

Diagnosis. *Size range (mm):* 35–56.

Body colouration: body light to dark brown or greenish brown.

Head: eyes ovoid, bearing a conical, apical process of variable size; occipital area with multiple granulations, vertex and juxtaocular tubercles developed and well-delimited by deep sutures in both sexes; lower frons with upper angle bifid.

Thorax: pronotum slender in both sexes, its surface is mostly smooth in males, finely granulated in females, usually with a few small but otherwise well-developed tubercles scattered over the prozona and distal-most portion of metazona; metazona 2.5–3x as long as prozona, lateral margins of metazona parallel, and supracoxal bulges poorly developed. Spination formula: F=4DS/15–16AvS/6–7PvS; T=18–19AvS/23–27PvS; Proximal, dorsal margin of forefemora with or without a small, lobe-like protrusion. Genicular lobes of mid- and hindfemora, rounded in males, triangular and pointy in females (Fig. 3.13u). Male: forewings brown, opaque or subhyaline, usually exhibiting hyaline regions both distally and/or along posterior margin,

proximal section of costal area broad, and widely curved proximally, without pre-apical sinuosity, and apical lobe absent (Fig. 3.13z); hindwings coloured as forewings, hyaline and colorless spots also present but restricted to distal portion only, apical lobe absent or vaguely indicated at most. Females: forewings opaque and narrow, apical lobe with serrated posterior border and often greatly elongated (Fig. 13aa); hindwings reduced, yellow-coloured, with brown markings on its distal-most cells, thus forming a reticulated pattern.

Abdomen: lateral margins of segments 5–6 with conspicuous extensions in both sexes (larger in females). Subgenital plate of males with a central notch. Cerci with their last cercomeres flattened, rounded or rectangular in shape, with a distal, median notch (thus appearing bilobed).

Remarks. In order to make the nomenclatorial procedures here introduced more transparent, I first present the history of the names *Plesiathanops* and *Mirathanops* separately, as follows:

i) ***Plesiathanops***. Chopard (1913) established *Plesiathanops* to accommodate his new species, *P. rehni* Chopard, 1913, from Argentina. In the same publication, Chopard also took the opportunity to incorporate one additional species, *Acanthops tuberculata* Saussure, 1870, from French Guiana, designating it as the type species of *Plesiathanops*. While Lombardo (2000a) correctly identified *P. rehni* as a valid species but as a member of *Decimiana* instead, the taxonomic status of *Plesiathanops tuberculata* remained unresolved until Roy (2002a) discovered and described the male. After assessing the morphological characters of both sexes, Roy (2002a) concluded that the enigmatic species *Plesiathanops tuberculata* was, after all, a member of *Acanthops*. As a result of this procedure, *Plesiathanops* became a synonym of *Acanthops*. Lombardo and Ippolito (2004) accepted Roy's action and incorporated *Acanthops tuberculata* in their phylogenetic analysis of the genus, where it formed a sister-species relationship with *A. occidentalis*, a newly described species. According to Lombardo and Ippolito (2004), these two species, in addition to *A. parva* and *A. erosula*, together formed a monophyletic subclade within the “*falcataria*” species group, which they denominated the “*tuberculata*” subgroup.

ii) ***Mirathanops***. Roy (2004) established *Mirathanops* on the basis of female specimens to accommodate two striking species from Ecuador and Peru, respectively: *M. lombardoi* and *M.*

poulaini. These two species were remarkable because of their relatively elongated body, the narrowness and length of their forewings, among other unique character states which clearly warranted their own generic placement. Rivera (2005) added a third species, *M. eseejja*, this time on the basis of both sexes, thus completing the generic description. The newly gained knowledge on the degree of variation in sexually dimorphic characters in *Miracanthops* also allowed the identification of *Acanthops occidentalis* Lombardo and Ippolito, 2004 as a member of *Miracanthops*, thus increasing the number of species to four: *lombardoi*, *poulaini*, *eseejja* and *occidentalis* (Rivera 2005).

The available information, in addition to my own examinations of type and non-type specimens, suggest that *Acanthops tuberculata* and *Miracanthops* spp. all represent the same lineage. Pronotum and wing morphology are all remarkably similar in males of *A. tuberculata*, *M. occidentalis* and *M. eseejja*, as documented respectively, in Roy (2002), Lombardo and Ippolito (2004) and Rivera (2005). Males of all three species share forewings with a wide costal area, as defined by the broadly curved costal vein, and the lack of a rounded, apical lobe. In *Acanthops* spp., the proximal undulation of the costal vein is followed by a smaller, secondary, preapical undulation, whereas the tip of both forewings terminate in a rounded or oval lobe. The female forewings of *M. lombardoi*, *M. poulaini* and *M. eseejja*, all exhibit a significant elongation and a remarkably long and rather straight apical lobe, although relatively shorter in *A. tuberculata* — the latter somewhat comparable in overall shape to those seen in some species of *Acanthops*. Whereas males of *Miracanthops* spp. and *A. tuberculata* can be difficult to distinguish from males of *Acanthops* spp., the more elongated and less-robust body type of the females, in addition to their distinctly elongated pronotum and forewings, clearly set these taxa apart from “true” *Acanthops*. Another revealing, yet overlooked character in the forewings of females, is the “serrated” posterior edge of its distal lobe. This character state is evident in *Miracanthops* spp., *A. tuberculata* and *A. erosula*. Although this character has not been tested in a phylogenetic context, it could represent a putative synapomorphy of the “*tuberculata*” subgroup of Lombardo and Ippolito (2004).

Summarizing, having an expanded sampling of both inter- and intraspecific variation, I concluded that *Acanthops tuberculata* represents a taxon distinct from *Acanthops* spp., showing

instead morphological affinities with *Miracanthops* spp.; thus, *Plesiakanthops* cannot be considered a synonym of *Acanthops* as Roy (2002c) proposed. Consequently, I herein propose the revalidation of *Plesiakanthops*, with *A. tuberculata* as its type species. On the basis of the Principle of Priority (ICZN, 1999), I also subsume *Miracanthops* Roy, 2004 under *Plesiakanthops* Chopard, 1913 as its new synonym. *Plesiakanthops* now includes all species of *Miracanthops* (*M. occidentalis*, *M. eseejja*, *M. lombardoi*, and *M. poulaini*) in addition to *A. tuberculata*. For further comparative analyses and descriptions see Roy (2002c, 2004), Lombardo and Ippolito (2004), and Rivera (2005).

Species checklist

Plesiakanthops eseejja (Rivera, 2005)* (**comb. n**) [PE]

Plesiakanthops lombardoi (Roy, 2004)* (**comb. n**) [EC]

Plesiakanthops occidentalis (Lombardo and Ippolito, 2004) (**comb. n**) [EC, PE]

Acanthops occidentalis Lombardo and Ippolito, 2004

Plesiakanthops poulaini (Roy, 2004) (**comb. n**) [PE]

Plesiakanthops tuberculata Saussure, 1870* [EG, GU]

Acanthops tuberculatus Saussure, 1870

=*Plesiakanthops (Acanthops) tuberculata* (Saussure, 1870)

Taxonomic notes. Lombardo and Ippolito (2004) provided morphology-based phylogenetic evidence for subdividing *Acanthops* into two species groups: “*falcata*” and “*falcataria*”. The “*falcataria*” group comprised three subgroups, one of them dubbed as “*tuberculata*”. The “*tuberculata*” clade was supported by shared similarities in the shape of the forewings of males. Within the “*tuberculata*” group, *P. occidentalis* and *P. tuberculata* were found to be sister species, altogether sister to *A. parva*, and finally *A. erosula* as sister of all three species. This interpretation is at odds with Rivera and Svenson (2016), who found strong molecular evidence of *A. erosula* as sister of *Pseudacanthops* + *Acanthops*, thus rendering *Acanthops* as paraphyletic. Although sampling within *Acanthops* in Rivera and Svenson (2016) was limited, all the available evidence suggests that, if the identity of *A. parva* is confirmed as a member of *Plesiakanthops* (i.e. upon discovery of the female), the “*tubercuata*” subgroup of Lombardo and Ippolito (2004) (excluding of *A. erosula*) would then correspond to my interpretation of

Plesiacanthops. This will have important nomenclatorial effects in the current taxonomic concept of *Acanthops*, such as the exclusion of *A. erosula*, and the consequently subdivision of *Acanthops* into at least two genera. Summarizing, the present evidence raises concerns about the monophyletic nature of *Acanthops* and warrants further investigation to clarify its affinities.

Finally, Lombardo and Ippolito (2004: 1092, Fig. 22C and 22F) depicted the male genitalia of *P. occidentalis* (cited under *Acanthops*) as part of its original description, but it was not specified whether it corresponded to the holotype or the paratype. I confirmed that this figure corresponds to the holotype from Limnocochoa, in the Ecuadorian Napo (F. Lombardo pers. com.).

Pseudacanthops Saussure, 1870 (Fig. 3.13ab-ag)

Pseudacanthops Saussure, 1870: 243. Type species: *Hymenopus caelebs* Saussure, 1869 (subsequent designation: Kirby, 1904).

Paracanthops Saussure, 1870: 243.

Distribution. Southern Mexico, south into the Amazon basin and the Guiana Shield.

Diagnosis. *Size range:* 35–50 mm.

Body colouration: males can be dark to light brown, sometimes with green, ochre or greyish tones; females can be green, yellowish or brown, sometimes in combination (mostly green when alive).

Head: vertex markedly convex, with a rectangular process in the postocellar region that bears small tubercles (Fig. 3.13ab–ac); eyes ovoid, with a conical spine-like process apically; lower frons with upper angle bifid.

Thorax: Pronotum of males slender and mostly smooth (Fig. 3.13ad), but robust and extensively covered with tubercles in females (Fig. 3.13ae); metazona 1.6–2x as long as prozona. Spination formula: F=4DS/13–15AvS/6PvS; T=16–22AvS/14–17PvS. Forecoxae slightly longer than metazona, with one or two spine-like tubercles proximally, the same are larger in females.

Forefemora mostly smooth in males, with multiple tubercles in females, both sexes bearing a dorsal, proximal lobe. Mid- and hind femora with a proximal lobe ventrally, the corresponding tibiae are narrower along their distal half, each with a median swelling (sometimes bearing a lobe), and a small, pre-apical lobe. Male: forewings light brown, often distally darkened, costal margin bi-sinuuous and with an apical, rounded lobe (Fig. 3.13af); hindwings similar to forewings. Female: forewings brownish green and opaque, with a long, falciform, apical lobe (Fig. 3.13ag); hindwings reduced, yellow in colour and with brown pigment within the cells, resulting in a reticulated pattern; apex with a small, triangular lobe.

Abdomen: Slightly flattened, with well-developed lateral lobes on segments 5–6, remaining segments with smaller lobes or with forked, filiform projection (all much more developed in females). Cerci with last cercomere flattened and bilobed.

Species checklist

Pseudacanthops caelebs (Saussure, 1869) [BE, GU, HO, MX]

Hymenopus caelebs Saussure, 1869

Pseudacanthops centralis Ippolito and Lombardo, 2013 [NI, PN]

Pseudacanthops clorindae Rivera and Lombardo, 2013* [BR (AM), PE]

Pseudacanthops huaoranianus Lombardo and Ippolito, 2013 [EC]

Pseudacanthops lobipes La Greca and Lombardo, 1997* [BO, BR, PE] – **Fig. 3.47f**

Pseudacanthops spinulosus (Saussure, 1870) [FG, GY, MX, VE]

Paracanthops spinulosa Saussure, 1870

Pseudacanthops spinulosa (Saussure, 1870)

3.5 Conclusions

The Acanthopoidea now includes eight families, 15 subfamilies, 18 tribes, 63 genera (including one *genus inquirenda*), and 316 species (including two *insertae sedis* species). The families Acanthopidae, Liturgusidae, Stenophyllidae, Aconstistidae and Photinaidae are all relatively well studied, whereas Angelidae and Coptopterygidae have not been previously reviewed in any

depth (but see Rehn 1913), despite their relatively prevalence in collections. All seven aforementioned families, together, include 31 genera and 178 species, whereas the Thespidae alone includes the same number of genera and 139 species. My examination of extensive holdings of this poorly known family revealed an important number of undescribed genera and species. Their small size, often dull colouration and less “charismatic” appearance, have likely played a part in making of the Thespidae among the least studied groups of Neotropical praying mantises. Given their wide range of ecological and morphological adaptations, a likely rich diversity of new forms is surely waiting to be uncovered. Recent investigations have also revealed numerous issues afflicting taxonomy and nomenclature in this family (Rivera *et al.* 2011, Agudelo and Rafael 2014, Agudelo and Rivera 2015, Scherrer 2014, Maldaner *et al.* 2015), many of which have been addressed in this study, in particular genus-group names. Moreover, additional cases of synonymy in the species-group are likely pending further study. Finally, the nonmonophyly of *Bantia*, *Musoniella*, *Acanthops* and *Microphotina* raises additional concerns about the taxonomic status of these taxa. More in-depth investigations are necessary to address these issues.

Taxonomic contributions published over the last quarter century have provided a clearer picture of the geographical distribution of Neotropical praying mantises. However, there remains a massive gap in our knowledge of basic biology and natural history of most Neotropical species. I have compiled most of the published biological information available on acanthopoids, including my own field observations, but further effort should be directed towards documenting relevant biological data at the time of collection. Of particular value is information about cryptic strategies and their associated behaviours, type of habitat and/or level within the vegetation where specimens live (e.g. undergrowth, bark), and time and method of collection (e.g. night collections with malaise traps). Registering and compiling this data will help accelerate the documentation of ecological associations of praying mantises, elusive insects whose secretive and generally sedentary lifestyle often prevent direct and sustained observation.

Tables and Figures for Chapter 3

Figure 3.1. Phylogeny and higher-level classification of the eight families of Acanthopoidea (from Rivera and Svenson 2016). The figures in parentheses represent the number of valid genera in each family as a result of this study.

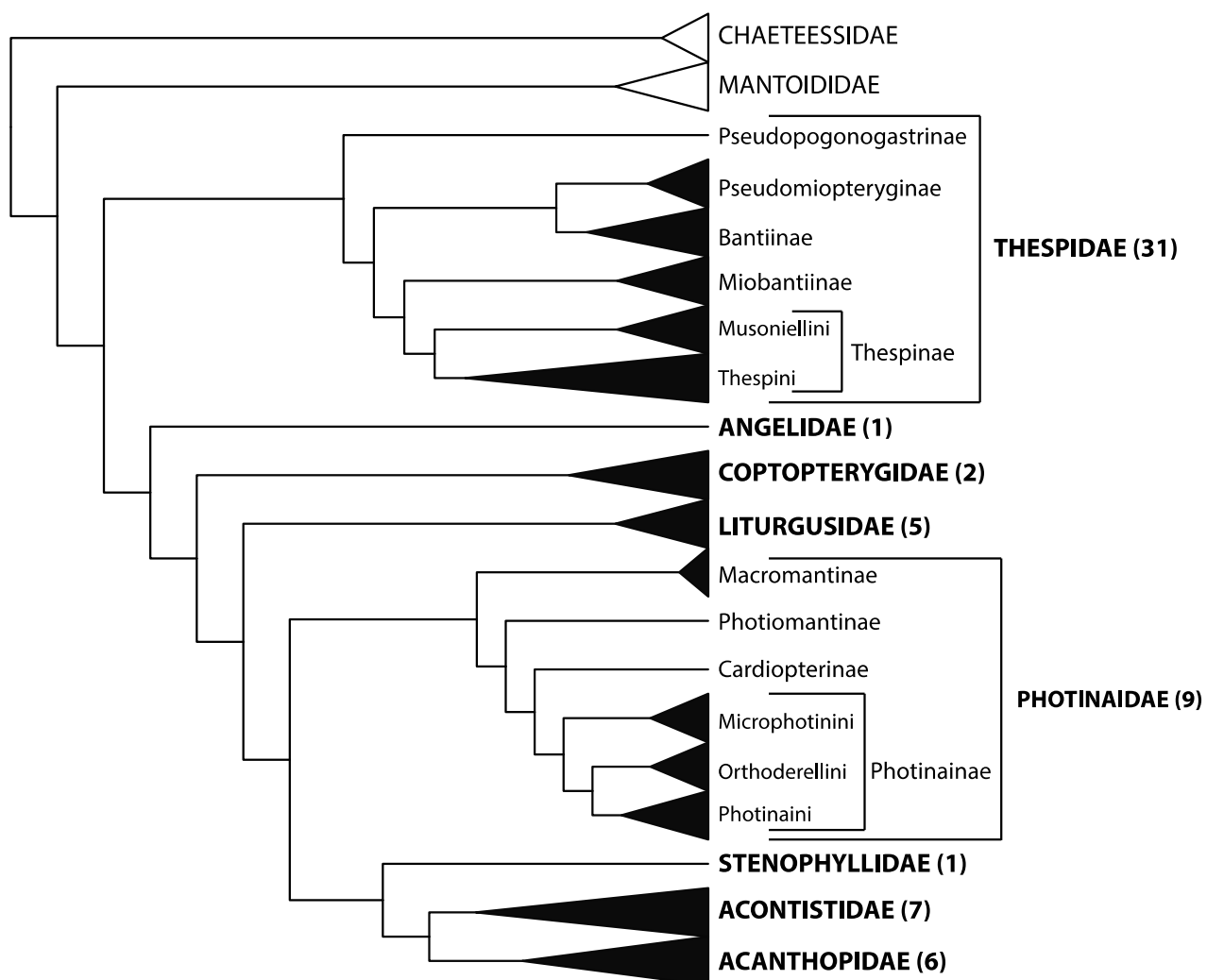


Figure 3.1

Figure 3.2. THESPIDAE - Pseudomiopteryginae: (a) *Pseudomiopteryx* sp., head and detail of vertex; (b) *Pseudomiopteryx* sp., head and detail of vertex; (c) *Pseudomiopteryx* sp., head and prothorax; (d) *Pseudomiopteryx* sp., pronotum; (e) *Pseudomiopteryx* sp., foretibial spines (f) *Pseudomiopteryx* sp., foretibial spines; (g); *Promiopteryx* sp., head; (h) *Promiopteryx* sp., pronotum. THESPIDAE - Bantiinae: (i) *Bantia weneri*, head; (j) *Bantia fusca*, head and prothorax; (k) *Bantia* sp. n., pronotum; (l) *Bantia fusca*, pronotum; (m) *Bantia* sp., pronotum; (n) *Bantia* spp., foretibiae. Abbreviations: AvS, anteroventral spines; PvS, posteroventral spines. All scale bars = 1 mm, except Fig. 2n = 0.5 mm (Fig. 2f not to scale).

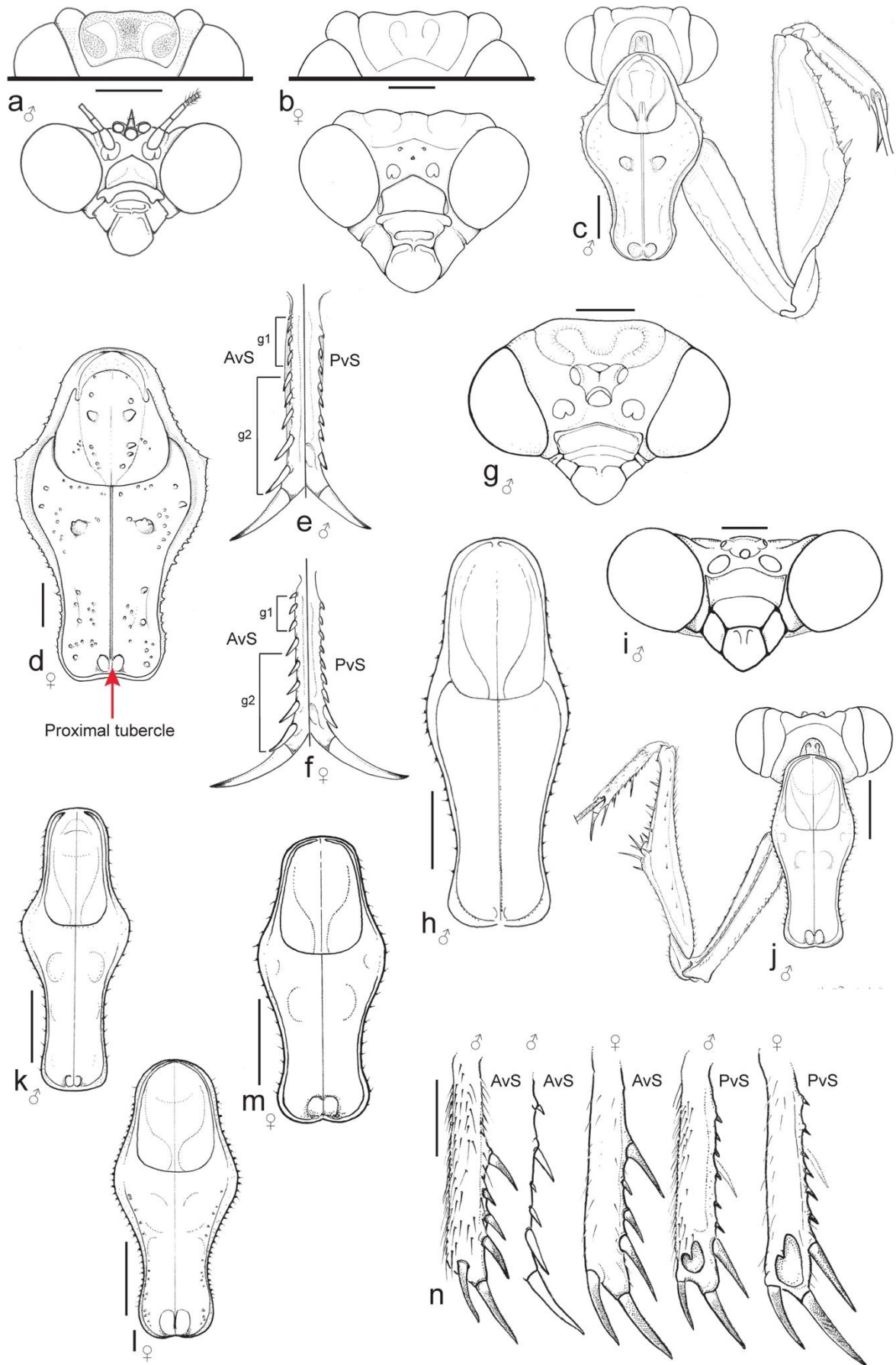


Figure 3.2

Figure 3.3. THESPIDAE – Bantiinae. (a) *Bantiella trinitatis*, head and detail of vertex; (b) *Bantiella trinitatis*, head and detail of vertex; (c) *Bantiella trinitatis*, head and prothorax; (d) *Bantiella trinitatis*, pronotum; (e) *Diabantia* sp., head and prothorax; (f) *Diabantia* sp., discoidal spines; (g) *Mantellias* sp., head; (h) *Mantellias* sp., head and prothorax; (i) *Mantillica nigricans*, pronotum; (j) *Mantillica nigricans*, foretibia; (k) *Thrinaconyx fumosa*, prothorax; (l) *Thrinaconyx fumosa*, pronotum; (m) *Thrinaconyx fumosa*, foretibiae. Abbreviations: AvS, anteroventral spines; DS, discoidal spines; PvS, posteroventral spines. All scale bars = 1mm, except Fig. 3m = 0.5 mm (Fig. 3f not to scale).

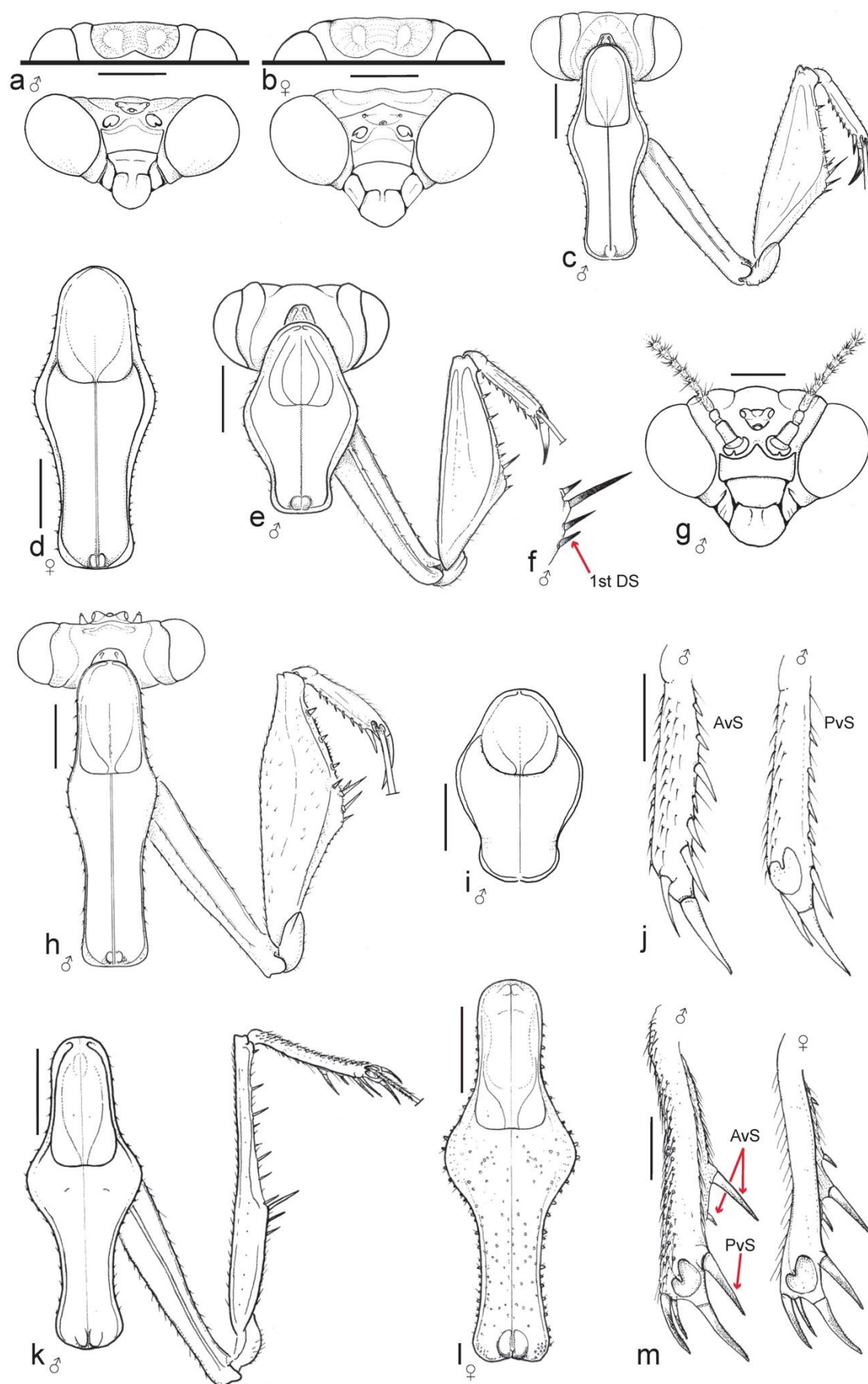


Figure 3.3

Figure 3.4. THESPIDAE - Miobantiinae: (a) *Anamiopteryx tuberculata*, head; (b) *Anamiopteryx grandis*, head; (c) *Anamiopteryx tuberculata*, pronotum; (d) *Anamiopteryx grandis*, pronotum; (d) *Anamiopteryx grandis*, pronotum; (f) *Chloromiopteryx mirim*, head and detail of vertex; (g) *Chloromiopteryx mirim*, head and detail of vertex; (h) *Chloromiopteryx thalassina*, head and prothorax; (i) *Chloromiopteryx mirim*, pronotum; (j) *Eumiopteryx* sp. n., head and detail of vertex; (k) *Eumiopteryx laticollis*, head and detail of vertex; (l) *Eumiopteryx* sp. n., head and prothorax; (m) *Eumiopteryx* sp. n., pronotum; (n) *Miobantia ciliata*, head and detail of vertex; (o) *Miobantia sulista*, head and detail of vertex; (p) *Miobantia sulista*, head and prothorax; (q) *Miobantia sulista*, pronotum; (r) *Miobantia* sp., foretibial spines; (s) *Paradiabantia* sp., pronotum; (t) *Paradiabantia* sp., discoidal spines; (u) *Paradiabantia* sp., detail of forewing membrane showing micro- and macrosetae. Abbreviations: AvS, anteroventral spines; DS, discoidal spines, PvS, posteroventral spines. All scale bars = 1mm, except Fig. 4e, 4s and 4u = 0.5 mm (Fig. 4t not to scale).

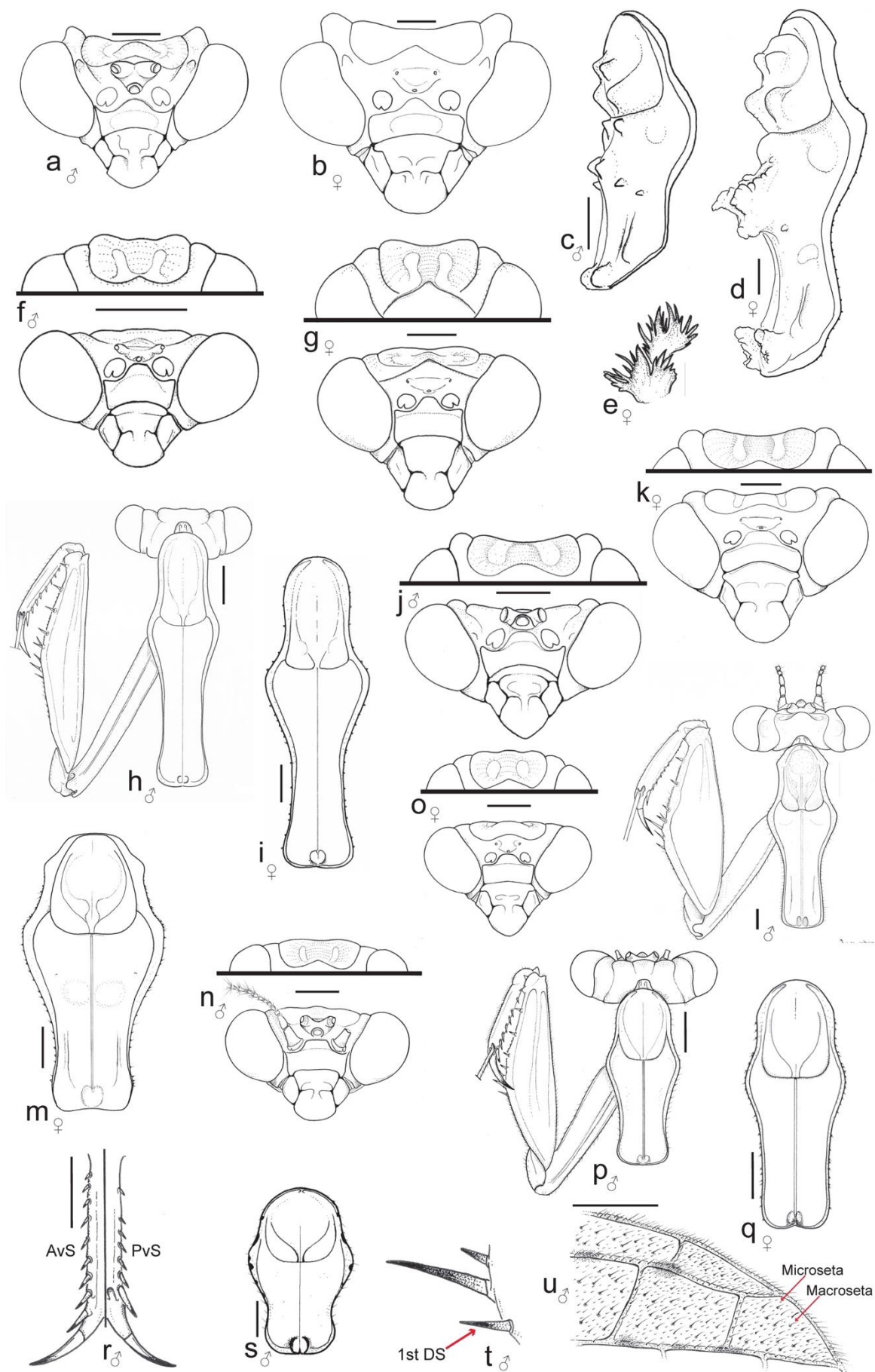


Figure 3.4

Figure 3.5. THESPIDAE – Thespinae, Musoniellini: (a) *Eumusonia intermedia*, pronotum; (b) *Eumusonia* sp., pronotum; (c) *Eumusonia intermedia*, pronotum; (d) *Eumusonia* sp., supraanal plate and cerci; (e) *Eumusonia* sp., supraanal plate and cerci; (f) *Leptomiopteryx dispar*, head and detail of vertex; (g) *Leptomiopteryx dispar*, head and detail of vertex; (h) *Leptomiopteryx dispar*, head and pronotum; (i) *Leptomiopteryx dispar*, pronotum; (j) *Leptomiopteryx dispar*, foretibiae; (k) *Musoniella argentina*, head and detail of vertex; (l) *Musoniella* sp., head; (m) *Musoniella argentina*, pronotum; (n) *Musoniella fragilis*, pronotum; (o) *Musoniella* sp., pronotum; (p) *Musoniella argentina* sp., supraanal plate and cerci; (q) *Musoniella* sp., supraanal plate and cerci; (r) *Pizaia seabrai*, head and detail of vertex (arrows points to distinct carina); (s) *Pizaia seabrai*, pronotum; (t) *Pizaia* sp., head and pronotum. Abbreviations: AvS, anteroventral spines; PvS, posteroventral spines. All scale bars = 1mm (Fig. 5p not to scale).

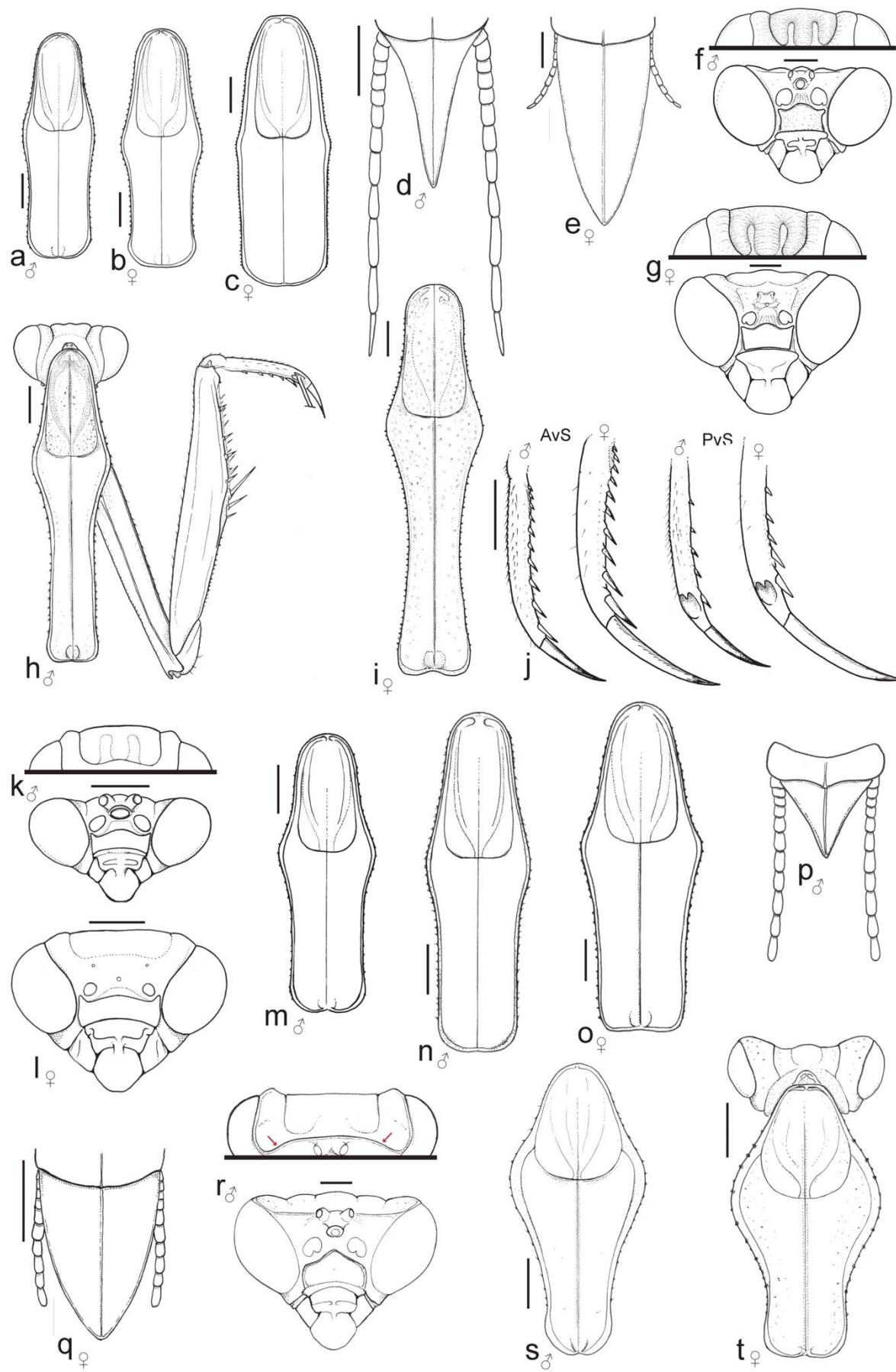


Figure 3.5

Figure 3.6. THESPIDAE – Thespinae, Thespini: (a) *Dugonyx maculosus*, detail of vertex; (b) *Dugonyx maculosus*, head and prothorax; (c) *Dugonyx maculosus*, head and pronotum; (d) *Dugonyx maculosus*, foretibial spines; (e) *Dugonyx maculosus*, ventral phallomere (ventral view); (f) *Galapagia solitaria*, detail of vertex; (g) *Galapagia solitaria*, pronotum; (h) *Galapagia solitaria*, detail of vertex; (i) *Galapagia solitaria*, pronotum; (j) *Galapagia solitaria*, foretibial spines; (l) *Liguanea pediodromia*, head and prothorax; (m) *Liguanea pediodromia*, foretibial spines; (n) *Macromusonia conspersa*, head; (o) *Macromusonia major*, head and pronotum; (p) *Musonia margarethae*, head; (q) *Musonia surinama*, head; (r) *Musonia surinama*, head and prothorax; (s) *Musonia margarethae*, pronotum; (t) *Musonia surinama*, pronotum; (u) *Musonia margarethae*, foretibial spines. Abbreviations: AvS, anteroventral spines; g1, group 1; g2, group 2; PvS, posteroventral spines. All scale bars = 1mm.

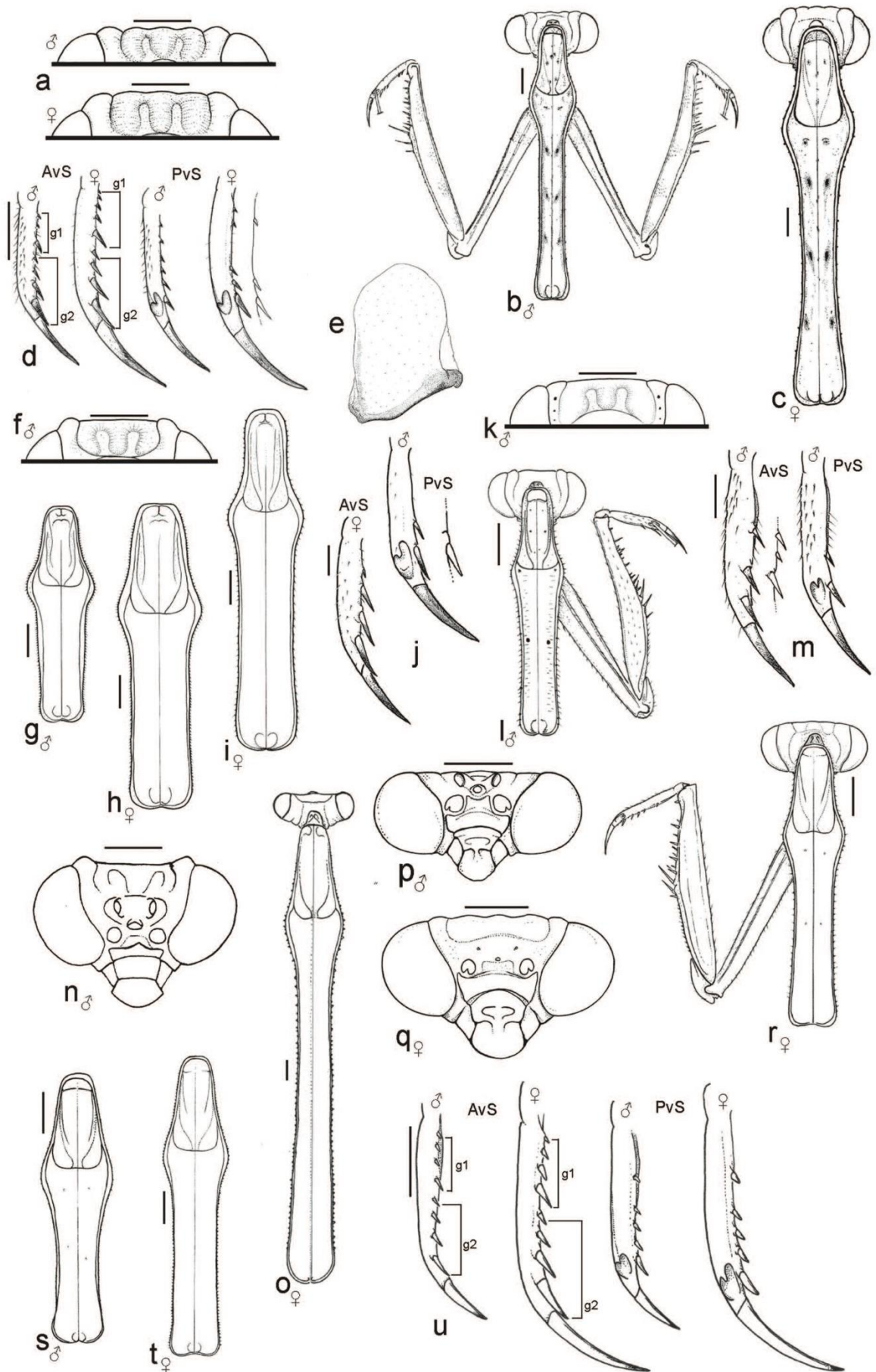


Figure 3.6

Figure 3.7. THESPIDAE – Thespiinae, Thespini: (a) *Musoniola* sp., detail of vertex; (b) *Musoniola* sp., pronotum; (c) *Musoniola* sp., head and prothorax; (d) *Oligonicella striolata*, head; (e) *Oligonicella mexicana*, pronotum; (f) *Oligonicella striolata*, prothorax; (g) *Oligonicella tessellata*, pronotum; (h) *Oligonicella striolata*, pronotum; (i) *Oligonicella mexicana*, pronotum; (j) *Oligonicella* spp., foretibial spines; (k) *Oligonyx* spp., detail of vertex; (l) *Oligonyx* sp., pronotum; (m) *Oligonyx* sp., foretibia; (n) *Oligonyx bidens*, foretibial spines; (o) *Piscomantis peruana*, detail of vertex; (p) *Piscomantis peruana*, head and pronotum; (q) *Piscomantis peruana*, head and prothorax; (r) *Piscomantis peruana*, distal portion of abdomen. Abbreviations: AvS, anteroventral spines; g1, group 1; g2, group 2; PvS, posteroventral spines. All scale bars = 1mm.

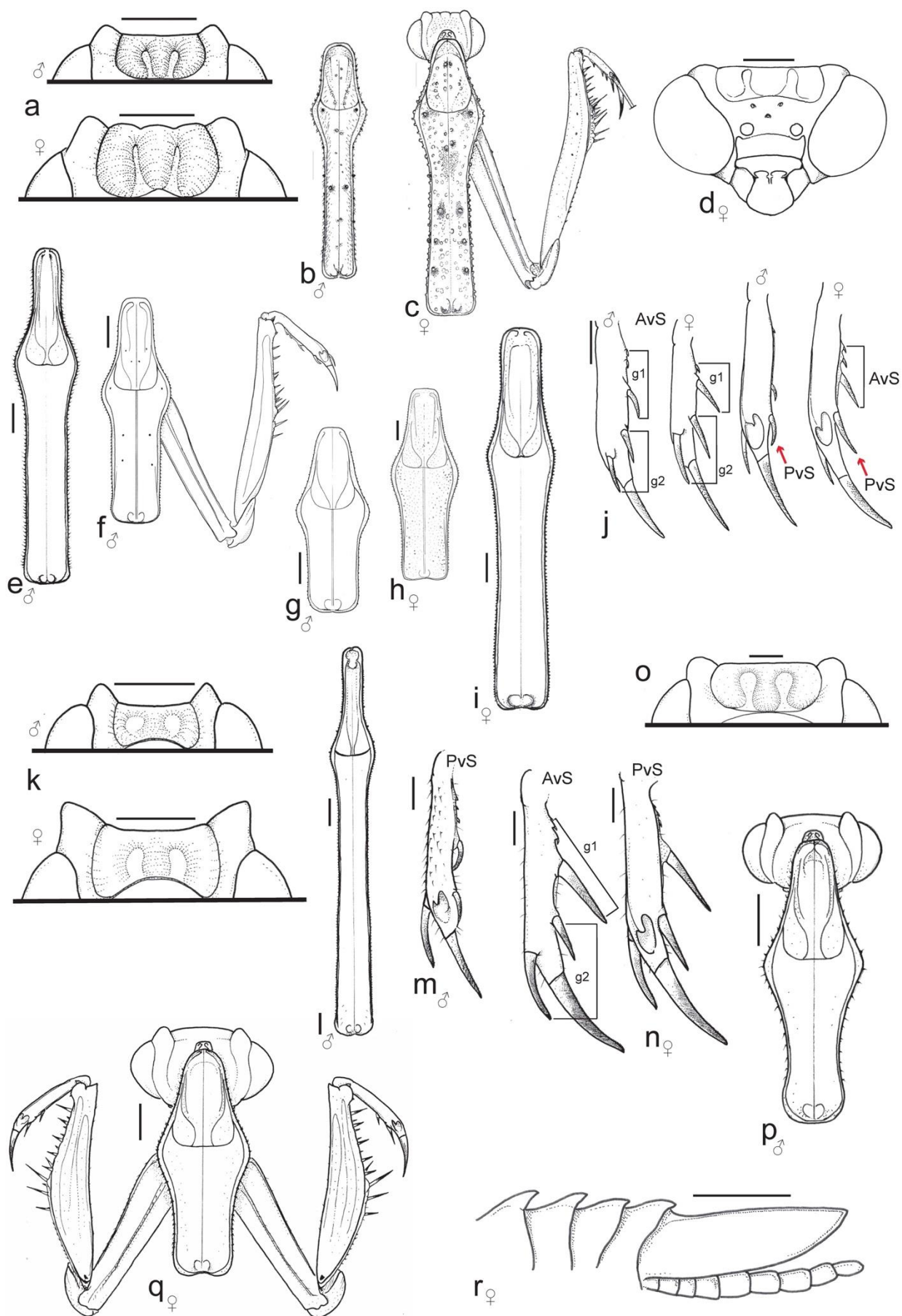


Figure 3.7

Figure 3.8. THESPIDAE - Thespinae, Thespini: (a) *Pseudomusonia lineativentris*, detail of vertex; (b) *Pseudomusonia lineativentris*, head and prothorax; (c) *Pseudomusonia lineativentris*, pronotum; (d) *Pseudomusonia lineativentris*, foretibial spines; (e) *Thespis cubensis*, head and detail of vertex; (f) *Thespis media*, head and detail of vertex; (g) *Thespis boliviana*, pronotum; (i) *Thespis media*, pronotum; (j) *Thespis* spp., foretibial spines; (k) *Thespis media*, supraanal plate and cerci (pilosity omitted in one side). All scale bars = 1mm.

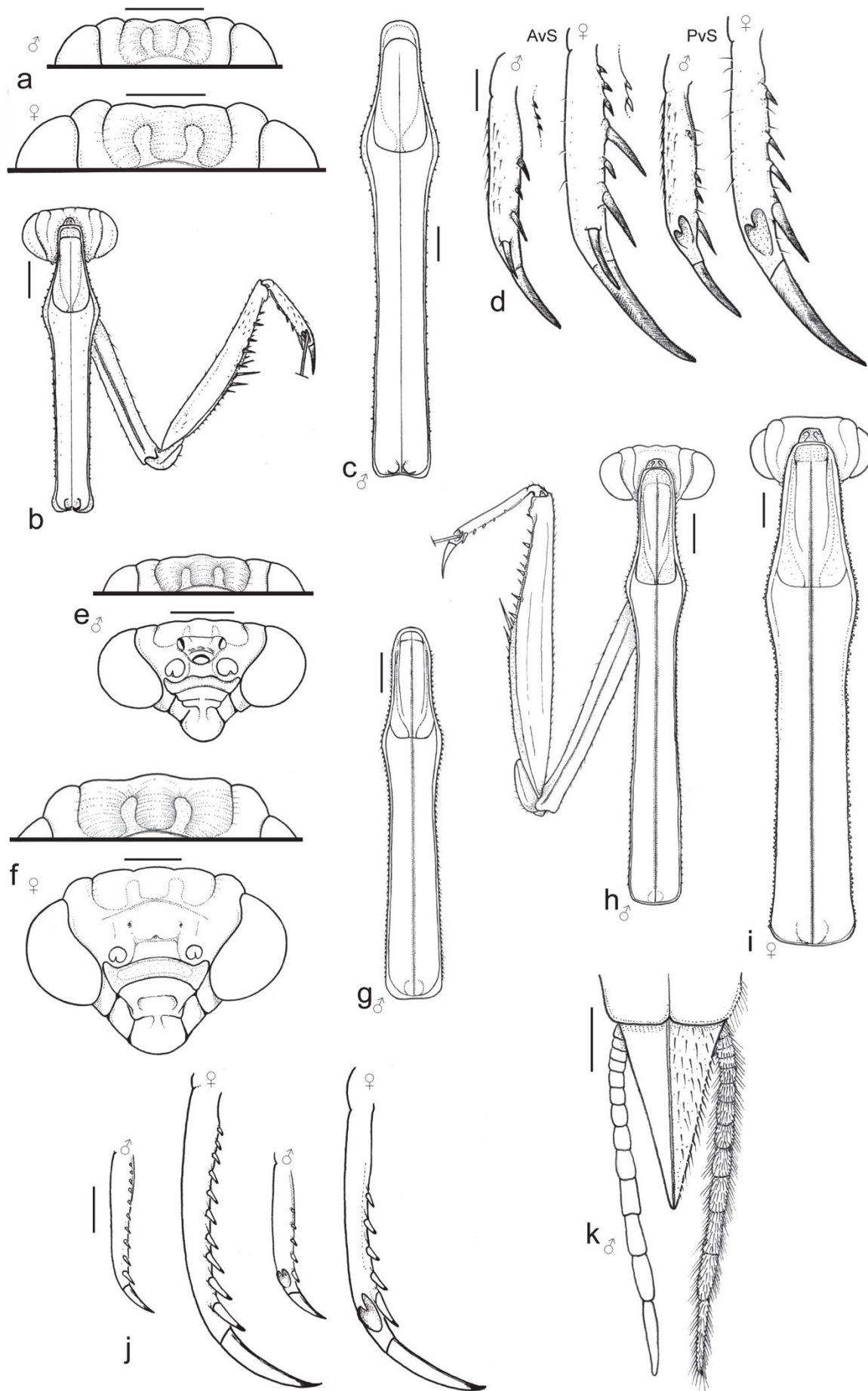


Figure 3.8

Figure 3.9. THESPIDAE - Thespinae, Thespini: (a) *Thesprotia graminis*, head and prothorax; (b) *Thesprotia* spp., foretibial spines; (c) *Thesprotiella* sp. n., detail of vertex; (d) *Thesprotiella* sp. n., detail of vertex; (e) *Thesprotiella peruana*, detail of vertex; (f) *Thesprotiella* sp. n., head and pronotum; (g) *Thesprotiella peruana*, pronotum, (h) *Thesprotiella peruana*, pronotum; (i) *Thesprotiella latens*, pronotum; (j) *Thesprotiella* sp. n., head and prothorax; (k) *Thesprotiella* spp., foretibial spines; (l) *Thesprotiella* sp. n., abdomen; (m) *Thesprotiella ceratophora*, abdomen; (n) *Thesprotiella tristani*, abdomen (redrawn from Rehn 1918). All scale bars = 1 mm, except Fig. 9k = 0.5 mm. (Fig. 9 l–n not to scale).

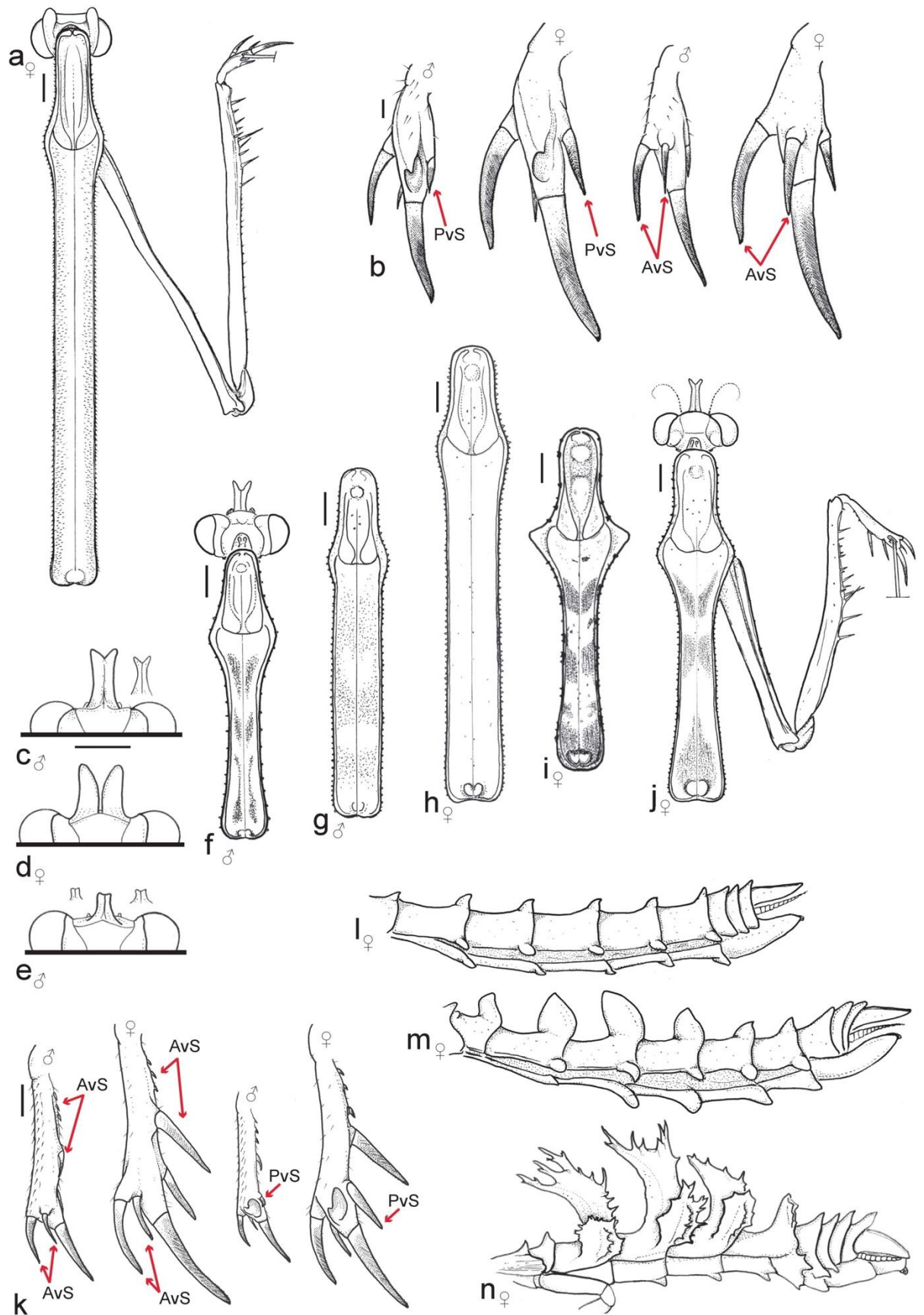


Figure 3.9

Figure 3.10. ANGELIDAE, COPTOPTERYGIDAE, LITURGUSIDAE and PHOTINAIDAE: (a) *Angela guianensis*, head; (b) *Angela guianensis*, pronotum; (c) *Angela guianensis*, pronotum; (d) *Angela armata*; (e) *Angela purpurascens*, hindwing; (f) *Angela guianensis*, hindwing; (g) *Angela miranda*, hindwing; (h) *Angela quinquemaculata*; (i) *Brunneria borealis*, head; (j) *Brunneria brasiliensis*; (k) *Coptopteryx gayi*; (l) *Hagiomantis superba*, pronotum; (m) *Hagiomantis superba*, forecoxa; (n) *Hagiomantis superba*, forefemur; (o) *Liturgusa maya*, dorsal habitus (redrawn from Svenson 2014b); (p) *Photiomantis planicephala*, pronotum; (q) *Photiomantis planicephala*; (r) *Cardioptera squalodon*, head and prothorax; (s) *Cardioptera brachyptera*, pronotum; (t) *Chromatophotina* sp. n., head; (u) *Chromatophotina* sp. n., head; (v) *Chromatophotina awajun*, pronotum (from Rivera, 2010a); *Chromatophotina awajun*, pronotum (from Rivera, 2010a); (x) *Microphotina* sp., head; (y) Genus A, pronotum; (z) Genus A, ventral phallomere and apofisi phalloide; (aa) *Orthoderella elongata*, head; (ab) *Orthoderella elongata*, pronotum; (ac) *Paraphotina* sp., pronotum; (ad) *Hicetia goeldiana*, pronotum; (ae) *Photina pilosella*, head and pronotum; (af) *Photina boliviana*, pronotum; (ag) *Photina brevis*, head and pronotum; (ah) *Photina occidentalis*. Scale bars = 1 mm (Fig. 10 e–h, z, ad not to scale).

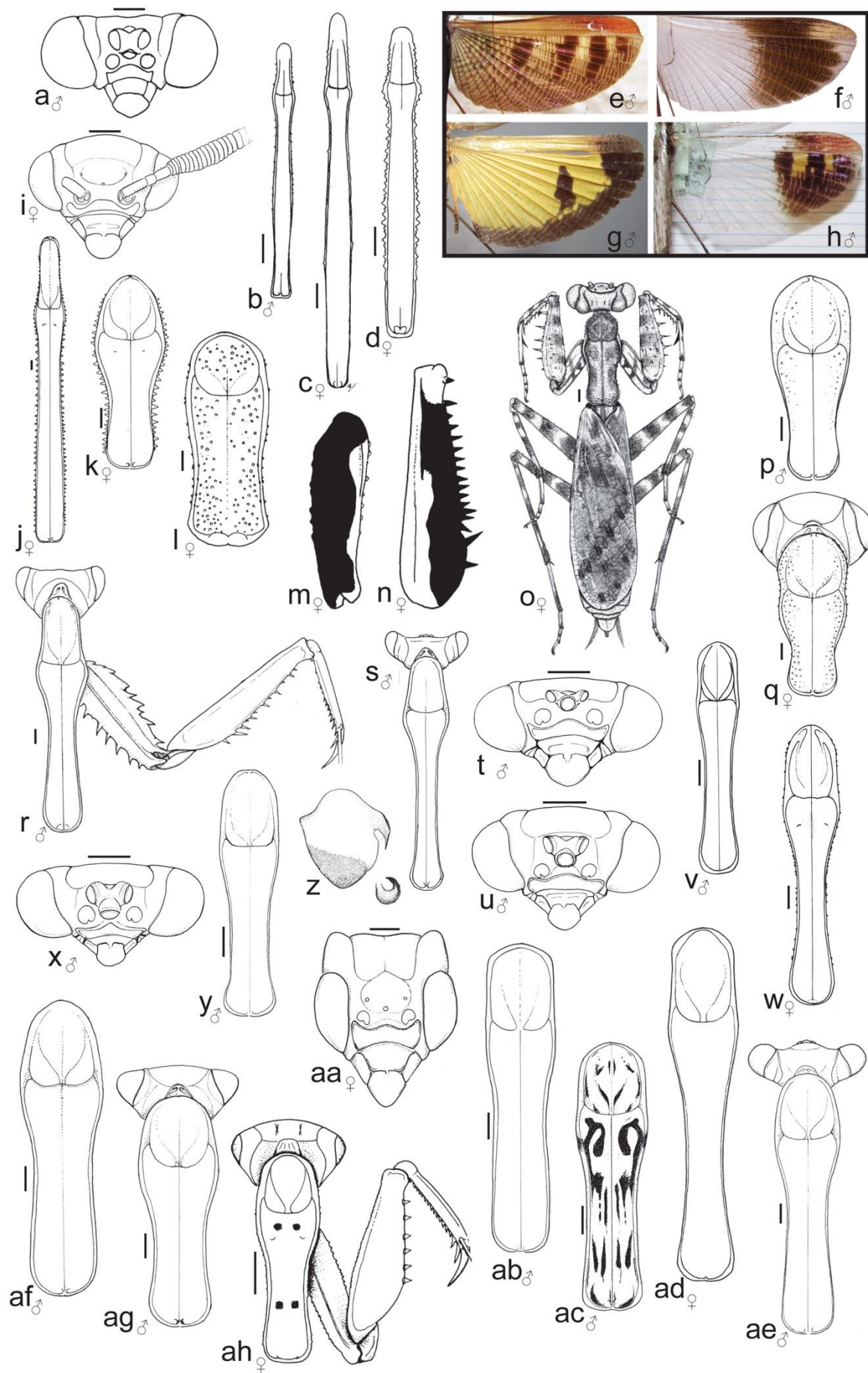


Figure 3.10

Figure 3.11. ACONTISTIDAE: (a) *Acontista amoenula*, head; (b) *Acontista amoenula*, head; (c) *Acontista amoenula*, pronotum; (d) *Acontista amoenula*; (e) *Acontista concinna*; (f) *Acontista concinna*, head and prothorax; (g) *Acontista cordillerae*, head; (h) *Acontista cordillerae*, head; (i) *Acontista cordillerae*, pronotum; (j) *Acontista cordillerae*, head and prothorax; (k) *Acontista cordillerae*, pronotum; (l) *Acontista cordillerae*; (m) *Acontista cordillerae*, pronotum; (n) *Acontista fraterna*, head; (o) *Acontista fraterna*, pronotum; (p) *Acontista fraterna*, head and prothorax; (q) *Acontista multicolor*, pronotum; (r) *Acontista multicolor*, head and prothorax; (s) *Acontista multicolor*, pronotum. Scale bars = 1mm.

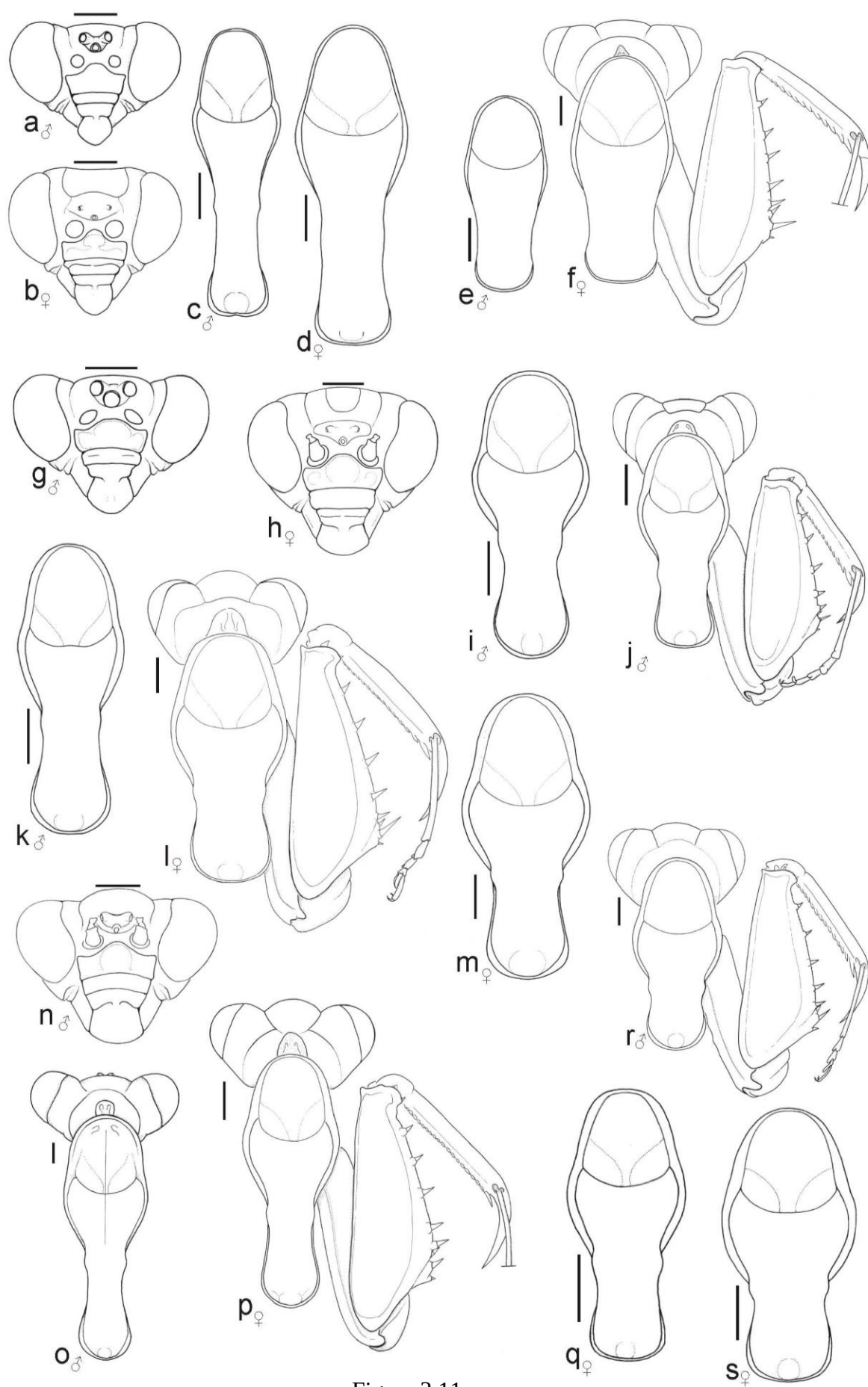


Figure 3.11

Figure 3.12. ACONTISTIDAE: (a) *Callibia diana*, head; (b) *Callibia diana*, head; (c) *Callibia diana*, mid femur; (d) *Metaphotina bimaculata*, head; (e) *Metaphotina bimaculata*, head; (f) *Metaphotina bimaculata*, head and pronotum; (g) *Metaphotina bimaculata*, pronotum; (h) *Metaphotina bimaculata*, pronotum; (i) *Metaphotina bimaculata*, pronotum; (j) *Metaphotina bimaculata*, pronotum; (k) *Raptrix perspicua*, head; (l) *Raptrix perspicua*, head; (m) *Raptrix perspicua*, head and prothorax; (n) *Raptrix westwoodi*, pronotum; (o) *Raptrix perspicua*, pronotum; (p) *Raptrix westwoodi*, pronotum; (q) *Tithrone roseipennis*, head; (r) *Tithrone roseipennis*, head; (s) *Tithrone roseipennis*, pronotum; (t) *Tithrone roseipennis*, pronotum. Scale bars = 1mm.

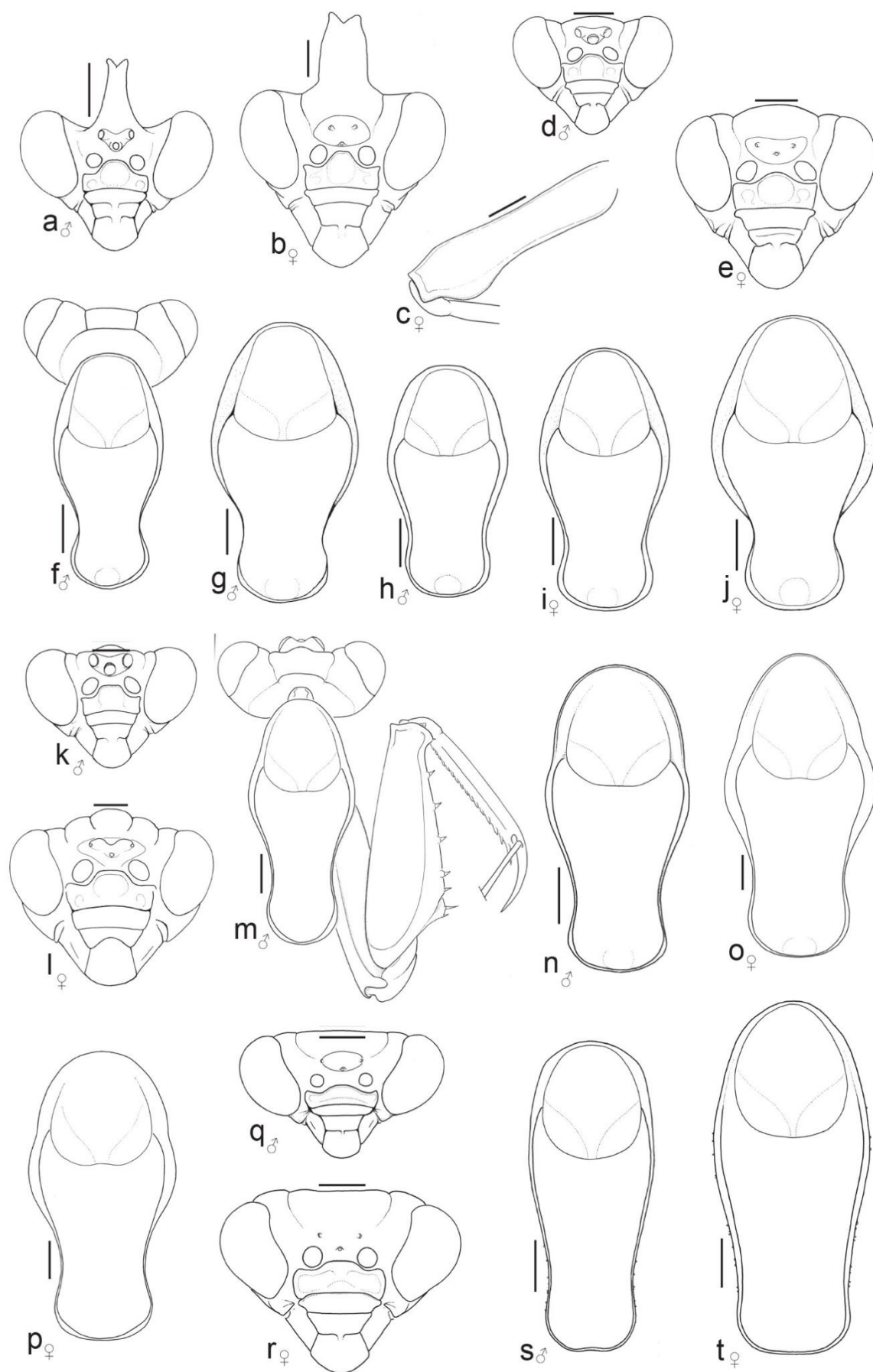


Figure 3.12

Figure 3.13. ACANTHOPIDAE: (a) *Acanthops fuscifolia*, head; (b) *Acanthops parafalcata*, head; (c) *Acanthops erosula*, head and prothorax; (d) *Acanthops royi*, pronotum; (e) *Acanthops falcata*, pronotum; (f) *Acanthops parafalcata*, detail of costal margin of forewing; (g) *Acanthops royi*, detail of costal margin of forewing; (h) *Acanthops erosa*, detail of costal margin of forewing; (i) *Decimiana bolivari*, detail of costal margin of forewing; (j) *Decimiana tessellata*, detail of costal margin of forewing; (k) *Metilia brunneri*, head; (l) *Metilia* cf. *amazonica*; (m) *Metilia adusta*, head; (n) *Metilia brunneri*, head and pronotum; (o) *Metilia brunneri*, pronotum; (p) *Metilia* sp., dorsal habitus; (q) *Metilia* sp., detail of costal margin of forewing; (r) *Metilia* sp., detail of apex of hindwing; (s) *Metilia* sp. n., detail of costal margin of forewing; (t) *Metilia* sp. n., detail of apex of hindwing; (u) *Plesiakanthops* sp., lower genicular lobe of mid femur; (v) *Acanthops parafalcata*, lower genicular lobe of mid femur; (w) *Acanthops erosula*, lower genicular lobe of mid femur; (x) *Metilia* sp. n., lower genicular lobe of mid femur; (y) *Decimiana tessellata*, lower genicular lobe of mid femur; (z) *Plesiakanthops eseejja*, detail of costal margin of forewing; (aa) *Plesiakanthops eseejja*, forewing; (ab) *Pseudacanthops lobipes*, head; (ac) *Pseudacanthops lobipes*, head; (ad) *Pseudacanthops lobipes*, head and prothorax; (ae) *Pseudacanthops lobipes*, head and pronotum; (af) *Pseudacanthops lobipes*, detail of costal margin of forewing; (ag) *Pseudacanthops lobipes*, detail of costal margin of forewing. Abbreviation: gl, genicular lobe. All scale bars = 2.5 mm, except 13a-d, k-m, u-y, ab-ac = 1mm.

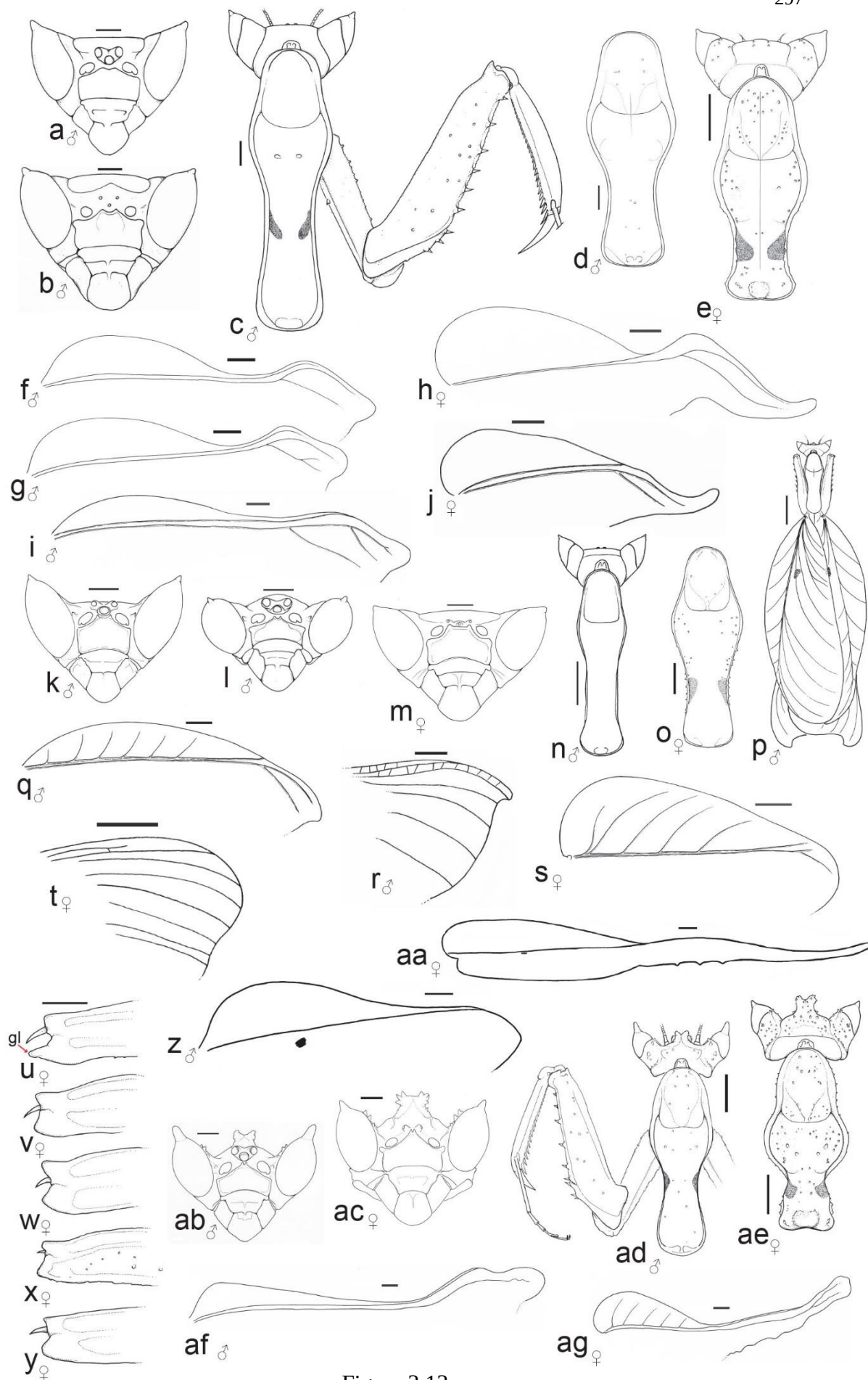


Figure 3.13

Figure 3.14. THESPIDAE (Pseudopogonogastrinae and Pseudomiopteryginae) - type specimens: (a) *Calopteromantis hebari* Terra, 1982* [now *Pseudopogonogaster hebari*], male holotype; (b) *Pseudomiopteryx bogotensis* Saussure, 1870, male holotype; (c) *Pseudomiopteryx columbica*, male holotype; (d) *Pseudomiopteryx decipiens*, male syntype (non holotype); (e) *Pseudomiopteryx decipiens*, female syntype; (f) *Pseudomiopteryx decipiens*, female syntype; (g) *Pseudomiopteryx guyanensis*, male syntype. Scale bar = 10 mm.



Figure 3.14

Figure 3.15. THESPIDAE (Pseudomiopteryginae) - type specimens: (a) *Pseudomiopteryx guyanensis*, female syntype; (b) *Pseudomiopteryx infuscata*, male syntype; (c) *Pseudomiopteryx infuscata*, male syntype; (d) *Pseudomiopteryx meridana*, male holotype; (e) *Pseudomiopteryx spinifrons*, male holotype. Scale bar = 10 mm.



Figure 3.15

Figure 3.16. THESPIDAE (Pseudomiopteryginae and Bantiinae) - type specimens: (a) *Promiopteryx fallax*, male holotype; (b) *Miopteryx granadensis* [*Promiopteryx granadensis*], male holotype; (c) *Mionicella chopardi* [now *Bantia chopardi*], male holotype; (d) *Bantia fusca*, male holotype; (e) *Bantia metzi*, male holotype; (f) *Bantia michaelesi*, male holotype. Scale bar = 10 mm.

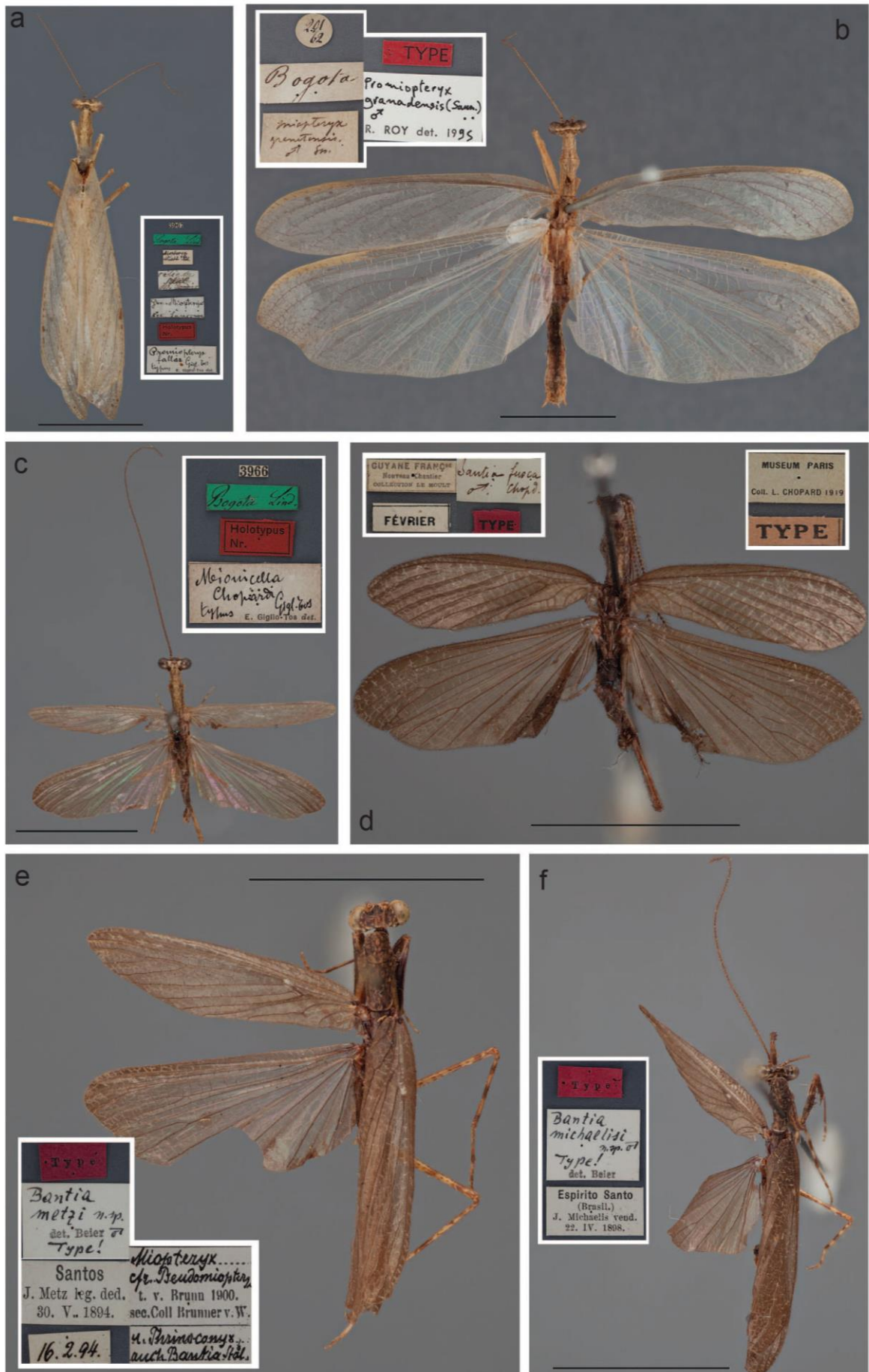


Figure 3.16

Figure 3.17. THESPIDAE (Bantiinae) – type specimens: (a) *Bantia marmorata*, female holotype; (b) *Musoniella nana* [now *Bantia nana*], male holotype; (c) *Bantiella pallida*, male holotype; (d) *Miopteryx simoni* [now *Bantiella simoni*], male holotype; (e) *Bantiella trinitatis*, male syntype; (f) *Bantiella trinitatis*, female syntype; (g) *Bantiella columbina* [n. syn. of *Bantiella trinitatis*], male holotype; (h) *Bantiella fusca* [n. syn. of *Bantiella trinitatis*], male syntype; (i) *Bantiella fusca* [n. syn. of *Bantiella trinitatis*], female syntype. Scale bar = 10 mm.

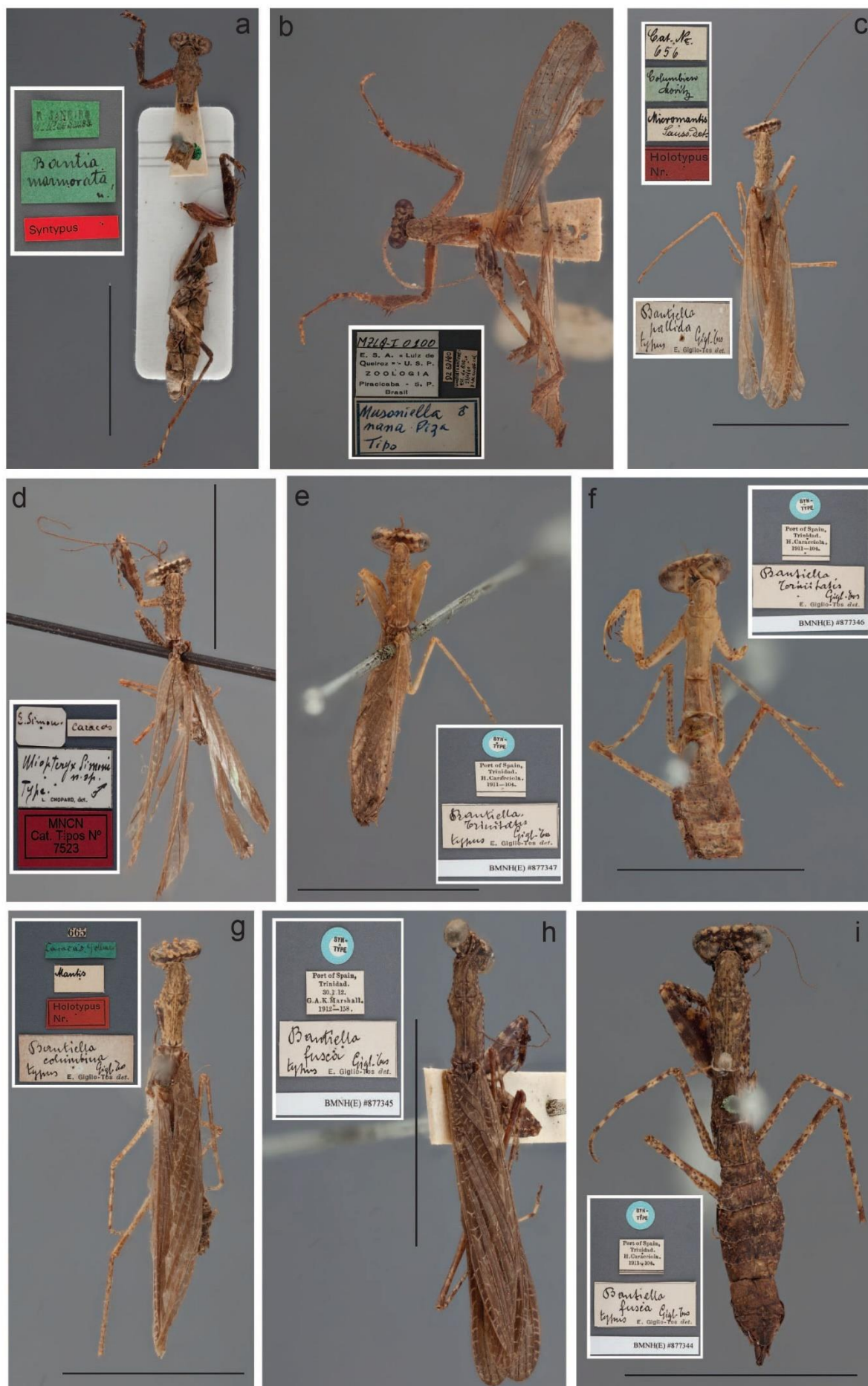


Figure 3.17

Figure 3.18. THESPIDAE (Bantiinae) – type specimens: (a) *Diabantia minima*, male holotype; (b) *Mantellias pubicornis*, male lectotype; (c) *Mantillica nigricans*, male holotype; (d) *Thrinaconyx fumosa*, male syntype; (e) *Thrinaconyx fumosa*, male syntype; (f) *Mantillica sialidea* [now *Thrinaconyx sialidea*], male holotype. Scale bar = 10 mm.



Figure 3.18

Figure 3.19. THESPIDAE (Miobantiinae) – type specimens: (a) *Anamiopteryx grandis*, male holotype; (b) *Anamiopteryx grandis*, female paratype; (c) *Trachymiopteryx tuberculata* (now *Anamiopteryx tuberculata*), male holotype; (d) *Emboicy mirim* [now *Chloromiopteryx mirim*], female holotype; (e) *Mantis (Photina) thalassina* [now *Chloromiopteryx thalassina*], lectotype male; (f) *Piracicaba bicentenaria* [now *Eumiopteryx bicentenaria*], male holotype; (g) *Eumiopteryx laticollis*, male holotype; (h) *Eumiopteryx laticollis*, female allotype. Scale bar = 10 mm.

Figure 3.20. THESPIDAE (Miobantiinae) – type specimens: (a) *Promiopteryx punctata* [now *Miobantia punctata*], (b) *Paradiabantia perparva*, male holotype; (c) *Eumusonia intermedia*, male syntype; (d) *Eumusonia viridis*, male holotype; (e) *Leptomiopteryx dispar*, male syntype; (f) *Leptomiopteryx dispar*, female syntype; (g) *Leptomiopteryx dispar*, male syntype. Scale bar = 10 mm.

Figure 3.21. THESPIDAE (Miobantiinae and Thespinae) – type specimens: (a) *Navimons amapaensis* [now *Leptomiopteryx dispar*], male holotype; (b) *Miopteryx argentina* [now *Musoniella argentina*], male syntype; (c) *Miopteryx laevithorax* [now *Musoniella laevithorax*], male holotype; (d) *Musoniella chopardi*, male holotype; (e) *Musoniella parva*, male holotype; (f) *Musoniella affinis*, male syntype; (g) *Musonia costalis* [now *Musoniella costalis*], male holotype; (h) *Eumusonia fragilis* [now *Musoniella fragilis*], male holotype; (i) *Musoniella longicauda*, male holotype. Scale bar = 10 mm.



Figure 3.21

Figure 3.22. THESPIDAE (Thespinae) – type specimens: (a) *Musoniella ipiranga*, male holotype; (b) *Promiopteryx stigmatica* [now *Musoniella stigmatica*], male holotype; (c) *Musoniella precaria* (n. syn. of *Musoniella stigmatica*), male holotype; (d) *Musoniella seabrai* [now *Pizaia seabrai*], male holotype; (e) *Galapagia amazonica* [now *Dougnonyx amazonicus*], female holotype; (f) *Galapagia amazonica* [now *Dougnonyx amazonicus*], female holotype, detail of head; (g) *Galapagia amazonica* [now *Dougnonyx amazonicus*], female holotype, pronotum. Scale bar = 10 mm.

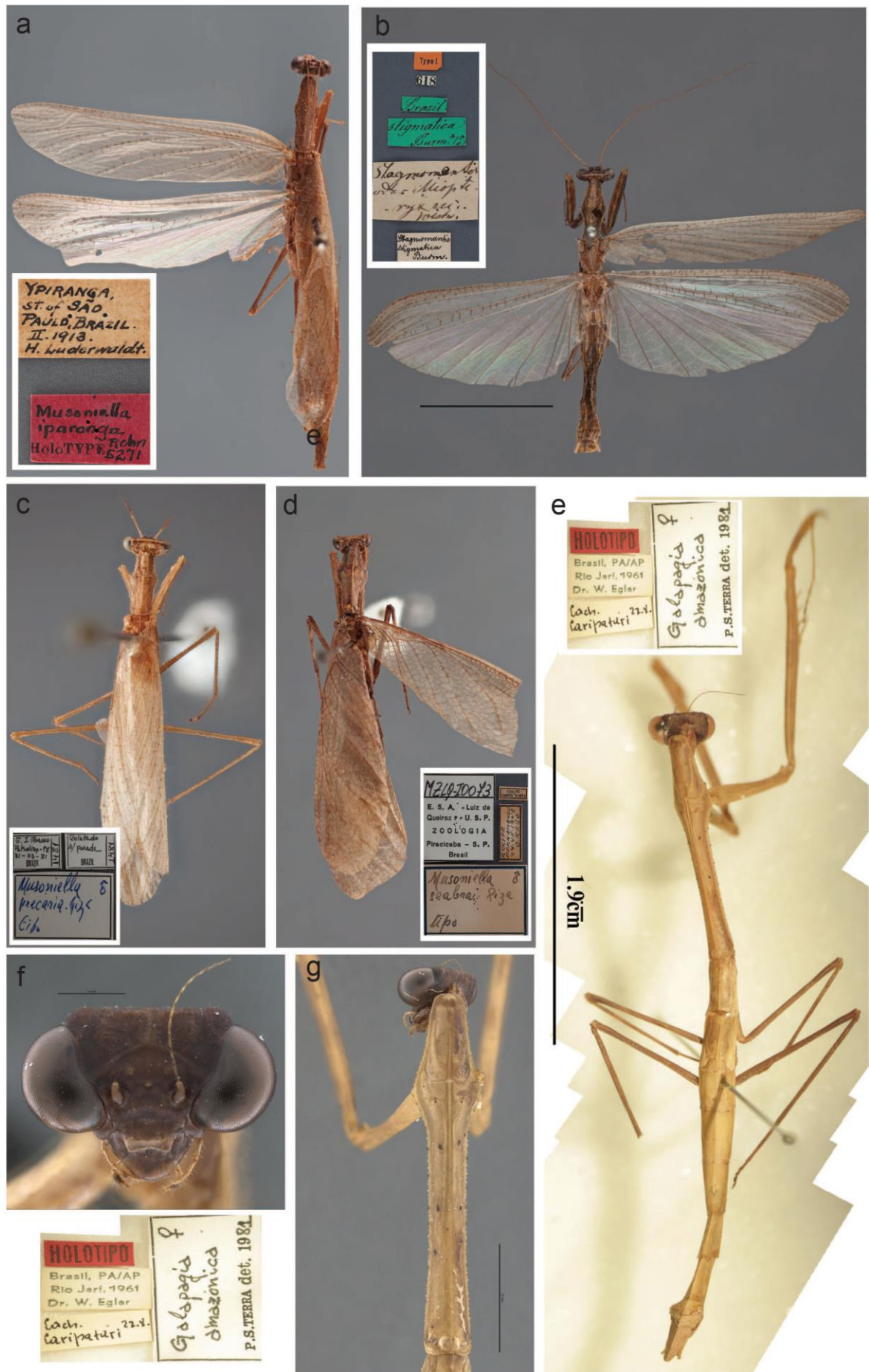


Figure 3.22

Figure 3.23. THESPIDAE (Thespinae) – type specimens: (a) *Mionyx maculosus* [now *Dougnonyx maculosus*], male syntype; (b) *Mionyx maculosus* [now *Dougnonyx maculosus*], male syntype; (c) *Mionyx maculosus* [now *Dougnonyx maculosus*], female syntype; (d) *Liguanea pediodromia*, female holotype; (e) *Liguanea pediodromia*, male paratype; (f) *Thespis conspersa* [now *Macromusonia conspersa*], male holotype; (g) *Musonia major* [now *Macromusonia major*], female syntype; (h) *Musonia major* [now *Macromusonia major*], male syntype. Scale bar = 10 mm.

Figure 3.24. THESPIDAE (Thespinae) – type specimens: (a) *Musonia femorata*, female holotype; (b) *Musonia fuscescens* [now *Musonia surinama*], male syntype; (c) *Musonia fuscescens* [now *Musonia surinama*], female syntype; (d) *Mionyx lineatus* [n. syn of *Musonia surinama*], female holotype; (e) *Mionyx fera* [now *Musoniola fera*], male holotype; (f) *Catamusonia minor* [now *Musoniola fera*], female syntype; (e) *Catamusonia minor* [now *Musoniola fera*], male syntype; *Musoniola vicina*, male holotype. Scale bar = 10 mm.



Figure 3.24

Figure 3.25. THESPIDAE (Thespinae) – type specimens: (a) *Oligonyx mexicanus* [now *Oligonicella mexicana*], male syntype; (b) *Oligonyx toltecus* [now *Oligonicella mexicana*], female holotype; (c) *Oligonyx punctulatus* [now *Oligonicella punctulata*], male holotype; (d) *Oligoxyx bollianus* [now *Oligonicella scudderi*], male syntype; (e) *Oligoxyx bollianus* [now *Oligonicella scudderi*], (f) female syntype; *Olygonyx striolatus* [now *Oligonicella striolata*], male syntype. Scale bar = 10 mm.



Figure 3.25

Figure 3.26. THESPIDAE (Thespinae) – type specimens: (a) *Oligonyx striolatus* [now *Oligonicella striolata*], male syntype(?); (b) *Oligonyx tesellatus* [now *Oligonicella tesellata*], male holotype; (c) *Spanionyx* (*Oligonyx*) *bicornis* [now *Oligonyx bicornis*], female holotype(?)/syntype(?); (d) *Harpagonyx gryps* [now *Oligonyx bicornis*], male syntype; (e) *Harpagonyx gryps* [now *Oligonyx bicornis*], male syntype(?); (f) *Spanionyx bidens* [now *Oligonyx bidens*], female holotype; (g) *Harpagonyx maya* [now *Oligonyx maya*], male holotype. Scale bar = 10 mm.

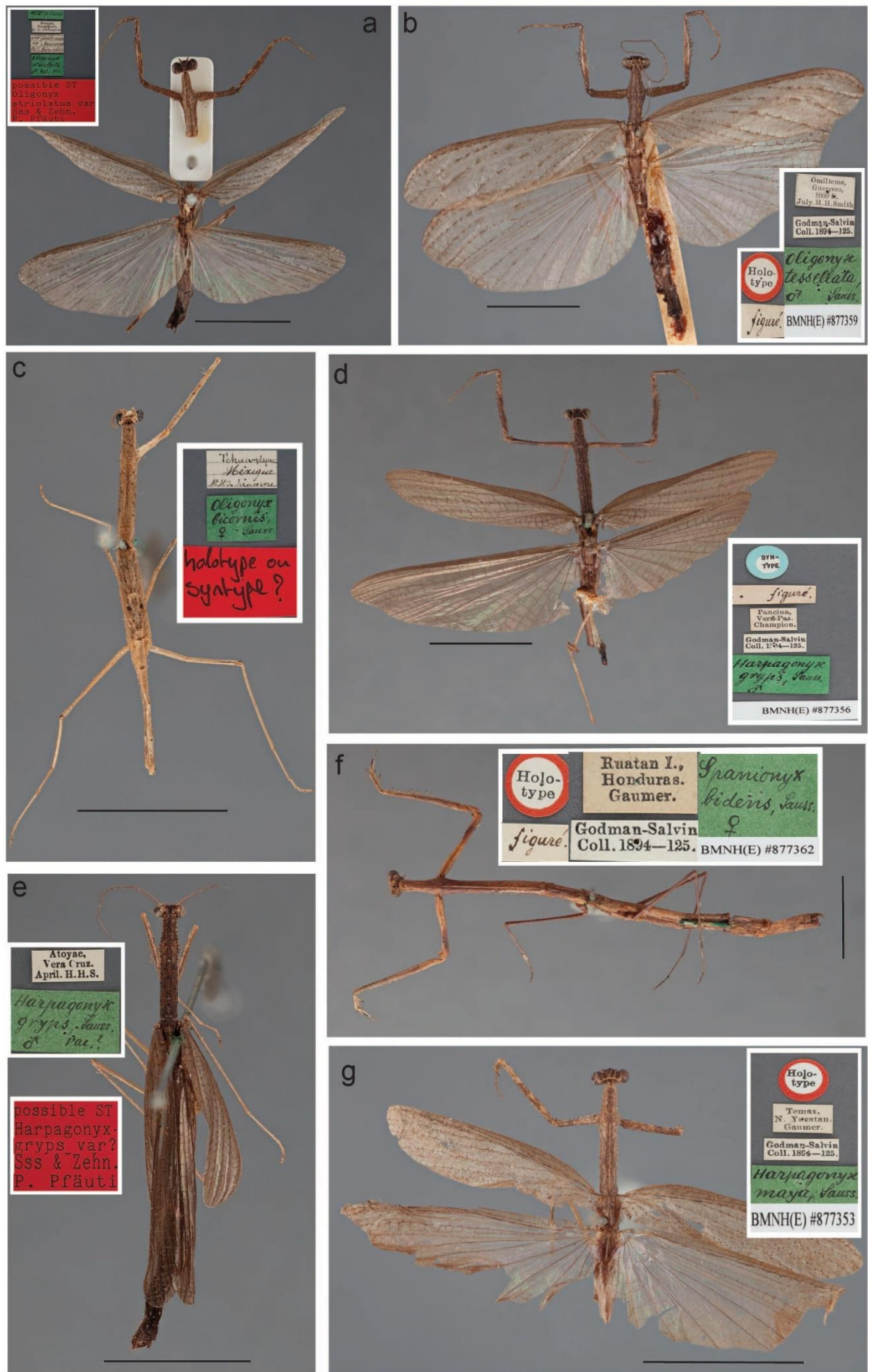


Figure 3.26

Figure 3.27. THESPIDAE (Thespinae) – type specimens: (a) *Galapagia peruana* [now *Piscomantis peruana*], female paratype; (b) *Mionyx saevus* [now *Pseudomusonia lineativentris*], male syntype; (c) *Mionyx saevus* [now *Pseudomusonia lineativentris*], male syntype; (d) *Thespis boliviana* [now *Musonia boliviana*], male holotype; (e) *Leptomiopteryx argentina* [n. syn. of *Thespis boliviana*], male holotype; (f) *Musonia cubensis* [now *Thespis cubensis*]. Scale bar = 10 mm.



Figure 3.27

Figure 3.28. THESPIDAE (Thespinae) – type specimens: (a) *Diamusonia media* [now *Thespis media*], female holotype; (b) *Thespis metae*, male holotype; (c) *Paramusonia seclusa* [now *Thespis seclusa*], male holotype; (d) *Thesprotia gigas*, female holotype; (e) *Thesprotia vidua* [now *Thesprotia infumata*], female holotype; (f) *Thesprotia insolita*, male holotype; (g) *Thesprotia macilenta*, male syntype; (h) *Thesprotia macilenta*, female syntype. Scale bar = 10 mm.



Figure 3.28

Figure 3.29. THESPIDAE (Thespinae) – type specimens: (a) *Thesprotia brasiliensis* [now *Thesprotia macilenta*]; (b) *Thesprotia maculata*, female holotype; (c) *Thesprotia pellucida*, male syntype (non holotype); (d) *Thesprotia pellucida*, male syntype; (e) *Thesprotia simplex*, male holotype; (f) *Carrikerella ceratophora* [now *Thesprotiella ceratophora*], female holotype; (g) *Carrikerella empusa* [now *Thesprotiella empusa*]; [h] *Spanionyx festae* [now *Thesprotiella festae*], male syntype (non holotype); (i) *Thesprotiella fronticornis*, male holotype. Scale bar = 10 mm.

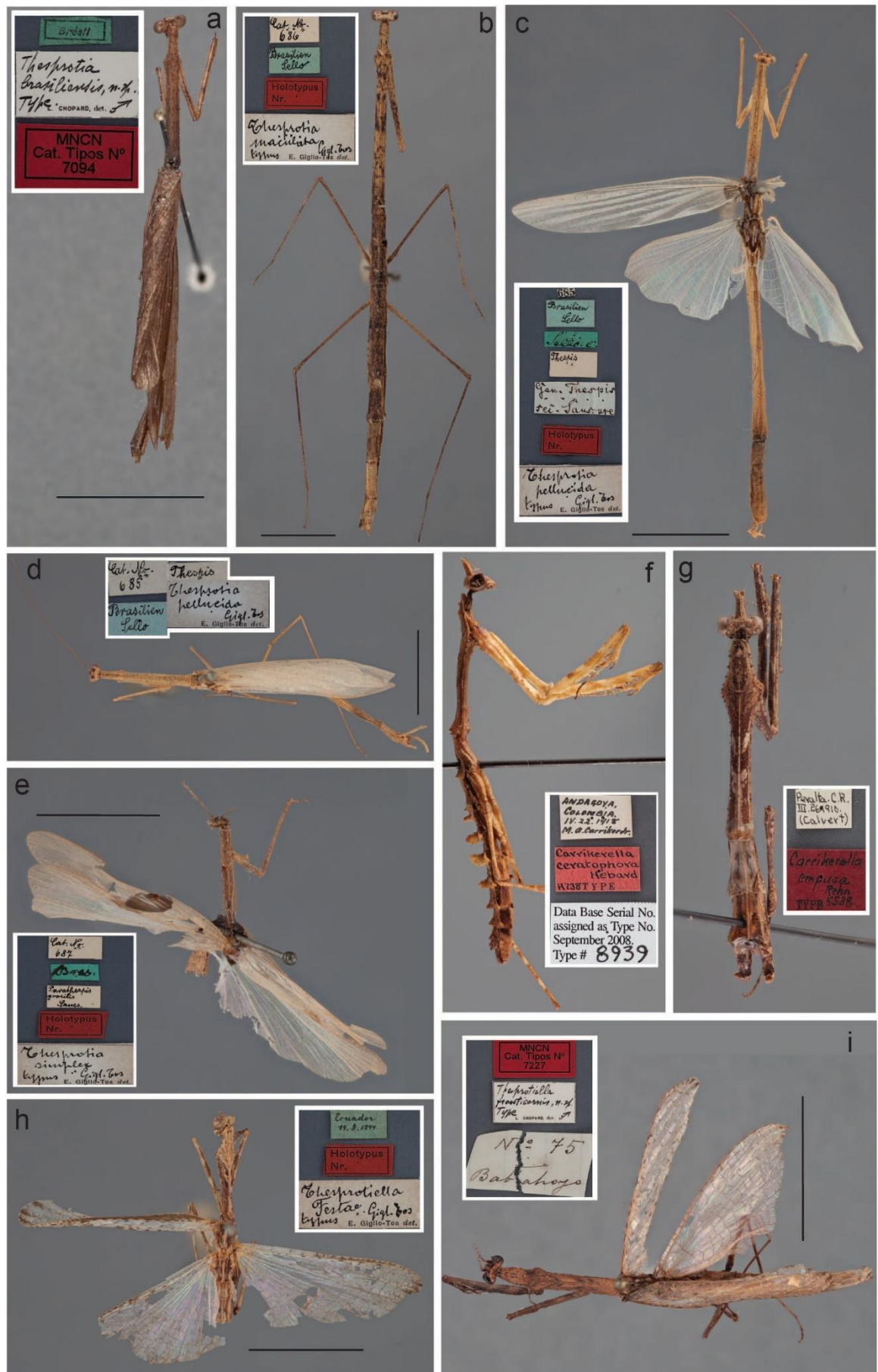


Figure 3.29

Figure 3.30. THESPIDAE (Thespinae) – type specimens: (a) *Oligonyx insularis* [now *Thesprotiella insularis*], male holotype; (b) *Oligonyx insularis* [now *Thesprotiella insularis*], female allotype; (c) *Pogonogaster tristani* [now *Thesprotiella tristani*], female holotype; (d) *Thesprotiella peruana*, male holotype. Scale bar = 10 mm.



Figure 3.30

Figure 3.31. LITURGUSIDAE and COPTOPTERYGIDAE – type specimens: (a) *Liturgusa superba* [now *Hagiomantis superba*], female holotype; (b) *Brunneria borealis*, female holotype; *Brunneria longa*, male syntype (non holotype); (c) *Brunneria longa*, female syntype (non allotype); (d) *Brunneria subaptera*, female holotype. Scale bar = 10 mm.



Figure 3.31

Figure 3.32. COPTOPTERYGIDAE – type specimens: (a) *Brunneria gracilis* [n. syn. of *Brunneria subaptera*], male syntype (non holotype); *Brunneria gracilis* [n. syn. of *Brunneria subaptera*], male syntype; (c) *Coptopteryx claraziana* [n. syn. of *Coptopteryx argentina*], male lectotype (here designated, see text); (d) *Coptopteryx cf. gayi*, female paralectotype (here designated) of *Coptopteryx claraziana* (see text); (e) *Coptopteryx viridis* [n. syn. of *Coptopteryx argentina*], male lectotype (here designated, see text); (f) unidentified Coptopterygidae, paralectotype (here designated, see text) of *Coptopteryx viridis*. Scale bar = 10 mm.



Figure 3.32

Figure 3.33. COPTOPTERYGIDAE – type specimens: (a) *Coptopteryx argentina*, female paralectotype (here designated, see text) of *Coptopteryx viridis*; (b) *Coptopteryx brevipennis*, male holotype; (c) *Coptopteryx constricta*, female holotype; (d) *Coptopteryx constricta*, male allotype; (e) *Coptopteryx rehni* [now *Coptopteryx gayi*], male syntype (non holotype); (f) *Coptopteryx minuta* [now *Coptopteryx gayi*], female holotype. Scale bar = 10 mm.



Figure 3.33

Figure 3.34. COPTOPTERYGIDAE – type specimens: (a) *Coptopteryx inermis*, female holotype; (b) *Coptopteryx parva*, female holotype; (c) *Coptopteryx pumila*, male holotype; (d) *Coptopteryx thoracica*, female holotype; (e) *Coptopteryx fallax* [n. syn. of *Coptopteryx thoracica*], male syntype (non holotype); (f) *Coptopteryx thoracica*, male allotype (note: incorrectly labelled as the paratype of *C. constricta*); (g) *Coptopteryx fallax* [n. syn. of *Coptopteryx thoracica*], female syntype (non allotype). Scale bar = 10 mm.

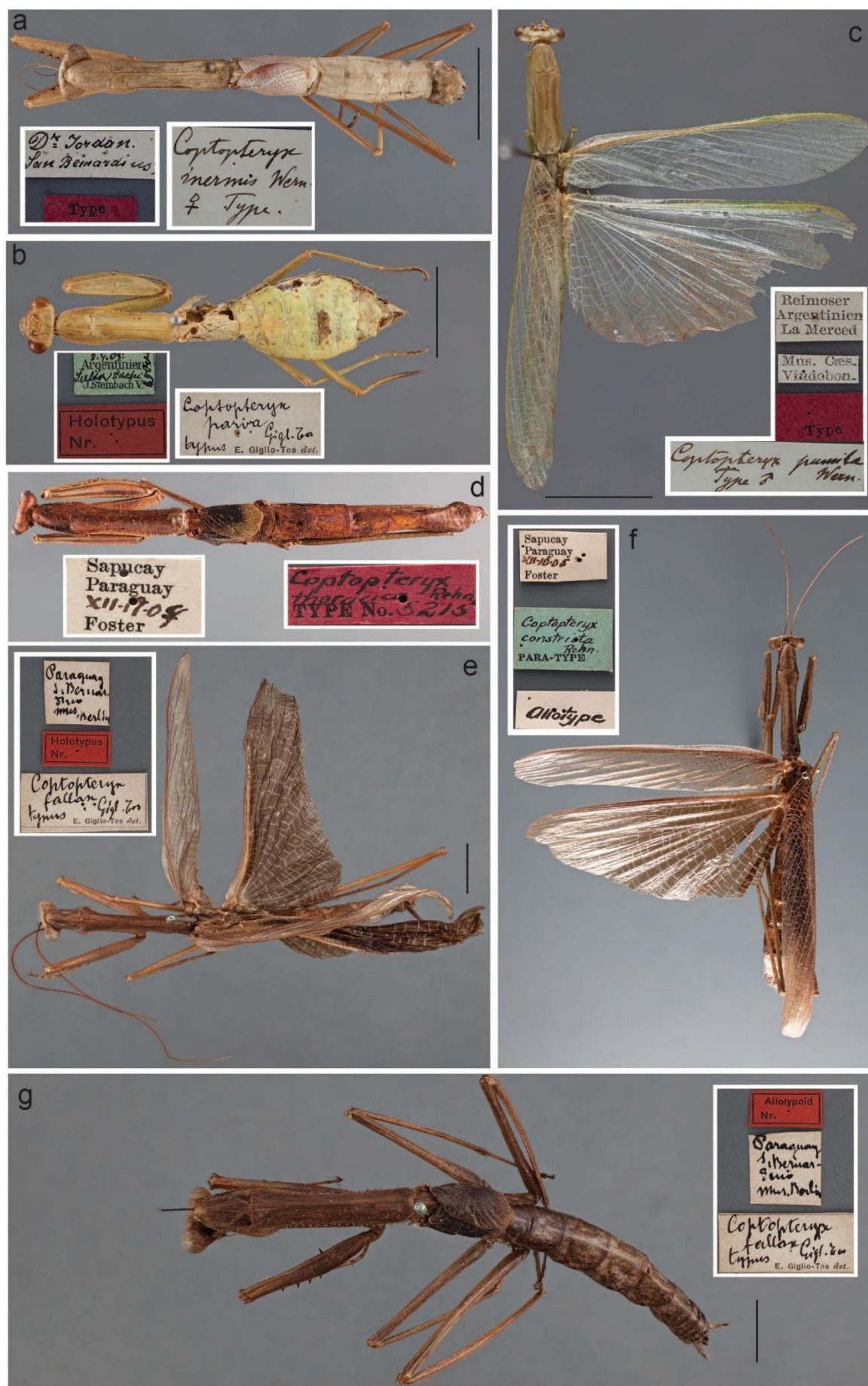


Figure 3.34

Figure 3.35. COPTOPTERYGIDAE – type specimens: (a) *Coptopteryx spinosa* [n. syn. of *Coptopteryx thoracica*], male lectotype (here designated, see text); (b) *Coptopteryx gayi*, female paralectotype (here designated, see text) of *Coptopteryx spinosa*; (c) *Coptopteryx thoracoides*, male syntype; (d) *Coptopteryx thoracoides*, male syntype. Scale bar = 10 mm.



Figure 3.35

Figure 3.36. PHOTINAIDAE (Cardiopterinae) – type specimens: (a) *Cardioptera brachyptera*, female syntype; (b) *Cardioptera brachyptera*, male syntype; (c) *Cardioptera minor*, female holotype; (d) *Cardioptera parva*, female allotype; (e) *Cardioptera parva*, male holotype; (f) *Cardioptera viridipennis*, male holotype. Scale bar = 10 mm.

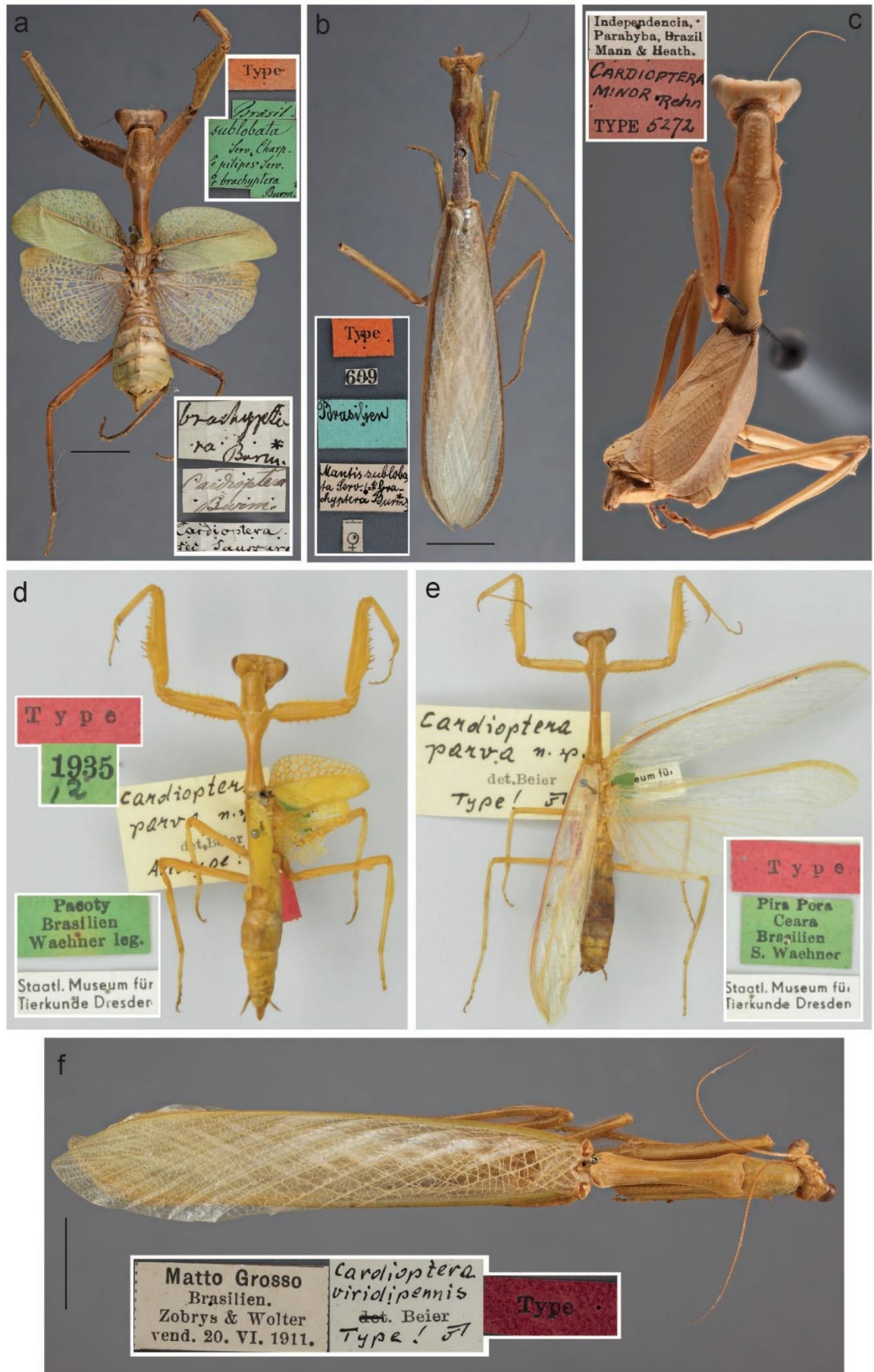


Figure 3.36

Figure 3.37. PHOTINAIDAE (Photinainae) – type specimens: (a) *Cardioptera reticulata* [now *Paraphotina reticulata*], male holotype; (b) *Paraphotina venezuelana*, male holotype; (c) *Hicetia goeldiana*, female holotype; (d) *Photina* (*Metriomantis*) *gracilicollis* [now *Hicetia goeldiana*], male syntype; (e) *Photina* (*Metriomantis*) *gracilicollis* [now *Hicetia goeldiana*], female syntype; *Parastagmatoptera amazonica* [now *Photina amazonica*], female holotype. Scale bar = 10 mm.



Figure 3.37

Figure 3.38. PHOTINAIDAE (Photinainae) – type specimens: (a) *Photina brevis*, male holotype; *Cardioptera cupido* [now *Photina cupido*], female holotype; *Metriomantis ovata* [now *Photina ovata*], holotype?; (d) *Metriomantis paraensis* [now *Photina paraensis*], female holotype; (e) *Metriomantis paraensis* [now *Photina paraensis*], female holotype; (f) *Photina pilosa*, male syntype. Scale bar = 10 mm.



Figure 3.38

Figure 3.39. PHOTINAIDAE (Photinainae) – type specimens: (a) *Metriomantis pilosella* [now *Photina pilosella*], male holotype; (b) *Metriomantis pilosella* [now *Photina pilosella*], male holotype; (c) *Mantis (Cardioptera) reticulata* [now *Photina reticulata*], female holotype; (d) *Metriomantis biramosa* [n. syn. of *Photina reticulata*], female syntype; (e) *Mantis (Photina) vitrea* [now *Photina vitrea*], holotype. Scale bar = 10 mm.

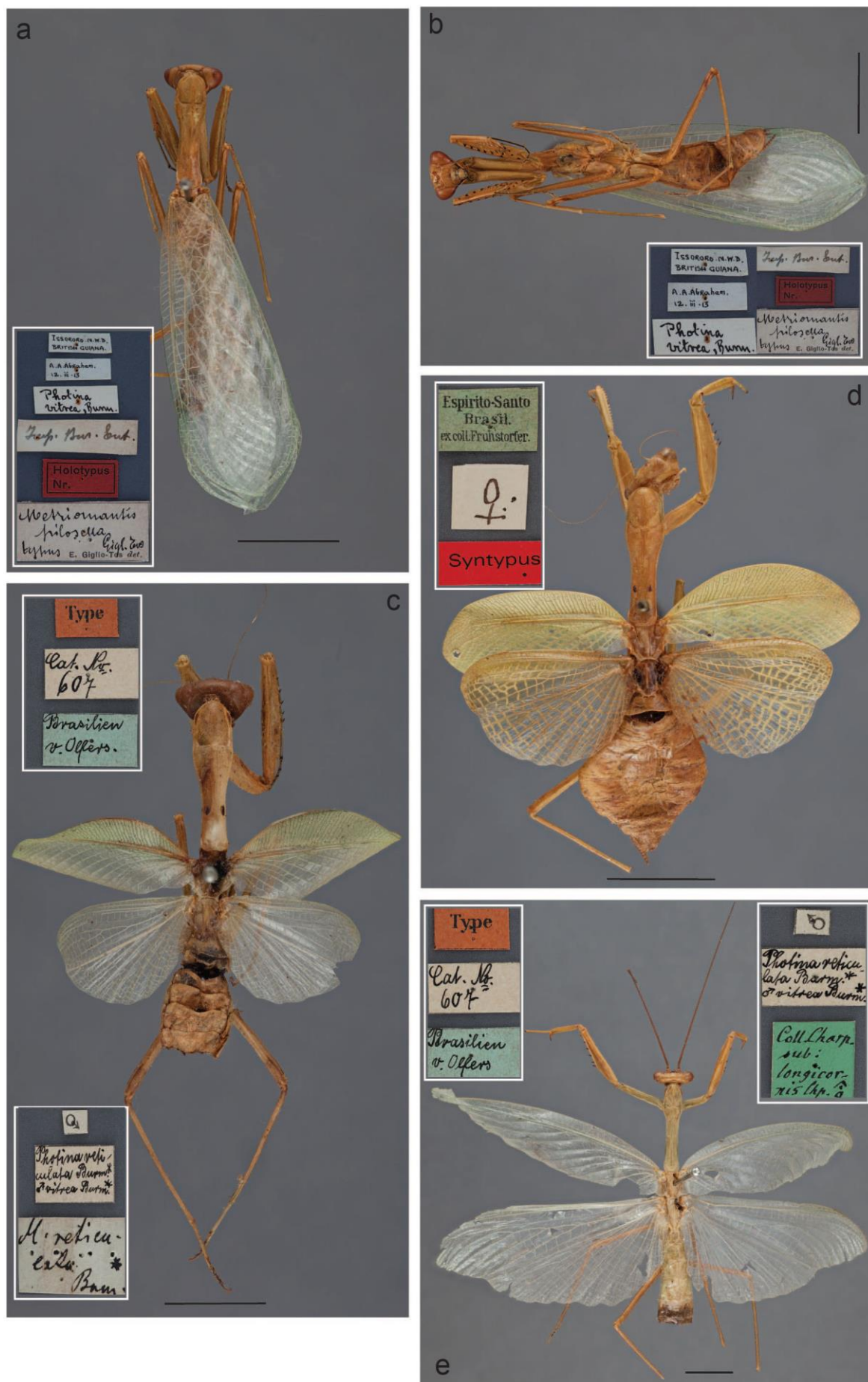


Figure 3.39

Figure 3.40. STENOPHYLLIDAE and ACONTISTIDAE – type specimens: (a) *Stenophylla cornigera*, male holotype; (b) *Stenophylla gallardi*, female holotype; (c) *Acontistella amazonica* [now *Acontista amazonica*], male holotype; (d) *Acontista amoenula*, female holotype; (e) *Acontistes semirufa* [now *Acontista amoenula*], male holotype; (f) *Acontista cayennensis*, female holotype. Scale bar = 10 mm.

Figure 3.41. ACONTISTIDAE – (a) *Acontista maroniensis* [n. syn. of *Acontista cayennensis*], male syntype; (b) *Acontista maroniensis* [n. syn. of *Acontista cayennensis*], female syntype; (c) *Acontista maroniensis* [n. syn. of *Acontista cayennensis*], female syntype; (d) *Mantis concinna* [now *Acontista concinna*], female lectotype; (e) *Acontista cordillerae*, male holotype; (f) *Acontista mexicana* [now *Acontista cordillerae*], female syntype; (g) *Acontista mexicana* [now *Acontista cordillerae*], female syntype. Scale bar = 10 mm.



Figure 3.41

Figure 3.42. ACONTISTIDAE – (a) *Acontista inquinata*, [now *Acontista cordillerae*], male syntype; (b) *Acontista inquinata*, [now *Acontista cordillerae*], male syntype; [c] *Acontista vitrea* [now *Acontista cordillerae*], male syntype; [d] *Acontista vitrea* [now *Acontista cordillerae*], male syntype; [e] *Acontista championi* [now *Acontista cordillerae*], male holotype (note: *championi* is a replacement name for *Acontista quadrimaculata* (Audinet Serville, 1839); see Marshall, 1976); (f) *Acontiothespis iriodes* [n. syn. of *Acontista cordillerae*], male holotype; (g) *Acontista paraensis* [now *Acontista paraensis*], female holotype. Scale bar = 10 mm.

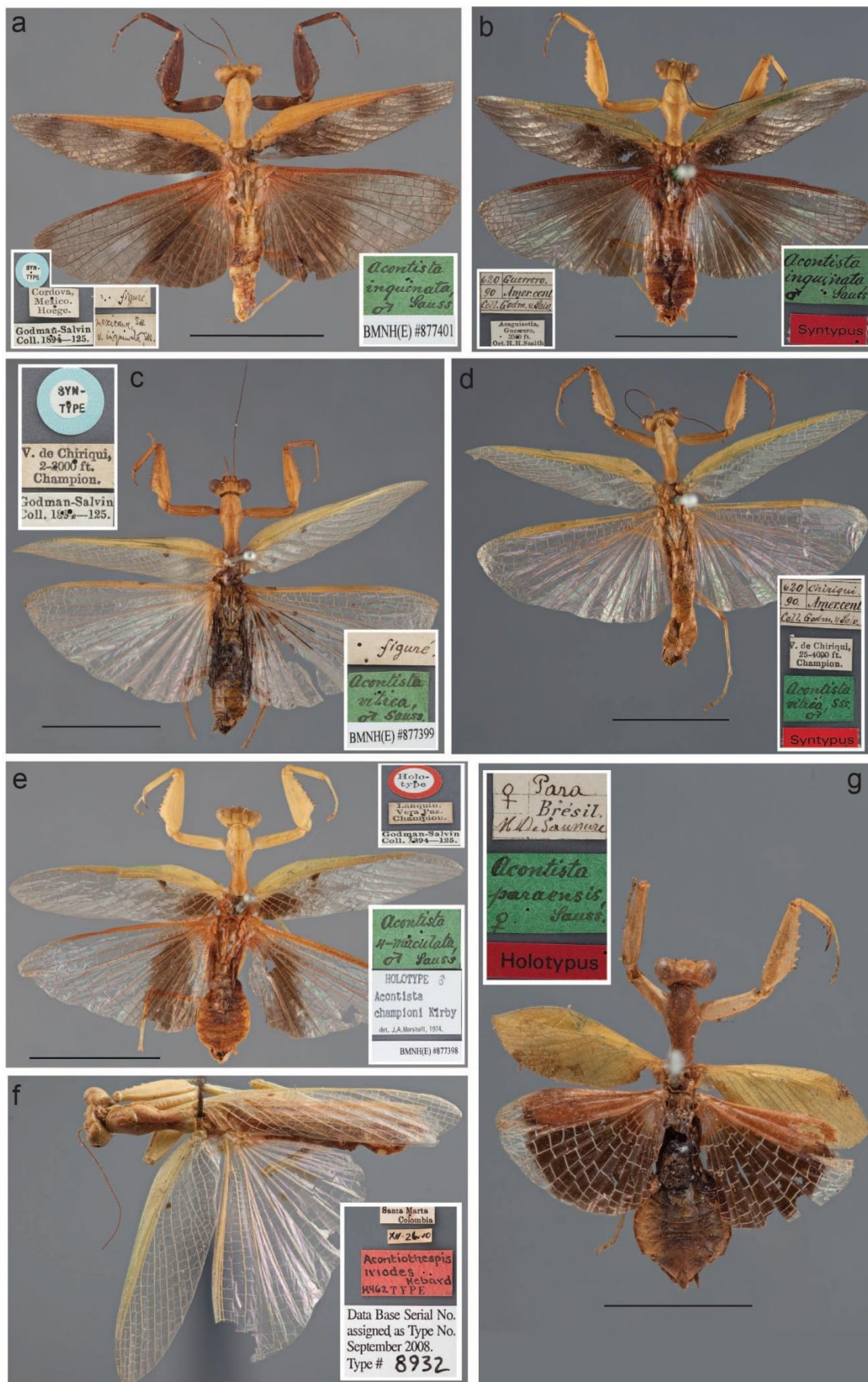


Figure 3.42

Figure 3.43. ACONTISTIDAE – (a) *Acontista fraterna*, female holotype; (b) *Acontiothespis ecuadorica* [n. syn. of *Acontista fraterna*], male holotype; (c) *Acontiothespis ecuadorica* [n. syn. of *Acontista fraterna*], female paratype; (d) *Acontistella violacea* [now *Acontista fraterna*], male holotype; (e) *Acontistella violacea* [now *Acontista fraterna*], male paratype; (f) *Acontista gracilis*, female syntype; (g) *Acontista gracilis*, male syntype. Scale bar = 10 mm.

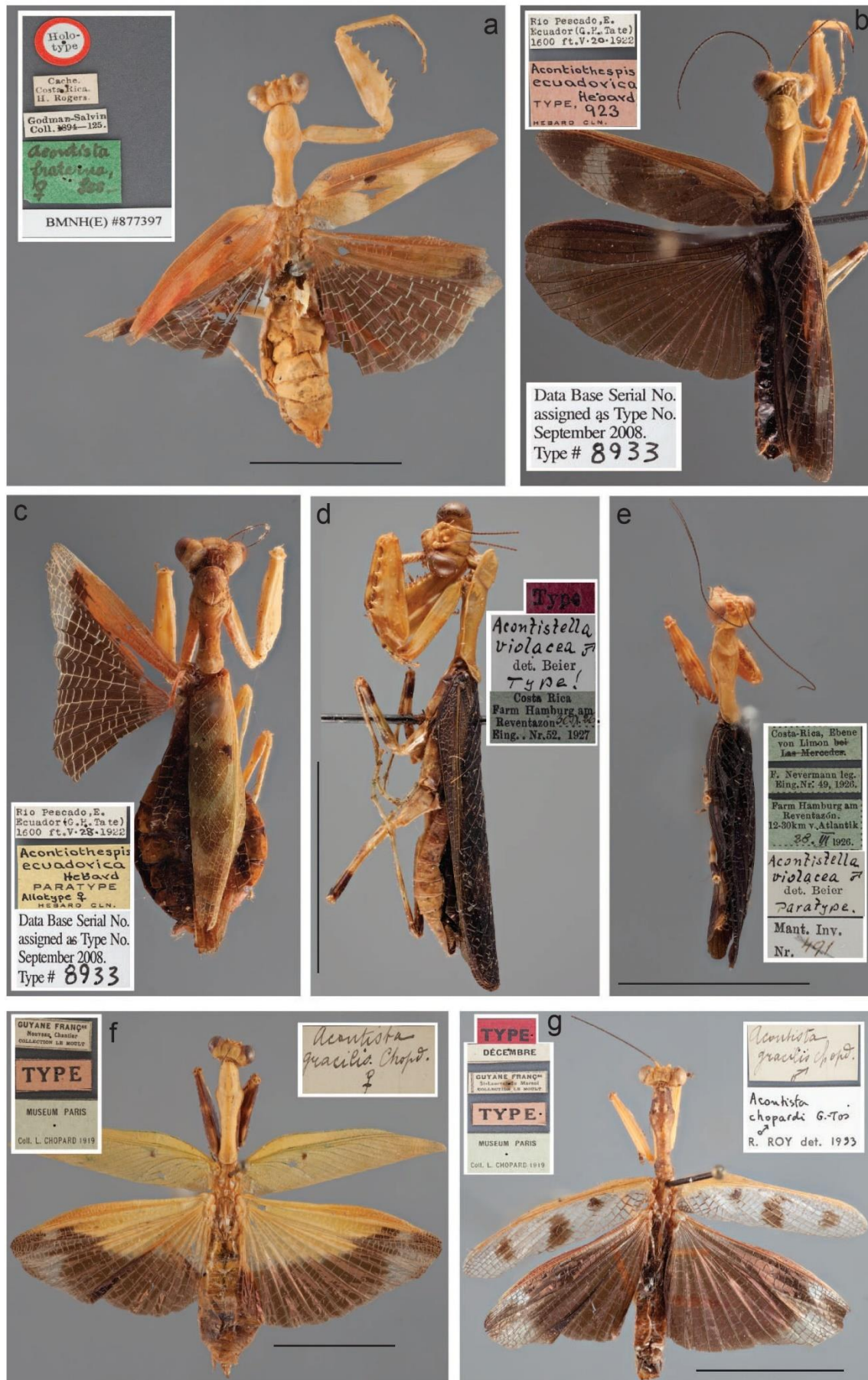


Figure 3.43

Figure 3.44. ACONTISTIDAE – (a) *Acontista minima*, male holotype; (b) *Acontista multicolor*, female holotype; (c) *Acontista bimaculata* [now *Metaphotina bimaculata*], male syntype; (d) *Acontista bimaculata* [now *Metaphotina bimaculata*], male syntype; (e) *Metaphotina piracicabensis*, male holotype; (f) *Acontista rehni* [now *Metaphotina rehni*], female holotype. Scale bar = 10 mm.



Figure 3.44

Figure 3.45. ACONTISTIDAE – (a) *Paratithrone royi*, male holotype; (b) *Paratithrone royi*, female allotype; (c) *Tithrone latipennis*, male holotype; (d) *Tithrone latipennis*, female allotype; (e) *Acontista roseipennis* [now *Tithrone roseipennis*], female holotype. Scale bar = 10 mm.

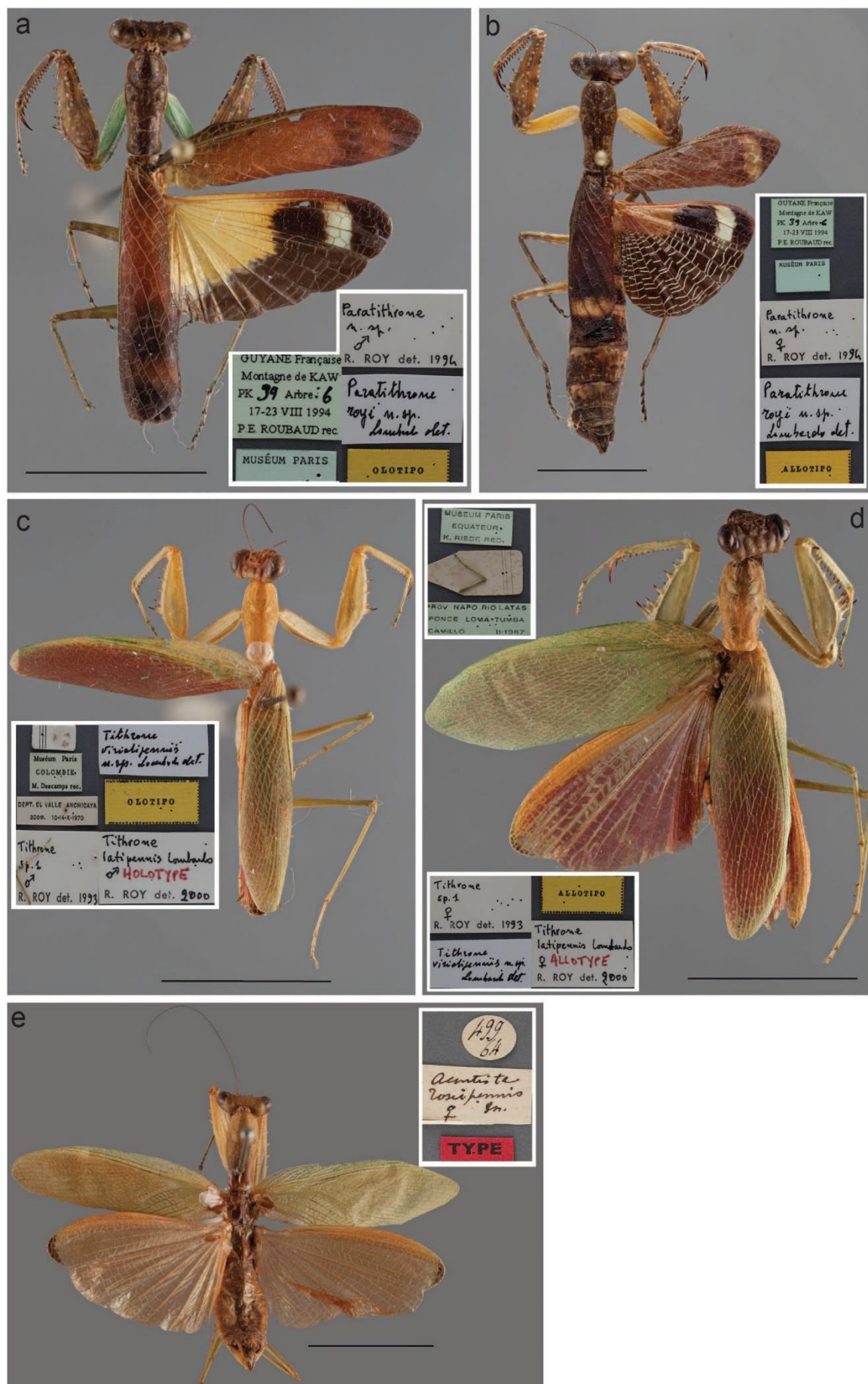


Figure 3.45

Figure 3.46. ACANTHOPIDAE – (a) *Acanthops bidens*, male holotype; (b) *Acanthops griffinii* [now *Acanthops falcata*], female syntype; (c) *Acanthops godmani*, male holotype; (d) *Acanthops parafalcata*, male paratype; (e) *Acanthops royi*, male holotype; (f) *Acanthops soukana*, female holotype; (g) *Acanthops bolivari* [now *Decimiana bolivari*], female syntype. Scale bar = 10 mm.

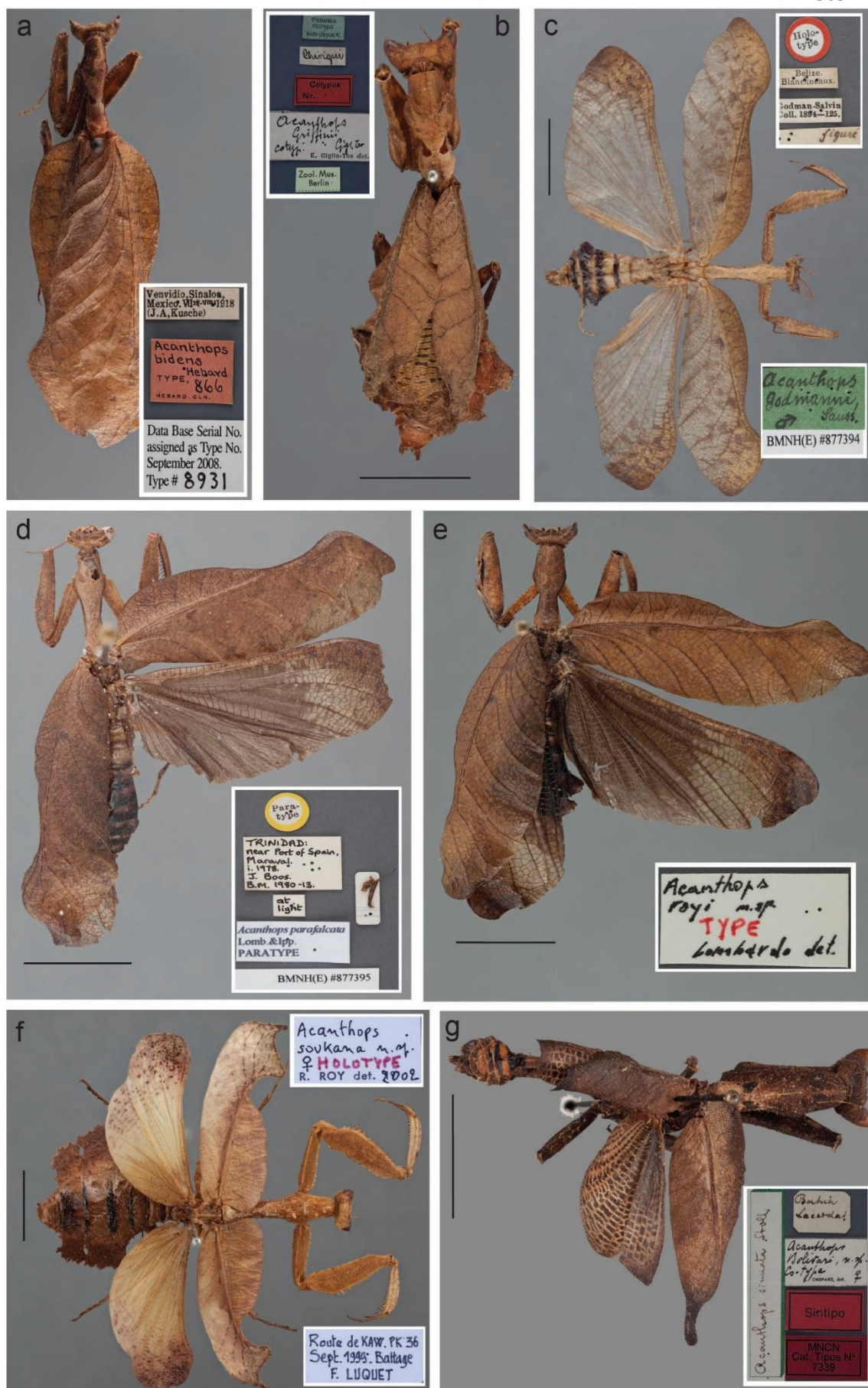


Figure 3.46

Figure 3.47. ACANTHOPIDAE – (a) *Acanthops bolivari* [now *Decimiana bolivari*], female syntype; (b) *Plesiacanthops rehni* [now *Decimiana rehni*], female holotype; (c) *Lagrecacanthops guyanensis*, female paratype; (d) *Acanthops amazonica* [now *Metilia amazonica*], male holotype; (e) *Acanthops brunnerii* (now *Metilia brunnerii*), male holotype; (f) *Pseudacanthops lobipes*, male holotype. Scale bar = 10 mm.



Figure 3.47

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Chapter 4

Historical biogeography of the Neotropical “polymorphic earless praying mantises”: emerging patterns from a biome-level analysis (Mantodea: Acanthopoidea)

4.1 Summary

The “polymorphic earless praying mantises” (PEPM) or Acanthopoidea, constitutes an heterogeneous monophyletic assemblage of eight families distributed in the neotropics, where they represent ~55% of mantodean biodiversity in that region. Despite recent developments in mantodean taxonomy and systematics, there have so far been no attempts to elucidate biogeographical pattern nor the historical processes responsible for assembling of praying mantis communities and their ecological associations. Using a combination of recently published phylogenetic datasets, time estimates of Dictyoptera evolution, and information from the fossil record, I present for the first time an explicit biogeographic hypothesis of the Acanthopoidea to provide insights on possible diversification drivers. The specific goals of this study are to: (I) investigate the evolutionary time-line of acanthopoids using contemporary patterns of biome-level distribution and ecological associations to reconstruct the likely habitat evolution of all Acanthopoidea clades, and (II) reconcile contemporary biogeography with current hypotheses about the evolution of Neotropical biodiversity in an ecological context. BEAST (Bayesian Evolutionary Analysis by Sampling Trees) and RASP (Reconstruct Ancestral State in Phylogenies) analyses suggested that the Acanthopoidea has existed since at least 139 My. Higher-level diversification transpired through the Cretaceous, whereas crown diversification of contemporary diversity spanned the Paleogene/Neogene (Cenozoic). Results of ancestral habitat reconstructions suggest that ancient early diversification of PEPM was most likely confined to mesic biomes, followed by relatively recent transition of certain lineages to more seasonal, drier, biomes throughout the Cenozoic, without reversals to mesic ones. I conclude that higher-level diversification of acanthopoids occurred when angiosperms came to dominate terrestrial environments, whereas contemporary diversity, patterns of distribution, and community assembling likely resulted from cyclical climatic fluctuations that spanned the Cenozoic, in agreement with the Refuge Hypothesis. The Niche Conservatism Hypothesis provides a possible explanation for early diversification and long-term

persistence of taxa in mesic biomes. Punctuated evolutionary shifts to xeric biomes probably occurred in relatively recent evolutionary time (Cenozoic). The implications of these findings for historical patterns of community assembly are discussed.

4.2 Introduction

The “polymorphic earless praying mantises” (PEPM) or Acanthopoidea, constitute an heterogeneous monophyletic assemblage of eight families distributed in the neotropics, where they represent ~55% of mantodean diversity in that region (Agudelo *et al.* 2007). Members of the Acanthopoidea exhibit a wide array of life history strategies (Rivera and Svenson 2016) and can be found in all kinds of ecological settings. Females of the dead-leaf and moss-mimicking Acanthopidae, for example, hang motionless on the undersides of branches, where they resemble curled, dry leaflets or clumps of moss (Edmunds and Brunner 1999, Lombardo *et al.* 2013). The *Orthoderella* (Photinaidae) have elongated, flattened bodies, and use their short legs to squirrel about the branches of trees and bushes (Brunner and Gandolfo 1990, Roy and Stiewe 2011). Some acanthopoid families exhibit a rather conservative morphology, such as the Liturgusidae or Neotropical bark mantises, an otherwise well-diversified and specialized group of forest-dwelling mantises (Svenson 2014). Others, like the Thespidae, have evolved a striking array of ecological and morphological specializations, and occur in diverse Neotropical microhabitats (e.g. bromeliads, lichen, grass, bark, ground) (Rivera and Svenson 2016).

Long neglected by systematists (Rivera 2010), the Neotropical praying mantis fauna has been the subject of increased research in recent years (Rivera and Svenson 2014). Although the taxonomy and systematics of the PEPM is currently in flux, knowledge about their geographic distribution is growing rapidly, with certain patterns becoming more apparent (Rivera and Svenson 2016). For example, the greatest diversity of acanthopoids occur exclusively in moist forest habitats, including the Pseudopogonogastrinae and Bantiinae (Thespidae), Macromantinae and Microphotinini (Photinaidae), and families Liturgusidae, Angelidae and Liturgusidae. Other lineages, including the Cardiopterinae, Orthoderellini and Photinaini (Photinaidae), Thespinae and Miobantinae (Thespidae), and the families Acanthopidae and Acontistidae all have representatives mainly in moist, but seasonal and variously xeric habitats throughout the neotropics. Members of the Coptopterygidae and Photiomantinae (Photinaidae) occur exclusively in xeric habitats. Despite these new revelations, no biogeographic hypotheses have so far been adduced to explain the present-day distribution of Neotropical praying

mantises.

In this study I combine the phylogenetic dataset of Rivera and Svenson (2016), with fossil information, and recently published time estimates of Dictyoptera evolution (Legendre *et al.* 2015) to reconstruct the biogeographical history of the Acanthopoidea. The general goal of this chapter is to provide insights about the ecological drivers of PEPM diversification. The specific goals of this chapter are to: (I) investigate contemporary patterns of distribution and ecological associations to reconstruct the likely ancestral habitat of all Acanthopoidea clades, and (II) reconcile contemporary biogeography with current hypotheses about the evolution of Neotropical biodiversity and its ecological context.

4.3 Material and Methods

4.3.1 Phylogeny and dating analysis

Divergence times within acanthopoids were estimated by implementing BEAST (Bayesian Evolutionary Analysis by Sampling Trees) v1.8.2 (Drummond and Rambaut 2007, Drummond *et al.* 2015a) on a subset of the original molecular dataset and phylogeny of Rivera and Svenson (2016). The original number of taxa in this phylogeny was reduced to a single individual per species or morphospecies, and incorporate the taxonomic changes introduced in Chapter 3. A total of 82 in-group terminals representing all higher-level taxa within the Acanthopoidea (i.e. 8 families, 15 subfamilies, 18 tribes, and 46 genera) were included, whereas outgroup sampling scheme was kept unmodified for adequate calibration (see Rivera and Svenson 2016). Divergence times were estimated under the relaxed molecular clock model, with uncorrelated lognormal distribution of substitution rate variation among lineages (Drummond *et al.* 2006), and under a birth-death process with incomplete sampling priors. Substitution model selected was GTR+G+I (4 gamma categories) for all 10 gene partitions (Rivera and Svenson 2016). Calibration points were defined using the software package BEAUti v1.8.2 (Drummond *et al.* 2015b), where I incorporated one primary and four secondary calibration points (Fig. 4.1, inset). The primary calibration point incorporates the fossil *Mantoida matthiasglinki* Zompro, 2005 (Zompro 2005). This specimen, attributed to the Mantoididae, is preserved in Baltic amber from Germany, dating from the Eocene Epoch (56–33.9 Mya) of the Paleogene Period. Although other putative members of Mantoididae are known from the Eocene (Grimaldi 2003,

Wieland 2013), the relatively more elongated body and different foreleg structure suggest a different generic placement (Zompro 2005). Nevertheless, *Mantoida matthiasglinki* exhibits other features that are characteristic of the Mantoididae, such as head with wide and globular vertex, protruding and exophthalmic compound eyes, and a short and square-shaped pronotum. Therefore, I here consider *M. matthiasglinki* a stem relative of the Mantoididae for calibrating the node Mantoididae + PEPM (Fig. 4.1). Secondary calibration points incorporate upper and lower dates estimated for members of the Dictyoptera, by Legendre *et al.* (2015): the split between Blattodea and Isoptera + *Cryptocercus* (283.2–263.6 Mya), crown Mantodea (203–164.3 Mya) and crown Isoptera (153.7–149.3 Mya) (Fig. 4.1). The age of the root (Dictyoptera) was set to 304.4 Mya (Legendre *et al.* 2015). I used default prior distributions for all parameters except for calibration points. Uncertainties in the primary fossil calibration were incorporated into the divergence dating analysis using a log-normal prior distribution with soft bounds (no zero offset), whereas normal distribution was used for secondary calibration points (Yang and Rannala 2006, Forest 2009). To reduce subjectivity in prior selection for calibration points, I parameterized values of mean and standard deviation for both the lognormal and normal distributions using ParameterSolver v3 (Cook *et al.* 2013) so that the upper and lower absolute dates represented 2.5 and 97.5% quantiles of the distributions. I performed several independent Markov chain Monte Carlo (MCMC) runs for 50 million generations, with trees sampled every 5000 generations. Tracer v1.6 (Rambaut *et al.* 2013) was used to check for convergence. I combined four well-converged runs using LogCombiner v1.8.2 (Rambaut and Drummond 2015a), discarding 20% of trees as burn-in. The final tree was compiled using TreeAnnotator v1.8.2 (Rambaut and Drummond 2015b), with FigTree v1.4.2 (<http://tree.bio.ed.ac.uk/software/figtree/>) used for tree editing and visualization of mean and 95% high posterior density (HPD) of each estimated node age.

4.3.2 Biogeographic analyses

To infer the most likely ancestral ranges, and reconstruct possible historical biogeography scenarios for acanthopoids, I implemented two different event-based approaches in RASP (Reconstruct Ancestral State in Phylogenies v3.2, Yu *et al.* 2015) using the ultrametric tree generated in BEAST: (I) Statistical dispersal-vicariance analysis (S-DIVA), and (II) Bayesian binary MCMC (BBM). The DIVA method estimates ancestral ranges through optimizing biogeographic events in a parsimony context, allowing for vicariance, but minimizing assumptions of dispersal and extinction. A limitation

of DIVA is that it does not take phylogenetic uncertainty into consideration. Optimization of ancestral ranges are performed along a fixed topological reconstruction that may not be unique (i.e. when two or more most parsimonious trees are inferred), thus resulting in nodes with multiple ancestral states (Ronquist, 1997). To overcome this limitation, S-DIVA implements a Bayesian approach by integrating the posterior distribution of a Bayesian MCMC sample of tree topologies (Yu *et al.*, 2010), simultaneously estimating the probability of alternative ancestral areas on a given node (Sanmartín *et al* 2008; Nylander *et al.*, 2008; Harris *et al* 2013). The BBM method implements Markovian chains to model dispersal (i.e. transitions) among two or more discrete states (i.e. biogeographic units) as a stochastic process, where the different states have different rates or probabilities. Because the focus of this study is to investigate large-scale biogeographical patterns of acanthopoids throughout the Neotropical Region, sampled taxa in the phylogeny were coded as being present in one or more of the following five biomes: A, Tropical and Subtropical Moist Broadleaf Forest; B, Tropical and Subtropical Dry Broadleaf Forest; C, Tropical and Subtropical Grassland, Savanna and Shrubland; D, Montane Grassland, Savanna and Shrubland; E, Desert and Xeric Shrubland. These subdivisions follow the 14-biome system of the world of Olson *et al.* (2001), and includes the contemporary distribution of acanthopoids within the Neotropical Region (see Dinerstein *et al* 1995 for biome definitions). Biomes reflect broader patterns of endemism in response to abiotic (local climatic variables) and biotic (plant formations and physiognomy) conditions (Woodward *et al*, 2004), resulting in differential assembly of insect communities that are functionally integrated to the "architecture" of each biome. Phylogenetic evidence from Rivera and Svenson (2016), in addition to distributional data from Chapter 3, suggest that praying mantis communities are distinct across biomes. Accordingly, using biomes as operational units as a starting point for biogeographic inference is biologically meaningful at the level of taxonomic resolution provided by the phylogeny (Fine 2015). For the S-DIVA analysis, a sample of 10,000 random trees were drawn from the 40,000 pool of trees from the posterior distribution obtained in BEAST. The maximum number of ancestral areas allowed at each node was set to 3, a value that correspond to the maximum number of biomes occupied by extant congeneric members of acanthopoids. For the BBM analysis I used the maximum clade credibility tree obtained in BEAST to compute two independent runs of 10 chains (temperature 0.1) for 50000 generations sampled every 100 generations using F81 + Γ (transition and among-site variation parameters empirically estimated). This is the most general and complex model implemented in RASP, thus providing maximum realism to biogeographic reconstructions (Sanmartín *et al.*, 2008). Both

biogeographical analyses were conducted using the same area coding.

4.4 Results

4.4.1 Phylogeny and dating analysis

Tree topology was well-supported overall (Fig. 4.1, Table 4.1). Only four nodes in the backbone of Photinae (Photinidae) had posterior probability (PP) values below 0.95, and of these, nodes 54 and 56 had PP values of 0.49 and 0.44 respectively. Overall topology was largely consistent with the maximum-likelihood tree of Rivera and Svenson (2016). The only discrepancy was with relationships among Photinae tribes, wherein the Orthoderellini was recovered as sister of the Microphophtinini + Photinaini. In Rivera and Svenson (2016), the Microphophtinini was sister to the Orthoderellini + Photinaini. Nevertheless, the three tribe (Orthoderellini, Microphophtinini and Photinaini) arrangement proposed by Rivera and Svenson (2016) found further support in this analysis. The BEAST analysis placed the origin of acanthopoids in the Lower Cretaceous, $\sim 139 (\pm 17)$ Mya, which corresponded with the origin of contemporary PEPM families. Rapid higher-level diversification followed through the Upper Cretaceous (100 Mya). The most recent common ancestor (MRCA) of most thespid subfamilies and the Liturgusidae originated towards the end of the Cretaceous — a time when most photinaids also experienced rapid diversification. The Paleogene Period (23–66 Mya) saw the origin and early diversification of almost all modern genera. Most modern thespid genera originated around the Paleocene/Eocene boundary (55 Mya) and continued diversifying through the end of the Paleogene. The MRCA of extant Thespini, Coptopterygidae, Photinae (Orthoderellini, Microphophtinini and Photinaini), Macromantinae, Acontistidae, Acanthopidae diversified relatively recently, well after the Paleocene/Eocene boundary (55 Mya), through the Oligocene (23–33 Mya) epoch and early Neogene (Miocene).

4.4.2 Ancestral area reconstruction

S-DIVA and BBM analyses predominantly recovered biome A as the ancestral biome type of extant acanthopoids lineages, with slight discrepancies in the S-DIVA analysis at node 78 (areas AB, AC and ABC all equiprobable, ~33%) and node 79 (areas A and ABC, ~50%). Both analyses (Fig. 4.2) suggested that early higher-level diversification in the Thespidae also occurred in biome A (nodes 36-39). Similarly, the MRCA of at least 4 subfamilies, namely Pseudomiopteryginae (node 2), Bantiinae (node 8), Miobantiinae (node 13) and Thespinae (node 36) was most likely associated to a type-A biome (although S-DIVA inferred A and C the most likely ancestral biome types for the MRCA of Miobantiinae). Pseudopogonogastrinae had only one terminal, and thus it could not be further addressed. The analyses also inferred the same biome A for the MRCA of at least the Liturgusidae, Photinaidae, Acontistidae and Acanthopidae (Angelidae and Stenophyllidae have only one terminal, and thus cannot be further addressed). S-DIVA inferred B and C as the most likely ancestral biome types for the MRCA of extant Coptopterygidae (node 42); the same node was ambiguous in the BBM analysis, although type-A biome had the highest probability among all possible reconstructions (~40% marginal probability), followed by B (~30%) and C (~18%). The BBM analysis inferred a single biome type (A) scenario through the early diversification of extant Photinaidae (nodes 54–58), whereas S-DIVA suggested two alternative reconstructions: AE (~54%) and AC (~45%) at nodes 57, and AE (100%) at node 55. The global analysis of RASP indicates that 66 of 80 nodes (82.5%) resolved A as the most likely ancestral biome type in at least one of the analyses. Both S-DIVA and BBM also suggest that a greater number of within-biome diversification (i.e., speciation) events (S-DIVA=51; BBM=53) occurred in a type-A biome, compared to B (6) and C (7); no such events were detected in D and E. This suggests that humid, Tropical and Subtropical forests have been important evolutionary grounds for acanthopoids. The remaining 14 nodes resolved B, C, D and E (individually or in combination) as the most likely ancestral biome types for the MCRA of extant, crown taxa. These include the MRCA of the following lineages: *Eumiopteryx* and *Anamiopteryx* (node 12), *Musoniella* (in part) and *Eumusonia* (node 14), *Thesprotia*, *Piscomantis*, *Oligonyx* and *Oligonicella* (node 32), *Paraphotina* and *Orthoderella* (node 47), sampled members of *Coptopteryx* (node 41), *Decimiana* (node 69), and at least some members of *Photina* (node 50). Type-A biome accounts for 77–81% (S-Diva and BBM combined) of dispersal events, having historically contributed most to biome types B and C, and to a lesser

extent E. Biome types B and C received more elements from other biomes than they contributed with, whereas D and E only received elements from other biomes. Neither analysis suggested that biome A received contributions from other biomes. Time-events curve implemented in RASP showed that, in both analyses, vicariance and dispersal events are mostly associated with nodes where ancestors transitioned from a humid, type-A biome, to more seasonal or xeric environments (biomes B–E). Finally, the analysis of the time to events curve implemented in RASP shows that, in both analyses, dispersal and vicariance events took place ca. 20–70 mya, largely corresponding to Paleogene times.

4.5 Discussion

While the sister-group relationships of Mantodea to all other extant Dictyoptera (i.e., Blattodea + Isoptera) is well established (Rivera and Svenson 2016), the origin and timeline of diversification of praying mantises remains little explored (Svenson and Whiting 2009). Estimates based on the fossil record suggest a possible origin of Mantodea around the Upper Jurassic (145–164 Mya), whereas the radiation that gave rise to contemporary taxa is estimated to have occurred during the Paleogene and Neogene Periods of the Cenozoic Era (66–2.5 Mya) (Vršanský 2002, Grimaldi 2003, Grimaldi and Engel 2005). The timeline of events revealed by this study, and detailed below, builds on the estimates of Legendre *et al.* (2015), although focused on the Acanthopoidea. Detailed discussion about early divergences within the Dictyoptera in relation to other dating studies (e.g. Misof *et al.* 2014) can be found therein.

4.5.1 Early evolution

Analysis in BEAST estimated the origin of crown Mantodea to be ca. 187 Mya, during the Lower Jurassic. The MRCA of Mantoididae and Acanthopoidea diverged around the Jurassic Cretaceous boundary (~145 Mya), whereas crown acanthopoids originated during the Lower Cretaceous (~140 Mya). Fossil evidence, although meagre, reveals that various stem-Mantodea lineages were present during the Cretaceous, suggesting that the origin of Mantodea likely extends further back into the Jurassic Period (Grimaldi, 2003). Current evidence supports this conclusion, as estimated dates of splitting events conform to the Mantodea fossil record and

phylogenetic evidence presented by Svenson and Whiting (2009). My dating analysis in BEAST places the backbone of the acanthopoid phylogeny, and higher-level diversification within the two most diversified families (the Thespidae and Photinaidae), mainly through the Cretaceous Period (60–139 Mya). This estimate for the early diversification of acanthopoids overlaps with the 79–126 My estimate given by Svenson and Whiting (2009), extending the early evolutionary timeline of acanthopoids by about 13 My. Evidence clearly suggests that the ancestors of all acanthopoid families and stem-Mantodea (Grimaldi 2003, Delclòs et al 2016) were already present through much of the Cretaceous Period.

The Cretaceous Period was a time of intense tectonic and climatological activity. The break-up of Gondwana was well under way, global climate was warmer and moister than today's, there were no polar ice caps, and tropical vegetation extended further into higher latitudes (Gale 2000, Novas 2009). Modern biomes and latitudinal zonification developed during this period, with angiosperms largely replacing gymnosperms (Friis et al 2010, Graham 2011). The rapid diversification of angiosperms represented a major turnover in higher taxonomic levels of plants, a change that was closely tracked by insects (Labandeira *et al.* 1994). In fact, all major clades of modern insects were already present by the end of the Cretaceous Period (Labandeira 1998, Lupia 2000, Szwedo and Nel 2015). Insect-plant association resulted in a major co-evolutionary dynamic of mutual exploitation that persists today (Grimaldi and Engel 2005). Svenson and Whiting (2009) suggested that modern praying mantises of the Artimantodea clade (i.e. the polymorphic earless praying mantises + Cernomantodea) may also have diversified in response to the diversification of angiosperms. Paleontological evidence strongly suggests a major turnover in insect species composition as new feeding ecologies evolved throughout the Cretaceous Period (Zherikhin 2002). Phylogenetic evidence points to the rise of angiosperms as a likely fundamental event that promoted rapid diversification of phytophagous insects, such as various groups of Lepidoptera (Kristensen and Skalski 1999, Wahlberg 2013), Coleoptera (Farrel 1998, Wilf et al 2000, McKenna *et al.* 2009), corbiculate bees (Grimaldi 1999), ants (Moreau 2006) true bugs (Li *et al.* 2012, Hwang and Weirauch 2012), and aphids (von Dohlen and Moran, 2000, Huang *et al.* 2012). All of these lineages originated and diversified during the time of early angiosperm radiation. The proliferation of phytophagous insects and the origin of new habitats presented ideal conditions for the morphologically labile praying mantises to adapt during a time when other (now extinct) polyneopterans were well-established in the predatory

niche, such as the †Manipulatoridae, †Raphidiomimidae (Blattodea), and †Alienopteridae (Alienoptera) (Vršanský 2002, Grimaldi and Ross 2004, Vršanský and Bechly 2015, Bai *et al.* 2016, see Legendre *et al.* 2015). Although the timing of events is tantalizing, there is no direct evidence of a correlation between the evolution of modern praying mantises and the rise of angiosperms. Nevertheless, the diversification of flowering plants unquestionably influenced the evolution and diversification in primary consumers, but possibly also of lineages at higher trophic levels (Zherikhin 2002). This is the case with parasitic Hymenoptera, which closely tracks the diversification of their lepidopteran ditrysian hosts throughout the Cretaceous (Whitfield 2003).

Given the wide range of ecomorphological adaptations in extant Mantodea, arguably the result of their intimate connection to flowering plants as their main hiding and hunting grounds (Svenson and Withing, 2009), a causal relationship between the rise of angiosperm and the early diversification of the Artimantodea (at least), cannot be ruled out.

4.5.2 The origin of contemporary diversity

The Paleogene Period (23–66 Mya) saw the origin of most contemporary genus-level clades. Modern thespid genera originated ca. the Paleocene/Eocene boundary (~55 Mya), continuing to diversify through the end of the Paleogene. Modern diversity of Thespini, Coptopterygidae, Photinae (Orthoderellini, Microphotinaini and Photinaini), Macromantinae, Acontistidae, and Acanthopidae originated after the Paleocene/Eocene boundary (55 My), continuing through the Oligocene (23–33 My) and early Neogene (Miocene).

The most conspicuous pattern during Paleogene and early Miocene times was the origin of clades adapted to seasonal and xeric biomes (biome types B–E). The rough bark-dwelling *Eumiopteryx*, *Anamiopteryx* (Thespidae, Miobantiinae), and open ground and lower vegetation-dwelling *Musoniella*, *Eumusonia*, diverged from Paleocene through mid-Eocene times. Ancestors of the dry-adapted *Metaphotina* and rainforest-dwelling *Raptrix* (Acontistidae) diverged by the mid-Eocene, as did certain members of the *Photina* (Photinidae). Somewhat later during the late Eocene, *Coptopteryx* and *Brunneria* (Coptopterygidae), and certain species

of the branch-dwelling *Paraphotina* and *Orthoderella* (Photinaidae, Orthoderellini), diverged from their respective common ancestors. These taxa mainly associate with savannah-like and dry-forest habitats. Various other genera such as *Thesprotia*, *Piscomantis*, *Oligonyx*, and *Oligonicella*, originated during the same time period, diversifying through the late Oligocene and forming a subclade within the Thespini. Sampled species pairs within the genera *Musonia*, *Decimiana* (Acanthopidae), *Coptopteryx* and *Eumiopteryx*, all associated with seasonal and/or xeric environments, diverged during late Oligocene to mid Miocene times. The timing of these events overlap with Oligocene and Miocene crown diversification within the Vatinae (Svenson *et al.* 2016) — a Neotropical lineage of the Cernomantodea (hearing mantises) that occurs mainly in type-A biomes, but a smaller number of species (mainly in *Pseudovates* and *Zoolea*) are represented in biomes B and C (Rivera 2001, Orofino *et al.* 2006, Roy and Ehrmann 2009, Svenson *et al.* 2016). A recent survey of the praying mantises of Rwanda (Tedrow *et al.* 2015) revealed a rich diversity of genera from multiple, non-monophyletic groups associated with biome C, most of which diversified between the Eocene and early Miocene according to estimates by Svenson and Whiting (2009). This coincides with the development of xeromorphic plants and grass-dominated communities in Africa (Jacobs *et al.* 2010, Linder 2014). Overlapping evolutionary shifts from mesic to xeric biomes in the Acanthopoidea and other Neotropical Cernomantodea (i.e. the hearing clade) taxa suggest that different groups of praying mantises adapted in similar ways, most likely in response to a common process.

4.5.3 Broad-scale patterns in Acanthopoidea historical biogeography and implication for community structure

Two main patterns emerged from this analysis: (1) early higher-level diversification occurred during the Cretaceous Period, followed by crown diversification of contemporary diversity throughout the Paleogene/Neogene, and (2) early diversification most likely occurred in tropical, humid, environments (type-A biome), followed by more recent transition in certain lineages to more seasonal and drier environments (biome types B–E), but without apparent reversal to the ancestral biome.

Early higher-level diversification during the Cretaceous Period, and genus-level diversification throughout the Paleogene, is a common pattern in the evolutionary history of many organisms, including land plants and insects (Whitfield and Kjer 2008, Fiz-Palacios *et al.* 2011). Tropical

conditions were widespread during Cretaceous times, with a cooling trend following the Paleocene/Eocene thermal maximum (~55 Mya). This marked the beginning of cyclical climatic events that persisted for another 60 My, encompassing most of the Cenozoic Era. These climatic fluctuations transformed the once predominately tropical and humid biome A of the early Eocene, to cooler and more seasonal biomes by the end of the Oligocene, setting climatic conditions for development of the savannah-like and dry forests habitats that became widespread by Neogene times (Jaramillo *et al.* 2009, Graham 2010). The fossil record, in addition to numerous phylogenetic studies, shows that this cyclical expansion and retraction of biomes during the Cenozoic, in response to alternating humid and dry climatic conditions at tropical latitudes (Milankovitch cycles), was fundamental for shaping contemporary patterns of species diversity in the Neotropical Region (e.g. Barreda and Palazzesi 2007, Almeida *et al.* 2011; revised in Haffer 1993, 2008 and Rull 2011).

Evidence points to the Cenozoic Era as a time of dynamic and episodic evolutionary shifts in habitat preferences among acanthopoids — consistent with the main postulate of the Refuge Hypothesis as outlined by Haffer (2008). Divergence of contemporary acanthopoid taxa adapted to seasonal and xeric environments coincides with a time when xeromorphic open plant formations and grass-dominated environments came to dominate the Neotropical landscape. Such habitats were present in South America since at least mid-Eocene times, becoming more extensive by the late Miocene (Barreda and Palazzesi 2007, Strömberg 2011). The analyses in RASP shows that basal and more derived dichotomies occurred within the type-A biome, and that both dispersal and vicariance played a role in transitions from mesic to xeric environments. The Refuge Hypothesis emphasizes allopatric speciation (a likely outcome of vicariance, but also of dispersal) as one of the main mechanisms of diversification, with refugia acting as both sources and sinks of species (Haffer 2008). RASP infers secondary shifts within and between biomes B–E, with a trend for moving towards increasingly drier habitats. However, no reversals towards the type-A biome exist. This suggests that tropical, humid, environments were the primary source of diversity, whereas seasonal and xeric ones acted as sinks, thus adding support to the view that tropics are the source of contemporary diversity in extratropical areas (Fine 2015). Nevertheless, the scattered occurrence of such transitions in the phylogeny, and the relative depauperate diversity of acanthopoids outside of biome A, suggest that habitat shifts from mesic to xeric habitats occur infrequently, whereas reversals from these to the ancestral

biome seem to be rare enough events to escape detection (at least, with this particular dataset). Although the reason for lack of reversals remains unclear, the Niche Conservatism Hypothesis (NC) provides a possible explanatory framework.

The NC refers to the propensity of species and clades to retain their niches and associated ecological traits over evolutionary time (Wiens and Graham 2005). According to Wiens *et al.* (2010), NC is a pattern resulting from many biotic and abiotic processes occurring at different evolutionary and temporal scales, as historical processes leading to speciation (i.e. vicariance, dispersal) link not only to large-scale geological events (e.g. continental drift), but also to small-scale, ecological ones (e.g. habitat fragmentation) (Wiens and Graham 2005). How much the latter explain a given biogeographic pattern depends on how species respond to the aforementioned processes, as not all are equally affected (Wiens and Donoghue 2004). NC provides a plausible explanation for latitudinal species diversity gradients, habitat-specific endemism, and the assembly of biological communities, and is, thus, a potential driver for observed patterns of species richness in biomes across the globe, including the Neotropical Region (e.g. Hawkins *et al.* 2006, Kennedy *et al.* 2014; revised in Rull *et al.* 2011). One version of NC — the Tropical Niche Conservatism Hypothesis — predicts a higher number of species in tropical biomes compared to seasonal and xeric ones, even when differential diversification rates among climatic zones are taken into consideration (Jansson *et al.* 2013). Evolutionary constraints of niche-related traits (e.g. physiological, behavioural, morphological) are believed to be the main reason species are prevented from moving partially, or completely, from their tropical zone of origin to extratropical habitats (Wiens and Donoghue, 2004).

The extant diversity of Acanthopoidea is concentrated largely in biome A. The relatively low number of lineages that successfully transitioned into seasonal and drier environments during relatively recent times, and the plethora of ecomorphological strategies observed in acanthopoids and many other praying mantis clades (e.g. Rivera and Svenson 2016), suggests that NC is the most likely explanation for the observed patterns (Wiens and Graham 2005). Further, mantodean genera exhibit remarkably conserved morphology among included species, but often exhibit marked intergeneric differences. This latter aspect is often reflected in praying mantis common names, as these tend to be inspired by microhabitat preferences and associated cryptic strategies (e.g. grass-mantises, lichen-mantises, bark-mantises, etc.) (Rivera and Svenson 2016). These

distinct and widespread morphological patterns, in addition to extensive ecomorphological convergence observed across the Mantodea, strongly suggest a pattern of NC.

A pattern of NC does not necessarily predict species' responses to environmental pressures, especially at large scales (e.g. climate change). However, it provides a framework to investigate what niche-related traits allow praying mantises to move from one environment to another. For example, some acanthopoid genera, such as *Eumiopteryx* (Thespidae), *Cardioptera*, *Photina*, *Paraphotina* and *Orthoderella* (Photinaidae), are present in both mesic and xeric biomes, where they occupy similar micro-habitats. These genera evidently have traits that allow their species to successfully thrive in different biomes. Elucidating what these niche-related traits are, whether morphological or physiological, and how they are functionally integrated into the habitat, could help shed light about the mechanisms underlying assembly of praying mantis communities in response to climate change.

4.6 Conclusions

A clearer picture of the biogeography and evolutionary history of Neotropical praying mantises is beginning to emerge. Evidence presented in this study suggests that the Acanthopoidea have existed at least since 139 My, with the origin of contemporary diversity most likely dating to Paleogene and Neogene times (i.e. the Cenozoic Era). Higher-level diversification within acanthopoids transpired during a time when angiosperms came to dominate the terrestrial environment, whereas contemporary diversity and distinct patterns of mantis distribution and community assembling likely originated as the result of cyclical climatic fluctuations during the Cenozoic Era, in agreement with the main postulate of the Refuge Hypothesis. The Niche Conservatism Hypothesis provides a possible explanation for early diversification and long-term persistence of taxa in mesic, type-A biomes, with punctuated evolutionary shifts to xeric, type B-E biomes during relatively recent (Cenozoic) evolutionary times. As our knowledge of praying mantis distribution and habitat preference increase, it will be possible to incorporate these data into species distribution models that, under a NC framework, will be helpful for determining which niche-related traits mediate contemporary community structure and thus biogeography.

The role of vicariance and dispersal in praying mantis diversification need to be examined more closely. The repeated range expansion and contraction of populations as a consequence of the climatic fluctuations during the Cenozoic Era, resulted in a continent-wide reshuffling of plant communities, ultimately defining contemporary biomes (Graham 1999). On one hand, these dynamics likely favoured periods of active speciation in allopatry through episodic vicariance and enhanced local processes, such as competition and other interspecific interactions (Kricher 2015). On the other, dynamic interconnectivity among expanding and contracting biomes could also have offered new ecological opportunities for dispersal and evolutionary habitat shifts in parapatry, as biome boundaries and habitat edges tend to be ecologically complex and are in constant flux (Cole 1960; Fagan *et al.* 1999, Costa 2003; Batalha-Filho *et al.* 2013; Thomé *et al.* 2016). Given their intimate connection to plants and their preference for specific plant physiognomies, it is not difficult to envision praying mantises sorting through the changing landscape, assembling into distinct communities as they tracked the habitat that best fitted their niche-related traits. Although my analyses suggest that both vicariance and dispersal played an important role in shaping broad, biome-level biogeography, the underlying mechanisms behind these processes and their role in community assembly warrants further investigation.

Exploring the tempo and mode of mantis diversification in other continents would provide a test for the interpretations advanced in the current study. Likewise, a broader species-level sampling, coupled with the much-needed assessment of the Cenozoic fossil record, will help narrow divergence time estimates across the Acanthopoidea. It will also reduce uncertainty in some of the nodal reconstructions presented here, allowing for ecological and evolutionary scenarios to be tested at various biogeographic scales (e.g., specific clades or communities). One question that remains unanswered is: do niche-related traits evolve before or after differentiation of habitat and climatic preferences? Similarly, how do differential physiognomies influence biome-specific community assemblages of praying mantis taxa at regional and continental scales? Elucidating the mechanisms behind differential species sorting will not only shed important new light about historical processes behind community assembly in praying mantises, but will also provide insights about ecomorphological convergence, a widespread characteristic of the order (Svenson and Whiting 2009).

Table 4.1. Estimated node ages (with upper and lower bounds) and values of posterior probability (PP) according to BEAST.

Node	Age [lower - upper]	PP	Node	Age [lower - upper]	PP	Node	Age [lower - upper]	PP	Node	Age [lower - upper]	PP
1	35.65 [22.89-51.39]	1	21	28.79 [19.38-39.65]	1	41	22.68 [15.11-32.33]	1	61	25.65 [16.49-35.91]	0.99
2	50.17 [35.09-66.93]	1	22	57.81 [44.95-71.62]	1	42	35.5 [24.43-48.68]	1	62	35.26 [25.83-45.76]	0.95
3	23.6 [14.42-35.11]	1	23	72.97 [59.97-86.95]	1	43	62.73 [43.06-81.92]	0.99	63	40.55 [30.87-50.91]	0.87
4	55.52 [40.6-71.02]	0.99	24	19.07 [12.1-27.5]	0.99	44	77.28 [57-96.81]	0.99	64	48.15 [35.16-56.06]	0.9
5	35.23 [22.91-49.15]	1	25	38.18 [26.42-50.4]	0.99	45	28.07 [16.51-41.96]	0.99	65	50.62 [40.58-61.46]	0.99
6	57.23 [43.05-72.2]	0.98	26	46.4 [35.52-57.83]	0.99	46	27.79 [17.42-40.74]	0.99	66	15.85 [9.35-23.37]	1
7	69.51 [54.22-84.55]	0.99	27	21.22 [13.15-30.14]	1	47	34.55 [23.16-47.57]	1	67	28.96 [20.38-38.68]	1
8	81.25 [66.81-96.4]	0.99	28	34.16 [22.59-46.03]	0.99	48	25.82 [16.49-36.44]	1	68	41.52 [31.26-52.17]	0.99
9	103.1 [86.9-119]	0.99	29	34.64 [25.98-44.6]	0.99	49	32.02 [22.23-42.47]	0.97	69	16.91 [10.08-25.68]	1
10	10.93 [6.11-17.1]	1	30	40.19 [31-50.06]	0.99	50	22.53 [13.67-32.53]	1	70	23.20 [15.12-33.25]	1
11	41.78 [27.01-59.73]	0.99	31	47.30 [37.53-57.83]	0.99	51	25 [16-35.5]	1	71	41.40 [31.24-51.95]	0.99
12	59.56 [41.27-78.3]	0.99	32	53.72 [43.33-64.28]	0.99	52	36.77 [27.42-47.45]	0.95	72	46.66 [36.48-57.26]	0.99
13	84.92 [66.81-103.56]	0.99	33	68.78 [57.3-81.15]	0.99	53	40.38 [30-51.35]	1	73	55.42 [44.5-66.16]	0.99
14	42.63 [28.33-57.83]	0.98	34	81.11 [68.61-94.88]	0.99	54	60.78 [49.41-72.37]	0.49	74	80.28 [67.32-94.92]	0.99
15	51.07 [36.78-67.19]	0.99	35	89.11 [76.19-102.38]	0.99	55	65.1 [53.62-77.55]	0.62	75	91.14 [77.8-106.33]	0.98
16	59.63 [44.21-75.93]	0.99	36	111.27 [96.36-127]	0.99	56	69.18 [56.86-81.64]	0.44	76	98.89 [85.06-113-48]	0.89
17	33.25 [21.33-46.4]	1	37	119.53 [103.49-135.06]	0.99	57	71.79 [59.21-84.89]	0.99	77	104.63 [90.55-119.79]	0.98
18	52.7 [39.62-67.37]	1	38	125.28 [109.23-141.23]	0.99	58	83.95 [69.96-98.3]	0.98	78	110.05 [94.89-125.89]	0.99
19	12.19 [7.04-18.7]	1	39	131.92 [115.88-148.55]	0.99	59	19.45 [12.06-28.74]	1	79	131.42 [114.18-148.18]	0.98
20	22.12 [12.87-33.93]	1	40	10.07 [5.62-15.76]	1	60	37.91 [27.84-48.56]	0.99	80	139.64 [129.19-164.17]	0.99

Figure 4.1. Estimated time of divergence of acanthopoids obtained in BEAST. Phylogeny is color coded according to family. Inset shows the calibrated nodes on a summary phylogeny: primary calibration (red star) and secondary calibrations (green stars) (see Material and Methods for specific dates and calibration strategy). Overlaying bars on branches represent 95% confidence intervals. Solid and dashed lines indicate taxa sampled from humid (A) and xeric biomes (B-E) respectively. Major paleoecological events (1-5) are placed in context within the phylogeny for quick reference. See Fig. 4.2 and Table 4.1 for values of posterior probabilities, age, and lower/upper bounds estimated for each node.

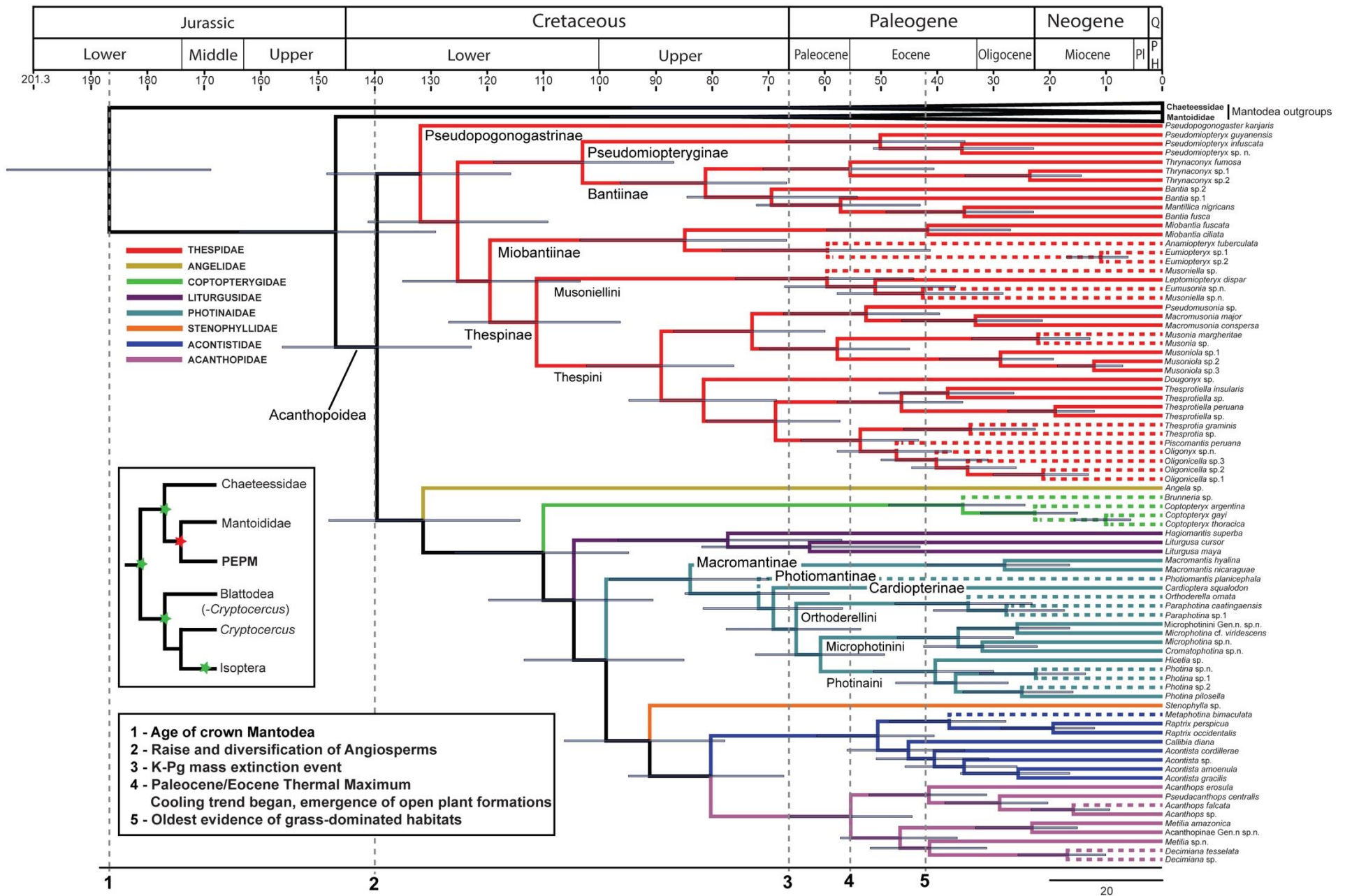
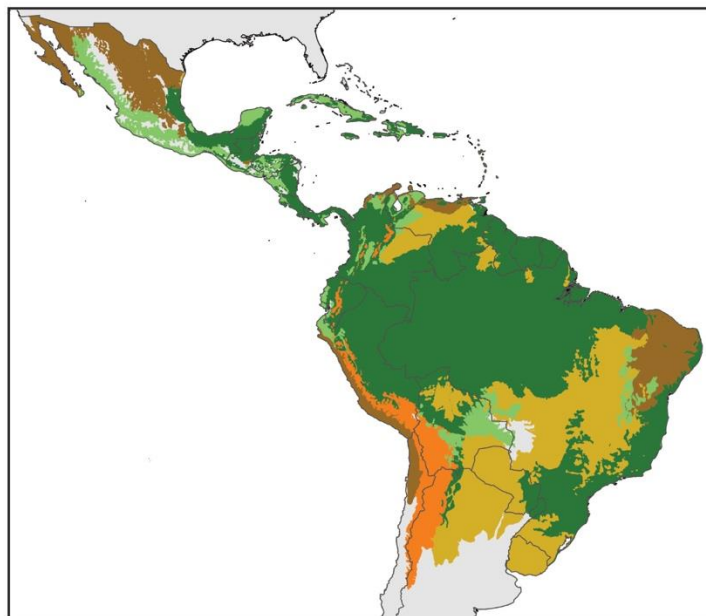


Figure 4.1

Figure 4.2. RASP output. The map on the left panel shows the distribution of biomes A to E, whereas the pie charts on the nodes of the phylogeny represent the ancestral areas inferred by S-DIVA (left pie) and BBM (right pie), where colors correspond to both coded (A-E) and/or inferred ancestral RASP areas – variation in pie size is meaningless. Distribution of sampled taxa are represented by colored squares next to each terminal in the tree. Each area within the pie charts represent the relative probabilities of ancestral habitats (states). Those reconstructions with probabilities lower than 0.2 were combined in the “remainder” category (black regions of pie charts). Red stars represent events (vicariance, dispersal) that had less than 0.5 of probability of occurrence. Time axis is in millions of years. Nodes are numbered in agreement to Table 4.1. Abbreviations: H. Holocene; P, Pleistocene; Pl, Pliocene; Q, Quaternary.



- A** Tropical and Subtropical Moist Broadleaf Forest
- B** Tropical and Subtropical Dry Broadleaf Forest
- C** Tropical and Subtropical Grasslands, Savannas and Schrublands
- D** Montane Grasslands and Savannas
- E** Desert and Xeric Schrublands

Ancestral RASP areas

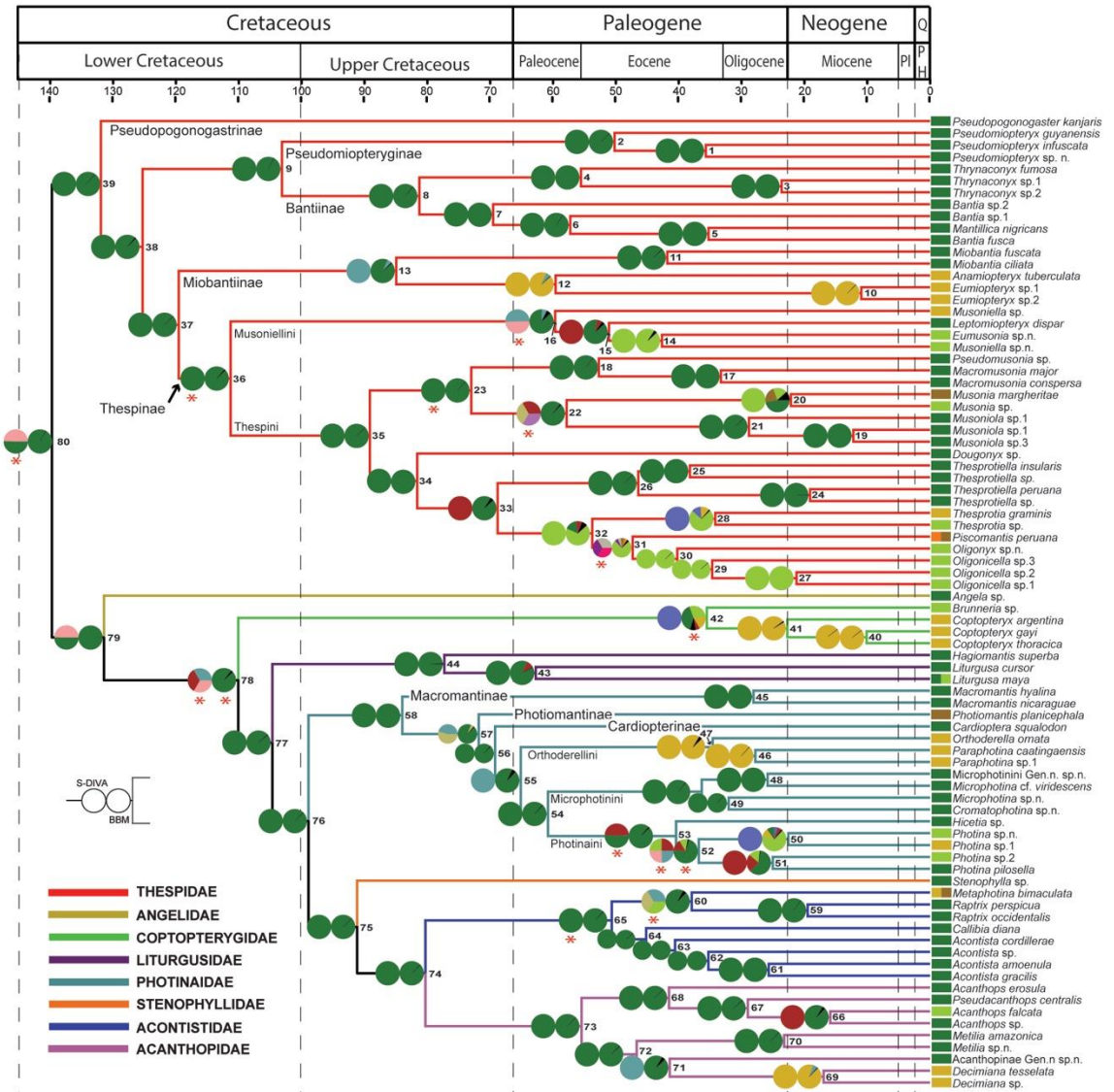


Figure 4.2

4.7 References

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Chapter 5

General Conclusions

This dissertation provides a solid taxonomic and phylogenetic foundation for the polymorphic earless praying mantises (PEPM), and includes the first biogeographic investigation of these organisms in an explicitly phylogenetic framework. The results are far reaching, as my dissertation not only sets the stage for more in-depth taxonomic and systematic studies of PEPM, but also for ecological and evolutionary research that use praying mantises as model organisms.

The application of molecular phylogenetic methods in Chapter 2 resulted in a complete reclassification of the PEPM. Now assigned superfamily status, the Acanthopoidea includes eight families, 15 subfamilies, 18 tribes and 62 valid genera (in addition to one *genus inquerenda*), totaling 315 species. New morphological concepts for taxon delineation necessitated recognition of new taxa and numerous nomenclatorial actions in order to develop a classification that reflects evolutionary affinities. The phylogenetic and taxonomic studies in Chapters 2–3 — in part complemented by a series of previously published studies (Rivera 2014, Agudelo and Rivera 2015, Rivera *et al.* 2011, Lombardo *et al.* 2013) — provide a robust platform from which to launch species-level taxonomic revisions. Special emphasis must now be directed towards little known taxa, particularly the Thespidae, a diverse group of microhabitat specialists that represent the next frontier of species discovery within the PEPM.

My dissertation also reveals the potential of male genitalia and egg case structure for future taxonomic and evolutionary studies. These character systems, one somatic and the other physiological, were revealed in Chapter 2 as having hitherto unrecognized potential to carry phylogenetic signal at various hierarchical levels. In fact, this is the first ever study of egg-case diversity in the Mantodea from a phylogenetic perspective — an otherwise sparsely documented aspect of praying mantis reproductive biology. An important question that remains is: why is egg case structure so well conserved at higher (e.g., family-level) taxonomic levels? Similarly, what are the selective pressures that drive variation in egg-case structure and their adaptive significance? The genitalic and cryptic egg case diversity revealed by this thesis raises numerous

additional questions, providing exciting new avenues of research for ecological and evolutionary studies of praying mantises.

Phylogenetic methods applied to historical biogeography (Chapter 4) provides new insights about the evolutionary timeline of PEPM. Likely processes and palaeoecological scenarios that explain patterns of PEPM diversity are consistent, both in tempo and in mode, to those documented in many other Neotropical taxa. This means that praying mantis diversification can be now investigated through the lens of climate change to test particular evolutionary scenarios using phylogenetic comparative methods. Further, integrating life history data with biogeographical and phylogenetic studies will help to investigate aspects of ecomorphological evolution and functionality — lines of research that are eminently suited to a diverse and charismatic group like praying mantises. In many respects, praying mantises are analogous to another well-known and broadly investigated group of organisms — *Anolis* lizards. The new classification proposed in my thesis will undoubtedly assist such endeavors because it serves as an information storage and retrieval system for biological data, fundamental to any higher-level scientific inquiry.

In summary, this thesis represents a major leap forward in our understanding of a large and fascinating lineage of Neotropical praying mantises. It also contributes towards the building of the long-neglected mantodean tree of life. Although other lineages of Neotropical praying mantids require attention, most notably the hearing mantises, this contribution will no doubt pave the way for future endeavours using similar methods. Only then will the exuberant (and underappreciated) diversity of praying mantises, and their varied ecological roles, become fully appreciated.

5.1. References

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Appendices

Appendix 1

Appendix 1 Chapter 2. List of voucher specimens, Mantodea voucher collection MN codes (Cleveland Museum of Natural History-CMNH) and Genbank accession numbers – those in bold were retrieved from Genbank (see Svenson & Whiting (2009)). Fields labelled as ‘<200bp’ correspond to sequences containing less than 200bp and thus could not be uploaded in Genbank; these sequences are available upon request. Microphotinini gen.n. sp.n. (MN271) was identified as *Microphotina* sp. in Svenson & Whiting (2009). Outgroup taxa are marked with * at the bottom of the table.

Species	Location	Voucher (CMNH)	12S	16S	18S	28S	COI	COII	H2A	H3	ND4	Wnt-1
<i>Pseudopogonogaster kanjaris</i>	Peru	MN497	-	KU320364	KU320468	KU320570	KU507715	KU528816	-	KU507915	KU508017	KU508090
<i>Pseudomiopteryx guyanensis</i>	Guyana	MN555	KU320279	KU320362	KU320464	KU320565	KU507711	-	-	KU507911	KU508013	-
<i>Pseudomiopteryx infuscata</i>	Panama	MN557	KU320280	KU320361	KU320463	KU320566	KU507710	-	-	KU507910	KU508012	KU508087
<i>Pseudomiopteryx</i> sp.n.	Peru	MN114	EF383219	EF383379	EF383539	EF383702	EF383862	EF383989	-	EF384118	FJ802528	FJ803016
<i>Pseudomiopteryx</i> sp.n.	Peru	MN550	KU320281	KU320363	KU320465	KU320567	KU507712	-	KU507813	KU507912	KU508014	KU508088
<i>Bantia fusca</i>	French Guiana	MN712	KU320201	KU320303	KU320386	KU320489	KU507637	-	-	KU507833	KU507937	KU508034
<i>Bantia</i> sp.1	Colombia	MN713	KU320202	KU320304	KU320387	KU320490	KU507638	-	-	KU507834	KU507938	KU508035
<i>Bantia</i> sp.2	Peru	MN115	FJ806017	FJ806192	FJ806379	FJ806578	FJ802796	FJ806906	-	FJ806742	FJ802529	-
<i>Bantia</i> sp.2	Peru	MN582	KU320203	KU320305	KU320388	KU320491	-	-	-	KU507835	KU507939	-
<i>Bantia</i> sp.2	Peru	MN583	KU320204	KU320306	KU320389	KU320492	KU507639	-	-	KU507836	KU507940	-
<i>Mantillica nigricans</i>	Brazil	MN495	KU320237	-	KU320421	KU320525	KU507670	-	-	KU507868	KU507971	KU508057
<i>Mantillica nigricans</i>	Brazil	MN496	KU320238	-	KU320422	KU320526	KU507671	-	<200 bp	KU507869	KU507972	KU508058
<i>Thrinaconyx fumosa</i>	Panama	MN086	EF383203	EF383363	EF383523	EF383685	EF383846	EF383973	-	EF384102	FJ802501	FJ802995
<i>Thrinaconyx</i> sp.1	Brazil	MN578	KU320291	KU320372	KU320477	KU320579	KU507723	-	KU507819	KU507924	KU508025	-
<i>Thrinaconyx</i> sp.2	Peru	MN216	EF383268	EF383427	EF383587	EF383753	EF383908	EF384038	KU507820	EF384165	FJ802621	-

<i>Thrinaconyx</i> sp.2	Peru	MN579	KU320292	KU320373	KU320478	KU320580	KU507724	-	KU507821	KU507925	KU508026	-
<i>Anamiopteryx tuberculata</i>	Brazil	MN714	KU320200	-	KU320385	KU320488	KU507636	-	-	KU507832	KU507936	KU508033
<i>Eumiopteryx</i> sp.1	Bolivia	MN560	KU320221	KU320320	KU320405	KU320510	KU507655	-	KU507748	KU507853	KU507957	KU508045
<i>Eumiopteryx</i> sp.2	Unespecified	MN445	KU320222	-	KU320406	KU320509	KU507656	<200 bp	KU507749	KU507854	KU507958	KU508046
<i>Miobantia ciliata</i>	Brazil	MN715	KU320250	KU320345	KU320434	KU320537	KU507683	-	-	KU507881	KU507984	-
<i>Miobantia fuscata</i>	Brazil	MN716	KU320251	KU320346	KU320435	KU320538	KU507684	KU528797	KU507783	KU507882	KU507985	-
<i>Miobantia fuscata</i>	Brazil	MN717	KU320252	KU320347	KU320436	-	KU507685	-	KU507784	KU507883	KU507986	-
<i>Eumusonia</i> sp.n.	Bolivia	MN123	EF383223	EF383383	EF383543	EF383706	EF383866	EF383993	KU507750	EF384122	FJ802536	-
<i>Eumusonia</i> sp.n.	Bolivia	MN570	KU320223	KU320321	KU320407	KU320511	KU507657	KU528780	KU507751	KU507855	KU507959	-
<i>Leptomiopteryx dispar</i>	French Guiana	MN718	KU320226	KU320324	KU320410	KU320514	KU507660	-	KU507755	KU507858	<200 bp	-
<i>Musoniella margheritae</i>	Peru	MN571	KU320255	KU320348	KU320439	KU320541	KU507688	KU528798	KU507786	KU507886	KU507989	KU508071
<i>Musoniella margheritae</i>	Peru	MN615	KU320256	-	KU320440	KU320542	KU507689	KU528799	KU507787	KU507887	KU507990	KU508072
<i>Musoniella</i> sp.	Brazil	MN137	EF383229	EF383389	EF383549	EF383712	EF383872	EF383999	KU507788	EF384128	FJ802550	-
<i>Musoniella</i> sp.	Bolivia	MN572	KU320257	KU320349	KU320441	KU320543	KU507690	KU528800	KU507789	KU507888	KU507991	-
<i>Musoniella</i> sp.n.	Bolivia	MN122	FJ806021	FJ806196	FJ806383	FJ806582	FJ802800	FJ806910	KU507790	FJ806746	FJ802535	-
<i>Musoniella</i> sp.n.	Bolivia	MN573	KU320258	KU320350	KU320442	KU320544	KU507691	-	KU507791	KU507889	KU507992	-
<i>Musoniella</i> sp.n.	Bolivia	MN574	KU320259	KU320351	KU320443	KU320545	KU507692	KU528801	KU507792	KU507890	KU507993	-
<i>Carrikerella</i> sp.	Costa Rica	MN148	EF383234	EF383394	EF383554	EF383717	EF383876	EF384004	KU507738	EF384133	FJ802558	FJ803040
<i>Carrikerella</i> sp.	Nicaragua	MN561	KU320211	KU320311	KU320395	KU320497	KU507645	KU528771	KU507739	KU507843	KU507947	KU508041
<i>Carrikerella</i> sp.	Nicaragua	MN562	KU320209	KU320309	KU320393	KU320498	KU507643	KU528769	KU507737	KU507841	KU507945	KU508039
<i>Carrikerella</i> sp.n.	Peru	MN563	KU320210	KU320310	KU320394	KU320499	KU507644	KU528770	<200 bp	KU507842	KU507946	KU508040
<i>Galapagia peruana</i>	Peru	MN494	KU320224	KU320322	KU320408	KU320512	KU507658	-	KU507752	KU507856	KU507960	KU508047

<i>Macromusonia conspersa</i>	Peru	MN126	FJ806023	FJ806198	FJ806385	FJ806584	FJ802802	FJ806912	KU507762	FJ806748	FJ802539	FJ803023
<i>Macromusonia conspersa</i>	Peru	MN567	KU320234	KU320332	KU320418	KU320522	KU507668	KU528786	KU507763	KU507865	KU507969	KU508055
<i>Macromusonia major</i>	French Guiana	MN566	KU320235	KU320333	KU320419	KU320523	KU507669	KU528787	KU507764	KU507866	-	KU508056
<i>Macromusonia major</i>	French Guiana	MN586	KU320236	-	KU320420	KU320524	-	-	KU507765	KU507867	KU507970	-
<i>Musonia</i> sp.	Colombia	MN721	KU320253	-	KU320437	KU320539	KU507686	-	KU507785	KU507884	KU507987	KU508070
<i>Musonia</i> sp.	Colombia	MN722	KU320254	-	KU320438	KU320540	KU507687	-	-	KU507885	KU507988	-
<i>Musoniola</i> sp.1	Nicaragua	MN588	KU320260	-	KU320444	KU320546	KU507693	-	KU507793	KU507891	KU507994	KU508073
<i>Musoniola</i> sp.2	Nicaragua	MN590	KU320261	KU320352	KU320445	KU320548	KU507694	-	KU507794	KU507892	KU507995	KU508074
<i>Musoniola</i> sp.3	Guatemala	MN438	KU320262	-	KU320446	KU320547	KU507695	-	KU507795	KU507893	KU507996	-
<i>Musoniola</i> sp.3	Nicaragua	MN591	KU320263	-	KU320447	KU320549	-	-	KU507796	KU507894	KU507997	KU508075
<i>Oligonicella</i> sp.1	Guatemala	MN441	KU320266	-	KU320450	KU320550	-	KU528803	KU507799	KU507897	KU508000	KU508077
<i>Oligonicella</i> sp.1	Guatemala	MN443	KU320267	-	KU320451	KU320551	KU507698	KU528804	KU507800	KU507898	KU508001	-
<i>Oligonicella</i> sp.2	Mexico	MN616	KU320264	KU320353	KU320448	KU320552	KU507696	KU528802	KU507797	KU507895	KU507998	KU508076
<i>Oligonicella</i> sp.3	Costa Rica	MN049	FJ805996	FJ806172	FJ806358	FJ806557	FJ802775	AY491314	-	FJ806724	FJ802464	FJ802967
<i>Oligonicella</i> sp.3	Guatemala	MN442	KU320265	-	KU320449	KU320553	KU507697	-	KU507798	KU507896	KU507999	-
<i>Oligonyx insularis</i>	Dominica	MN724	KU320268	-	KU320452	KU320554	KU507699	KU528805	KU507801	KU507899	KU508002	KU508078
<i>Oligonyx insularis</i>	Dominica	MN725	KU320269	KU320354	KU320453	KU320555	KU507700	KU528806	KU507802	KU507900	-	KU508079
<i>Oligonyx insularis</i>	Dominica	MN726	KU320270	KU320355	KU320454	KU320556	KU507701	KU528807	KU507803	KU507901	KU508003	KU508080
<i>Oligonyx</i> sp.n.1	Nicaragua	MN444	KU320271	-	KU320455	KU320557	KU507702	KU528808	KU507804	KU507902	KU508004	-
<i>Oligonyx</i> sp.n.2	Dom. Republic	MN592	KU320272	KU320356	KU320456	KU320558	KU507703	KU528809	KU507805	KU507903	KU508005	KU508081
<i>Pseudomusonia</i> sp.	Colombia	MN719	KU320282	-	KU320466	KU320568	KU507713	KU528814	KU507814	KU507913	KU508015	-
<i>Pseudomusonia</i> sp.	Colombia	MN720	KU320283	-	KU320467	KU320569	KU507714	KU528815	KU507815	KU507914	KU508016	KU508089

<i>Thesprotia graminis</i>	USA	MN058	EF383182	EF383342	EF383502	EF383662	EF383826	EF383952	-	EF384081	FJ802473	FJ802975
<i>Thesprotia</i> sp.	Bolivia	MN593	KU320289	KU320370	KU320475	KU320577	KU507721	KU528822	KU507817	KU507922	KU508023	KU508094
<i>Thesprotia</i> sp.	Bolivia	MN604	KU320290	KU320371	KU320476	KU320578	KU507722	KU528823	KU507818	KU507923	KU508024	-
<i>Thesprotiella peruana</i>	Peru	MN269	FJ806076	FJ806251	FJ806441	FJ806640	FJ802854	FJ806965	-	FJ806802	FJ802665	FJ803116
Thespini Gen.n. sp.n.	Brazil	MN723	-	-	KU320474	KU320576	KU507720	-	KU507816	KU507921	-	-
<i>Angela</i> sp.	Bolivia	MN353	KU557266	KU557271	KU557275	KU557280	KU557247	-	KU507729	KU557258	KU557254	-
<i>Angela</i> sp.	Bolivia	MN355	KU557267	KU557272	KU557276	KU557281	-	-	KU507730	KU557259	-	-
<i>Brunneria</i> sp.	Bolivia	MN124	EF383224	EF383384	EF383544	EF383707	EF383867	EF383994	KU507731	EF384123	FJ802537	-
<i>Brunneria</i> sp.	Argentina	MN362	KU557268	-	KU557277	KU557282	KU557248	KU557251	KU507732	KU557260	KU557255	KU557263
<i>Coptopteryx argentina</i>	Argentina	MN609	KU320216	KU320316	KU320400	KU320504	KU507650	KU528775	KU507741	KU507848	KU507952	-
<i>Coptopteryx argentina</i>	Argentina	MN610	KU320217	KU320317	KU320401	KU320505	KU507651	KU528776	KU507742	KU507849	KU507953	-
<i>Coptopteryx brevipennis</i>	Peru	MN545	KU320218	KU320318	KU320402	KU320506	KU507652	KU528777	KU507743	KU507850	KU507954	-
<i>Coptopteryx gayi</i>	Chile	MN136	EF383228	EF383388	EF383548	EF383711	EF383871	EF383998	KU507744	EF384127	FJ802549	-
<i>Coptopteryx thoracica</i>	Argentina	MN612	KU320219	KU320319	KU320403	KU320507	KU507653	KU528778	KU507745	KU507851	KU507955	-
<i>Hagiomantis superba</i>	Peru	MN215	EF383267	EF383426	EF383586	EF383752	EF383907	EF384037	KU507753	EF384164	FJ802620	-
<i>Liturgusa cursor</i>	Nicaragua	MN618	KU320227	KU320325	KU320411	KU320515	KU507661	-	KU507756	KU507859	KU507962	KU508049
<i>Liturgusa cursor</i>	Nicaragua	MN620	KU320228	KU320326	KU320412	KU320516	KU507662	KU528782	-	KU507860	KU507963	-
<i>Liturgusa maya</i>	Nicaragua	MN617	KU320229	KU320327	KU320413	KU320517	KU507663	KU528783	KU507757	KU507861	KU507964	KU508050
<i>Liturgusa maya</i>	Peru	MN619	KU320230	KU320328	KU320414	KU320518	KU507664	KU528784	<200 bp	KU507862	KU507965	KU508051
<i>Liturgusa maya</i>	Mexico	MN621	KU320231	KU320329	KU320415	KU320519	KU507665	KU528785	KU507758	KU507863	KU507966	KU508052
<i>Macromantis hyalina</i>	Peru	MN608	KU320232	KU320330	KU320416	KU320520	KU507666	-	KU507759	KU507864	KU507967	KU508053
<i>Macromantis nicaraguae</i>	Costa Rica	MN144	EF383230	EF383390	EF383550	EF383713	EF383873	EF384000	KU507760	EF384129	FJ802554	FJ803036

<i>Macromantis nicaraguae</i>	Costa Rica	MN606	KU320233	KU320331	KU320417	KU320521	KU507667	<200 bp	KU507761	-	KU507968	KU508054
<i>Photiomantis planicephala</i>	Brazil	MN761	KU320276	KU320358	KU320460	KU320562	KU507707	KU528812	<200 bp	KU507907	KU508009	KU508084
<i>Cardioptera squalodon</i>	French Guiana	MN178	EF383248	EF383407	EF383567	EF383733	EF383889	EF384018	KU507733	EF384145	FJ802584	-
<i>Cardioptera squalodon</i>	French Guiana	MN539	KU320206	KU320307	KU320391	KU320494	KU507640	KU528768	KU507734	KU507838	KU507942	KU508036
<i>Cardioptera squalodon</i>	French Guiana	MN540	KU320207	-	-	KU320495	KU507641	-	KU507735	KU507839	KU507943	KU508037
<i>Cardioptera squalodon</i>	French Guiana	MN541	KU320208	KU320308	KU320392	KU320496	KU507642	-	KU507736	KU507840	KU507944	KU508038
<i>Chromatophotina</i> sp.n.	Peru	MN602	KU320214	KU320314	KU320398	KU320502	KU507648	KU528773	<200 bp	KU507846	KU507950	KU508042
<i>Chromatophotina</i> sp.n.	Peru	MN603	KU320215	KU320315	KU320399	KU320503	KU507649	KU528774	-	KU507847	KU507951	KU508043
<i>Microphotina</i> cf. <i>viridescens</i>	Bolivia	MN357	KU557269	KU557273	KU557278	KU557283	KU557249	KU557252	KU507781	KU557261	KU557256	KU557264
<i>Microphotina</i> sp. n.	French Guiana	MN599	KU320249	KU320344	KU320433	KU320536	KU507682	KU528796	KU507780	KU507880	KU507983	KU508069
Microphotinini Gen.n sp.n.	French Guiana	MN271	EF383300	FJ806252	FJ806443	FJ806641	EF383940	EF384071	KU507782	EF384196	FJ802667	-
<i>Orthoderella caatingaensis</i>	Brazil	MN727	KU320273	-	KU320457	KU320559	KU507704	-	-	KU507904	KU508006	-
<i>Orthoderella ornata</i>	Argentina	MN060	EF383183	EF383343	EF383503	EF383663	EF383827	EF383953	KU507806	EF384082	FJ802475	FJ802976
<i>Orthoderella ornata</i>	Bolivia	MN129	FJ806025	FJ806200	FJ806387	FJ806586	FJ802804	FJ806914	-	FJ806750	FJ802542	FJ803026
<i>Paraphotina</i> sp.	Bolivia	MN356	KU557270	KU557274	KU557279	KU557284	KU557250	KU557253	KU507807	KU557262	KU557257	KU557265
<i>Hicetia</i> sp.	Brazil	MN502	KU320225	KU320323	KU320409	KU320513	KU507659	KU528781	KU507754	KU507857	KU507961	KU508048
<i>Metriomantis pilosella</i>	French Guiana	MN536	KU320243	KU320338	KU320427	KU320530	KU507676	KU528790	KU507774	KU507874	KU507977	KU508063
<i>Metriomantis pilosella</i>	French Guiana	MN538	KU320244	KU320339	KU320428	KU320531	KU507677	KU528791	KU507775	KU507875	KU507978	KU508064
<i>Metriomantis</i> sp.n.	Bolivia	MN542	KU320245	KU320340	KU320429	KU320532	KU507678	KU528792	KU507776	KU507876	KU507979	KU508065
<i>Metriomantis</i> sp.n.	Bolivia	MN543	KU320246	KU320341	KU320430	KU320533	KU507679	KU528793	KU507777	KU507877	KU507980	KU508066
<i>Metriomantis</i> sp.n.	Bolivia	MN598	KU320247	KU320342	KU320431	KU320534	KU507680	KU528794	KU507778	KU507878	KU507981	KU508067
<i>Metriomantis</i> sp.n.	Bolivia	MN613	KU320248	KU320343	KU320432	KU320535	KU507681	KU528795	KU507779	KU507879	KU507982	KU508068

<i>Photina</i> sp.	Bolivia	MN295	FJ806095	FJ806268	FJ806462	FJ806662	FJ802869	FJ806984	KU507808	FJ806823	FJ802690	FJ803139
<i>Photinella</i> sp.	Brazil	MN498	KU320274	-	KU320458	KU320560	KU507705	KU528810	KU507809	KU507905	KU508007	KU508082
<i>Photinella</i> sp.	Brazil	MN499	KU320275	KU320357	KU320459	KU320561	KU507706	KU528811	KU507810	KU507906	KU508008	KU508083
<i>Stenophylla</i> sp.	Bolivia	MN352	KU320288	KU320369	KU320473	KU320575	-	KU528821	-	KU507920	KU508022	-
<i>Acontista amoenula</i>	Peru	MN596	KU320193	-	KU320378	KU320482	KU507629	-	-	KU507825	KU507929	-
<i>Acontista bimaculata</i>	Bolivia	MN111	EF383217	EF383377	EF383537	EF383700	EF383860	EF383987	-	EF384116	FJ802525	FJ803014
<i>Acontista bimaculata</i>	Bolivia	MN513	KU320194	KU320297	KU320379	KU320483	KU507630	KU528763	<200 bp	KU507826	KU507930	KU508030
<i>Acontista bimaculata</i>	Bolivia	MN514	KU320195	KU320298	KU320380	KU320484	KU507631	KU528764	<200 bp	KU507827	KU507931	KU508031
<i>Acontista cordillerae</i>	Nicaragua	MN508	KU320196	KU320299	KU320381	KU320485	KU507632	KU528765	-	KU507828	KU507932	-
<i>Acontista cordillerae</i>	Nicaragua	MN510	KU320197	KU320300	KU320382	KU320486	KU507633	KU528766	KU507728	KU507829	KU507933	KU508032
<i>Acontista gracilis</i>	French Guiana	MN511	KU320198	KU320301	KU320383	KU320487	KU507634	KU528767	-	KU507830	KU507934	-
<i>Acontista</i> sp.	Peru	MN517	KU320199	KU320302	KU320384	-	KU507635	-	-	KU507831	KU507935	-
<i>Callibia diana</i>	Brazil	MN728	KU320205	-	KU320390	KU320493	-	-	-	KU507837	KU507941	-
<i>Raptrix occidentalis</i>	Peru	MN520	KU320284	KU320365	KU320469	KU320571	KU507716	KU528817	-	KU507916	KU508018	-
<i>Raptrix occidentalis</i>	Peru	MN521	KU320285	KU320366	KU320470	KU320572	KU507717	KU528818	-	KU507917	KU508019	KU508091
<i>Raptrix perspicua</i>	Guyana	MN203	FJ806057	FJ806233	FJ806421	FJ806623	FJ802836	FJ806948	-	FJ806784	FJ802608	FJ803069
<i>Raptrix perspicua</i>	French Guiana	MN522	KU320286	KU320367	KU320471	KU320573	KU507718	KU528819	-	KU507918	KU508020	KU508092
<i>Raptrix perspicua</i>	French Guiana	MN525	KU320287	KU320368	KU320472	KU320574	KU507719	KU528820	-	KU507919	KU508021	KU508093
<i>Acanthops erosula</i>	Bolivia	MN527	KU320190	KU320294	KU320375	KU320479	KU507626	KU528760	KU507726	KU507822	KU507926	KU508028
<i>Acanthops falcata</i>	Venezuela	MN526	KU320191	KU320295	KU320376	KU320480	KU507627	KU528761	KU507727	KU507823	KU507927	KU508029
<i>Acanthops falcata</i>	Venezuela	MN549	KU320189	KU320293	KU320374	-	-	KU528759	KU507725	-	-	KU508027
<i>Acanthops</i> sp.	Venezuela	MN529	KU320192	KU320296	KU320377	KU320481	KU507628	KU528762	-	KU507824	KU507928	-

<i>Decimiana</i> sp.	Unespecified	MN112	EF383218	EF383378	EF383538	EF383701	EF383861	EF383988	KU507746	EF384117	FJ802526	FJ803015
<i>Decimiana tessellata</i>	Argentina	MN436	KU320220	-	KU320404	KU320508	KU507654	KU528779	KU507747	KU507852	KU507956	KU508044
<i>Metilia amazonica</i>	French Guiana	MN532	KU320239	KU320334	KU320423	KU320527	KU507672	-	KU507769	KU507870	KU507973	KU508059
<i>Metilia</i> sp.n.	Costa Rica	MN146	EF383232	EF383392	EF383552	EF383715	EF383874	EF384002	KU507770	EF384131	FJ802556	FJ803038
<i>Metilia</i> sp.n.	Nicaragua	MN534	KU320240	KU320335	KU320424	KU320528	KU507673	KU528788	KU507771	KU507871	KU507974	KU508060
<i>Metilia</i> sp.n.	Guatemala	MN535	KU320241	KU320336	KU320425	KU320529	KU507674	KU528789	KU507772	KU507872	KU507975	KU508061
<i>Metilia</i> sp.n.	Costa Rica	MN614	KU320242	KU320337	KU320426	-	KU507675	-	KU507773	KU507873	KU507976	KU508062
<i>Pseudacanthops centralis</i>	Nicaragua	MN530	KU320277	KU320359	KU320461	KU320563	KU507708	-	KU507811	KU507908	KU508010	KU508085
<i>Pseudacanthops centralis</i>	Nicaragua	MN531	KU320278	KU320360	KU320462	KU320564	KU507709	KU528813	KU507812	KU507909	KU508011	KU508086
Acanthopidae Gen.n sp.n.	Bolivia	MN311	FJ806107	FJ806282	FJ806478	FJ806680	FJ802881	FJ806999	-	FJ806838	FJ802705	FJ803149
<i>Chaeteessa</i> sp.1*	Nicaragua	MN622	KU320213	KU320313	KU320397	KU320501	KU507647	KU528772	KU507740	KU507845	KU507949	-
<i>Chaeteessa</i> sp.2*	Brazil	MN482	KU320212	KU320312	KU320396	KU320500	KU507646	-	-	KU507844	KU507948	-
<i>Mantoida</i> sp.1 *	French Guiana	MN179	EF383249	EF383408	EF383568	EF383734	EF383890	EF384019	KU507766	EF384146	FJ802585	-
<i>Mantoida</i> sp.2*	Bolivia	MN110	FJ806015	FJ806190	FJ806377	FJ806576	FJ802794	FJ806904	-	FJ806740	FJ802524	-
<i>Mantoida</i> sp.3*	Costa Rica	MN009	EF383155	EF383315	EF383475	EF383637	EF383799	AY491275	KU507767	AY491333	FJ802424	FJ802929
<i>Mantoida</i> sp.4 *	French Guiana	MN180	FJ806044	FJ806220	FJ806408	FJ806610	FJ802822	FJ806935	KU507768	FJ806771	FJ802586	-
<i>Blatta orientalis</i> *	-	BL097	-	FJ806140	FJ806323	FJ806520/1	FJ802746	FJ806875	-	-	FJ802410	FJ802919
<i>Cryptocercus kyebangensis</i> *	-	BL115	FJ805967	FJ806141	FJ806324	FJ806522	-	FJ806876	-	-	FJ802411	-
<i>Gyna capucina</i> *	-	BL048	FJ805966	FJ806138	FJ806321	FJ806517	FJ802744	FJ806873	-	-	FJ802408	-
<i>Supella longipalpa</i> *	-	BL138	-	FJ806144	FJ806327	FJ806525/6	FJ802748	FJ806878	-	FJ806721	FJ802414	FJ802920
<i>Macrotermes subhyalinus</i> *	-	IS095	FJ805973	FJ806150	FJ806334	FJ806532	FJ802752	FJ806885	-	-	-	-
<i>Mastotermes darwiniensis</i> *	-	IO9	EF383150	EF383310	EF383470	EF383632	EF383795	EF383948	-	-	-	-

Appendix 2

Appendix 2 Chapter 3. Examined material. See Material and Methods for collection accronyms.

THESPIDAE

PSEUDOPOGONOGASTRINAE. *Pseudopogonogaster hebardi* - ECUADOR: 1 male (holotype of *Calopteromantis hebardi*), Cerro de Bulán, Valle del Paute (F. Campos leg.) (ANSP), 1 female, Cerro de Bulán (F. Campos leg.) (note: topotypical to *Calopteromantis hebardi*) (ANSP), 1 male & 2 females, Loja, Jardín Botánico “Reinaldo Espinosa” (2152m), 4°2,284’S; 79°11,957’W, 1.II.2001 (C. Rasmussen leg.) (UNALM), 1 female, Loja, Saraguro, (2700m) 03° 37’ 18” S; 79° 14” 36” W, 7.IV.2007 (G. Onore leg.) (UNALM), 1 female, Loja, Saraguro (3100m), 20.III.1997 (G. Onore leg.) (UNALM). *Pseudopogonogaster kanjaris* - PERU: 1 male (holotype) & 7 females (paratypes), Lambayeque, Ferreñafe province, Kañaris district, Mountain Forest of Kañaris, 06° 05’ 32.220” S; 79° 16’ 0.234” W, 3000 m, 1.V.2010 (J. Rivera & E. Castro leg.) (UNALM). *Pseudopogonogaster mirabilis* - ECUADOR: 1 female (holotype), Balzapampa (R. Haensch S.) (MIZ), 2 females, Cotopaxi, La Otonga, 1000m., 5.X.1996 & 13.II.1997 (M. Ayala leg.) (UNALM) (note: topotypical to *Calopteromantis otongica*). *Pseudopogonogaster* sp.n. - PERU: 1 female, Chanchamayo, 2000m (ANSP).

PSEUDOMIOPTERYGINAE. *Pseudomiopteryx amazonensis* - BRAZIL: 1 male (holotype, code: MZLQ-I0094), Amazonas, Manaus, 26.II.1958 (Elias & Roppa leg.) (ESALQ), 1 female (code: 1984), Amazonas, Manaus, 04.II.[19]92 (A.A. Lise leg.) (id. as *Anamiopteryx tuberculata* by L. Jantch) (MCTP); 1 male (code: 0730), Amazonas, Manaus, 13.I.1978 (A.E. Ratcliffe leg.) (id. as *Pseudomiopteryx infuscata* by L. Jantsch) (MCTP). *Pseudomiopteryx bogotensis* - COLOMBIA: 1 male (holotype), Bogotá (MNHN), 5 males, St. Antonio, 2000m 8.IV–14.VI.1908 (id. as *Pseudomiopteryx bogotensis* by M. Hebard) (USNM). *Pseudomiopteryx columbica* - COLOMBIA: 1 male (holotype) (ZMB). *Pseudomiopteryx decipiens* – COLOMBIA: 1 male (syntype, mislabeled as holotype) “N. Granad.” (Goud. Leg.) (ZMB), 2 females (syntypes), Tolima, 1300m (SDEI). *Pseudomiopteryx cf. festae* - ECUADOR: 1 male, Imbabura, Lita 600m, 00° 50’ 24” N; 78° 27’ 18” W, 19.I.2002 (J.E. Iglesias leg.) (UNALM), 1

male, Pastaza, 13.3Km NW Puyo, rd to H Snta. Barbara, 1150m Rio Pindo, 9.IV.1990 (S.J. Weller, P. Batra & M.J. Ryan leg.) (USNM), 1 male, Past. Puyo (16 Kms. W.) at “blacklite”, 3.II.1976 (Spangler *et al.*, leg.) (USNM). ***Pseudomiopteryx guyanensis*** - FRENCH GUIANA: 1 male (syntype) (MNHN), St. Laurent du Maroni (Le Moullet leg.) (MNHN), 1 female (syntype), St. Laurent du Maroni (Le Moullet leg.) (MNHN), 1 male, St. Laurent du Maroni –I-80 (J. Boudinot, P. Thiaucourt & H. de Toulgoët leg.) (MNHN). ***Pseudomiopteryx infusata*** - COSTA RICA: 1 male, Turrialba, VII.15–19.[19]65 (P.J. Spangler leg.) (USNM), 1 female, Finca La Selva nr. Puerto Viejo, Sapapiqui Dist., Heredia Prov., 22–31.VII.1969, DHJ 80m (D.H. Jansen leg.) (USNM), 1 male (code: 1969), Pr. Puntarenas, Río Catarata, 5.5 km S El Brujo on Hy to Zona Sur, 120m, 6.VII.1992 (Contreras & Munoz leg.) (id. as *Pseudomiopteryx infusata* by L. Jantsch) (MCTP), 1 male & 1 female, Piedras Negras (Schild & Burgdorf leg.) (USNM); GUATEMALA: 1 male (syntype), Las Mercedes 300ft (Champion leg.) (MHNG), 2 males, Suchitepequez W. Samayac, 11.VI.1966 (Flint & Ortiz leg.) (USNM), 4 males, Suchitepequez, Cuyotenango, 10–20.VI.1966 (Flint & Ortiz leg.) (USNM), 1 male, Moate Grande, 29.V.1925 (E.G. Smith leg.) (USNM); HONDURAS: 1 female, “36 6 193” on banana debris, 16.III.1936 (Galveston leg.) (USNM), 1 female, X.1938 (Galv. “1786”) (USNM); MEXICO: 2 males, Ver[acruz], Córdoba, 20–30.VIII.[19]66 (A.B. Lau leg.) (USNM); PANAMA: 1 male (syntype), V[olcán] de Chiriqui 2500–4000ft (*sensu* Saussure & Zehntner, 1894), 1894 (Champion leg.) (BMNH), 3 males, Boquette, 16–17.VII.1967 (O.S. Flint Jr. leg.) (USNM). ***Pseudomiopteryx meridana*** - VENEZUELA: 1 male (holotype) (MHNG). ***Pseudomiopteryx spinifrons*** – BRAZIL(?): 1 male (MHNG) (id. as *Pseudomiopteryx spinifrons* by R. Roy). ***Pseudomiopteryx*** spp. (possibly new species) - BOLIVIA: 3 males, Dept. Beni, Rio Mamore at mouth of Rio Ibare, 16–17.VIII.1965 (J.K. Bouseman leg.) (CMNH); PERU: 1 male, 1 female, Cusco, La Convencion, Distr. Echárte (male at light), 11.V.1998, (several collectors) (UNALM), 38 males, Madre De Dios, Tambopata, (Malaise Trap), IV.2000–I.2001, (A. de la Cruz leg.) (UNALM), 1 male 1 female, Posada Amazonas, 28–29.IX.2001 (M. Deza leg.) (UNALM); 1 female, Madre De Dios, Manu, X.2000, (J. Santisteban leg.), 1 male, Madre De Dios, Boca Río La Torre (300 m), 27.VII.1980, (G. Lamas leg.) (MUSM), 1 male (code: ROM863562), Madre de Dios, Manu Natl. Pk. Cocha Cashu Biol. Stat. Trail 1, 22–27.VIII.1986 (D.C. Darling leg.) (ROM). ***Promiopteryx fallax*** - COLOMBIA: 1 male (holotype), Bogotá (ZMB). ***Promiopteryx granadensis*** - COLOMBIA: 1 male (holotype of *Miopteryx granadensis*), Bogotá (MNHN).

Promiopteryx sp. - VENEZUELA: 1 male, Aragua, Rancho Grande, 1000m, 20.V.1975 (Salcedo & Clavijo leg.) (id. as *Promiopteryx fallax* by P.S. Terra) (MZUSP), 1 male, Rancho Grande, 1100m, 25.IV.1949 (F. Fernandez leg.) (id. as *Promiopteryx fallax* by P.S. Terra) (MZUSP).

BANTIINAE. *Bantia chopardi* - COLOMBIA: 1 male (holotype of *Mionicella chopardi*), Bogotá (ZMB). ***Bantia fusca*** - FRENCH GUIANA: 1 male (holotype), Nouveau Chantier, “Février” (Le Moullet leg.) (MNHN). ***Bantia metzi*** - BRAZIL: 2 males (holotype and paratype), São Paulo, Santos, 30.V.1894 (J. Metz leg.) (holotype in ZMUH, paratype in NHMW). ***Bantia michaelisi*** - BRAZIL: 3 males (holotype and paratypes), Espirito Santo, 22.IV.1898 (J. Michaelis leg.) (holotype and one paratype in ZMUH, additional paratype in NHMW). ***Bantia marmorata*** - BRAZIL: 1 female (syntype), Rio de Janeiro (MHNG). ***Bantia nana*** - BRAZIL: 1 male (holotype), Rio de Janeiro, Umberto Antunes, 23.I.1960 (D. Lacombe leg.) (ESALQ). ***Bantia weneri*** - PERU: 1 male (code: ROM107013), Madre de Dios, Manu Natl. Pk., Cocha Cashu Biol. Station 11°55’S; 77°18’W 380m 18–22.VIII.1986 (D.C. Darling leg.) (ROM). ***Bantia*** sp. - BRAZIL: 1 male, Rio de Janeiro, Itatiaia 700m, 18.II.1946 (BM), 1 male, Mato Grosso, Organ. Mt., IV.1935 (P. Sandig leg.) (USNM). ***Bantia*** spp. (likely new species) – BOLIVIA: 1 female, Tumupasa, 1921 (WmMMann leg.) (USNM), 1 male “Dec. 1921” (USNM); PERU: Madre de Dios, Tambopata Wldlf. Res. 30Km SW Pto. Maldonado, 12°50’S; 69°20’W, 290m, 1–14.X.1982 (J.J. Anderson leg.) (USNM); VENEZUELA: 1 male, Bolivar ptari-tepui, 30mi. N. Kavanayén 1800m., 17–19.VIII.1970 (R.E. Dietz IV leg.) (USNM). ***Bantiella pallida*** - COLOMBIA: 1 male (holotype) (ZMB). ***Bantiella simoni*** - VENEZUELA: 1 male (holotype of *Miopteryx simoni*), Caracas (E. Simon leg.) (MNCN). ***Bantiella trinitatis*** - GUYANA: 1 female, Kartabo, Bartica Distr. (W. Beebe leg.) (AMNH), 1 male, Kartabo, VIII.1922 (M.D. Haviland leg.) (BM); TRINIDAD & TOBAGO: 1 male (syntype) and 1 female (syntype), Trinidad Island, Port of Spain, 1911–104 (H. Caracciola leg.) (BM), 1 male & 1 female (syntypes of *Bantiella fusca*), Trinidad Island, Port of Spain, 1911–104 (H. Caracciola leg.) (BM), 11 males & 3 females, Trinidad Island, Arima Valley (various dates between III.1953–IV.1952) (B.W.I. leg.) (USNM), 1 male (code: ROM872017), Trinidad Island, St. George Co., Arima River, 10°41’N; 61°17’W, 3.VI.1987 (M.L. Pickles leg.) (ROM), 1 female (code: ROM872023), Trinidad Island, St. George Co. Cocoa estate 1 mi. N. Paxvala on Saddle Rd., 7.VI.1987 (M.L. Pickles leg.) (ROM), 1 male, Trinidad Island, Maracas, 8.XI.1921 (Busck leg.) (USNM) 1 male, Tobago Island, St. John Prov., Hermitag River Bridge Charlottesville, in

malaise trap, 12–21.III.1979 (D. Hardy & W. Rowe leg.) (USNM), 1 female, Tobago Island, St. Paul Prov. Roxborough 6.5 miles N. on Bloody Bay road (USNM), 1 male & 1 female, Tobago Island nr. Bloody Bay, Bloody Bay River (in secondary rainforest), 24.X.1977 (BM); VENEZUELA: 1 male (holotype of *Bantiella columbina*), Caracas (ZMB), 1 male, Bolivar, 83Km S. El Dorado, 900m, 28.VI.1984 (D.S. Bogar leg.) (USNM). *Bantiella* sp. n. - VENEZUELA: 1 male & 1 female, Bolivar, Cerro Guiaquinima tepui camp 1: 63°37'W; 5°56'N, 1000m, II.1990 (D. Grimaldi leg.) (AMNH). *Diabantia minima* - COLOMBIA: 1 male (holotype) (ZMB). *Diabantia* sp (possibly new species) - VENEZUELA: 3 males, Puerto Colombia, VIII.1988 (J. Anderson leg.) (USNM). *Mantellias pubicornis* - BRAZIL: 1 male (lectotype), Pará (OUMNH); FRENCH GUIANA: piste Dégrad Corrèze pk 0.1 (at light), 27.XII.1996 (P. Kinol leg.) (MNHN). *Mantellias* sp. - PERU: 1 male & 1 female (juv.), Madre de Dios, canopy-fogging project (T. Erwin leg.) (USNM), 1 male, Huánuco, “Cuchara”, Rio Huallaga 450m (F. Woytkosky leg.) (NHMW). *Mantillica nigricans* - BRAZIL: 1 male (holotype), Amazonas, upper Amazon, Fonte Boa (Bates leg.) (OUMNH). *Thrinaconyx brunneri* - VENEZUELA: 1 male, Aragua, Rancho Grande 1200m, 18.VII.1981 (M. Descamps leg.) (MNHN) (id. as *Thrinaconyx* sp. by R. Roy), 1 male, Aragua, Rancho Grande, 10–21.II.1969 (Duckworth & Dietz leg.) (USNM). *Thrinaconyx fumosa* - COSTA RICA: 1 male, Turrialba, 13–17.III.1965 (S.S. & W.D. Duckworth leg.) (USNM), 1 male, Turrialba, 19.VI.1951 (O.L. Cartwright leg.) (USNM), 1 female, Heredia, La Selva Biol. Station nr. Puerto Viejo de Sarapiquí, 18.II.1985 (L. Herman leg.) (AMNH); PANAMA: 1 male (syntype), V[olcán] Chiriqui 2–3000 ft., 1894–125, Godman-Salvin (Champion leg.) (BM), 1 male (syntype), Volc[án] Chiriqui 25–4000 ft. Godman-Salvin (Champion leg.) (MHNG), 1 male, El Valle, 15.VII.1967 (O. S. Flint leg.) (USNM), 1 male & 1 female, Barro Colorado, Gatun Lake Pan., 20.VI–17.VII.1923 (R.C. Shannon leg.) (USNM), 2 males, Barro Colorado Isl[and], 5.IV–20.VI.1965 (S.S. & W.D. Duckworth leg.) (USNM), 1 male, C[anal] Z[one] pipeline road, Rio Frijoles, 6–7.VII.1967 (Flint & Ortiz leg.). *Thrinaconyx sialidea* - BRAZIL: 1 male (holotype of *Mantillica sialidea*), Ega [Amazonas] (Bates leg.) (OUMNH). *Thrinaconyx* spp. (possibly new species) - COLOMBIA: 1 male, Anchicaya, Valle del Cauca 200–300m, 27.IV.1969 (M. Salazar *et al.* leg.) (MNHN) (id. as *Thrinaconyx fumosus* by R. Roy); ECUADOR: 2 males, Quito, 1923 (R. Benoist leg.) (MNHN), 1 male, Past[aza], Puyo, in malaise trap, 1–7.II.1976 (Spangler *et al.* leg.) (USNM), 1 male, Esmeraldas, Mayronga 100m, 25.X.1993 (G. Onore leg.) (CM); PERU: 1

male, Cuzco, Parq[ue]. National Manu, Pillcopata a Paucartambo 39Km 1782m, 13°04'40.2"; S; 71°33'25.5"W, 24.II.2009 (J. Barhent leg.) (MNHN), 1 male, Junín, San Ramón, Pampa Hermosa, vicinity of Zona Reservada P[ampa] H[ermosa], XII.2008 (C. Carrera leg.) (UNALM).

MIOBANTIINAE. *Anamiopteryx borellii* – BOLIVIA: 1 male, Villa Montes, X.[19]30 (Dr. Eisentarut leg.) (ZMB) (id. as *Anamiopteryx "borelli"* by F. Lombardo); BRAZIL: 1 female, Mato Grosso, Chapada (nr. Cuiabá), "No." (Carnegie Mus. Exp.) (ANSP) (id. as *A. tuberculata* by P.S. Terra). *Anamiopteryx grandis* - BRAZIL: 1 male (holotype), Espiritu Santo, 6.III.1898 (H. Fruhstorfer leg.) (ZMUH), 1 female (paratype), Espiritu Santo, 22.IV.1898 (J. Michaelis leg.) (ZMUH), 1 female, Espiritu Santo, Rés. de Sooretama, IBAMA, 19° 03' 20.7" S; 40° 08' 49" W (Sta 1), "forêt semi-décidue" 44km NNE Linhares, 15.XI.2000 (P. Grandcolas & R. Pellens leg.) (MNHN) (id. as *Anamiopteryx grandis* by R. Roy). *Anamiopteryx tuberculata* - BRAZIL: 1 male (holotype of *Trachymiopteryx tuberculata*), Goiás (ANSP), 11 males, Dist[rito] Fed[eral], Planaltina, 1000m "Cerrado", mal[aise]. trap, 24.IX–25.X.1985 (S.E. Miller leg.) (USNM); PARAGUAY: 1 male, Horqueta, 45 Km. E of Rio Paraguay, 23°N; 57°W, 11.XII.1983 (A. Schulze leg.) (ANSP). *Chloromiopteryx mirim* - BRAZIL: 1 female (holotype of *Emboicya mirim*), São Paulo, Praia do Iporanga, 11.X.1959 (LT.F. leg.) (MZUSP), 1 female, São Paulo, E.B. Boraceia, 23.IV.1962 (Rabello leg.) (MZUSP), 1 female, São Paulo, E.B. Boraceia, 12.II.1958, (H.A. Camargo leg.) (MZUSP), 1 female, São Paulo, Salesopolis, E.B. Boraceia, "coleta noturna" [at night], 24–28.III.2011, (Biff, Kawada & Rodrigues leg.) (MZUSP), 1 male, São Paulo, E.B. Boraceia, 1.IV.1910 (L. Travessos Filho leg.) (MZUSP), 1 male, São Paulo, Ubatuba, Pq. Est. Serra do Mar, Nucleo Picinguaba [malaise] 800m, I–III.2006 (E.F. Santos & C.P. Scott-Santos leg.) (MZUSP), 1 female, São Paulo, Salesopolis, Station Biologique de Boraceia 800m (at night, on vegetation) forêt de brume", 26.X.1992 (L. Desutter-Grandcolas leg.) (MNHN). *Chloromiopteryx modesta* - BRAZIL: 1 male (holotype of *Metathespis modesta*), Rio de Janeiro, Petropolis (alto da cerra), 2.II.1960, D.Z.105.60 (Mielks-Herbert leg.) (ESALQ). *Chloromiopteryx thalassina* - BRAZIL: 1 male (lectotype of *Mantis thalassina*), Santa Catharina, "Eschsch." Cat. No. 617 (labeled as *Stagmomantis thalassina* Burm.) (ZMB), 1 male, "St. Cathar." [Santa Catharina], Eschsch, Cat. No. 617 (labeled as *Stagmomantis thalassina* Burm.) (ZMB). *Chloromiopteryx* spp. (likely new species) - BRAZIL: 1 male, Rio Grande do

Sul, Porto Alegre, São Francisco de Paula, 25.III.85, (G.S. Carvalho leg.) (MCTP), 1 female, São Paulo, Campos do Jordão, 3.I.1948 (F. Lane leg.) (MZUSP), 1 female, São Paulo, Campos do Jordão 1650m (F. Lane leg.) (MZUSP), 1 male, São Paulo, Eug. Lefevre, 1200 m, C[ampos] de Jordão, 21.II.1963 (J. Guimaraes *et al.*, leg) (MZUSP). ***Eumiopteryx bicentenaria*** - BRAZIL: 1 male (holotype of *Piracicaba bicentenaria*), São Paulo, Piracicaba, campus ESALQ (A. Quintana leg.) (ESALQ), 1 female (labeled as the “allotype” *Piracicaba bicentenaria* but not part of the type series), São Paulo, Itu, VIII.1973 (ESALQ). ***Eumiopteryx laticollis*** - BOLIVIA: 1 female (allotype), Prov. Sara, 750m (J. Steinbach leg.) (ZMB); PARAGUAY: 1 male (holotype) (ZMB), 1 Depto. Amambay, Parq. Nac. Cerro Cora 22° 39’ S; 56°01’ W, 7–10.IV.1986 (M. Pogue & M. Solis leg.) (USNM). ***Eumiopteryx terrai*** - BRAZIL: 1 male (holotype of *Calopteromantis terrai*, code: 2046), Mato Grosso, CoopHEMA, Cuiabá, 18.III.1990 (E. Fachim leg.) (MCTP). ***Eumiopteryx* spp.** (likely new species) - BRAZIL: 1 male & 1 female, Mato Grosso, Uirapuru usine Alcomat 14° 16’ 4’’ S; 59° 14’ 1’’ W, 20.I.1996 (A. Foucart leg.) (MNHN); FRENCH GUIANA: Montagne de Kaw, PK 39 Arbre 6, 17–23.VIII.1994 (P.E. Roubaud leg.) (id. as *Miobantia aptera* by R. Roy) (MNHN); PERU: 1 male, San Martín, Moyobamba, Hera, 4.VI.1947 (F. Woytkowski leg.) (AMNH), 1 female, Huánuco, Tingo María, campus UNAS, [on cacao tree] 9° 25’ S; 76° 00’ W, 14.V.2005 (G. Joyo leg.) (UNALM).

Miobantia aptera - BRAZIL: 1 male, Espiritu Santo, Rés de Linhares, O/RD 19° 08’ S; 40° 03’ W (Sta 2), 06.X.1999, 40km NNE Linhares “forêt semi-décidue”, “Mata Alta” (P. Grandcolas & R. Pellens leg.) (MNHN). ***Miobantia ciliata*** - BRAZIL: 2 males, Santa Catarina, Corupa (Hansa Humboldt), III/XI.1946 (A. Maller leg.) (AMNH), 1 male, no data (USNM). ***Miobantia immanis*** - BRAZIL: 1 male (paratype), Espiritu Santo, Linhares, 25.VI.1990 (J.S. Santos leg.) (id. as *Miobantia rustica* by L. Jantsch) (MCTP). ***Miobantia cf. punctata*** - PARAGUAY: 5 males, Depto. Paraguari 26.2km SE Ybycuí, Parq. Nac. Ybycuí 26° 07’ S; 56° 47’ W, 15–18.III.1986 (M. Pogue & M. Solis leg.) (USNM), 1 male, 5km SE Ybycuí, 13.I.1983 (E.G. Riley leg.) (AMNH). ***Miobantia sulista*** - ARGENTINA: 1 male, Mis[iones], San Ignacio, 4–5.IV.1974 (C. & M. Vardy leg.) (BMNH), 2 females, Misiones, env. De San Ignacio, Villa Lutecia, 1910 (E.R. Wagner) (MNHN), 1 male, Misiones, Iguazú, XI.1945 (Duret leg.) (MNHN), 1 male, Misiones, Ao. Piray Mini W Dos Hermanas, 23.XI.1973 (O.S. Flint Jr. Leg.) (USNM); BRAZIL: 1 male (holotype, code: 0776), Rio Grande do Sul, Porto Alegre, 03.II.1985 (L. Jantsch leg.) (id. as *Miobantia ciliata* by L. Jantsch) (MCTP), 1 male (paratype, code: 1504), Rio Grande do Sul,

Porto Alegre, 25.XI.1988 (A.L. Azevedo leg.) (id. as *Miobantia phryganea* by L. Jantsch) (MCTP), 1 female (paratype, code: 1505), Rio Grande do Sul, Garabi, 3–12.I.1989 (E. Lerner leg.) (id. as *Miobantia aptera* by L. Jantsch) (MCTP), 1 male & 1 female (paratype, code: 0469), Porto Alegre, São Borja 8.VII.1980 (A.A. Lise leg.) (id. as *Miobantia aptera* by L. Jantsch) (MCTP), 1 male, Rio Grande do Sul, Sapiranga, 26.VIII.1979 (S. Moraes leg.) (id. as *Miobantia phryganea* by L. Jantsch) (MCTP), 1 male, Rio Grande do Sul, Viamão, 28.X.1989 (M. Coitinho leg.) (id. as *Miobantia phryganea* by L. Jantsch) (MCTP), 1 male, Rio Grande do Sul, Caixas do Sul, 20.X.1993 (R.A. Lise leg.) (id. as *Promiopteryx punctata* by L. Jantsch) (MCTP), 1 male, Paraná, Caviuna, XII.1945 (A. Maller leg.), 2 males, Santa Catharina, Nova Teutonia 300–500m, XI.1968 & 25.XII.1961 (F. Plaumann leg.) (USNM), 1 male, Santa Catharina, Rio Vermelho, 1945 (A. Maller leg.) (AMNH). ***Paradiabantia perparva*** - BRAZIL: 1 male (holotype), São Paulo, Monte Mor, 9.X.1963 (Messias leg.) (ESALQ), 1 male (paratype), São Paulo, Pirassununga, 5.IV.1968 (R.P.L. Carvalho leg.) (ESALQ). ***Paradiabantia*** spp. (possibly new species) - BRAZIL: 1 male, Mato Grosso 12°50'S; 51°47'W, 5.X.1968 (O.W. Richards leg.) (BMNH), 3 males, Mato Grosso do Sul, Maracaju, II.1937 (R.C. Shannon leg.) (USNM), 1 male, Rio Grande do Sul, Rio Pardo, 8.IV.1984 (A. Tavares leg.) (id. as *Diabantia perparva* by L. Jantsch) (MCTP).

THESPINAE - MUSONIELLINI - *Eumusonia intermedia* - ARGENTINA: 1 male, Chaco de Santa Fé, Las Garzas (along the shore of Las Garzas river), 1903, (E.-R. Wagner leg.) (MNHN); BRAZIL: 2 males (syntypes), Rio Grande do Sul, Santa Maria (ESALQ), 1 male, Rio Grande do Sul, C. de Canoa, 12.XI.1990 (A.A. Lise leg.) (id. as *Leptomiopteryx dispar* by L. Jantsch) (MCTP), 1 female, Rio Grande do Sul, Tapes, 8.VI.1985 (M.Z. Pinzon leg.) (id. as *Eumusonia viridis* by L. Jantsch) (MCTP), 1 female, Rio Grande do Sul, Porto Alegre, 29.IV.1982 (M. Feijó leg.) (id. as *Eumusonia viridis* by L. Jantsch) (MCTP). PARAGUAY: 3 males, Sapucay, 25.I–13.II.1901 (W,T. Foster leg.) (USNM) (id. as *Musonia livida*), 1 male, Puerto Pinasco, 30.IX.1920 (A. Wetmore leg.) (USNM), 1 male, Paraguari Dept., Ybycuí (10 KmE) in Ybycuí National Park, 12–24.IV.1980 (P.J. Spangler leg.) (USNM). ***Eumusonia viridis*** - BRAZIL: 1 male (holotype), [São Paulo], Salto Grande (ZMB). ***Eumusonia*** spp. (possibly new species) - ARGENTINA: 1 male, Villa Guillermina, F.C.S.Fe., 5.V.1925 (K.J. Hayward leg.) (BMNH); BOLIVIA: 1 male, Beni, San Antonio, XI.1921 (W. Mann leg.) (USNM), 1 male, Santa Cruz, Prov. Valle Grande, 3Km S of Valle Grande, 30.IV.1999 (P. Bettella leg.) (BMNH), 6 males,

Sara, 450m. (J. Steinbach leg.) (ANSP); BRAZIL: 1 male, Marenhao, Ilha de Balsas reg.[at light], 19–20.VIII.1963 (Westminster School Exp. to Brazil leg.) (BMNH), 1 male, Rio Grande do Sul, Pelotas, 29.X.1961 (C.M. Biezanko leg.) (BMNH), 1 male, Mato Grosso, Chapada dos Parecis, Campos de Julho, 20.XII.1998 (M. Lecor leg.) (USNM), 1 male, Mato Grosso, Reg. de Uirapuru, Chapada dos Parecis, 24.III.1997 (M. Lecoy leg.) (MNHN), 1 female [Mato Grosso] Corumba (ANSP); PARAGUAY: 1 male, Dpto. Concepción, Ao. Tagatiya-mi 22°39'S 57°32'W, 3-5.IV.1986 (M. Pogue & M. Solis leg.) (USNM), 1 male, Dpto. Amambay, Parq. Nac. Cerro Cora, 22°39'S 56°01'W, 7-10.IV.1986 (M. Pogue & M. Solis leg.) (USNM). ***Leptomiopteryx dispar*** - BRAZIL: 1 male (holotype of *Navimons amapaensis*), Amapá, Serra do Navio (I.C.O.M.I.), II–IV.1961 (J.O.M. Carvalho leg.) (ESALQ); FRENCH GUIANA: 2 males (syntypes), 2 females (syntypes), St. Laurent du Maroni (Le Moullet leg.) (2 syntypes of each sex at ANSP and MNHN), 1 male, Arataye, Affl. Approuagues, 8 Km NE pied Saut Parare (chasse de jour), 28.V.1988 (L. Desutter leg.) (MNHN), 1 female, Arataye, Affl. Approuagues, aval du Saut Parare (chasse de jour), 8.VII.1988 (L. Desutter & P. Grandcolas leg.) (MNHN).

Musoniella affinis - BRAZIL: 1 male (syntype; note: labeled as “type” but not explicitly designated as such in the original description), São Paulo, Rib[erão] Preto, Guarani, XI.1957 (Zamith-Maniconi leg.) (ESALQ), 1 male (syntype), São Paulo, Riberão Preto, Rio Tamanhué, 27.IV.1954 (Duret leg.) (USNM), 1 male (syntype), Minas Gerais, Carmo do Rio Claro, VI.15.1942 (Carvalho leg.) (USNM); ***Musoniella argentina*** - ARGENTINA: 1 male (syntype? of *Miopteryx argentina*), Corrientes (MHNG), 1 male, Santiago del Estero, Rio Salado, 1909 (E.R. Wagner leg.) (MNHN), 2 males, Villa Ana, F.C.S.F., II.1924 (K.J. Hayward leg) (id. as *Musoniella argentina* by M. Beier) (BMNH), 2 males, Entre Rios, Pronunciamiento, III-IV.1964 (Walz leg.) (ANSP); BRAZIL: 1 male, Minas Gerais, Carmo do Rio Claro, VI.15.1942 (Carvalho leg.) (USNM), 1 male, São Paulo, Riberão Preto, Rio Tamanhué, 27.IV.1954 (Duret leg.) (MNHN); PARAGUAY: 1 male, Asunción, 10.XII.1954 (J. Munro leg.) (Smith.), 1 male, Dpto. Cordillera, Caacupé, 12.VIII.1981 (R.D. Cave leg.) (USNM), 1 male, Dpto. Cordillera, Caacupé, 15.XII.1980 (R.D. Cave leg.) (USNM), 1 male, Villa Rica, XII.1921 (P. Jorgensen leg.) (ANSP), 1 male (no data) (ANSP). ***Musoniella costalis*** - BRAZIL: 1 male (holotype of *Musonia costalis*; code: 8972), Goyaz (ANSP). ***Musoniella fragilis*** - BRAZIL: 1 male (holotype of *Eumusonia fragilis*) (A.M. Parko leg.) (ESALQ), 3 males, Mato Grosso, Reg. de Uirapuru, cpd. Dos Parecis, 9–14.IV.1997 (M. Lecoy leg.) (MNHN). ***Musoniella ipiranga*** -BRAZIL: 1

male (holotype), São Paulo, Ypiranga, II.1913 (H. Luderwaldt leg.) (ANSP). ***Musoniella laevithorax*** - PARAGUAY: 1 male (holotype of *Musoniella laevithorax*) (Lauffer leg.) (MNCN). ***Musoniella longicauda*** - BRAZIL: 1 male (holotype), Minas Gerais, Poços de Caldas, Morro do Ferro, 8.IX.1966 (J. Becker *et al.* leg.) (ESALQ), 1 male, Minas Gerais, Carmo do Rio Claro, 15.VI.1942 (Carvalho leg.) (USNM). ***Musoniella parva*** - PARAGUAY: 1 male (holotype of *Musoniella parva*), Panadero, 26.X.1896 (Jerrman leg.) (ZMUH). ***Musoniella precaria*** - BRAZIL: 1 male (holotype), Pernambuco, Petrolima, 31.III.1981 (G.J. Moraes leg.) (ESALQ), 1 male, Pernambuco, Taquaritinga do Norte, 17.VII.1990 (Amedegnato & Poulain leg.) (MNHN). ***Musoniella stigmatica*** - BRAZIL: 1 male (holotype of *Promiopteryx stigmatica*) (ZMB). ***Pizaia seabrai*** - BRAZIL: 1 male (holotype of *Musoniella seabrai*), Pernambuco, Recife, 24.VII.1952 (O. Regow leg.) (ESALQ). ***Pizaia*** sp. - BRAZIL: 2 males (pre-adults) & 1 female, Pernambuco, Recife, Parque Zoobotanico, 5.VI.1963 (Vanzo leg.) (MZUSP).

THESPINAE - THESPINI. ***Dougonyx amazonicus*** - BRAZIL: 1 female (holotype of *Galapagia amazonica*) PA[Pará]/AP[Amapá], Rio Jarí, cach[oeira] Caripaturi, 22.VIII.1961 (Dr. W. Egler leg.). ***Dougonyx maculosus*** - FRENCH GUIANA: 1 male (lectotype of *Mionyx maculosus* - here designated, see text for details), St. Laurent du Maroni (Le Moult leg.) (MNHN), 1 female (paralectotype of *Mionyx maculosus*), St-Jean du Maroni (Le Moult leg.) (MNHN), 1 female, Camp. Arataye, 28.IV.1997 (A. Francois leg.) (MNHN), 1 male, Oyapock (MNHN); GUYANA: 1 male, Essequibo R[iver], Moraballi creek, 9.IX.1929 (Oxf[ord] Univ[ersity] Exped[itio]n leg.) (id. as *Leptomiopteryx dispar*) (MNHN). ***Dougonyx*** spp. - BRAZIL: 1 male, [Amazonas] Rio Purus, **Hyutanahan (Carnegie Mus. Exp.; S.M Klages leg) (ANSP) (id. as *Thespis* sp. by P.S. Terra); PERU: 1 male, Loreto, Iquitos, 5.I.1975 (P. Sánchez leg.) (UNALM). ***Galapagia solitaria*** - ECUADOR (Galapagos Islands): 3 females, Baltra Island 30m, 24.I.1989, arid zone night beating (S. Peck leg.) (USNM), 1 male & 2 females, Baltra Island, 1948 (K.Vinton leg.) (USNM), Baltra Island, 1 male, 30m. 24.I.1989 (arid zone, nightbeating) (S. Peck leg.), 1 male & 1 female, Baltra Island, at airport (intercepted in continental shipment), 09.IX.1999 (R. Cevallos leg.) (CDRS), 1 female, Charles Island, Post Office Bay, 1925 (Wollenboeck Galapagos Exp.) (ANSP) (id. as *Galapagia solitaria* by M. Hebard), 2 females, Española Island, Bahia Manzanillo, 23.IV–2.V.1992, gen coll. & MV light (S Peck & J. Cook leg.) (USNM), 2 males & 2 females, Fernandina Island, 10Km NE Cabo Hammond, 400m, 6–10.V.1991, edge of valley, forest FIT (S. & J. Peck leg.) (USNM), 1 juv. Floreana Island, 1991 (L. Baert leg.) (USNM), 1**

male, Genovesa Island (Tower Island), 6.VI.1968 (A. Mott.) (BM1969-352) (BM), 1 female (lectotype), S. Albemarle [Isabela] I[island] (U.S.N.M.-Acc. 26662) (USNM), 1 female, Isabela Island, Cerro Azul, 1991 (L. Baert leg.) (USNM), 1 female, Isabela Island, Villamil, beating *Scalesia*, arid z[one] 5m, 9.III.1989, (S. Peck leg.) (USNM), 1 female, Isabela Island NE rim Alcedo, 1100m, 21–25.VI.1991, Shrub forest, gen. colln. (S. Peck leg.) (USNM), 3 females, Isabela Island, 6Km NE Tagus Cove, 600m V. Darwin, 16.V.1992 arid z[one] mv. Light & gen. colln. (J. Cook & S. Peck leg.) (USNM), 1 male & 1 female, San Cristóbal Island, Laguna El Junco, 05.XII.2002 (G. Estévez leg.) (CDRS), 1 male, Santa Cruz (Indefatigable) Island [no date or collector data] (AMNH), 2 males, Santa Cruz Island, 1.7Km N. Sta. Rosa, 1–30.V.1001, 550m *Scalesia* [forest], mal[aise]-FIT (S. & J. Peck leg.) (USNM), 1 male, Santa Cruz Island, Los Gemelos, *Scalesia* forest, 610, 1–28.II.1989, FIT (Peck & Sinclair leg.) (USNM), 1 juv., Santa Cruz Island, Media Luna, 408m, 5.VII.1992 (L. Baert leg.) (USNM), 1 male (pre-adult), Santa Cruz [Indefatigable] Island, Conway Bay, IV.1923 (no collector data) (USNM), 1 male, Santa Cruz (Indefatigable), Conway Bay, IV.1923 (no collector data) (AMNH), 2 males, Santa Cruz (Indefatigable) Island, Conway Bay, 13–17.III.1935 (Crocker exp. leg.) (AMNH), 1 male, Santa Cruz, Media Luna, 21.I.1989 (Abedrabbo leg.) (CDRS), 1 male, Santa Fé Island, S. Coast (no date) (A. Laurie & Y. Lubin leg.) (CDRS), 1 male, Isabella Island, Volcán Alcedo, 10.V.1998 (V. Cruz leg.) (CDRS). ***Liguanea pediodromia*** - JAMAICA: 1 female (holotype, code: 5917), Liguanea Plain, Kingston, 5.VII.1920 (R. & H. leg.) (ANSP), 1 male (allotype, code: 5917) Flamstead, 10.IX.1921 (C.C. Gowdey leg.) (ANSP), 1 male, W.I. St. Andrew, Elizabeth Ave., III.1965 (R. Farr leg.) (USNM) (id. as *Liguanea pediodromia* by A.B. Gueney), 1 male, Mona [Island], 25.II.1974 (B. Heineman leg.) (AMNH). ***Macromusonia conspersa*** - BRAZIL: 1 male, (holotype of *Thespis conspersa*) (MNHN); PERU: 1 female (green morph), Loreto, vic. Yurimaguas, 3Km S on Rio Shanushu, 1.VI.1964 (R.M. Starw leg.), 1 female, San Martín, Río Abiseo, Quebrada “Timón” (380 m), 9IX.1999, (L. Beck leg.) (UNALM), 1 female, Tarapoto, Yurimaguas Km. 20 “Biodiversidad” 0634/7620 950 masl, IX–X.2001, (Rasmussen & Ríos leg.) (UNALM), 1 male, San Martín, Hera, 15 Km SE Moyobamba 860m, 3.VII.1947 (F. Woytkowski leg.) (AMNH), 1 female, San Martín, Mishqui-Yacu (1200-1600 m), VIII.1947, (P. Hocking leg.) (MUSM); 1 female, Huánuco, Parque Nacional Tingo María, nr. “cueva de la lechuzas” (on herbaceous vegetation), 4.IV.1998 (J. Rivera leg.) (UNALM), 1 female, Madre De Dios, Manu, 10.VI.1995 (UNALM), 1 male, Huanuco, Tingo Maria, XII.1949 (H.A. Allard leg.),

1 female, R[eserva] N[acional] Tambopata (Lago Sandoval), 1.X.2000, (R. Acosta leg.) (UNALM), 1 female, Tambopata-Lago Sandoval (on trail ground) 13.X.2001 (R. Acosta leg.) (UNALM), 1 female Río Tambopata (300 m) (UNALM). **Macromusonia major** – BRAZIL: 1 male, Pará, 1947 (A. Miles Moss leg.) (BM); COLOMBIA: 1 male, Amazonas, Rio Igara-Parana 30Km aval La Chorrera, VII–VII.1974 (M. Descamps leg.) (MNHN); FRENCH GUIANA: 1 male, St. Laurent du Maroni (no date) (Le Moullet leg.), 1 male (syntype of *Musonia major*) & 1 female (syntype of *Musonia major*) (MHNG); GUYANA: 1 female, Pied Saut Parare, 12.VIII.1977 (M. Descamps leg.) (MNHN), 2 males & 2 females, Kartabo, Bartica Distr. (W. Beebe leg.) (AMNH); PERU: 1 male, Loreto, Maynas, Panguana III 3°54,672'S, 7°37,37674'W, 8–11.V.2001 (K. Rasmussen leg.) (UNALM), 1 male, Ucayali, Yarinacocha, 13.XI.1964 (P. Hocking leg.) (MUSM). **Musonia margarethae** - CHILE: 1 male, Tarapacá, Región Lluta [=Río Lluta, Arica prov.], 10.II.1994 (L. Peña leg.) (ANSP) (id. as *Musoniella* sp. by P.S. Terra); ECUADOR: 1 male, Guayaquil, 11.II.1945 (E.A. Chopin leg.) (USNM), 1 male & 1 female, Manabí, Rocafuerte/Tosagua, 80°22'W, 0°52'S, 31.VIII–01.IX.1997 (Amédégnato & Poulain leg.) (MNHN) (id. as *Musoniella argentina* by R. Roy), 1 male, Azuay, nr. Camilo Ponce, 52.4m, 2°58.905S, 79° 42.786W, at light, 9.X.2008 (Rivera & Pinault leg.) (UNALM); PERU: 3 males, Tumbes, VIII.2000, (P. Castillo, leg.), 11.III.1992 (UNALM), 1 female, Puerto Pizarro (on ground with grass), 31.V.2001, (S. Kelez leg.) (UNALM), 1 male, Piura, 25.VII.1968, (O. Velarde leg.) (UNALM), 1 male, Lambayeque, Olmos, 30.V.1999, (Valencia leg.) (UNALM), 1 female, Mochumí, 21.I.2001, (A. Albújar leg.) (UNALM), 1 female, La Libertad, Trujillo, 8.IX.1983, (J. Amaya leg.) (UNALM), 1 male, Moche, 11.VIII.1996, (R. Castillo leg.) (UNALM), 1 male, Ancash, Casma, 18.VI.2000, (M. Sánchez leg.) (UNALM), 1 male, Lima, La Molina, campus UNALM, 14.I.2000, (C. Vergara leg.) (UNALM), 1 male, campus UNALM, 20.XI.1999, (J. Rivera leg.) (UNALM), 1 female, Lima, Pueblo Libre, 5.II.1962 (R. García leg.) (MUSM), 1 male, Lima, Chosica, 10.VII.1972, (B. Risco leg.) (UNALM), 1 male, Lima, Santa Eulalia, 15.I.1996, (I. Hidalgo) (MUSM), 1 male, Lima, Cieneguilla, 10.V.2000, (H. Quiros) (UNALM), 1 male & 1 female, Lima, Chorrillos, Pantanos de Villa (on herbaceous vegetation), VI.1999, (C. Véliz leg.), 1 female (juv.), Lima, Chorrillos, Pantanos de Villa nr. “Las Garzas” Club, (running on open ground with patches of grass), IV.2002, (J. Rivera leg.) (UNALM), 1 female, Lima, Cañete, Lunahuaná, 13.IV.2001 (Tello leg.) (UNALM), 2 males, Ica, Chinchá, II.1989, (J. Herrera leg.) (UNALM), 1 male, Ica, 24.IV.1990, (C. leg.) (UNALM), 1 male, Ica,

Fundo “Arrabal”, Subtanjaya (zona urbano-rural), 350m, 24.III.2003, (J. Cervantes leg.) (UNALM), 1 male & 1 female, Tacna, La Yerada, asentamiento 5 y 6, fundo “La Universidad” 250m, on clumps of grass, IV.2007 (R. Alanoca leg.) (UNALM). ***Musonia surinama*** - BARBADOS: 1 male, 25.12.1980 (B. Brown leg.) (UNALM); FRENCH GUIANA: 1 female (syntype of *Mionyx fuscesens*), St.-Laurent du Maroni (Le Moult leg.) (MNHN), 1 male (syntype of *Mionyx fuscesens*), St.-Jean du Maroni (Le Moult leg.) (MNHN), 1 female (holotype of *Mionyx lineatus*), St.-Laurent du Maroni (Le Moult leg.) (MNHN), 1 male, Macouria, lumière, 27.VII.1998 (A. Faille leg.) (MNHN); GUYANA: 1 male, Kartabo, Bartica distr. (W. Beebe leg.) (AMNH); NICARAGUA: 1 female (holotype of *Musonia femorata*), Chontales, Janson, Godman-Salvin coll. 1894–125 (BM); PANAMA (Note: all specimens possible referable to *Musonia femorata*): 1 female, Tocumen, 29.I.1953 (F. Blanton leg.) (USNM), 2 males, Alajuelo, 10.IV.1911 (A. Busck leg.) (USNM), 1 female, Gamboa, Canal Zone, pipeline rd., malaise trap, VII.1967 (W.W. Wirth leg.) (USNM), 1 female, Coiba, 1924 (L.E. Cheesman leg.) (BM), 1 female, Corozal, Canal Zone, 12.II.1911 (A. Busck leg.) (USNM); VENEZUELA: 12 males, Guarico, Hato Masaguaral (44 Km S Calabozo), 20–28.V.1985 (Menke & Carpenter leg.) (USNM), 1 male, Miranda 2.5 KM E Carenero, 26.III.1982 (G.F. & J.F. Hevel leg.) (USNM), 1 male, Sucre, Cumaná, 30.VI.1978 (J. Maldonado C. leg.) (USNM), 1 male, Maracay, 16.IX.1952 (A. Villiers leg.) (MNHN); TRINIDAD & TOBAGO: 1 female, Trinidad Island, St. Augustine, VI.1952 (B.W.I. leg.) (USNM) (id. as *Musonia surinama* by J.A.G. Rehn), 4 males, Brasso, 19–26.VII.1930 (G. Belmontes leg.) (ANSP). ***Musonia*** sp. n. - DOMINICAN REPUBLIC: 2 males, RD-069 Km 54 Rd. Azua-Barahona, Barahona Prov., 70 m, 18°21.871'N 71°09.080'W, 24.XI.2002 (D. Perez, B. Hierro, H. Andújar leg.) (USNM), 1 male, RD-064 Parque Mirador del Norte, Santo Domingo Prov., 96 m, 18°31.387'N 69°55.497'W, 21.XI.2002, (leg. D. Perez, R. Bastardo, B. Hierro leg.) (USNM), 1 male, Km18 Carretera Duarte, 18.II.2003 (R.H. Bastardo leg.) (USNM), 1 female, RD-200, ~2 Km W San Rafael del Yuma, nr. vertedero, 75m, 18°26.276'N 68°41.520'W (D. Perze & R. Bastardo leg.) (USNM). ***Musoniola fera*** - COSTA RICA: 1 male (holotype of *Harpagonyx carlottae*, code: 7053 U.S.N.M.), San Carlos (Schild & Burgdorf leg.) (USNM), 1 male, Punta Arenas, 1957 (C.W. Palmer leg.) (AMNH), 1 male (holotype of *Catamusonia minor*), Ebene von Limon, Las Mercedes 150–300m, 20–30Km vom Atlantik, 10.II.1922 (F. Nevermann leg.) (ZMUH), 1 female (paratype of *Catamusonia minor*), Ebene von Limon, bei Las Mercedes, Farm Hamburg am Reventazón, 10–30m, 12–30Km vom

Atlantik 25.VI.1922 (F. Nevermann leg.) (ZMUH), 1 female, Farm Hamburg (in Bananal), 18.XII.1936 (F. Nevermann leg.) (ZMUH) (id. as *Macromusonia minor* by M. Beier); GUATEMALA: 1 male, Quiriguá (no date) (W.P. Cockerell leg.) (USNM), 2 males, Cayuga, III-V.1915 (W.M. Schaus leg.) (USNM), 1 male, Alta Verapaz, Trece Aguas, VI.1901 (USNM); PANAMA: 1 male (holotype of *Mionyx fera*), V[olcán] de Chiriqui, 25–4000 ft. Goldman-Salvin (1894) (Champion leg.) (BM). ***Musoniola vicina*** - NICARAGUA: 1 male (holotype), Matagalpa (Dr. Rothschild leg.) (ZMB). ***Musoniola* spp.** - PANAMA: 1 female, Barro Colorado, 1954 (Dr. Gregoire leg.) (MNHN); NICARAGUA: 1 female, Zelaya, Cerro Saslaya 13°44'N, 85°01'W, IV.1996, 700m (Maes & Hernandez leg.) (MNHN). ***Oligonicella mexicana*** - MEXICO: 1 male (syntype of *Oligonyx mexicanus*), Presido, 1894 Godman-Salvin (Forrer leg.) (BM), 1 female (holotype of *Oligonyx toltecus*), Guerrero, Chilpancingo 4600ft, VIII.1894 Godman-Salvin (H.H. Smith leg.) (BM), 1 male, Nayarit, San Blas, 3.VII.1964 (P.J. Spangler leg.) (USNM), 1 male, Nayarit, 27Mi S. Tepic, 28.VII.1964 (W.J. Gertsch & J.A. Woods leg.) (AMNH), 1 female, Taxco, 1.XI.1974 (R. Gregg leg.) (FMNH), 1 male, Sinaloa, Mazatlan, 1–2.VII.1964 (P.J. Spangler leg.), 1 male, Cuautla, 28.X.1922 (E.G. Smyth leg.) (USNM), 1 male, Villa Juarez, 14.V.1931 (R.A. Robles leg.); UNITED STATES OF AMERICA: 1 male, Texas, Hidalgo Co., Bentsen St. Pk., 100 ft. nr. Mission, 8.VII.1963 (H. Clench leg.) (CM), 1 male, Texas, Cameron County, Audubon Sabal Palm Sanctuary, 20.X.1997 (R.A. Androw leg.) (CM), 1 male, Arizona, Santa Cruz Co. 15.9Km WNW Nogales, White Rock Campground, Peña Blanca lake MV/UV light sheet, 3800', 31°23'37"N; 111°05'20"W, 21.VII.2000 (R.A. Androw *et al.* leg.) (CM), 1 male, Arizona, Sta. Rita Mountains, 26.VII.1925 (USNM). ***Oligonicella punctulata*** - MEXICO: 1 male (holotype of *Oligonyx punctulatus*), Guerrero, Dos Arroyos, 1000ft, IX.1894 Godman-Salvin (H.H. Smith leg.) (BM). ***Oligonicella cf. punctulata*** - MEXICO: Oaxaca, Pocutla, La Concordia (C.V. Morton & E. Makrinius leg.) (USNM). ***Oligonicella scudderi*** - UNITED STATES OF AMERICA: 1 male, Arkansas, Scorr Co., Ouachita National Forest, 12Km WSW Waldron 24°51'N; 94°16'W, 14.IX.2000 (R. Rockwell leg.) (CM), 1 male, Florida, Columbia Co. I-10 & Rt-441, 30.VII.1982 (S. Gross leg.) (USNM), 1 male, Alachua Co. Gainesville (al light), 6.IX.1983 (S.W. Gross leg.) (USNM), 1 male, Kansas, Oswego, Labette Co. (at night lights), 27.IX.1969 (G.F. Hevel leg.) (USNM), 1 male, Kansas, Oswego, Labette Co. (at white light), 14.IX.1968 (G.F. Hevel leg.) (USNM), 1 male, Oklahoma, Paine Co. 1962 (J.F. Reinert leg.) (USNM), 1 male, Texas, Dallas (Boll leg.) (USNM), 1 male (syntype of *Oligonyx*

bollianus), Texas, Dallas (MHNG), 1 female (syntype of *Oligonyx bollianus*), Texas (Boll leg.) (MHNG), 3 males, Texas, Blanco Co. Pedernales Falls State Park, 9.X.1985 (J.E. Rawlins leg.) (CM), 1 male, Texas, Georgetown, fall-1936 (USNM), 1 male, Texas, Travis Co. 1Km SE Bee Cave, 11–18.X.1985 (J. Rawlins & R. Davidson leg.) (CM), 1 male, Texas, Bastrop Co., 5Km N. Smithville, 13.X.1985 (J.E. Rawlins leg.) (CM). ***Oligonicella striolata*** - GUATEMALA: 1 male (syntype of *Olygonyx striolatus*), Dueñas (G.C. Champion leg.) (MNHG), 1 male (syntype of *Olygonyx striolatus*), S. Gerónimo, 1894 Godman-Salvin (Champion leg.) (BM). ***Oligonicella tessellata*** - MEXICO: 1 male (holotype), Guerrero, Omilteme, 8000ft, VII.1894 Godman-Salvin (H.H. Smith leg.) (BM), 4 males, Sinaloa, Venvidio, 16.VI–24.VII.1918 (J.A. Kusche leg.) (ANSP), 1 female (juv.), Michoacán, 11.VI.1899 (S.N & M.C. Rhoads. Leg) (ANSP). ***Oligonicella cf. tessellata*** - MEXICO: 1 male, Mecatan, Nayarit, 800', 23.V.1949 (G.M. Bradt leg.). ***Oligonicella*** spp. MEXICO: 2 males, Nayarit, 18Km SW Compostela, (at black light) 20.VII.1974 (M.E.&P.D. Perkins leg.) (USNM), 1 male, Oaxaca, Tehuantepec, 15.VIII.1972 (G.F.&S. Hevel leg.) (USNM); NICARAGUA: 1 male, Granada, Volcán Mombacho, San Joaquín#1, malaise trap, 15.XI.1998 (J.M. Maes) (MNH), 1 female, Granada, Volcán Mombacho Bosque Seco (malaise trap), 30.VIII.1998 (J.M. Maes leg.) (MNH), 1 female (code: GSMC003374), Granada, Reserva Silvestre Privada Domatila 11.709000°N; 85.953500°W, 68m. 6–9.X.2010 (G.J. Svenson) (CMNH), 1 female, Chinandega, El Jicarito (s/*Jatropha* sp.) 27.XII.1994 (C. Grimm leg.) (MNH); EL SALVADOR: Vol[cán] Conchagua, 27–29.V.1958 (O.L. Cartwright leg.) (USNM), 1 male, San Salvador, 21.VI.1958 (O.L. Cartwright leg.) (USNM). ***Oligonyx bicornis*** - MEXICO: 1 juv. female (holotype), Tehuantepec (MHNG), 1 male (syntype of *Harpagonyx gryps*), Pancina, Veracruz, 1894 Godman-Salvin (Champion leg.) (BM), 1 male (syntype of *Harpagonyx gryps* var?), Atoyac, Veracruz (April) (MHNG), 1 female, Motzorongo, II.1892 (ANSP) (id. as *Oligonyx bicornis* by M. Hebard), 1 male, Córdoba, 24–28.IX.1966 (A.B. Lau leg.) (USNM), 1 male, Chiapas (L. Hotzen leg.) (USNM). ***Oligonyx bidens*** - COSTA RICA: 1 female, Turrialba (Schild & Burgdorf leg.) (USNM) (id. as *Spanionyx bidens* by J.A.G. Rehn); GUATEMALA: 1 female, Secanquim, IV.1904 (O.F. Cook leg.) (USNM); HONDURAS: 1 female (holotype of *Spanionyx bidens*), Roatán I[sland], 1894 Godman-Salvin (Gaumer leg.) (BM), 1 female, Charleston, 10.VIII.1936 (USNM) (id. as *Oligonyx bidens* by A.B. Gurney); MEXICO: 1 female, Colonia Yucatán, Yucatán, 25.VIII.1952 (J.&D. Pallister leg.) (AMNH). ***Oligonyx maya*** - MEXICO: 1 male

(holotype of *Harpagonyx maya*), N. Yucatán, Temax, 1894 Godman-Salvin (Gaumer leg.) (BM), 1 male, Yucatán, Merida, 20–30.VII.1964 (P.J. Spangler leg.) (USNM). ***Oligonyx* sp.** - MEXICO: 1 male, Chiapas, Tuxtla Gutierrez (at zoo), on bush, 20.X.1990 (L. Desutter leg.) (MNHN), 1 male, Chiapas, Sumidero 20Km NE Tuxtla Gutierrez, 18.VIII.1966 (J. & R. Ovie leg.) (AMNH), 1 male, Tapachula, V.1924 (W.M. Mann leg.) (USNM). ***Piscomantis peruana*** - PERU: 3 males (paratypes of *Galapagia peruana*), “Peru”, 1924 (NHMW), 1 male, Ancash, Cajacay, Rio Fortaleza 2750m, 6.III.1956 (Weyrauch leg.) (IFML) (id. as *Galapagia peruana* by M. Beier), 2 males & 2 females, Lima, Huarochirí (Surco), Carretera Central, 2000m 7.I.2001 (Rivera & Oyague leg.) (UNALM), 3 males, Lima, Huarochirí (Surco), Carretera Central, 2000m, 3.II.2003 (Rivera & Montgomery leg.) (UNALM), 1 male, Lima, Huarochirí (Surco), S.J. de Surco, road to Palacala, 10.VII.2008 (C. Mendoza leg.) (UNALM), 2 males, Lima, Verrugas, 7.IV–19.V.1928 (R.C. Shannon leg.) (USNM), 1 female, Ica, Pampa de Ocas, Fundo “Nueva Esperanza” (“cerca de campo de alfalfa”), IV.2001, (A. Tapia leg.) (UNALM). ***Pseudomusonia lineativentris*** - COLOMBIA: 1 female (juv.) Antioquia, Andagoya, 22.IV.1918 (M.A. Carriker Jr. leg.) (ANSP); PANAMA: 1 male (syntype of *Mionyx saevus*), David, Chiriqui (Champion leg.) (MHNG), 1 male & 2 females, Canal Zone, Paris Field, Cristobal, 7.VII.1920 (M. Hebard leg.) (ANSP). ***Thespis boliviana*** - ARGENTINA: 1 male (holotype of *Leptomiopteryx argentina*), Villa Ana, F.C.S. Fe, II.1926 (K.J. Hayward leg.) (BM), 2 males, Prov. Santiago de Estero, bords du Rio Salado, 1909 (E.R. Wagner leg.) (MNHN), 1 male, Prov. Santiago de Estero, Colonia Dora, 15–26.XI.1979 (C.M. Vardy leg.) (BM), 1 male, Santiago del Estero, btw Santiago & Rio Hondo, 23.III.1979 (H.R. Roberts leg.) (ANSP), 1 male, Chaco de Santa Fé, Las Garzas, bord du Rio Las Garzas, 20Kil. A l/o D’Campo. 1903 (E.-R. Wagner leg.) (MNHN); BOLIVIA: 1 male (holotype of *Musonia boliviana*), “Bolivia” (BM), 1 male, Prov. Cordillera, Santa Cruz, Palmar de las Islas, 2-10.VIII.2006 (T. Vidaurre leg.) (CMNH), 1 male, Prov. del Sara, 450m (J. Steinbach leg.) (ANSP). ***Thespis cubensis*** - BAHAMAS: 1 male, Grand Bahama Isl., West End, 12.V.1953 (E.B. Hayden & L. Giovannoli leg.) (AMNH); CAYMAN ISLANDS: 2 males & 2 females, 17.VII–26.VIII.1938 (C.B.Lewis & G.H. Thompson leg.) (BM); CUBA: 1 male (syntype), “Cuba” (MHNG), 1 male, Cayamas, (E.A. Schwarz leg.) (USNM), 1 female, 12.5Km. S. of Pinar Rio, 12–23.IX.1913 (no collector data) (id. as *Paramusonia cubensis* by M. Hebard 38’) (AMNH), 1 male, Pinar d. Rio, 27.III.1900 (Palmer & Riley leg.) (USNM), 3 males, nr Casa Harvard, Soledad Cienfuegos, VII.1933 (N.A. Weber leg.)

(ANSP), 1 male, Soledad, Sta. Clara, 4.VIII.1932 (Bates & Fairchild leg.) (ANSP). ***Thespis media*** – TRINIDAD & TOBAGO: 1 female (paratype of *Diamusonia media* [not a syntype *sensu* Marshall, 1975]), Trinidad Island, 1903 (P. Rendall leg.) (BM), 4 males & 3 females, Moruga area, II–III.1986 (G.B. Popov leg.) (BM); VENEZUELA: 1 female (holotype of *Diamusonia media*), La Guayra (Otto leg.) (ZMB), 1 male, Barrancas, 17.V.1985 (M. Lanotte leg.) (MNHN), 1 male, Bolivar 7.6Km SE Guasipati, 22.III.1982 (G.F.&J.F. Hevel leg.) (USNM), 4 males, Guarico, Hato Masaguaral, 45Km. S. Calabozo 8.57N, 67.58W, Savanna #12, 75m., uv Light, 8–15.VII.1989 (M. Epstein & M. Deza leg.) (USNM), 2 males, Guarico, Hato Masaguaral, 45Km. S Calabozo 8.57N, 67.58W, at Light in gallery forest, 1.VIII.1989 (M. Epstein leg.) (USNM), 4 males, Guarico, Hato Masaguaral (44Km. S Calabozo), 20–28.V.1985 (Menke & Carpenter leg.) (USNM), 1 male, La Iguana, Llanos, IX.1985 (M. Lamotte leg.) (MNHN). ***Thespis metae*** - COLOMBIA: 1 male (holotype, code: 8950), Villavicencio “150” (from A. Maria), X.1919 (ANSP). ***Thespis seclusa*** - ARGENTINA: 1 male (holotype of *Paramusonia seclusa*), San Luis, Alto Pencosa 660m, 20.XII.1908 (P. Jorgensen leg.).

Thesprotia brasiliensis - BRAZIL: 1 male (holotype, code: 7094), “Brasil” (MNCN). ***Thesprotia caribea*** - ANTIGUA & BARBUDA: 1 female (holotype, code: USNMMENT0087396) & 1 male (paratype, code: USNMMENT00873968), Antigua W.I., VI.1918 (D. Stoner leg.) (USNM).

Thesprotia gigas - BRAZIL: 1 female (holotype), “Brazil” (ZMB). ***Thesprotia graminis*** - UNITED STATES OF AMERICA: 1 male (syntype of *Oligonyx graminis*), Florida (ANSP), 8 males, Florida, Archbold Research Station, Highlands Co., 31.III–5.IV.1962 (D.C. Ferguson leg.) (USNM), 1 male, Florida, Gainesville, Alachua Co., 14.IV.1982 (S.W. Gross leg.) (USNM), 2 females, Florida, Stemper, 6.III–6.VIII.1911 (G. Krautwurm leg.) (CM), 2 males, Georgia, Chatham Co., 7.8Km N Richmond Hill, 255', 32°00'34"N; 81°17'03"W at Motel lights (R. Androw leg.) (CM). ***Thesprotia infumata*** - BRAZIL: 1 male (syntype of *Thesprotia filiformis*) “America Meridional” (Yurine leg.) (MNHN). ***Thesprotia insolita*** - COSTA RICA: 1 male, (holotype), La Fortuna, betw. Cervantes & Pacayas (at light), 1430m, III.1906 (P. Biolley leg.) (ANSP). ***Thesprotia macilenta*** - BRAZIL: 1 male (syntype), “Brasil” (MHNG), 1 female (syntype), Rio de Janeiro (MHNG), 1 female (holotype of *Thesprotia vidua*) (MHNG).

Thesprotia maculata - BRAZIL: 1 female (holotype), “Brasilien” (ZMB). ***Thesprotia pellucida*** - BRAZIL: 2 males (holotype & paratype), “Brasilien” (ZMB). ***Thesprotia simplex*** - BRAZIL: 1 male (holotype), “Bras.” (ZMB). ***Thesprotia subhyalina*** - BRAZIL: 1 male (holotype of

Oligonyx subhyalina), “Brasil” (MNHN). ***Thesprotia*** sp. n. - PERU: 2 males, Cusco, Macchu Pichu, 29.I.1952, 2300m. (F. Monros leg.) (FMNH). ***Thesprotia*** spp. - BRAZIL: 2 males, São Paulo, Piracicaba, 6.XII.1964 (W.E. & C.A. Triplehorn leg.) (USNM), 1 male & 1 female, Goiás, Jatai (id. as *T. filiformis*) (MNHN), 1 female, Maranhão, Betw. São Domingo & Loreto, 6.IX.1963 (Westminster school Exp. to Brazil leg.) (BM). ***Thesprotiella ceratophora*** - COLOMBIA: 1 male (holotype of *Carrikerella ceratophora*, code: 8939), Andagoya, 22.IV.1918 (M.A. Carriker Jr. leg.) (ANSP). ***Thesprotiella empusa*** - COSTA RICA: 1 female (holotype of *Carrikerella empusa*, code: 5538), Peralta, 26.III.1910 (Calvert leg.) (ANSP), 1 male, Heredia, Est. Biol. La Selva, 50–150m, 10°26’N, 84°01’W, X.1995 (INBio-OET leg.) (BM). ***Thesprotiella festae*** - ECUADOR: 1 male (holotype of *Spanionyx festae*), “Ecuador”, 19.VIII.1899 (ZMB), 1 male (holotype of *Thesprotiella similis*), “Ecuador”, 30.VII.1899 (ZMB). 1 male, Pichincha, Naranjillo Valley, nr Mindo, 1450m, 27.VI.1987 (M. Cooper leg.) (BM). ***Thesprotiella fronticornis*** - ECUADOR: 1 male (holotype), Babahoyo (MNCN). ***Thesprotiella insularis*** - DOMINICA: 3 males, S. Chiltern, 5.XI–10.XII.1964 (P.J. Spangler leg.) (USNM), 1 male, Trois Pitons, 22.X.1964 (P.J. Spangler leg.) (USNM), 2 males, 1.3 mi. E of Pont Casse, 12.V–27.IV.1964 (O.S. Flint leg.) (USNM), 1 male, St. Paul Springfield State 2.5Km ENE Canefield 15°21’N; 61°22’W, 450m, 11–18.VI.1991 (J.E. Rawlins & S.A. Thompson leg.) (CM), 1 male, I.1999 (S.P. Montgomery leg.) (ANSP) (id. as *Oligonyx* sp. by P.S. Terra); GUADELOUPE: 1 male & 1 female (holotype and allotype of *Oligonyx insularis*), Duclos, Petit-Bourg, 11.V.1962 (J. Bonfils leg.) (MNHN), 1 male, Massif de la Soufriere, La Citerne, 19.I.1966 (J. Bonfils leg.) (MNHN). ***Thesprotiella latens*** - COLOMBIA: 1 female (holotype of *Pogonogaster latens*, code: USNMENT00873971), Cauca, Rio Aguatal 2000m, XI.1908 (USNM). ***Thesprotiella peruana*** - ECUADOR: 1 female, Napo, Rio Tiputini, Pindo, Mandaripanga 76°44’W, 0°43’S, 30.IX–15.X.1997 (Amédégnato & Poulain leg.). PERU: 1 male (holotype), Marcapata, 5.II.1902 (Dr. O. Staudinger leg.) (ZMUH), 1 male, Loreto, Rio Ucayali, Yarinacocha, 8.XI.1953 (P. Hocking leg.) (FMNH), 1 male, Middle Rio Ucayali, 8.XI.1913 (AMNH). ***Thesprotiella tristani*** - COSTA RICA: 1 female (holotype of *Pogonogaster tristani*, code: 5353), La Palma, 1600m, V.1906 (J.F. Tristan leg.) (ANSP). ***Thesprotiella*** sp. - COSTA RICA: 1 female (juv.) San José, 1928 (M. Valerio leg.) (USNM). ***Thesprotiella*** spp. (possibly new species) - COLOMBIA: 1 female, Caquetá, Yuruyaco, 73K SW Florencia, 8.II.1979 (M. Cooper leg.) (BM); PERU: 1 male, Yurac 67mi E. of Tingo Maria, 16.XI.1954 (E.I. Schlinger &

E.S. Ross leg.) (CAS); VENEZUELA: 1 male, Ar. Rancho Grande 1100m, 24–31.X.1966 (SS & WD Duckworth leg.) (USNM).

ANGELIDAE

Angela armata - PERU: 1 female, Huánuco Tingo María, 19.VI.1999 (L. Gil leg.) (UNALM), 1 female, Huánuco, Bosque [Reservado], U[niversidad] N[acional] A[graria][de la] S[elva], 5.XII.1999 (M. Sobera leg.) (UNALM), 1 female, Huánuco, Puerto Inca, Río Pachitea, 1930 (P. Martin leg.) (MUSM). ***Angela guianensis*** - BOLIVIA: 1 male, Guanay, Corioco, XI.1996 (F. Lombardo leg.) (UNALM); COLOMBIA: 1 male (holotype of *Thespoides bolivari*, No. 7086), Cauca (Oberthür leg.) (MNCN); FRENCH GUIANA: 1 male (syntype of *Thespis infusata*, No. 7273; note: cited as “type” in Rivera, 2014), Nouveau Chantier (Le Moult leg.) (MNCN), 1 male (syntype of *Thespis infusata*; specimen No. 12592), St.-Jean du Maroni (Le Moult leg.) (ANSP); PERU: 14 males & 1 female, Cusco, La Convención, Distrito de Echárate, various dates within 1998 (2 males collected by R. Acosta, the rest by multiple collectors) (UNALM & MUSM), 1 male, Huánuco, Tingo María, 25.VI.2002 (M. Jiménez leg.) (UNALM), 1 female, Huánuco, Tingo María 17.VIII.1998 (A. Ruiz leg.) (UNALM), 2 males, Loreto, Iquitos, carretera Nauta, II.2000 (I. Vásquez leg.) (UNALM), 1 male, Madre de Dios, Boca, Río La Torre, 300m, 23.X.1982 (G. Lamas leg.) (MUSM), 1 male, Madre de Dios, Tambopata, 300m, 15.V.2002 (J. Grados leg.) (UNALM), 1 male, Madre de Dios, R[eserva] N[acional] Tambopata (Lago Sandoval), “trampa de luz”, 29.IX.2002 (J. Rivera leg.) (UNALM), 1 female, Madre de Dios, Parque Nac[ional] Manu (Pantiacolla), 24.X.2000 (J. Santisteban leg.) (UNALM), 1 female, San Martín, Tarapoto-Yurimaguas, Km. 20 “Biodiversidad” 0634/7620 950masl, IV-VI.2001 (C. Rasmussen leg.) (UNALM). ***Angela maxima*** - FRENCH GUIANA: 1 female (holotype of *Thespis maxima*), Saint Laurent du Maroni, November (year unknown) (Le Moult leg.) (MNHN). ***Angela minor*** - VENEZUELA: 1 male (holotype of *Thespis minor*), La Guayra (ZMB). ***Angela miranda*** - MEXICO: 1 male (holotype of *Thespis miranda*) (no additional data) (MNHN). ***Angela peruviana*** - PERU: 1 male (holotype of *Thespis peruviana*), Madre de Dios, 500m (Garlepp V. leg.) (ZMB). ***Angela subhyalina*** - PERU: 1 male, Cusco, La Convención, distrito de Echárate (at light), 22.VI.1998 (multiple collectors) (MUSM).

COPTOPTERYGIDAE

Brunneria borealis - UNITED STATES OF AMERICA: 1 female, Florida, Orlando, 9.IX.1958 (A.G. Selhime leg.) (USNM), 1 female, Florida, Lee Co., 20mi SE of Ft. Myers, 17.VIII.1968 (G.F. Hevel leg.) (USNM), 1 female, Georgia, Macón, 20.X.1922 (no collector data) (USNM), 1 female, Georgia, edge Okefenokee swp. 10mis. Waycross, 16.VIII.1964 (Gurney leg.) (USNM), 1 female, Georgia, Savannah, 7.X.1965 (C.E. Metz leg.) (USNM), 1 female, Georgia, Clarke Co., 1.X.1939 (C. Stelling leg.) (USNM), 1 female, Hawaii, O'ahu Island, Waahila Ridge, 27.IV.1974 (NB & JMA leg.) (USNM), 1 female, North Carolina, Raleigh, 8.X.1929 (B.B. Fulton leg.) (USNM), 1 female, North Carolina, Fort Bragg, 18.X.1967 (J.D. Birchim) (USNM), 1 female, Texas (holotype) (A. Agassiz leg.) (ANSP), 2 female, Texas, Forth Worth, 1929 (J. Bell leg.) (USNM). ***Brunneria brasiliensis*** - ARGENTINA: 1 female, Carcaraña (no date nor collector) (USNM); PARAGUAY: 1 male, 4 females, Sapucay (no date) (W.T. Foster leg.) (USNM) (id. as *Brunneria brasiliensis*). ***Brunneria longa*** - BRAZIL: 1 female (allotype), Mato Grosso (Rhode leg.) (ZMB); BOLIVIA: 1 male (holotype), Prov. Sara (J. Steinbach leg.) (ZMB), 8 male, 10 females, Prov. Sara, various times between 1912 and 1922 (J. Steinbach) (ANSP) (note: topotypical specimens from original collecting event); PARAGUAY: 1 male, Dpto. Concepción, Ao. Tagatiya-mi 22°39'S 57°32'W, 3–5.IV.1986 (M. Pogue & M. Solis leg.) (USNM), 1 male, Dpto. Cordillera, Caacupe, 2.VII.1980 (R. Cave leg.) (USNM). ***Brunneria subaptera*** - ARGENTINA: 1 female (holotype), [Buenos Aires], Bahía Blanca (G. Claraz leg.) (MHNG), 4 males, Buenos Aires, Sa. de la Ventana, R. Sauce Grande, 23–25.II.1968 (D.A. Nickle leg.) (USNM), 1 male, Buenos Aires, 4Km. S of Libano (on rd. to Reserva), 20.I.1982 (D.A. Nickle leg.) (USNM), 2 females, Buenos Aires, Hwy 76, 28 Km E Tornquist, 21.I.1982 (D.A. Nickle leg.) (USNM); BRAZIL: 1 male (syntype of *Brunneria gracilis*), Paraná, 1907 (E. Dukinfield Jones leg.) (BM), 1 male, Rio Grande do Sul, Porto Alegre, 11.V.1979 (L. Jantsch leg.) (id. as *Brunneria subaptera* by L. Jantsch) (MCTP), 1 male, Rio Grande do Sul, Tramandaí, 11.III.1985 (L. Jantsch leg.) (id. as *Brunneria subaptera* by L. Jantsch) (MCTP), 1 male, Rio Grande do Sul, Porto Alegre, 11.V.1980 (M. Ribero leg.) (id. as *Brunneria subaptera* by L. Jantsch) (MCTP); URUGUAY: 1 male (syntype [not holotype] of *Brunneria gracilis*), Montevideo (ZMB). ***Brunneria* spp** - VENEZUELA: 1 male, Bolivar, Auyan Tepui, 12.XII.1998 aich. 1000m. (G. Tavakelian leg.) (MNH), 1 female, Jusepín, VIII.1995 (M. Lamolte leg.) (MNH). ***Coptopteryx argentina*** - ARGENTINA: 1 male (syntype of *Coptopteryx affinis*), “Argentina”, 1904 (O.W. Thomas leg.) (BM), 1 male (syntype of *Coptopteryx affinis*),

Buenos Aires (ZMB), 1 female (syntype of *Coptopteryx affinis*) (ZMB), 1 male (lectotype of *Coptopteryx claraziana* - here designated, see text for details), Bahía Blanca (S. Claraz leg.) (MHNG), 1 female, Buenos Aires, San Fernando, III.1964 (or 1968) (J. Daguerre leg.) (USNM), 1 male, Buenos Aires, IV.1922 (USNM), 1 female, Mendoza, Chacras de Coria, 936m. 5.IV.1907 (P. Jorgensen leg.) (ANSP) (id. as *Coptopteryx argentina*), 2 males, Misiones, I–II.1911 (P. Jorgensen leg.) (ANSP), 1 male, Chaco, Resistencia, 8.IV.1918 (P. Jorgensen leg.) (ANSP), 1 male (lectotype of *Coptopteryx viridis* - here designated, see text for details) Salta, Cachi, 2500, 2.4.[19]05 (J. Steinbach V. leg.) (ZMB); BRAZIL: 1 female (holotype of *Paraphotina precaria*) [Paraná] Curitiba (ESALQ), 2 males (holotype & paratype of *Coptopteryx gigliotosi*), Rio Grande do Sul, Santa Maria, 1972 (Tarragó leg.) (ESALQ); PARAGUAY: 2 males & 2 females, Sapucay, 28.I–13.II.1905 (Foster leg.) (ANSP).

Coptopteryx brevipennis - PERU: 1 male (holotype), [Ayacucho], Huanta, 2400m, XII.1947 (Weyrauch leg.) (NHMW), 1 male, Ayacucho, 7.VI.2007 (I. Vilca leg.) (UNALM), 1 female, Ayacucho, 27.III.1996, (A. Jeri leg.) (UNALM), 2 males, 1 female, Ayacucho, Huamanga, Dist[rito] Quinua, Mituqasa, 2800m, 30.I.2001 (SENASA leg.) (UNALM). ***Coptopteryx constricta*** - ARGENTINA: 1 male (allotype) & 1 female (holotype), Prov. Mendoza, Chacras de Coria, 936m, 4–19.IV.1907 (P. Jorgensen leg.) (ANSP), 1 male & 1 female, Mendoza, 767m., 20.IV.1908 (P. Jorgensen leg.) (ANSP). ***Coptopteryx gayi*** - ARGENTINA: 1 female (holotype of *Brachypteromantis bonariensis*), Buenos Aires (M. Viana leg.) (ESALQ), 1 female (paralectotype of *Coptopteryx spinosa*), Buenos Aires (Schimpf leg.) (ZMB), 1 male (holotype of *Coptopteryx rehni*), Mendoza, Santa Rosa, 6.III.1901 (ZMB), 1 female, Buenos Aires Prov., Hwy76, Alvaro Barros (by Arroya Tapalqué), 20.I.1982 (D.A. Nickle leg.) (USNM), 1 female, Buenos Aires Prov. Hwy76, 28Km E Tornquist, 21.I.1982 (D.A. Nickle leg.) (USNM), 1 female, Mendoza, Potrerillos, 1368m., 29.XII.1909 (P. Jorgensen leg.) (ANSP) (id. as *C. gayi*), 1 female, Neuq[ué], Rio Limay, Neuquen, 17–18.II.1978 (C.M. & O.S. Flint Jr. leg.) (USNM); BRAZIL: 1 female, Rio Grande do Sul, no date, (Ihering leg.) (ANSP), 1 female, Mendoza, Chacras de Coria, 936m., 6.III.1908 (P. Jorgensen leg.) (ANSP); CHILE: 3 male & 2 females, [Santiago de Chile] Rinconada de Maipú, 25.III.1959 (USNM), 1 male, Santiago [de Chile], 1922 (F.C. Joseph leg.), 1 female (holotype of *Coptopteryx minuta*), “Chili”, 1904 (R.M. Middleton leg.) (BM); URUGUAY: 1 female, Las Piedras, 3.III.1963 (A.C. leg.) (FMNH). ***Coptopteryx gymnopyga*** - BRAZIL: 1 female (holotype of *Mantis gymnopyga*), Pará (Lieber leg.) (ZMB).

Coptopteryx inermis - PARAGUAY: 1 female (holotype), San Bernardino (D. Jordan leg.) (NHMW). ***Coptopteryx parva*** - ARGENTINA: 1 female (holotype), Cachí, 3.IV.1905 (J. Steinbach leg.) (ZMB). ***Coptopteryx pusilla*** - ARGENTINA: 1 male (holotype), La Merced (Reimoser leg.) (NHMW), 1 male, Santiago del Estero, 50-80 Km NE La Banda, rt5, 160m, 22.III.1979 (H.R. Roberts leg.) (ANSP). ***Coptopteryx thoracica*** - ARGENTINA: 1 male (lectotype of *Coptopteryx spinosa*), 1 female, Buenos Aires (Schimpf leg.) (ZMB), 1 male, Formosa, 20.XII.1917 (P. Jorgensen leg.) (ANSP); PARAGUAY: 1 female (holotype), Sapucay, 19.XII.1905 (Foster leg.) (ANSP), 1 male (allotype), Sapucay, 10.XII.1905 (Foster leg.) (ANSP), 1 male (paratype), Sapucay, 20.XII.1905 (Foster leg.) (ANSP), 1 male (paratype), Sapucay, I-VII.1905 (Foster leg.) (ANSP), 2 females (paratypes), Sapucay, 19.XII.1905 (Foster leg.) (ANSP), 1 male (holotype of *Coptopteryx fallax*) & 1 female (allotype of *Coptopteryx fallax*), San Bernardino (ZMB). ***Coptopteryx thoracoides*** - URUGUAY: 1 male (syntype), Montevideo (ZMUH), 1 male (syntype), Montevideo (SDEI).

LITURGUSIDAE

Hagiomantis superba - PERU: 1 female, Huánuco, Tingo María, 26.IV.2002 (V. Lecca leg.) (UNALM), 1 female, Huánuco, Tingo María, 12.IV.1998 (A. Ruiz leg.) (UNALM), 1 male, Huánuco, Tingo María, 25.IV.2002 (A. de Fina leg.) (UNALM), 1 female, Puerto Inca, Río Pachitea (P. Martin leg.) (MUSM), 1 female (holotype of *Liturgusa superba*), Loreto, Yurimaguas (ZMG). ***Liturgusa lichenalis*** - PERU: MADRE DE DIOS: 1 female, Tambopata (malaise trap), I.2001 (several collectors) (UNALM), 2 females, "Peru" (UNALM). ***Liturgusa nubeculosa*** - PERU: 1 female, Amazonas, P[uer]to de Comainas. 2-13.IX.1999, (L. Beck leg.) (UNALM), 2 females, Junín, Chanchamayo, La Merced, S[an]ta Isabel (Río Toro), XI.2000 & 26.II.2001, (E. Oyague leg.) (UNALM), 1 female, Cusco, La Convención, distrito de Echárte (Shivankoreni), 23.X.1998 (several colectores) (UNALM), 1 male, Huánuco, Tingo María, 29.III.1996 (A. Jeri leg.), 1 female, Huánuco, Tingo María, 15.III.1993, (J. Vilca leg.) (UNALM), 1 female, Loreto, Iquitos, 30.VIII.1996 (A. Guerra leg.) (MUSM), 1 female, Loreto, Iquitos 30.VII.1989 (P. Hocking leg.) (MUSM), 1 male, Loreto, S[an]ta Sofía, Río Utoquinia, 29-31.VIII.1974 (G. Lamas leg.) (MUSM), 1 female, Madre de Dios, R[eserva] N[acional] Tambopata, Lago Sandoval (on a trunk), 2.X.2002 (J. Rivera leg.) (UNALM), 1 female, Madre de Dios, Tambopata, Posada Amazonas (Puerto Maldonado) (on external walls of a house),

25.IX.2001 (M. Deza leg.) (UNALM), 1 male, 15 Km. E[ste] P[uer]to Maldonado (200),
 7.II.1990 (M. Medina leg.) (UNALM), 1 male, San Martín, Bosque Quinllal (n[ea]r Picota),
 8.VI.2002 (C. Rasmussen leg.) (UNALM), 1 female, Ucayali, Yarinacocha (Río Ucayali),
 18.IX.1946 (J. Schunke leg.) (MUSM). *Liturgusa maya* - PERU: 1 female, Cajamarca,
 Tambillo, 1877 (A. Raimondi leg.) (MUSM), 1 female, Cusco, Quillabamba, 3.VII.1996 (W.
 Catalán leg.) (UNALM), 1 male & 1 female, Cusco, La Convención, distrito de Echárte, 1998
 (several collectors) (UNALM), 1 male, Junín Satipo, 16.V.1999 (J. Laos leg.) (UNALM), Madre
 de Dios: 1 female, Tambopata, Posada Amazonas, 27.IX.2001 (M. Deza leg.) (UNALM),
 Tumbes: 1 male & 1 female, Papayal, 2000 (P. Castillo leg.) (UNALM), Ucayali: 1 female,
 Yarinacocha (Río Ucayali), 26.VIII.1951 (P. Hocking leg.) (MUSM), 1 female, Pucallpa,
 XII.1946 (UNALM); ECUADOR: Cotopaxi: 4 females, La Maná (secondary forest),
 7.VIII.2003, (Rivera & Montgomery leg.) (UNALM).

PHOTINAIDAE

MACROMANTINAE - *Macromantis hyalina* - GUYANA: 1 male, Demerara (no date) (S.
 Harris leg.) (FMNH); PERU: 1 male, Tingo María, IV–V.1952 (P. Araoz leg.) (USNM); PERU:
 1 male, Amazonas, San Martín, 15.IX.1971, (N. Chui leg.) (UNALM), 1 male, Huánuco, Tingo
 María, 29.II. 1967, (Morales leg.) (UNALM), 1 male, Huánuco, Tingo María, 24.IV.1999,
 (Bueno leg.) (UNALM), 1 male, Junín, San Ramón, 14.VII.2002, (R. García leg.) (UNALM), 1
 male, Loreto, Iquitos, Carretera Nauta (Km. 5), II.2000, (I. Velásquez leg.) (UNALM), 1 male,
 Madre de Dios, Tambopata, 30Km SSW Pto. Maldonado, 3–18.XI.1979 (Hepner & Pearson leg.)
 (ANSP). *Macromantis nicaraguae* - BELIZE: 1 male, San Antonio, V.1931 (J.J. White leg)
 (ANSP); GUATEMALA: 1 male, Volcan Sta. María, “June” (Schaus & Barnes leg.) (USNM), 1
 male, Cayuga, VI.1915 (W. Schaus leg.) (USNM); NICARAGUA: 1 male, Edén, 17.V.1922
 (Wharton, Huber leg.) (ANSP); PANAMA: 1 male, Barro Colorado Island, Canal Zone [at light],
 19.I.1959 (H.S. Dybas leg.) (FMNH), 1 male, Barro Colorado Island, Canal Zone, 9.II.1959 (no
 collector) (FMNH); PERU: 1 male, Huánuco, Huamincha 1600m. 26.III.1946 (AMNH), 2 males,
 Cusco, La Convención, Mangoriari, 12°21'S; 73°02'W, 1500m, X–XI. 2002, (J. Grados leg.)
 (MUSM), 1 male, Cusco, La Convención, Comerciato, 1247/7322, 1350m, XI.2002, (J. Grados
 leg.) (MUSM), 1 male, Junín, La Merced, 5.XII.1998, (Ortega leg.) (UNALM). *Macromantis*
ovalifolia - PERU: 1 male, Loreto, Yagua Indian village, headwaters of Loreto-Yacu, 24–

26.IV.1970 (B. Malkin leg) (FMNH). *Macromantis saussurei* - FRENCH GUIANA: 1 male (holotype), piste Coralie, pk2, 23.III.1988 (L. S  n  caux leg.) (MNHN), GUYANA?: 1 female (paratype) (MNHN).

PHOTIOMANTINAE - *Photiomantis planicephala* - BRAZIL: 1 female & 1 male (holotype and allotype of *Metriomantis planicephala*), Rio Grande do Norte, Cear   Mirim (W.M. Mann leg.) (ANSP), 1 male, Bahia, on the hwy to Barreiras, 22.XI.1974 (C.W. Castleton leg.) (USNM), 1 female, [Bahia], Fortaleza, X.1943 (B. MacCreary leg.) (USNM), 1 male, [Rio Grande do Norte], Cear   (F. Rocha leg.) (USNM) (id. as *Metriomantis biramosa*) (USNM), 1 male, [Rio Grande do Norte], Natal, 1944 (USNM), 1 female, Natal, Paramirim Field, 20.III.1945 (M.T. Dalmat leg.) (USNM).

CARDIOPTERINAE - *Cardioptera brachyptera* - BRAZIL: 1 male (syntype of *Mantis* (*Cardioptera*) *brachyptera*), “Brasilien” (ZMB), 1 female (syntype of *Mantis* (*Cardioptera*) *brachyptera*), “Brasil” (ZMB), 1 male, Esp  rito Santo, 25.VII.1998 (MCTP). *Cardioptera minor* - BRAZIL: 1 male, Bah  a, Bonfim, 30.XI.1907 (Carnegie Mus. Exp.) (ANSP), 2 females (holotype & paratype) Para  ba, Independ  ncia (no date) (Mann & Heath leg.) (ANSP), 1 female (juv.) Para  ba, Independ  ncia (no date) (Mann & Heath leg.) (ANSP). *Cardioptera squalodon* - FRENCH GUIANA: 2 males (codes: GSMC003339 & GSMC003340), (H. Romack leg.) (CMNH), 1 male (code: GSMC003792), Kaw Mountain res. 1Km W Amazonia 290m, 22–31.I.2012 (H. Romack leg.) (CMNH), 1 male, Road to Kaw, PK32, 6.VIII.1996, mercury vapor light (D. Pollock & F. Hovore leg.) (CM); GUYANA: 1 male (code: ROM905737), Rupununi dist., Kurupukari, Essequibo R[iver] E. side, at black light on rd. Rainf. 200”, 7.X.1990 (B. Hubley leg) (ROM). *Cardioptera viridipennis* - BRAZIL: 2 males (holotype & paratype), Mato Grosso, 20.VI.1911 (Zobrys & Wolmer vend.) (holotype-ZMUH; paratype-NHMW). *Cardioptera* sp. - BRAZIL: 3 males, Bahia, rt.116, at Minas Gerais border, 800m., 17-19.XI.1974 (Roberts & Carbonell leg.) (ANSP), 1 female, Bahia, Toca da Onca, 27.VI.1915 (USNM).

PHOTINAINAE - MICROPHOTININI - *Chromatophotina awajun* - PERU: 2 males (holotype & paratype) & 1 female (paratype), Amazonas, Bagua, Aramango District (Centro Poblado Numparque), 978m, 5  26’20.2’’S; 78  23’37.5’’W, XII.2006 (J. Layme leg.) (UNALM).

Chromatophotina cofan - ECUADOR: 1 female (holotype), Pr[ovince] Sucumbios, Coca/L[ago] Agrio, San Pedro de los Cofanes, 415m, 0°08'S; 77°52', 22/23.IX.1997 (C. Amégdégnato & S. Poulain leg.) (MNHN). ***Chromatophotina*** spp (likely new species) - PERU: 2 males, Loreto, Madre Selva Biological Station, -3.62096, -72.24744, 10–17.II.2013 (G. Svenson leg.) (CMNH), 1 male & 1 female, Peru: Madre de Dios (USNM). ***Microphotina viridescens*** - FRENCH GUIANA - 1 male (holotype of *Photina viridescens*; note: likely an intersexual specimen), Nouveau Chantier (Le Moullet leg.) (genital preparation no. 3381) (MNHN). ***Microphotina vitripennis*** - FRENCH GUIANA: 2 males, Piste de St. Ellie, nr Sinnamary, N55.29747 W53.05205, 8–10.V.2011, elev.40m (at light) (J. Rivera leg.) (CMNH). ***Microphotina*** sp. n. BOLIVIA: 1 male (code: GSMC000063), La Paz Dept., Ituralde Prov., San Miguel, 14° 30.602'S, 67° 29.555'W, 24–30.IX.2007 (K.B. Miller *et al.* leg.); PERU: 1 male, Madre de Dios (USNM). ***Microphotinainae*** gen.1 sp.1 (likely new genus and species) - FRENCH GUIANA: 2 males (code: GMSC004114, additional male lacking code), Kaw Mountain reserve, vicinity of Patawa, 30.IV–1.V.2011, N.456888 W52.21388, elev.282m (at light) (J. Rivera leg.) (CMNH), 1 male, Logging rd. S of Regina, 4.V.2001, N4.6217, W52.15288, elev.51.8m (at light) (J. Rivera leg.) (CMNH), 1 male, (H. Romack leg.) (CMNH). ***Microphotinainae*** gen.2 sp.1 (likely new genus and species) - FRENCH GUIANA: 1 male (code: GSMC004115), Logging rd. S of Regina, 4.V.2001, N4.6217, W52.15288, elev.51.8m (at light) (J. Rivera leg.).

PHOTINAINAE - ORTHODERELLINI - Orthoderella delucchii - PERU: 1 male (holotype), Cusco 5 Km. N Río Camisea, San Martín 3-camp, 474m. 11°47'S 72°42'W, 16.II.1997, (at light, multiple collectors) (MUSM), 1 male (paratype), same data as holotype but 12.III.1997 (UNALM), 2 males, Junín, Chanchamayo Prov., San Ramón, Fundo Génova, 11°05'24"S; 75°20'15"W, 1000m, 29.VIII.2009 (F. Lira leg.) (UNALM). ***Orthoderella elongata*** - ARGENTINA: 1 male (holotype), NC N. of Córdoba, W of Ascichinga, Sierra de Córdoba N., 7.I.2010 (Snizek leg.) (MNHN), 1 female (allotype), Chaco de Santa Fé, "Las Garzas", bords du rio Las Garzas, 25Kil a l'o d'ocampo, 1903, E.-R. (Wagner leg.); BRAZIL: 1 female, Mato Grosso, Santa Rosa de Descalvados, 8Km. S. of Carne de Vaca, 25.VII.1931 (J.A.G. Rehn leg.) (ANSP), 1 male, Mato Grosso, near Porto Esperança, 10.IX.1931 (J.A.G. Rehn leg.) (ANSP). ***Orthoderella major*** - BRAZIL: 1 male, Paraná, Caviúma, XI.1945 (A. Maller leg.) (AMNH), 1 male (lectotype of *Tithrone major*; code: MZLQ-I0075), São Paulo, Piracicaba, IV.1951 (ESALQ), 2 males & 1 female: São Paulo, Piracicaba (no additional data) (ANSP), 6 males,

Goiás (no additional data) (ANSP). ***Orthoderella ornata*** - ARGENTINA: 1 male (holotype), Chaco de Santiago del Estero, Rio Salado, “January” (male genital preparation No.3658) (MNHN) (id. as *Orthoderella ornata* by R. Roy) BOLIVIA: 3 males, 14 females, Prov. Del Sara, 450m, 1922 (J. Steinbach leg.) (ANSP); PARAGUAY: 1 male, Horqueta, 45Km E of Rio Paraguay, 57W, 23N, 9.I.1934 (A. Schulze leg.) (ANSP) (id. as *Orthoderella ornata* by F. Lombardo), 1 male, Villa Rica, III.1922 (P. Jorgensen leg.) (ANSP). ***Orthoderella* sp. n.** - VENEZUELA: 1 male, Vacaryzal (Carnegie Mu. Exp.) (ANSP). ***Paraphotina caatingaensis*** – BRAZIL: 1 male (paratype), Bahía, Pilão Arcado, Brejo Zacarias armadilha luminosa, 10° 07’248’’S, 42° 53’ 589’’W, 25.XI.2008 #45921 (Silva-Neto, A. Menezes, & E., Mota, E. leg.) (MZUSP). ***Paraphotina cf. caatingaensis*** - BRAZIL: 1 male & 1 female, Mato Grosso, Chapada dos Guimarães, 17.X.2009 [lab reared] (F. Fernandes leg.) (MZUSP). ***Paraphotina insolita*** - BRAZIL: 1 male (holotype of *Guaraunia insolitum*), Amazonas, Rio Branco, 30.VIII.1924 (Bequaert leg.) (ANSP); ***Paraphotina reticulata*** - BRAZIL: 1 male (holotype of *Cardioptera reticulata*), [Rio Grande do Norte], Natal (MHNG); VENEZUELA: 1 male (holotype of *Paraphotina venezuelana*), Maracay, V.1936 (P. Vogl leg.) (ZSM), 7 males, Guarico, Hato Masaguaral (44 Km. S Calabozo), 1–19.V.1985 (Menke & Carpenter leg.) (USNM), 2 males, Guarico, Hato Masaguaral (44 Km. S Calabozo), 8.57N, 67.58W, Gal[le]ryFor[est]#4 75m at light, 12–13.IV.1998 (M. Epstein & R. Blahnik leg.) (USNM), 1 male, Barin. 15 Km SW Barinitas, 25.II.1969 (Duckworth & Dietz leg.) (USNM).

PHOTINAINAE - PHOTINAINAE - ***Hicetia goeldiana*** - BRAZIL: 1 female (holotype), [Rio de] Janeiro (MHNG) (id. as *Hicetia goeldiana* by R. Roy, 2004), 1 male & 1 female (syntypes of *Photina (Metriomantis) gracilicollis*), São Paulo, Bosque da Saúde, 23.XI.1926 (ZMUH), 1 male, Salesópolis, Est. Bio. de Boracéia, 27–29.XII.1948 (L. Travassos & E. Rabello leg.) (MZUSP). ***Hicetia* sp.** - BRAZIL: 1 male & 1 female, Espírito Santo, 22.IV.1898 (J. Michaelis leg.) (ZMUH). ***Photina amazonica*** - BRAZIL: 1 female (holotype of *Parastagmatoptera amazonica*), Amazonas (ZMB). ***Photina brevis*** - PARAGUAY: 1 male (holotype of *Photina brevis*; code: 8981), Sapucay, 5.III.1905 (Foster) (ANSP), 1 male (paratype of *Photina brevis*), Sapucay, 3.X.1905 (Foster leg.) (ANSP), 3 males (note: topotypical specimens), Sapucay, 1901 (W.T. Foster leg.) (USNM). ***Photina boliviana*** - PERU: Madre De Dios: 9 males & 5 females, Zona Reservada Tambopata, Lagos Sandoval (wasai trail), 3.X.2000, (R. Acosta leg.) (UNALM), 1 male, Tambopata Jungle Lodge (225m.a.s.l.), 12°49’S; 69°24’W, 15–19.IX.2001,

(C. Rasmussen leg.) (UNALM), 2 males, Junín, Chanchamayo, La Merced, Santa Isabel (Río Toro), 22.VII.2000, 1120 m. (E. Oyague leg.) (UNALM), 1 male, Junín, Chanchamayo, 20.XI.2008 (R. Ramos leg.) (UNALM), 1 male, Huánuco Tingo María, 31.V.1999, (K. Lozano leg.) (MUSM), 1 male, Cusco, La Convención, distrito de Echaráte, 9.II.1998, (general collection) (MUSM). ***Photina cupido*** - BRAZIL?: 1 female (holotype of *Metriomantis cupido*) (Jurine leg.) (MNHG). ***Photina occidentalis*** - PERU: 1 female, Loreto, 1 Km. NO margen derecha Río Blanco (4°21'S 73°9'O), 15.VIII.2000, (S. Kelez leg.) (UNALM). ***Photina ovata*** - FRENCH GUIANA: 1 female (either syntype or holotype, see Roy & Cuche, 2008), Cayenne (Prudhomme leg.) (MNHG). ***Photina paraensis*** - BRAZIL: 1 female (holotype of *Metriomantis paraensis*), Pará (Lieber leg.) (ZMB). ***Photina pilosa*** - FRENCH GUIANA: 1 male (syntype), Nouveau Chantier (Le Moullet leg.) (male genital preparation No. 3493) (MNHN), 1 male (syntype; code: 12591), St.-Laurent du Maroni (Le Moullet leg.) (ANSP); VENEZUELA: 1 male, Alto Orinoco, Union Orinoco, Ugueto TFA, 14.X.1951 (USNM) (id. as: *Metriomantis pilosa* “compares well with co-type [i.e. syntype] at Phil[Adelphia] A.B.G.58”). ***Photina pilosella*** - BRAZIL: 1 male, [Amazonas] Ig[arape]. Taramazinho, 46Km N. Manaus, 6.II.1979 (O.S. Flint Jr. leg.) (USNM); FRENCH GUIANA: 1 male, vicinity of Cacao, N4.568; W52.46648, elev. 47.5m (at light), 2–3.V.2011 (J. Rivera leg.) (CMNH); GUYANA: 1 male (holotype of *Metriomantis pilosa*), Issosoro N.W.D., 12.III.1913 (A.A. Agram leg.) (ZMB); VENEZUELA: 4 males, T.F. Amaz. Cerro de la Neblina, base camp, 0°50'N; 66°9'44"W, 140m, 20.II–10.III.1984 (D. Davis & T. McCabe leg.) (USNM), 1 male, Territorio Amazonas, Cerro de la Neblina, base camp, 140m, 11.II.1984 (Rozen & Stupakoff leg.) (AMNH). ***Photina reticulata*** - BRAZIL: 1 female (holotype?) (Olfers) (ZMB), 1 female (syntype of *Metriomantis biramosa*), Espiritu Santo (Fruhstorfer leg.) (MNHG). ***Photina vitrea*** - ARGENTINA: 2 males, M[isio]nes Ao. Piray Guazú No., San Pedro, 22.X.1973 (O.S. Flint Jr. leg.) (USNM); BRAZIL: 1 male (holotype of *Mantis (Photina) vitrea*), “Brasil” (ZMB), 1 intersexual individual (holotype of *Mantis longicornis*; note: appearing female - see Agudelo 2014) “Brasil” (ZMB), 4 males, Rio de Janeiro, “3-4”, 1935 (P. Sandig leg.) (USNM), 2 males, Rio de Janeiro, Itatiaya, 700m, 3.XII.1936 (BM), 1 intersexual individual (holotype of *Photina gracilis*; note: appearing female but with full male genitalia - see Agudelo 2014), Santa Catharina, Theresopolis (Fruhstorfer leg.) (ZMB), 1 female (holotype of *Photina laevis*) Santa Catharina, Theresopolis (Fruhstorfer leg.) (ZMB), 1 male, Santa Catharina, Porto Belo, 30.X.1983 (E. Corseuil leg.) (MCTP) (id. as

Metriomantis pilosella by L. Jantsch), 2 males & 1 female, Santa Catharina, Corupa (Hansa Humboldt), X.1944 (A. Maller leg.) (AMNH), 1 female, Santa Catharina, Corupa (Hansa Humboldt), X.1945 (A. Maller leg.) (AMNH), 1 female, Santa Catharina, Corupa (Hansa Humboldt), X.1946 (A. Maller leg.) (AMNH), 1 female, Rio Grande do Sul, Cacequi, 8.IV.1978 (L. Tavares leg.) (MCTP) (id. as *Cardioptera parva* by L. Jantsch), 1 female, Paraná, Caviuna, X.1946 (A. Maller leg.) (AMNH), 1 male, Paraná, Caviuna, III.1945 (AMNH), 4 males, Rio de Janeiro, III–IV.1935 (P. Sandig leg.) (USNM), 2 males, Rio de Janeiro, Itatiaya, 700m, 3.XII.1936 (BM), 2 males, S[anta] Cath[arina], Corupa (Hansa Humboldt), X.1944 (A. Maller leg.) (AMNH), 1 female, S[anta] Cath[arina], Corupa (Hansa Humboldt), XI.1945 (A. Maller leg.) (AMNH), 1 male, S[anta] C[atharina], Porto Belo, 30.X.1983 (E. Corseuil leg.) (MCTP) (id. as *Metriomantis pilosella* by L.J. Jantsch leg.). ***Photina*** sp (likely new species) - BOLIVIA: 2 males (codes: GSMC003418 & GSMC003413), Dpto. Santa Cruz, nr Buena Vista 17°39.100'S; 063°35.136', 390m, 14.XI.2009, at MV/UV lights in cut trees (G.J. Svenson leg.) (CMNH), 1 female, Sta. Cruz de la Sierra, Barrio Jardín Latino (500m), II–III.2003 (I. Vaccari leg.) (UNALM); BRAZIL: 1 male, Mato Grosso, 12°50'S; 51°47'W, at light, 12.X.1968, R.S. & R.G.S. Exp. BM 1968-260 (O.W. Richards) (BM), 1 male, [Minas Gerais], Sabara, Bello Horizonte Rio das Velhao (1932) (A.G.N. Chalmes leg.) (BM).

STENOPHYLLIDAE

Stenophylla gallardi – FRENCH GUIANA: 1 female (holotype), montagne dès Singes, pres de Kourou, 150m “suspendue sous une feuille morte à environ 1.5m du sol”, 10.XI.1997 (J.-Y. Gallard leg.) (MNHN), 1 male (juv.) (paratype), Arataye affl. Approuagues, “aval du saut Parare, abattage de jour arbre n°9”, 13.VII.1988 (L. Desutter & P. Grandcolas leg.) (MNHN).

Stenophylla lobivertex - PERU: 1 female, Rio Ucayali, 19.I.1929 (H. Bassler leg.) (AMNH) (id. as *Stenophylla lobivertex* by M. Stiewe), 2 males, Madre de Dios, Manu, Cuenca Rio los Amigos, alrededores del [vicinity of] CICRA, 20.IX.2005 & II–III.2006 (M. Deza leg.) (UNALM), 1 male, Rio Santiago, 11.VII.1930 (H. Bassler leg.) (AMNH) (id. as *Stenophylla lobivertex* by M. Stiewe). ***Stenophylla cornigera*** - BRAZIL: 2 males, Santa Catarina, Corupa, Hansa Humboldt I.1945 & XI.1946 (A. Maller leg.) (AMNH) (id. as *Stenophylla cornigera* by M. Stiewe). ***Stenophylla*** sp. - PERU: 1 female, [Ucayali] Pucallpa, Divina Montaña, 16.IX.2005 (T. Kothe & M. Breitsameter leg.) (MSMC).

ACONTISTIDAE

Acontista amazonica - BRAZIL: 1 male (holotype of *Acontistella amazonica*), Alto Amazonas (Staundinger leg.) (NHMW). ***Acontista amoenula*** - BRAZIL: 1 female (*non* male) (holotype of *Acontista semirufa*), Ega [= Teffe, in the Brazilian Amazon], (BM) 1 female, Amazonas, Tefé, II.1925 (H. Bassler leg.) (AMNH); ECUADOR: 1 male, Orellana, Estación Científica Yasuni, to UV & merc. Vap. Lights, 17-18.X.2003 (D. Robacker, W. Warfield & M.H. Evans leg.) (FMNH), 1 male, Napo, Tiputini Biodiversity Stn. Vic. Yasuni National Park, 0°38'S, 76°10'W, 14-18.II.1999 (D.C. Darling leg.) (ROM); PERU: 1 female (holotype), Pebas (Amazonas) (ZMG), 1 female, Loreto, Iquitos, I-II.2005 (no collector) (MNHN), 1 female, Madre de Dios, Río Tambopata, 30km SW Pto. Maldonado, 290m. 12°50'S, 69°17'W, no date (T. Erwin leg.) (USNM), 1 male, Madre de Dios: Parque Nacional Bahuaja-Sonene, Puesto de Control San Antonio, Rio Heath, 21-22.VIII.2003 (D. Lucsic leg.) (UNALM); SURINAM: 1 female, "Surinam", 4-14.VIII.1980 (Coebiti leg.) (BM). ***Acontista cayennensis*** - FRENCH GUIANA: 1 female (holotype), Cayenne (MHNG), 1 male (syntype of *Acontista maroniensis*), Nouveau Chantier (Le Moult leg.) (MNHN), 1 female (syntype of *Acontista maroniensis*), Passoura "Octobre" (Le Moult leg.) (MNHN), 1 female (syntype of *Acontista maroniensis*, code: 12590), Nouveau Chantier (Le Moult leg.) (ANSP), 1 male & 2 females, Cayenne, I-II.1917 (Carnegie Mus. Exp. leg.) (ANSP). ***Acontista concinna*** - BRAZIL: 1 female (holotype of *Mantis concinna*), "Brasília" (ZSM), 1 female, Minas Gerais, Viçosa, 30.IX.1930 (E.J. Hambleton leg.) (id. as *Acontista eximia*) (USNM), 1 male, Minas Gerais, Organ Mountain, 1.V.1935, (P. Sandij leg.) (USNM), 1 female, Paraná, Caviúna, X.1945, (A. Maller leg.) (AMNH), 1 female, Paraná, Caviúna, X.1946, (A. Maller leg.) (AMNH), 1 male, Paraná, Caviúna, II.1946, (A. Maller leg.) (AMNH), 1 male, Paraná, Caviúna, I.1946, (A. Maller leg.) (AMNH), 1 male, Santa Catarina, Corupa, Hansa Humboldt, XI.1946 (A. Maller leg.) (AMNH), 1 male, Santa Catarina, Corupa, Hansa Humboldt, V.1946 (A. Maller leg.) (AMNH), 1 male, Nova Teutônia, 300-500m, 27°11'B. 52°23'L, 1986, (F. Plumann leg.) (USNM), 1 male, São Paulo, Ribeirão Pires, 28.VII.1961, (N.L.H. Crauss leg.) (USNM), 1 male, Sao Paulo, Sítio Bananal, 11.III.1959 (T. Halik leg.) (USNM), 1 male, Porto Flor. (TC, São Paulo), 21.I.1959 (T. Halik leg.) (USNM), 1 male, Rio de Janeiro, Petropolis, 24-27.X.1919 (Cornell U. Exp.) (ANSP) (id. as *Acontiothespis concinna* by J.A.G. Rehn), 1 female, Rio de Janeiro, Itatiaia, 700m, 25.III.1927, (J.F. Zikén leg.) (BM), 1 female, Rio de Janeiro, Corcovado, 12.VII.1915 (no collector) (USNM); URUGUAY: 1

female, Montevideo, III.1940 (no collector) (USNM). *Acontista cordillerae* - COLOMBIA: 1 male (holotype of *Acontiothespis iriodes*, code: 8932), Santa Marta, 26.XII.1910 (ANSP), 1 female, Chocó, Pangala, Río San Juan, 9–11.VI.1969 (B. Malkin leg.) (FMNH), 1 male, Chocó, Pangala, 400Km. of Palestina, Río San Juan, 12–27.I.1971 (Malkin & Burchard leg.) (FMNH), 6 male & 4 females, Magdalena, Río Frío, 26.III–23.VIII.1925 (F.W. Walker leg.) (ANSP), 1 male, Magdalena, Sevilla, 19.V.1925 (F.W. Walker leg.) (ANSP); COSTA RICA: 1 male, Farm Hamburg, Reventazon, 1.IX.1932 (NHMW), 1 male, Heredia, nr. Puerto Viejo, O.T.S, 11.I.1970 (at light) (M. Kosztarab leg.) (USNM), 1 male Turrialba, 2–5.XI.1967 (E.L. Todd leg.) (USNM), 1 male, Punta Arenas, 5 km. south of San Vito 1280m, 05.VI.1973 (G. Ekis leg.) (USNM), 1 male, Punta Arenas, Playa Carate, Peninsula de Osa, 3 Km S of Corcovado, 2 m., V.2002 (S. Kelez leg.) (UNALM), 1 female, Turrialba (no date) (Schild & Burgdorf leg.) (USNM), 1 male & 1 female, Piedras Negras, (no date) (Schild & Burgdorf leg.) (id as *Acontiothespis cordillerae vitrea* by J.A.G. Rehn:) (USNM), 1 male, Costa Rica: San José, 22.XII.1988 (A.S. Menke leg.) (USNM); ECUADOR: 1 male, Pacific slope (no date) (Mrs. E.W. Rover leg.) (USNM), 1 male, Balzapampa, 700m, 29.V.1938, (Wm. Clarke-macIntyre leg.) (FMNH), 1 male, Bucay (no date) (C. Campos leg.) (ANSP), 1 male, Esmeraldas, El Placer, 670m. 0052N, 7833W, 1–17.VIII.1987 (G.S. Glenn leg.) (ANSP), 1 male, Balzapampa, 700m, 29.V.1938 (Clarke-MacIntyre leg.) (FMNH); GUATEMALA: 2 females, Polochic River, 06.X.1934 (G.P. leg.) (id. as *Acontiothespis cordillerae vitrea* by J. Rehn:) (USNM), 1 female, San Sebastian, Huehuetenango, 1925 (L. Thiel. leg.) (USNM), 1 male, Cayuga, IV.1915 (WmSchaus leg.) (USNM); MEXICO: 1 female (syntype of *Acontista mexicana*), Veracruz, Atoyac, 1894 Godman-Salvin (Schumann leg.) (BM), 1 female, Veracruz, Córdoba, X.1965 (A.B. Lao leg.) (USNM), 1 female, Veracruz, Coyame, Lake Catemaco, 1–15.VII.1963 (D.R. Whitehead leg.) (USNM), 1 female (syntype of *Acontista mexicana*), Córdoba, 1894 Godman-Salvin (Hoegel leg.) (MHNG), 1 male (syntype of *Acontista mexicana* var. *inquinata*), “Cordova”, 1894 Godman-Salvin (Hoegel leg.) (BM), 1 female, Veracruz, Córdoba, 28.VII.1965 (Flint & Ortiz leg.) (USNM), 1 male (syntype of *Acontista mexicana* var. *inquinata*), Guerrero, Acaguizotla 3500ft, X.1894 Godman-Salvin (H.H. Smith leg.) (MHNG), 1 male, Oaxaca, Valle Nacional, 31.VI.1966 (A.B. Lao leg.) (USNM), 1 female, S.L.P., 31.6 miles north of Tamazunchale, 22.VIII.1967 (G.F. Hevel leg.) (USNM), 1 male, Tabasco, Villa Hermosa, X.1954 (N.L.H. Krauss leg.) (USNM), 1 male (holotype of *Acontista championi*; note: *A. championi* is a *nomen*

novum proposed by Kirby in 1904 for *Acontista mexicana* var *quadrimaculata* Saussure & Zehntner, 1894 not *Mantis quadrimaculata* Audinet Serville, 1839), Verapaz, Lanquin, Godman-Salvin (Champion leg.) (BM); PANAMA: 1 male (syntype of *Acontista vitrea*), V[olcán] de Chiriqui, 2–2000ft, 1894 Godman-Salvin (Champion leg.) (BM), 1 male (syntype of *Acontista vitrea*), V[olcán] de Chiriqui, 25–4000ft, 1894 Godman-Salvin (Champion leg.) (MHNG), 1 female, Canal Zone, 1.75 Km. NNW of Frijoles, 1.VI.1972 (R. H. Pine leg.) (USNM); EL SALVADOR: 1 female, José Quezada, 1972 (G. Gordh leg.) (id. as by *Acontiothespis cordillerae cordillerae* by M. Hebard) (USNM), 1 female, Santa Tecla, 21.I.1954, (P.A.B. leg.) (USNM), 1 female, José Quesada, 1972 (G. Gordh leg.) (USNM). ***Acontista eximia*** - BRAZIL: 1 female (syntype of *Callimantis eximia*), Pará (BM), 1 female (holotype of *Acontista paraensis*), Pará (MHNG), 1 male, Pará (Baker leg.) (ANSP), 2 males & 1 female, Pará, Igarapé-Açu, 19.I.1912 (H.S. Parish leg.) (ANSP), 1 male, Pará, Benevides, X.1918 (S.M. Klages leg.) (ANSP). ***Acontista fraterna*** - COSTA RICA: 1 female (holotype of *Acontista fraterna*), Cache, 1894 Godman-Salvin (H. Rogers leg.) (BM), 2 males, Carrillo, VII–IX.1903 (ANSP) (id. as *Acontistella fraterna* by J.A.G. Rehn), 1 male (holotype of *Acontistella violacea*) Farm Hamburg al Reventazón, 30.IX.1926 (ZMUH), 1 male (paratype of *Acontistella violacea*), Ebene von Limon, Farm Hamburg am Reventazón, 12–30Km v. Atlantik, 28.III.1926 (F. Nevermann leg.) (NHMW), 1 female, Tucurrique (Schild & Burgdorf leg.) (ANSP) (id. as *Acontistella fraterna* by J.A.G. Rehn), 2 females, Turrialba, 23–31.V.1951 (O.L. Cartwright leg.) (USNM); ECUADOR: 1 male (holotype of *Acontiothespis ecuadorica*, code: 8933), Rio Pescado (E. Ecuador) 1600ft, 20.V.1922 (G.H. Tate leg.) (ANSP), 1 female (allotype of *Acontiothespis ecuadorica*, code: 8933), Rio Pescado (E. Ecuador) 1600ft, 28.V.1922 (G.H. Tate leg.) (ANSP), 1 male, Provincia Carchi, 4Km SSW of Chical nr. Q. San José of Rio Blanco, 1650m.4-7.VIII.1988 (Glenn, Pedersen & Wechsler leg.) (ANSP) (id. as *Acontiothespis* sp. by F. Lombardo), 1 male, Prov. Pichincha rd. along S. Domingo, 2400m, 15.VIII.1969 (P.B. Wygodzinsky leg.) (CAS). ***Acontista gracilis*** - FRENCH GUIANA: 1 female (syntype), Nouveau Chantier (Le Moults leg.) (MNHN), 1 male (syntype), St.-Laurent du Maroni, “Décembre” (Le Moults leg.) (MNHN). ***Acontista minima*** - COLOMBIA: 1 male (holotype) (ZMB). ***Acontista multicolor*** - GUADELOUPE: 1 female (holotype), “Guadeloupe” (MNHN); TRINIDAD & TOBAGO: 1 female, Trinidad Island, Caparo, VII.1913 (S.M. Klages leg.) (ANSP) (id. as *Acontiothespis multicolor* by M. Hebard), 1 female, Trinidad Island, Saint Augustine, II.1954 (N.L.H. Krauss

leg) (USNM), 1 female, Trinidad Island, St. Augustine BWI, 22.X.1935 (J. Davies leg.) (ANSP); VENEZUELA: 1 female, Carabobo Est. 7Km E of Guacara, 28.I.1964 (W.H.C. leg.) (USNM), 1 female, Edo. Zulia Guasare Coal Camp., 45Km. W of Carrasquero, Sierra Perija 80-140m, 19-25.VIII.1981 (J. Hepp leg.) (USNM), 1 male, Lara 20 Km. east of Carora, 24.VI.1976 (A. S. Menke & D. Vincent leg.) (USNM), 1 females, El Limon, AR, 450m., 22.XII.1956 (M. Galvez leg.) (ANSP), 2 males, Sucre, Cariaco, 27.VI.1968 (J. Maldonado leg) (USNM). **Acontista** sp. - PERU: 1 male, Huánuco, Tingo María, 18.VIII.1998, (A. Ruiz leg.) (UNALM), 1 male, 31.X.1999, Huánuco, Tingo María (L. Gil leg.), 1 male, Loreto, Paso “El Boquerón”, 27.XII.1963, (P. Hocking leg.) (MUSM), 2 males, Pasco, Pozuzo (Seren) (reared from egg case), 3.VI.2008 (J. Echevarria leg.) (UNALM), 1 male, Pasco, Pozuzo, SESO, 13.X.2008 (C. Mendoza leg.) (UNALM), Pasco, P.N. Yanachaga Chemillen, sector Huanpal, 10°11’00”S; 75°34’28”W, 1000m, 12.VIII.2010 (Chunga leg.) (UNALM). **Callibia diana** - PERU: 1 female, Middle Ucayali River, 26.II.1931 (H. Bassler leg.) (AMNH), 1 female, Upper Río Huallaga, 8.X.1929 (H. Bassler leg.) (AMNH), 1 male, Huánuco, Tingo María Monson Valley, 10.X.1954 (E.I. Schlinger & E.S. Ross leg.) (CAS), 1 female, Huánuco, Tingo María, 17.VI.2000 (D. Ruiz leg.) (UNALM), 1 male & 1 female, Pasco, Pozuzo (reared from egg case), 16.X.2008 (C. Mendoza leg.) (UNALM), 1 female, Huánuco, Tingo María, 17.VI.2000, (D. Ruiz leg.) (UNALM), 1 female, Junín, La Merced, Pichanaki, 74°46’W; 10°51’S, 1200m, 2.VII.2005 (J. Malpartida leg.) (UNALM), 1 male (juv.), Junín, Satipo, 22.VI.2002 (no collector) (UNALM), 1 male, Satipo, Jauja Prov., III.1944 (P. Paprzycki leg.) (AMNH), 1 female, Cusco, La Convención, Sahuayacu (800m), 1996, (R. Casafranca leg.) (UNALM), 1 female (juv.), Cusco, Manu, Cocha Cashu, trail 2.450 (cerca al río), 400m., X.2000, (D. Davidson leg.) (UNALM). **Callibia** sp. - BRAZIL: 1 female, Utinga 31.III.1962 (H.A. Hanson leg) (USNM). **Metaphotina bimaculata** - BRAZIL: 1 male (syntype of *Acontista bimaculata*) (MHNG), “Brésil”, 1 male, Goias, high plateau, X.1945 (Payne leg.) (USNM). **Metaphotina piracicabensis** - BRAZIL 1 male (holotype of *Metaphotina piracicabensis*), [São Paulo], Piracicaba, V.1959 (Adiel leg.) (ESALQ), 1 female, [São Paulo] Piracicaba, X.1979, (A. Zamith leg.) (ESALQ) (note: labeled as the “allotype” of *Metaphotina piracicabensis* by Piza; however, the female remains undescribed as this specimen is not part of the type series). **Metaphotina rehni** - PARAGUAY: 1 female (holotype of *Acontista rehni*, code: 8930), Sapucay, (25.II.1905), (W.T. Foster leg.) (ANSP), 2 males & 3 females, Sapucay, (III–X, no year), (W.T. Foster leg.) (USNM), 1 male, Sapucay,

27.I.1901, (W.T. Foster leg.) (id as *A. bimaculata*) (USNM). ***Metaphotina* spp.** - ARGENTINA: 1 male & 1 female, Entre Ríos, Pronunciamiento, III–IV.1964 (Walz leg.) (ANSP), 1 male, Zapallar, 31.V.1942, (H.L. Parker leg.) (USNM); BRAZIL: 1 male, Ceará, Fortaleza, IX.1943 (D. MacCreary leg.) (USNM), 1 female, Mato Grosso, Corumba, (no further data) (USNM), 1 male, Rio Grande do Sul, Pelotas, 8.IV.1954 (C.M. Biezanko leg.) (USNM), 1 female, Rio Grande do Sul, Porto Alegre, 09.XI.1992, (labeled as *Acontista brevipennis* by L.J. Jantsch, in 1993) (MCTP), 1 male, Rio Grande do Sul, Viamão, 29.X.1983 (L. Capelli leg.) (MCTP) (id. as *Acontiothespis brevipennis* by L.J. Jantsch), 1 male, Rio Grande do Sul, Pelotas, 1964, (C.M. Bizenko leg.) (USNM); BOLIVIA: 1 female (code: GMSC003360), Sta. Cruz, Buena Vista, 17°30.867'S; 63°40.245'W, 334m, 13.XI.2009, caught at MV or UV light near river (G.J. Svenson leg.) (CMNH), 1 female (code: GMSC003361), Sta. Cruz nr Samaipata nr. Achira camp. 18°09.714S; 63°48.934W, 1347m, (general collection, on vegetation), 12.XI.2009 (G.J. Svenson leg.) (CMNH), 1 male (code: GMSC003362), Sta. Cruz, Buena Vista, nr. Treefall MVL, 17°39.100'S; 63°35.136'W, 390m, 14.XI.2009 (G.J. Svenson leg.) (CMNH), 1 female, Sta. Cruz, Ayacucho, 13-14.V.1969 (P. & .P. Spangler leg.) (USNM), 1 male, Santa Cruz, 14.X.1954 (Munro leg.) (USNM), 1 male, Santa Cruz, Ichilo, Buena Vista, 400m., 2.X.1994, M.V. Light, (R. Word leg.) (CM), 1 male, Santa Cruz, 14.X.1954 (Munro leg.) (USNM), 1 female, Concepción, Ao. Tagatiya, 22°39'S, 57°32'W, 3-5.IV.1986, (M. Pogue & M. Solis leg.) (USNM); PARAGUAY: 1 male, Amambay, Cerro Cora Nat. Park, 22°39'S, 56°1', 7-10.IV.1986 (M. Pogue & M. Solis leg.) (USNM), 1 male, 25.I.1901 (labeled as *A. bimaculata*) (USNM), 1 male, Areguá, 14.XI.1980 (D. Lowrie leg.) (USNM), 1 male, Paraguari, Ybycuí, 10km east in Ybycuí Nat. Park, 12–24.IV.1980 (P.J. Spangler leg.) (USNM). ***Paratithrone royi*** – FRENCH GUIANA: 1 male (holotype) & 1 female (allotype), Montagne de Kaw, PK 39 Arbre 6, 17–23.VIII.1994 (P.E. Roubad leg.) (MNHN). ***Raptrix occidentalis*** - PERU: 1 male, Cusco, Distrito de Echárata, 25.IV.1998 (R. Acosta leg.) (UNALM), 2 males & 1 female, Huánuco, Tingo María (Universidad Nacional Agraria de la Selva), 16.II.1992 (L. Gil leg.), 17.IV.1999, (L. Gil leg.), 15.X.2000, (Gómez leg.) (UNALM), 1 female, Tingo Maria, IV-V.1952 (P. Araoz leg.) (USNM), 1 male, Junín, Chanchamayo, Pampa Hermosa (Nueva Italia) (at light), 21.XII.2000, (Rivera & Oyague leg.) (UNALM), 3 males, Loreto, Boquerón del Padre Abad, 470m, 7-26.VII.1946 (F. Woytkowski leg.) (AMNH), 1 male, Loreto, Ucayali, Río Yarinacocha, 1.I.1954 (P. Hocking leg.) (FMNH), 2 males, Madre de Dios, Tambopata, Lago Sandoval, (at light), 1.X.2002, (J.

Rivera leg.), 1 male, Parque Nacional Bahuaja-Sonene, San Antonio, Rio Heath (21.VIII.2003) (D. Lucsic leg.) (UNALM). ***Raptrix perspicua*** - FRENCH GUIANA: 3 males (codes: GMSC004089, GMSC004079, GMSC004078), St. Jean du Maroni, N5.34232 W54o7148, 6–7.V.2011 elev.19.5m (at Light) (J. Rivera leg.) (CMNH), 6 males (codes: GMSC004077, GMSC004080, GMSC004075, GMSC004074, GMSC004073, GMSC004087), Vicinity of Cacao, 2–3.V.2011, N4.568 W52.46648, elev. 47.5m (at light) (J. Rivera leg.) (CMNH), 2 males (codes: GMSC004082, GMSC004088), Logging rd. S of Regina, 4.V.2001, N4.6217, W52.15288, elev.51.8m (at light) (J. Rivera leg.) (CMNH), 2 males (codes: GMSC004085, GMSC004083), Piste de St. Ellie, nr Sinnamary, N5.5.29747 W53.05205, 8–10.V.2011, elev. 40m (at light) (J. Rivera leg.) (CMNH), 1 male (code: GMSC004086), Kaw Mountain reserve, vicinity of Patawa, 30.IV/1.V.2011, N.456888 W52.21388, elev. 282m (at light) (J. Rivera leg.) (CMNH); GUYANA: 1 male (code: ROM905037), Rupununi Dist. Kurupukari, Essequibo R, E. side, blklight on road rainf. 200', 7.X.1990 (B. Hubley leg.) (ROM); VENEZUELA: El Dorado Km.107, Santa Helena BO 520m, 16.VIII.1957 (F. Fernandez & C.J. Rosales leg.) (ANSP).

Raptrix westwoodi - COLOMBIA: 1 male (syntype of *Acontista westwoodi*), “Colombia” (MHNG) (id. as *Raptrix westwoodi* by Lombardo & Marletta), 1 female (syntype of *Acontista westwoodi*; note: Lombardo & Marletta (2004) erroneously referred to this specimen as the holotype, otherwise the type series is made of at least 4 syntypes, two at BM and two at MHNG, see Roy & Cuche (2008)), “Colombie” (MHNG) (id. as *Raptrix westwoodi* by Lombardo & Marletta), 1 male, Valle Medellín, VIII–IX.1945, in fallow ground (F.L. Gallego leg.) (USNM), 1 male, Valle Medellín, IX.1944 (F.L. Gallego leg.) (USNM), 1 male, Valle Medellín, X.1946 (in fallow ground) (F.L. Gallego leg.) (labeled as *Raptrix perspicua*) (USNM), 1 female, “Colombia?” 11°8'S; 75°17'W, XI.1908 (C. Schunke leg.) (id. as *Acontista westwoodi*) (USNM). ***Tithrone latipennis*** - COLOMBIA: 1 male (holotype), Dept. El Valle, Anchicaya 300m 10–14.X.1970 (M. Descamps leg.) (MNHN); ECUADOR: 1 female (allotype), Prov. Napo, Rio Latas (“Ponce Loma Tumba Camillo”), II.1987 (K. Riede leg.) (MNHN). ***Tithrone roseipennis*** - FRENCH GUIANA: 1 female, Monte de Cacao, 3.IX.1976 (S. Boucher leg.) (MNHN); GUYANA: 1 female (holotype of *Acontista roseipennis*), “Guyana” (MNHN); TRINIDAD & TOBAGO: 1 male, Arima Valley, 12.V.1951 (W.I. Beebe leg.) (USNM), 1 male & 2 females, Trinidad Island, Arima Valley, 23.II–4.IV.1952 (W.I. Beebe leg.) (USNM), 1 male, Trinidad Island: Simla, Arima Valley, 1–5.II.1966 (S.S. & W.D. Duckworth leg.) (USNM), 1

male, Trinidad Island: St. Andrew, Aripo savanna, edaphic savanna, 8.I.1982 (O. Pellmyr leg.) (USNM), 1 female, Trinidad Island, Brickfield, 15.III.1947 (F. Wonder leg.) (FMNH); VENEZUELA: 1 male, Rancho Grande nr Maracay, I–IV.1946 (no collector data) (AMNH).

ACANTHOPIDAE

Acanthops bidens - MEXICO: 1 male (holotype; code: 8931), Sinaloa, Venvidio, 28.VIII.1918 (ANSP). ***Acanthops centralis*** - PANAMA: 1 male, Zona del Canal, Barro Colorado Island [at light], 18.VII.1980 (R. Silberglied & A. Aiello leg.) (USNM). ***Acanthops elegans*** - GUATEMALA: 3 males, Cayuga (one from IV.1915, and two from “may”) (Schaus & Barnes leg.) (USNM); MEXICO: 1 male (holotype), Guerrero, Acahuizotla, 750m. (MNHN) (id as *Acanthops elegans* by R. Roy). ***Acanthops erosa*** - BRAZIL: 1 female, Santarem, VI.1919 (S.M. Klages leg.) (CM); FRENCH GUIANA: 1 female, Maroni River, “july” (Wm. Schaus leg.) (USNM). ***Acanthops erosula*** - BOLIVIA: 1 male (code: GSMC003377), Santa Cruz nr. Buena Vista 17°39.100’S; 63°35.136W, 390m, 14.XI.2009 [at MV/UV lights in cut trees] (G.J. Svenson leg.) (CMNH); ECUADOR: 1 male, Napo, prov. Limoncocha, 8.VI.1977 (D.L. Vincent leg.) (USNM), 1 female, Guayaquil, [no date] (E.W. Rorer leg.) (USNM); PERU: 1 female, San Martín, Tarapoto, 21.IV.2000 (A. Cerna leg.) (UNALM) - ***Acanthops falcata*** - COLOMBIA: 1 female, Valley of Medellín, Antioquia, 7.XII.1940 (F.L. Gallego leg.) (USNM), 1 female, Casabe, 16.X.1957 (D.S.Bos leg.) (USNM); PANAMA: 1 female (paratype of *Acanthops griffinni*), Chiriqui (Rolle-Heyne leg.) (ZMB), 1 female, Alajuela, 28.V.1912 (A. Busck leg.) (id. as *Acanthops falcata* by M. Hebard) (USNM), 1 female, Tocumen, 5.I.1953 (R.P. Blanton leg.) (USNM); VENEZUELA: 1 female, El Valle, 1.X.1938 (C.H. Ballou leg.) (USNM), 1 male, AR, Rancho Grande 1100m, 8.VI.1965 (R.W. Poole leg.) (USNM), 1 female, Tama Tamma, upper Orinoco, 23.I.1930 (E.S. Hoit. Leg.) (USNM). ***Acanthops falcataria*** - BRAZIL: 1 male, Rio de Janeiro, 1935 (P. Sandig leg.) (USNM), 2 males, Santa Catarina, Corupa (Hansa Humboldt), III.1946 (A. Maller leg.) (AMNH), 1 male, Guapi, 3.III.1935 (P. Sandig leg.) (USNM). ***Acanthops fuscifolia*** - FRENCH GUIANA: 1 male, St. Laurent du Maroni, “june” (W.M.Schaus leg.) (USNM), 1 male, Cayenne, II.1904 (W.M.Schaus leg.) (USNM). ***Acanthops godmani*** - BELIZE: 1 male (holotype), “Belize”, 1894 Godman-Salvin (Blancaneaux leg.) (BM). ***Acanthops parafalcata*** - TRINIDAD & TOBAGO: 1 male (paratype) [Trinidad Island], near Port Spain, Maraval (at light), I.1978 (J. Boss leg.) (BM), 1 male, Trinidad Island, Balandra,

BWI, V.1955 (F.D. Bennett leg.) (USNM) (id. as *Acanthops falcata*) (USNM), 1 male, Trinidad Island, Arima Valley, BWI, 1.III.1951 (no collector) (USNM), 1 female, Trinidad Island, [no date] (A. Busck leg.) (USNM). ***Acanthops royi*** - ECUADOR: 1 male (holotype), Napo, Yasuni E.C.Y. 250m, 10.II.1997 (Dandrade leg.) (MNHN); PERU: 1 male, Huánuco, Tingo Maria (UNALM), 2 males, Madre de Dios, Tambopata-CICRA, 2010 (M. Deza leg.) (UNALM), 1 male, Madre de Dios, Tambopata [canopy fogging project], 1982 (T. Erwin leg.) (USNM). ***Acanthops soukana*** - FRENCH GUIANA: 1 female (holotype), Route de Kaw. Pk36, “Battage” IX.1999 (F. Luquet leg.) (MNHN). ***Decimiana bolivari*** - BRAZIL: 1 female (syntype of *Acanthops bolivari*), Bahia (MNCN), 1 male, Bahia, Fortaleza, VII.1945 (M.T. Dalmat leg.) (USNM) (id. as *Acanthops brunnerii* by A.B.G.) (USNM), 1 male, Bahia, Salvador, 18.III.1985 (V.V. Luz leg.) (id. as *Acanthops falcataria* by L.J. Jantsch leg.) (MCTP), 1 female, Ceará [no date] (F. Rocha leg.) (USNM); PARAGUAY: 1 female (syntype of *Acanthops bolivari*), “Paraguay central” (Oberthür Leg.) (MNCN), 1 female, Cordillera dpto., Caacupe, 17.II.1981 (R.D. Cavo leg.) (USNM), 1 male, Sapucay [no date] (W.T. Foster leg.) (USNM). ***Decimiana rehni*** - ARGENTINA: 1 female (holotype of *Plesiakanthops rehni*), Gran Chaco (MNHN), 1 male, Córdoba, Capilla del Monte, 3.XII.1997 (F. Lombardo leg.) (UNALM). ***Decimiana tessellata*** - BRAZIL: 1 male, Goiás, Mineiros, nr. P.N. das Emas, VII–VIII.2003 (S. Astete leg.) (UNALM), 2 males, Paraná, Arapoti, XII.1944 (A. Maller leg.) (AMNH), 1 male, Santa Catarina, Corupa (Hansa Humboldt), VIII.1946 (A. Maller leg.) (AMNH); PARAGUAY: 2 females, Sapucay, 15.II.1901 & 2.X.1903 (W.T. Foster leg.) (AMNH), 1 female, Sapucay “Nov.” (W.T. Foster leg.) (USNM), 1 female, Cacupé [no date], “44-1624” (H.L. Parker leg.) (USNM). ***Lagrecacanthops guyanensis*** - GUYANA: 1 male (paratype), New River mark 82, 1300–2000ft, 1–12.V.1938 (C.A. Hudson leg.) (BM). ***Metilia amazonica*** - BRAZIL: 1 male (holotype), Amazonas, Gurupá-Almeirim, 29.V.1927 (Zerny leg.) (NHMW); FRENCH GUIANA, 2 males (code: GSMC004116), Piste de St. Ellie, nr Sinnamary, N5.29747 W53.05205, 48m [at light], 8–10.V.2011 (J. Rivera leg.) (CMNH). ***Metilia adusta*** - PERU: 1 male (holotype of *Acanthops adusta*), Iquitos (Amazonas), IV.9.2.(1) (ZMG), 1 male, Junín, Chanchamayo, 9.III.1985 (G. Arellano leg.) (UNALM), 1 male, Junín, Chanchamayo, 1.IX.1986 (G. Arellano leg.) (UNALM), 1 female, Pozuzo, reared from egg case (C. Mendoza leg.) (UNALM). ***Metilia cf. brunnerii*** - BRAZIL: 1 male & 1 female, Tefé, XI.1924 (H. Bassler leg.) (AMNH). ***Metilia* sp. (c.f. *amazonica*)** - PERU: 1 male, San Martín, Uchiza, 5.IX.1999 (M. Viera leg.) (UNALM), 1 male,

Huánuco, Tingo María, IV.V–1952 (P. Araoz leg.) (USNM). *Metilia* spp. (likely undescribed species) - BRAZIL: 1 male, Brazil: Belém, 1973 (Sanved) (USNM); PANAMA: 1 male, Barro Colorado, Canal Zone, 14.II.1964 (no collector data) (USNM), 1 female, Barro Colorado, Canal Zone, 13.VII.1979 (C. Philips leg.) (USNM); PERU: 1 female, Huánuco, Tingo María, 4.II.2002 (V. Sánchez leg.) (UNALM), 1 male, Huánuco, Tingo María, IV.V–1952 (P. Araoz leg.) (USNM), 1 male, Madre de Dios, R.N. Tambopata - Lago Sandoval, 20.IX.2000 (R. Acosta leg.) (UNALM), 1 female, Lower Rio Marañón, 13.VIII.1924 (H. Bassler leg.) (AMNH).

Plesiacanthops eseejja - PERU: 1 male (holotype), Madre de Dios, R[eserva] N[acional] Tambopata (lago Sandoval), 31.X.2000 (R. Acosta leg.) (UNALM), 1 female (paratype), Madre de Dios, Puerto Maldonado, Tambopata (vicinity of Posada Amazonas), 28.IX.2001 (M. Deza leg.) (UNALM). *Plesiacanthops occidentalis* - PERU: 1 male, Madre de Dios, Tambopata, canopy-fogging project, 1992 (T. Erwin leg.) (USNM). *Plesiacanthops lombardoi* - ECUADOR: 1 female (holotype), Puerto Napo, 13.VIII.1984 (MNHN), 1 female, Napo, Shangri-la lodge [on vegetation], 15.IX.2008 (J. Rivera & L. Pinault leg.) (UNALM). *Plesiacanthops tuberculata* - FRENCH GUIANA: 1 male, Mgnes de la Trinité, Inselberg II/station météo., 20.IV.1998 (A. François leg.) (MNHN), 1 female, Les Nouragues (F1 ex ovo), 8.X.2002, 4°5'N; 52°41'W, 19.X.2003 (C. Schwarz leg.) (MNHN). *Pseudacanthops caelebs* - MEXICO: 1 male (holotype), Orizaba (MHNG). *Pseudacanthops clorindae* - BRAZIL: 1 male (paratype), Amazonas, IPIXUNA, Rio Gregorio, Lago Grande (07°10'11.7"S; 70°49'10.3"W (J.A. Rafael *et al.* leg.) (INPA); PERU: 2 males (paratypes), Cuzco, distrito de Echarate, 5.VIII.1998 (R. Acosta leg.) (UNALM). *Pseudacanthops lobipes* - BOLIVIA: 1 male (holotype), Region Chapare 400m (Zischka leg.) (ZSM), 1 male (code: GMSC000254), Dpto. Santa Cruz, Reserva Natural Potrerillo del Guenda, 17°40.281'S; 63°27.451'W, 400m, 3–9.XI.2009 at MV.UV lights, gen coll. (G.J. Svenson leg.) (CMNH). PERU: 1 male, [Loreto]. Rio Tapiche, 9.XII.1923 (AMNH), 1 female, Madre de Dios, Tambopata-Pakitz, canopy fogging project, 28.X.1982, Terra Firme forest (T. Erwin, leg.) (USNM), 1 male, Manu, Pakitz 11°56'S; 71°18'W 250m, 8-9.X.1989 nr campgr. MV light#7 (D. Adamski & M. Epetein leg.) (USNM), 1 male, Pasco, Oxapampa, P.N. Yanachaga-Chemillen, zona Huampal 1000m. (resting on a tree branch), VIII.2010 (J. Rivera leg.) (UNALM). *Pseudacanthops spinulosus* - FRENCH GUIANA: 1 male (code: GMSC3375), [no additional data] (H. Romak leg.) (CMNH), 1 male (code: GSMC003722), Kaw Mtn. Res. 290m, 1Km W Amazonia, 04.34.08-52.12.37, 1–11.II.2008 (H. Romack leg.) (CMNH);

MEXICO: 1 male, Veracruz, Los Tuxtlas area "Las Cabanas", 3-15.V.1981 (C.M. & S. Flint Jr. leg.) (USNM).